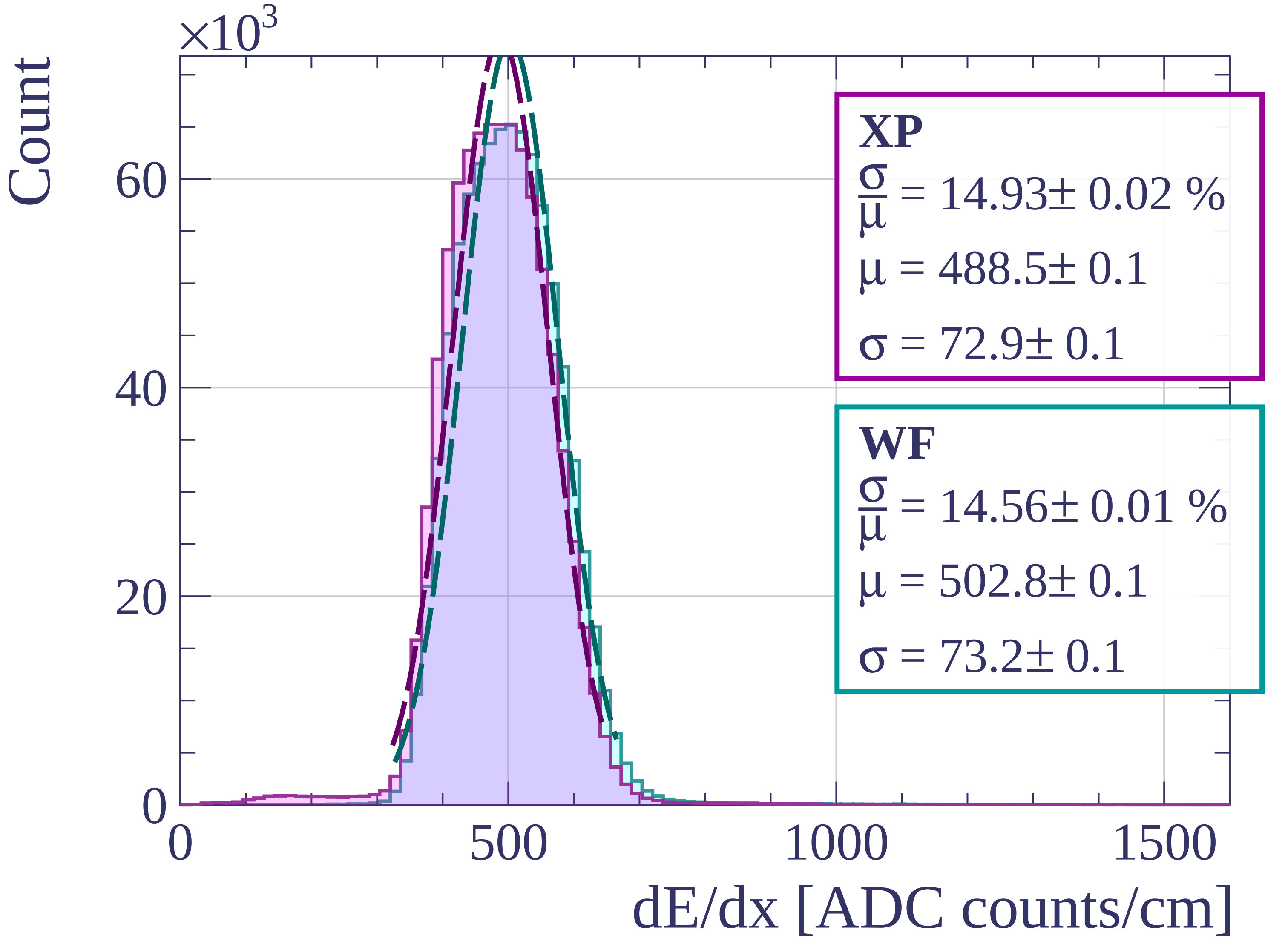
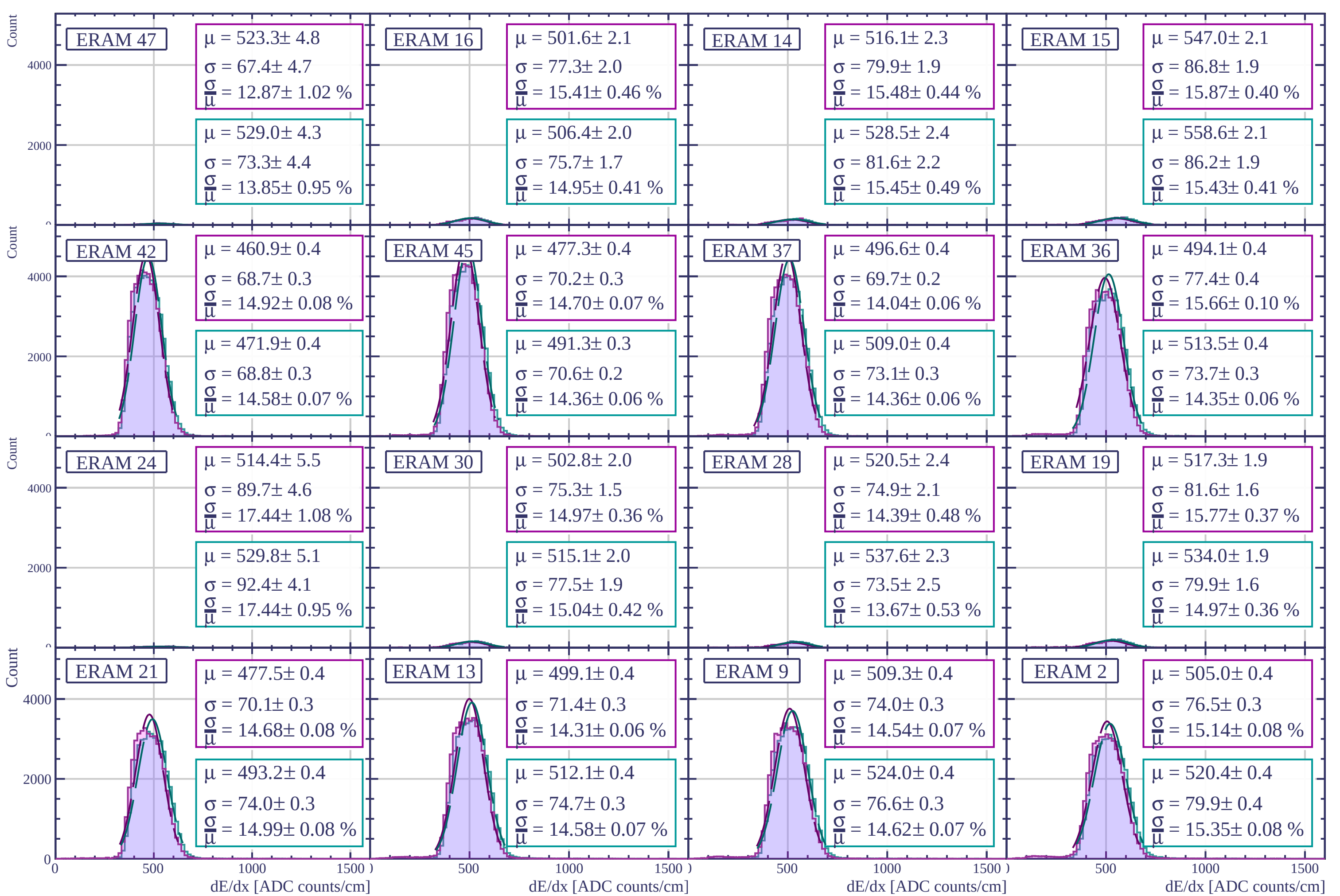
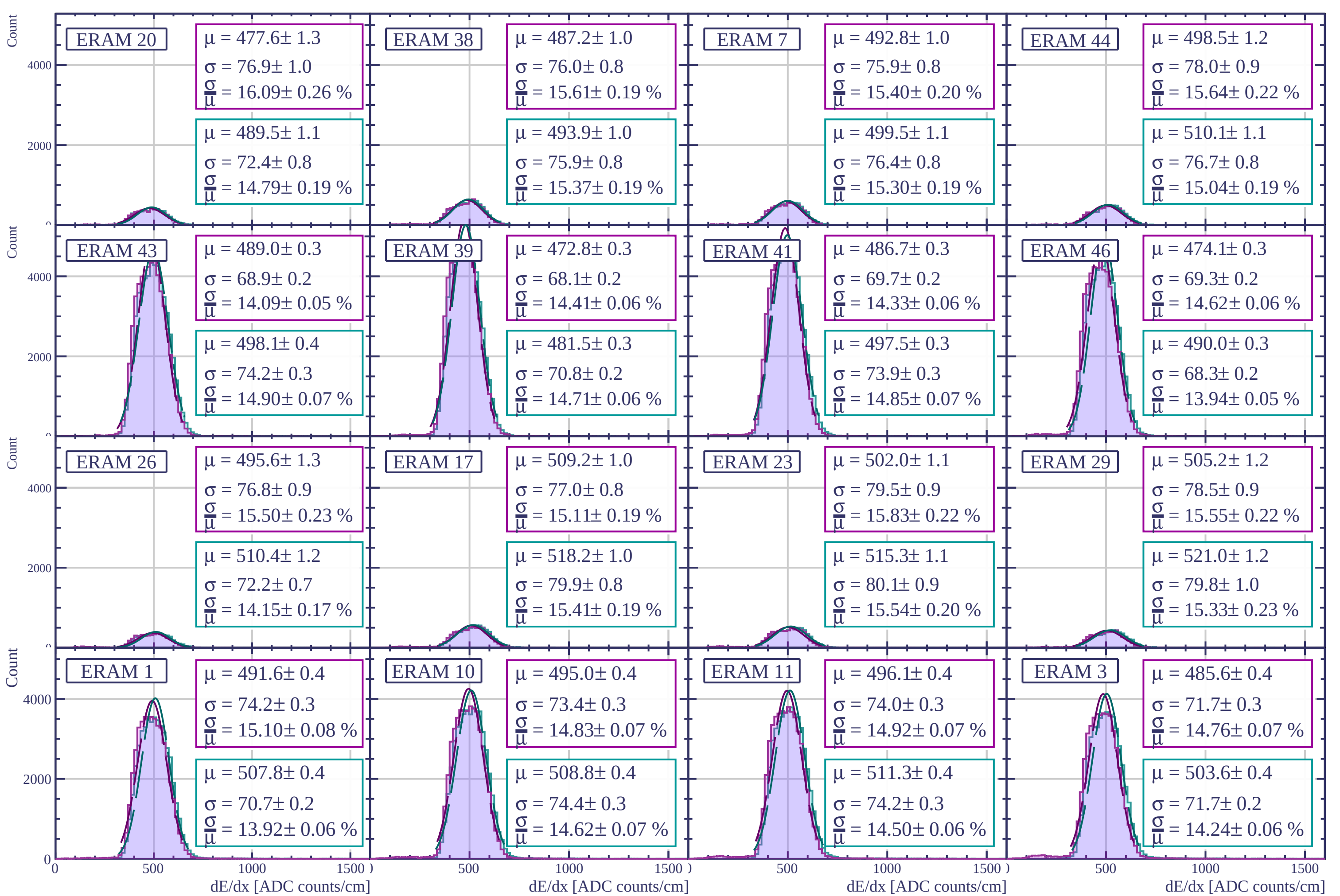
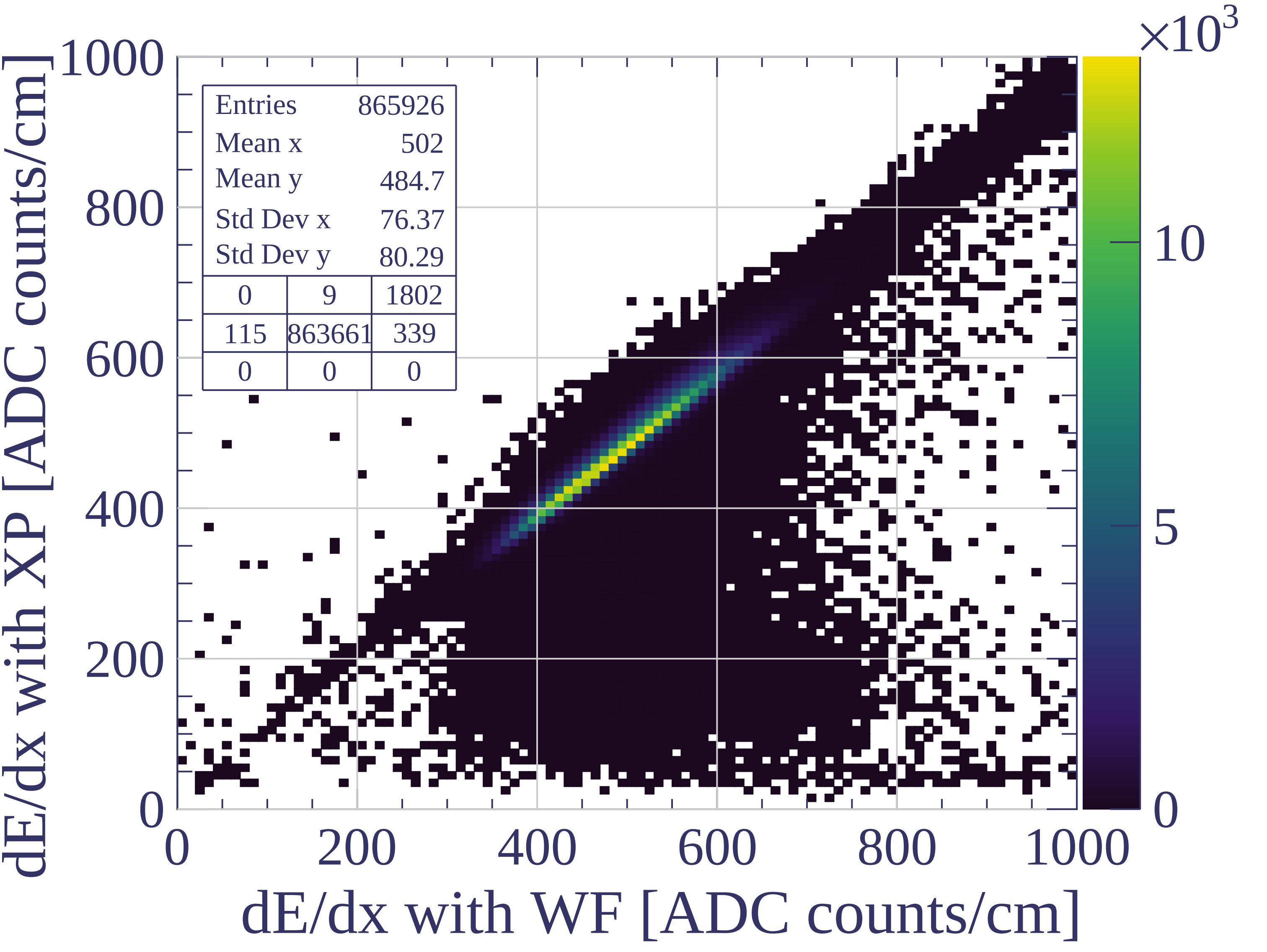
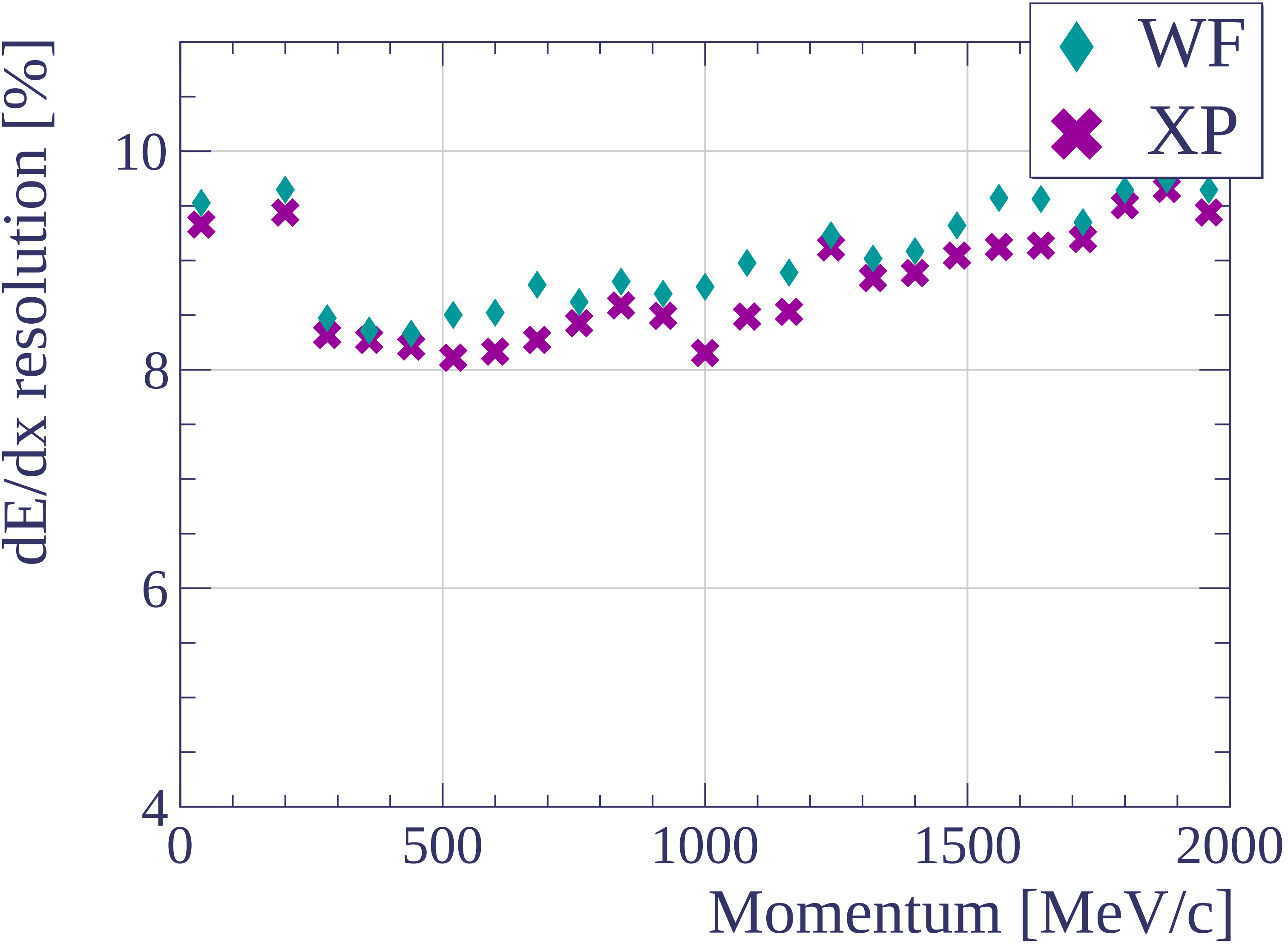


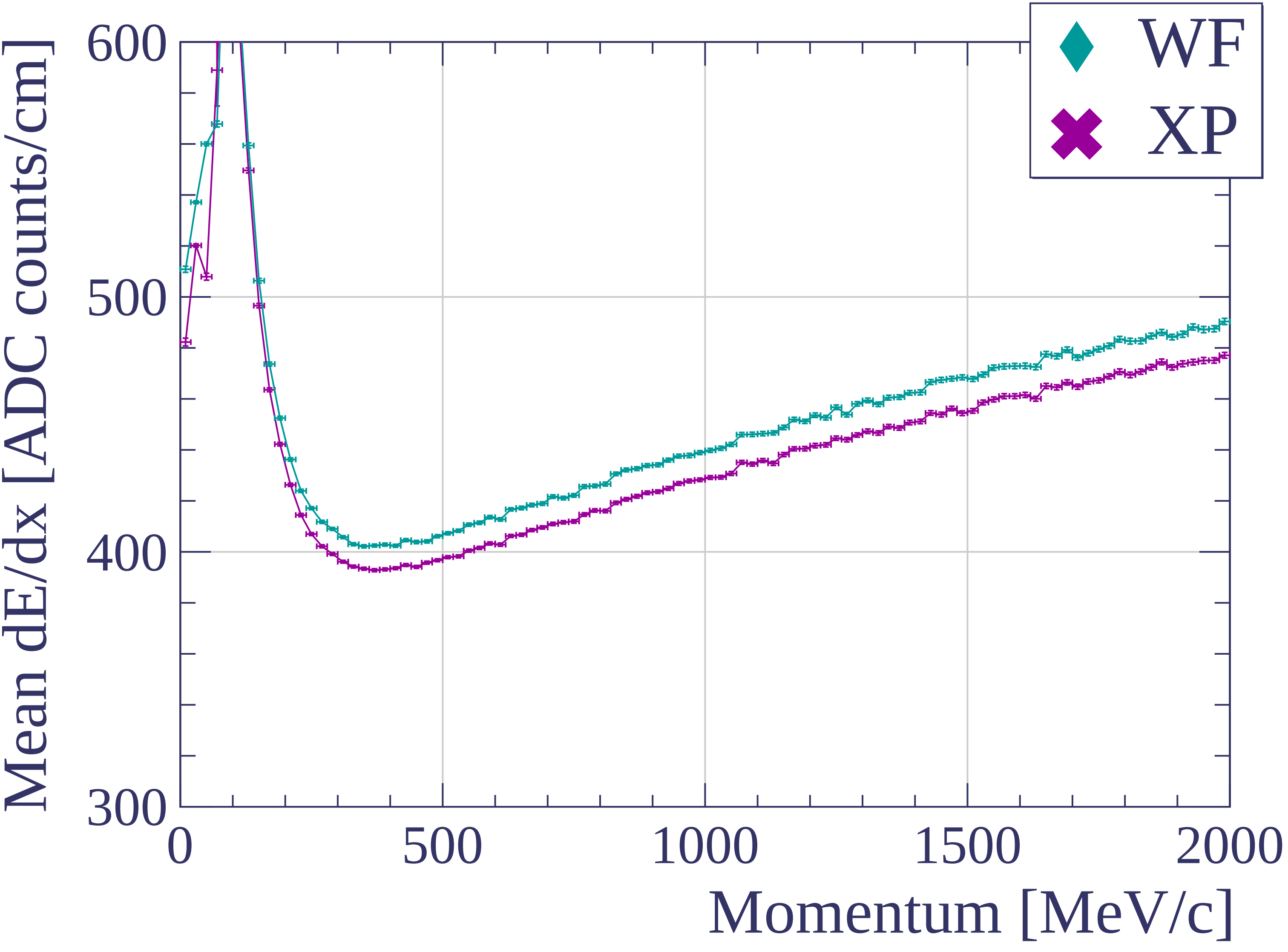
Count
×10^3
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 14.93± 0.02 %
μ = 488.5± 0.1
σ = 72.9± 0.1
WF
σ/μ = 14.56± 0.01 %
μ = 502.8± 0.1
σ = 73.2± 0.1


ERAM 47
μ = 523.3± 4.8
σ = 67.4± 4.7
σ/μ = 12.87± 1.02 %
μ = 529.0± 4.3
σ = 73.3± 4.4
σ/μ = 13.85± 0.95 %
ERAM 16
μ = 501.6± 2.1
σ = 77.3± 2.0
σ/μ = 15.41± 0.46 %
μ = 506.4± 2.0
σ = 75.7± 1.7
σ/μ = 14.95± 0.41 %
ERAM 14
μ = 516.1± 2.3
σ = 79.9± 1.9
σ/μ = 15.48± 0.44 %
μ = 528.5± 2.4
σ = 81.6± 2.2
σ/μ = 15.45± 0.49 %
ERAM 15
μ = 547.0± 2.1
σ = 86.8± 1.9
σ/μ = 15.87± 0.40 %
μ = 558.6± 2.1
σ = 86.2± 1.9
σ/μ = 15.43± 0.41 %
ERAM 42
μ = 460.9± 0.4
σ = 68.7± 0.3
σ/μ = 14.92± 0.08 %
μ = 471.9± 0.4
σ = 68.8± 0.3
σ/μ = 14.58± 0.07 %
ERAM 45
μ = 477.3± 0.4
σ = 70.2± 0.3
σ/μ = 14.70± 0.07 %
μ = 491.3± 0.3
σ = 70.6± 0.2
σ/μ = 14.36± 0.06 %
ERAM 37
μ = 496.6± 0.4
σ = 69.7± 0.2
σ/μ = 14.04± 0.06 %
μ = 509.0± 0.4
σ = 73.1± 0.3
σ/μ = 14.36± 0.06 %
ERAM 36
μ = 494.1± 0.4
σ = 77.4± 0.4
σ/μ = 15.66± 0.10 %
μ = 513.5± 0.4
σ = 73.7± 0.3
σ/μ = 14.35± 0.06 %
ERAM 24
μ = 514.4± 5.5
σ = 89.7± 4.6
σ/μ = 17.44± 1.08 %
μ = 529.8± 5.1
σ = 92.4± 4.1
σ/μ = 17.44± 0.95 %
ERAM 30
μ = 502.8± 2.0
σ = 75.3± 1.5
σ/μ = 14.97± 0.36 %
μ = 515.1± 2.0
σ = 77.5± 1.9
σ/μ = 15.04± 0.42 %
ERAM 28
μ = 520.5± 2.4
σ = 74.9± 2.1
σ/μ = 14.39± 0.48 %
μ = 537.6± 2.3
σ = 73.5± 2.5
σ/μ = 13.67± 0.53 %
ERAM 19
μ = 517.3± 1.9
σ = 81.6± 1.6
σ/μ = 15.77± 0.37 %
μ = 534.0± 1.9
σ = 79.9± 1.6
σ/μ = 14.97± 0.36 %
ERAM 21
μ = 477.5± 0.4
σ = 70.1± 0.3
σ/μ = 14.68± 0.08 %
μ = 493.2± 0.4
σ = 74.0± 0.3
σ/μ = 14.99± 0.08 %
ERAM 13
μ = 499.1± 0.4
σ = 71.4± 0.3
σ/μ = 14.31± 0.06 %
μ = 512.1± 0.4
σ = 74.7± 0.3
σ/μ = 14.58± 0.07 %
ERAM 9
μ = 509.3± 0.4
σ = 74.0± 0.3
σ/μ = 14.54± 0.07 %
μ = 524.0± 0.4
σ = 76.6± 0.3
σ/μ = 14.62± 0.07 %
ERAM 2
μ = 505.0± 0.4
σ = 76.5± 0.3
σ/μ = 15.14± 0.08 %
μ = 520.4± 0.4
σ = 79.9± 0.4
σ/μ = 15.35± 0.08 %
Count
0
2000
4000
0
500
1000
1500
dE/dx [ADC counts/cm]

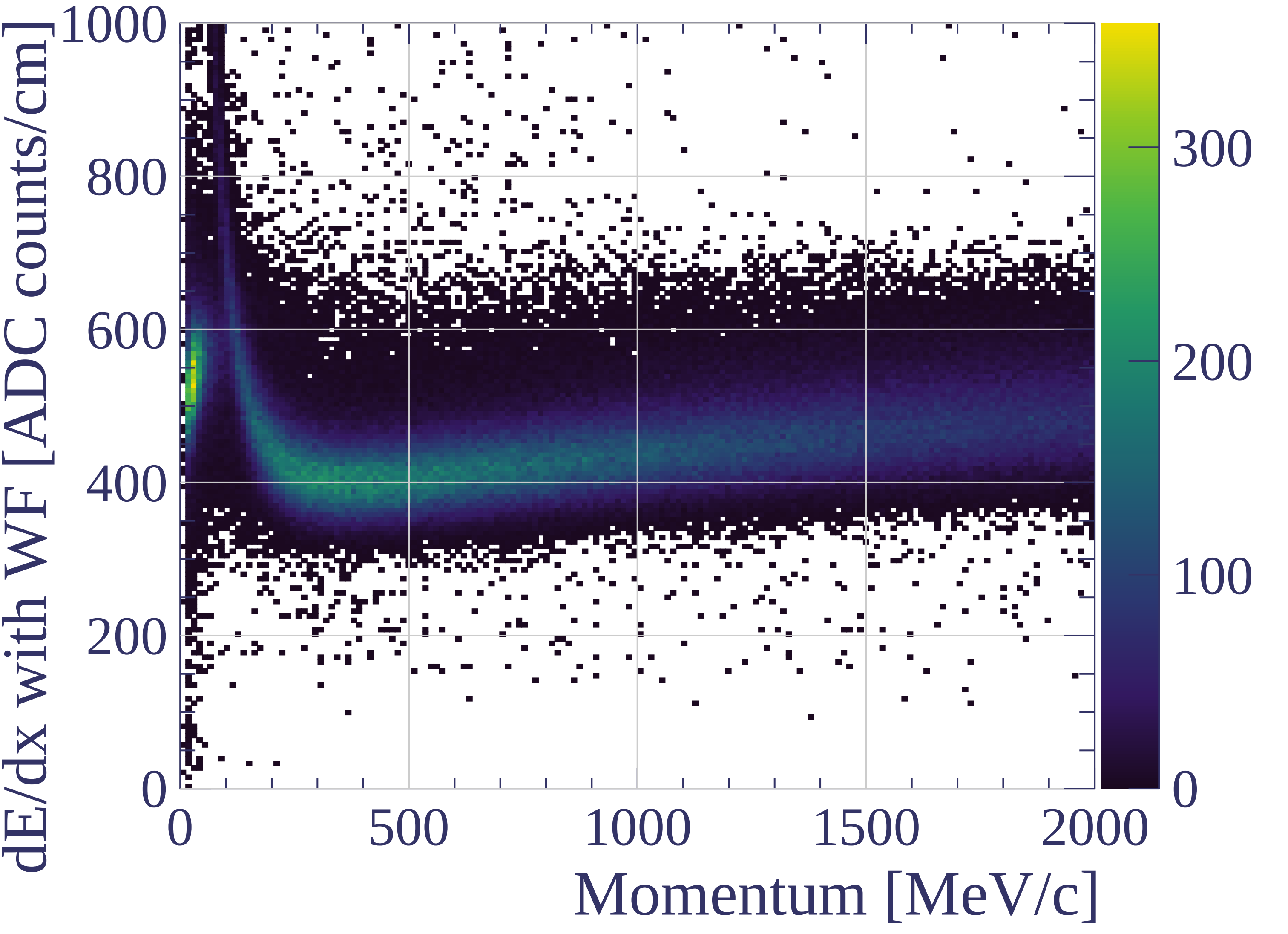

Count
ERAM 20
μ = 477.6± 1.3
σ = 76.9± 1.0
σ/μ = 16.09± 0.26 %
μ = 489.5± 1.1
σ = 72.4± 0.8
σ/μ = 14.79± 0.19 %
ERAM 38
μ = 487.2± 1.0
σ = 76.0± 0.8
σ/μ = 15.61± 0.19 %
μ = 493.9± 1.0
σ = 75.9± 0.8
σ/μ = 15.37± 0.19 %
ERAM 7
μ = 492.8± 1.0
σ = 75.9± 0.8
σ/μ = 15.40± 0.20 %
μ = 499.5± 1.1
σ = 76.4± 0.8
σ/μ = 15.30± 0.19 %
ERAM 44
μ = 498.5± 1.2
σ = 78.0± 0.9
σ/μ = 15.64± 0.22 %
μ = 510.1± 1.1
σ = 76.7± 0.8
σ/μ = 15.04± 0.19 %
ERAM 43
μ = 489.0± 0.3
σ = 68.9± 0.2
σ/μ = 14.09± 0.05 %
μ = 498.1± 0.4
σ = 74.2± 0.3
σ/μ = 14.90± 0.07 %
ERAM 39
μ = 472.8± 0.3
σ = 68.1± 0.2
σ/μ = 14.41± 0.06 %
μ = 481.5± 0.3
σ = 70.8± 0.2
σ/μ = 14.71± 0.06 %
ERAM 41
μ = 486.7± 0.3
σ = 69.7± 0.2
σ/μ = 14.33± 0.06 %
μ = 497.5± 0.3
σ = 73.9± 0.3
σ/μ = 14.85± 0.07 %
ERAM 46
μ = 474.1± 0.3
σ = 69.3± 0.2
σ/μ = 14.62± 0.06 %
μ = 490.0± 0.3
σ = 68.3± 0.2
σ/μ = 13.94± 0.05 %
ERAM 26
μ = 495.6± 1.3
σ = 76.8± 0.9
σ/μ = 15.50± 0.23 %
μ = 510.4± 1.2
σ = 72.2± 0.7
σ/μ = 14.15± 0.17 %
ERAM 17
μ = 509.2± 1.0
σ = 77.0± 0.8
σ/μ = 15.11± 0.19 %
μ = 518.2± 1.0
σ = 79.9± 0.8
σ/μ = 15.41± 0.19 %
ERAM 23
μ = 502.0± 1.1
σ = 79.5± 0.9
σ/μ = 15.83± 0.22 %
μ = 515.3± 1.1
σ = 80.1± 0.9
σ/μ = 15.54± 0.20 %
ERAM 29
μ = 505.2± 1.2
σ = 78.5± 0.9
σ/μ = 15.55± 0.22 %
μ = 521.0± 1.2
σ = 79.8± 1.0
σ/μ = 15.33± 0.23 %
ERAM 1
μ = 491.6± 0.4
σ = 74.2± 0.3
σ/μ = 15.10± 0.08 %
μ = 507.8± 0.4
σ = 70.7± 0.2
σ/μ = 13.92± 0.06 %
ERAM 10
μ = 495.0± 0.4
σ = 73.4± 0.3
σ/μ = 14.83± 0.07 %
μ = 508.8± 0.4
σ = 74.4± 0.3
σ/μ = 14.62± 0.07 %
ERAM 11
μ = 496.1± 0.4
σ = 74.0± 0.3
σ/μ = 14.92± 0.07 %
μ = 511.3± 0.4
σ = 74.2± 0.3
σ/μ = 14.50± 0.06 %
ERAM 3
μ = 485.6± 0.4
σ = 71.7± 0.3
σ/μ = 14.76± 0.07 %
μ = 503.6± 0.4
σ = 71.7± 0.2
σ/μ = 14.24± 0.06 %
4000
2000
0
500
1000
1500
dE/dx [ADC counts/cm]

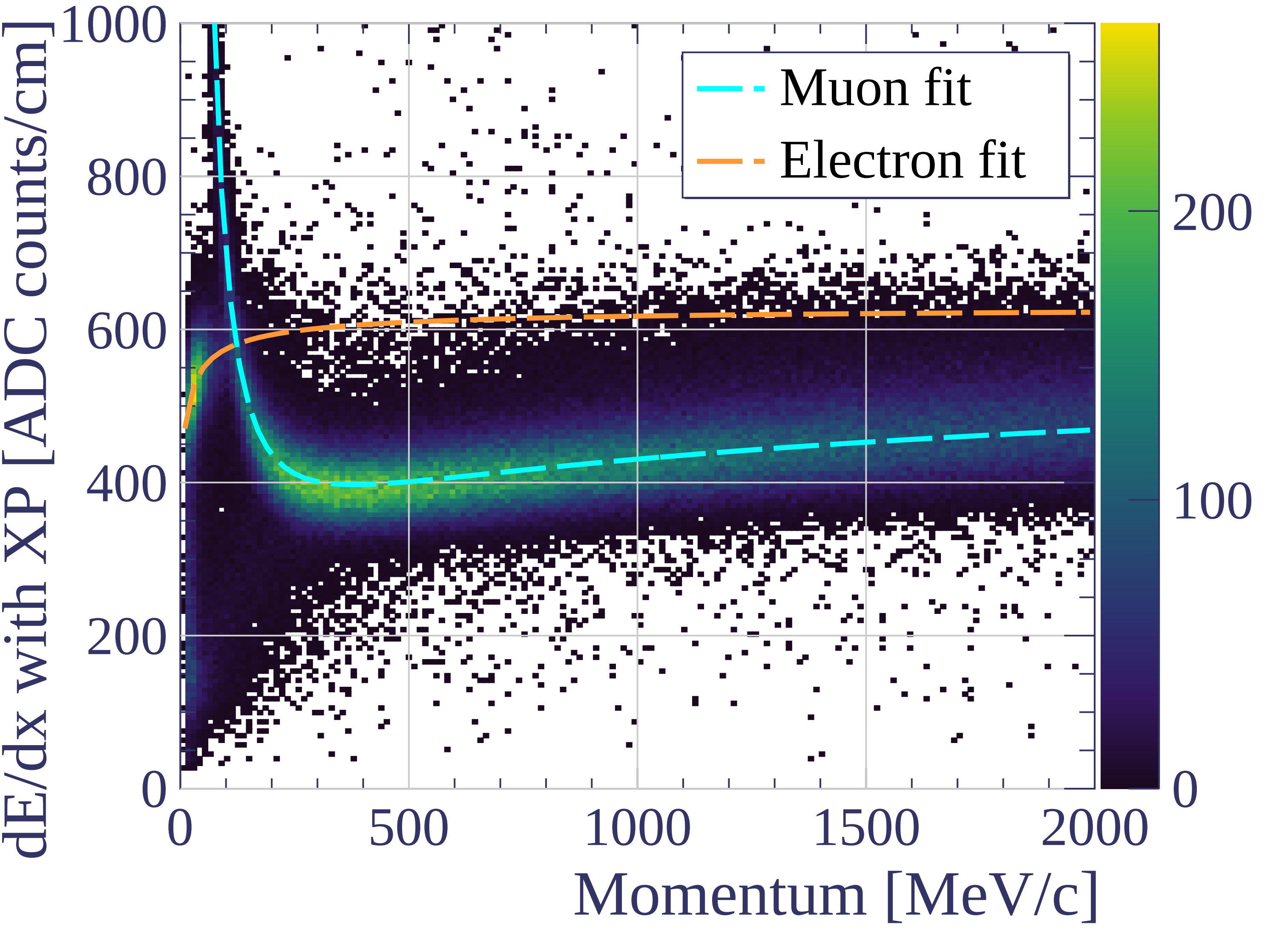

Entries 865926
Mean x 502
Mean y 484.7
Std Dev x 76.37
Std Dev y 80.29
0 9 1802
115 863661 339
0 0 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
×10^3
0
5
10

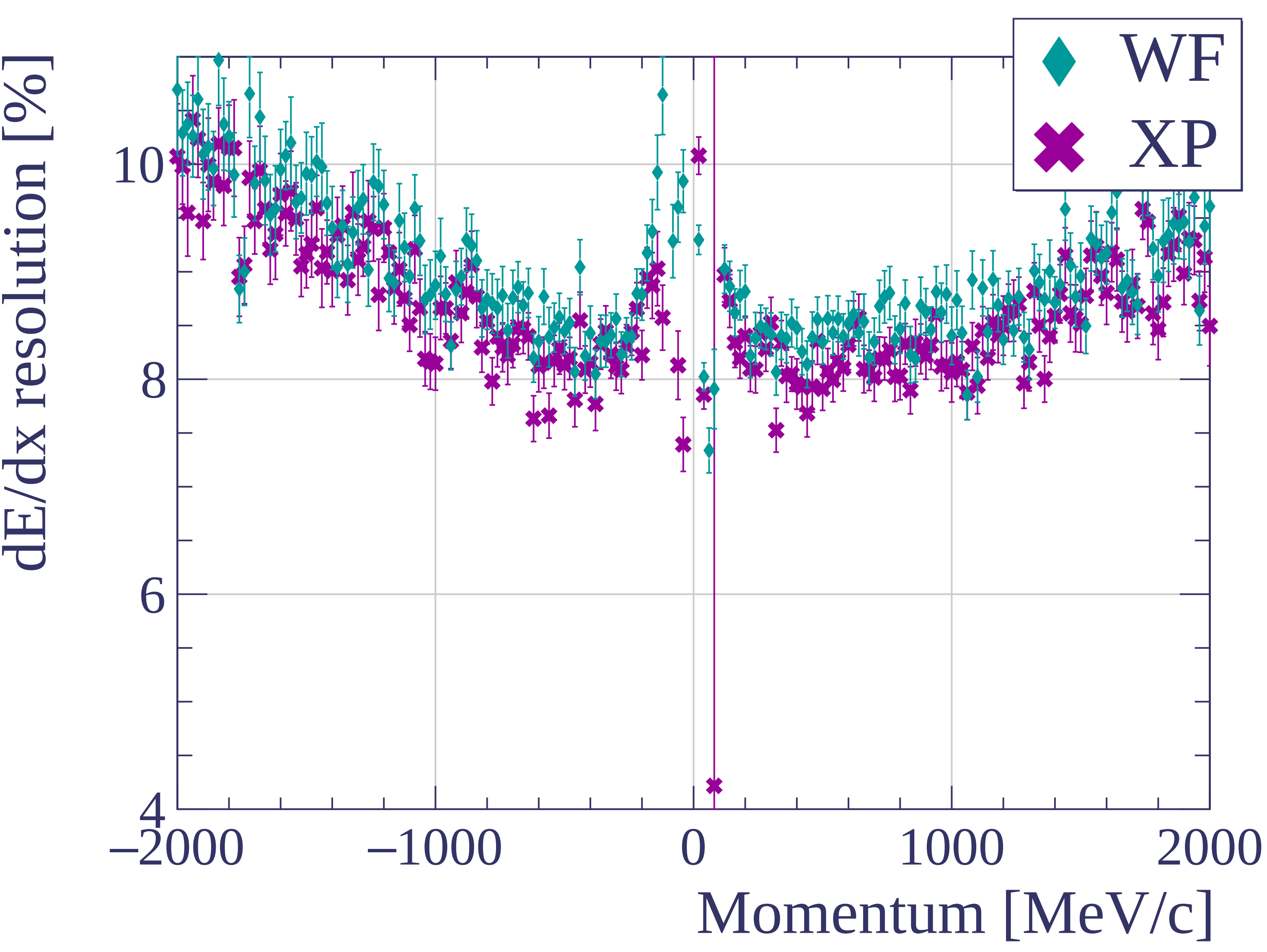

WF
XP
dE/dx resolution [%]
Momentum [MeV/c]
4
6
8
10
0
500
1000
1500
2000



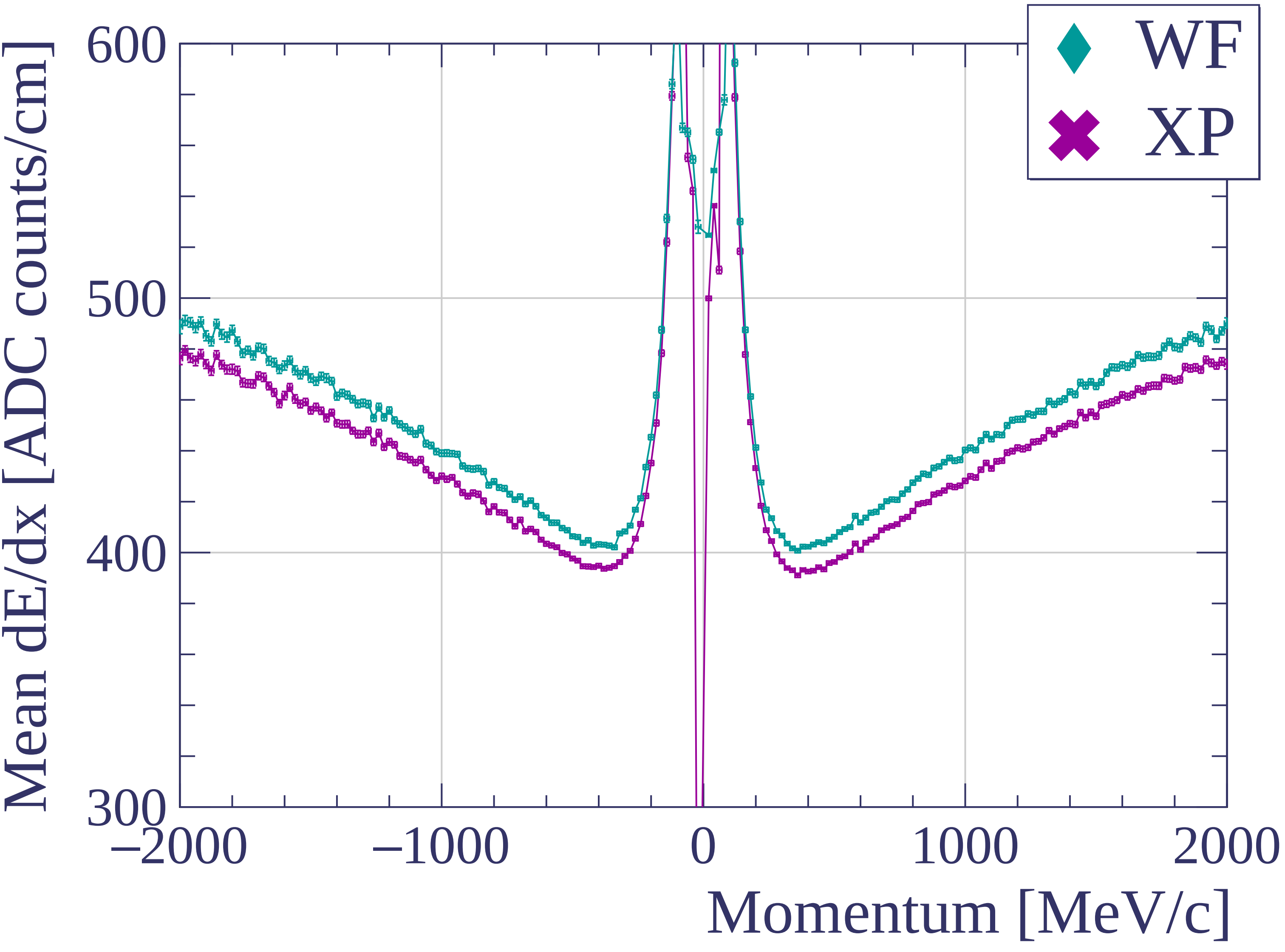
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
0
500
1000
1500
2000
Momentum [MeV/c]

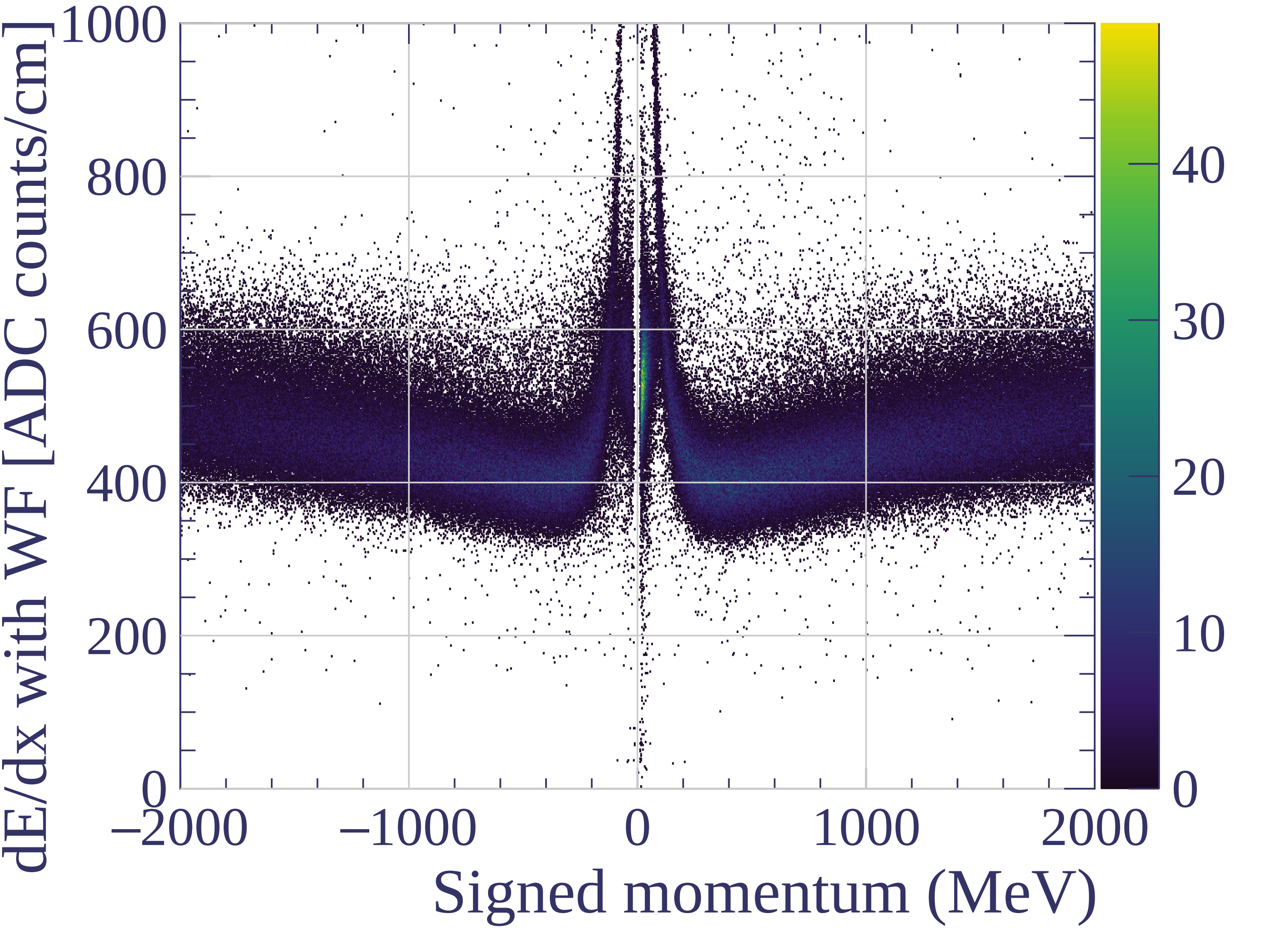

dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
300
200
100
0


Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1500
2000
100

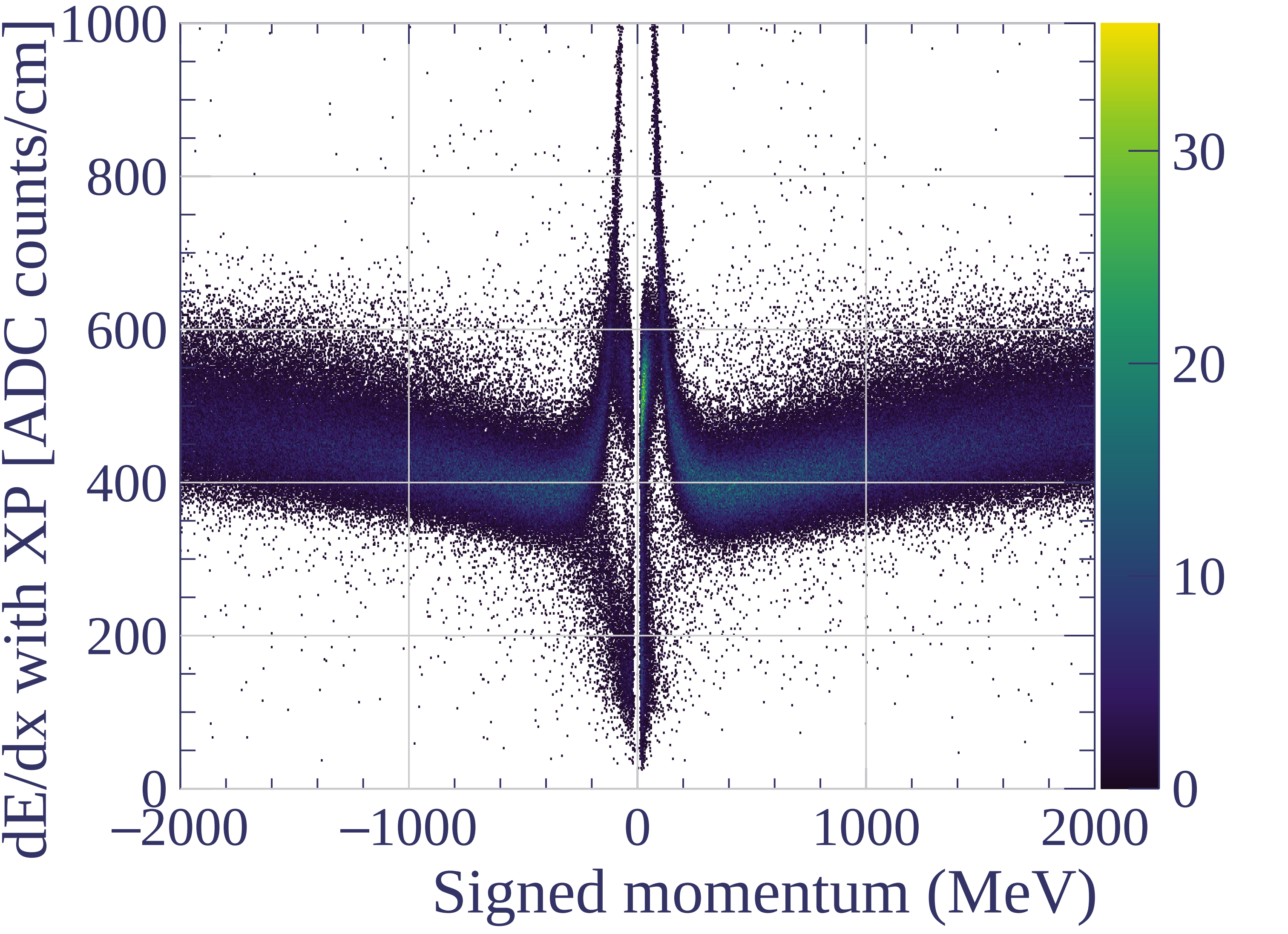


WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]

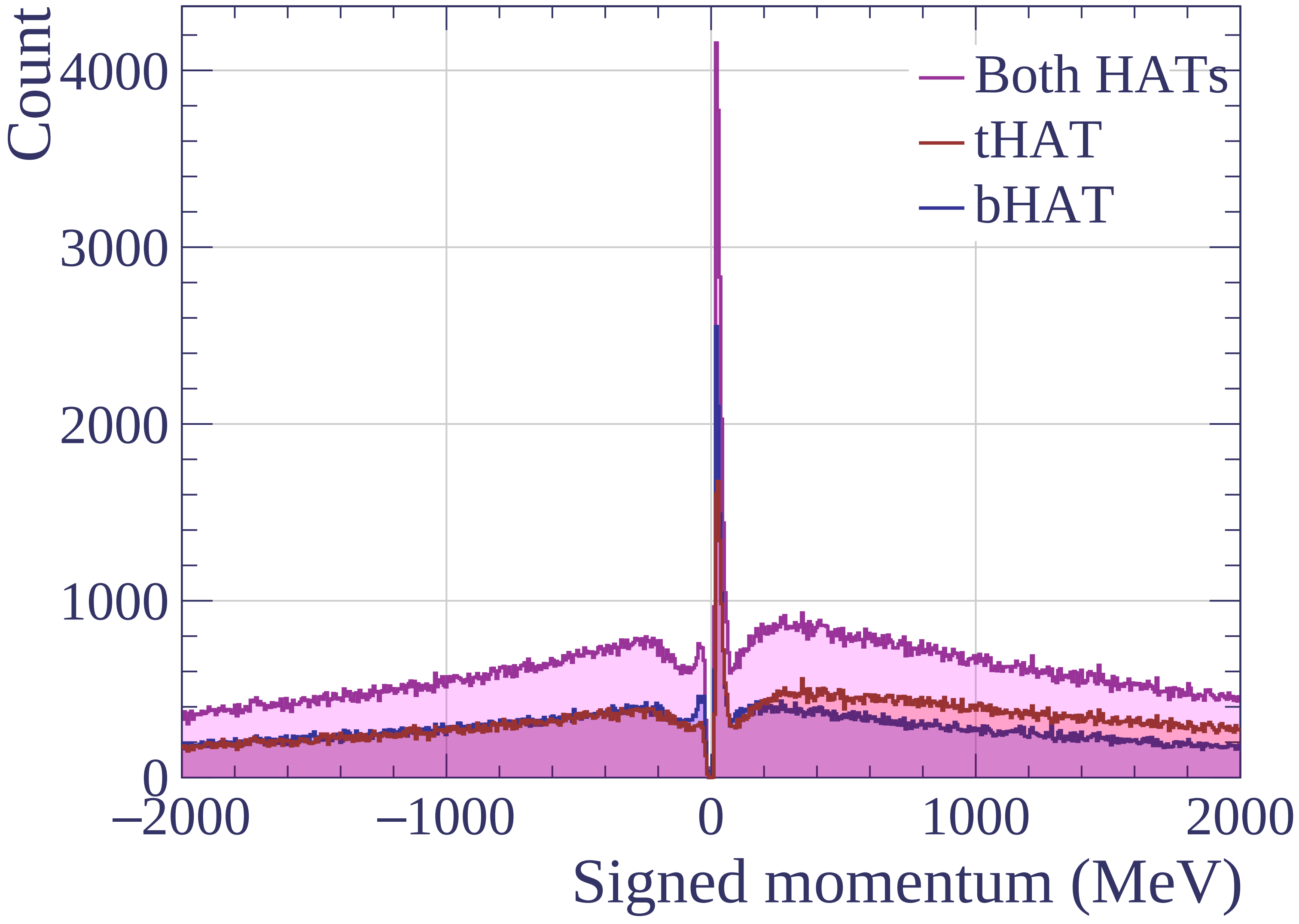

WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−2000
−1000
0
1000
2000
Momentum [MeV/c]



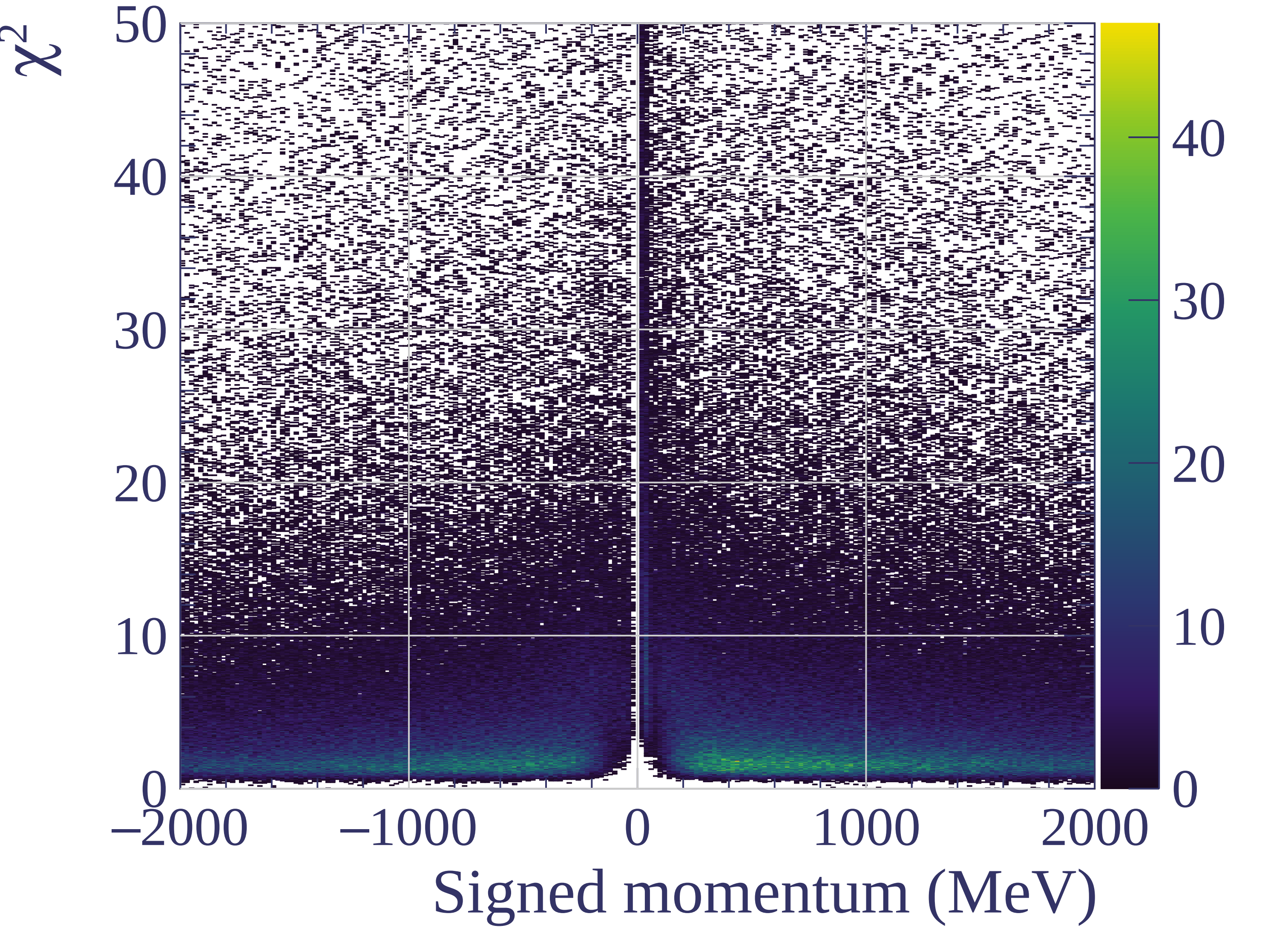
dE/dx with WF [ADC counts/cm]
Signed momentum (MeV)
0
200
400
600
800
1000
−2000
−1000
0
1000
2000
0
10
20
30
40

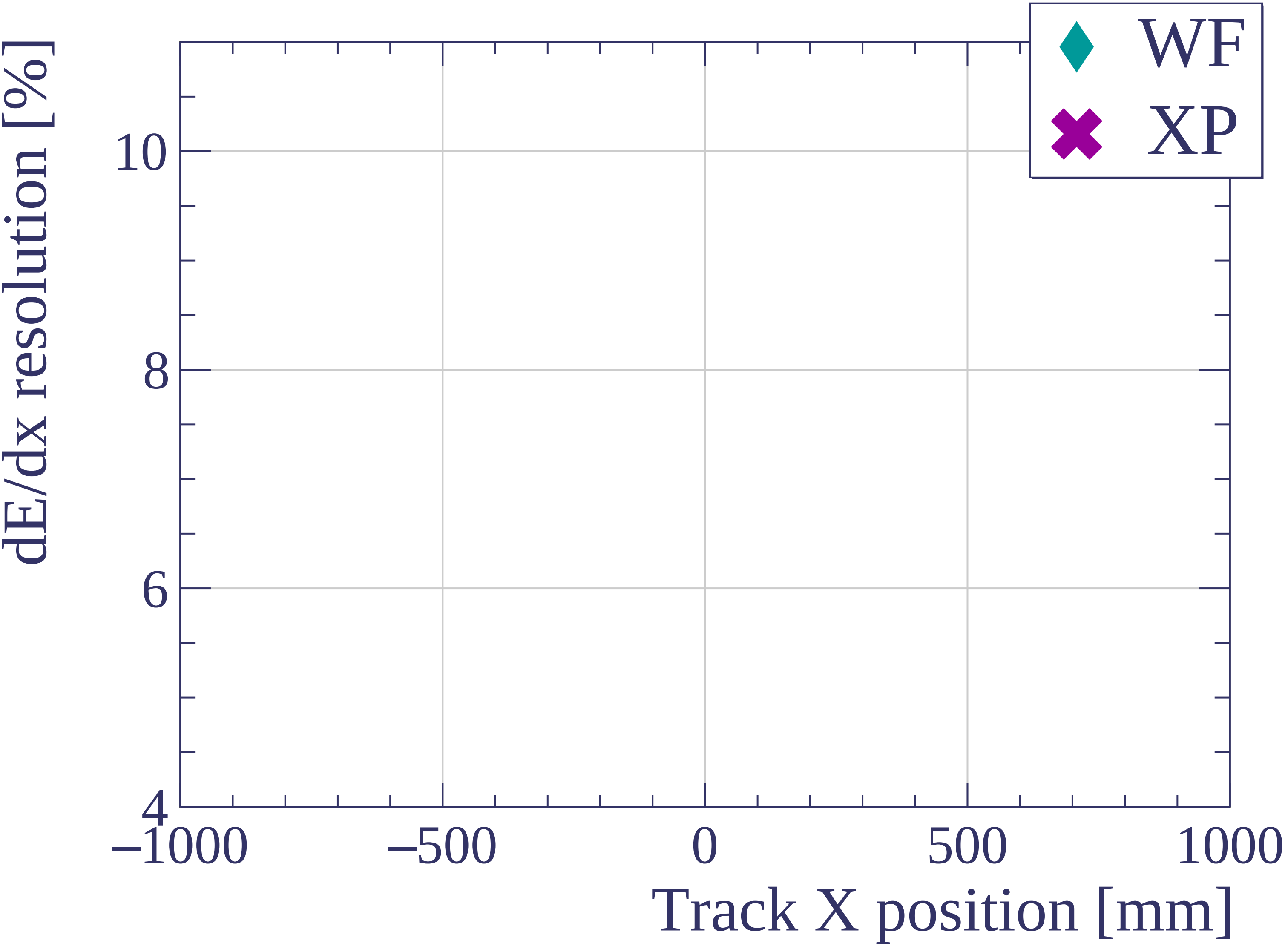


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
30
20
10
0

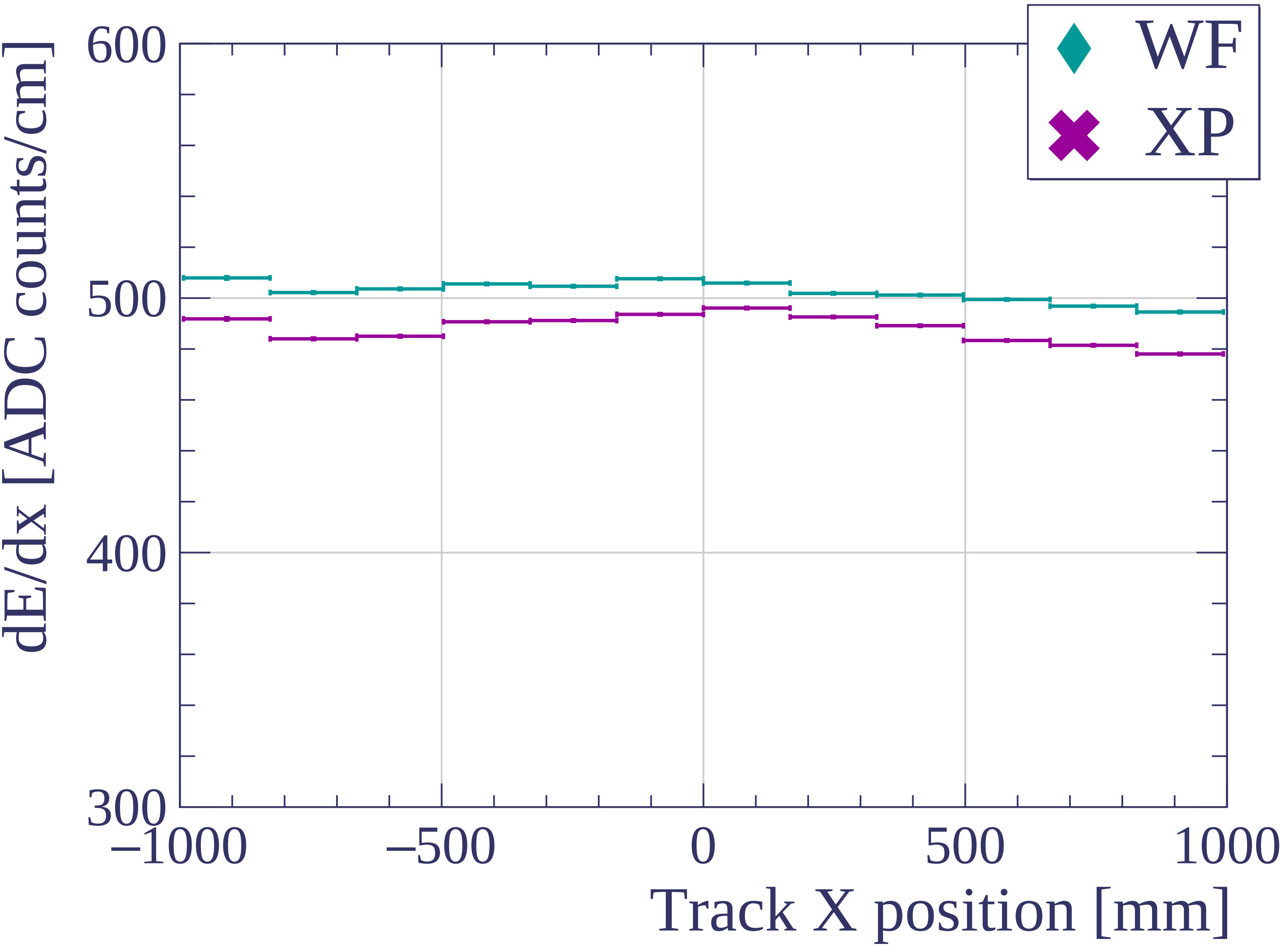


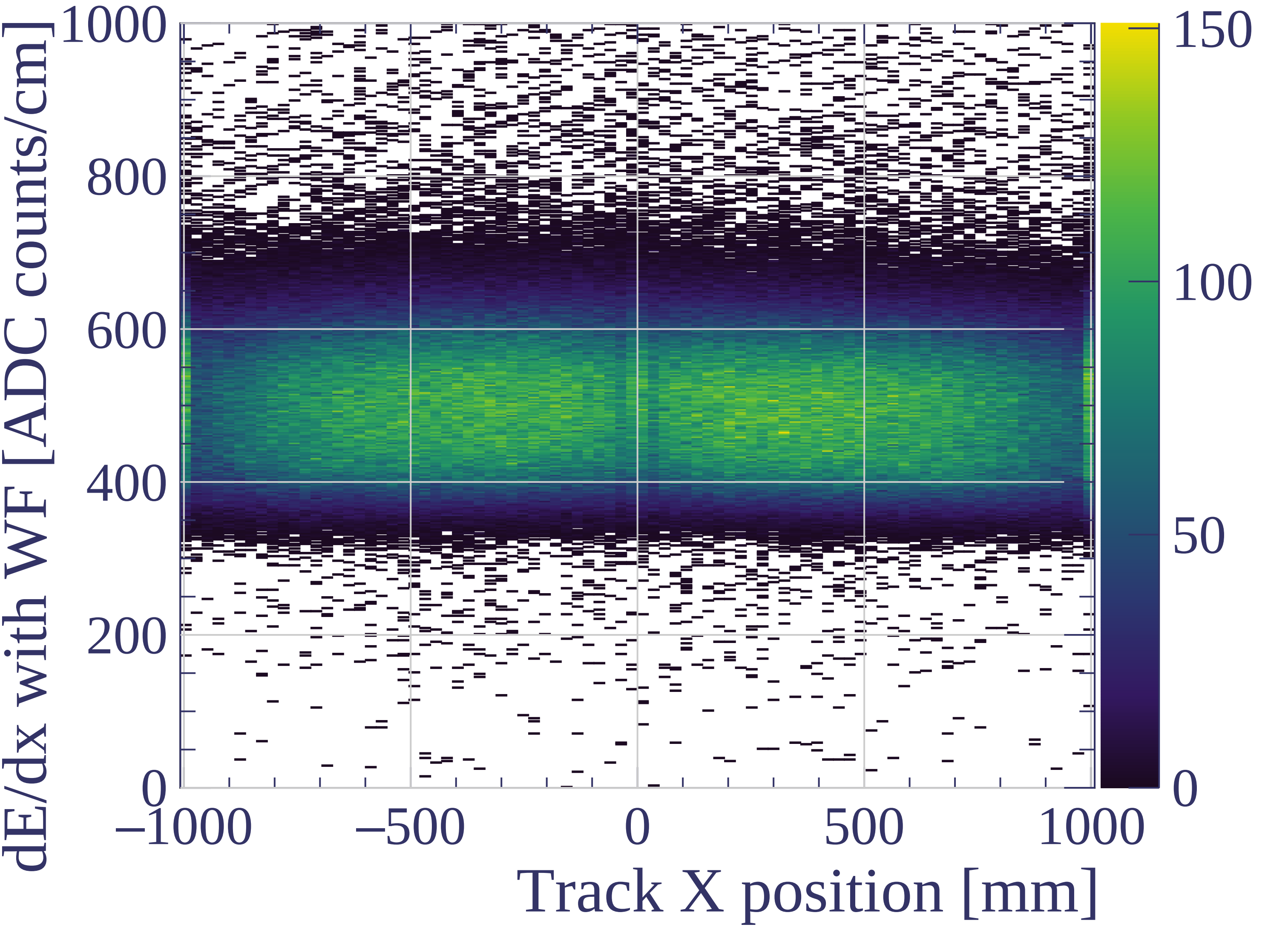
Count
4000
3000
2000
1000
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
Both HATs
tHAT
bHAT

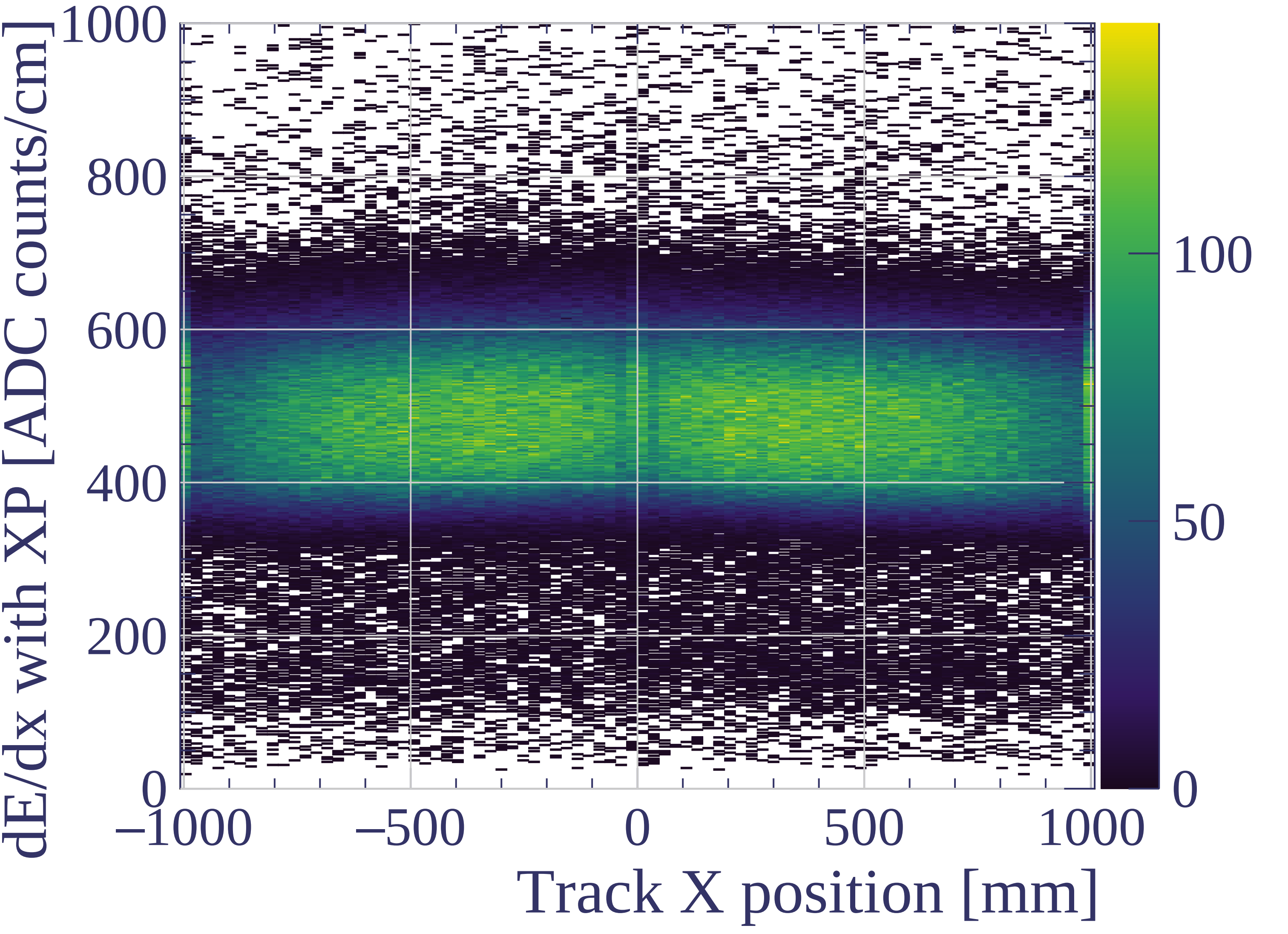

χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
40
30
20
10
0



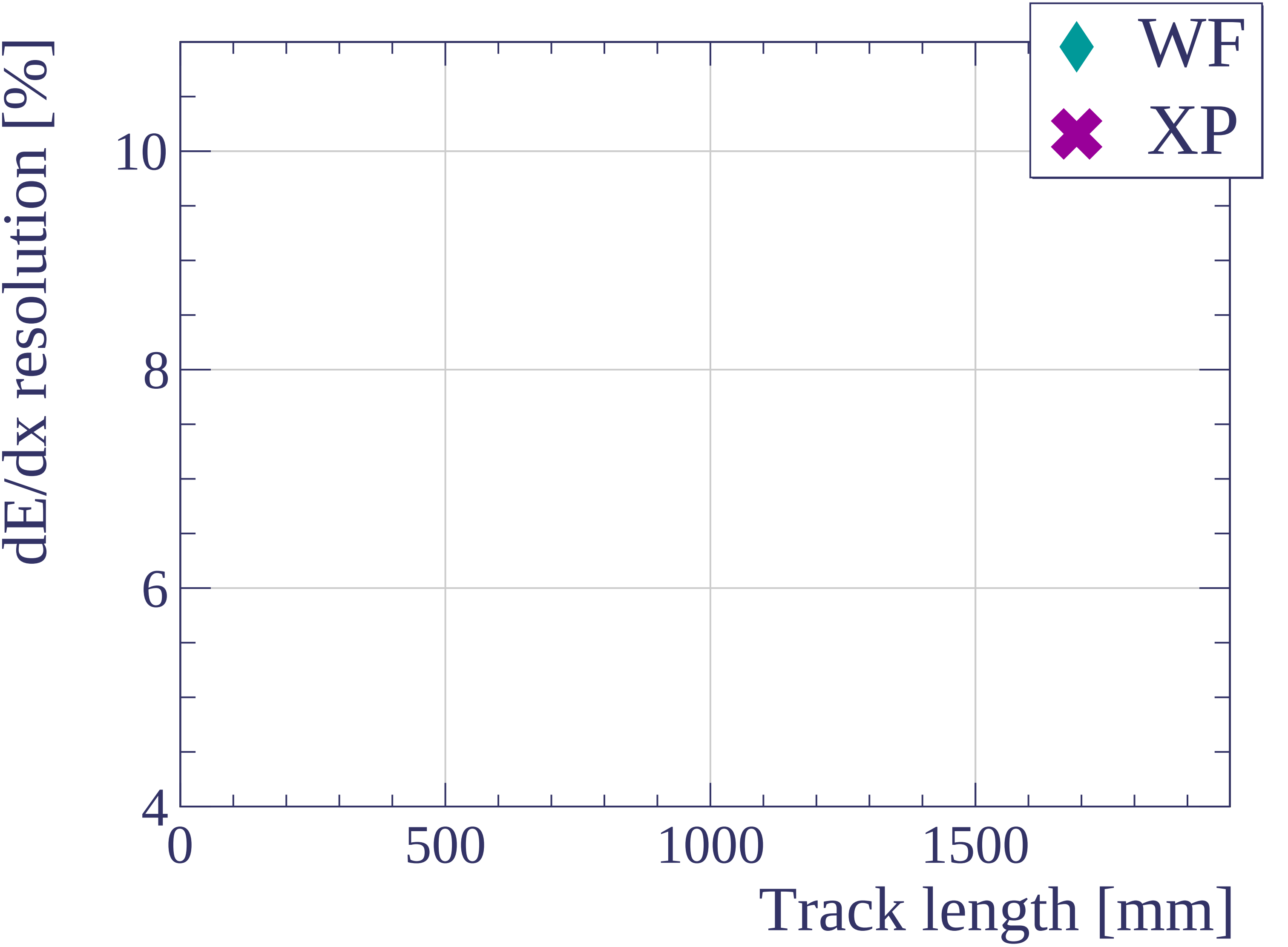
WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]


WF
XP
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]

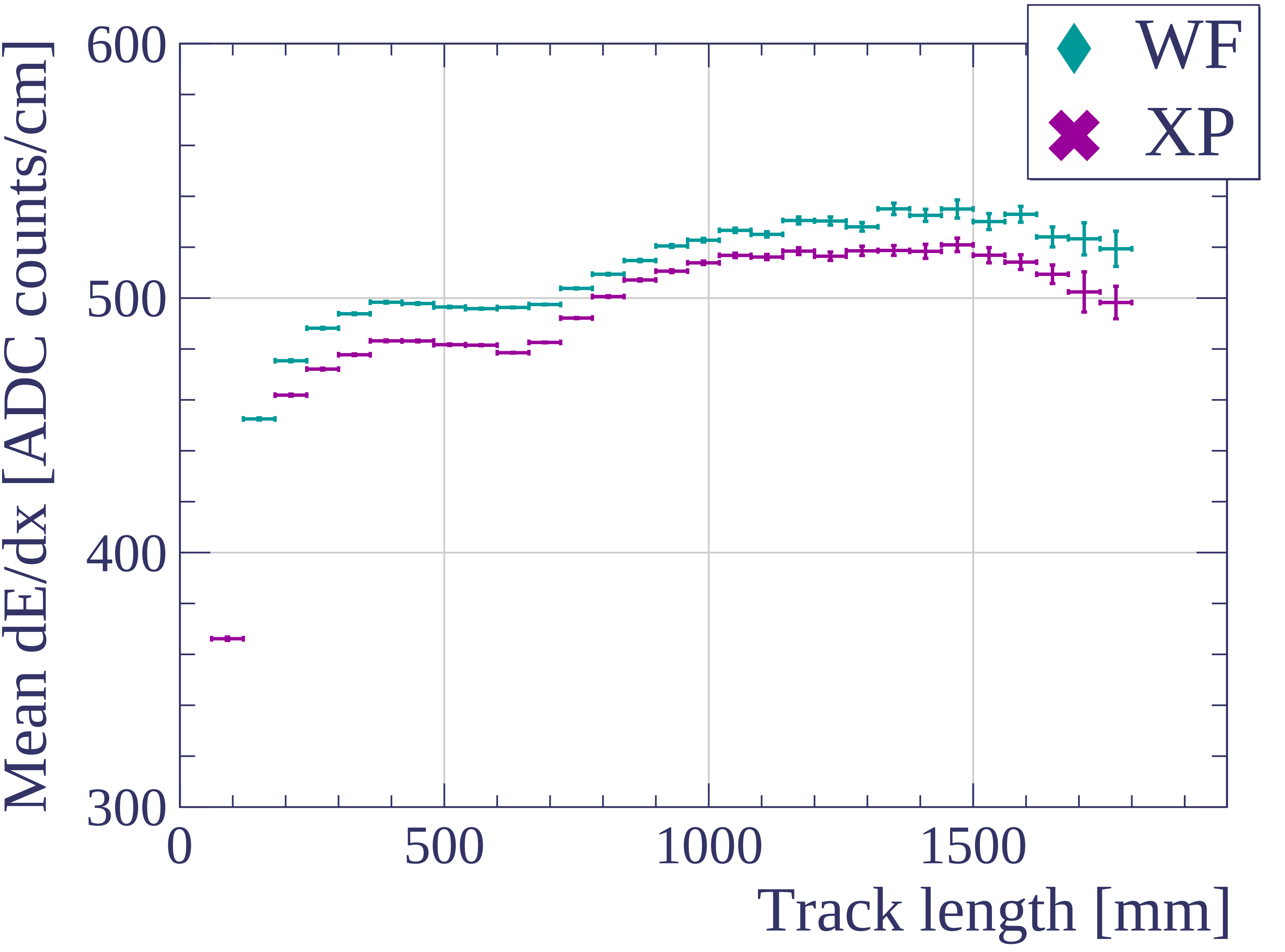

dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
150
100
50
0

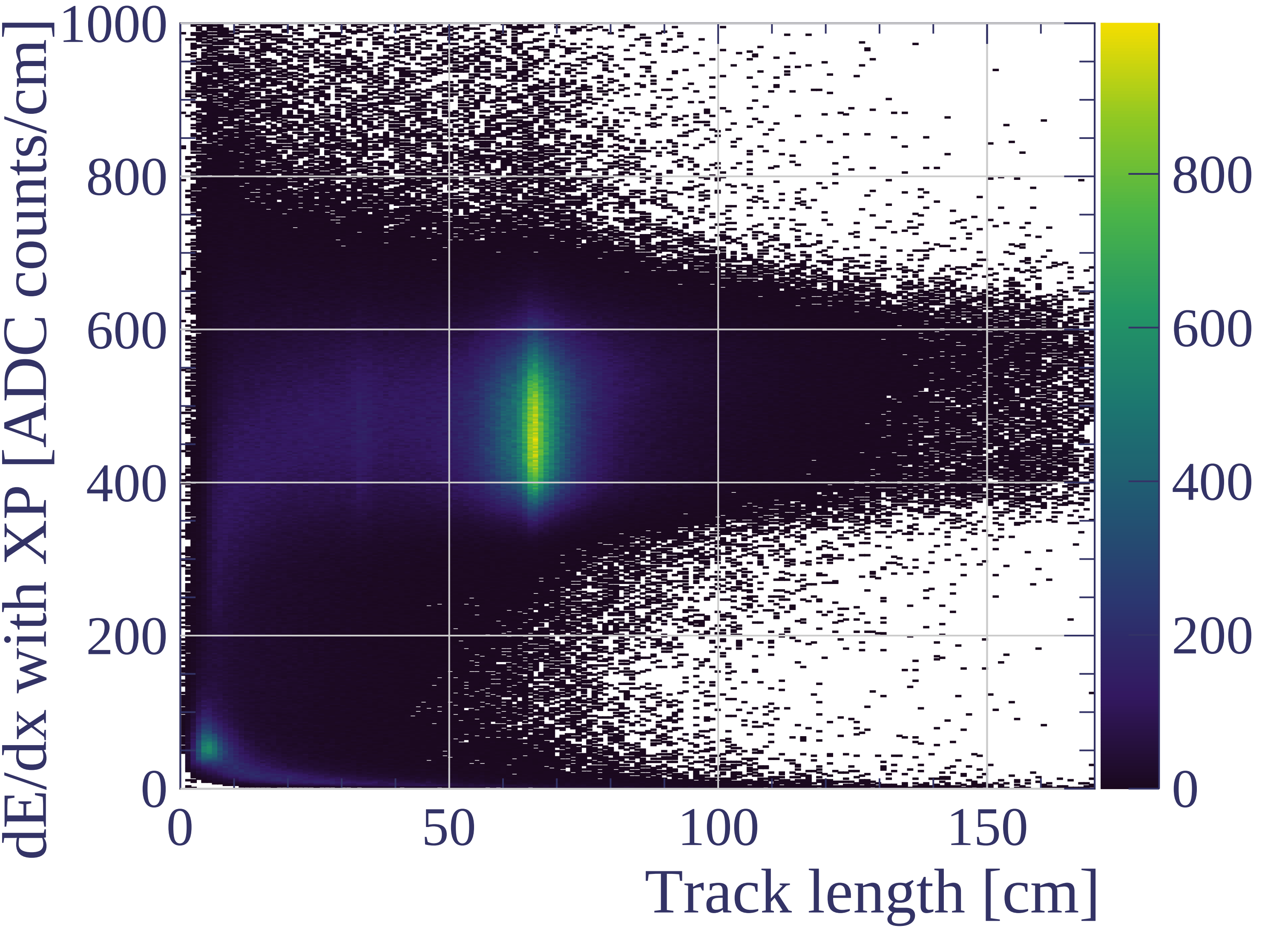


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
100
50
0

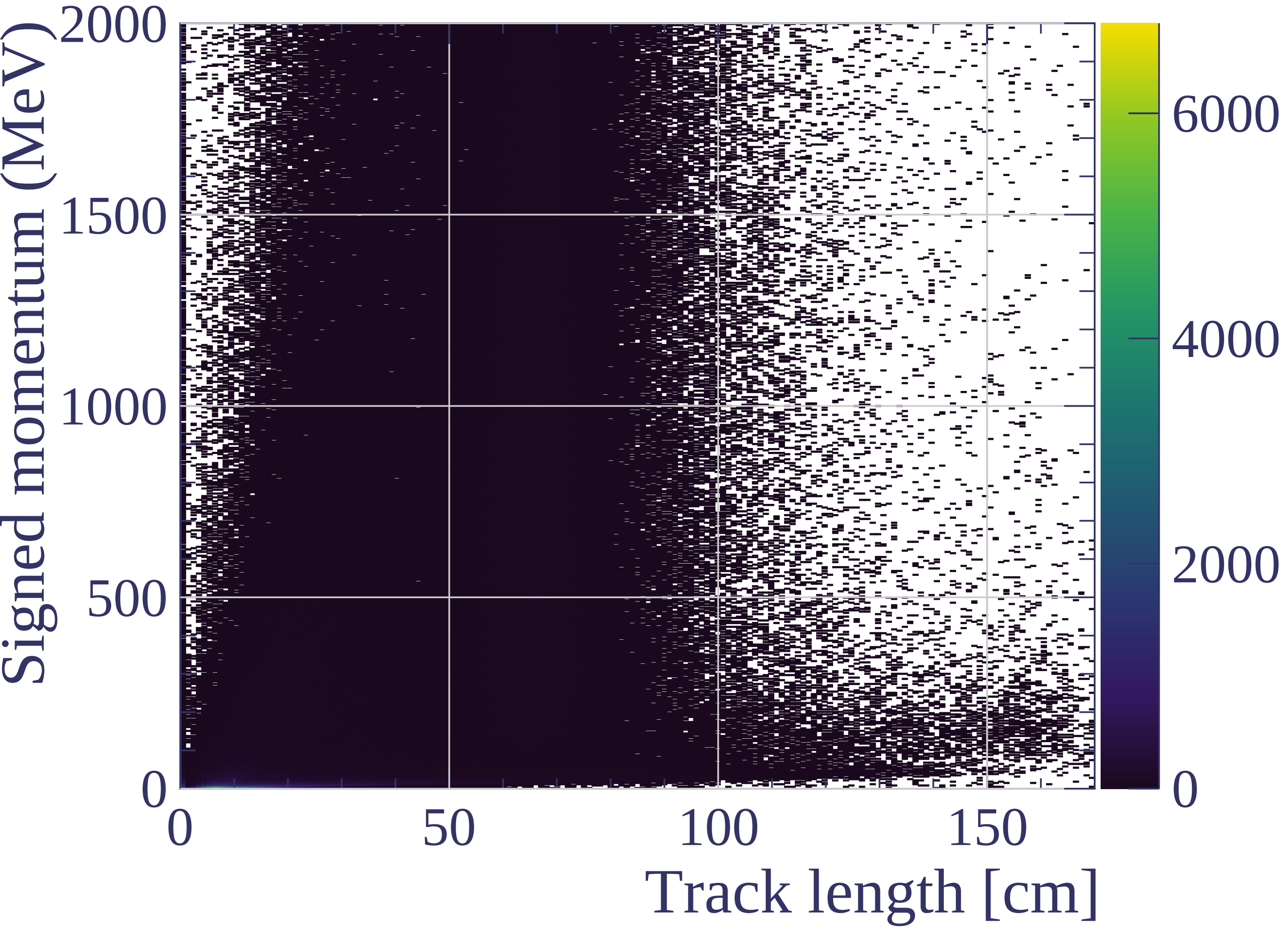


dE/dx resolution [%]
10
8
6
4
0
500
1000
1500
Track length [mm]
WF
XP

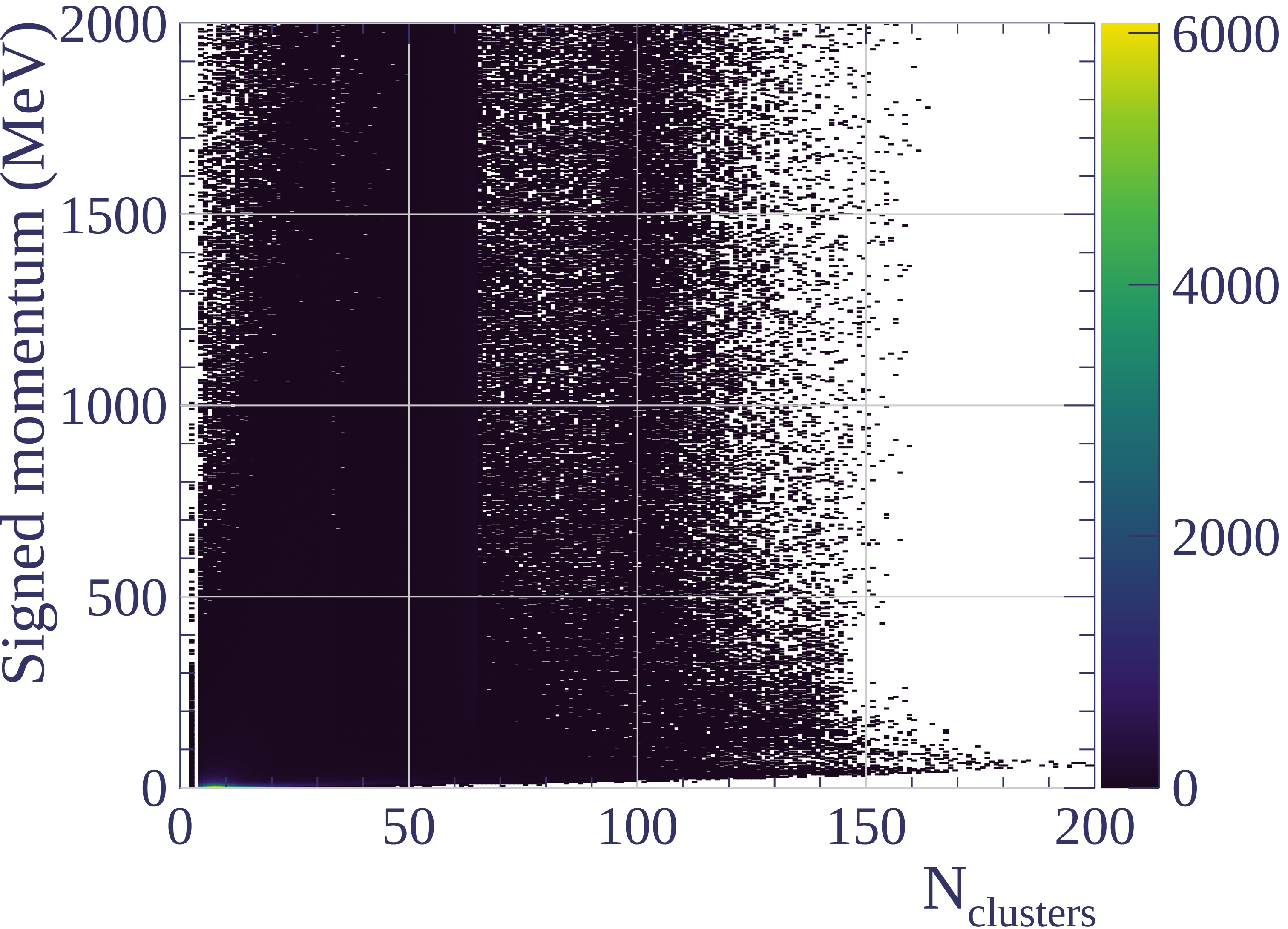

WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
500
1000
1500
Track length [mm]



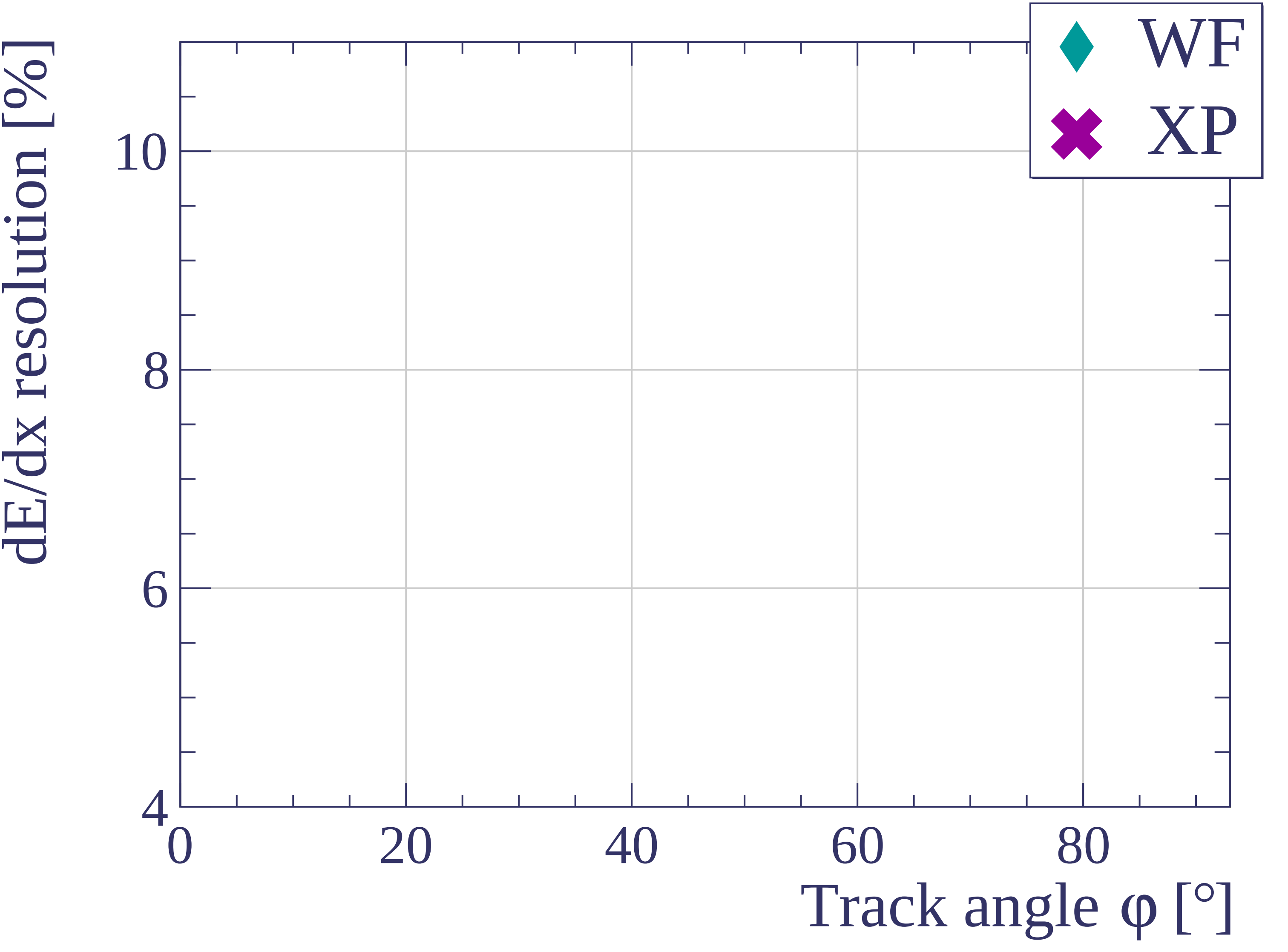
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
800
600
400
200
0



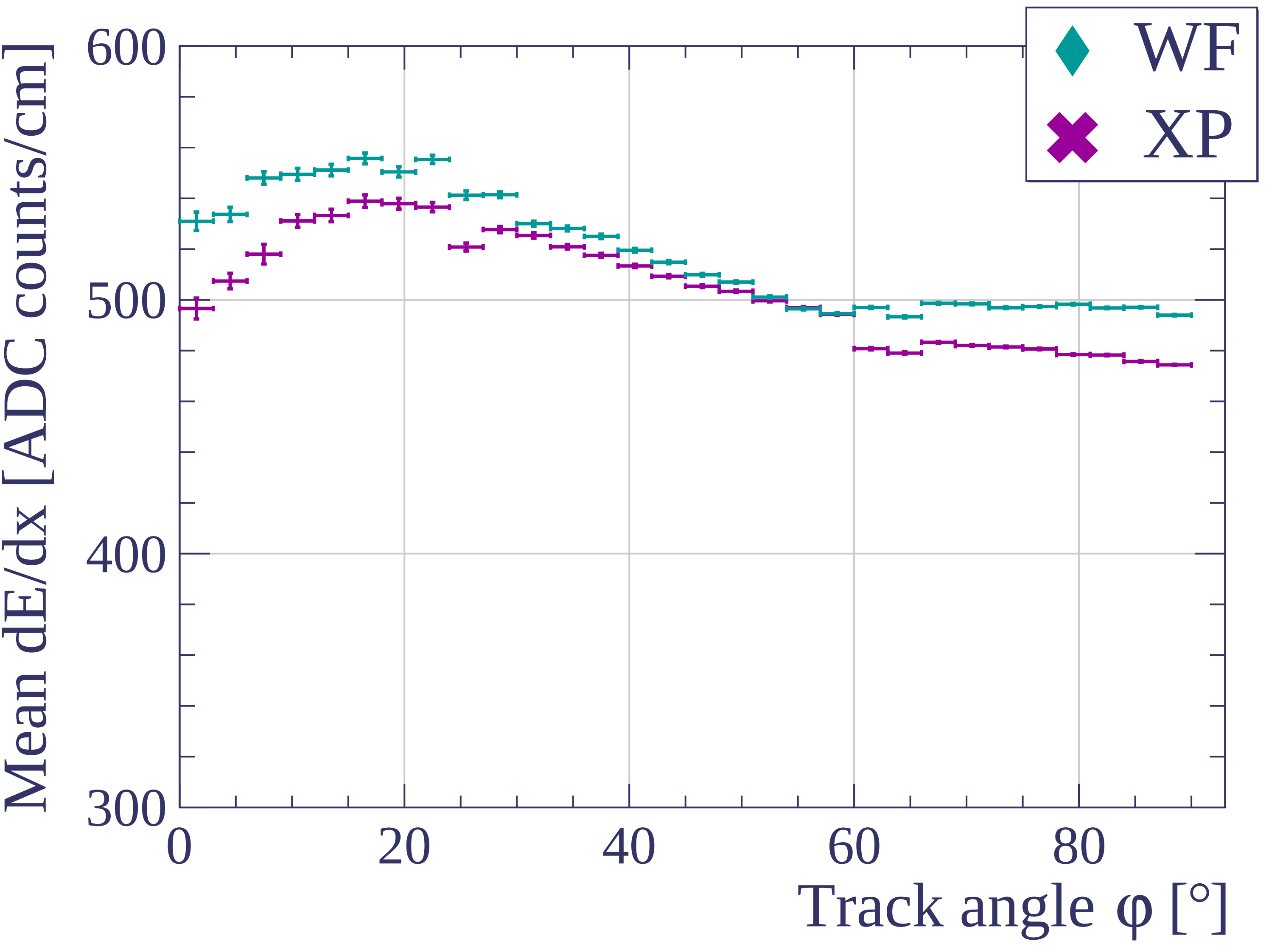
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
6000
4000
2000
0



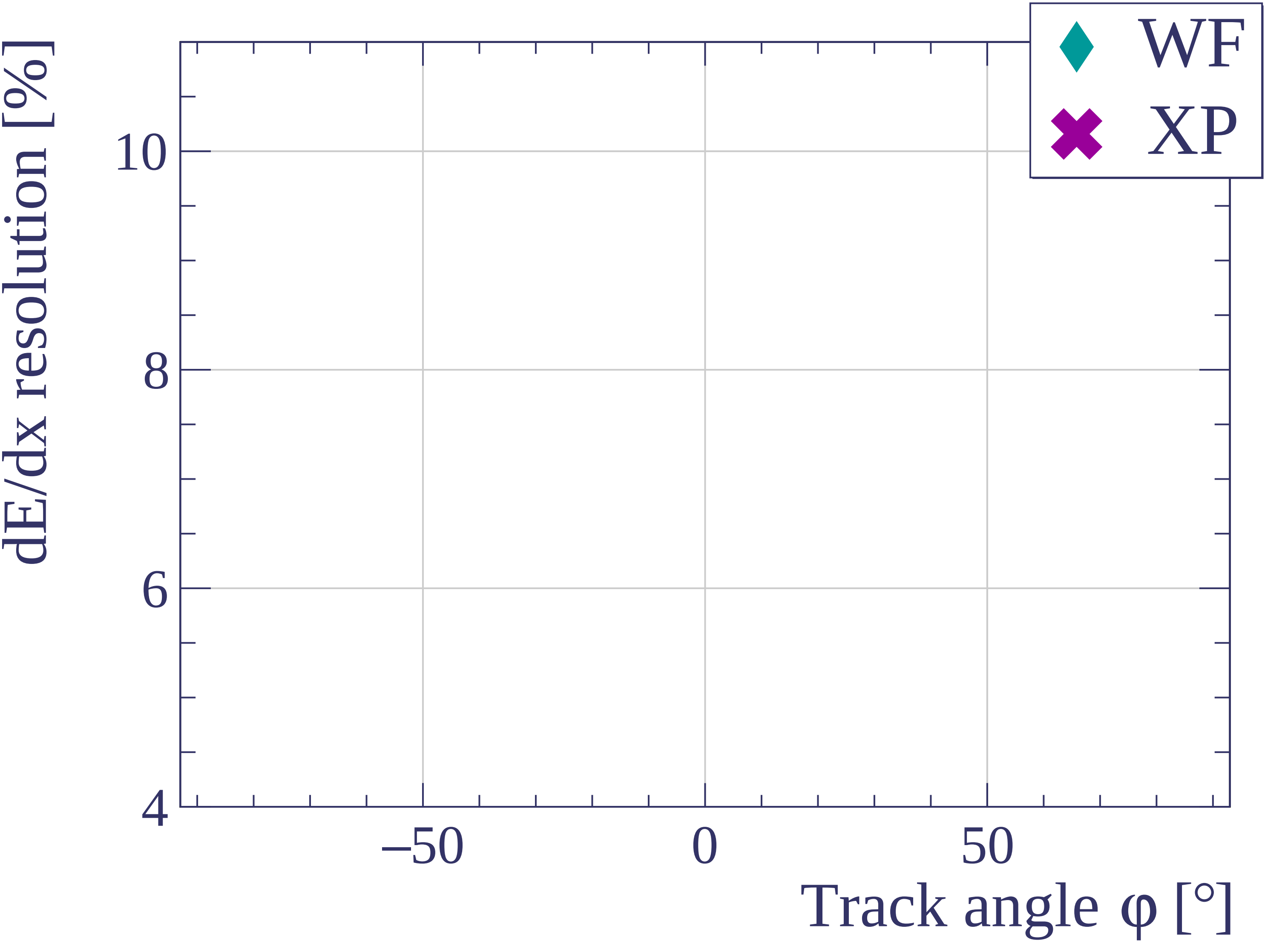
Signed momentum (MeV)
0
500
1000
1500
2000
0
50
100
150
200
$N_{clusters}$
0
2000
4000
6000



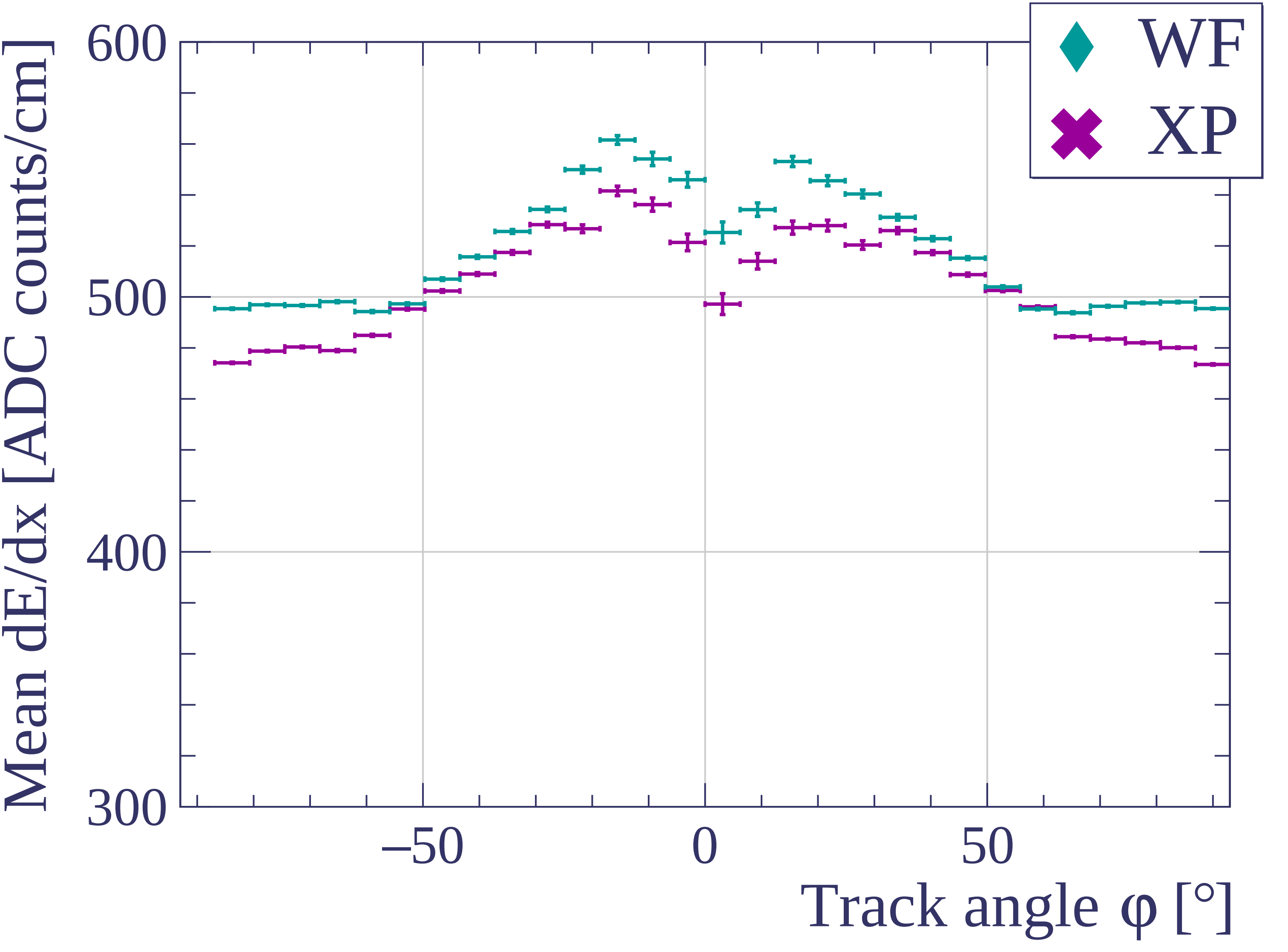
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle φ [°]
WF
XP



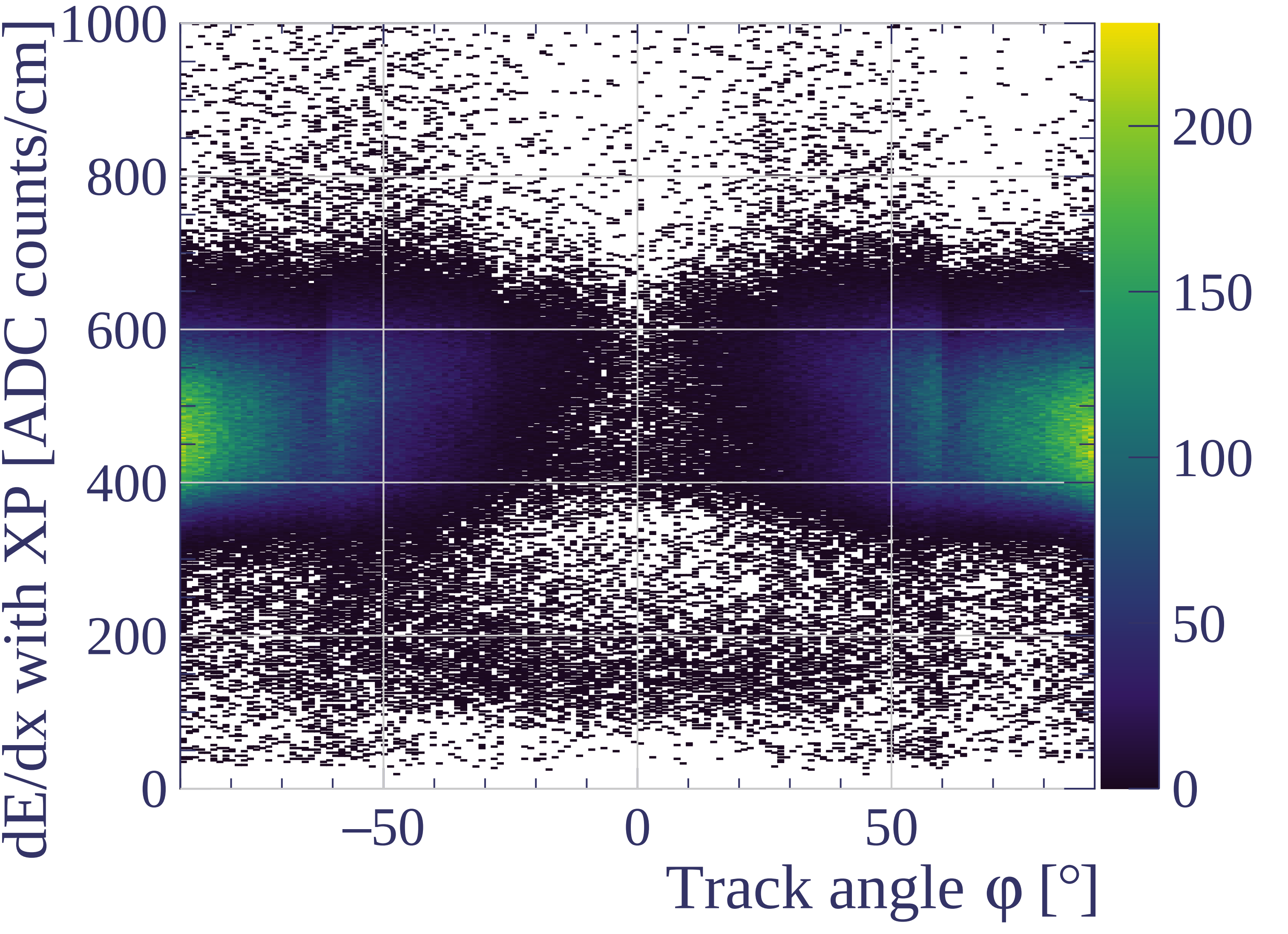
WF
XP
600
500
400
300
0
20
40
60
80
Mean dE/dx [ADC counts/cm]
Track angle φ [°]

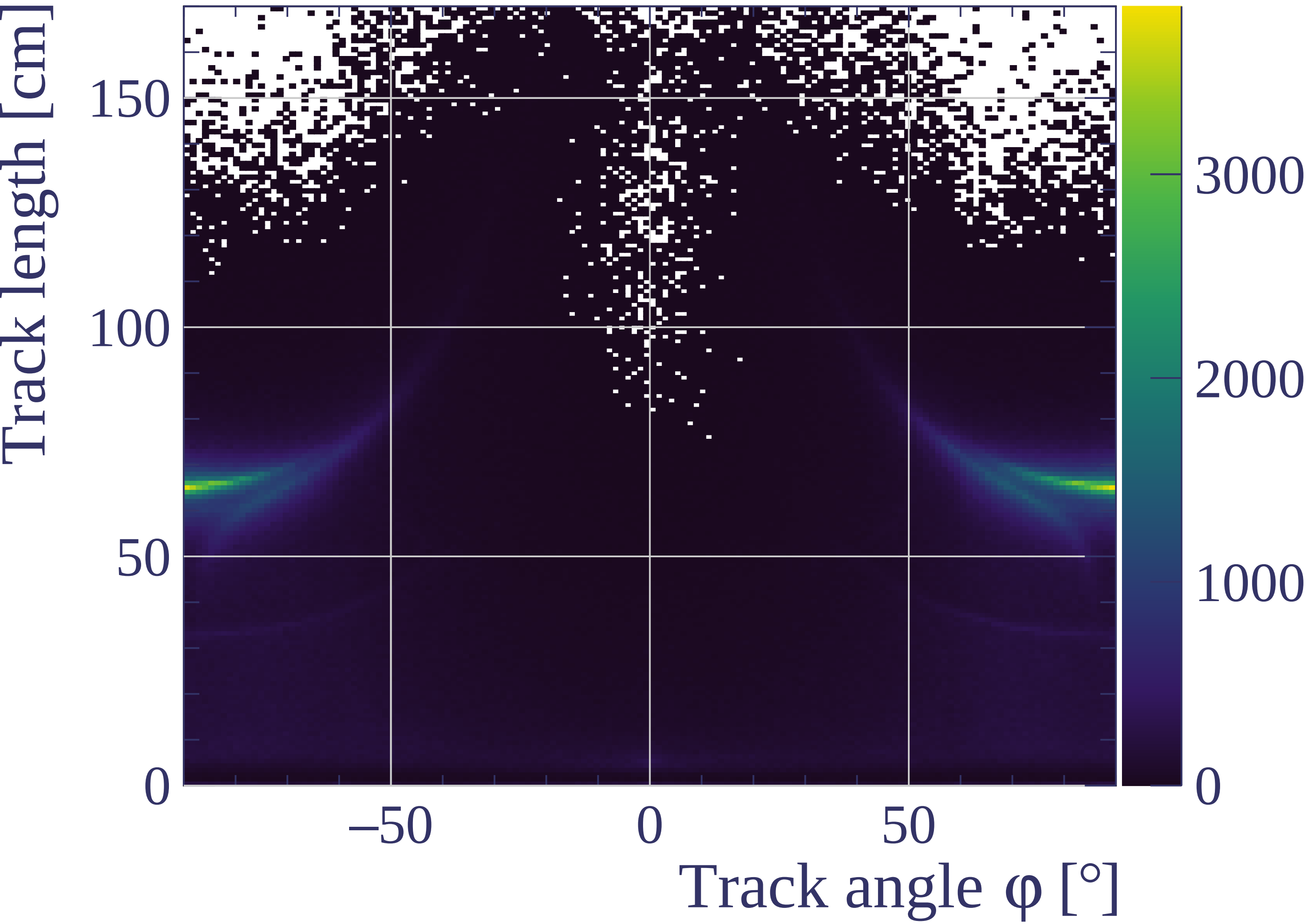


WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]

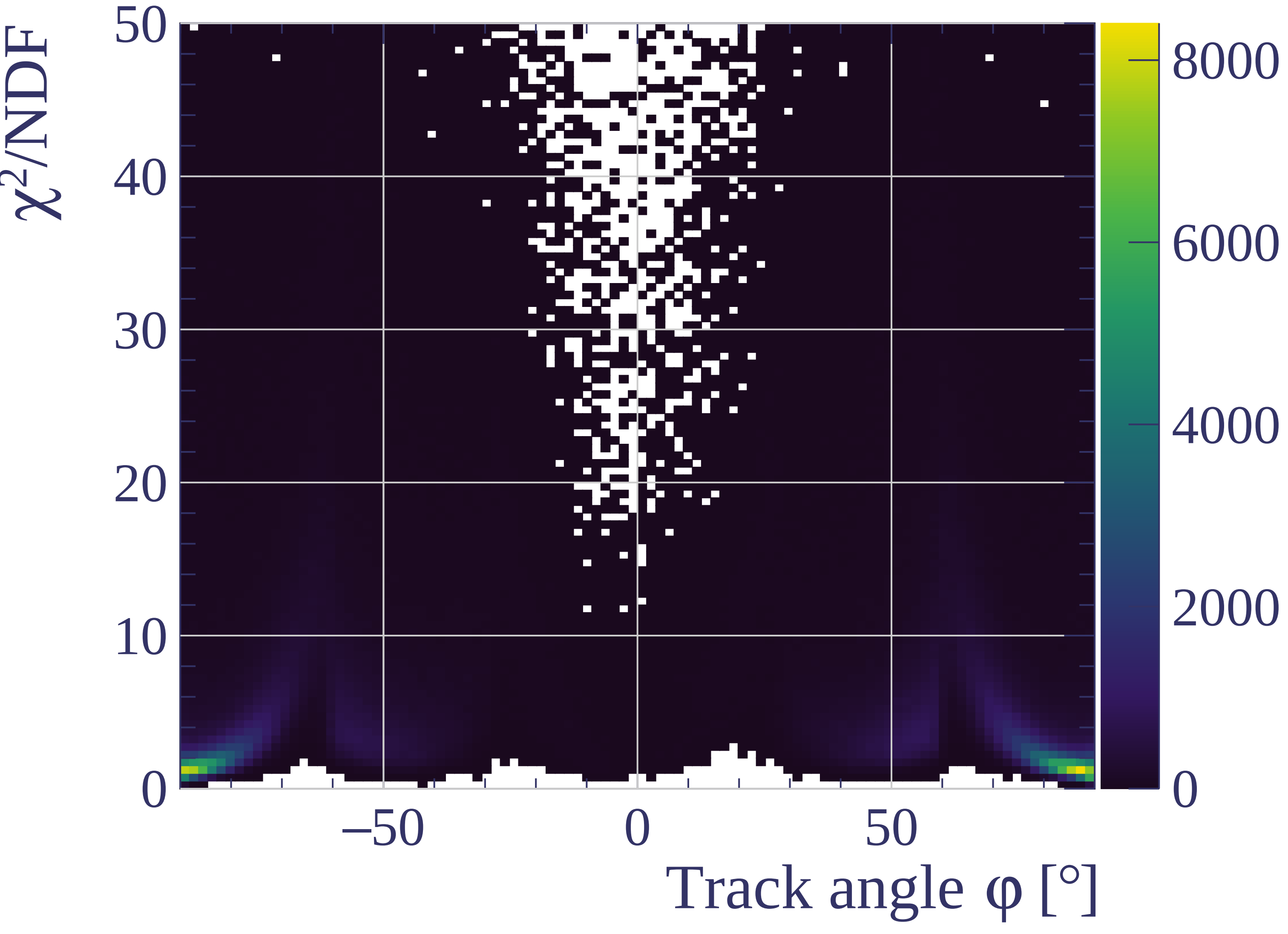


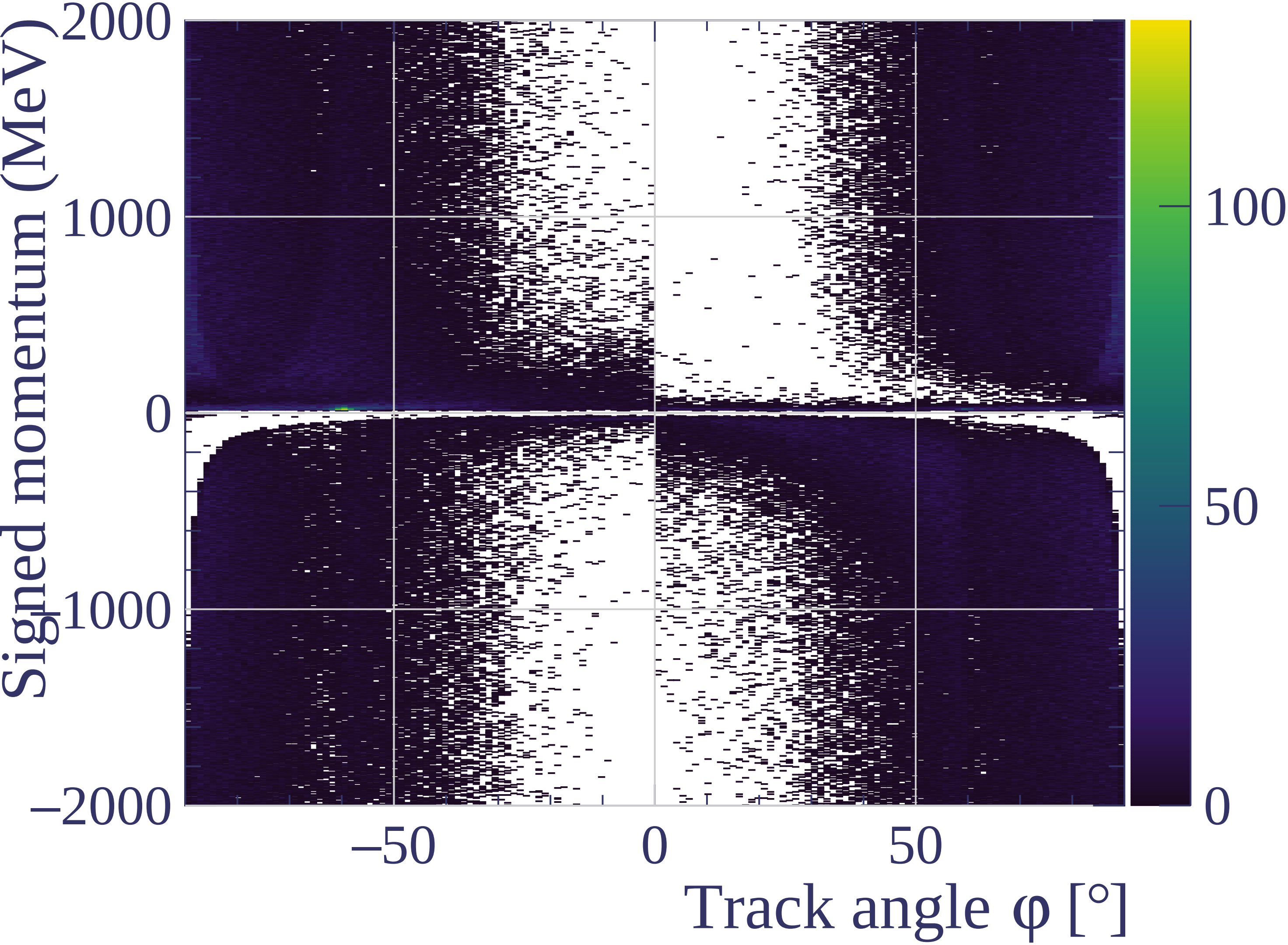
WF
XP
600
500
400
300
−50
0
50
Mean dE/dx [ADC counts/cm]
Track angle φ [°]


dE/dx with XP [ADC counts/cm]
Track angle φ [°]
1000
800
600
400
200
0
−50
0
50
200
150
100
50
0

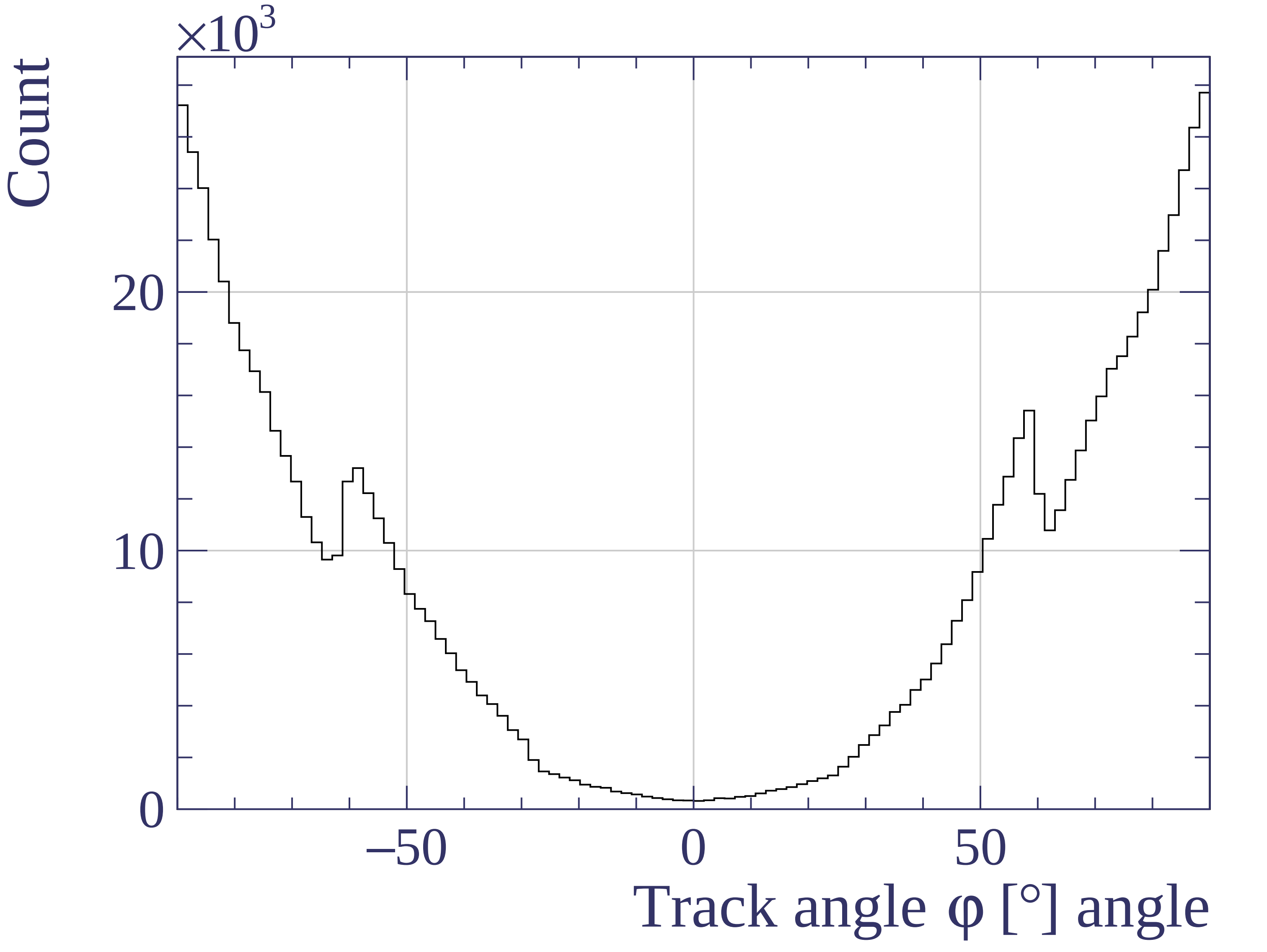


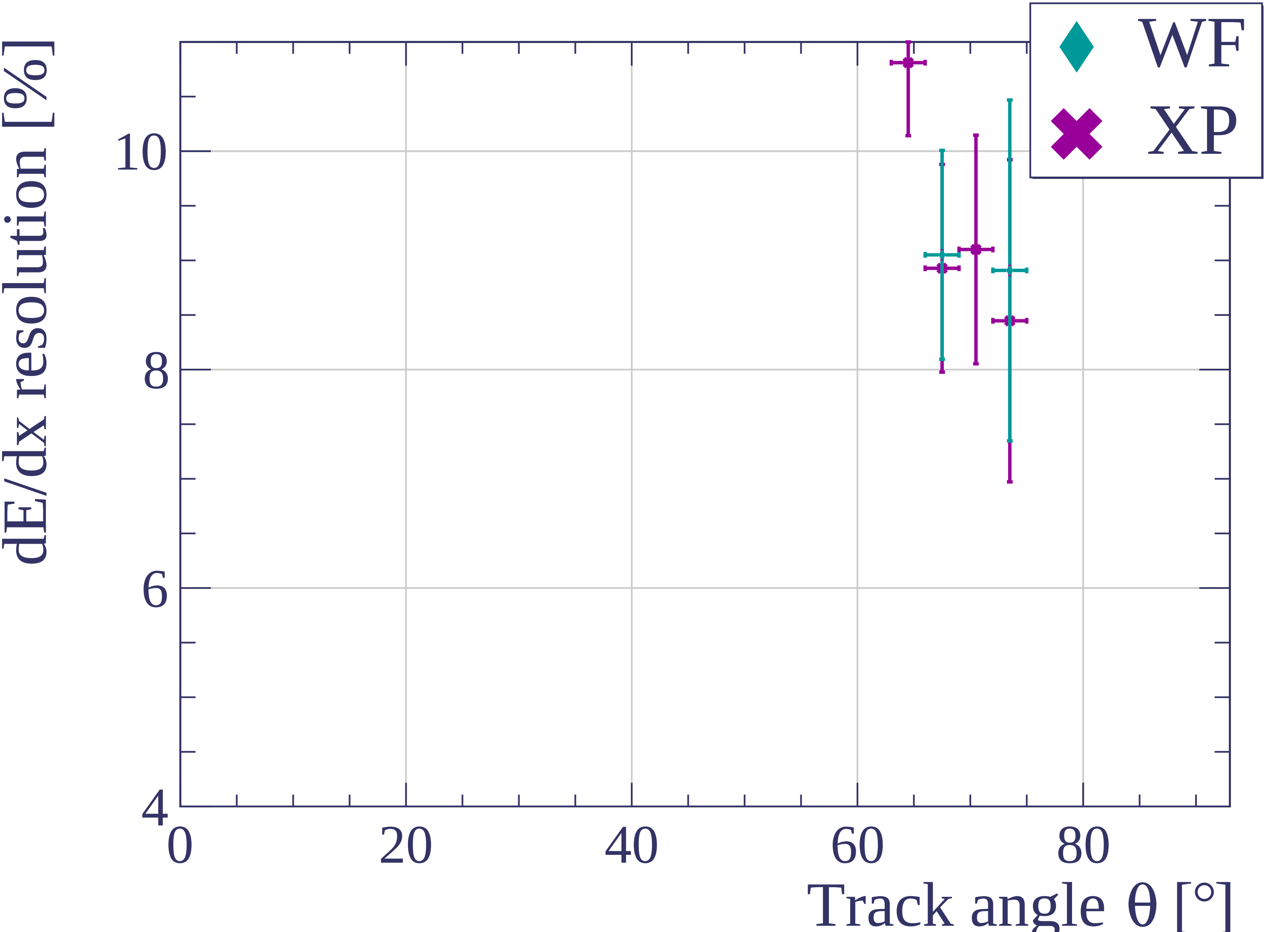
Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
3000
2000
1000
0

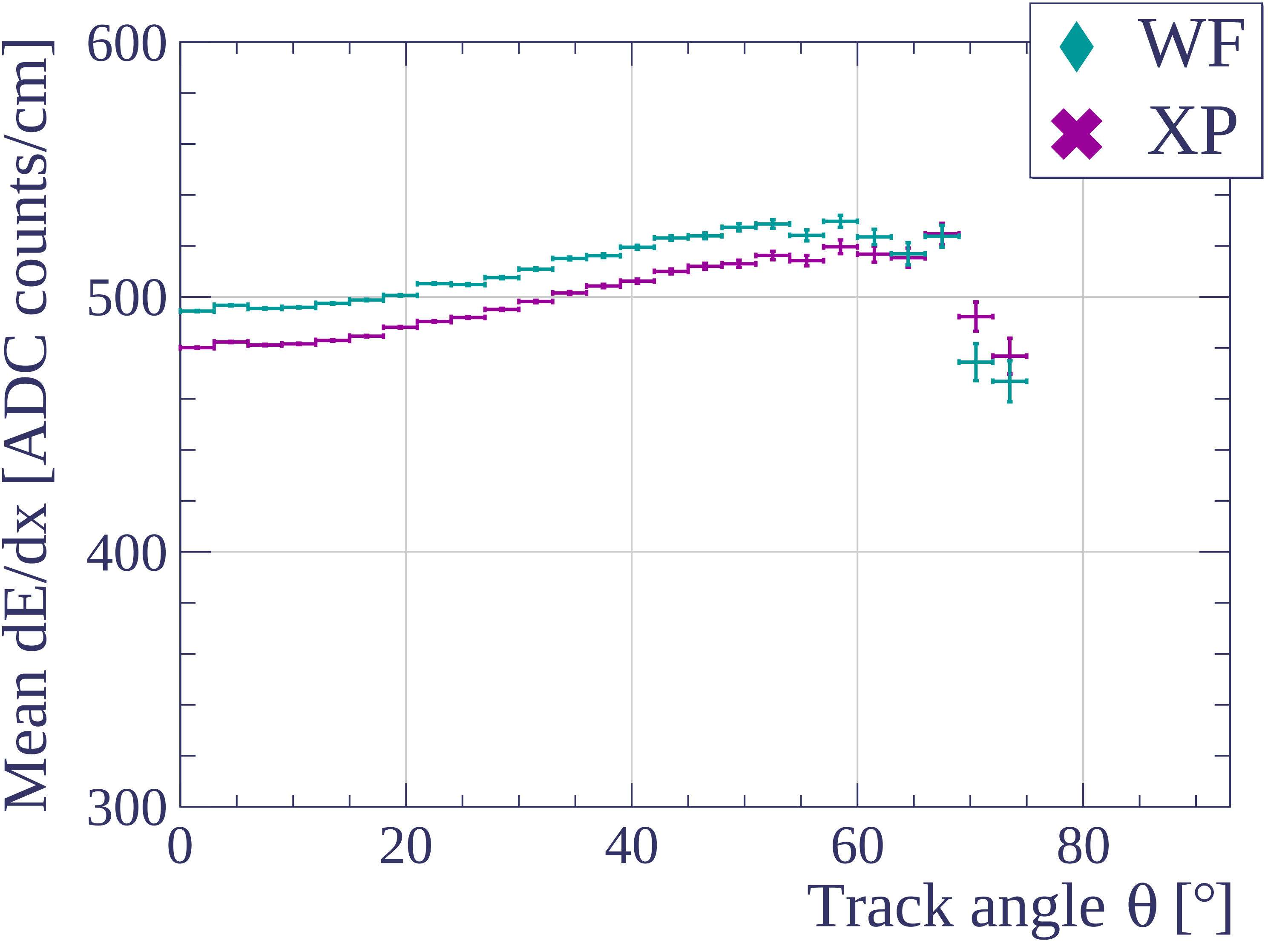

χ^2/NDF
0
10
20
30
40
50
−50
0
50
Track angle φ [°]
0
2000
4000
6000
8000



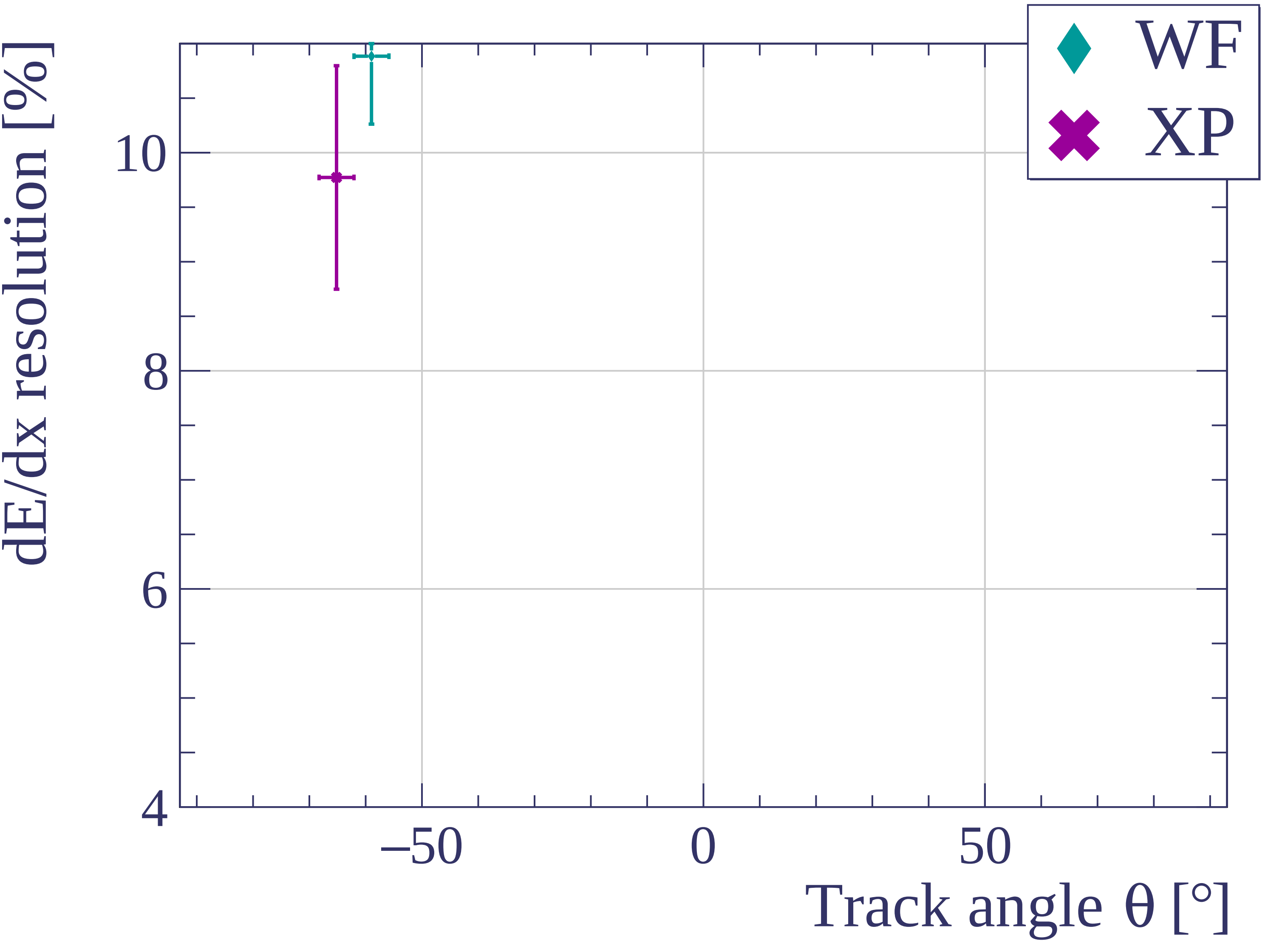
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
0
50
100


×10^3
Count
0
10
20
−50
0
50
Track angle φ [°] angle

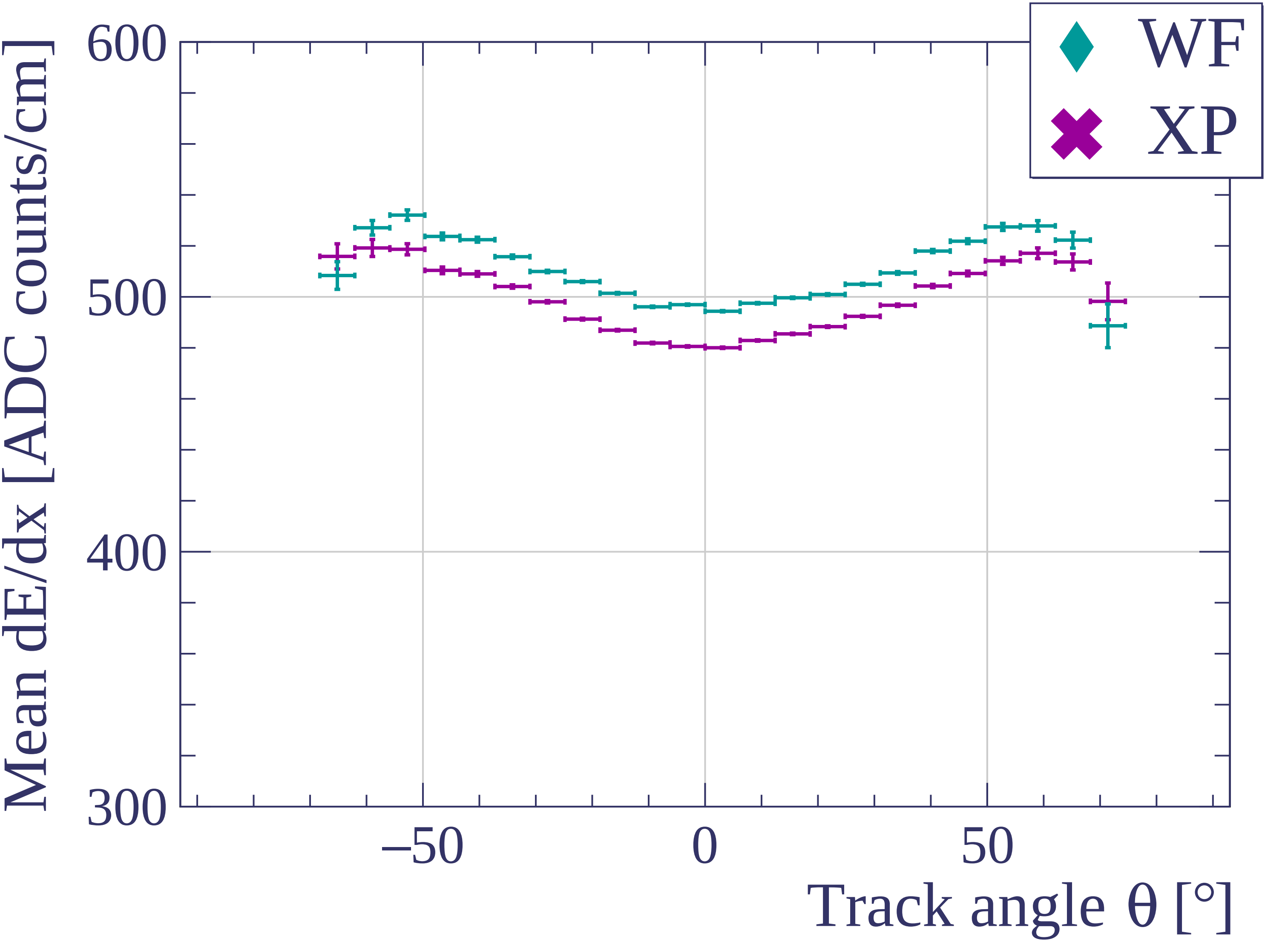

WF
XP
dE/dx resolution [%]
4
6
8
10
0
20
40
60
80
Track angle θ [°]



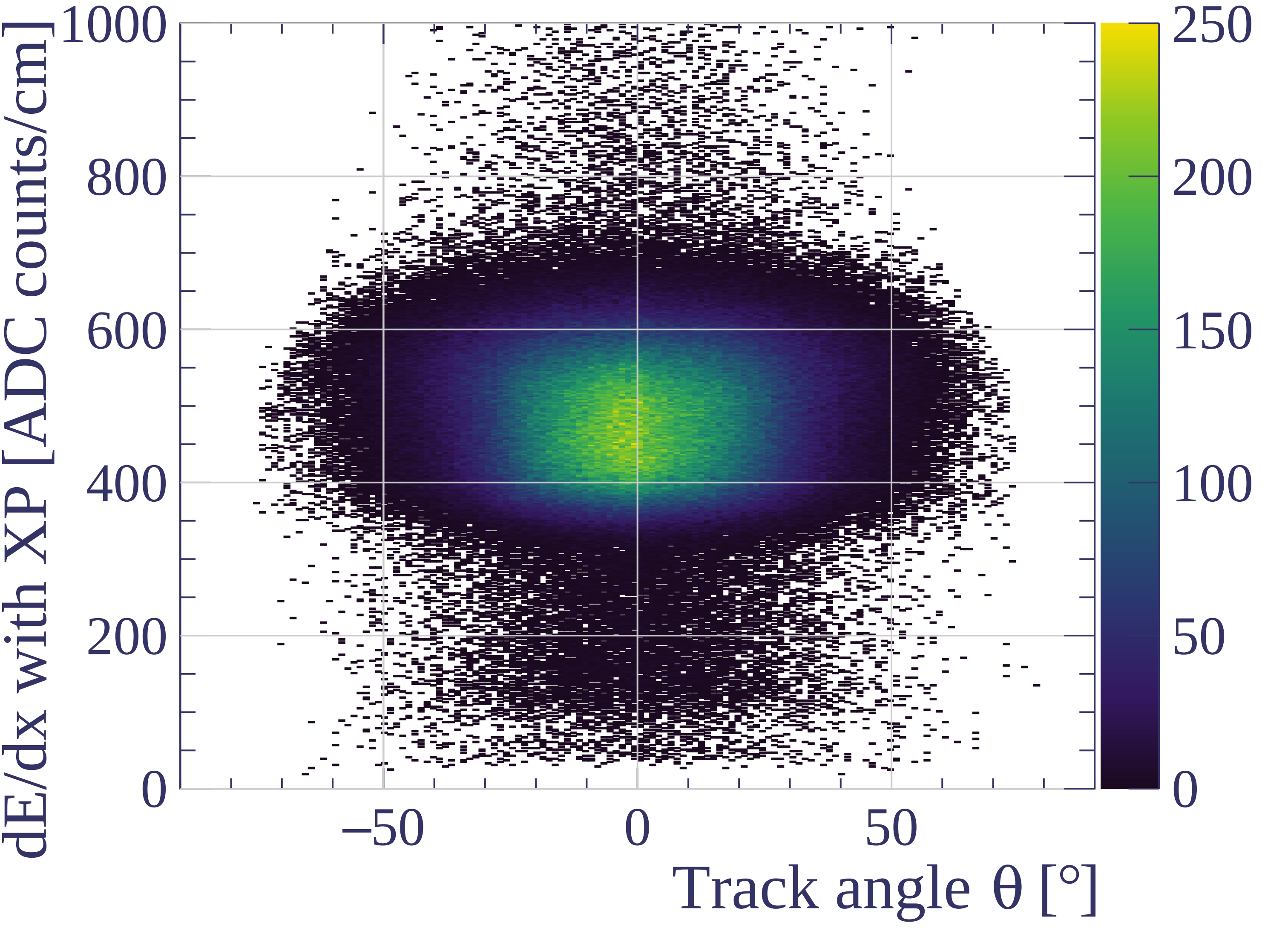
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle θ [°]
WF
XP



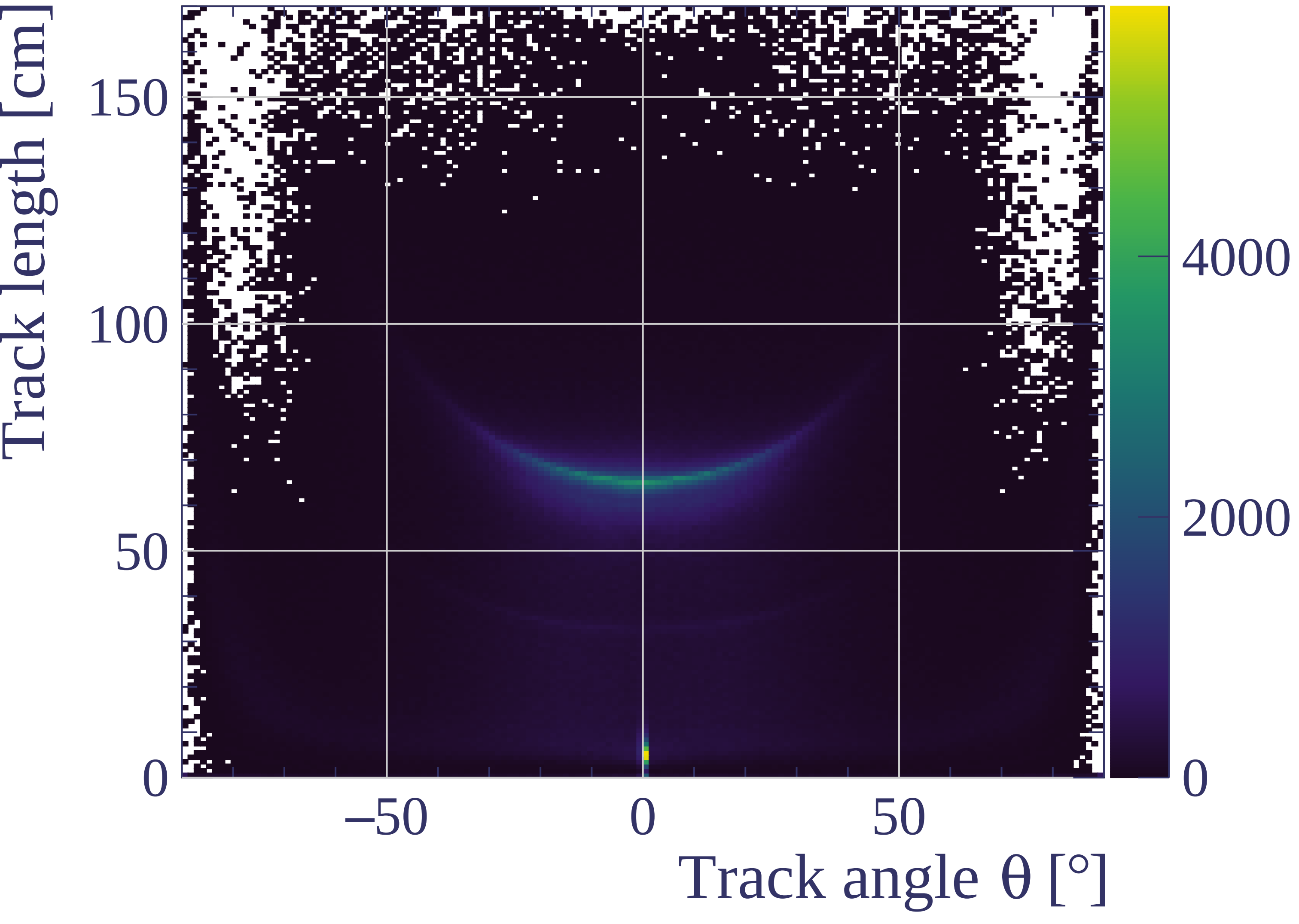
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]



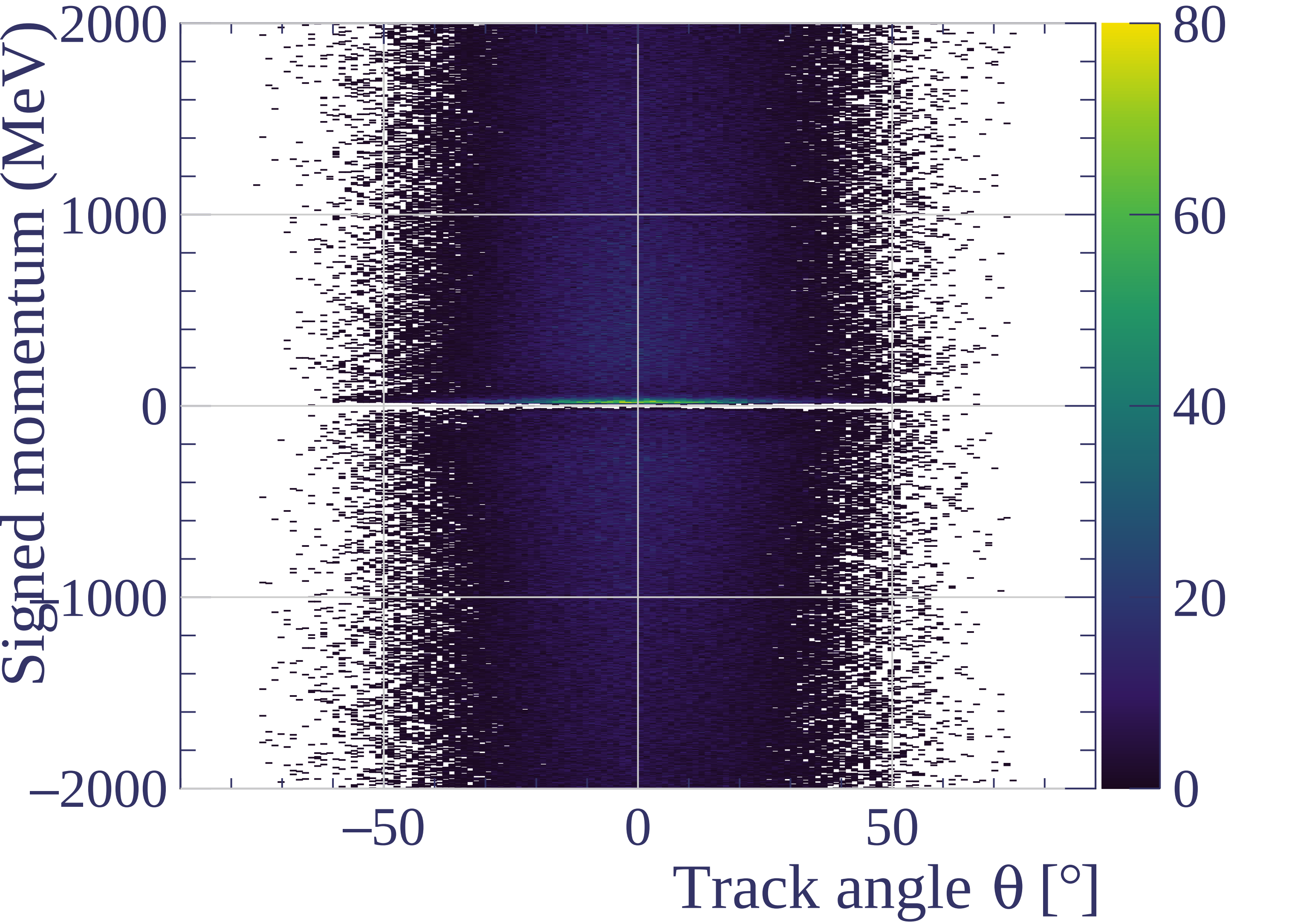
WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle θ [°]



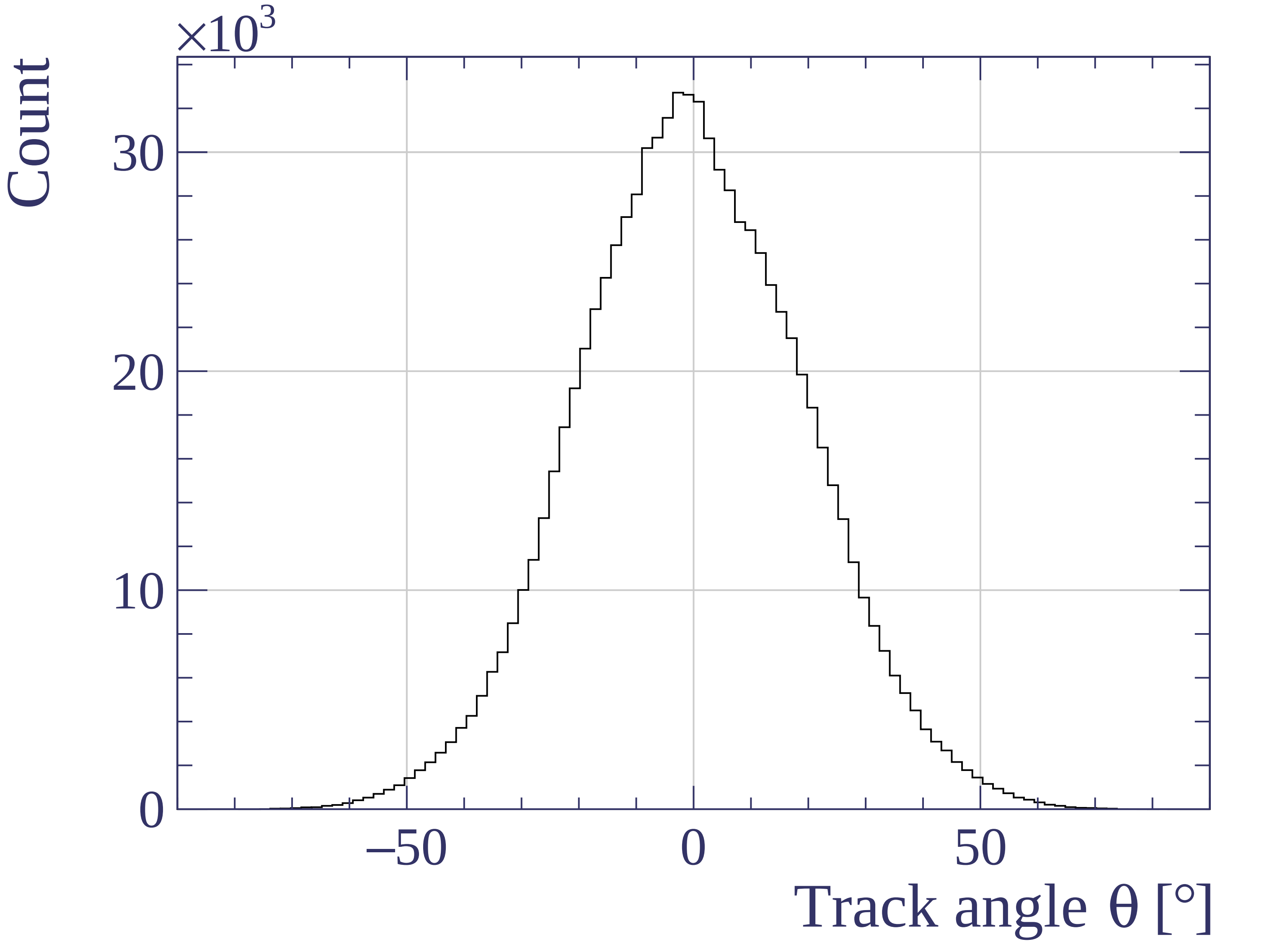
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
250
200
150
100
50
0



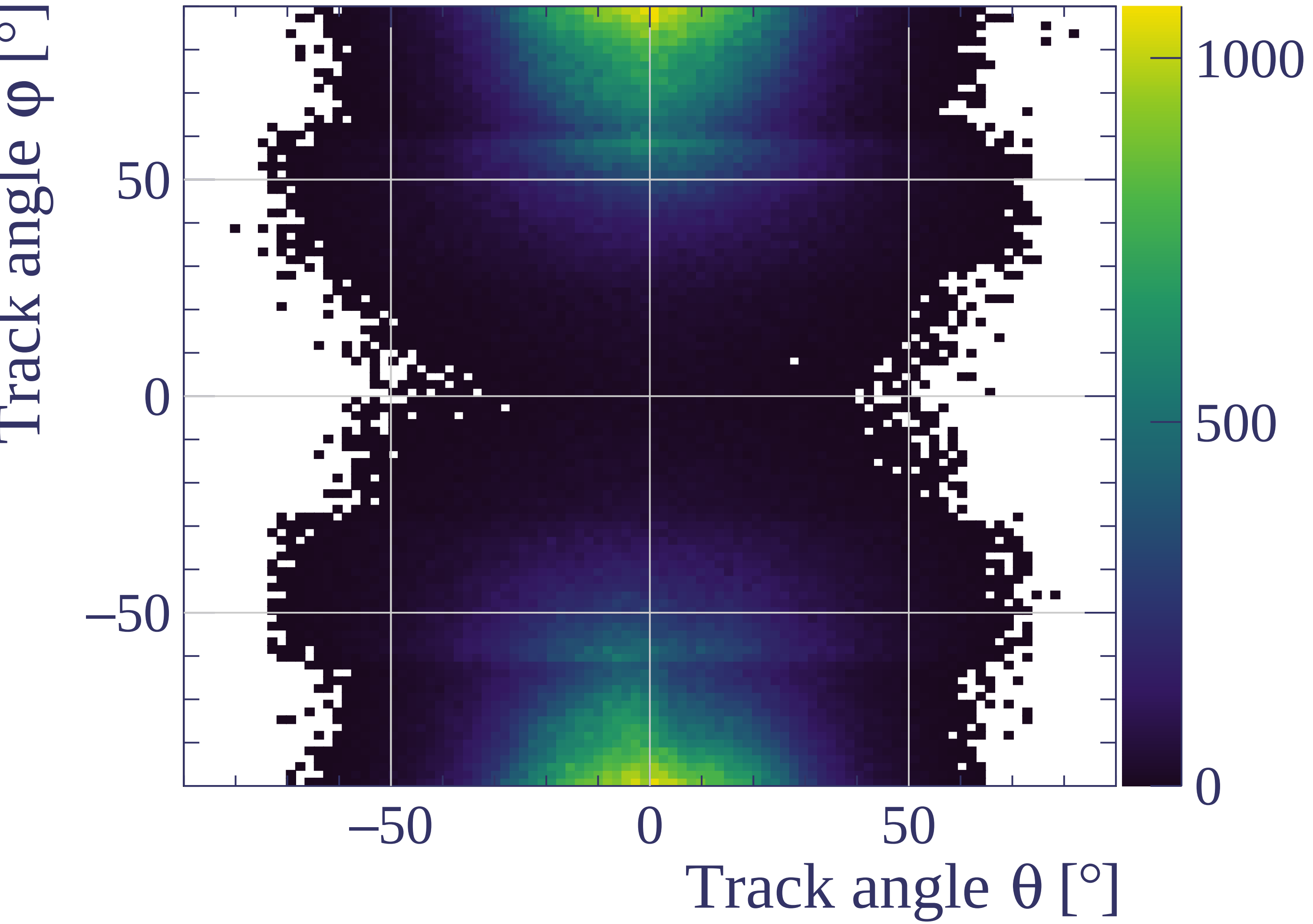
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
4000
2000
0

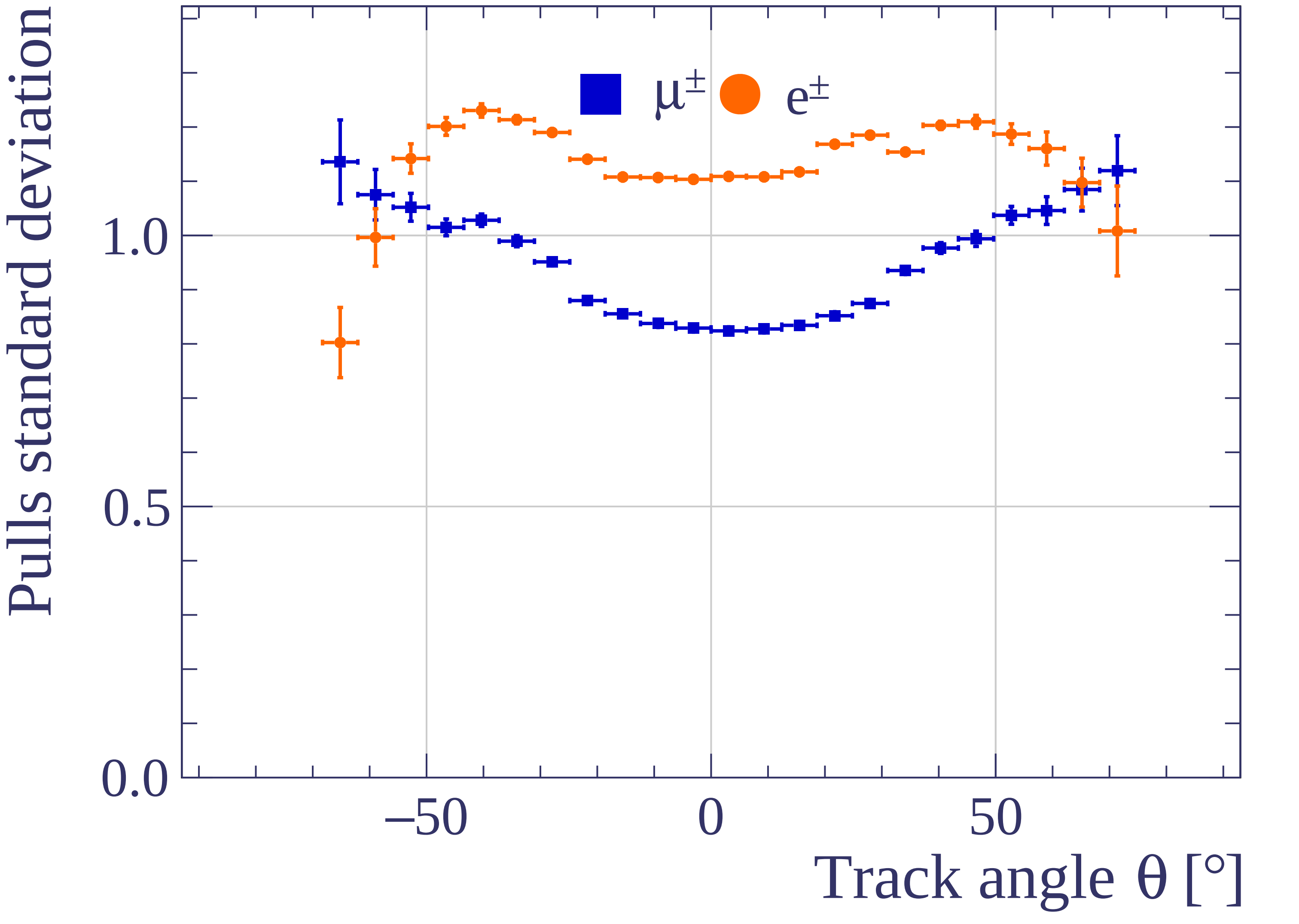


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
80
60
40
20
0

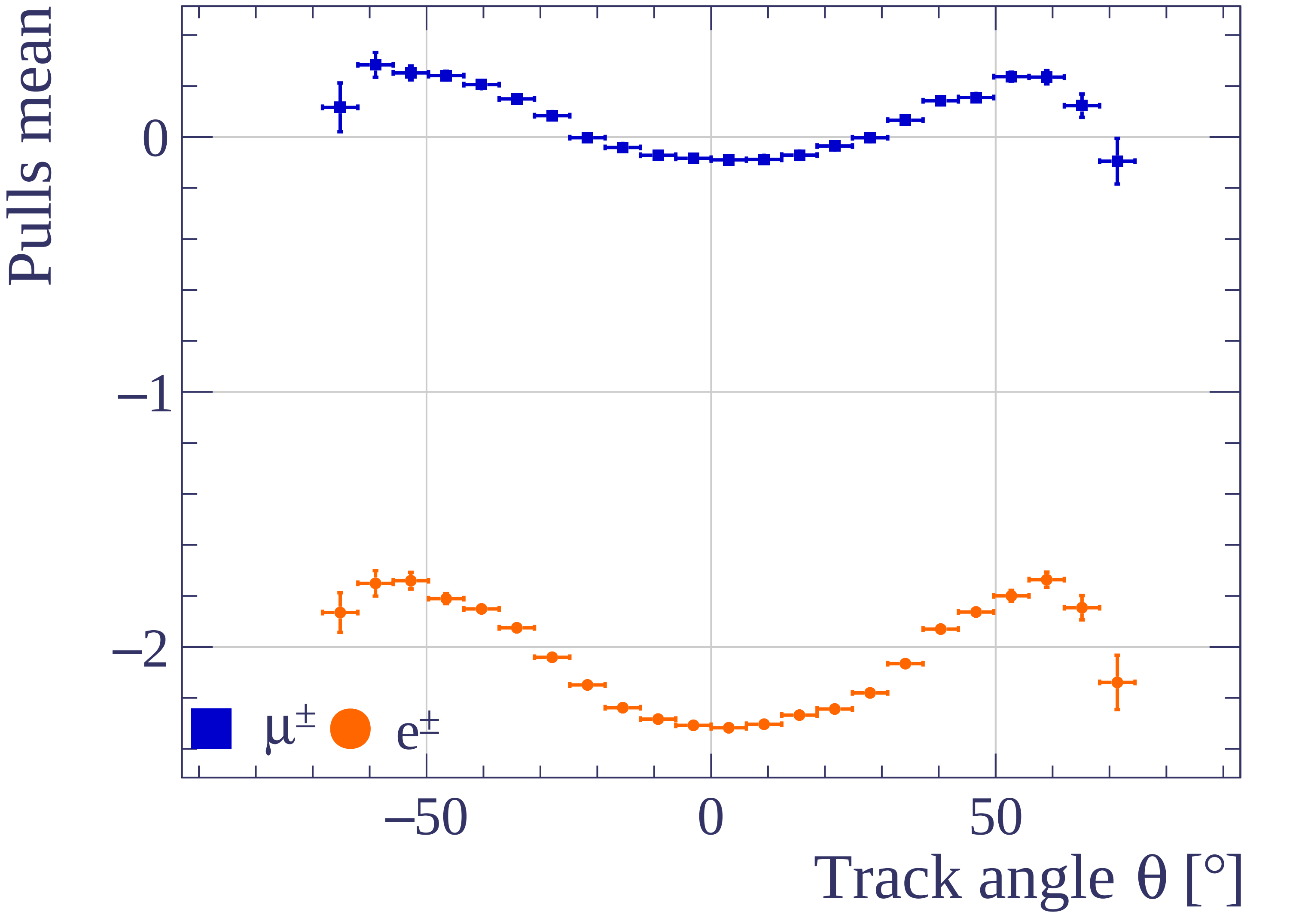


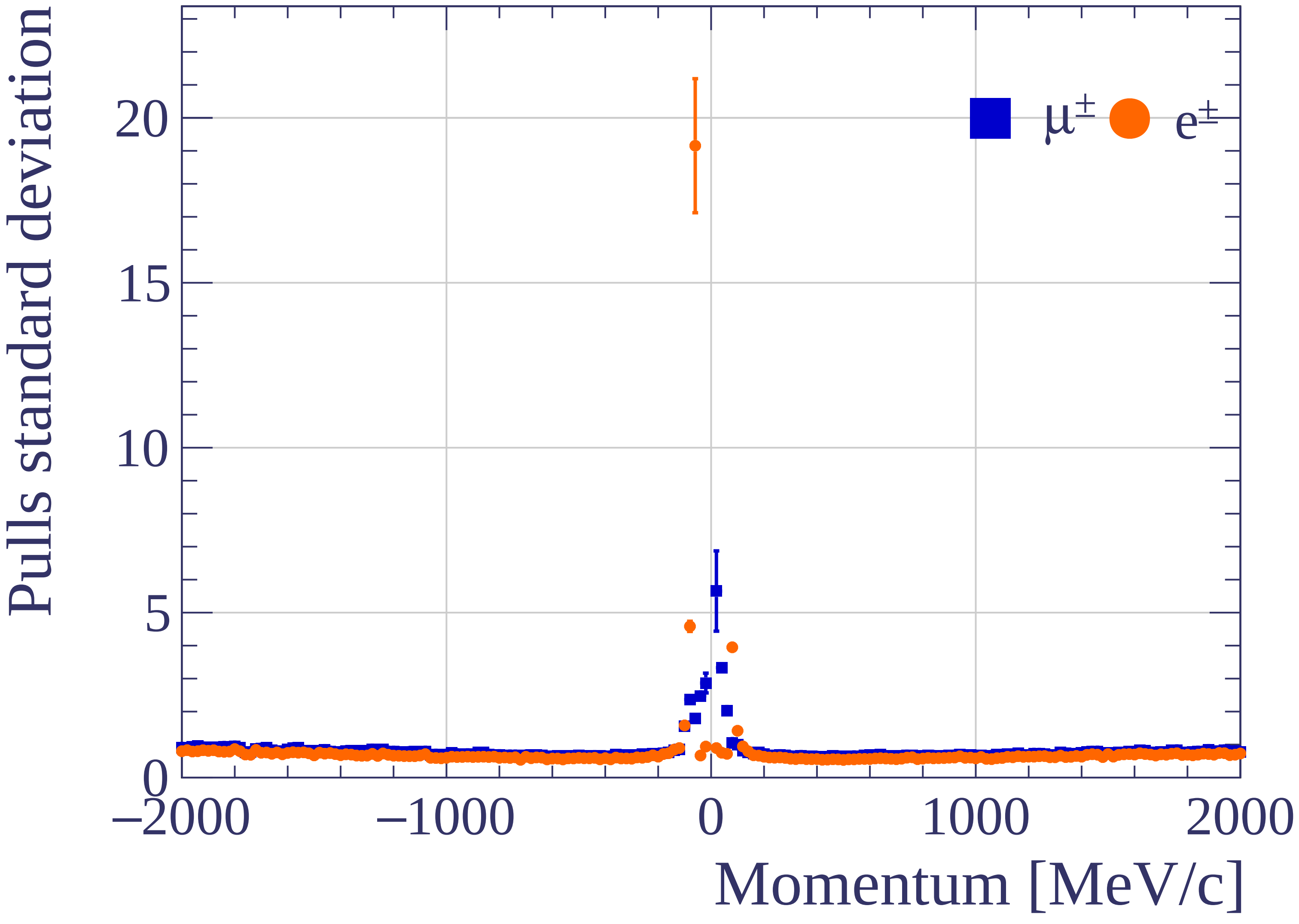
×10^3
Count
30
20
10
0
−50
0
50
Track angle θ [°]


Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
1000
500
0

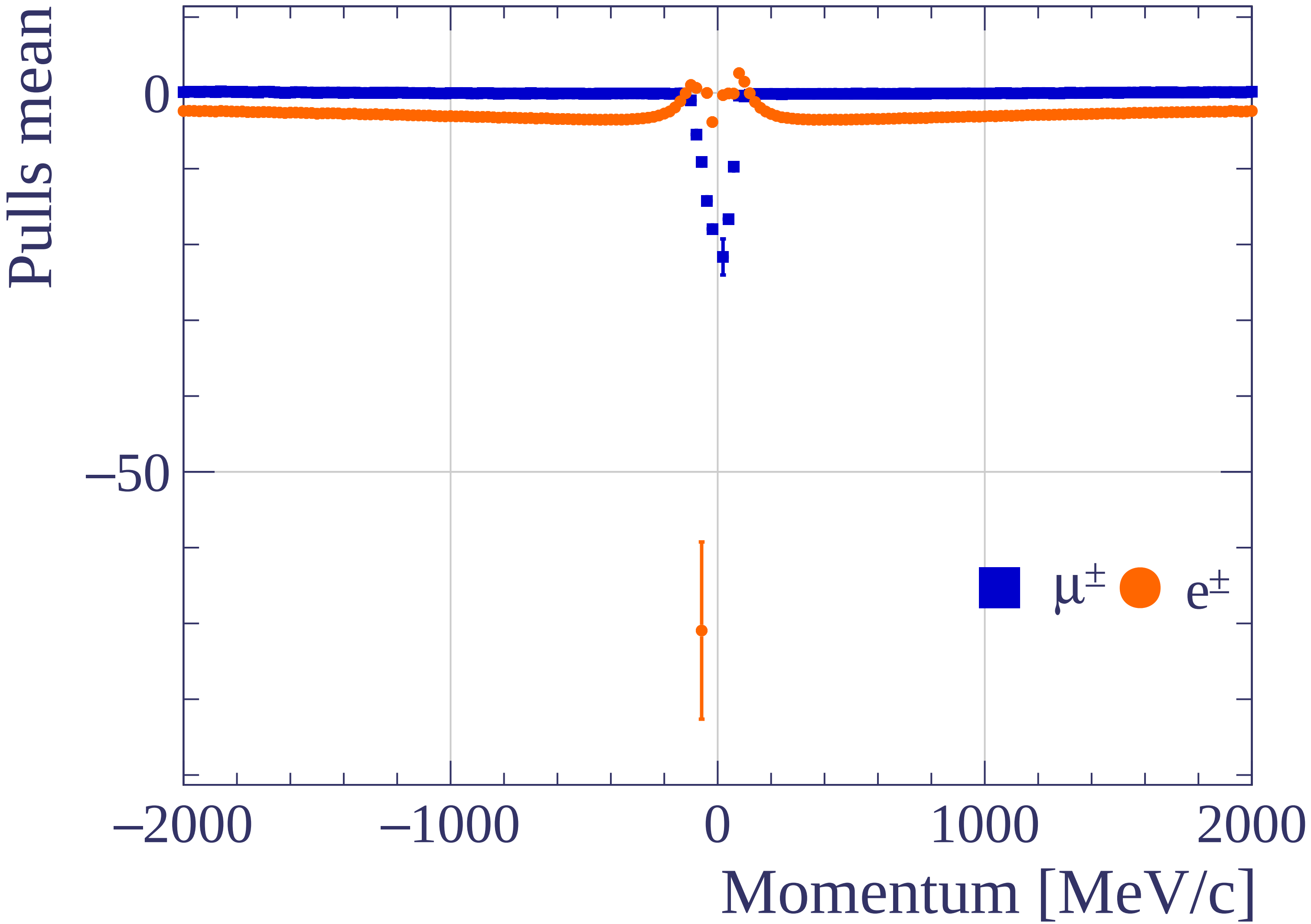


Pulls standard deviation
μ±
e±
1.0
0.5
0.0
−50
0
50
Track angle θ [°]

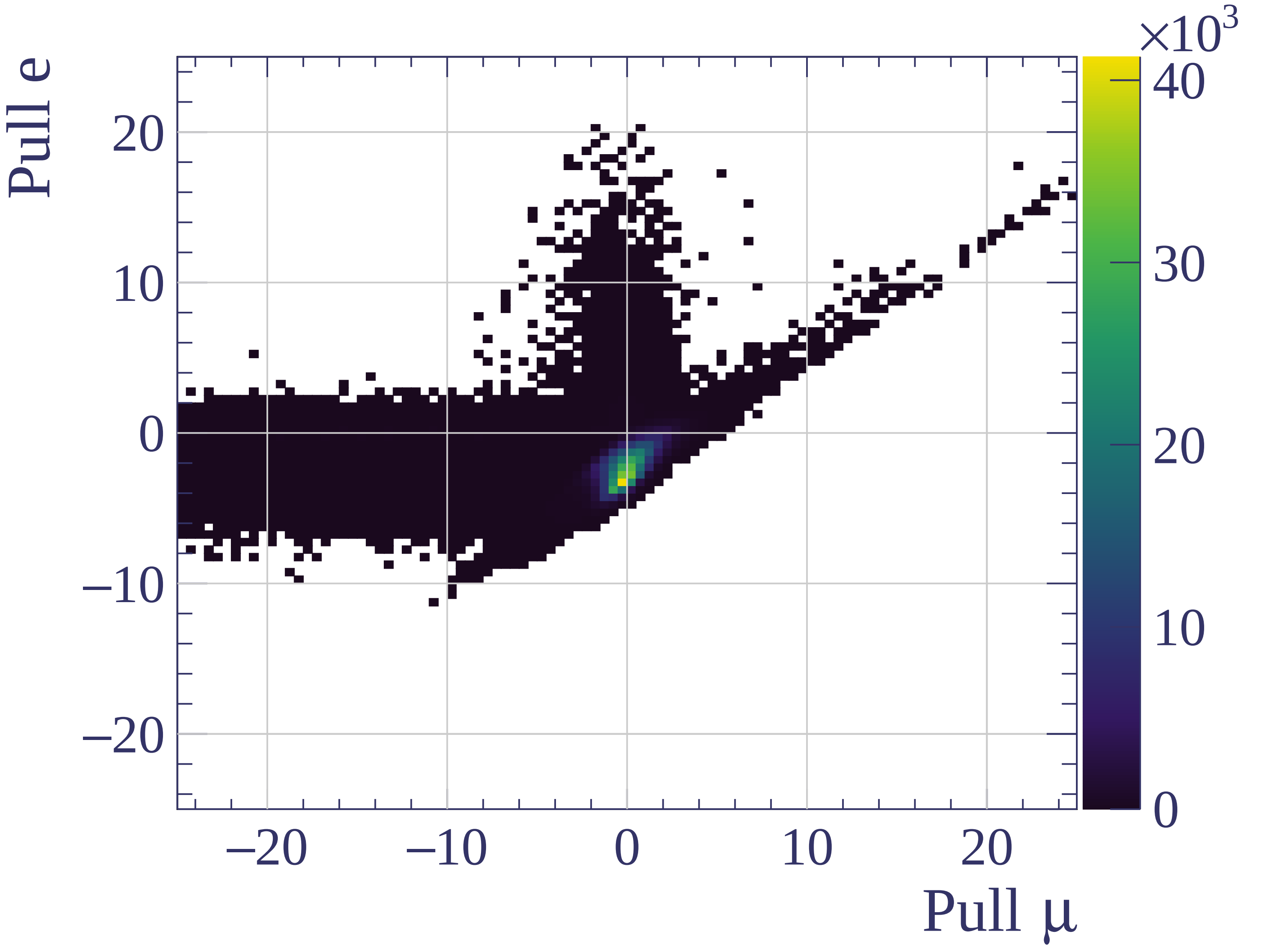

Pulls mean
0
−1
−2
−50
0
50
Track angle θ [°]
μ±
e±

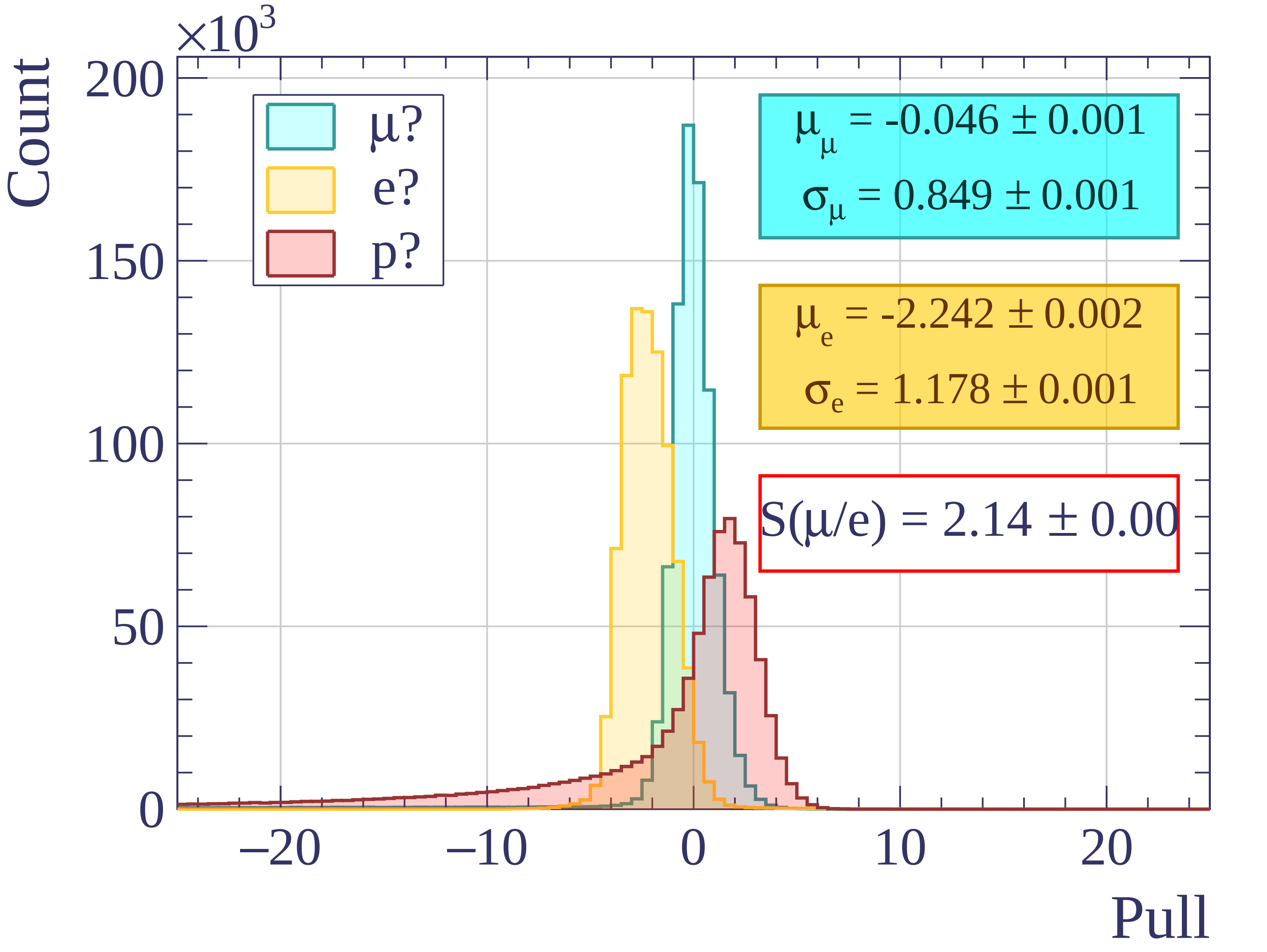


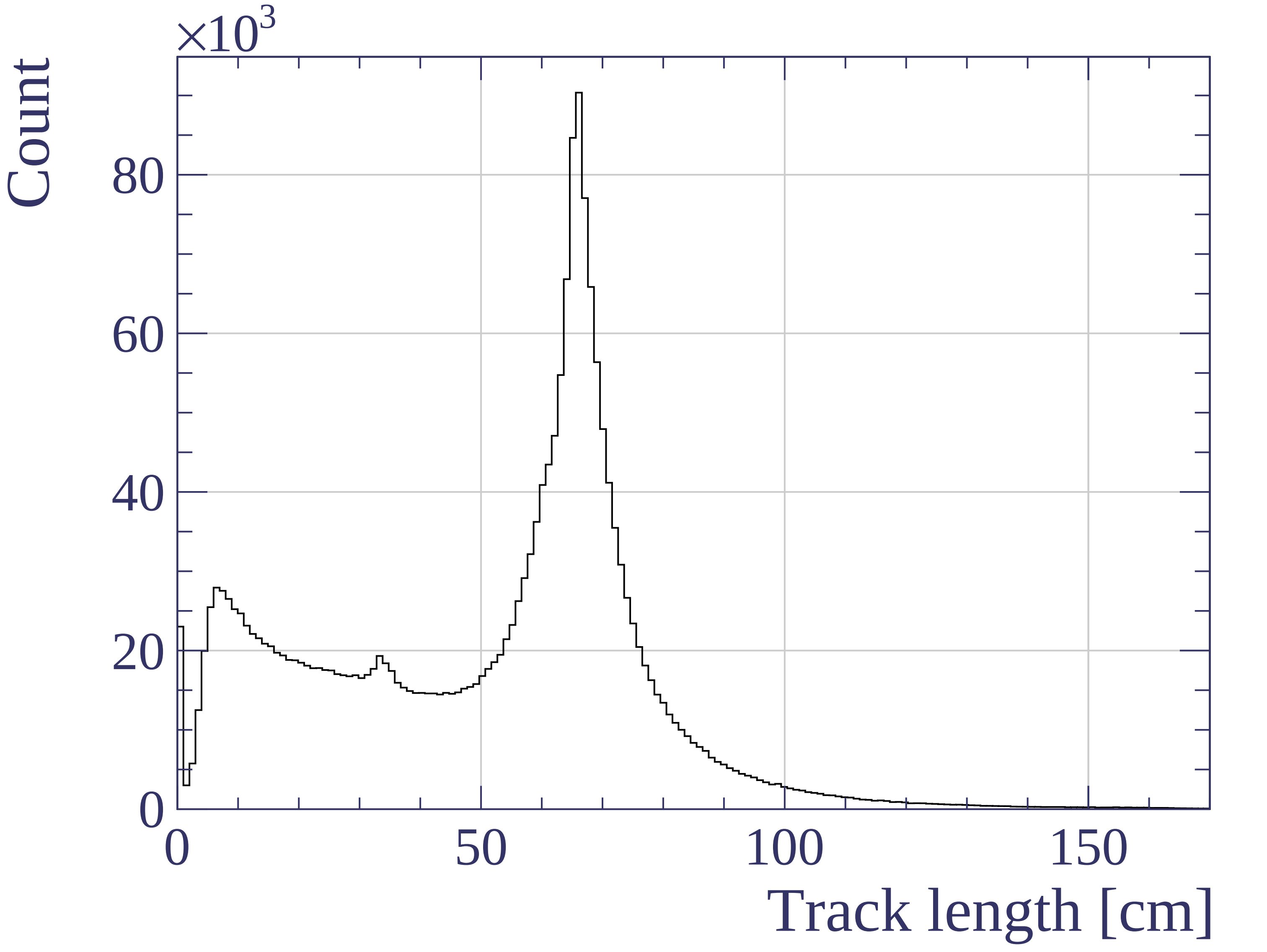
Pulls standard deviation
Momentum [MeV/c]
$\mu^\pm$
$e^\pm$
0
5
10
15
20
−2000
−1000
0
1000
2000



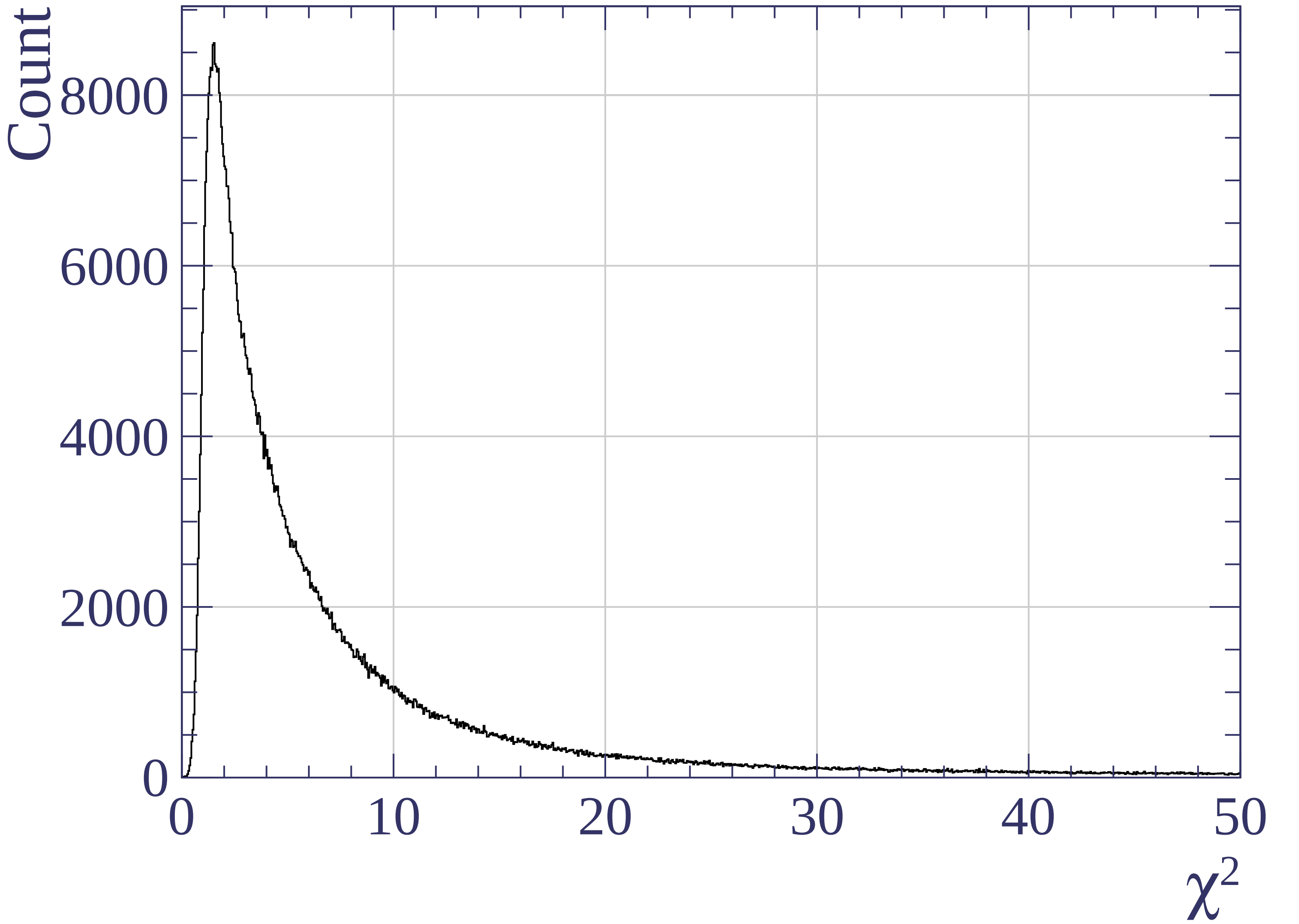
Pulls mean
0
−50
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$


Pull e
Pull μ
20
10
0
−10
−20
−20
−10
0
10
20
×10^3
40
30
20
10
0

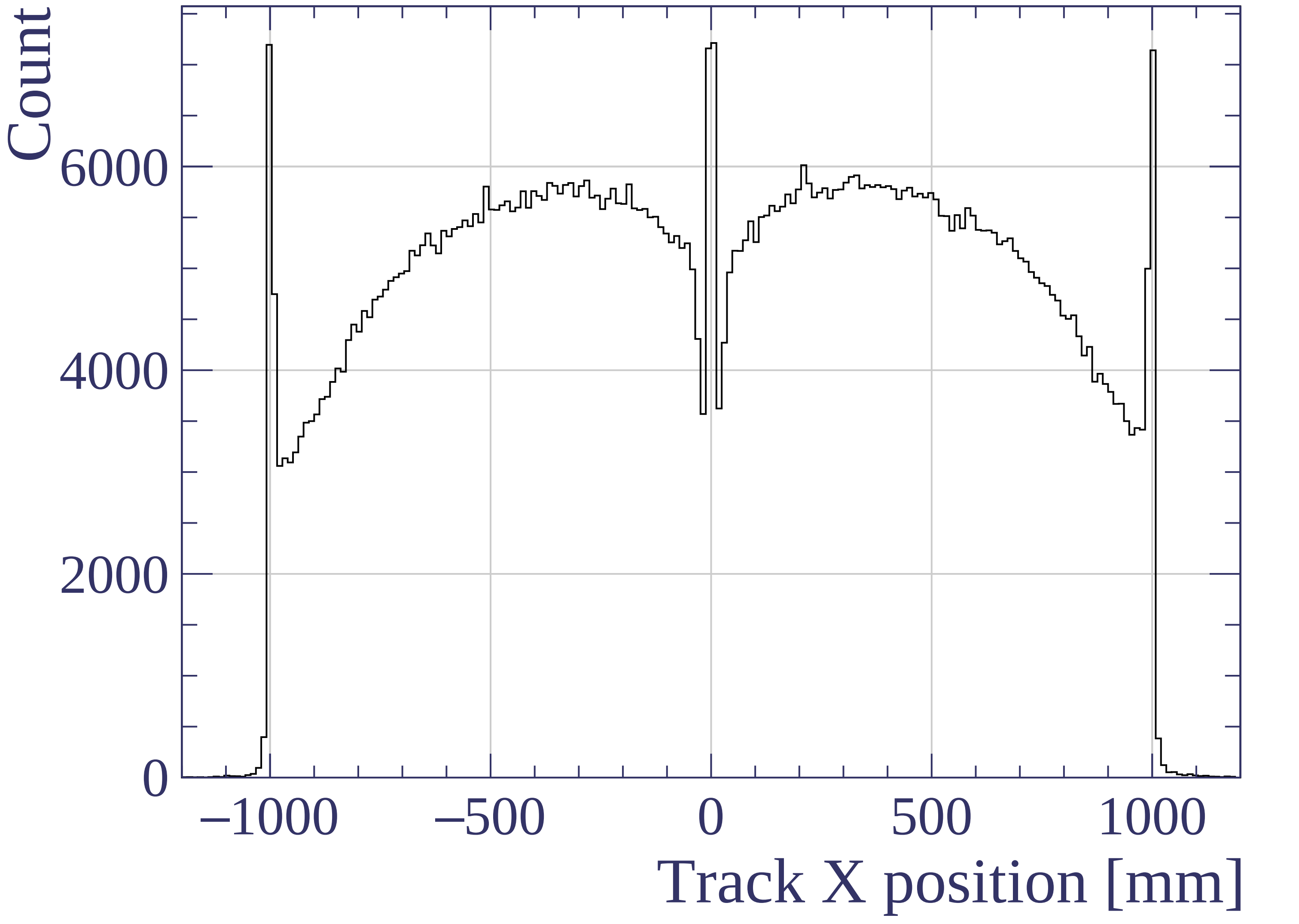

×10^3
Count
200
150
100
50
0
μ?
e?
p?
μ_μ = -0.046 ± 0.001
σ_μ = 0.849 ± 0.001
μ_e = -2.242 ± 0.002
σ_e = 1.178 ± 0.001
S(μ/e) = 2.14 ± 0.00
−20
−10
0
10
20
Pull

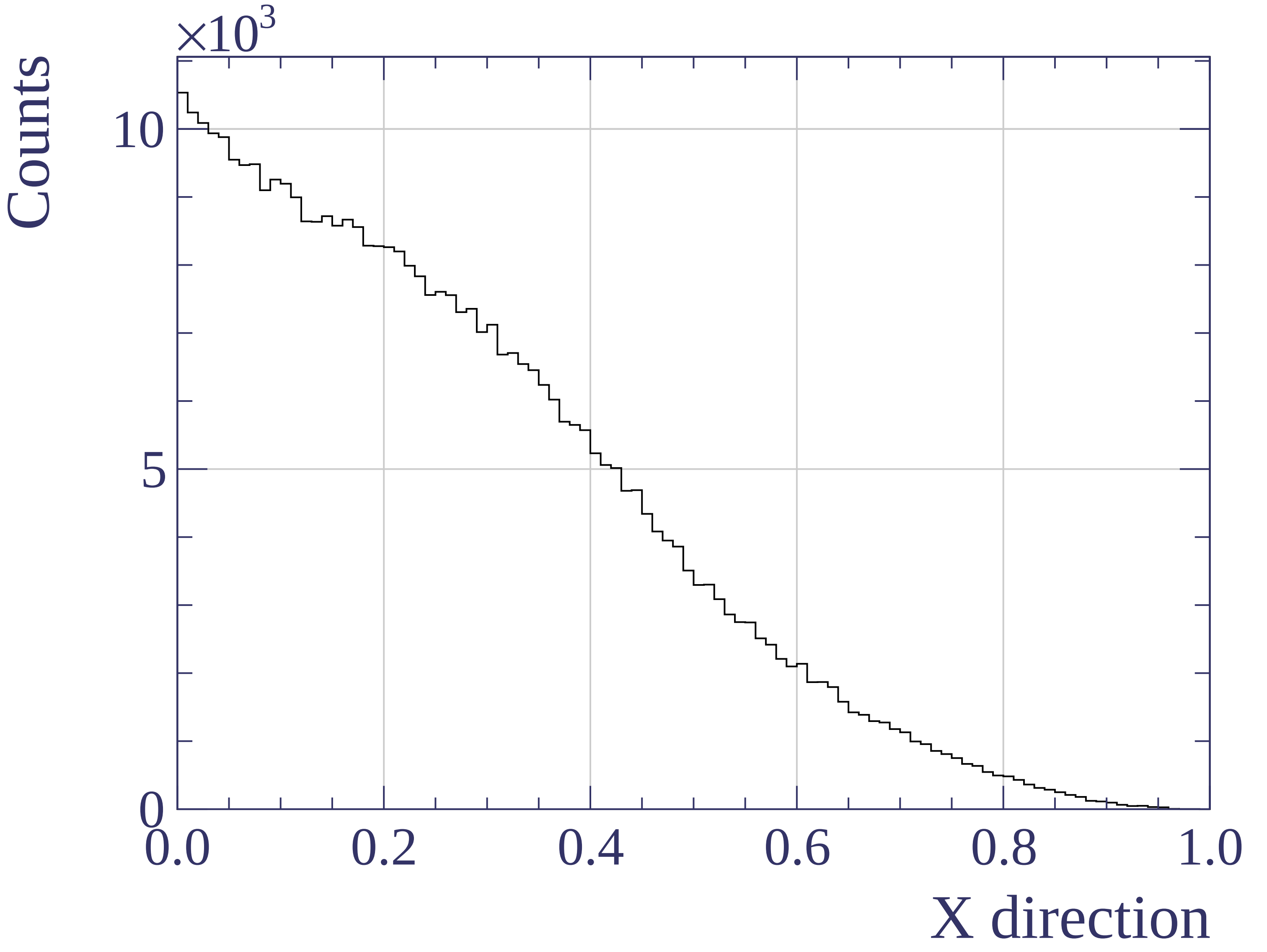


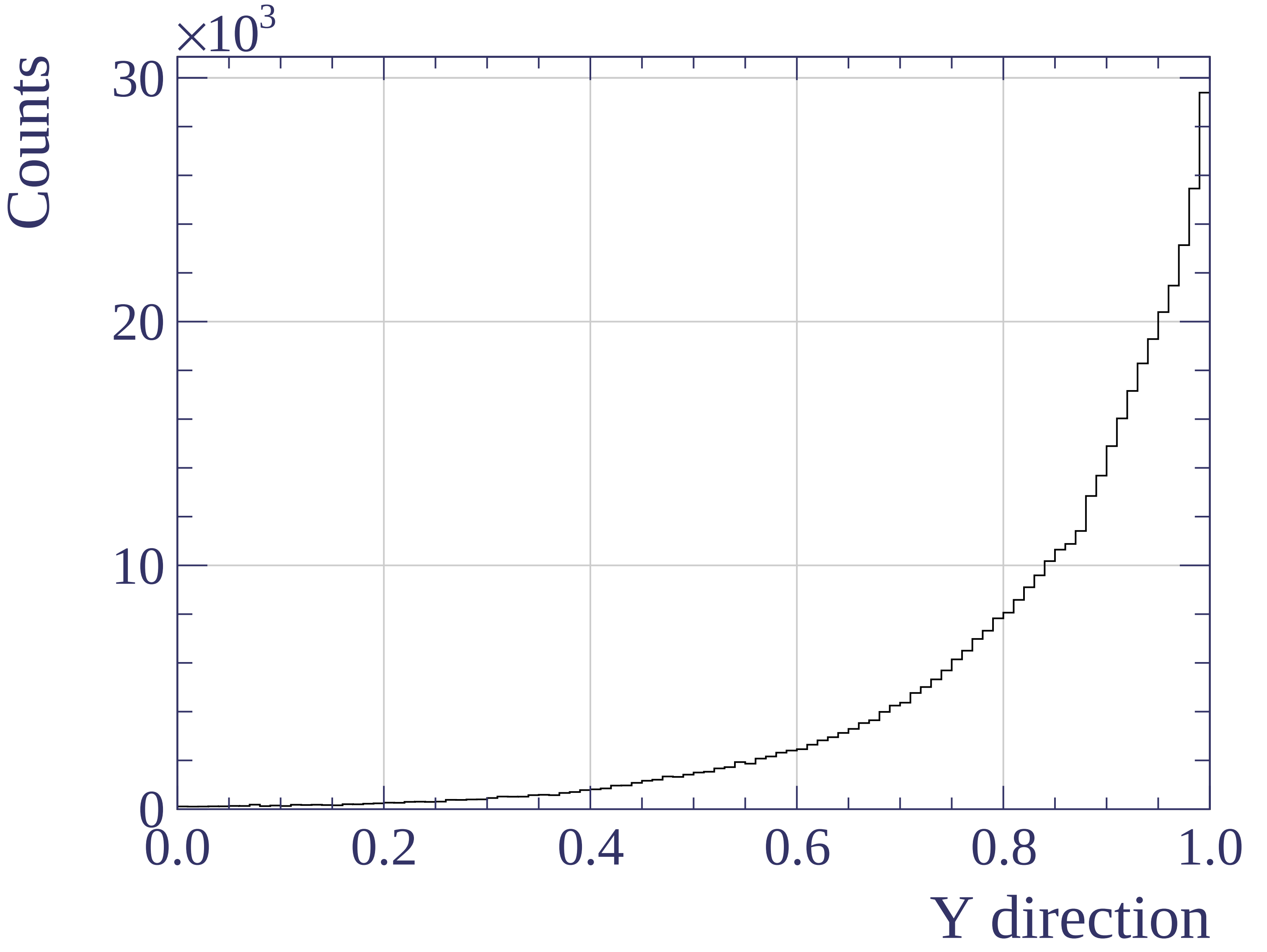
×10³
Count
80
60
40
20
0
0
50
100
150
Track length [cm]

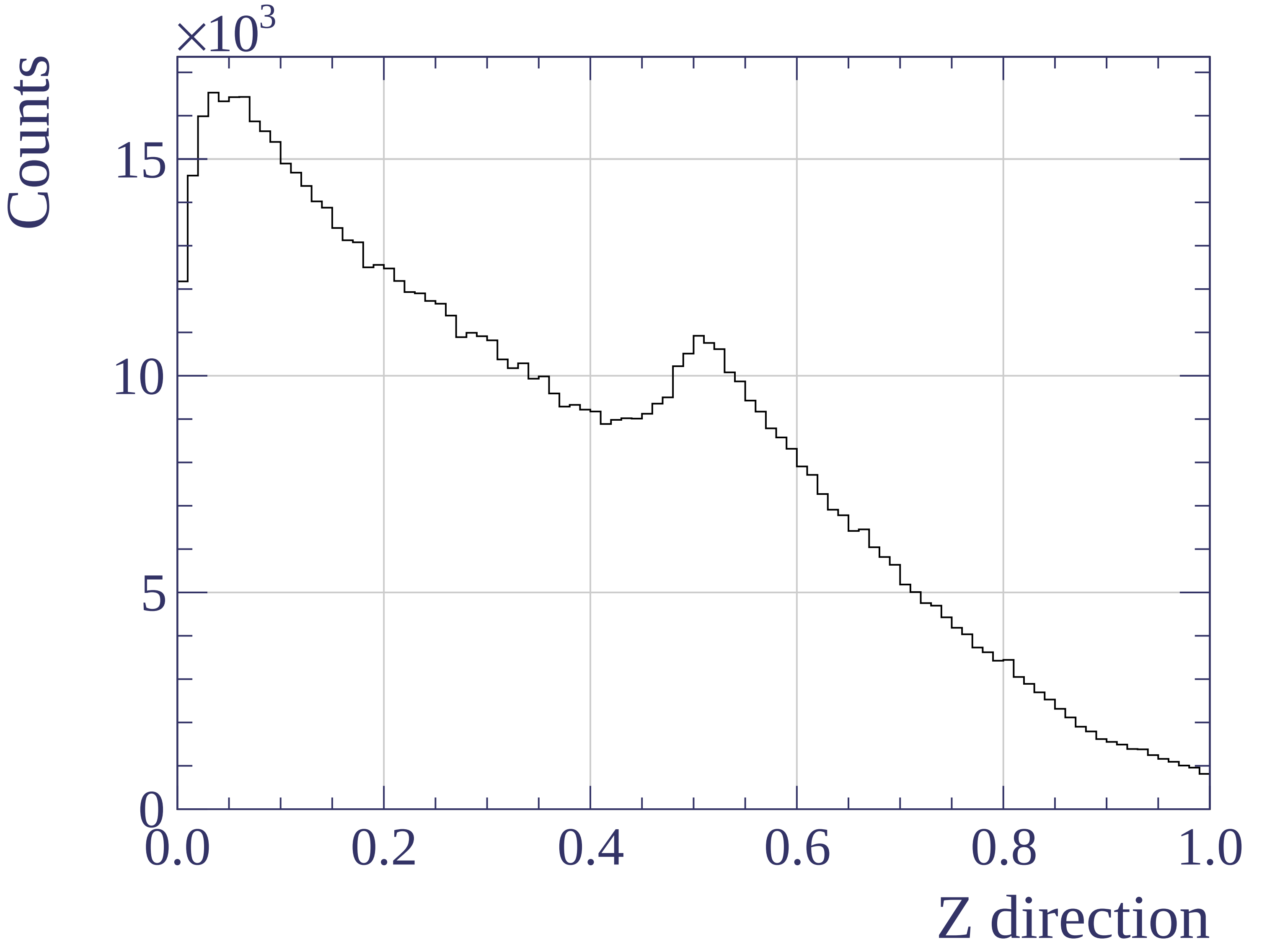


Count
8000
6000
4000
2000
0
0
10
20
30
40
50
χ^2

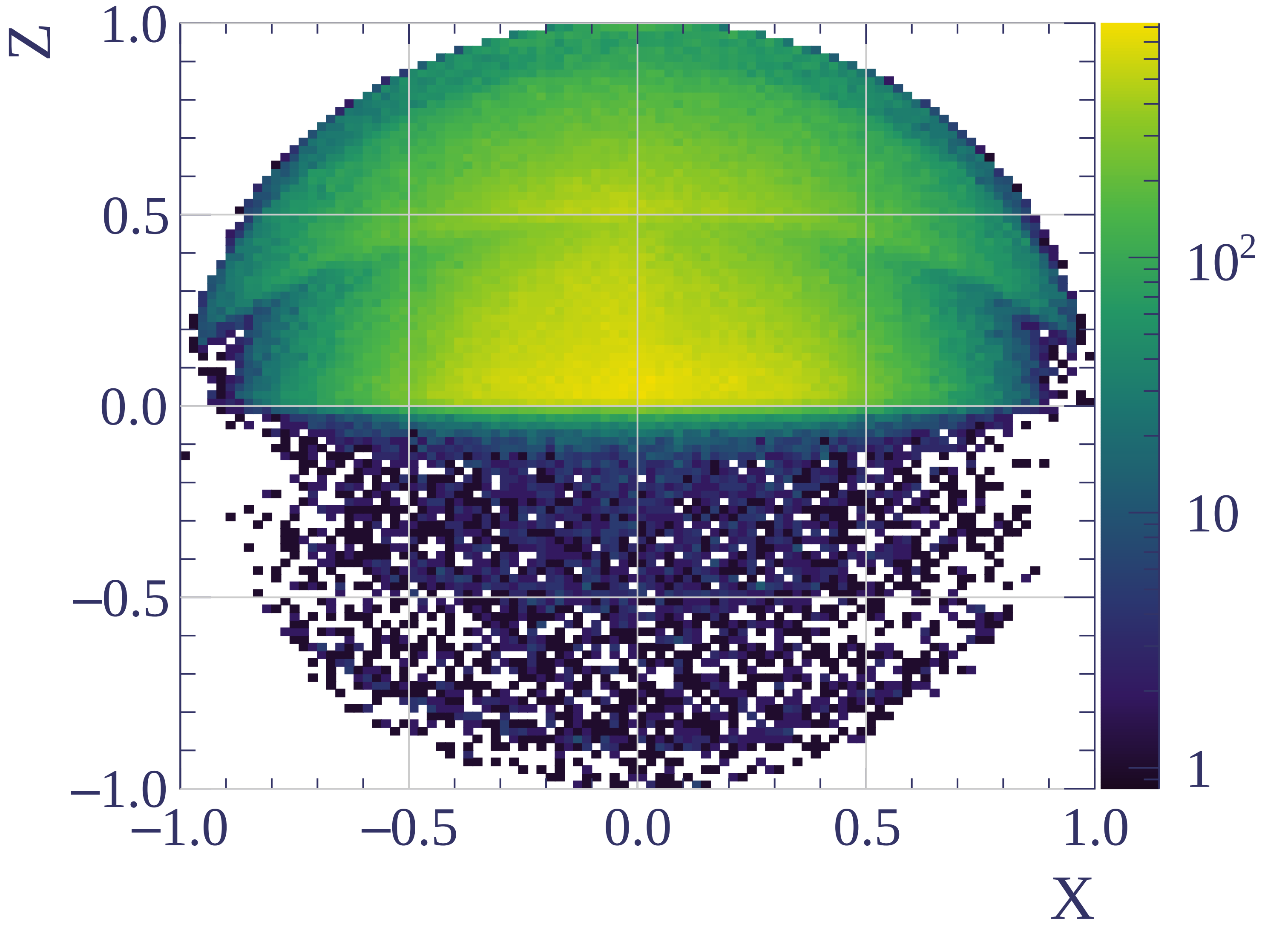

Count
0
2000
4000
6000
−1000
−500
0
500
1000
Track X position [mm]


×10^3
Counts
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

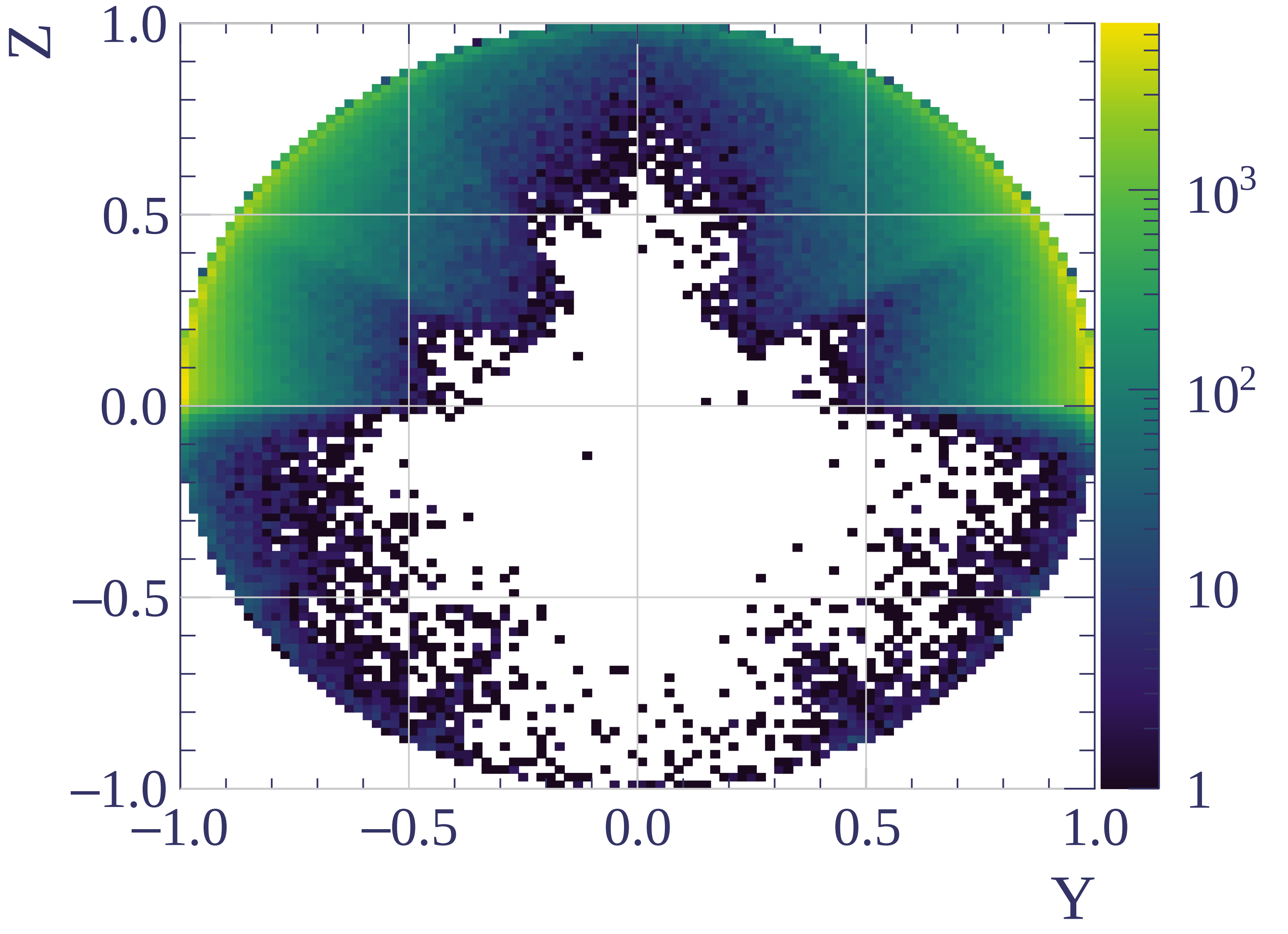

×10^3
Counts
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



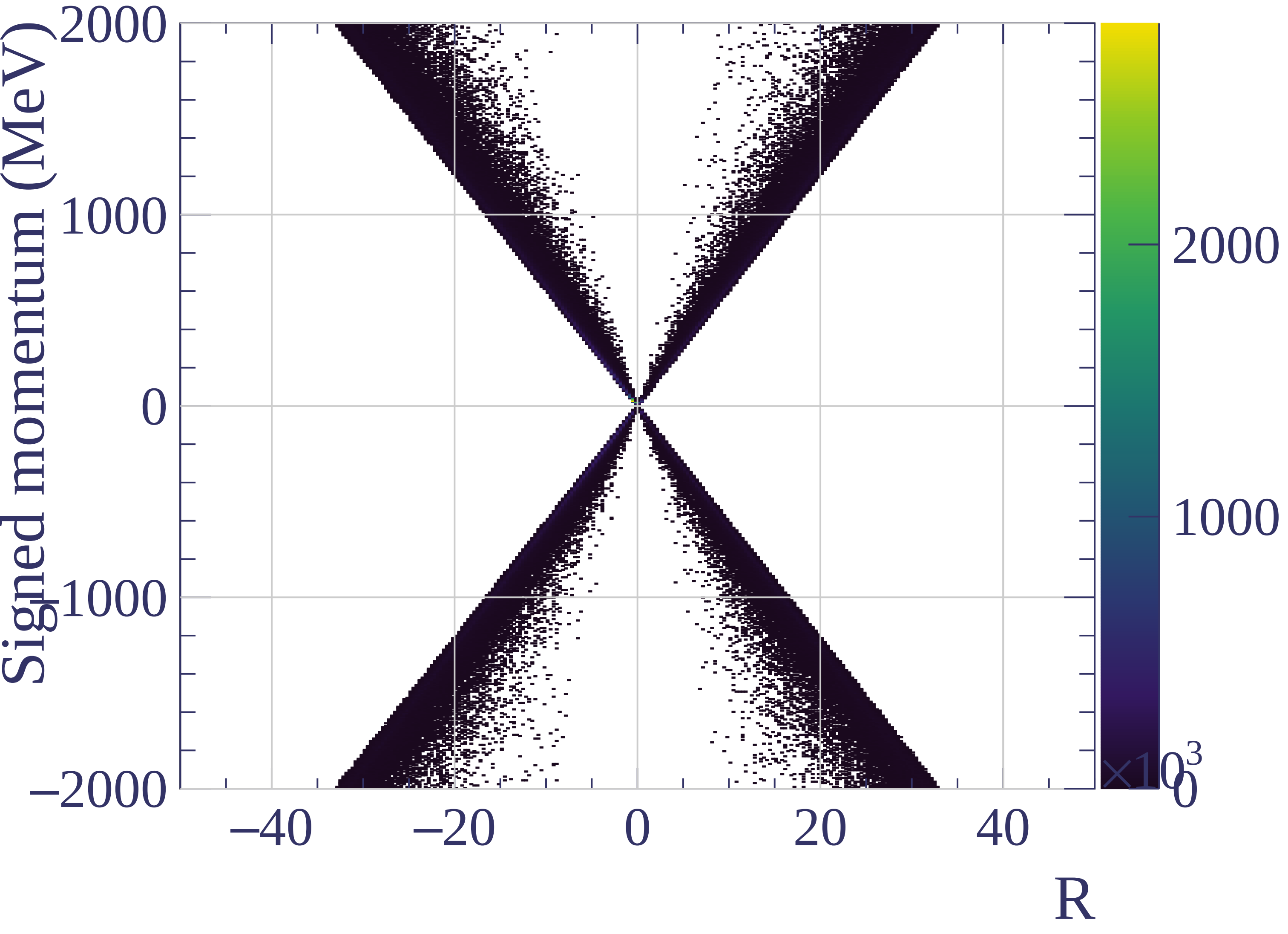
×10^3
Counts
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



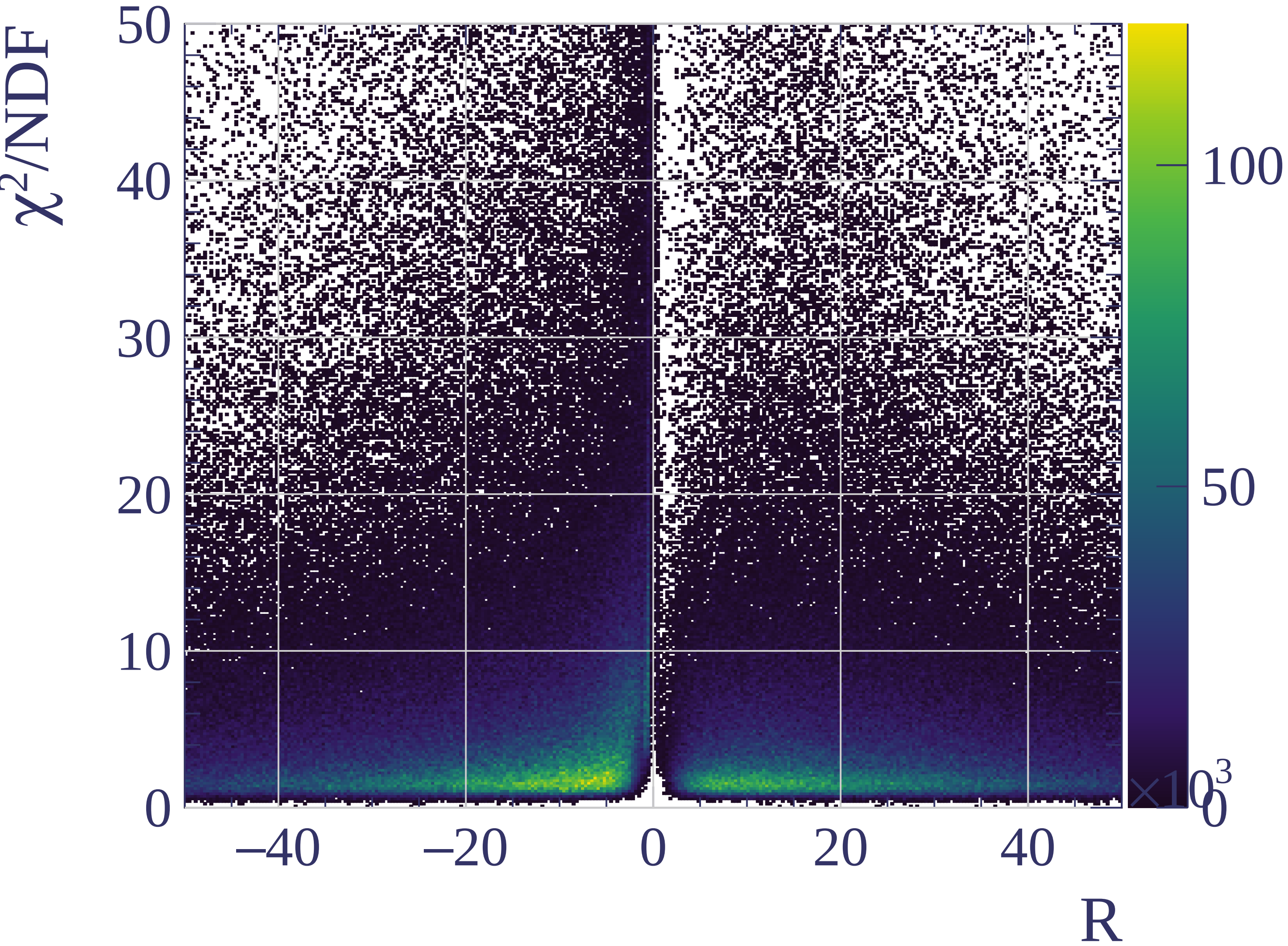
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^2
10
1



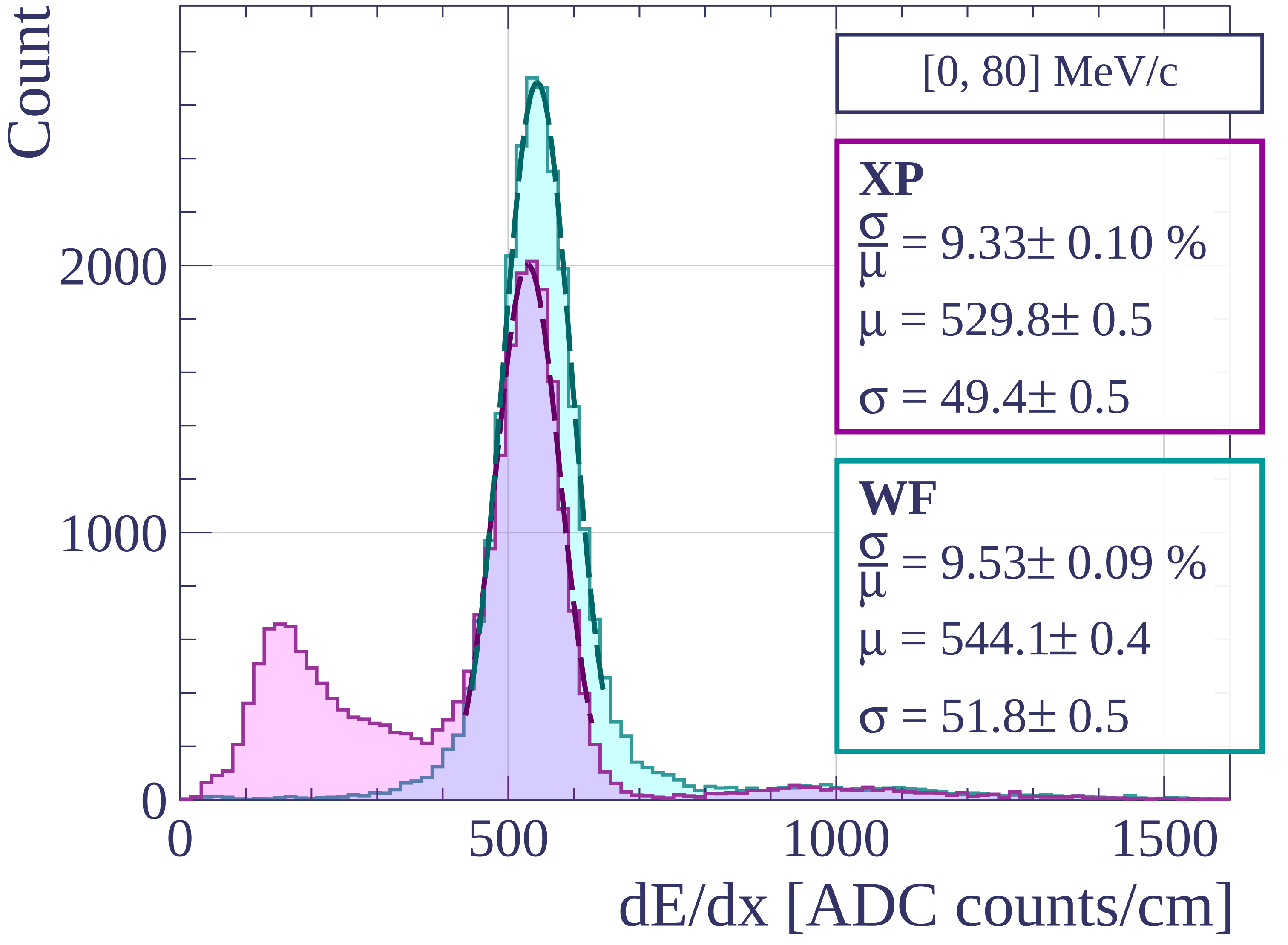
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^3
10^2
10
1



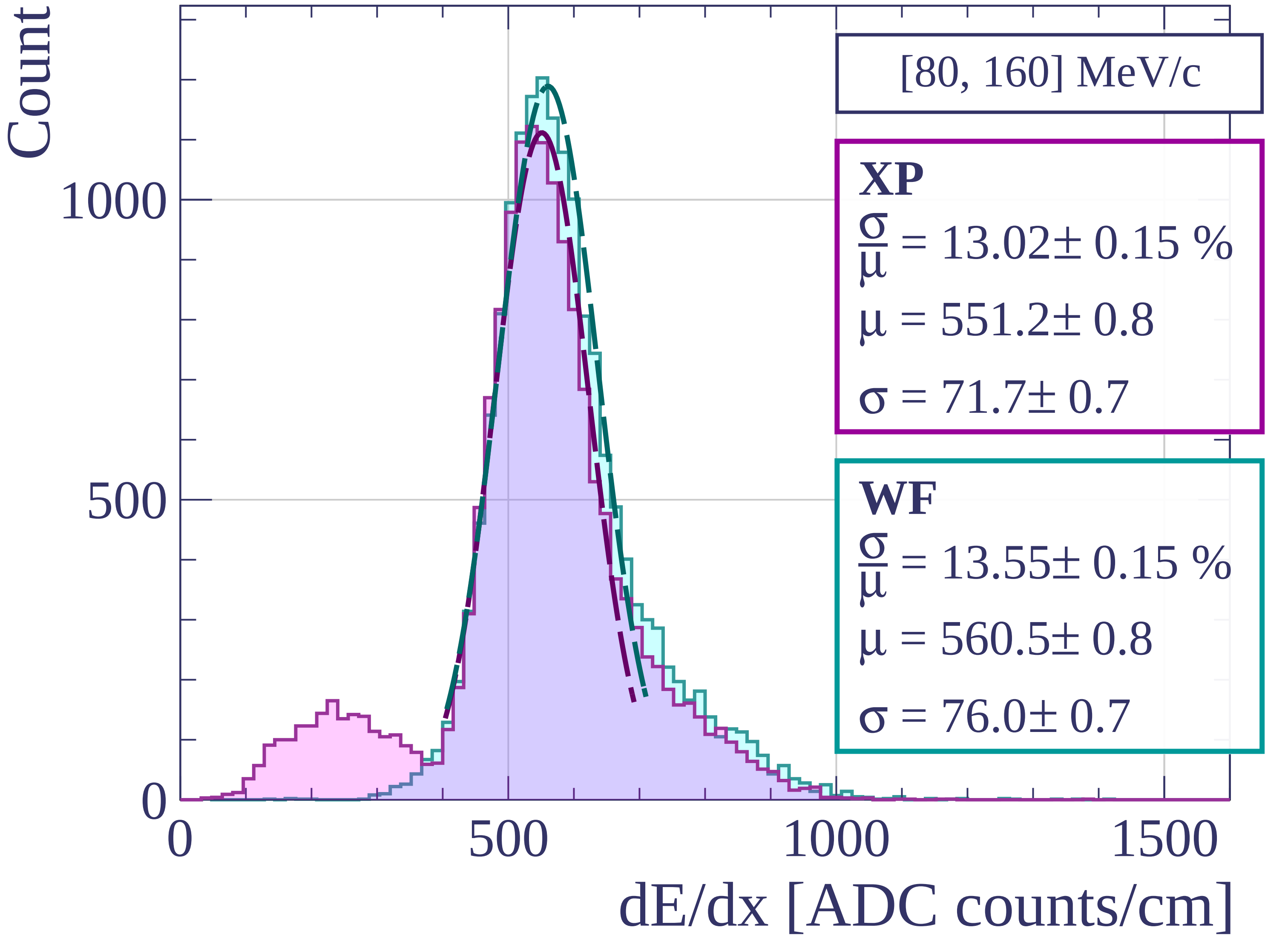
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
2000
1000
0
×10³



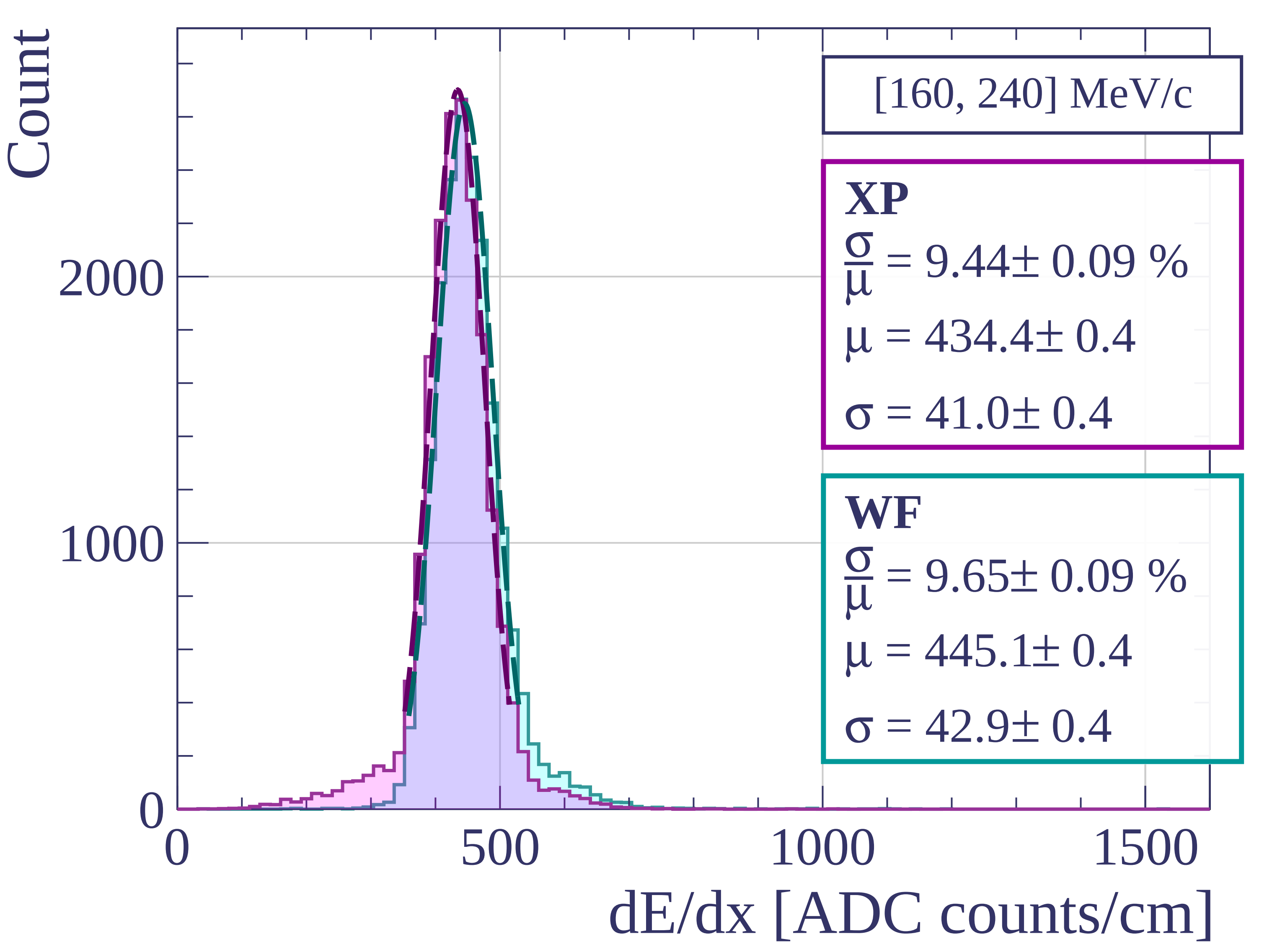
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
100
50
0
$\times 10^3$



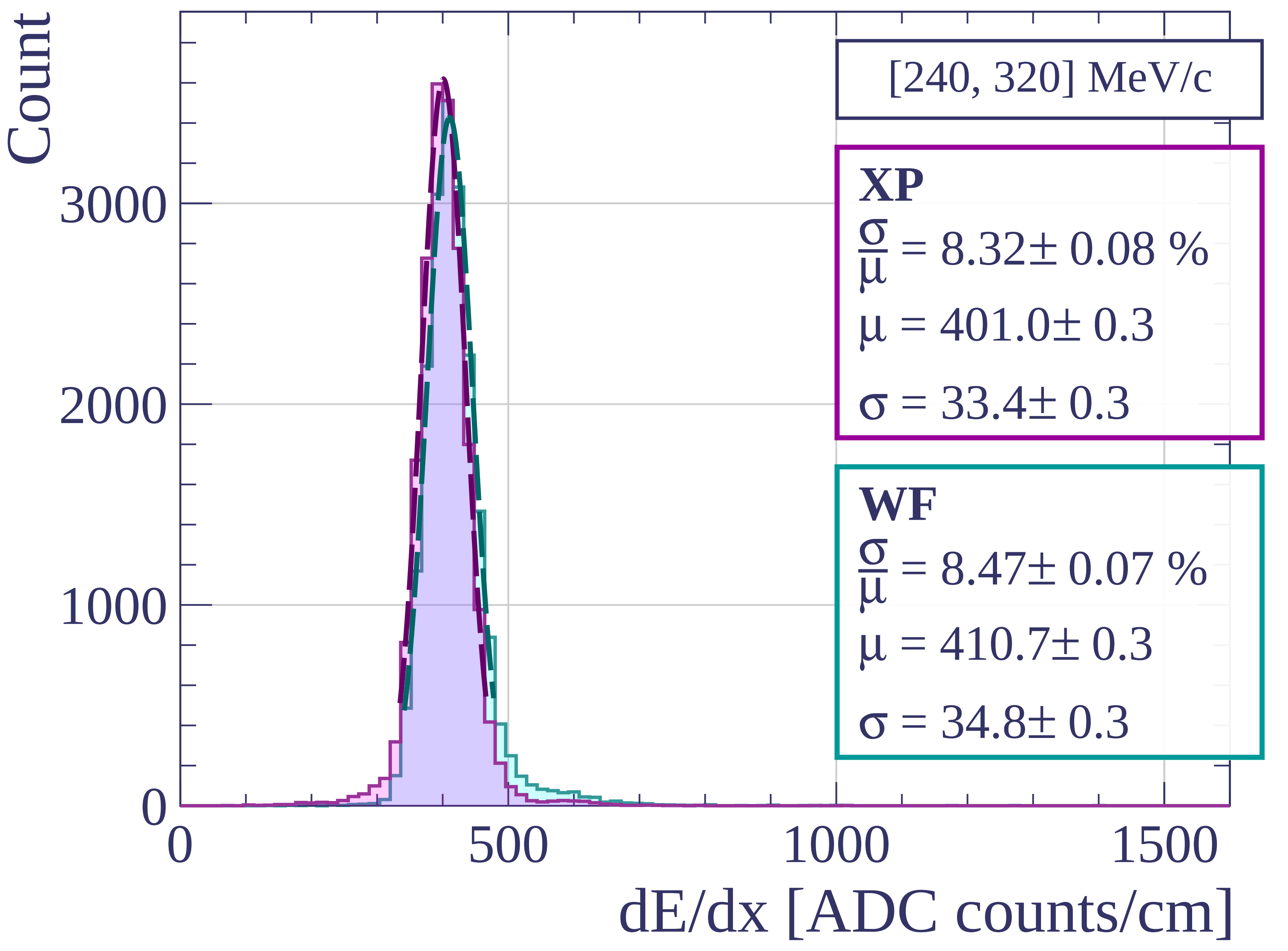
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.33± 0.10 %
μ = 529.8± 0.5
σ = 49.4± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.53± 0.09 %
μ = 544.1± 0.4
σ = 51.8± 0.5



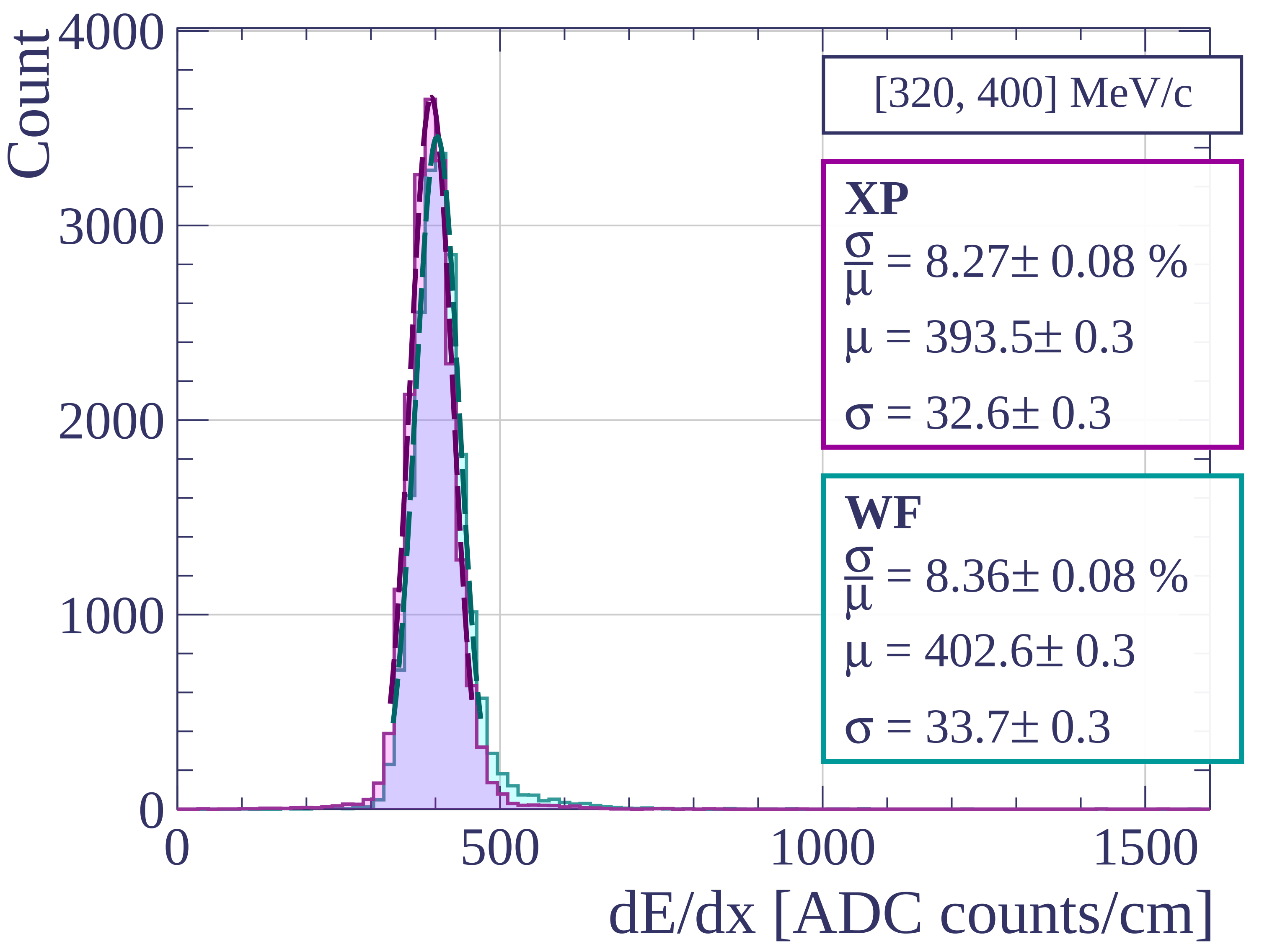
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 160] MeV/c
XP
σ/μ = 13.02± 0.15 %
μ = 551.2± 0.8
σ = 71.7± 0.7
WF
σ/μ = 13.55± 0.15 %
μ = 560.5± 0.8
σ = 76.0± 0.7



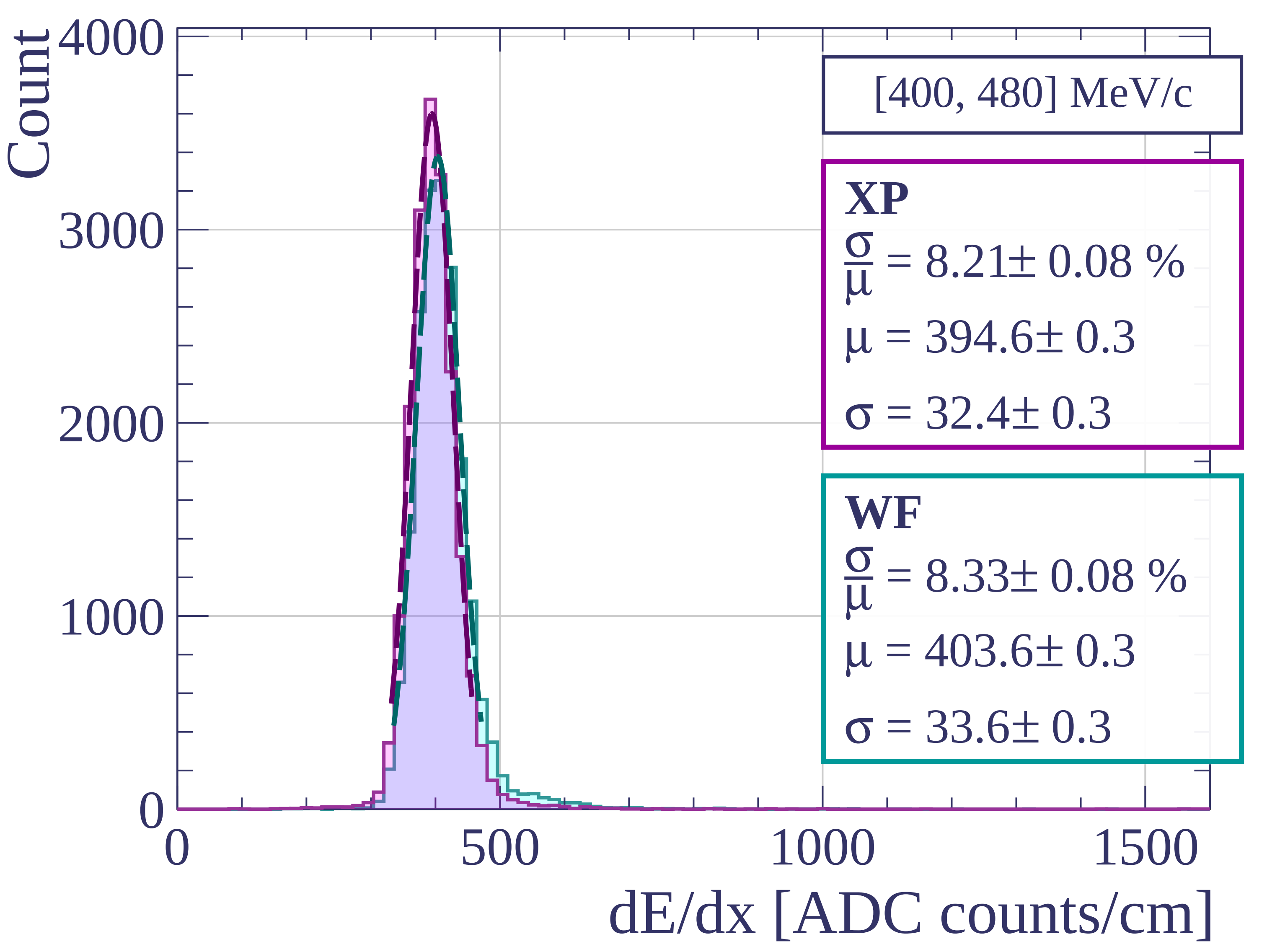
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.44± 0.09 %
μ = 434.4± 0.4
σ = 41.0± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.65± 0.09 %
μ = 445.1± 0.4
σ = 42.9± 0.4



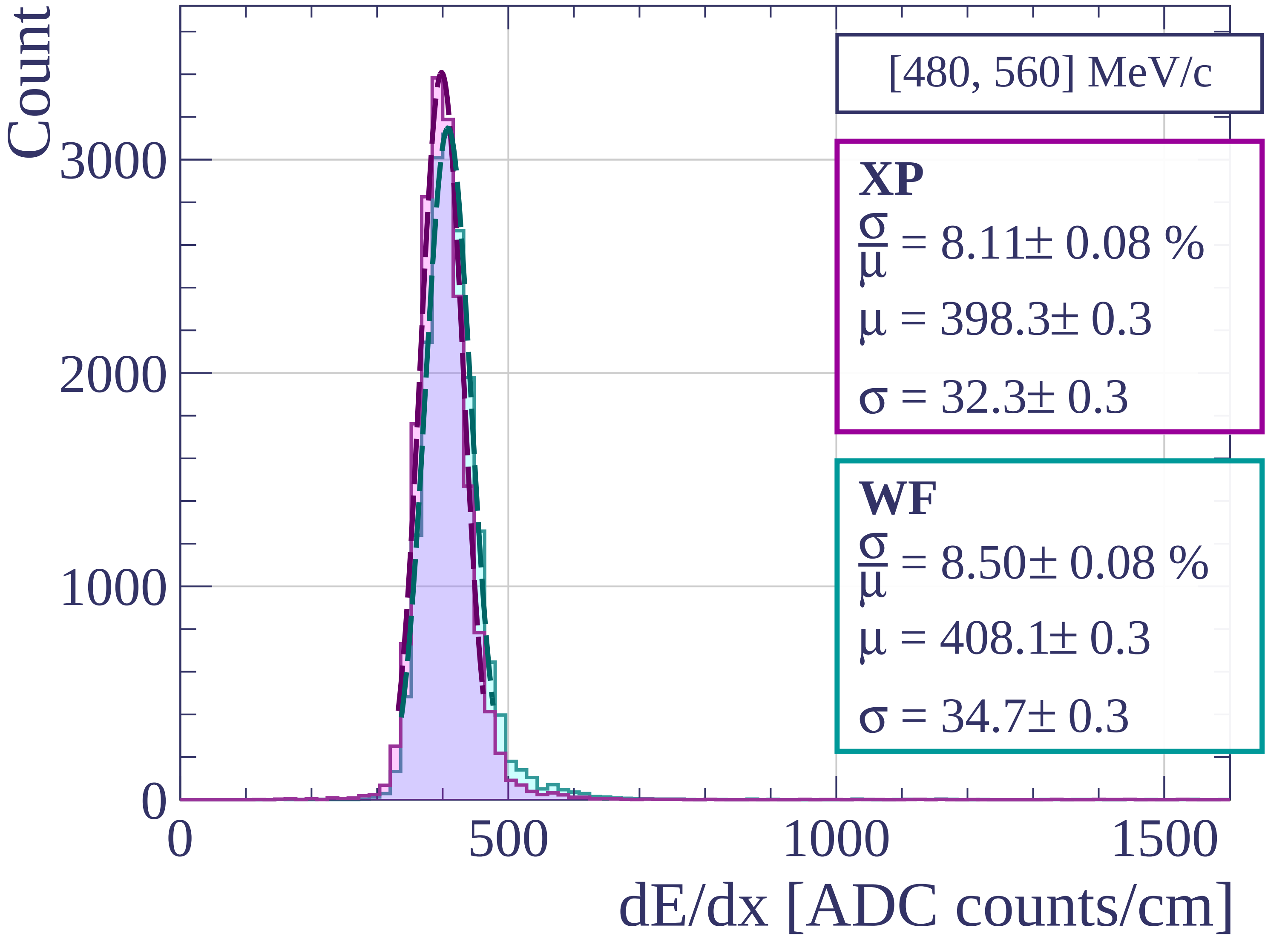
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 320] MeV/c
XP
σ/μ = 8.32± 0.08 %
μ = 401.0± 0.3
σ = 33.4± 0.3
WF
σ/μ = 8.47± 0.07 %
μ = 410.7± 0.3
σ = 34.8± 0.3



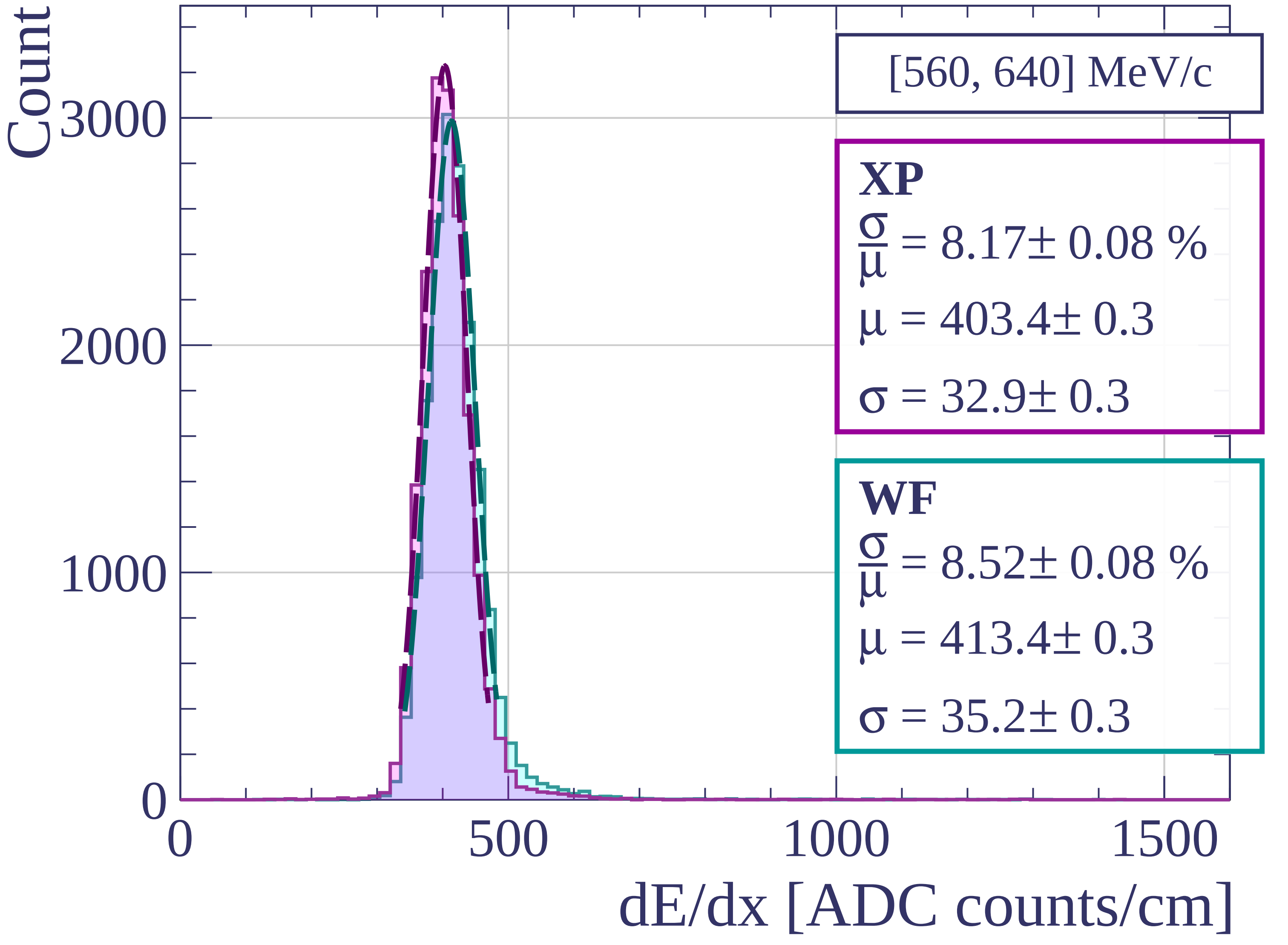
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.27± 0.08 %
μ = 393.5± 0.3
σ = 32.6± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.36± 0.08 %
μ = 402.6± 0.3
σ = 33.7± 0.3



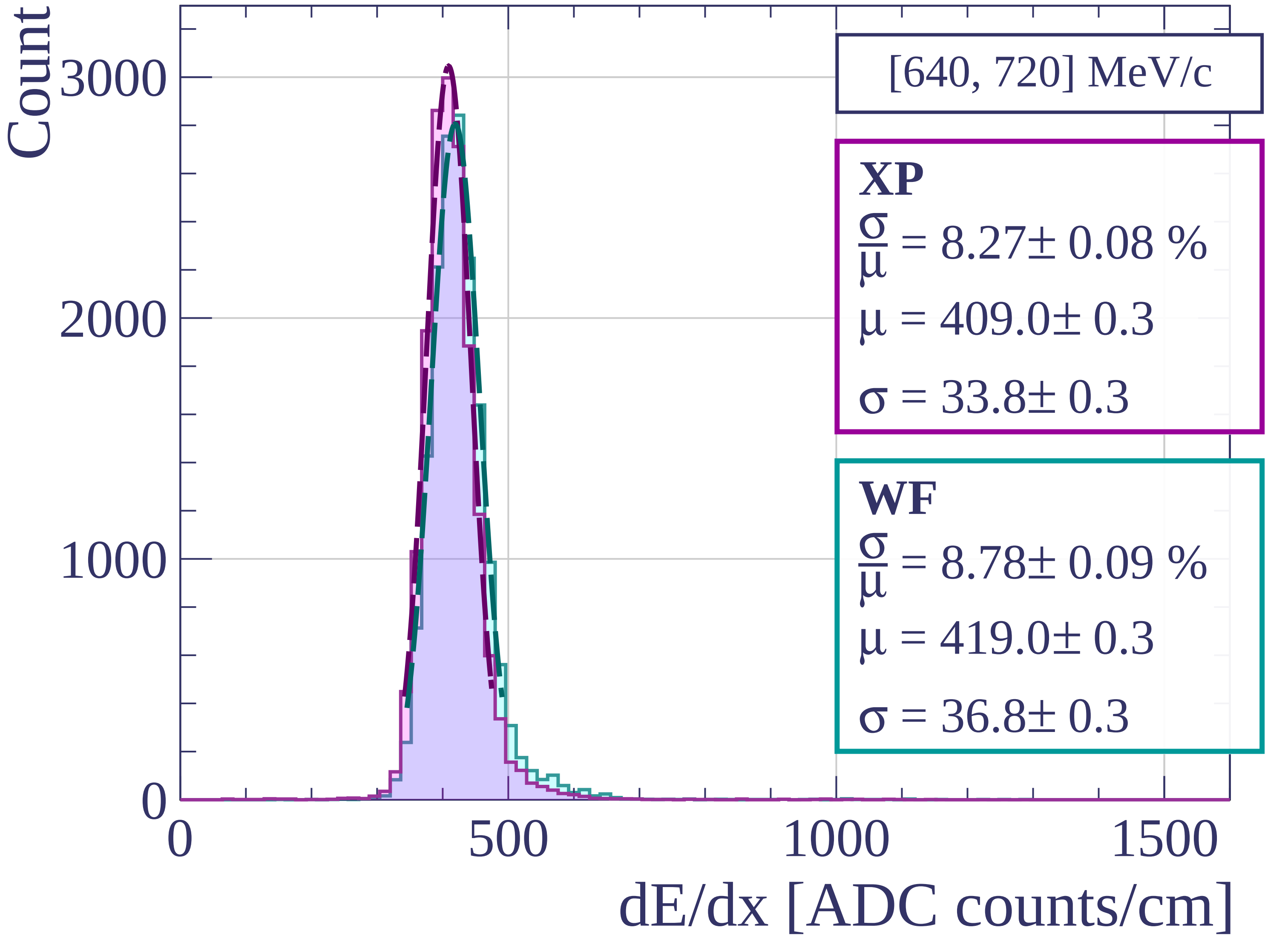
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 480] MeV/c
XP
σ/μ = 8.21± 0.08 %
μ = 394.6± 0.3
σ = 32.4± 0.3
WF
σ/μ = 8.33± 0.08 %
μ = 403.6± 0.3
σ = 33.6± 0.3



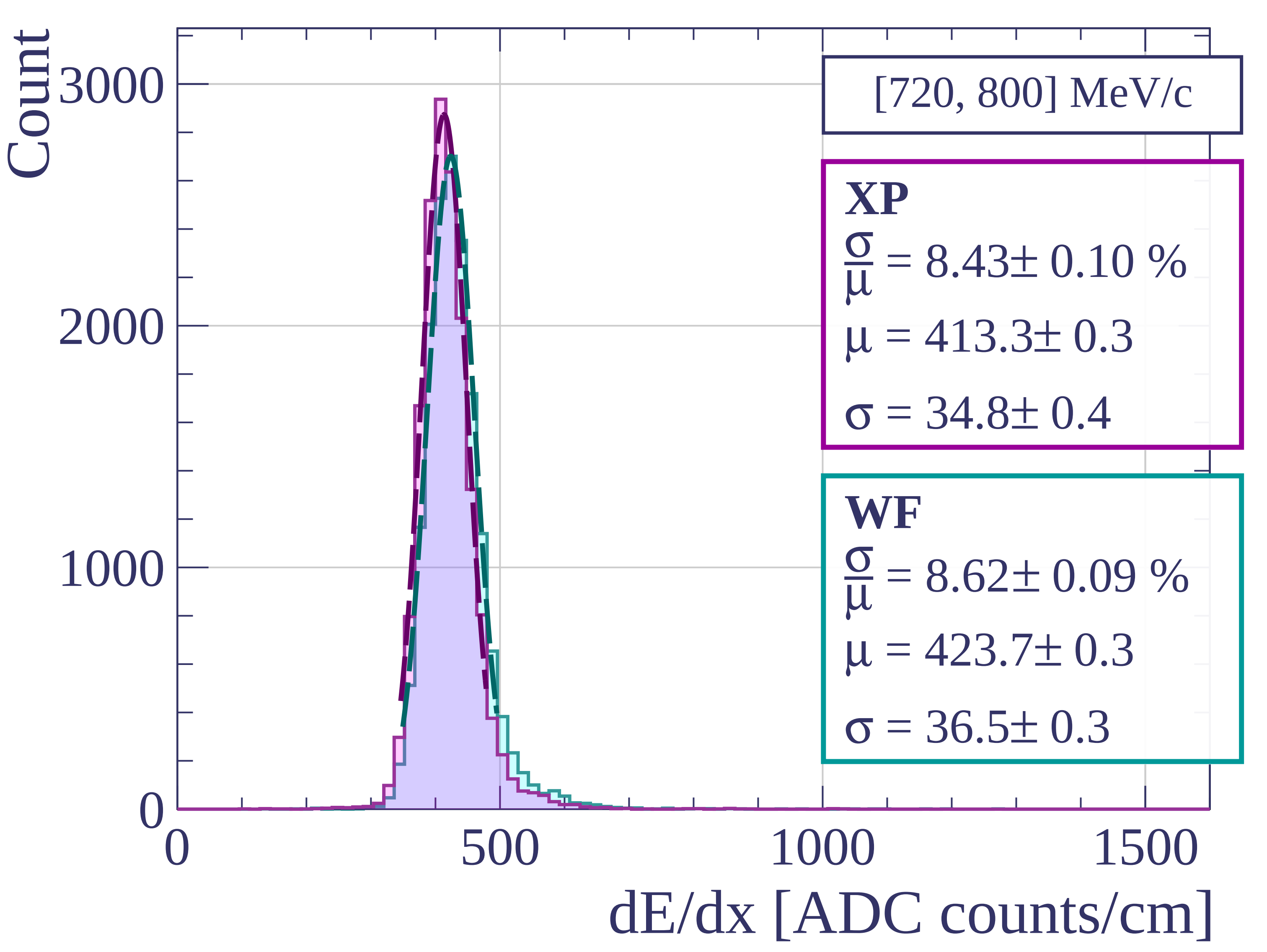
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.11± 0.08 %
μ = 398.3± 0.3
σ = 32.3± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.50± 0.08 %
μ = 408.1± 0.3
σ = 34.7± 0.3



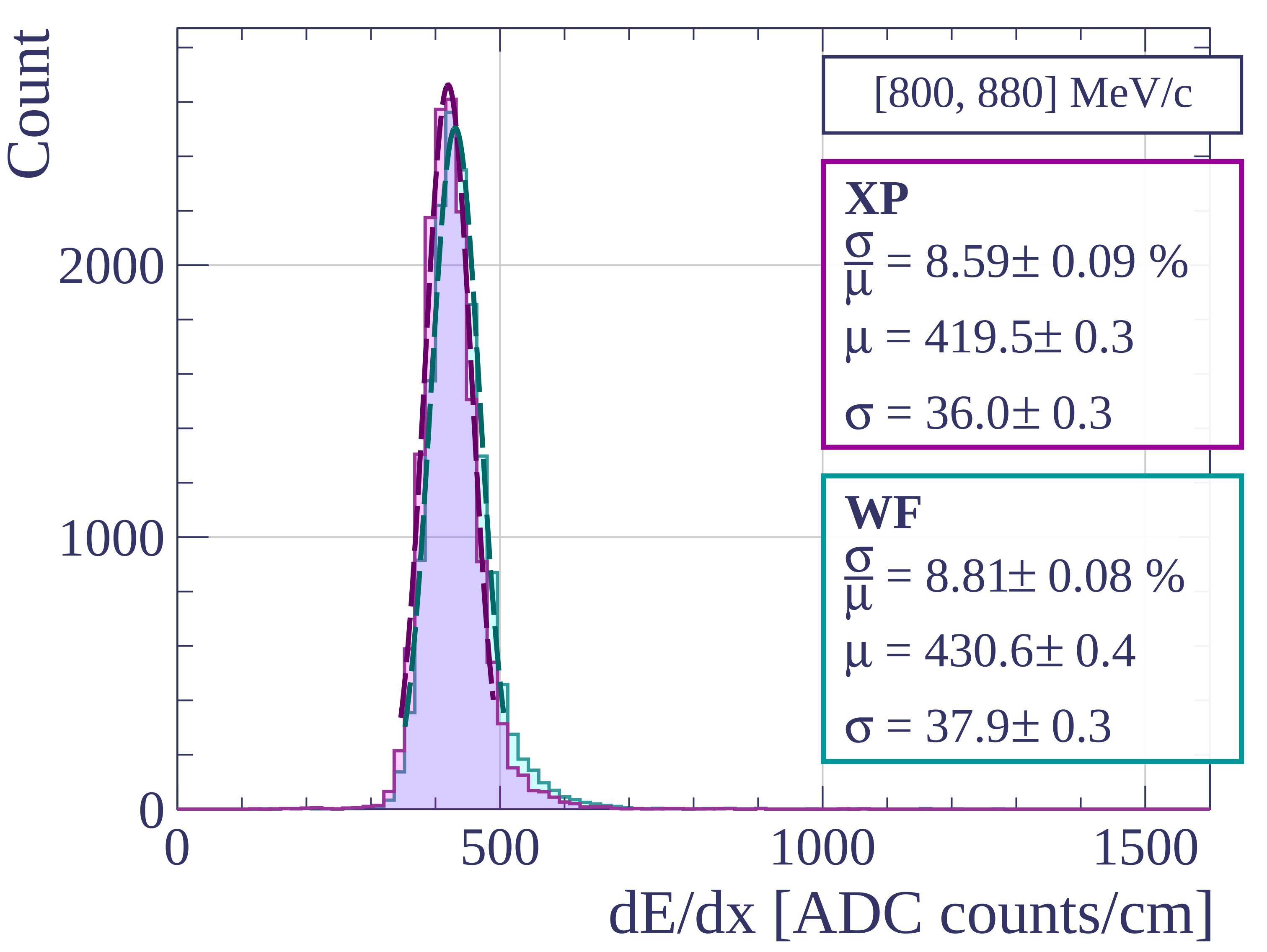
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 640] MeV/c
XP
σ/μ = 8.17± 0.08 %
μ = 403.4± 0.3
σ = 32.9± 0.3
WF
σ/μ = 8.52± 0.08 %
μ = 413.4± 0.3
σ = 35.2± 0.3



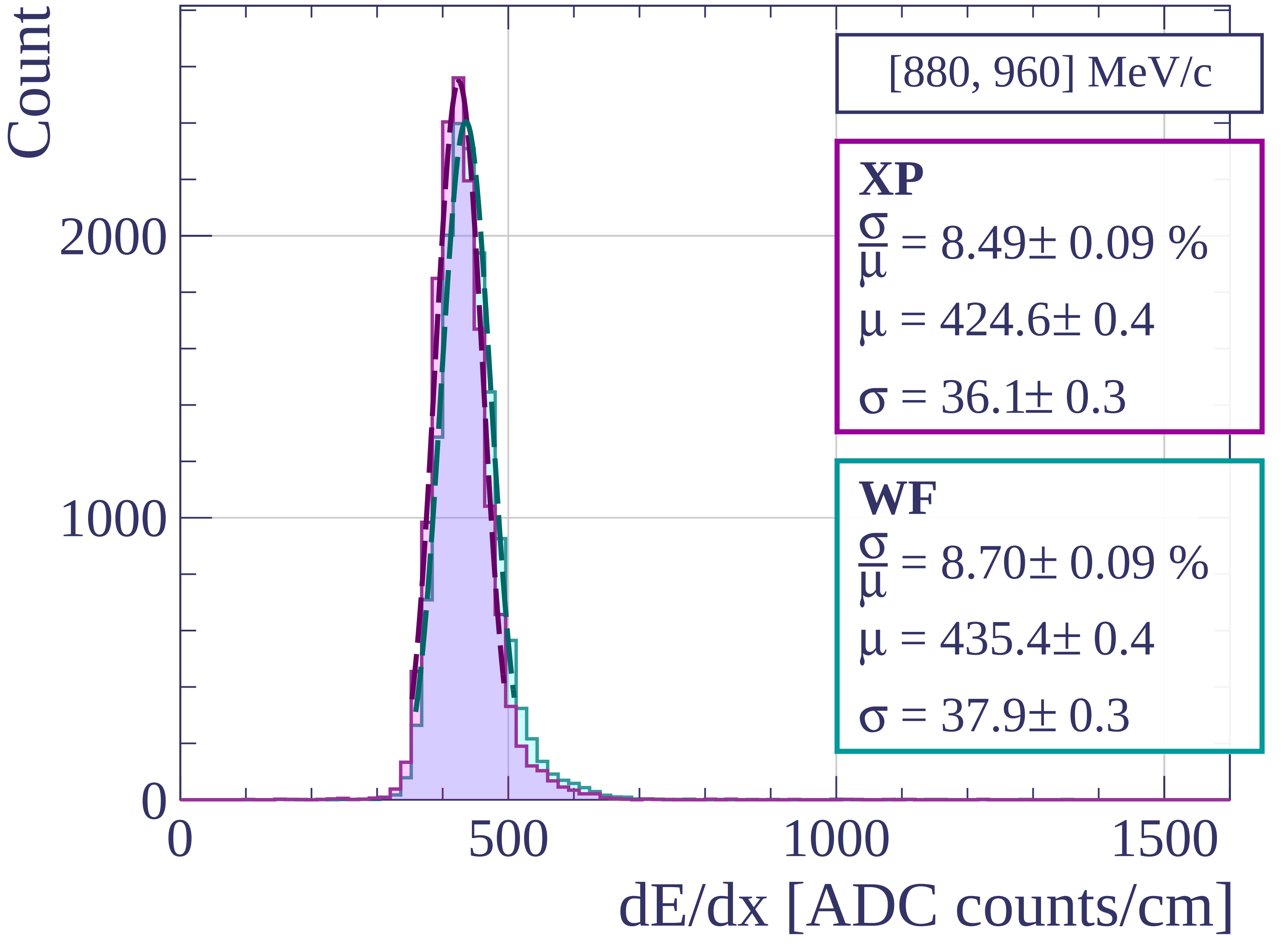
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 720] MeV/c
XP
σ/μ = 8.27± 0.08 %
μ = 409.0± 0.3
σ = 33.8± 0.3
WF
σ/μ = 8.78± 0.09 %
μ = 419.0± 0.3
σ = 36.8± 0.3



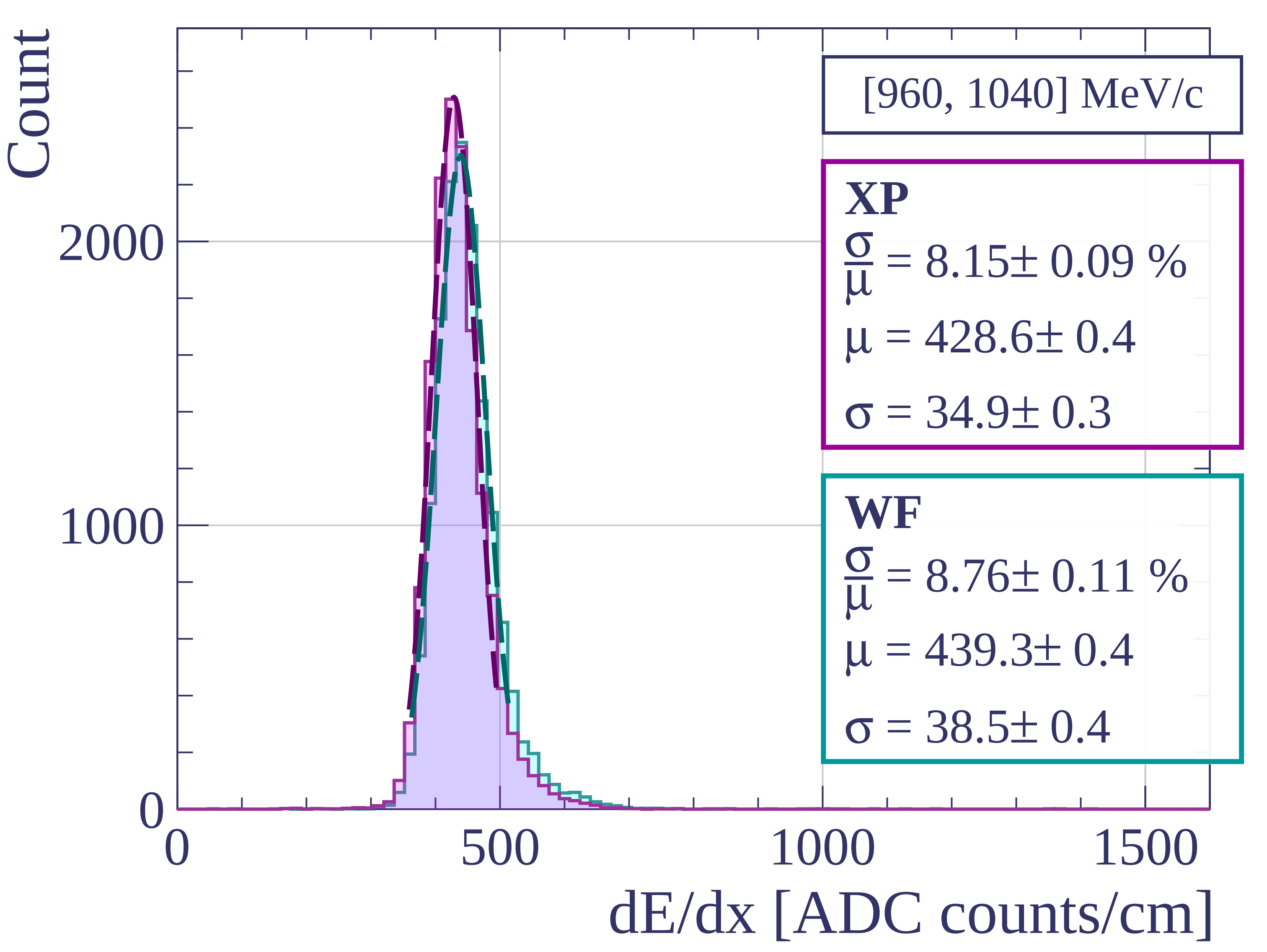
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.43± 0.10 %
μ = 413.3± 0.3
σ = 34.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.09 %
μ = 423.7± 0.3
σ = 36.5± 0.3



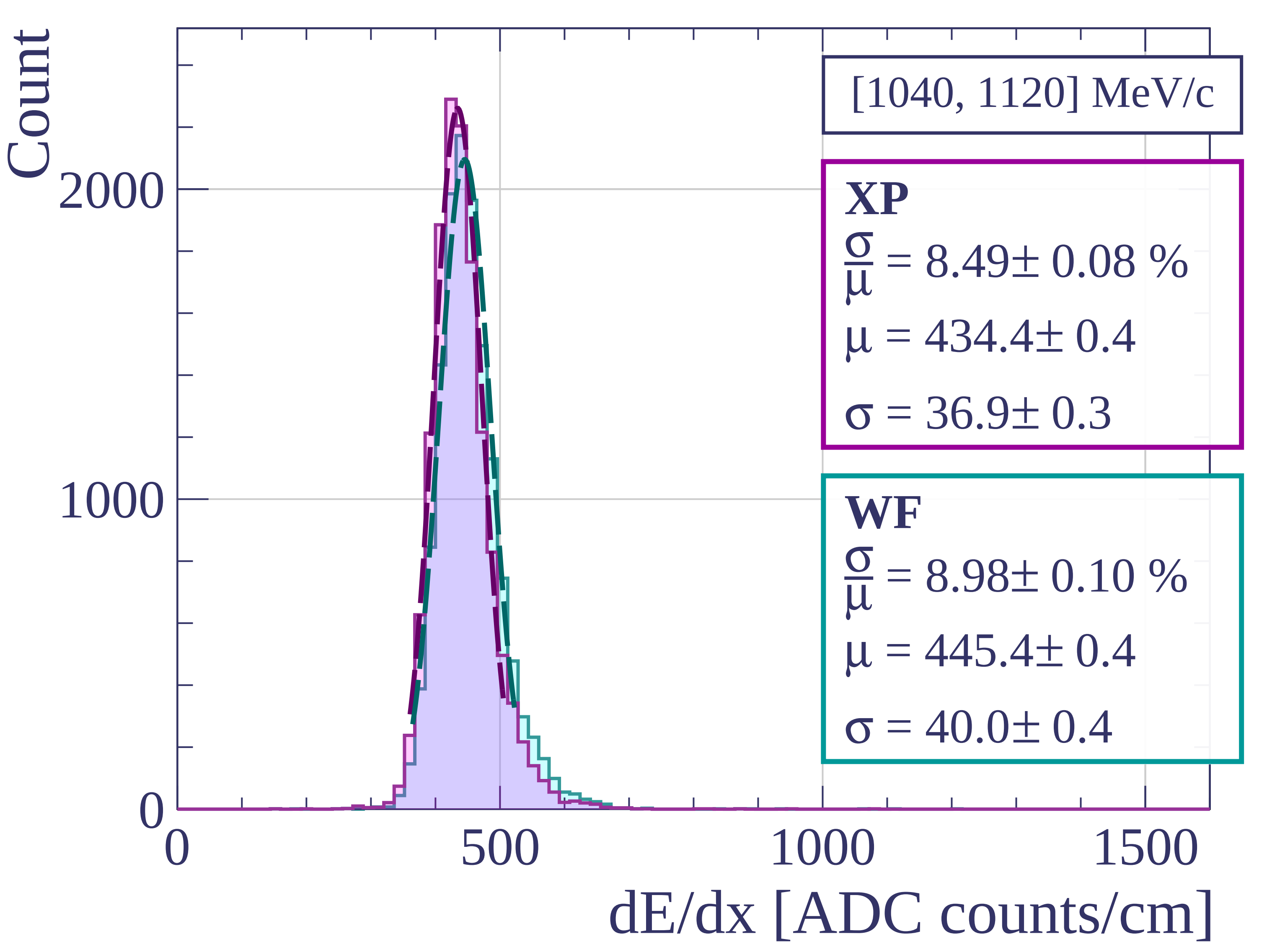
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 880] MeV/c
XP
σ/μ = 8.59± 0.09 %
μ = 419.5± 0.3
σ = 36.0± 0.3
WF
σ/μ = 8.81± 0.08 %
μ = 430.6± 0.4
σ = 37.9± 0.3

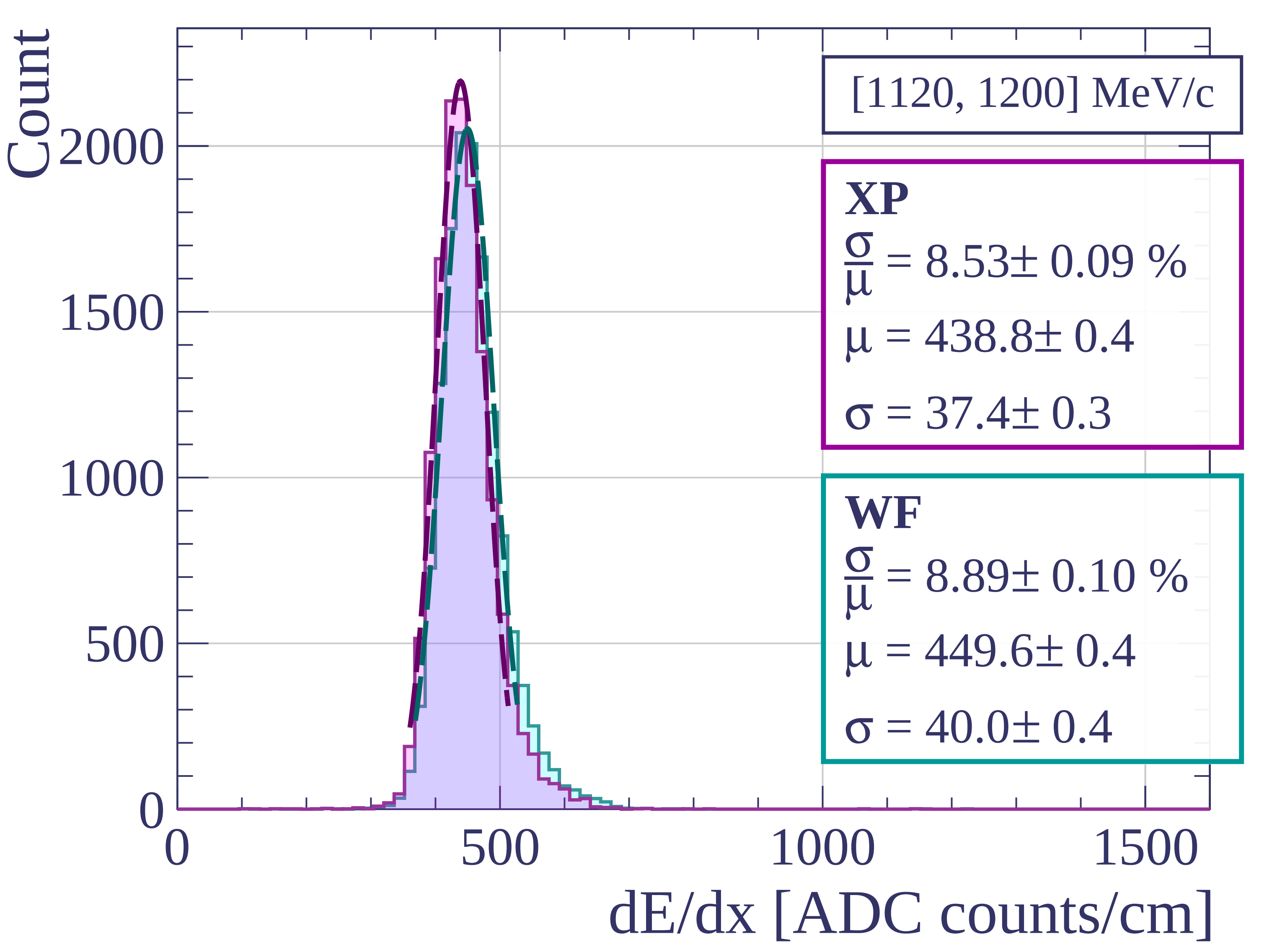


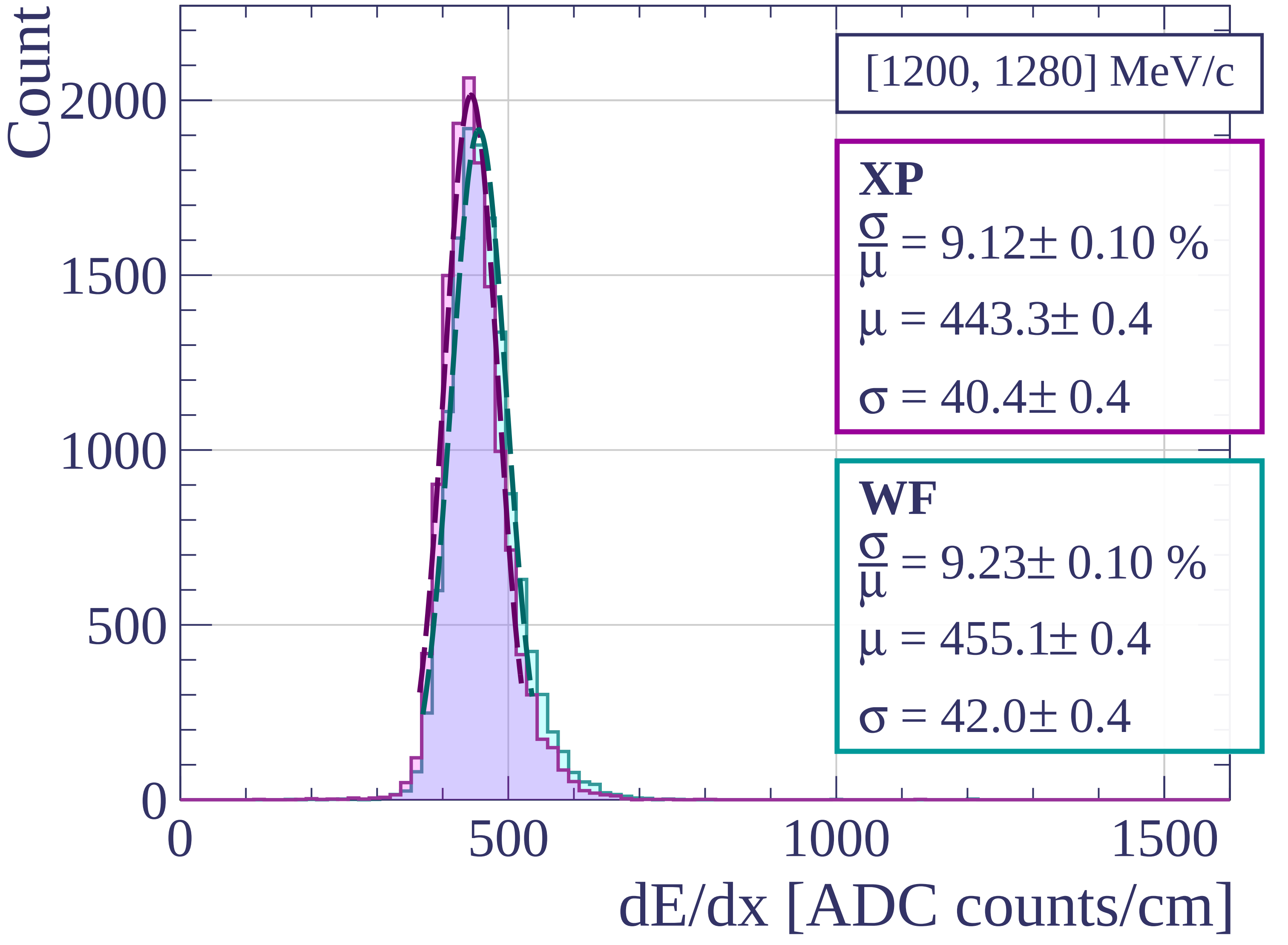
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.49± 0.09 %
μ = 424.6± 0.4
σ = 36.1± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.70± 0.09 %
μ = 435.4± 0.4
σ = 37.9± 0.3

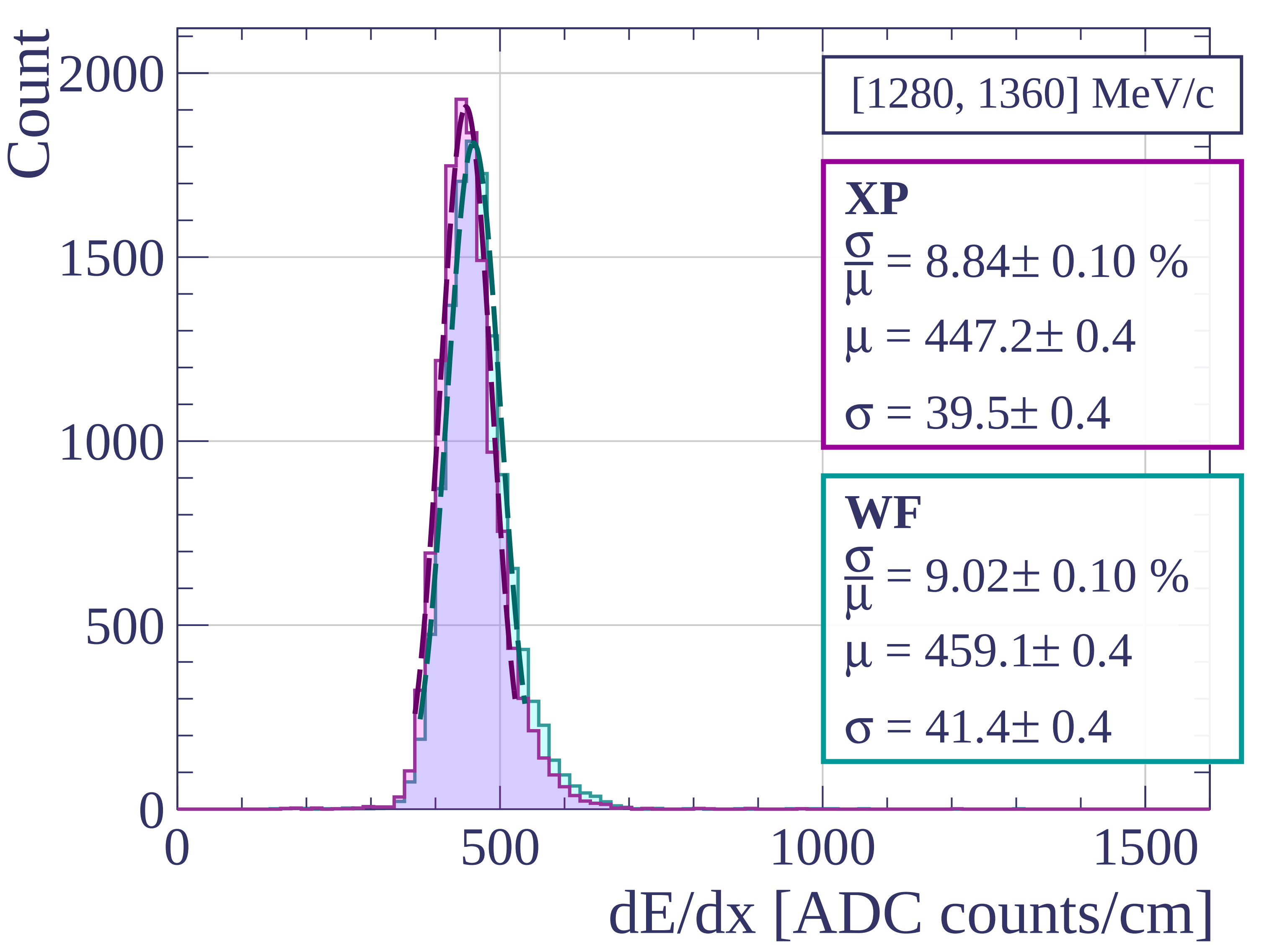


Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.15± 0.09 %
μ = 428.6± 0.4
σ = 34.9± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.76± 0.11 %
μ = 439.3± 0.4
σ = 38.5± 0.4

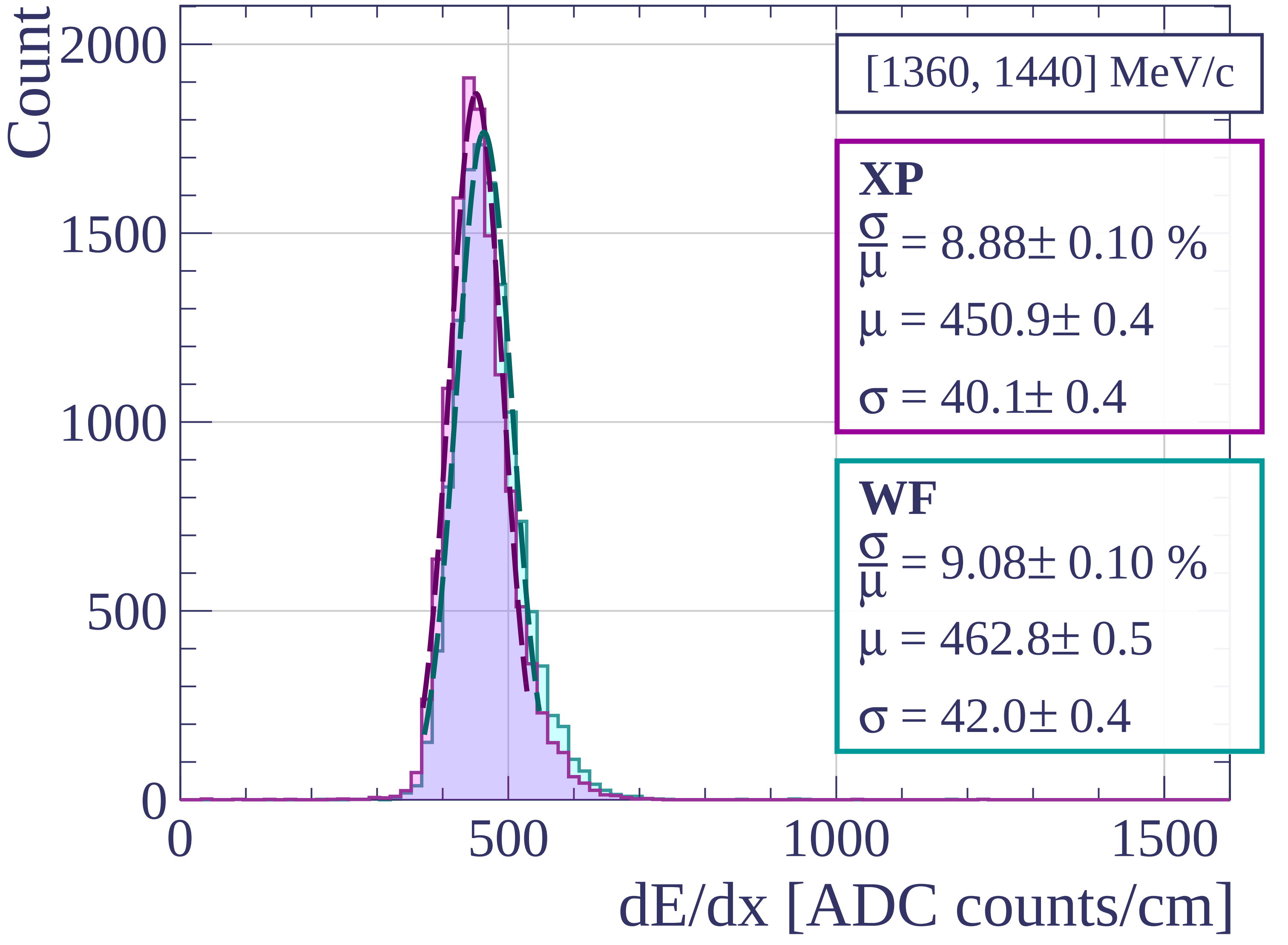

Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.49± 0.08 %
μ = 434.4± 0.4
σ = 36.9± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.98± 0.10 %
μ = 445.4± 0.4
σ = 40.0± 0.4

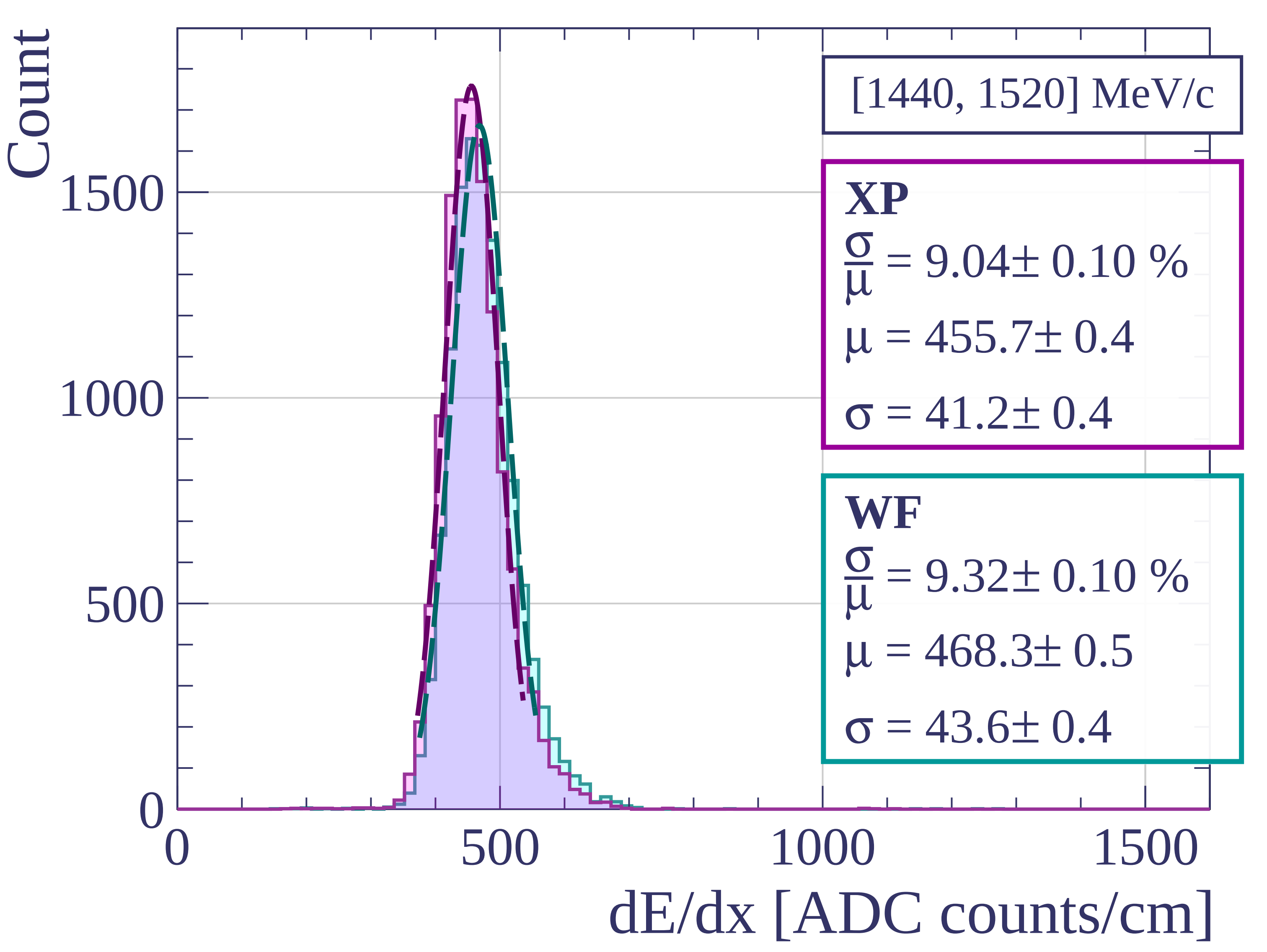

Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.53± 0.09 %
μ = 438.8± 0.4
σ = 37.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.89± 0.10 %
μ = 449.6± 0.4
σ = 40.0± 0.4

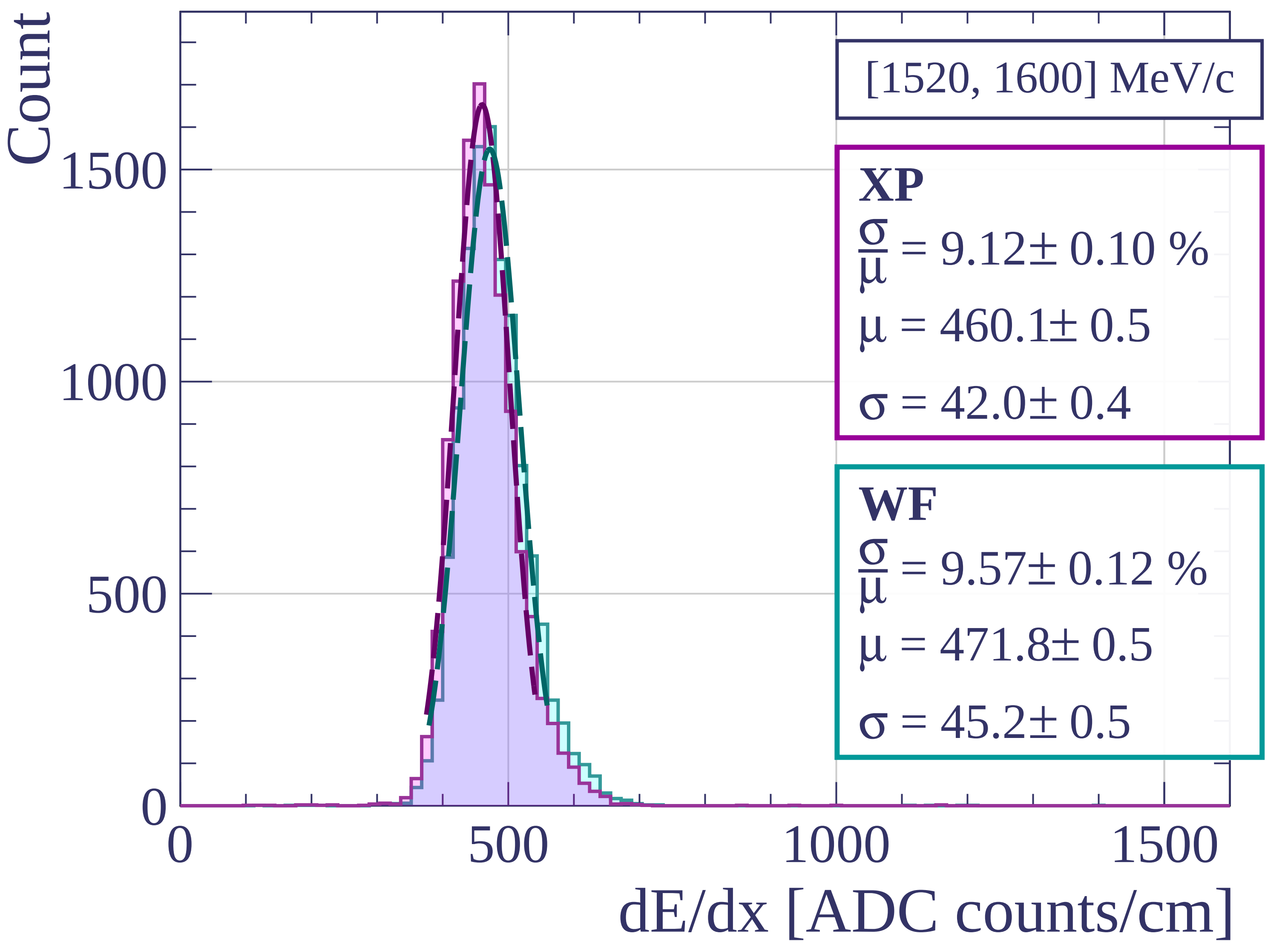

Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1280] MeV/c
XP
σ/μ = 9.12± 0.10 %
μ = 443.3± 0.4
σ = 40.4± 0.4
WF
σ/μ = 9.23± 0.10 %
μ = 455.1± 0.4
σ = 42.0± 0.4



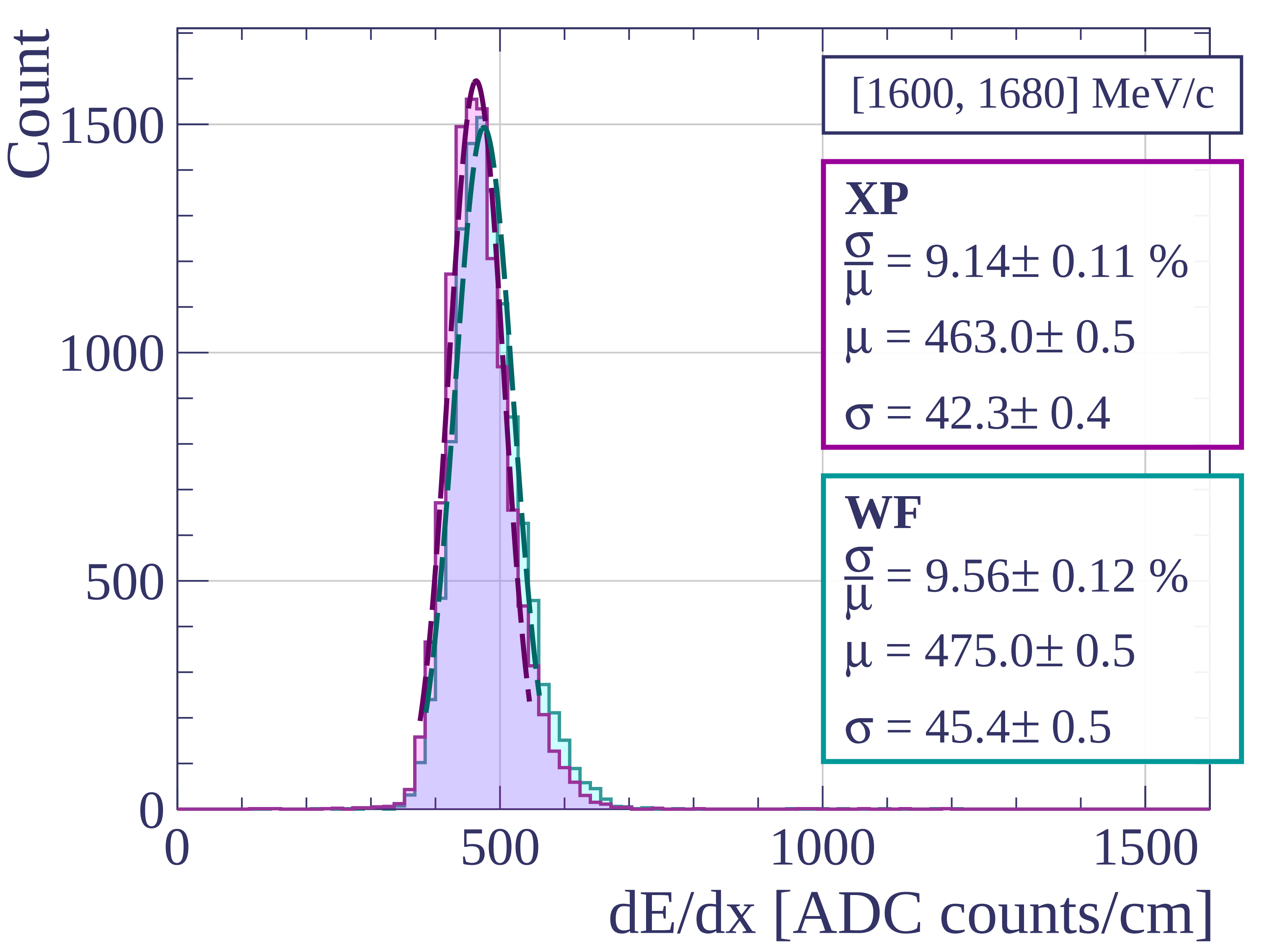
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.84± 0.10 %
μ = 447.2± 0.4
σ = 39.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.02± 0.10 %
μ = 459.1± 0.4
σ = 41.4± 0.4


Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.88± 0.10 %
μ = 450.9± 0.4
σ = 40.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.08± 0.10 %
μ = 462.8± 0.5
σ = 42.0± 0.4

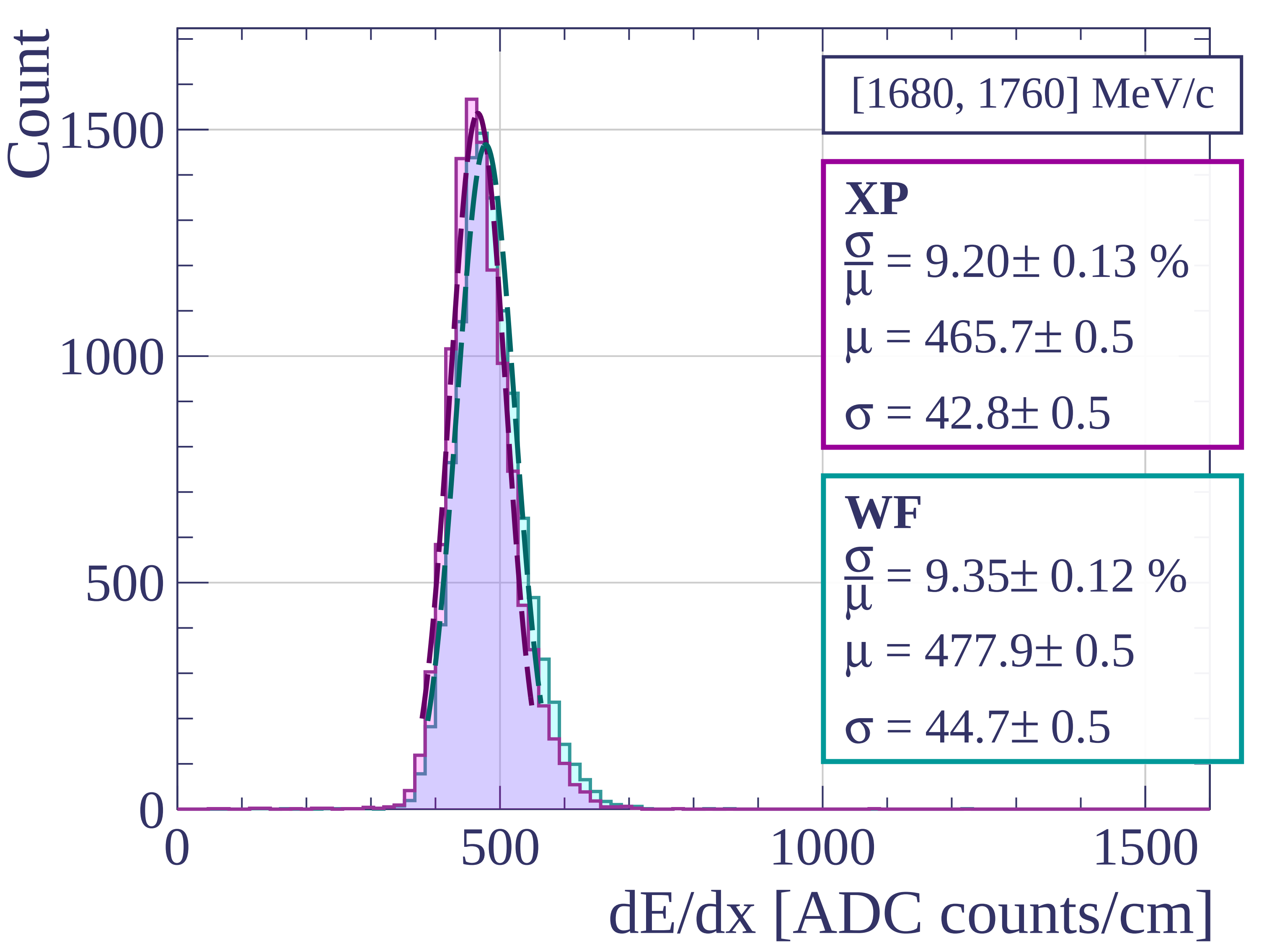

Count
0
500
1000
1500
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.04± 0.10 %
μ = 455.7± 0.4
σ = 41.2± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.32± 0.10 %
μ = 468.3± 0.5
σ = 43.6± 0.4



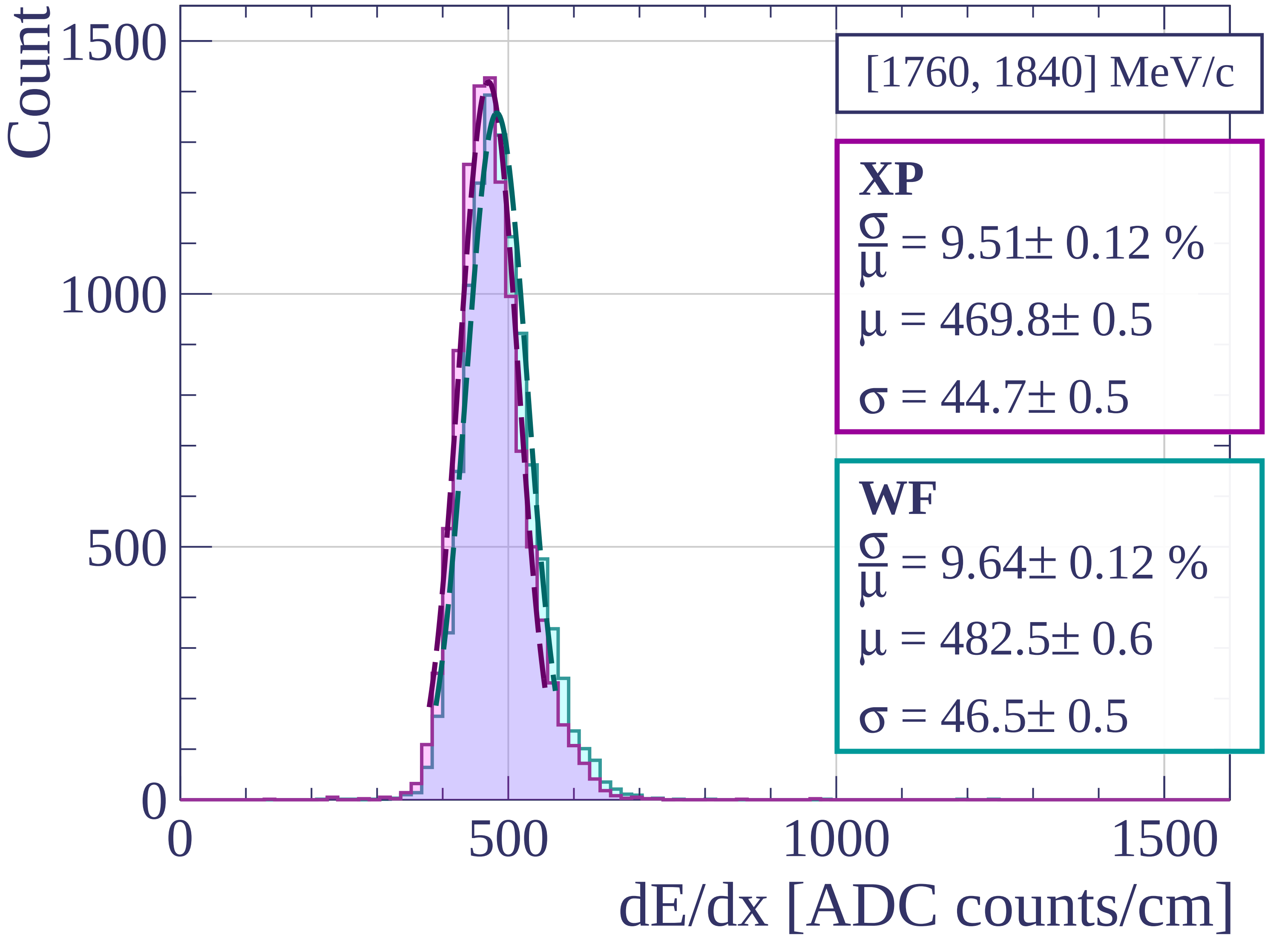
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
[1520, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.12± 0.10 %
μ = 460.1± 0.5
σ = 42.0± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.57± 0.12 %
μ = 471.8± 0.5
σ = 45.2± 0.5



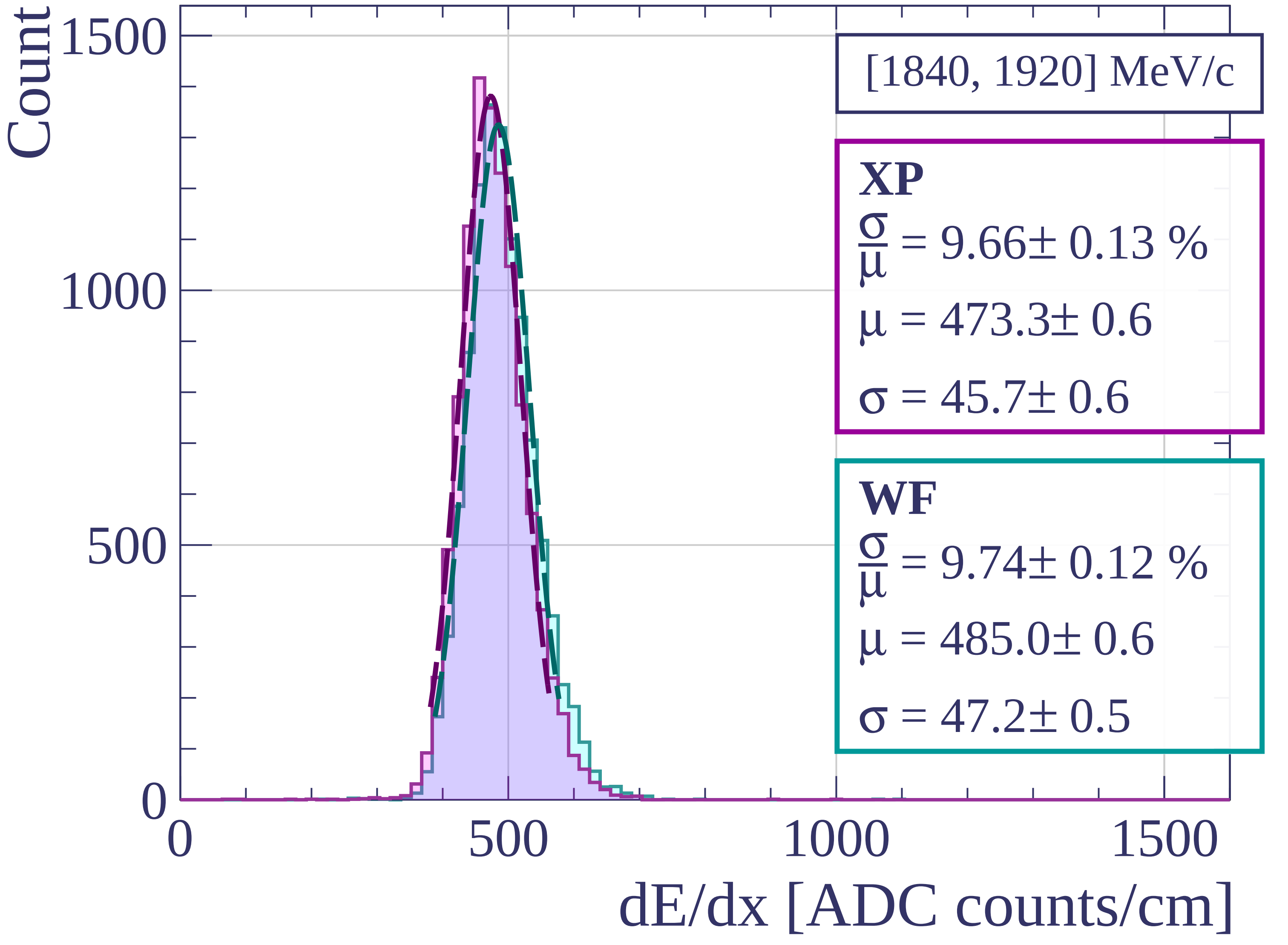
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1680] MeV/c
XP
σ/μ = 9.14± 0.11 %
μ = 463.0± 0.5
σ = 42.3± 0.4
WF
σ/μ = 9.56± 0.12 %
μ = 475.0± 0.5
σ = 45.4± 0.5



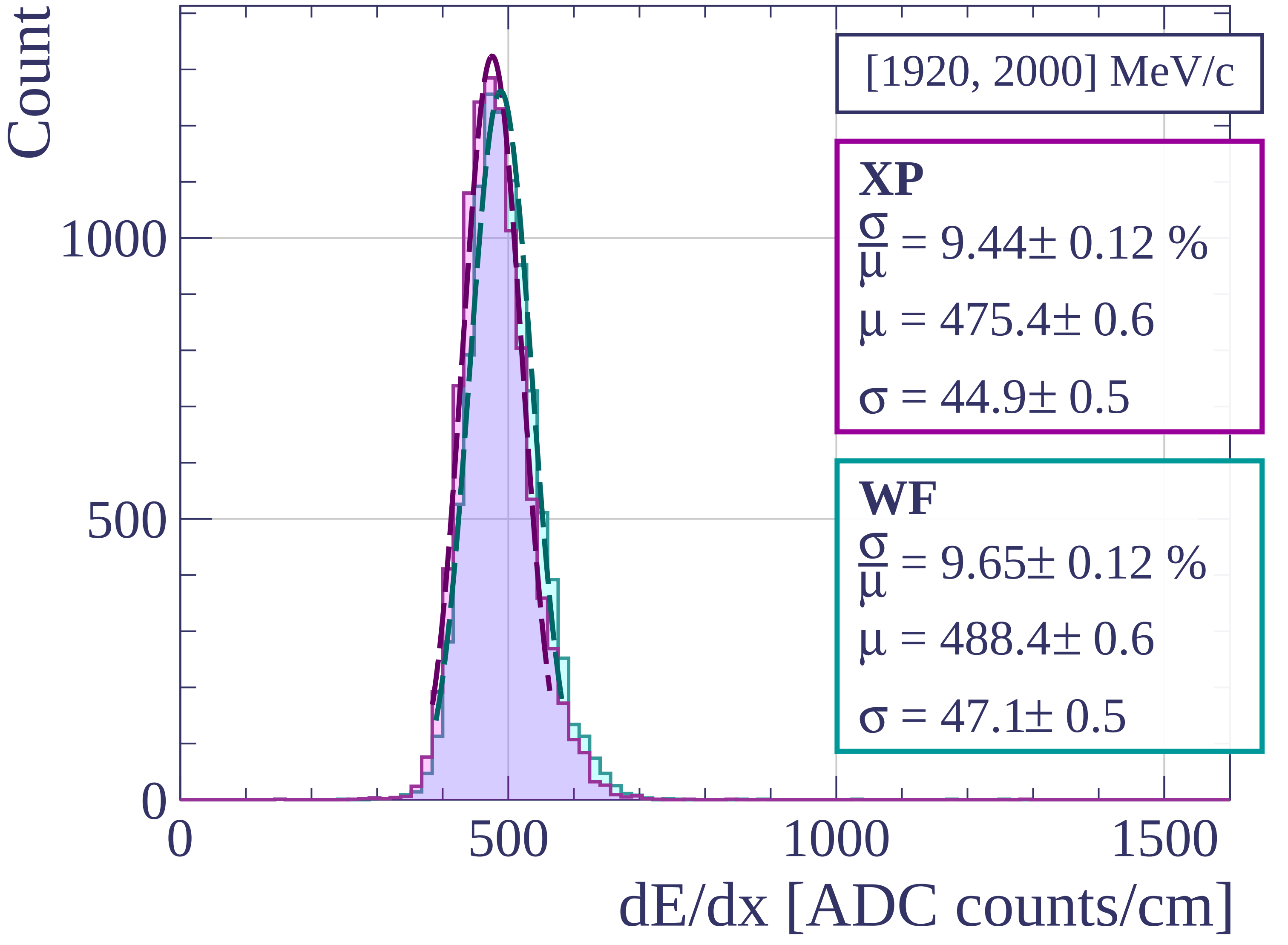
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1760] MeV/c
XP
σ/μ = 9.20± 0.13 %
μ = 465.7± 0.5
σ = 42.8± 0.5
WF
σ/μ = 9.35± 0.12 %
μ = 477.9± 0.5
σ = 44.7± 0.5



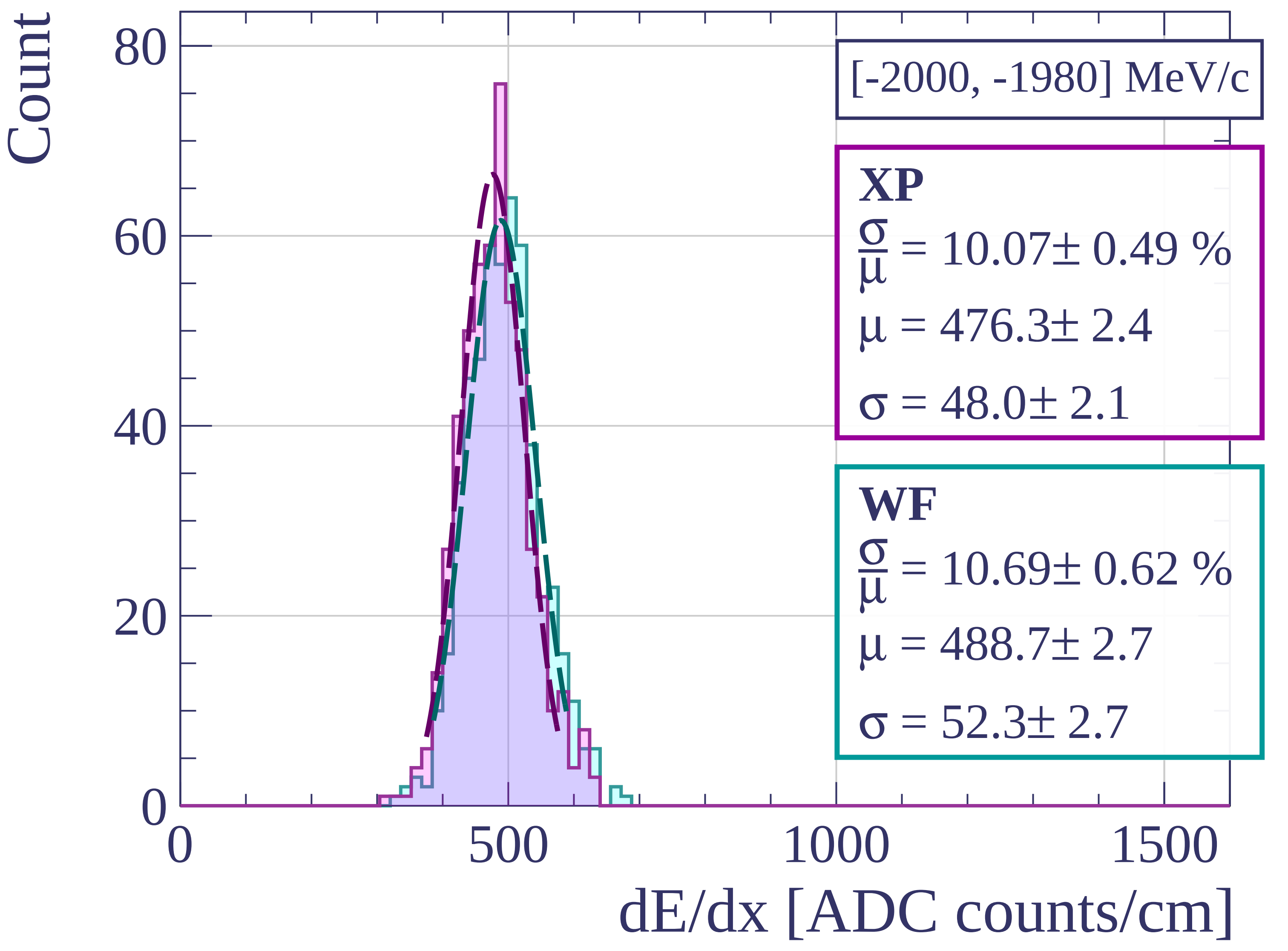
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1840] MeV/c
XP
σ/μ = 9.51± 0.12 %
μ = 469.8± 0.5
σ = 44.7± 0.5
WF
σ/μ = 9.64± 0.12 %
μ = 482.5± 0.6
σ = 46.5± 0.5



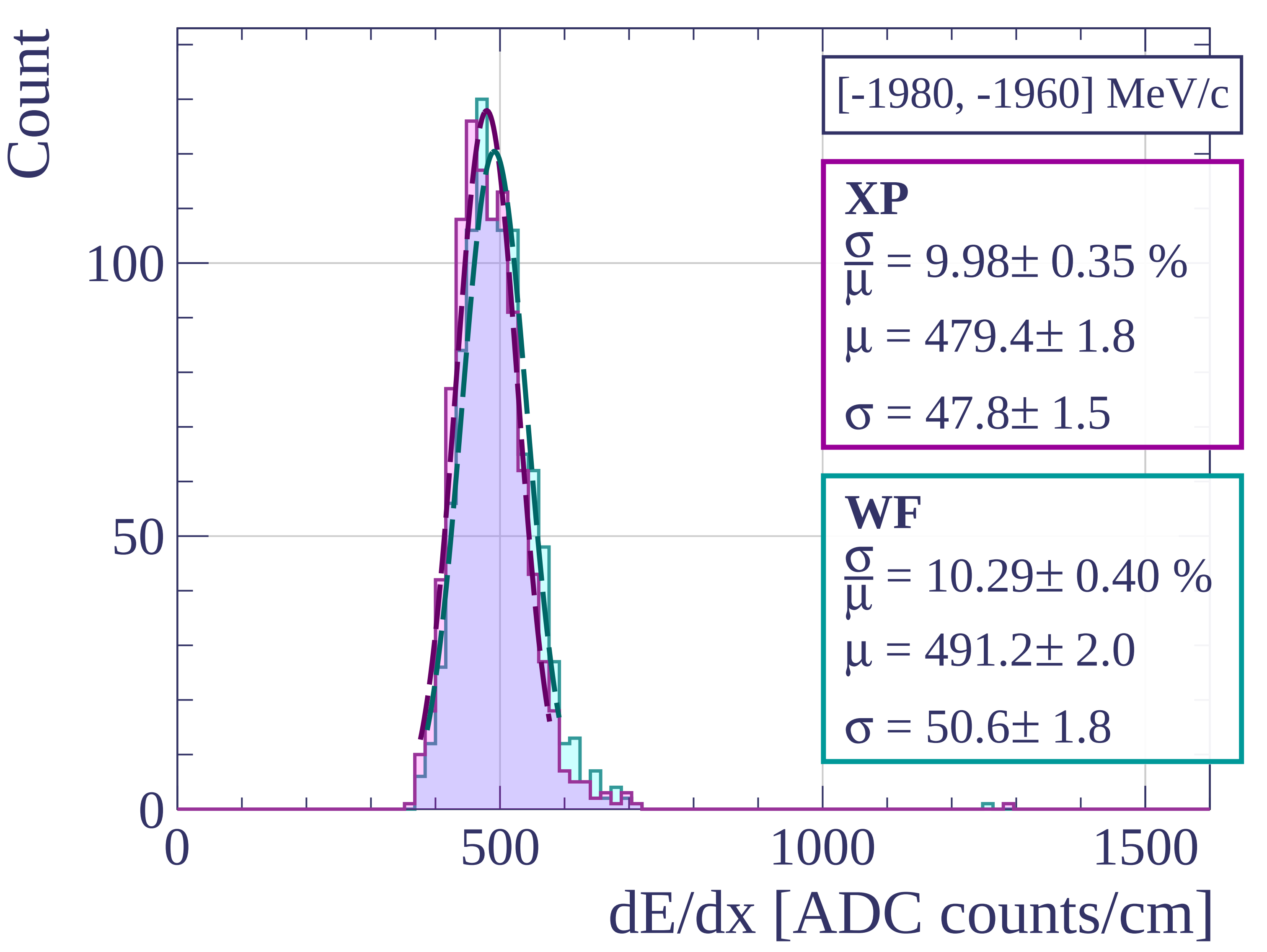
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1920] MeV/c
XP
σ/μ = 9.66± 0.13 %
μ = 473.3± 0.6
σ = 45.7± 0.6
WF
σ/μ = 9.74± 0.12 %
μ = 485.0± 0.6
σ = 47.2± 0.5

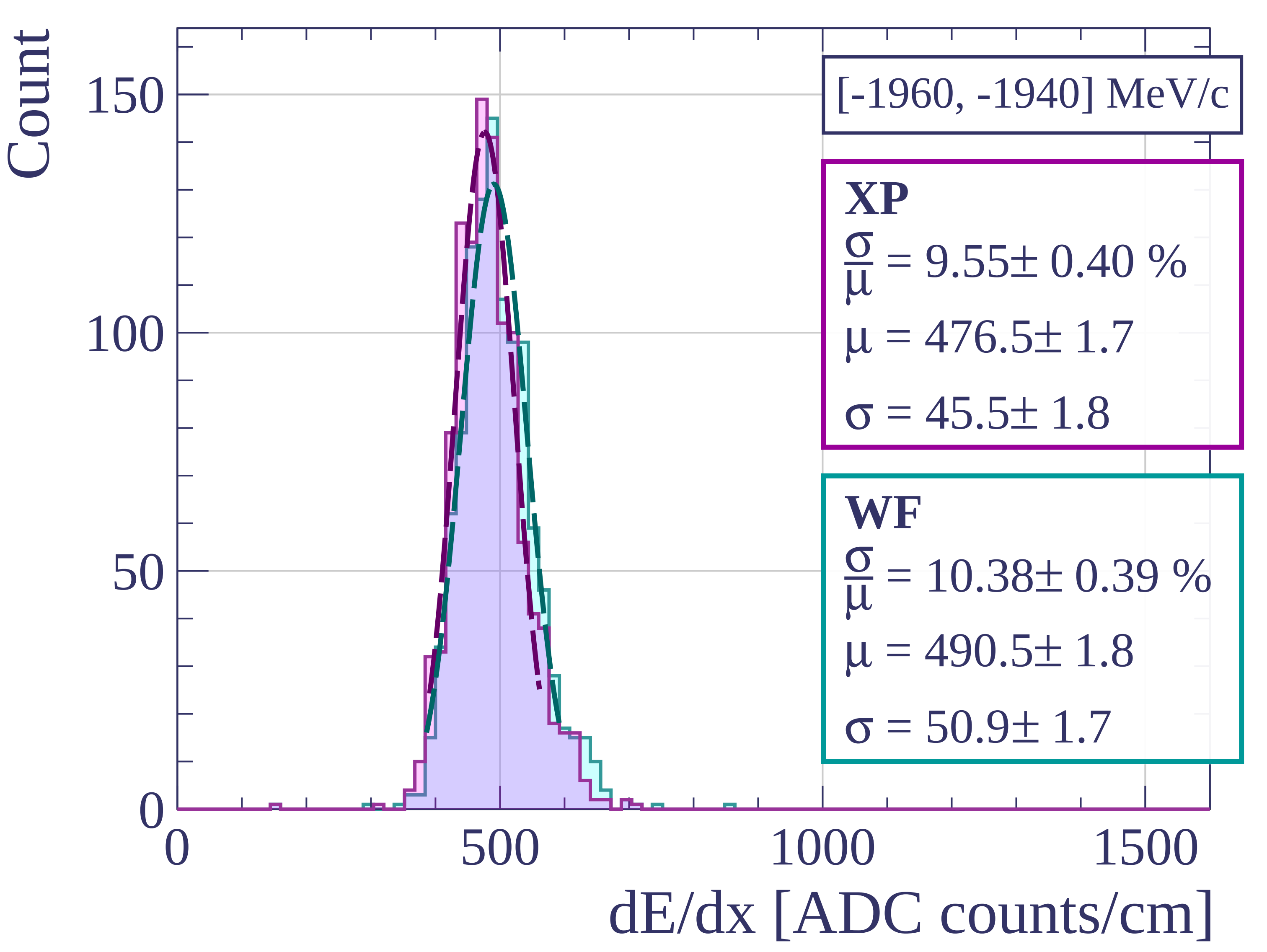


Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.44± 0.12 %
μ = 475.4± 0.6
σ = 44.9± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.65± 0.12 %
μ = 488.4± 0.6
σ = 47.1± 0.5

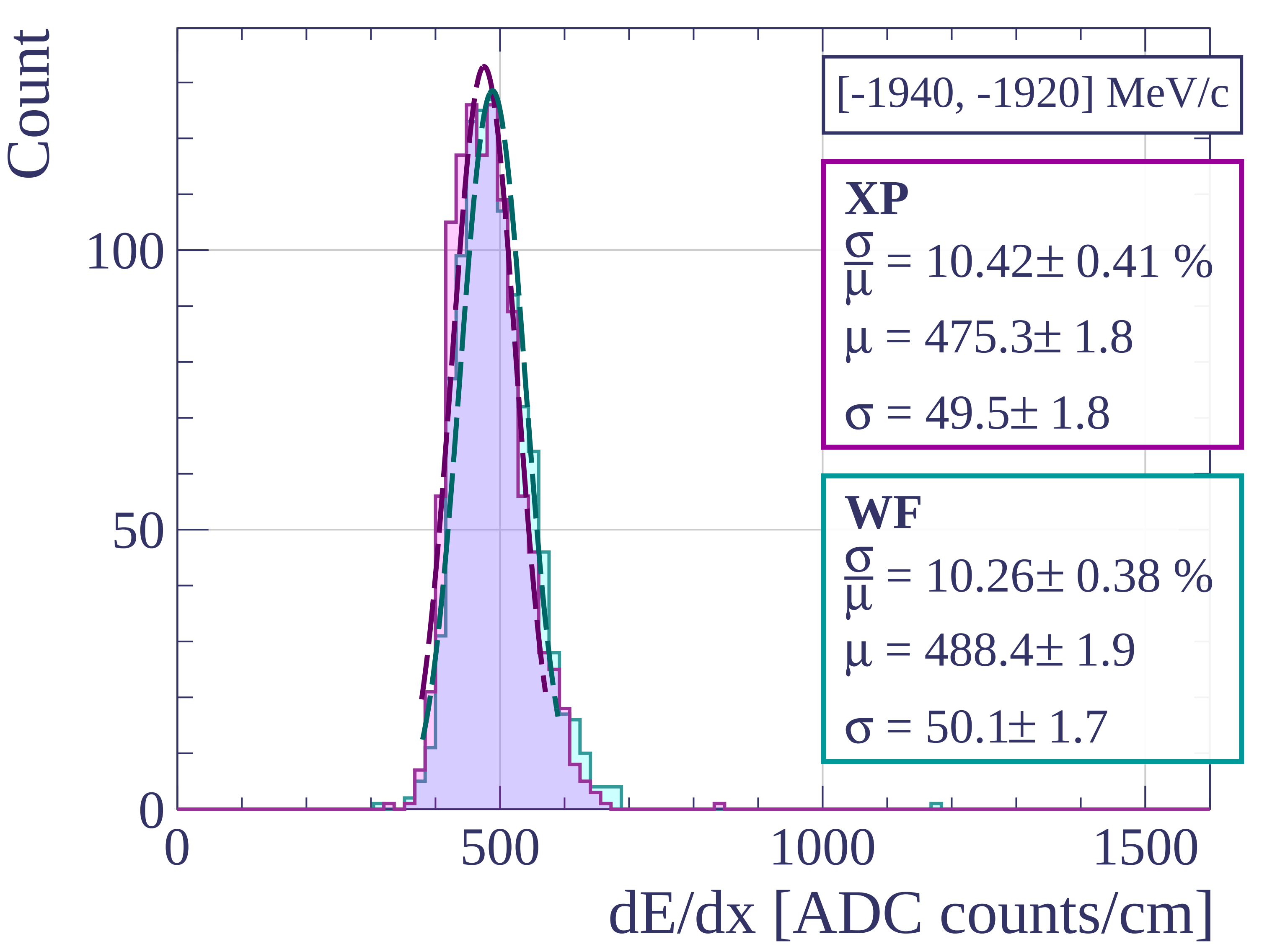


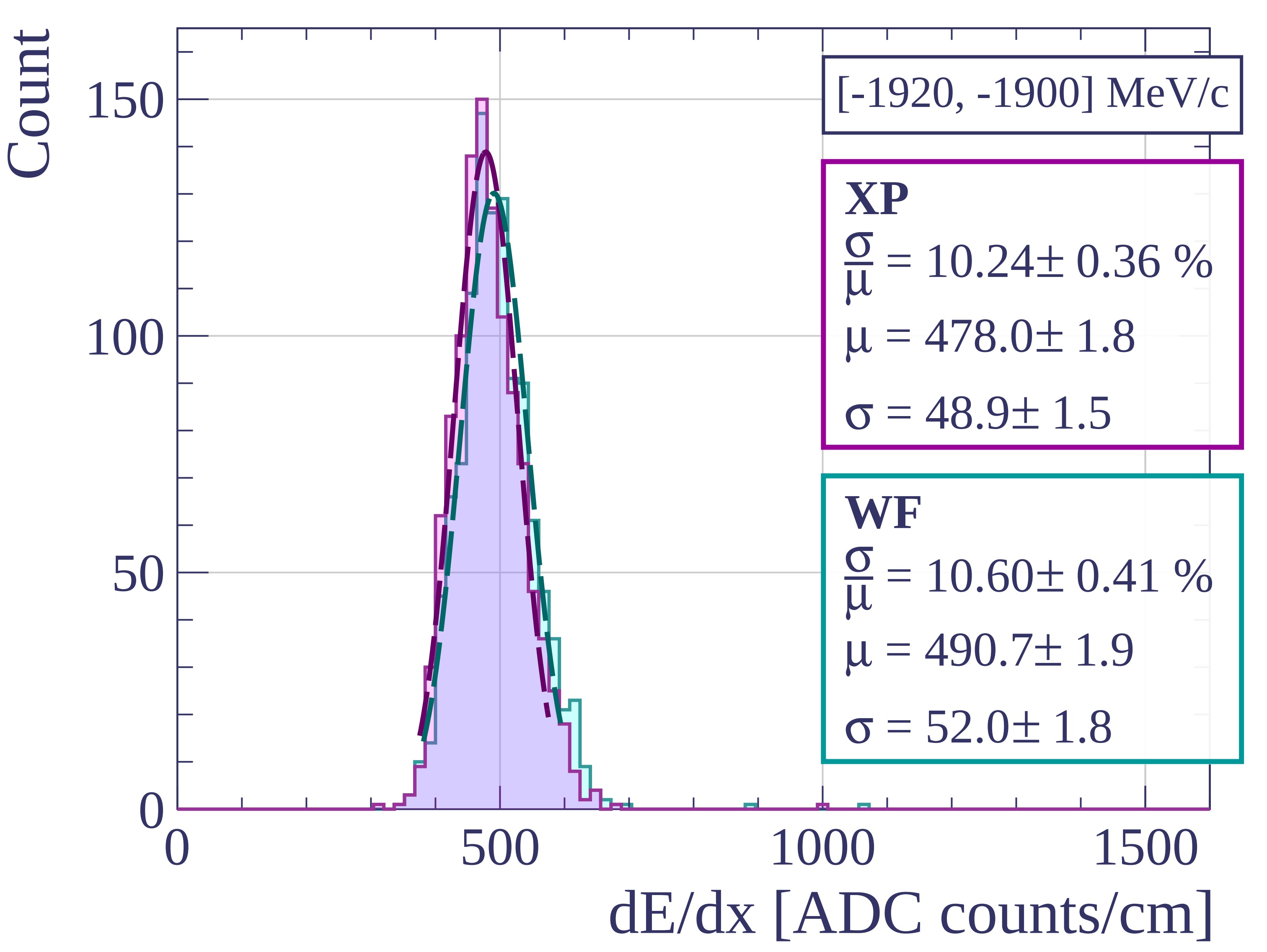
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-2000, -1980] MeV/c
XP
σ/μ = 10.07± 0.49 %
μ = 476.3± 2.4
σ = 48.0± 2.1
WF
σ/μ = 10.69± 0.62 %
μ = 488.7± 2.7
σ = 52.3± 2.7


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1980, -1960] MeV/c
XP
σ/μ = 9.98± 0.35 %
μ = 479.4± 1.8
σ = 47.8± 1.5
WF
σ/μ = 10.29± 0.40 %
μ = 491.2± 2.0
σ = 50.6± 1.8

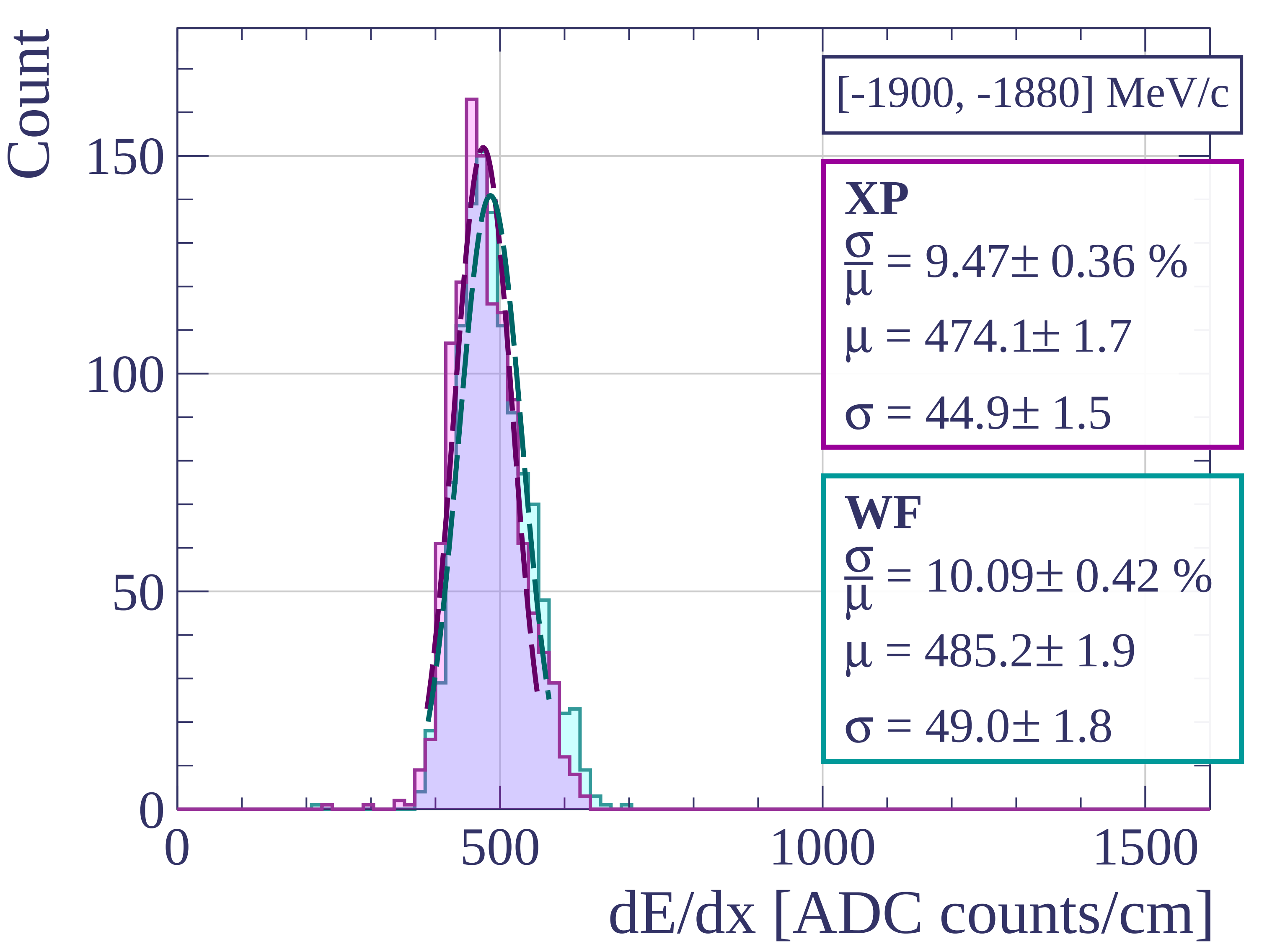


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1960, -1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.55± 0.40 %
μ = 476.5± 1.7
σ = 45.5± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.38± 0.39 %
μ = 490.5± 1.8
σ = 50.9± 1.7

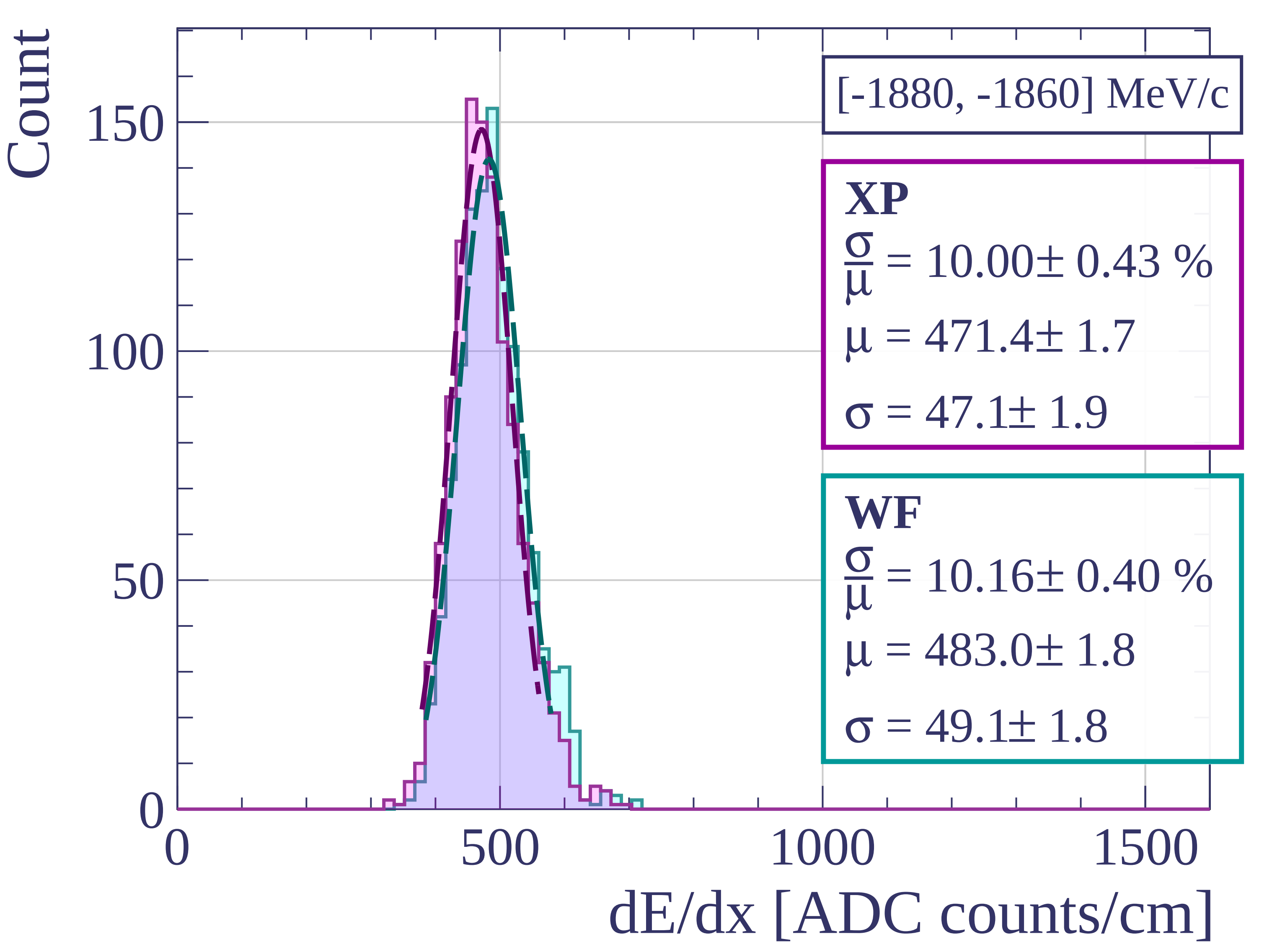

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1940, -1920] MeV/c
XP
σ/μ = 10.42± 0.41 %
μ = 475.3± 1.8
σ = 49.5± 1.8
WF
σ/μ = 10.26± 0.38 %
μ = 488.4± 1.9
σ = 50.1± 1.7



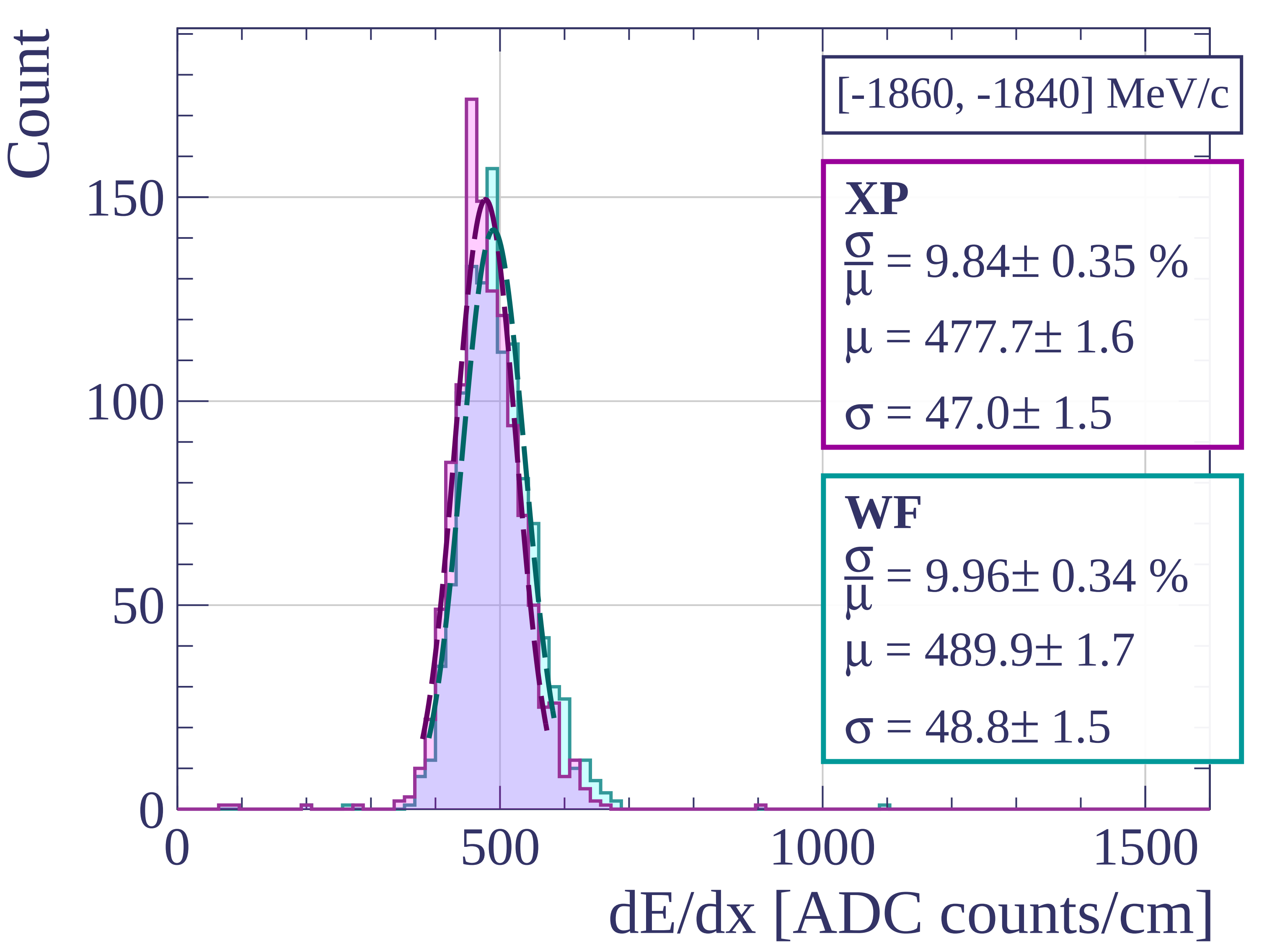
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1920, -1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.24± 0.36 %
μ = 478.0± 1.8
σ = 48.9± 1.5
WF
$\frac{\sigma}{\mu}$ = 10.60± 0.41 %
μ = 490.7± 1.9
σ = 52.0± 1.8

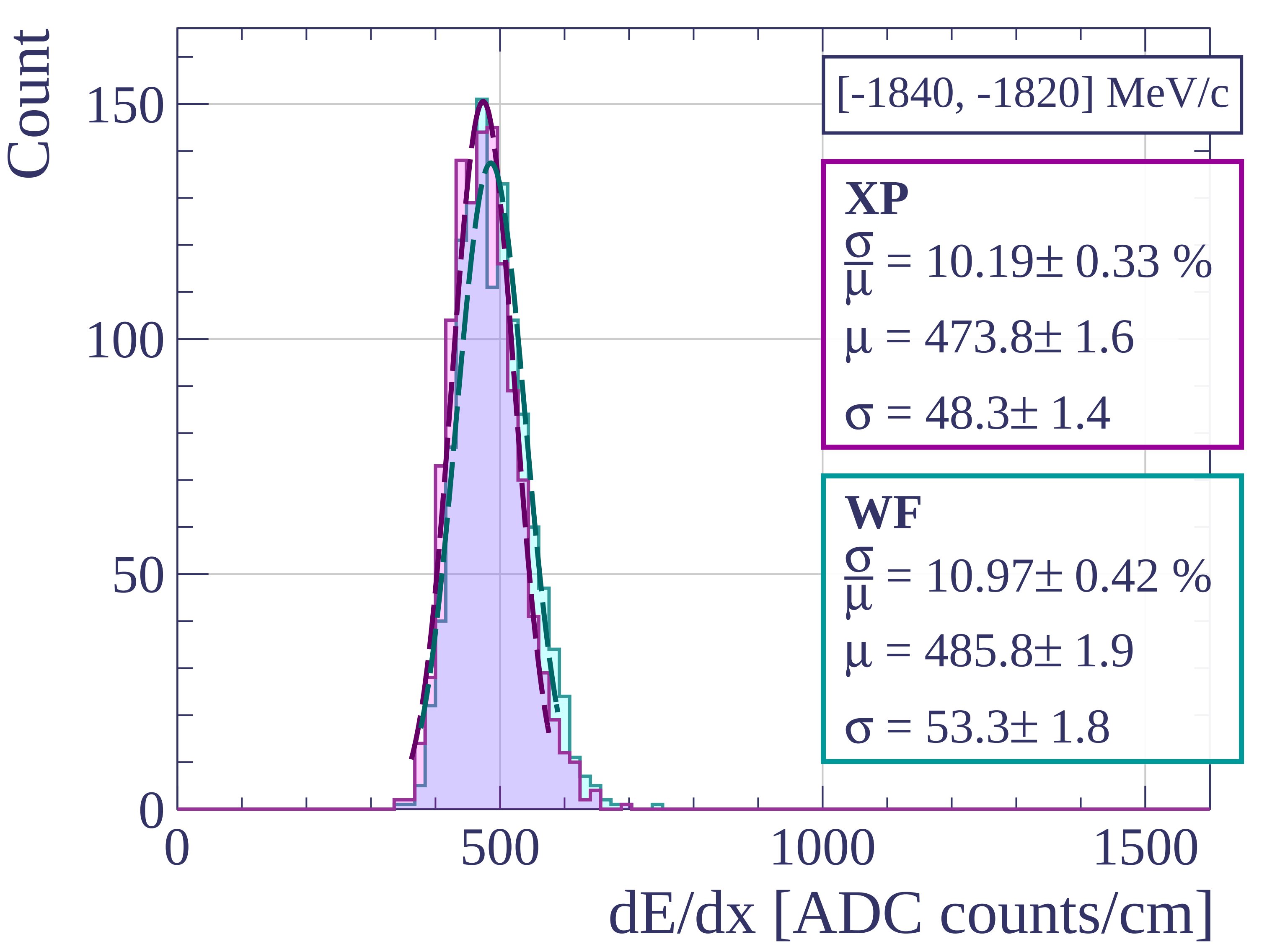


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1900, -1880] MeV/c
XP
σ/μ = 9.47± 0.36 %
μ = 474.1± 1.7
σ = 44.9± 1.5
WF
σ/μ = 10.09± 0.42 %
μ = 485.2± 1.9
σ = 49.0± 1.8

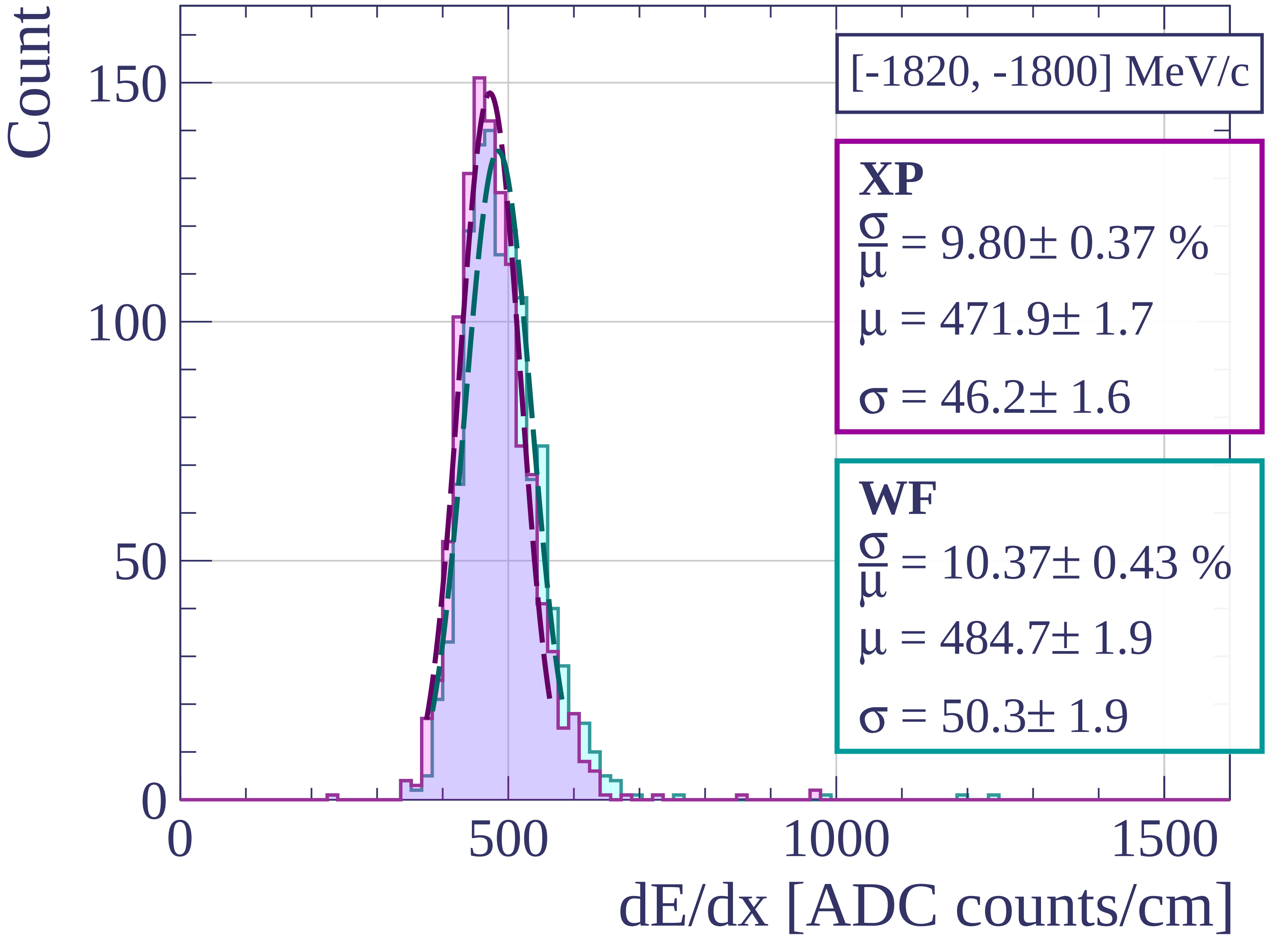


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1880, -1860] MeV/c
XP
σ/μ = 10.00± 0.43 %
μ = 471.4± 1.7
σ = 47.1± 1.9
WF
σ/μ = 10.16± 0.40 %
μ = 483.0± 1.8
σ = 49.1± 1.8

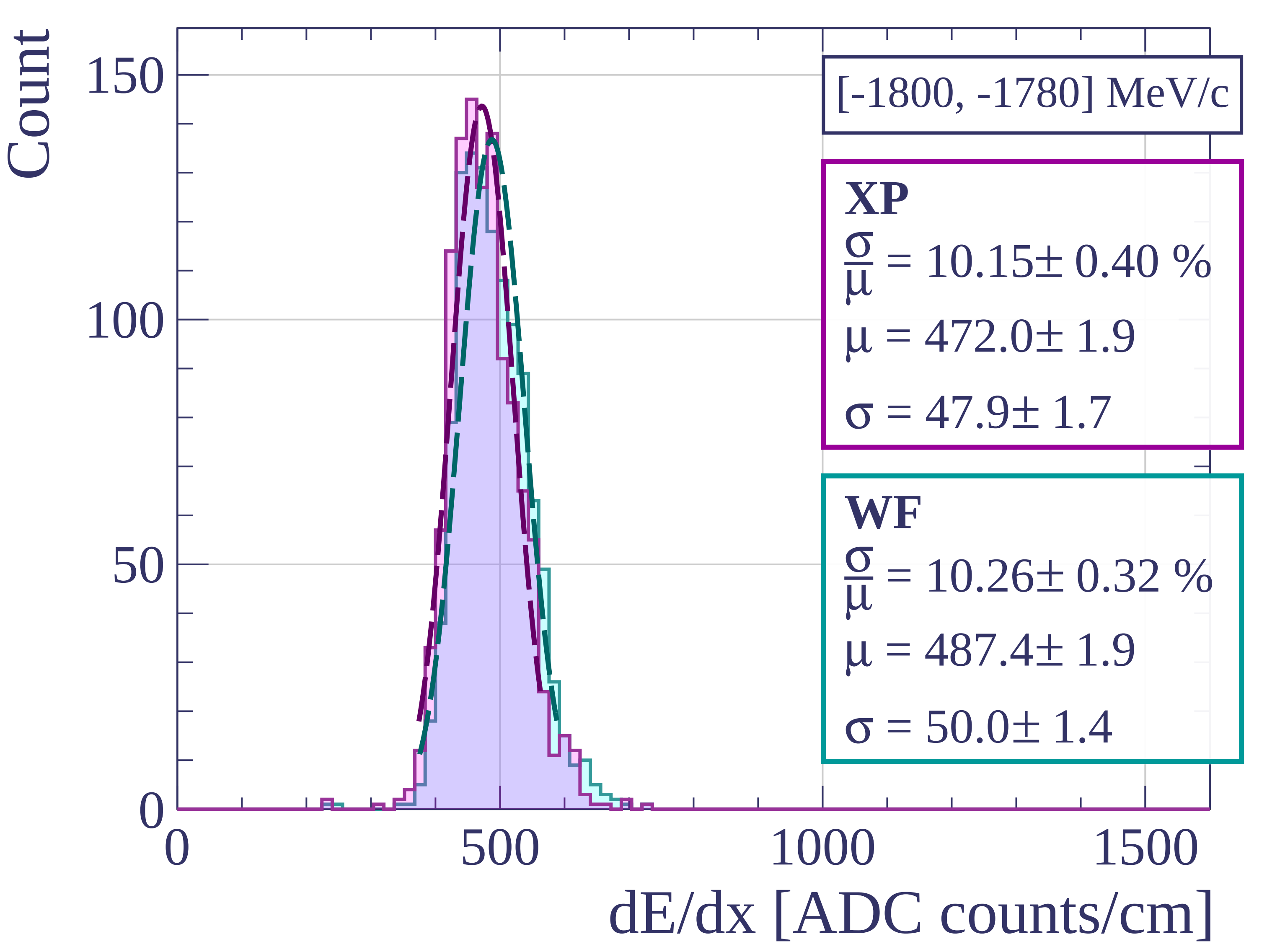

Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1860, -1840] MeV/c
XP
σ/μ = 9.84± 0.35 %
μ = 477.7± 1.6
σ = 47.0± 1.5
WF
σ/μ = 9.96± 0.34 %
μ = 489.9± 1.7
σ = 48.8± 1.5



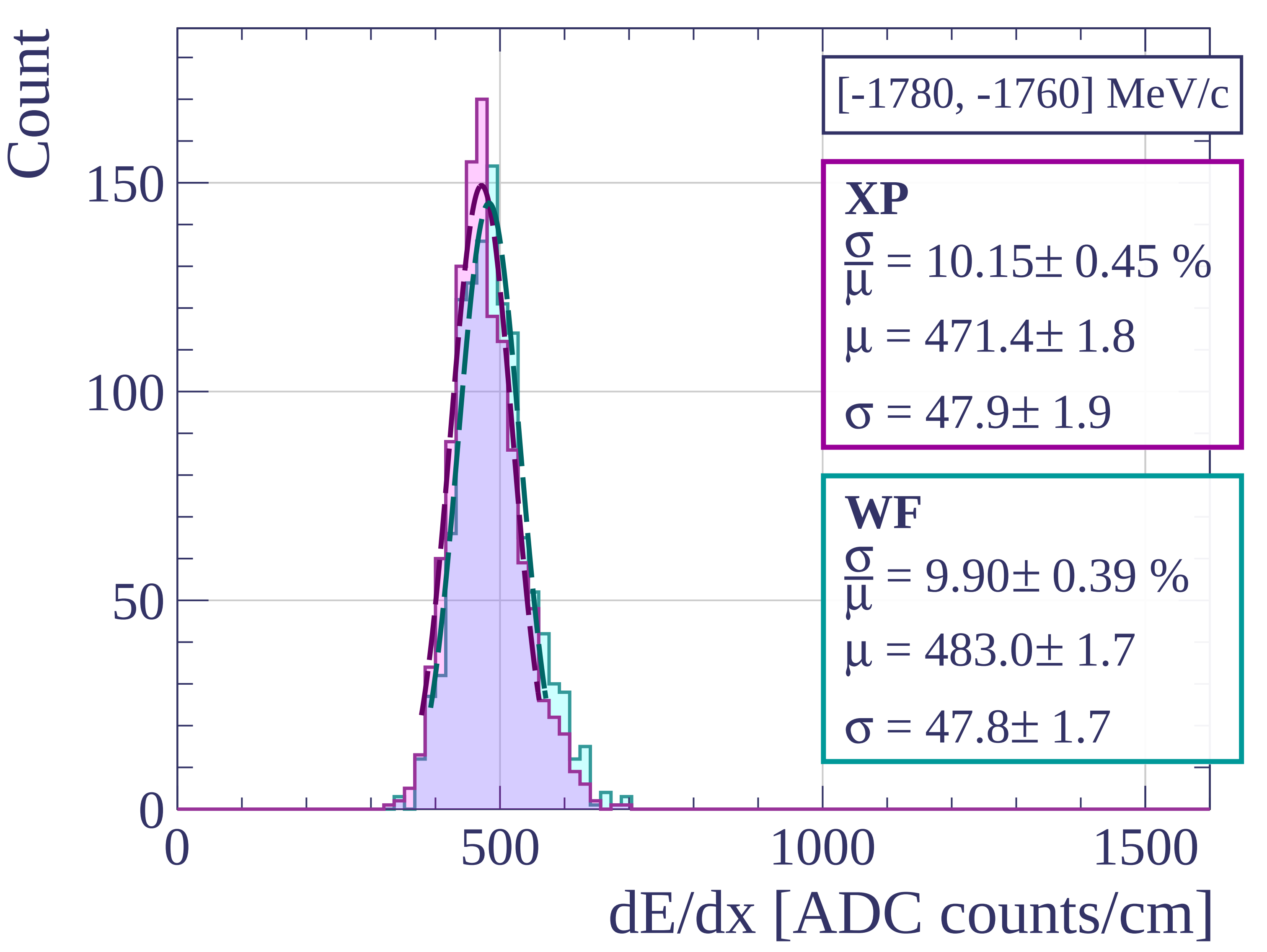
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1840, -1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.19± 0.33 %
μ = 473.8± 1.6
σ = 48.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 10.97± 0.42 %
μ = 485.8± 1.9
σ = 53.3± 1.8

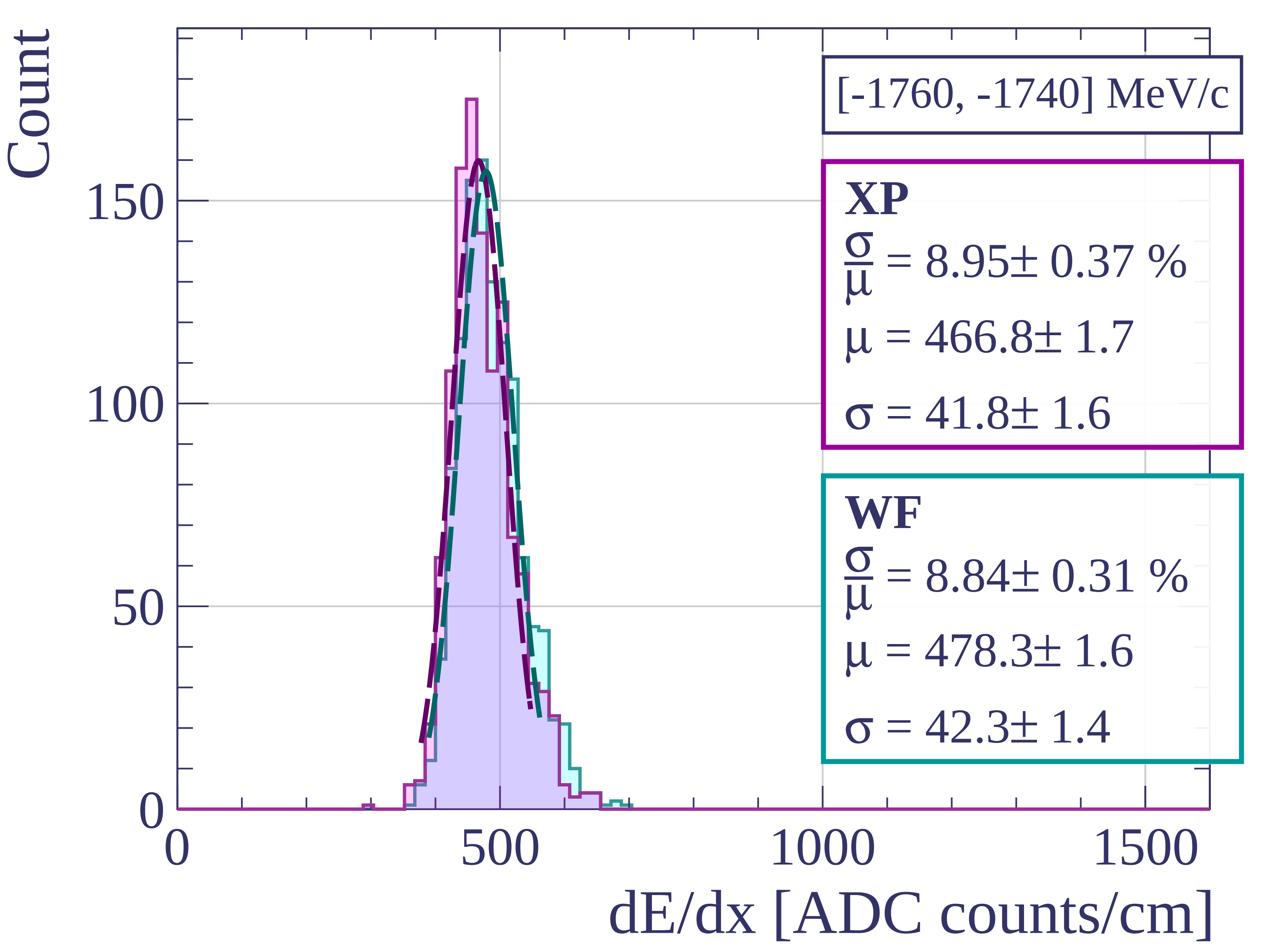


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1820, -1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.80± 0.37 %
μ = 471.9± 1.7
σ = 46.2± 1.6
WF
$\frac{\sigma}{\mu}$ = 10.37± 0.43 %
μ = 484.7± 1.9
σ = 50.3± 1.9

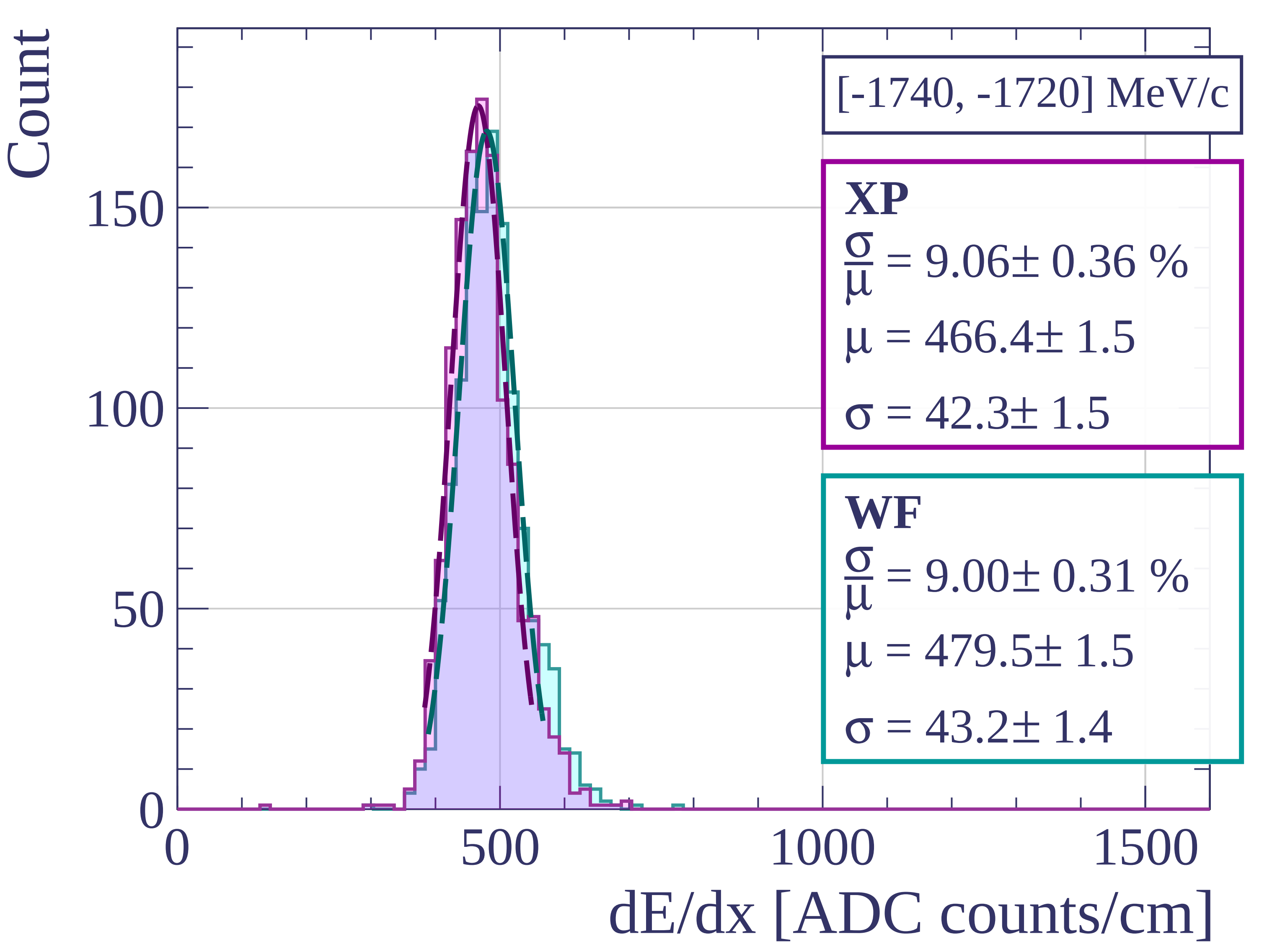


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1800, -1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.15± 0.40 %
μ = 472.0± 1.9
σ = 47.9± 1.7
WF
$\frac{\sigma}{\mu}$ = 10.26± 0.32 %
μ = 487.4± 1.9
σ = 50.0± 1.4

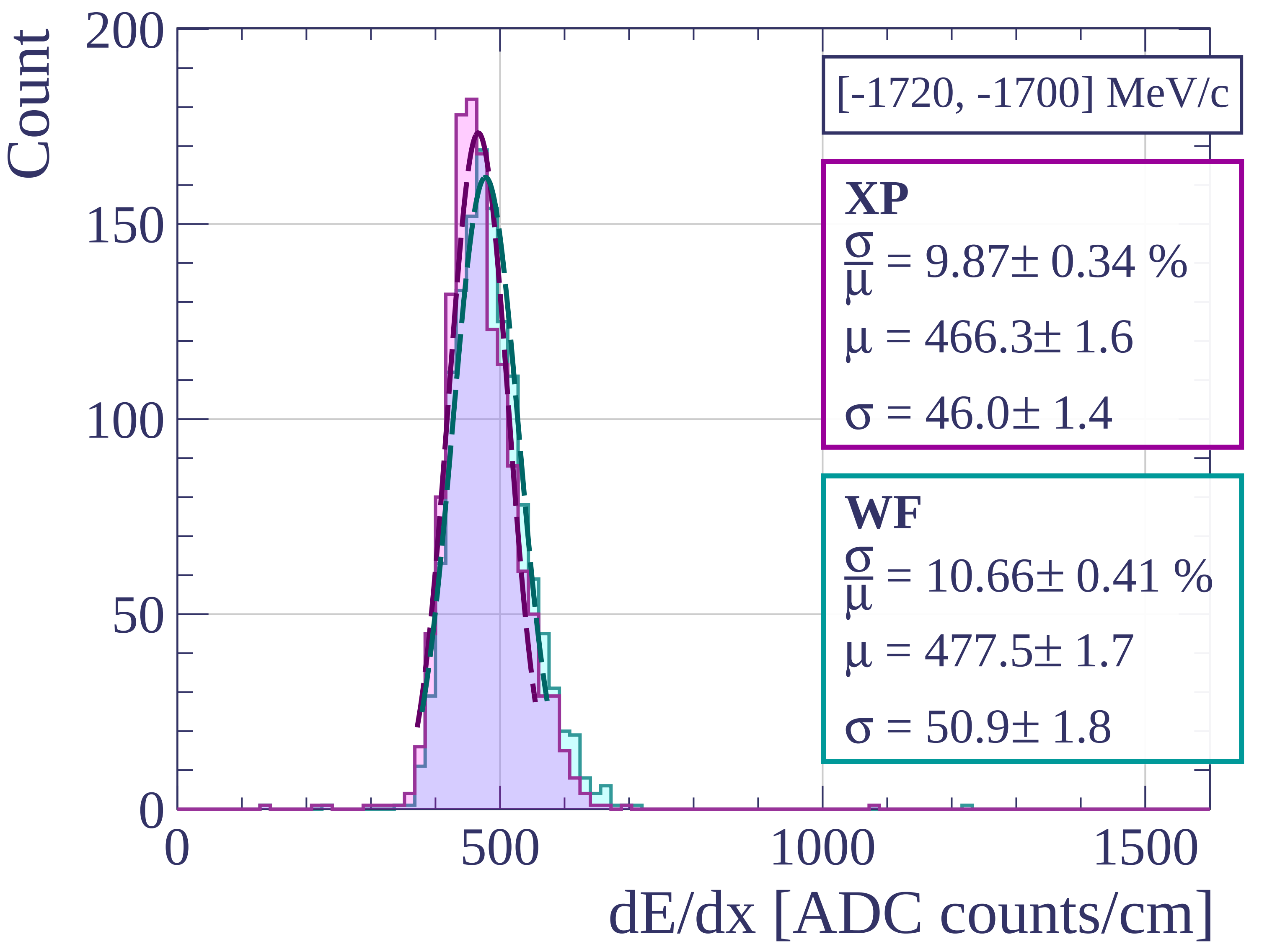

[-1780, -1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.15± 0.45 %
μ = 471.4± 1.8
σ = 47.9± 1.9
WF
$\frac{\sigma}{\mu}$ = 9.90± 0.39 %
μ = 483.0± 1.7
σ = 47.8± 1.7
Count
0
50
100
150
0
500
1000
1500
dE/dx [ADC counts/cm]



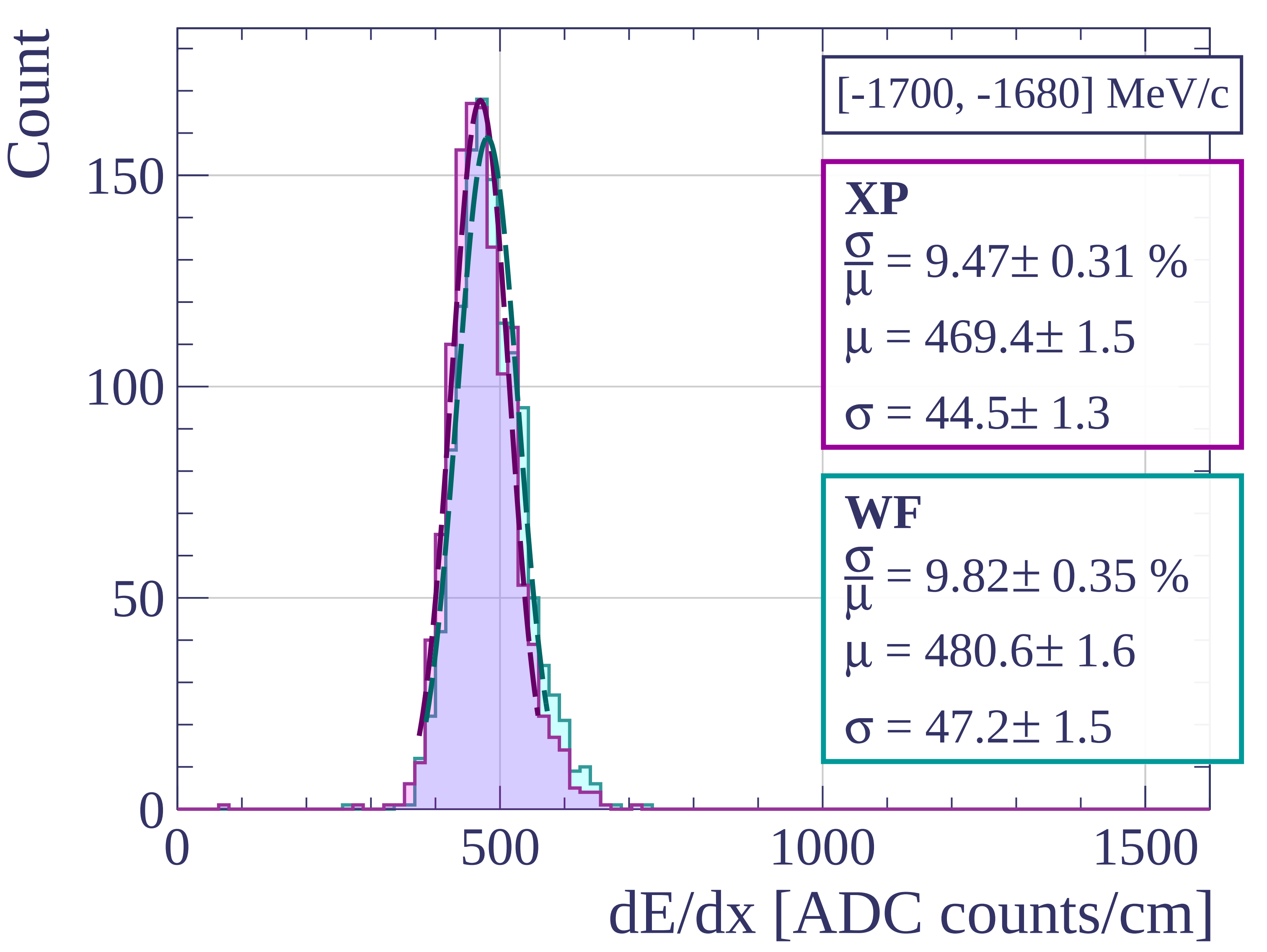
Count
dE/dx [ADC counts/cm]
[-1760, -1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.95± 0.37 %
μ = 466.8± 1.7
σ = 41.8± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.84± 0.31 %
μ = 478.3± 1.6
σ = 42.3± 1.4
0
500
1000
1500
50
100
150

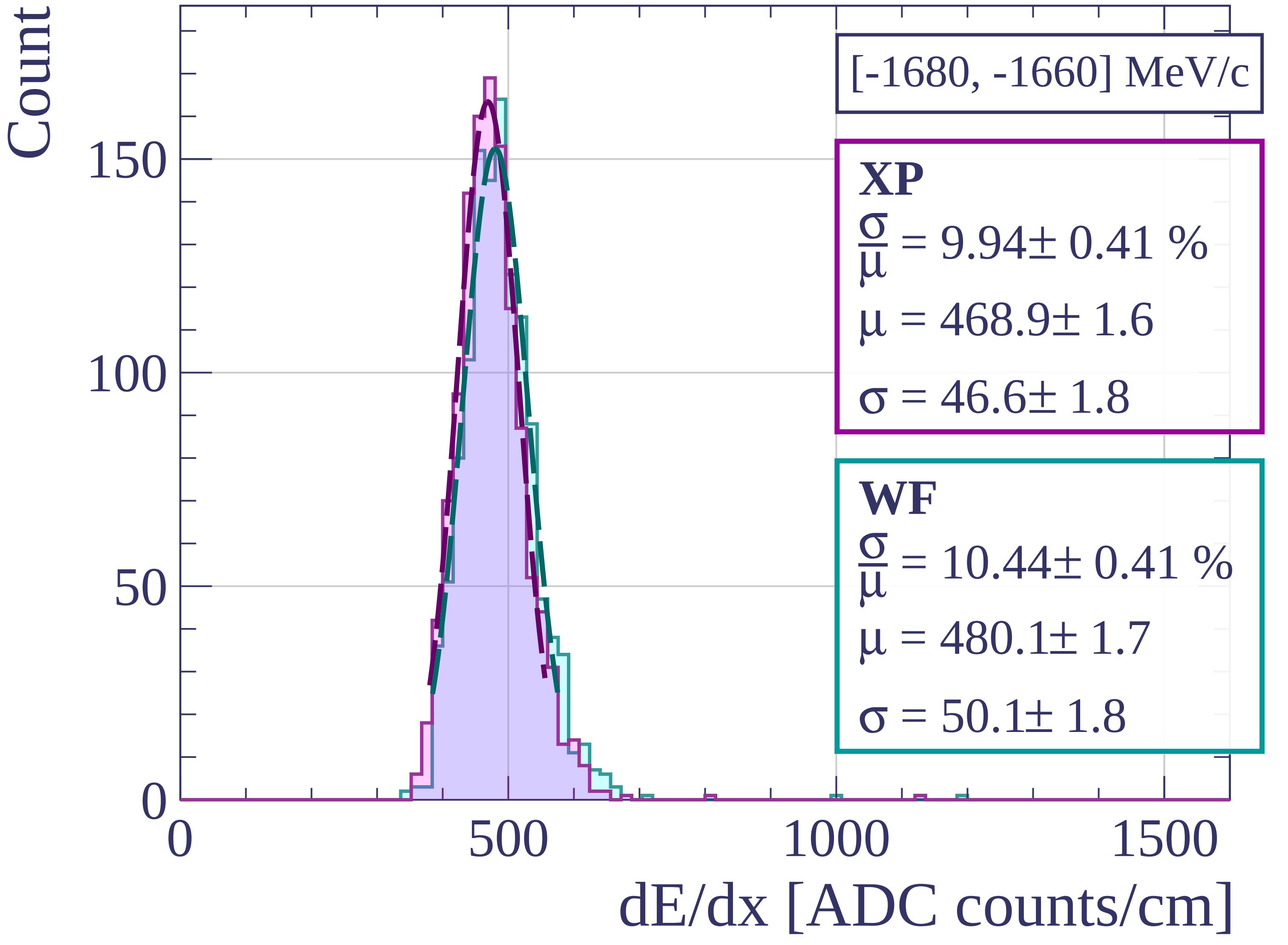


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1740, -1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.06± 0.36 %
μ = 466.4± 1.5
σ = 42.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.00± 0.31 %
μ = 479.5± 1.5
σ = 43.2± 1.4

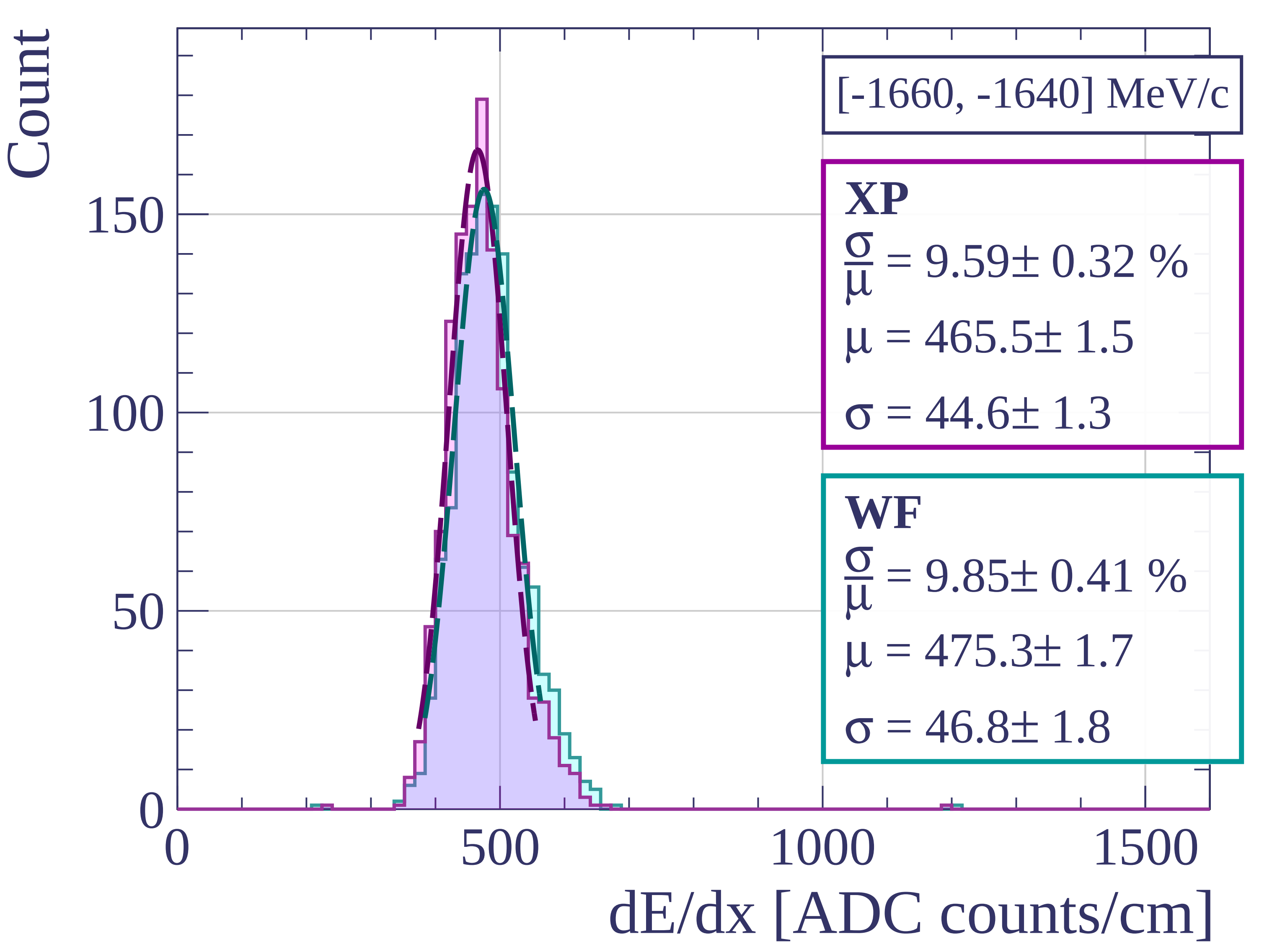


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1720, -1700] MeV/c
XP
σ/μ = 9.87± 0.34 %
μ = 466.3± 1.6
σ = 46.0± 1.4
WF
σ/μ = 10.66± 0.41 %
μ = 477.5± 1.7
σ = 50.9± 1.8

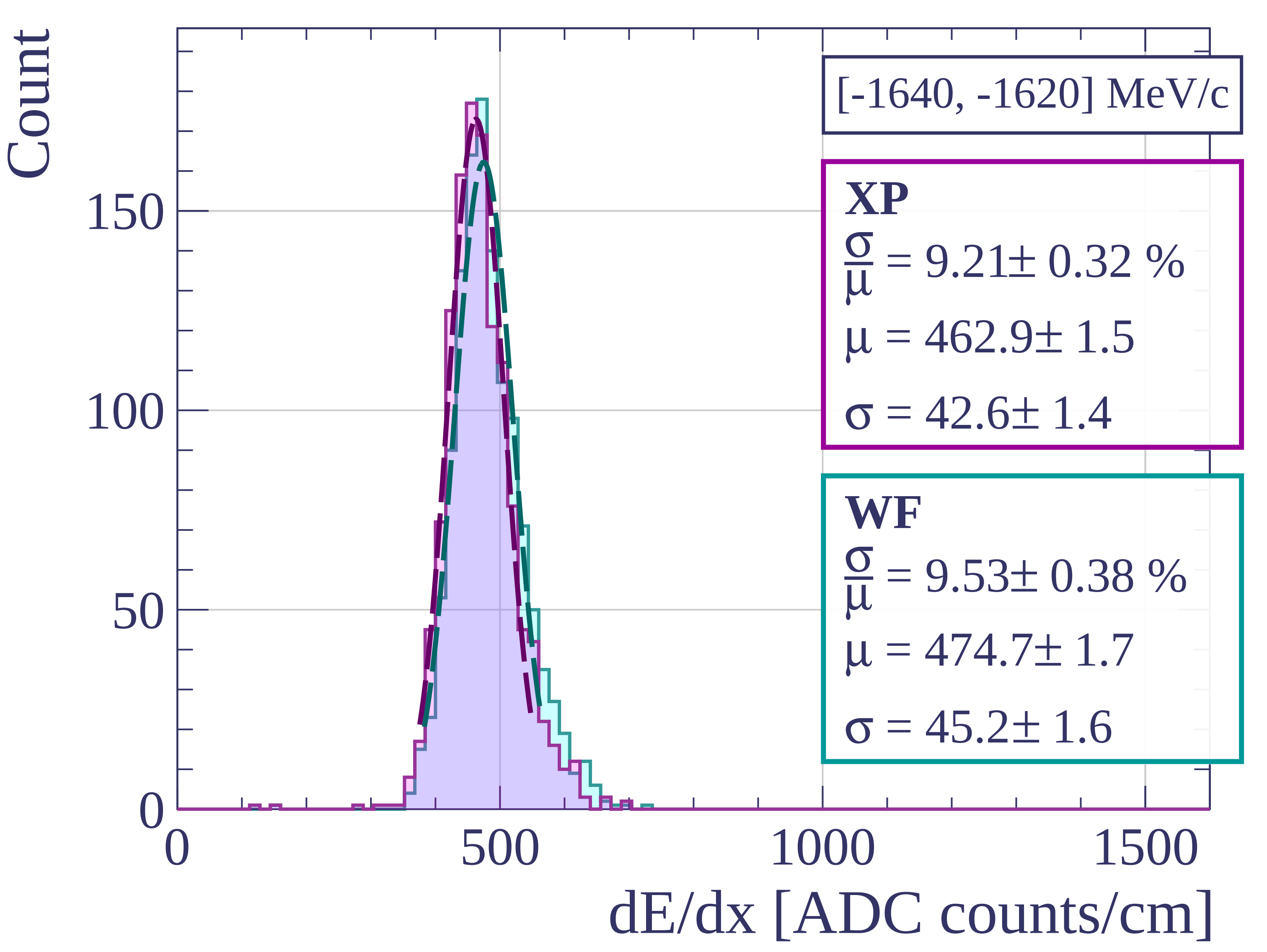

Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1700, -1680] MeV/c
XP
σ/μ = 9.47± 0.31 %
μ = 469.4± 1.5
σ = 44.5± 1.3
WF
σ/μ = 9.82± 0.35 %
μ = 480.6± 1.6
σ = 47.2± 1.5



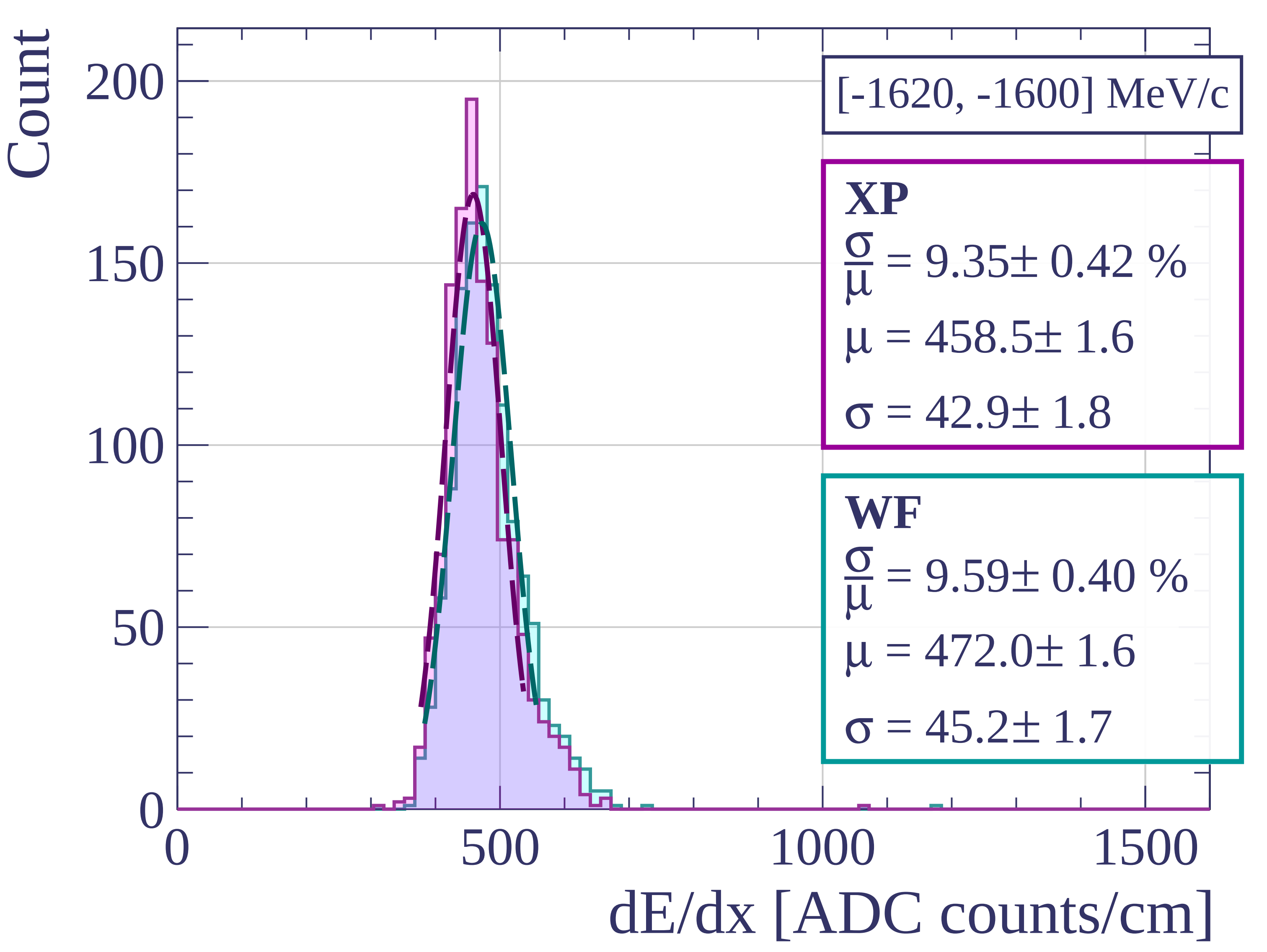
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
50
100
150
[-1680, -1660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.94± 0.41 %
μ = 468.9± 1.6
σ = 46.6± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.44± 0.41 %
μ = 480.1± 1.7
σ = 50.1± 1.8



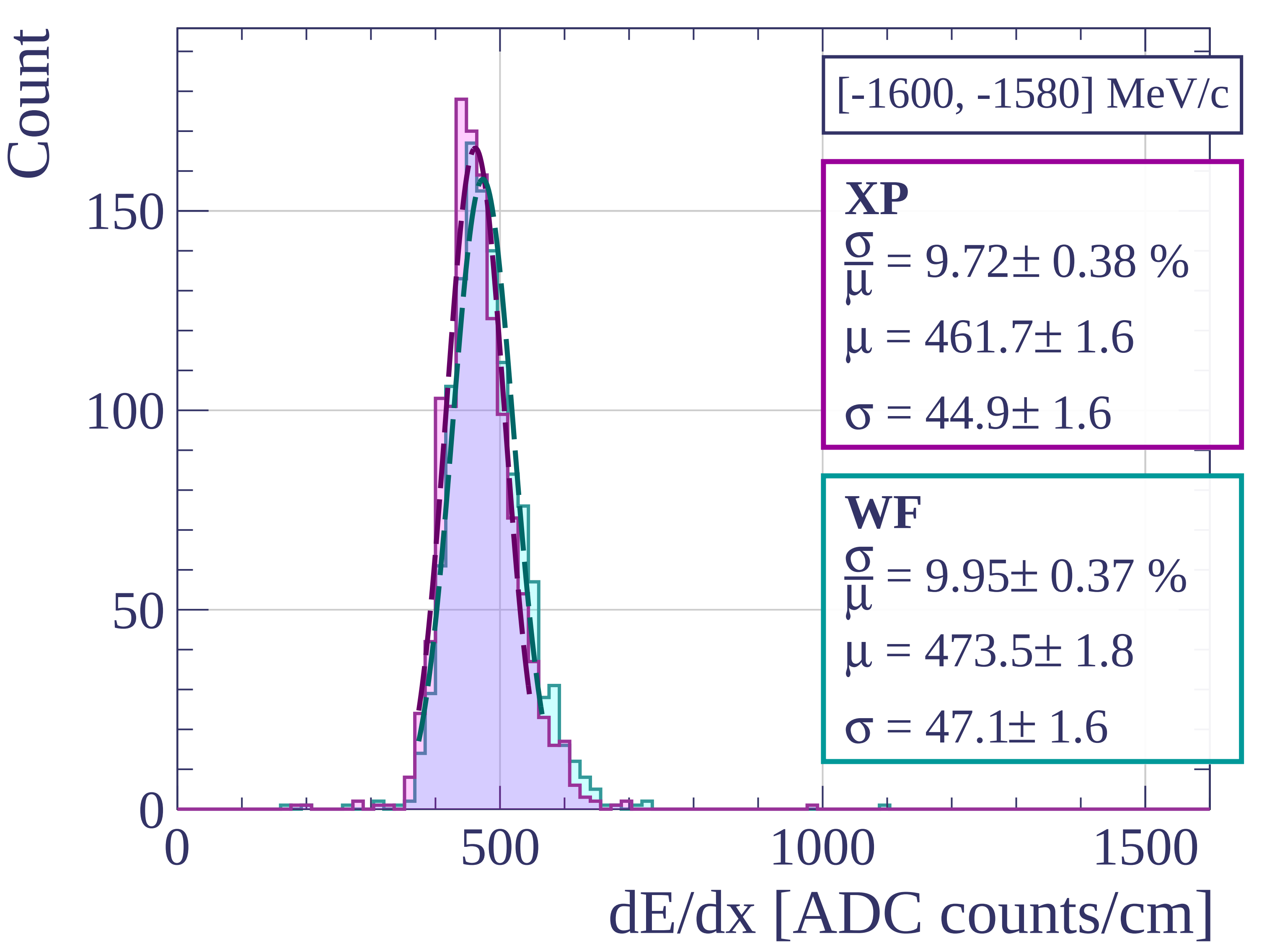
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1660, -1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.59± 0.32 %
μ = 465.5± 1.5
σ = 44.6± 1.3
WF
$\frac{\sigma}{\mu}$ = 9.85± 0.41 %
μ = 475.3± 1.7
σ = 46.8± 1.8



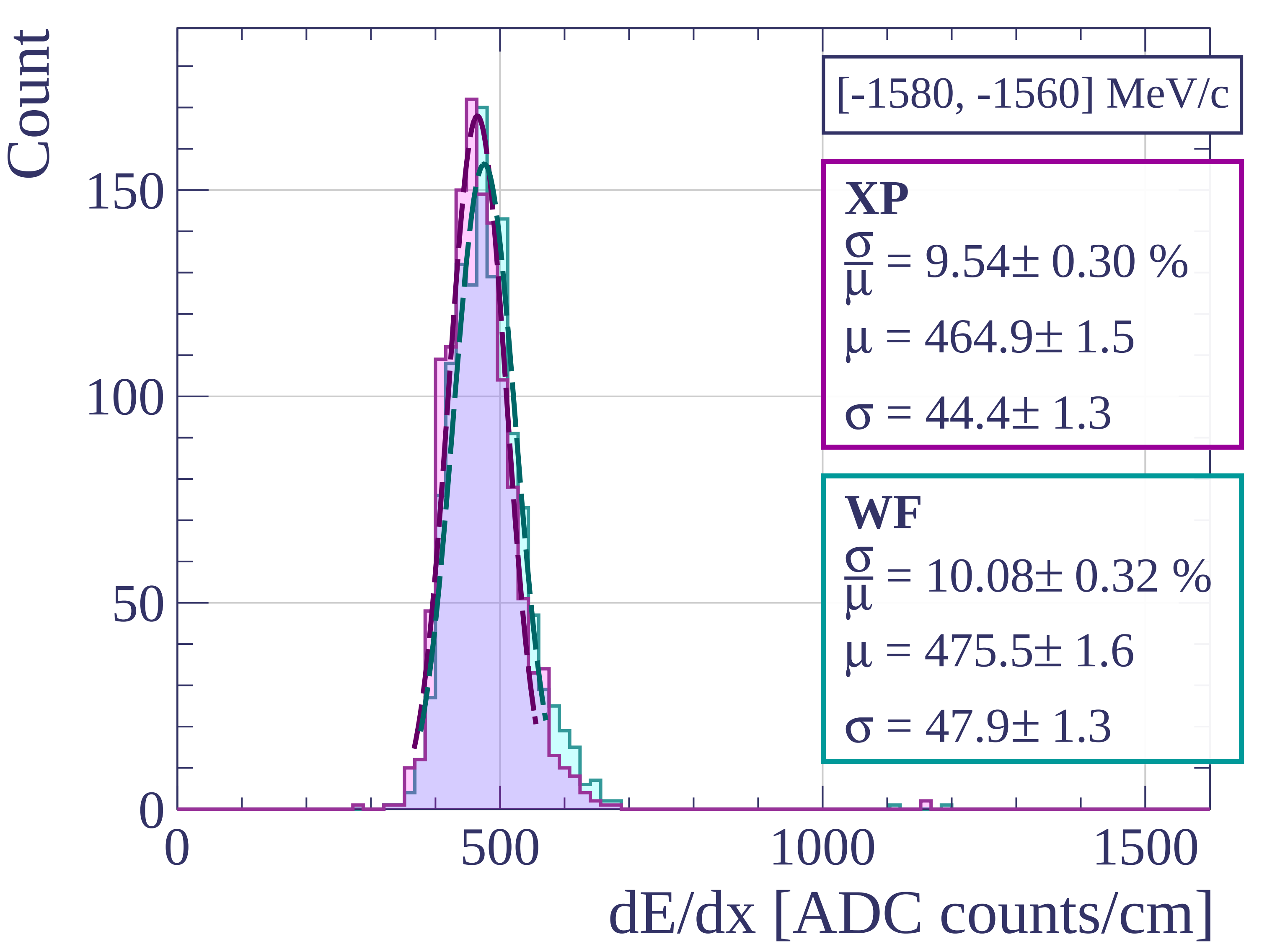
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1640, -1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.21± 0.32 %
μ = 462.9± 1.5
σ = 42.6± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.53± 0.38 %
μ = 474.7± 1.7
σ = 45.2± 1.6

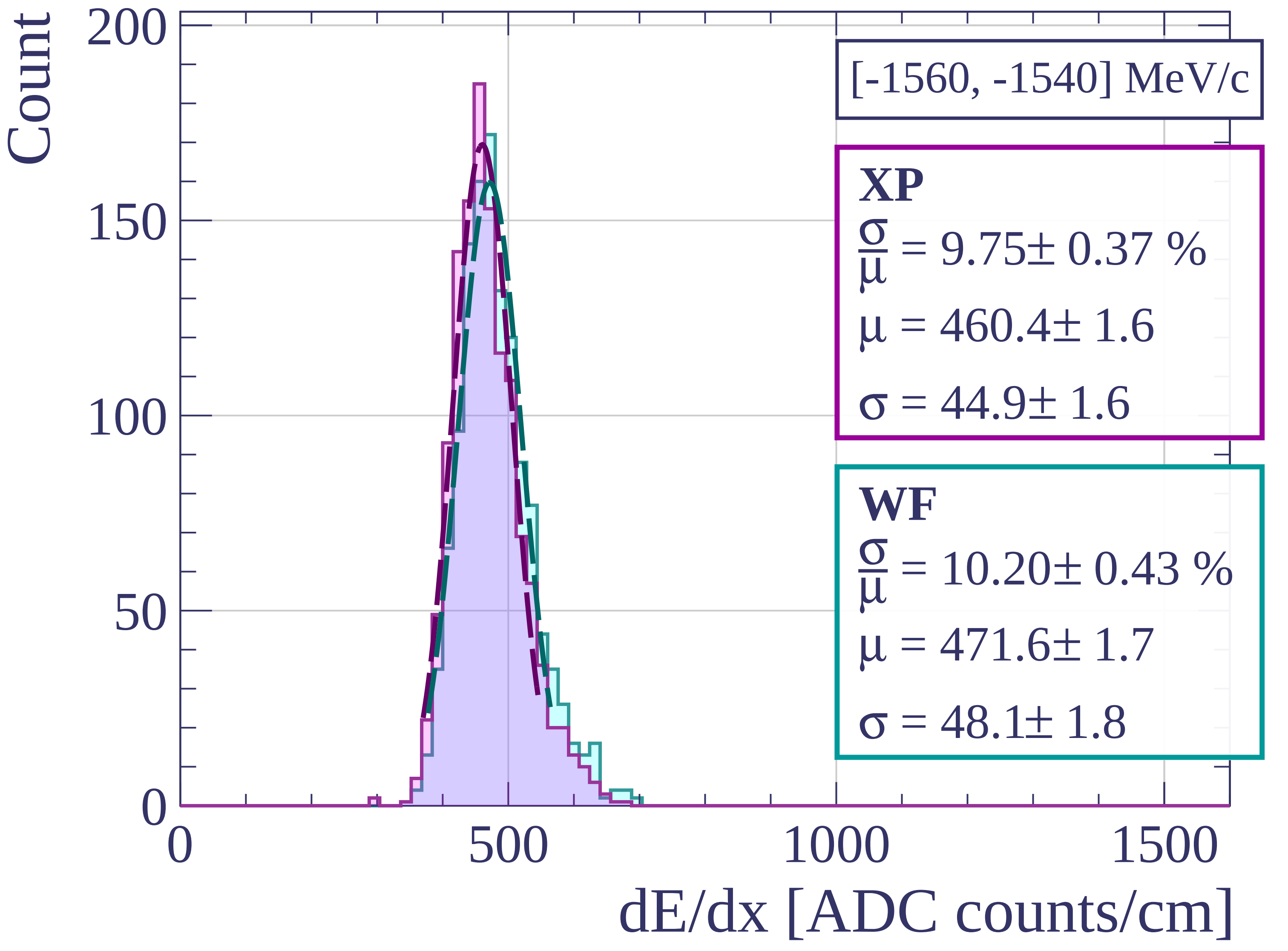


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1620, -1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.35± 0.42 %
μ = 458.5± 1.6
σ = 42.9± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.40 %
μ = 472.0± 1.6
σ = 45.2± 1.7

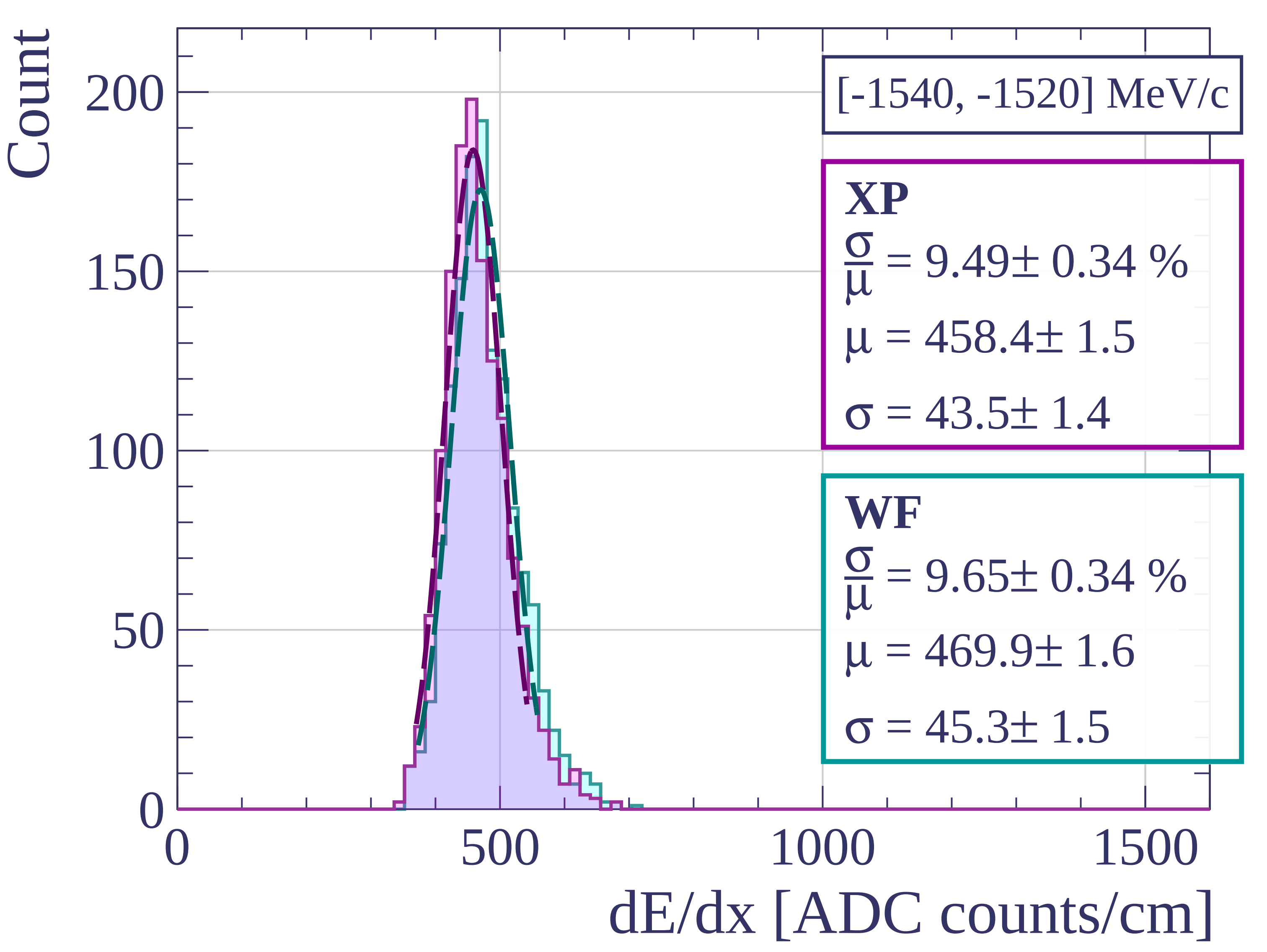


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1600, -1580] MeV/c
XP
σ/μ = 9.72± 0.38 %
μ = 461.7± 1.6
σ = 44.9± 1.6
WF
σ/μ = 9.95± 0.37 %
μ = 473.5± 1.8
σ = 47.1± 1.6

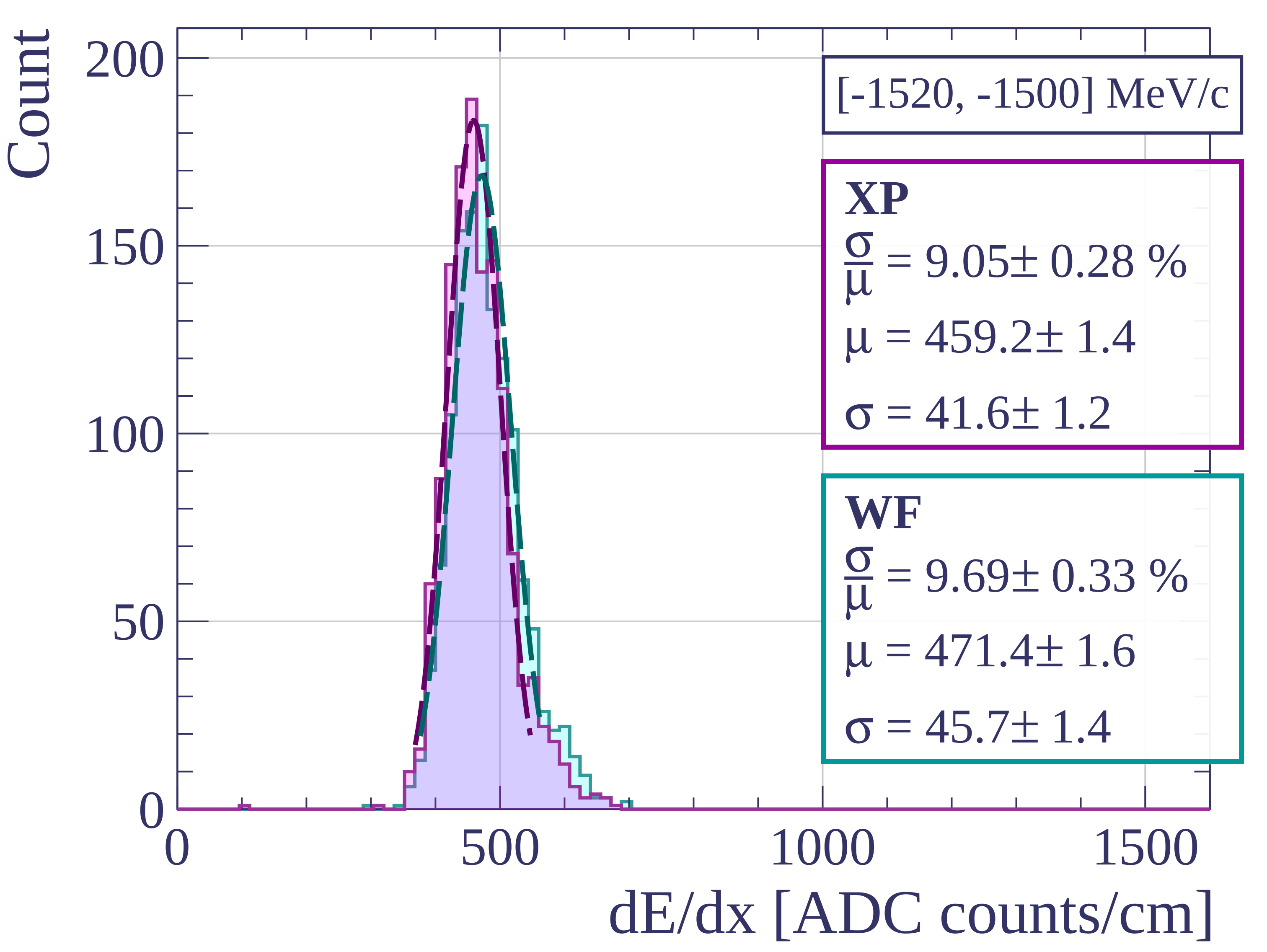

Count
dE/dx [ADC counts/cm]
[-1580, -1560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.54± 0.30 %
μ = 464.9± 1.5
σ = 44.4± 1.3
WF
$\frac{\sigma}{\mu}$ = 10.08± 0.32 %
μ = 475.5± 1.6
σ = 47.9± 1.3
0
50
100
150
0
500
1000
1500



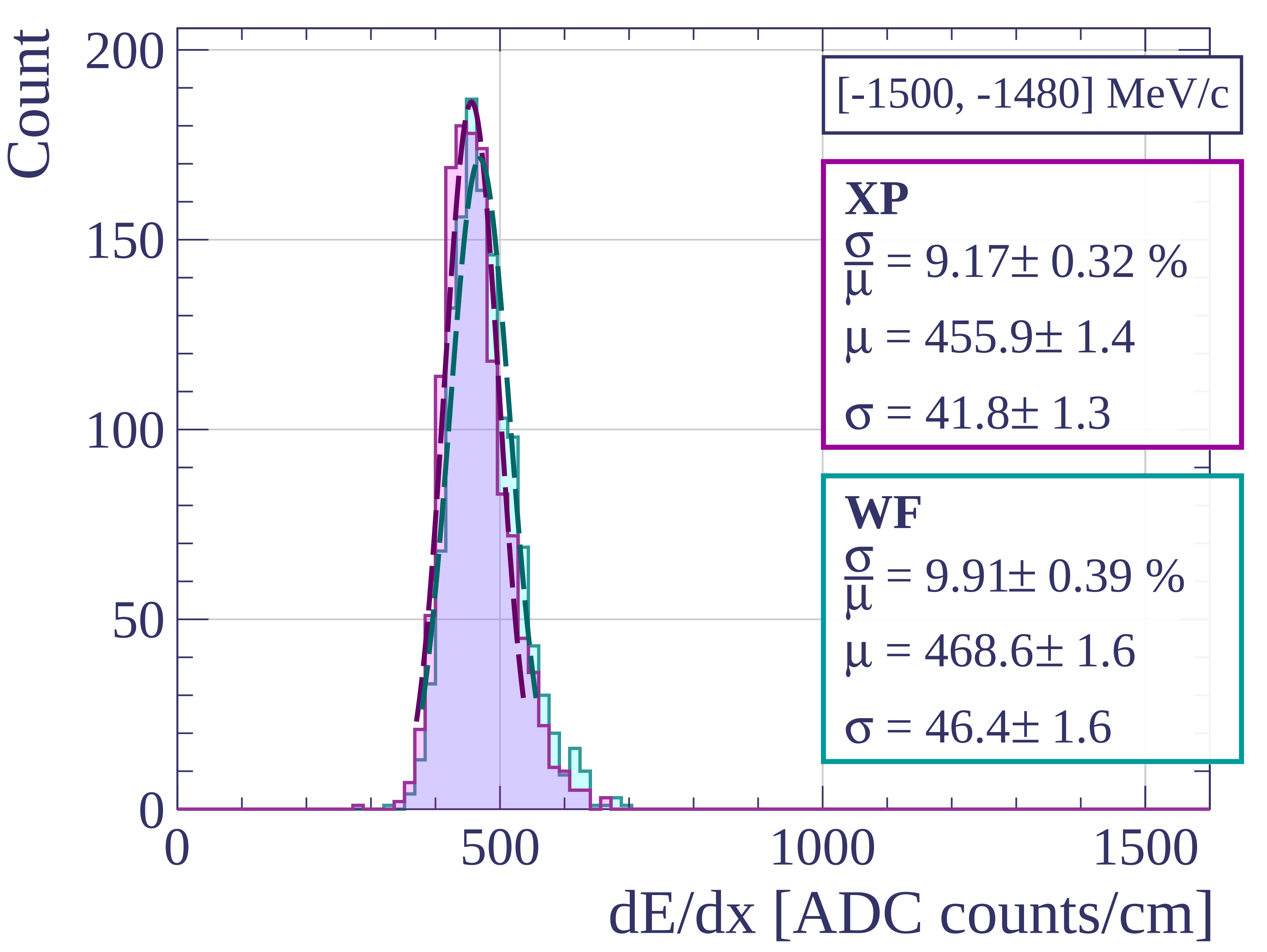
Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
[-1560, -1540] MeV/c
XP
σ/μ = 9.75± 0.37 %
μ = 460.4± 1.6
σ = 44.9± 1.6
WF
σ/μ = 10.20± 0.43 %
μ = 471.6± 1.7
σ = 48.1± 1.8

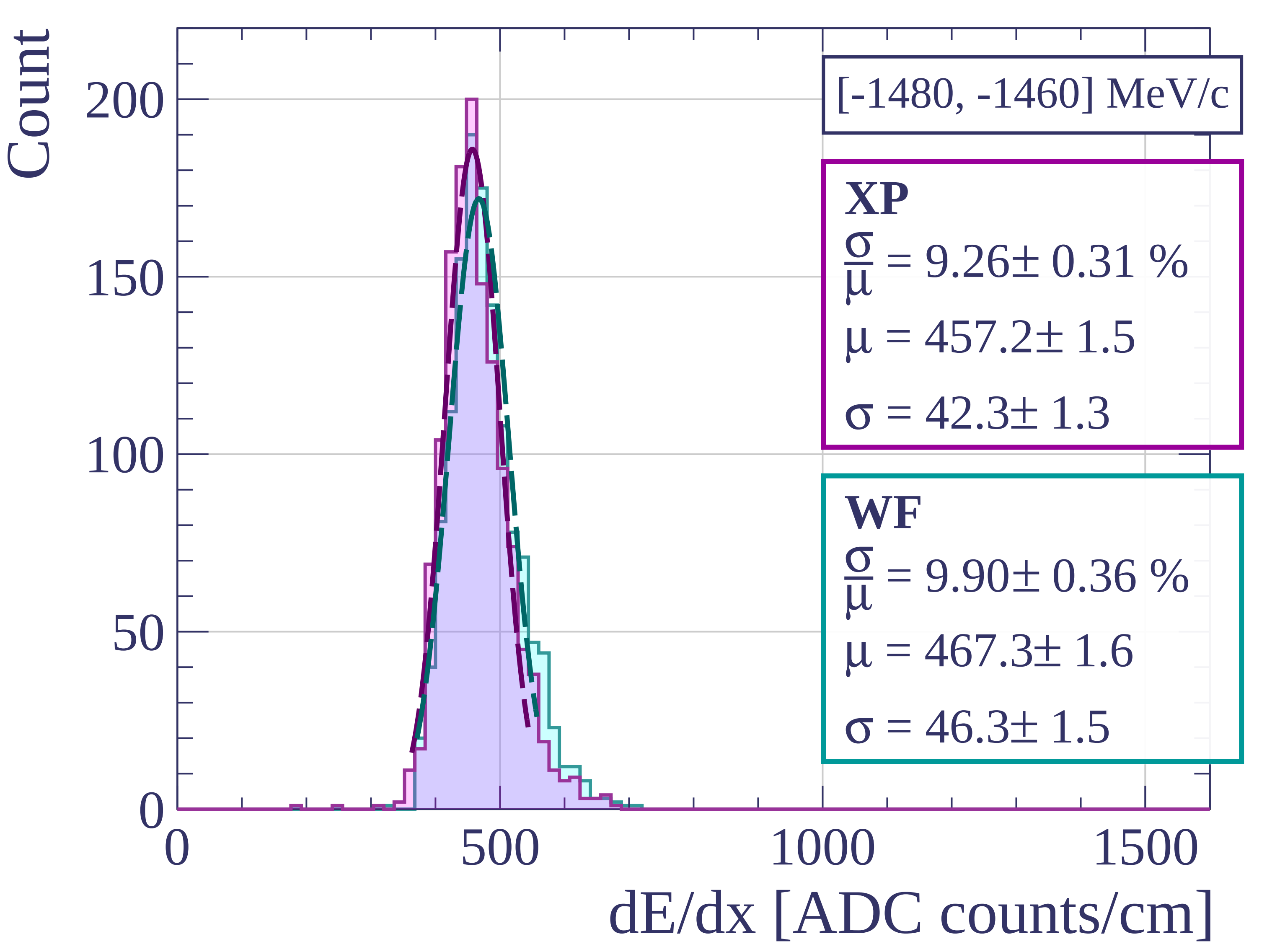


Count
dE/dx [ADC counts/cm]
[-1540, -1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.49± 0.34 %
μ = 458.4± 1.5
σ = 43.5± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.65± 0.34 %
μ = 469.9± 1.6
σ = 45.3± 1.5

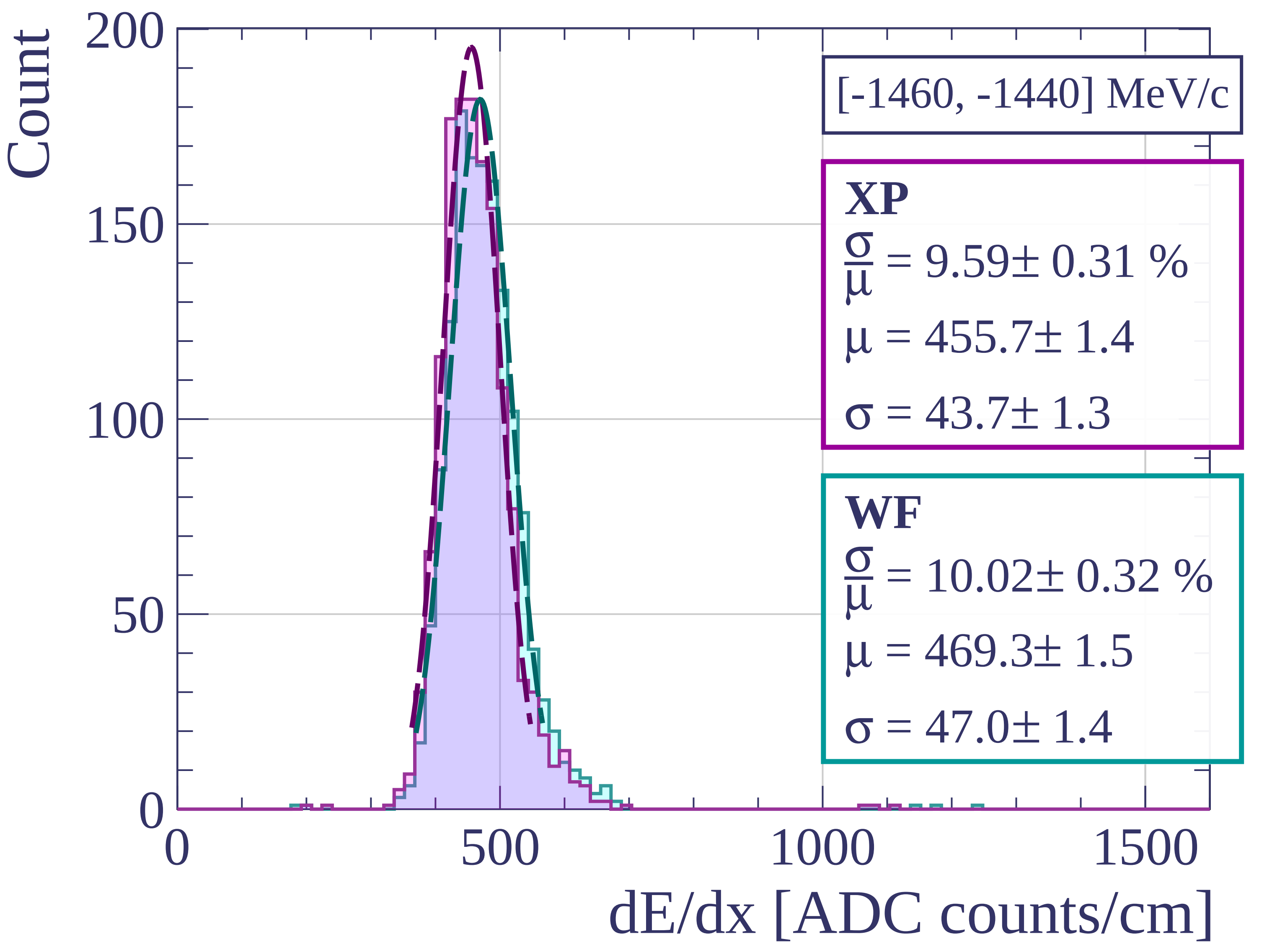


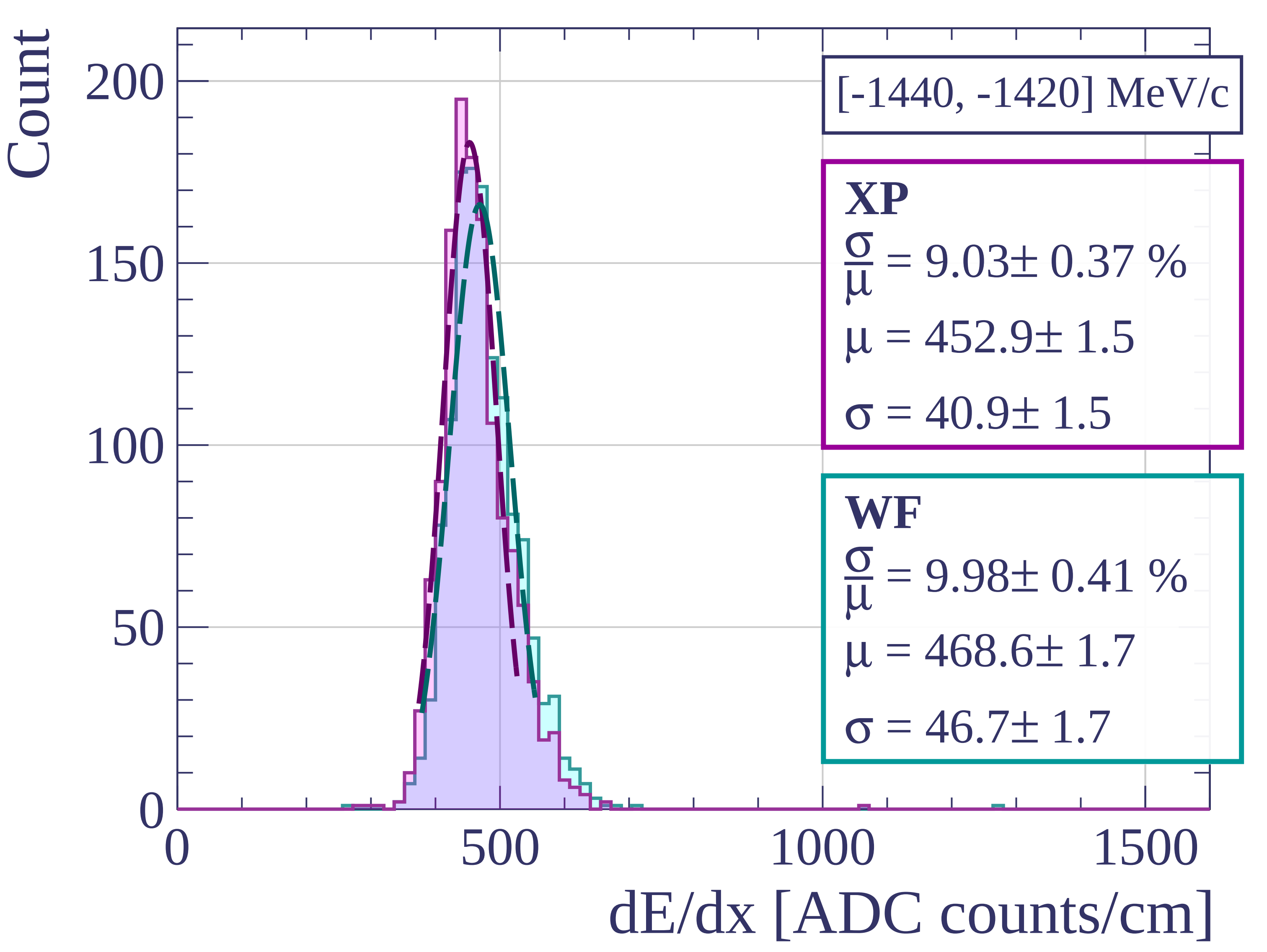
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1520, -1500] MeV/c
XP
σ/μ = 9.05± 0.28 %
μ = 459.2± 1.4
σ = 41.6± 1.2
WF
σ/μ = 9.69± 0.33 %
μ = 471.4± 1.6
σ = 45.7± 1.4


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1500, -1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.17± 0.32 %
μ = 455.9± 1.4
σ = 41.8± 1.3
WF
$\frac{\sigma}{\mu}$ = 9.91± 0.39 %
μ = 468.6± 1.6
σ = 46.4± 1.6

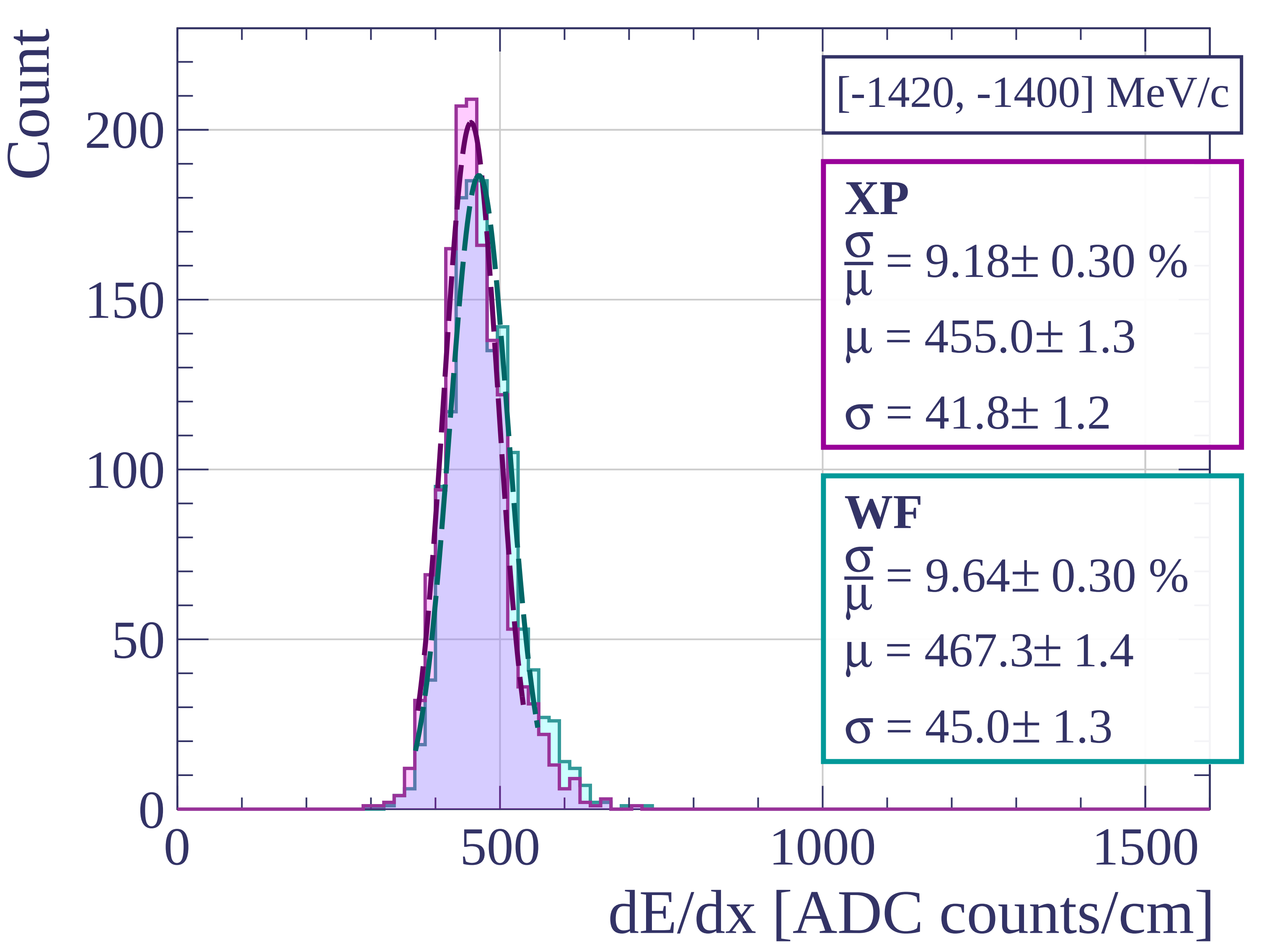


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1480, -1460] MeV/c
XP
σ/μ = 9.26± 0.31 %
μ = 457.2± 1.5
σ = 42.3± 1.3
WF
σ/μ = 9.90± 0.36 %
μ = 467.3± 1.6
σ = 46.3± 1.5

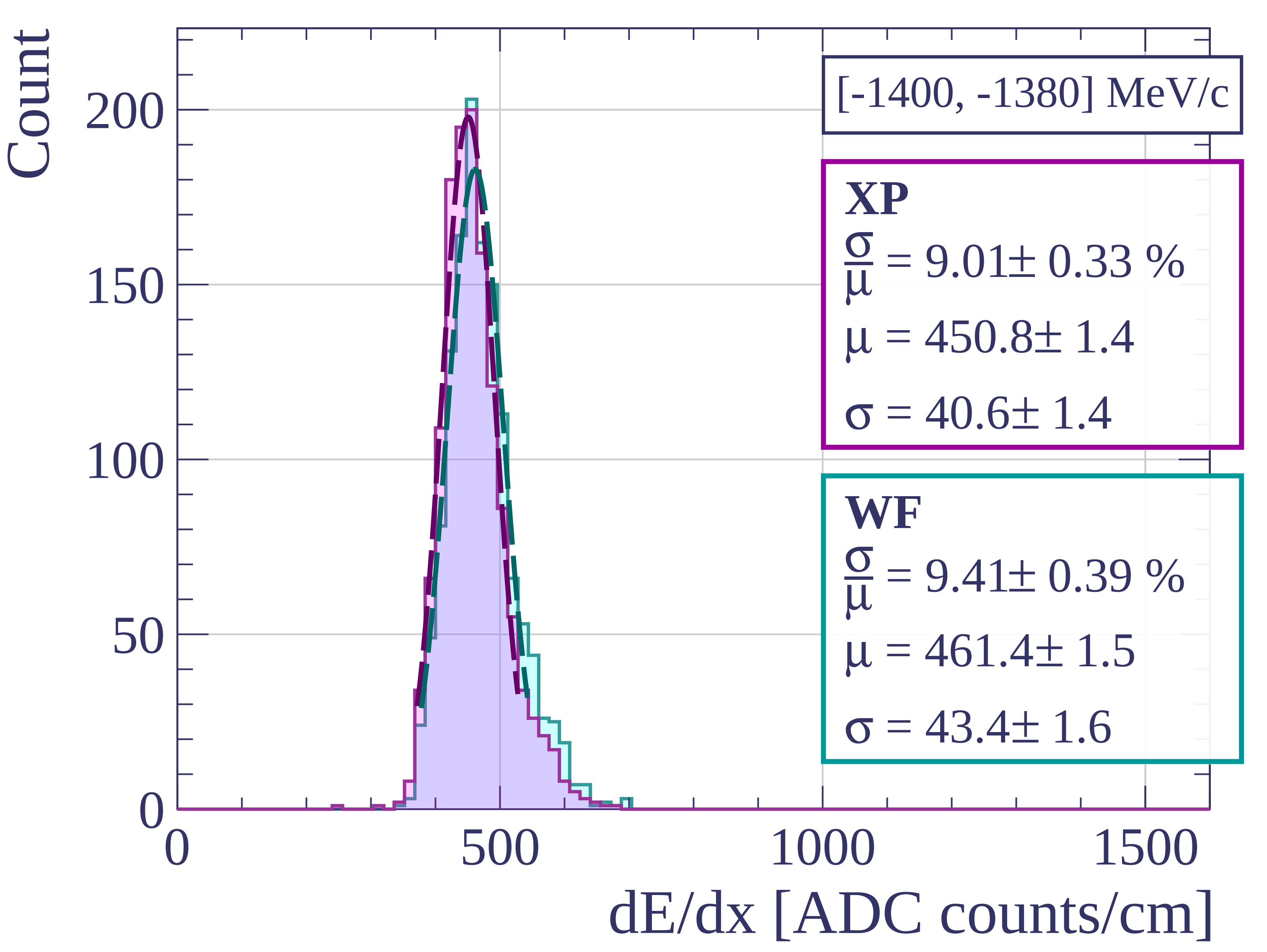

Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1460, -1440] MeV/c
XP
σ/μ = 9.59± 0.31 %
μ = 455.7± 1.4
σ = 43.7± 1.3
WF
σ/μ = 10.02± 0.32 %
μ = 469.3± 1.5
σ = 47.0± 1.4

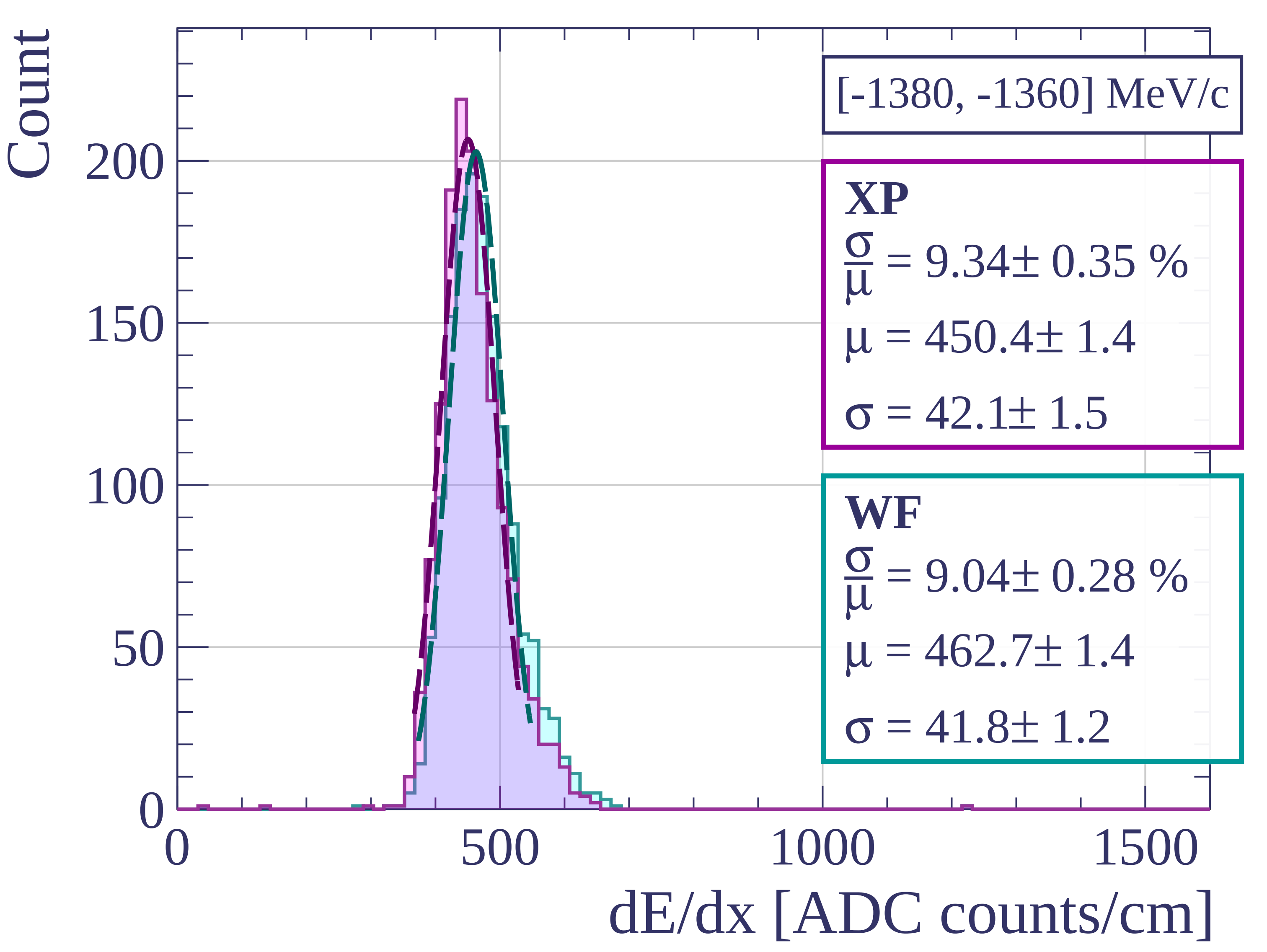


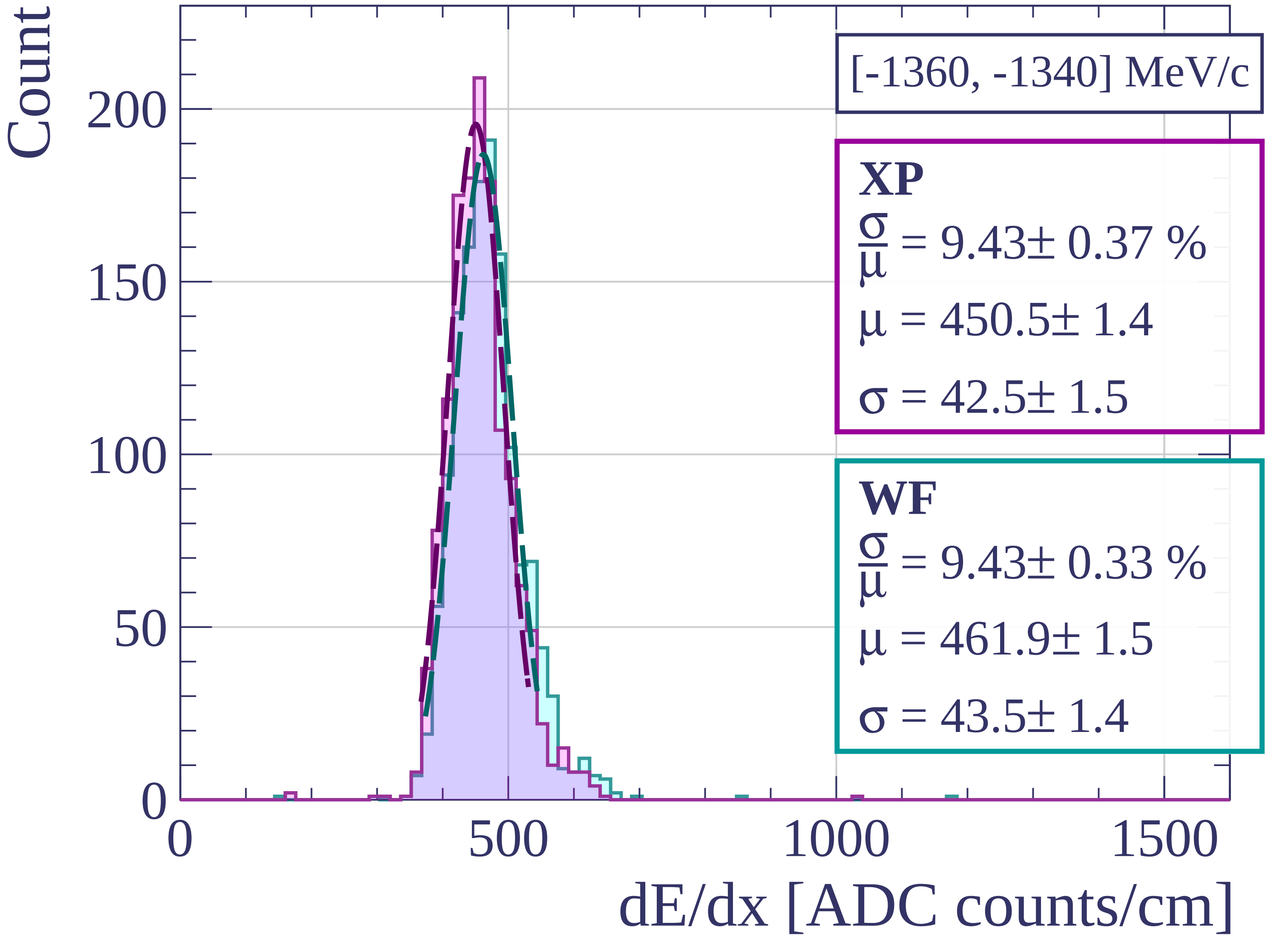
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1440, -1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.03± 0.37 %
μ = 452.9± 1.5
σ = 40.9± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.98± 0.41 %
μ = 468.6± 1.7
σ = 46.7± 1.7



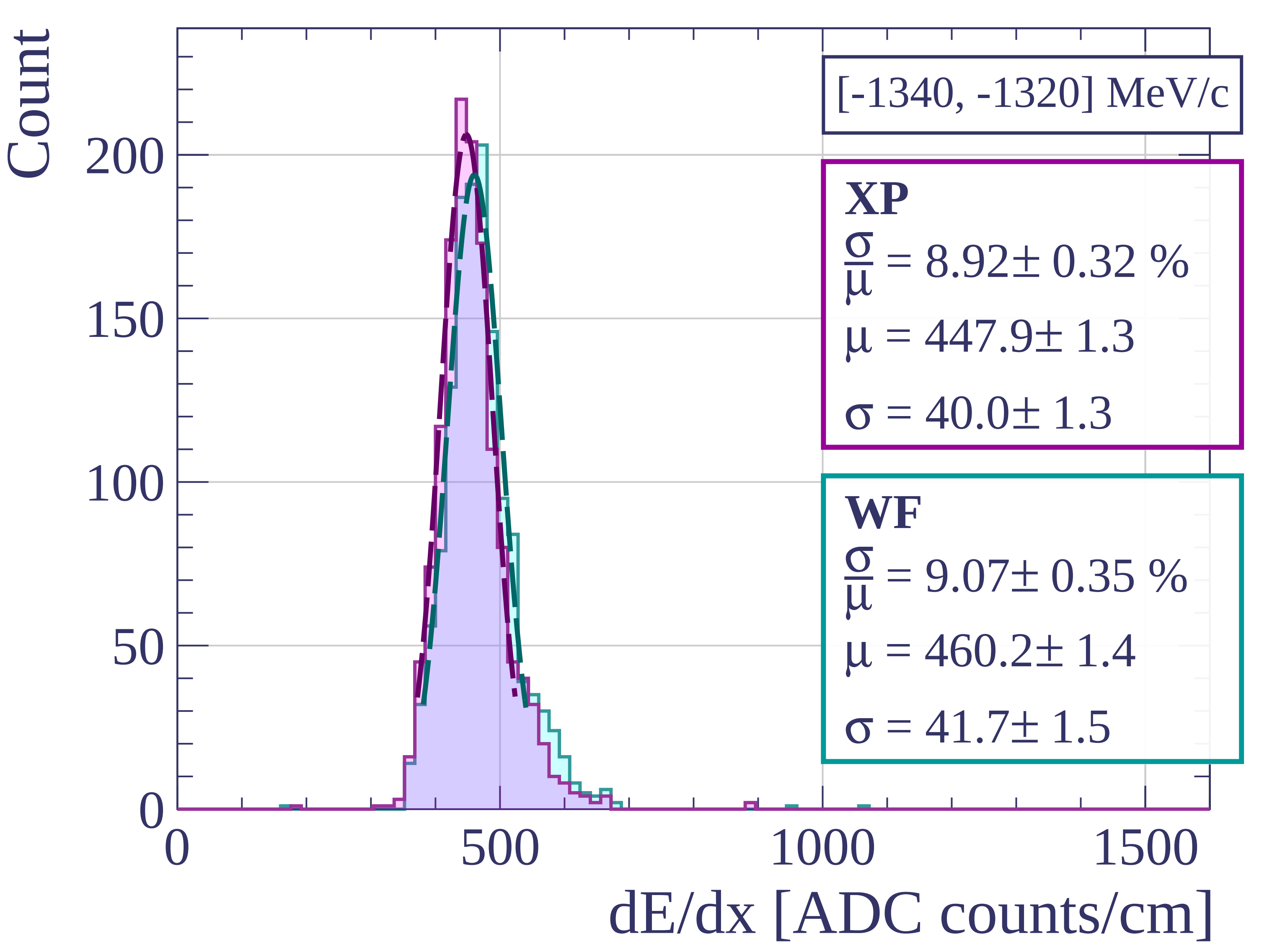
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1420, -1400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.18± 0.30 %
μ = 455.0± 1.3
σ = 41.8± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.64± 0.30 %
μ = 467.3± 1.4
σ = 45.0± 1.3

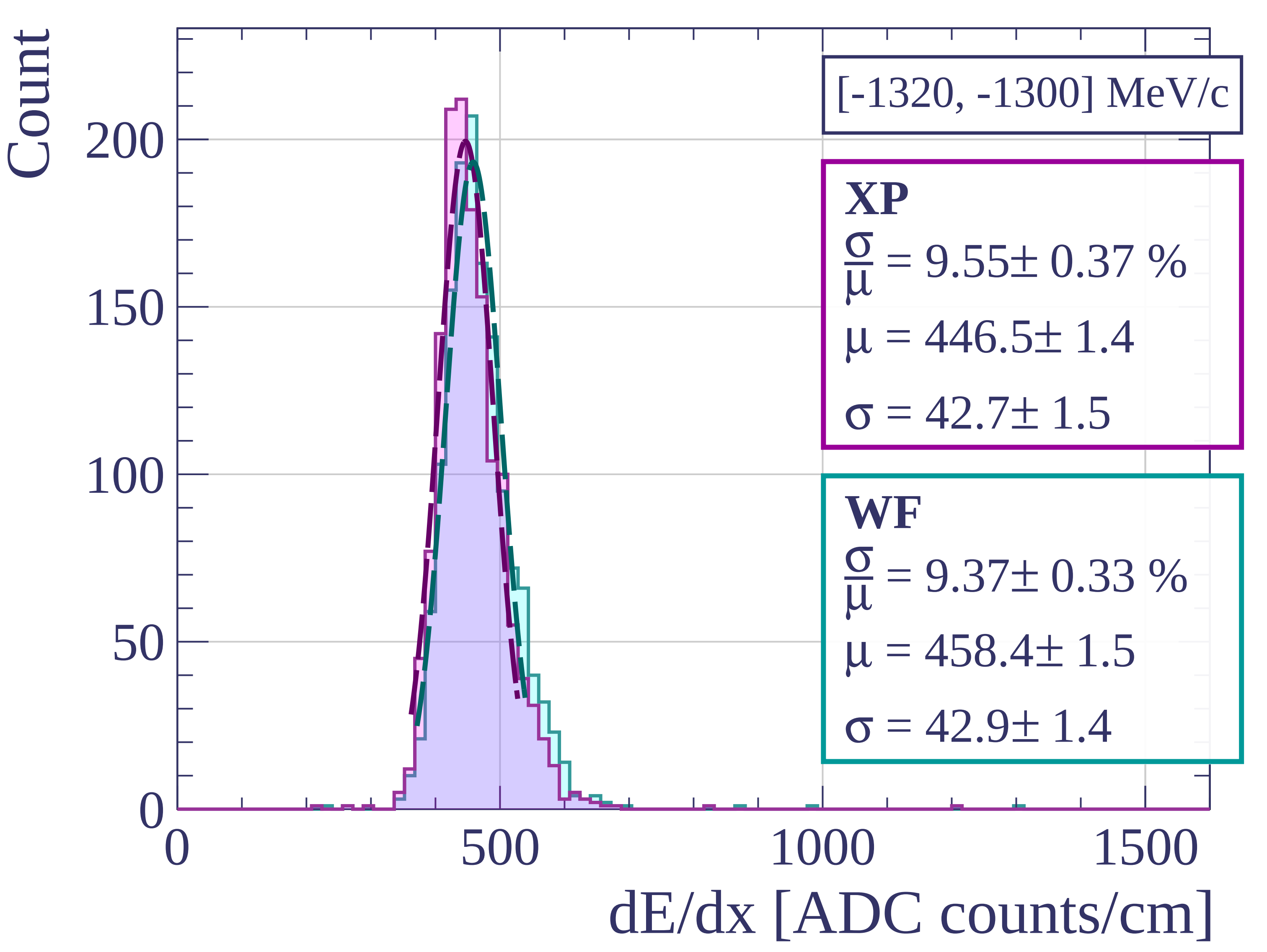

Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1400, -1380] MeV/c
XP
σ/μ = 9.01± 0.33 %
μ = 450.8± 1.4
σ = 40.6± 1.4
WF
σ/μ = 9.41± 0.39 %
μ = 461.4± 1.5
σ = 43.4± 1.6


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
50
100
150
200
[-1380, -1360] MeV/c
XP
σ/μ = 9.34± 0.35 %
μ = 450.4± 1.4
σ = 42.1± 1.5
WF
σ/μ = 9.04± 0.28 %
μ = 462.7± 1.4
σ = 41.8± 1.2

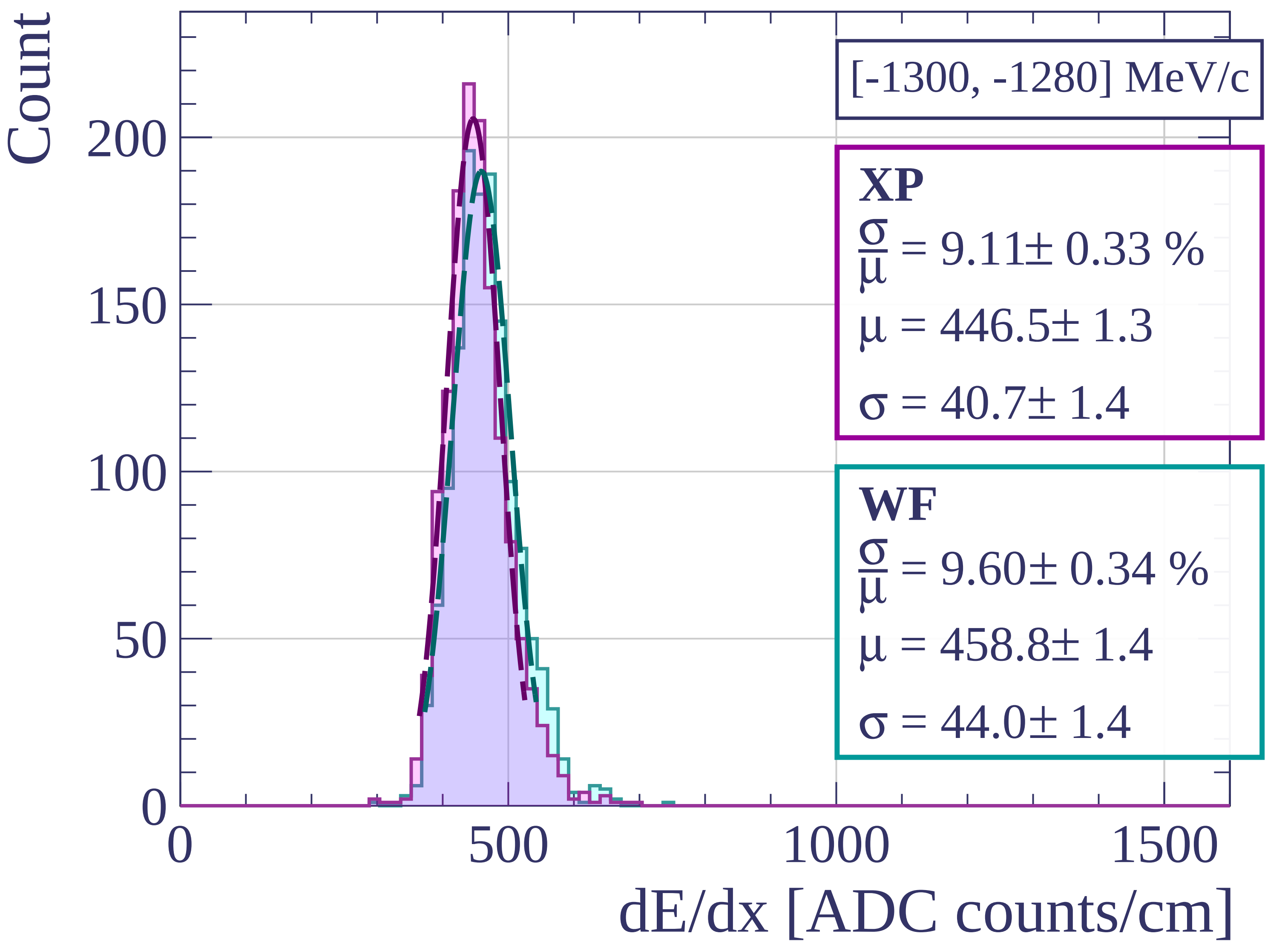


Count
dE/dx [ADC counts/cm]
[-1360, -1340] MeV/c
XP
σ/μ = 9.43± 0.37 %
μ = 450.5± 1.4
σ = 42.5± 1.5
WF
σ/μ = 9.43± 0.33 %
μ = 461.9± 1.5
σ = 43.5± 1.4
0
50
100
150
200
500
1000
1500

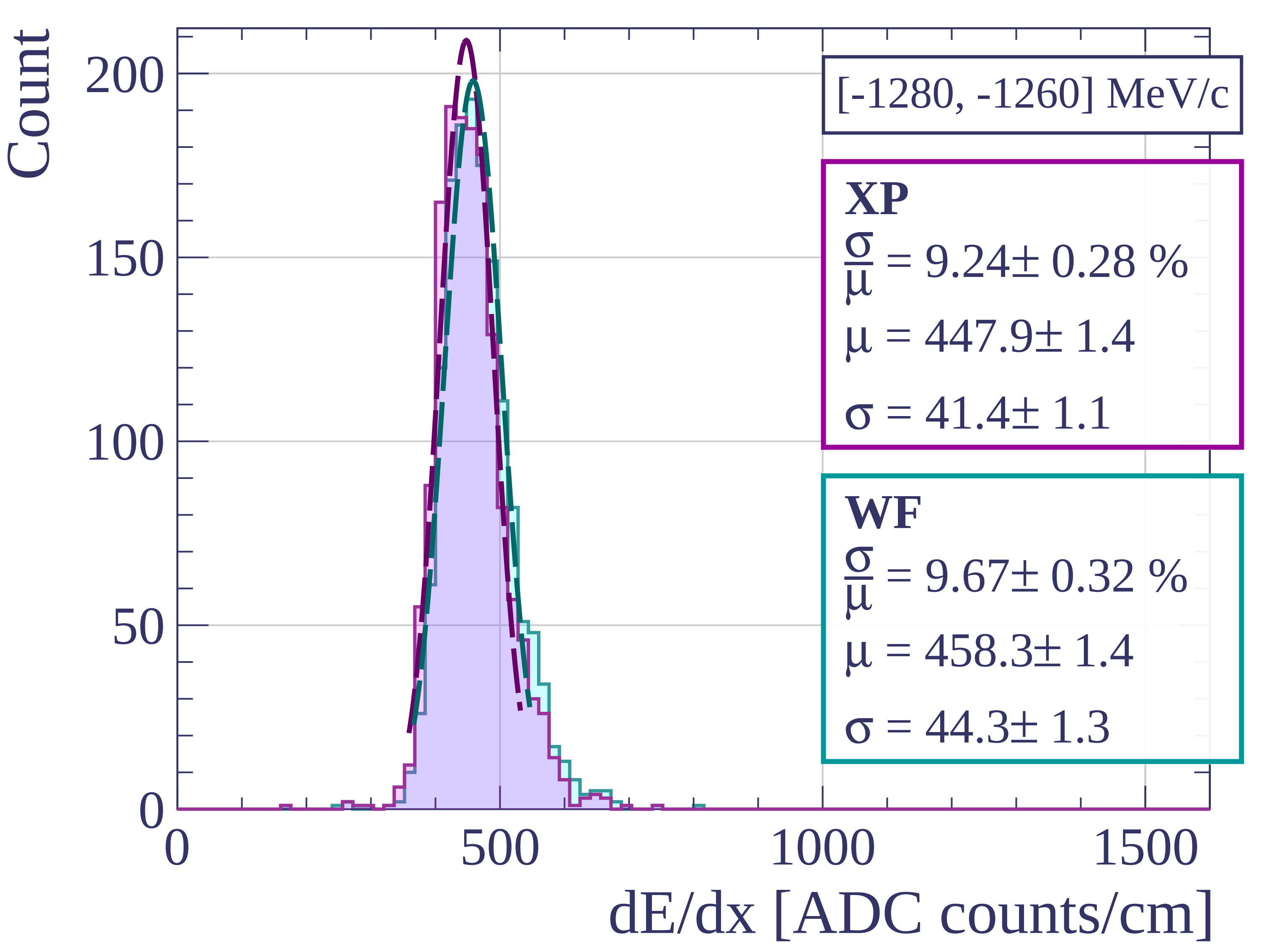

Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1340, -1320] MeV/c
XP
σ/μ = 8.92± 0.32 %
μ = 447.9± 1.3
σ = 40.0± 1.3
WF
σ/μ = 9.07± 0.35 %
μ = 460.2± 1.4
σ = 41.7± 1.5



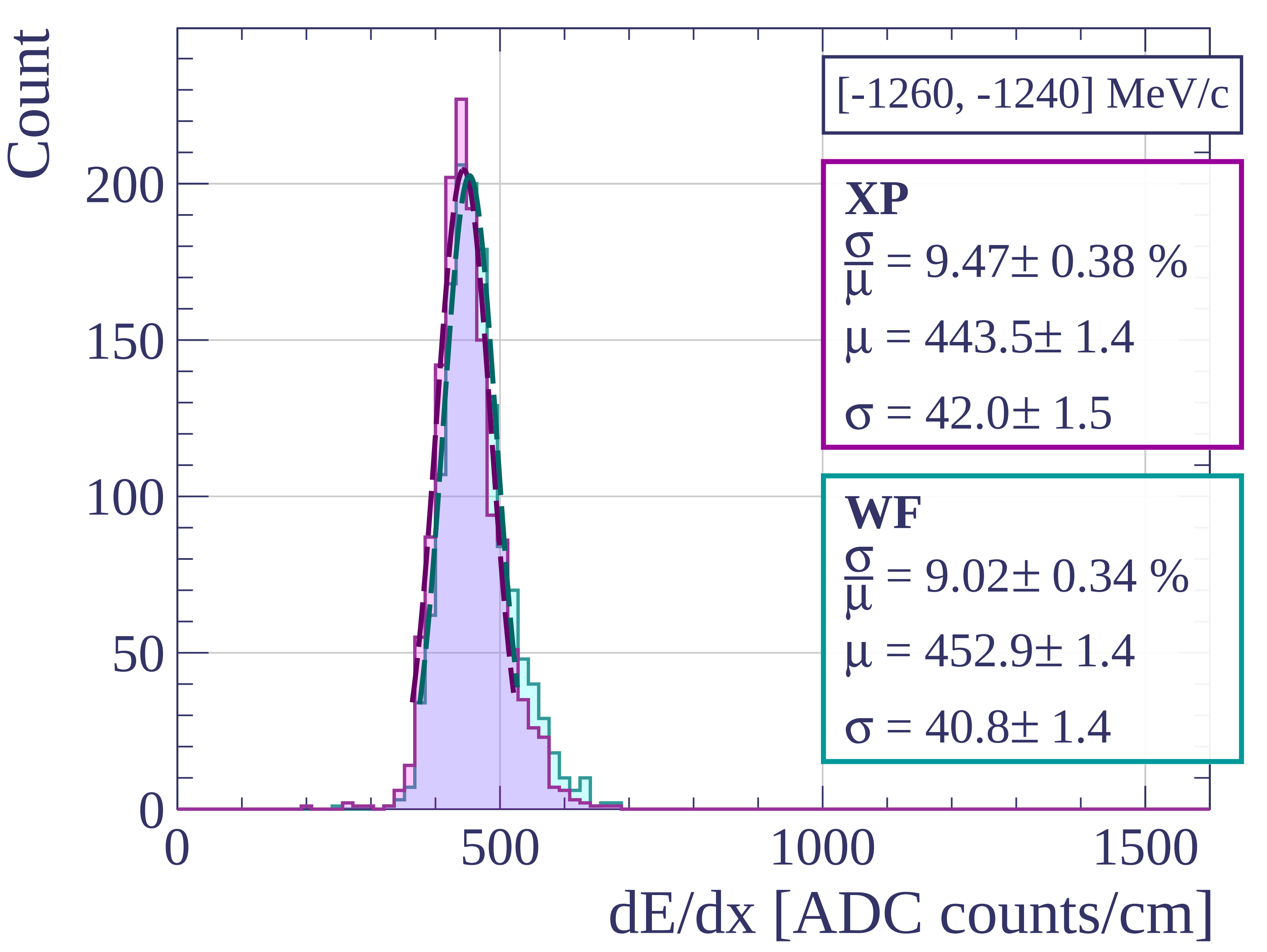
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1320, -1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.55± 0.37 %
μ = 446.5± 1.4
σ = 42.7± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.37± 0.33 %
μ = 458.4± 1.5
σ = 42.9± 1.4



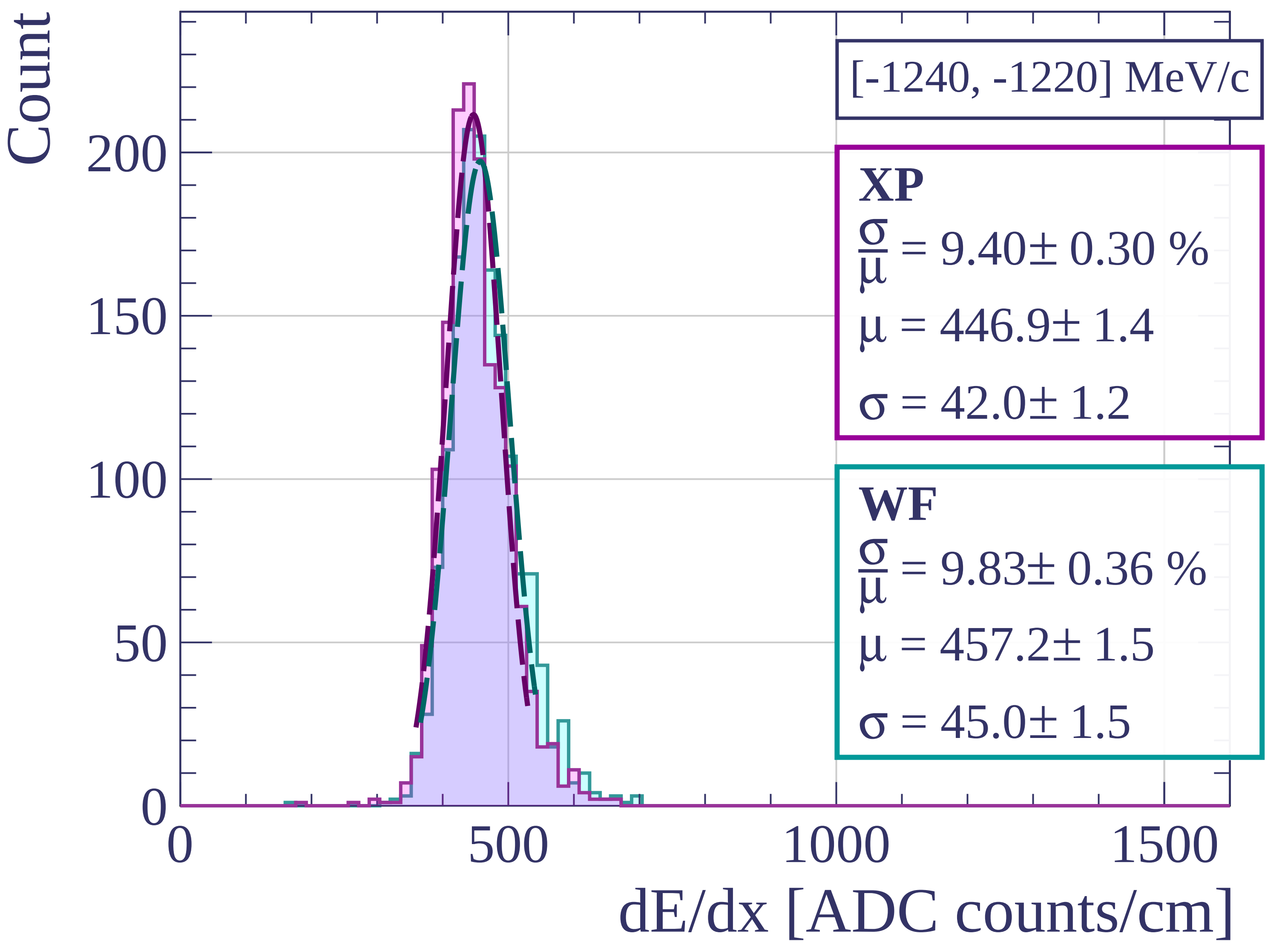
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1300, -1280] MeV/c
XP
σ/μ = 9.11± 0.33 %
μ = 446.5± 1.3
σ = 40.7± 1.4
WF
σ/μ = 9.60± 0.34 %
μ = 458.8± 1.4
σ = 44.0± 1.4



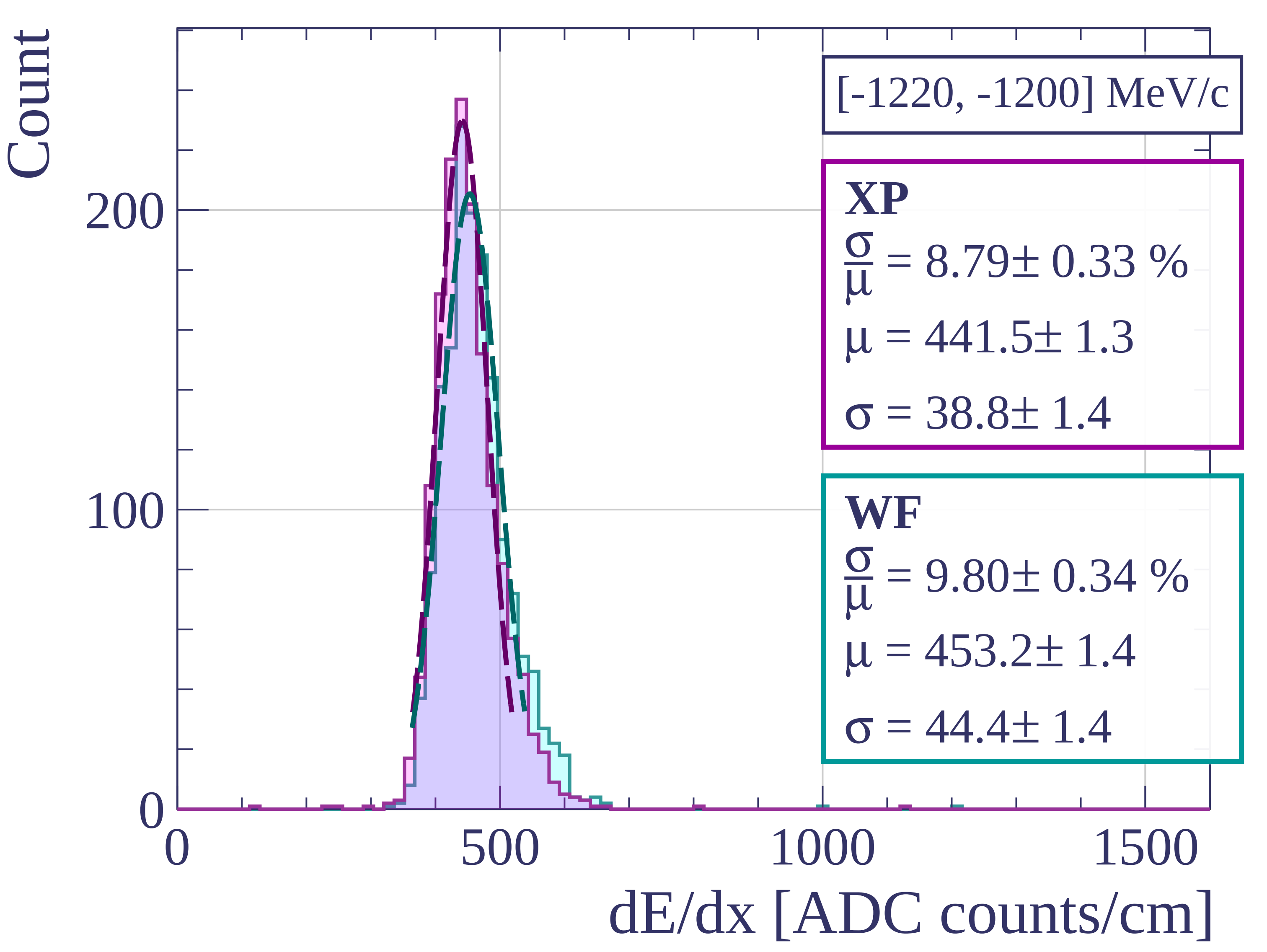
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1280, -1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.24± 0.28 %
μ = 447.9± 1.4
σ = 41.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.67± 0.32 %
μ = 458.3± 1.4
σ = 44.3± 1.3



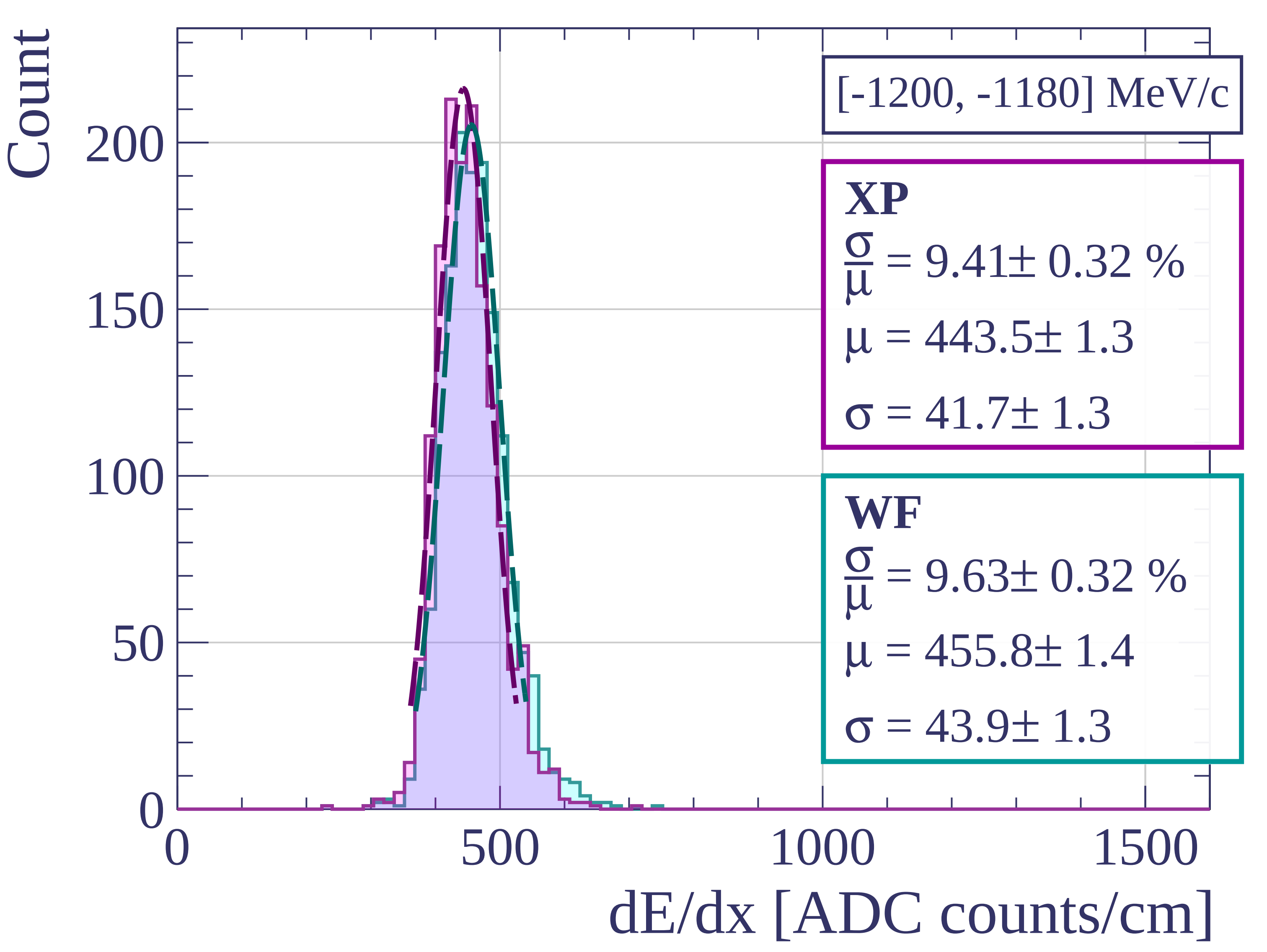
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1260, -1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.47± 0.38 %
μ = 443.5± 1.4
σ = 42.0± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.02± 0.34 %
μ = 452.9± 1.4
σ = 40.8± 1.4



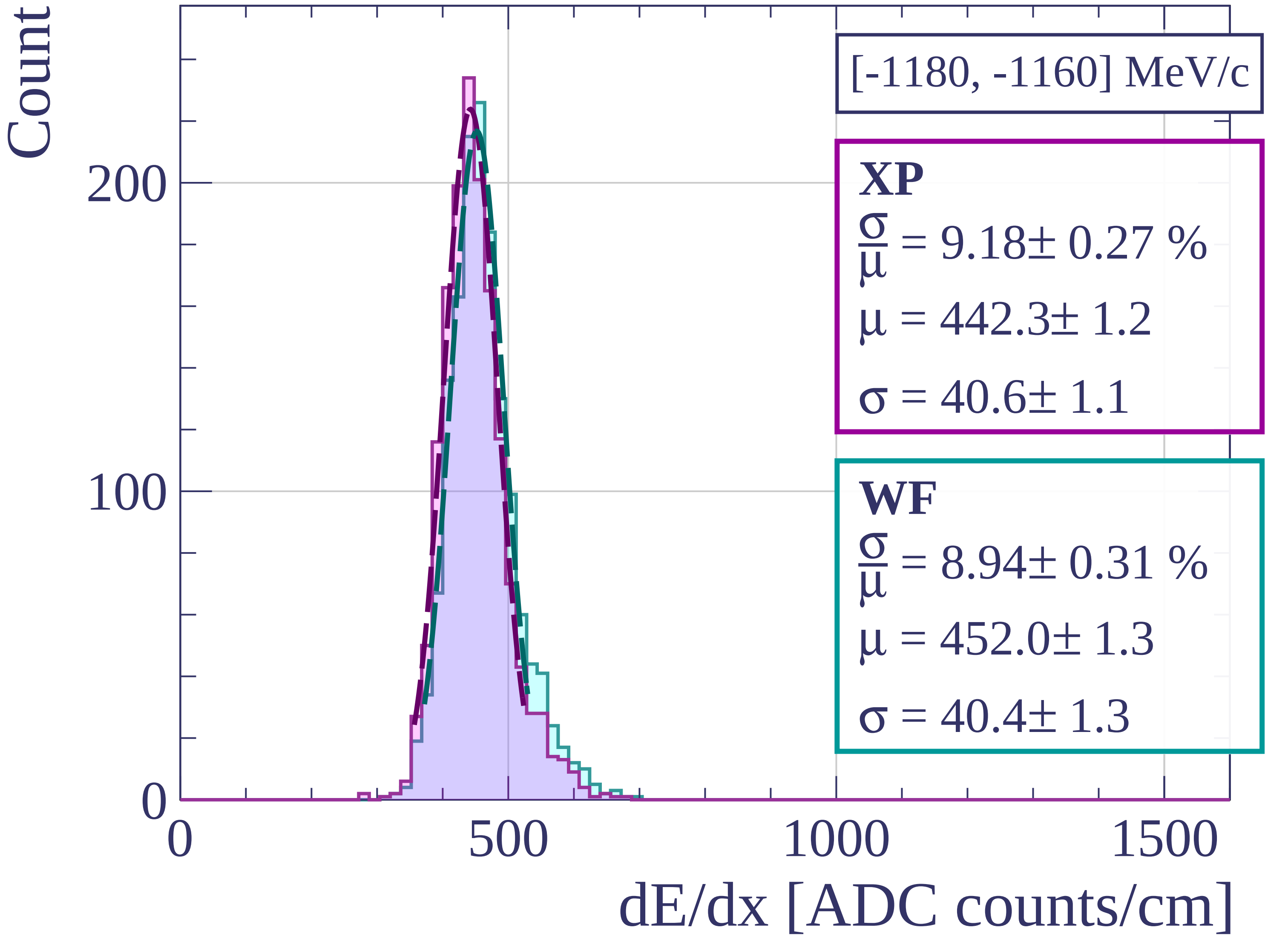
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1240, -1220] MeV/c
XP
σ/μ = 9.40± 0.30 %
μ = 446.9± 1.4
σ = 42.0± 1.2
WF
σ/μ = 9.83± 0.36 %
μ = 457.2± 1.5
σ = 45.0± 1.5

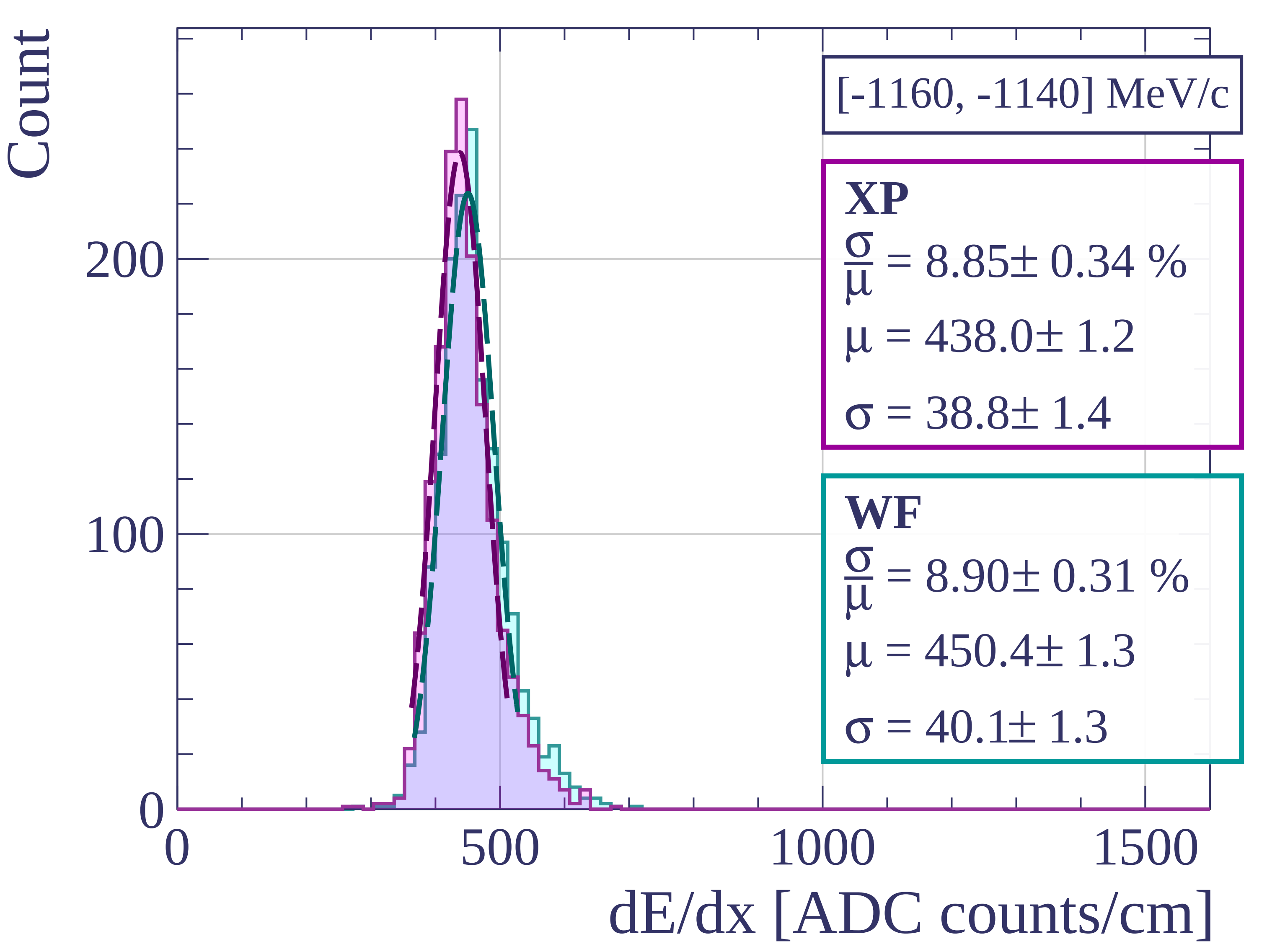


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1220, -1200] MeV/c
XP
σ/μ = 8.79± 0.33 %
μ = 441.5± 1.3
σ = 38.8± 1.4
WF
σ/μ = 9.80± 0.34 %
μ = 453.2± 1.4
σ = 44.4± 1.4

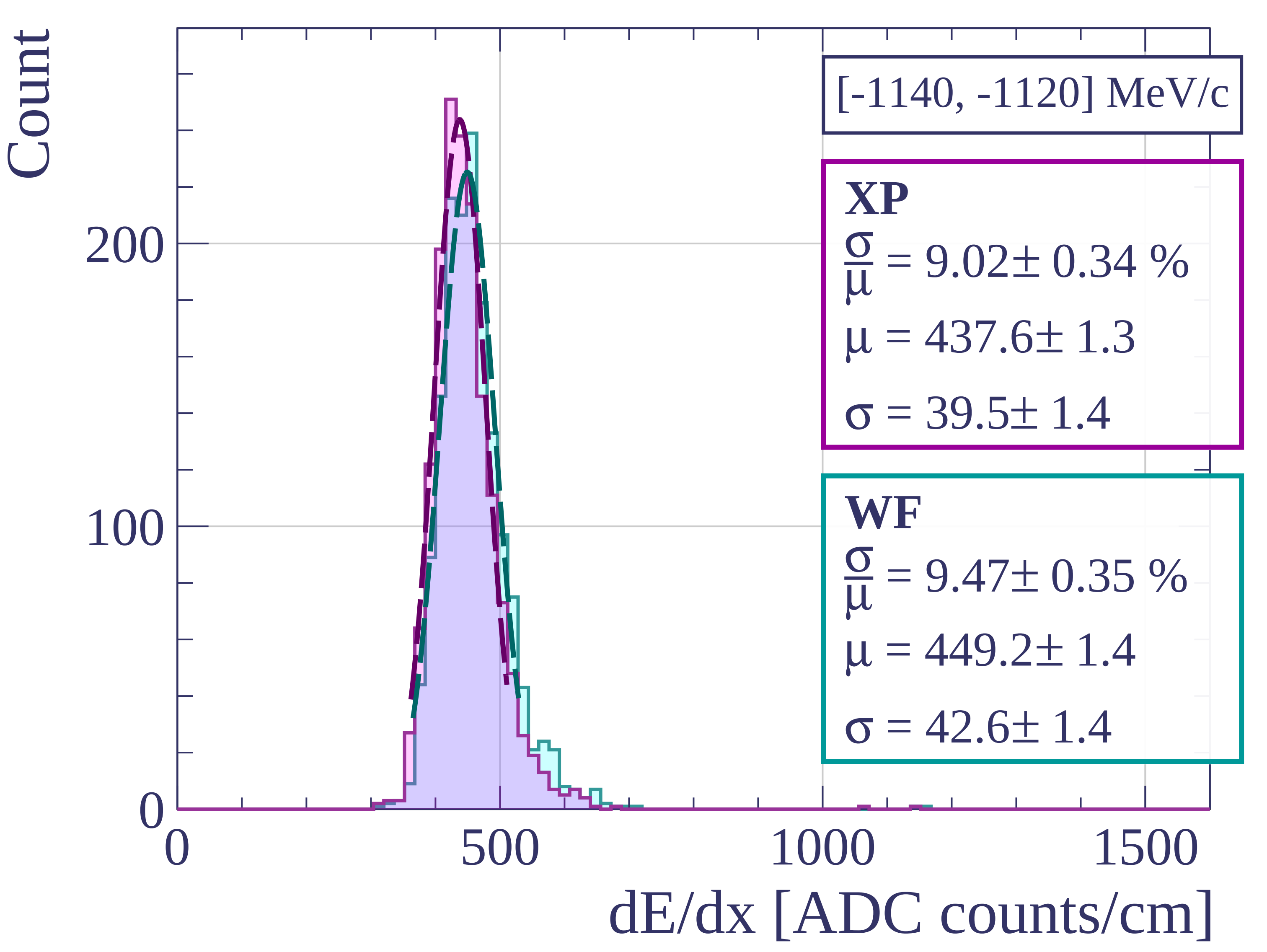


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1200, -1180] MeV/c
XP
σ/μ = 9.41± 0.32 %
μ = 443.5± 1.3
σ = 41.7± 1.3
WF
σ/μ = 9.63± 0.32 %
μ = 455.8± 1.4
σ = 43.9± 1.3

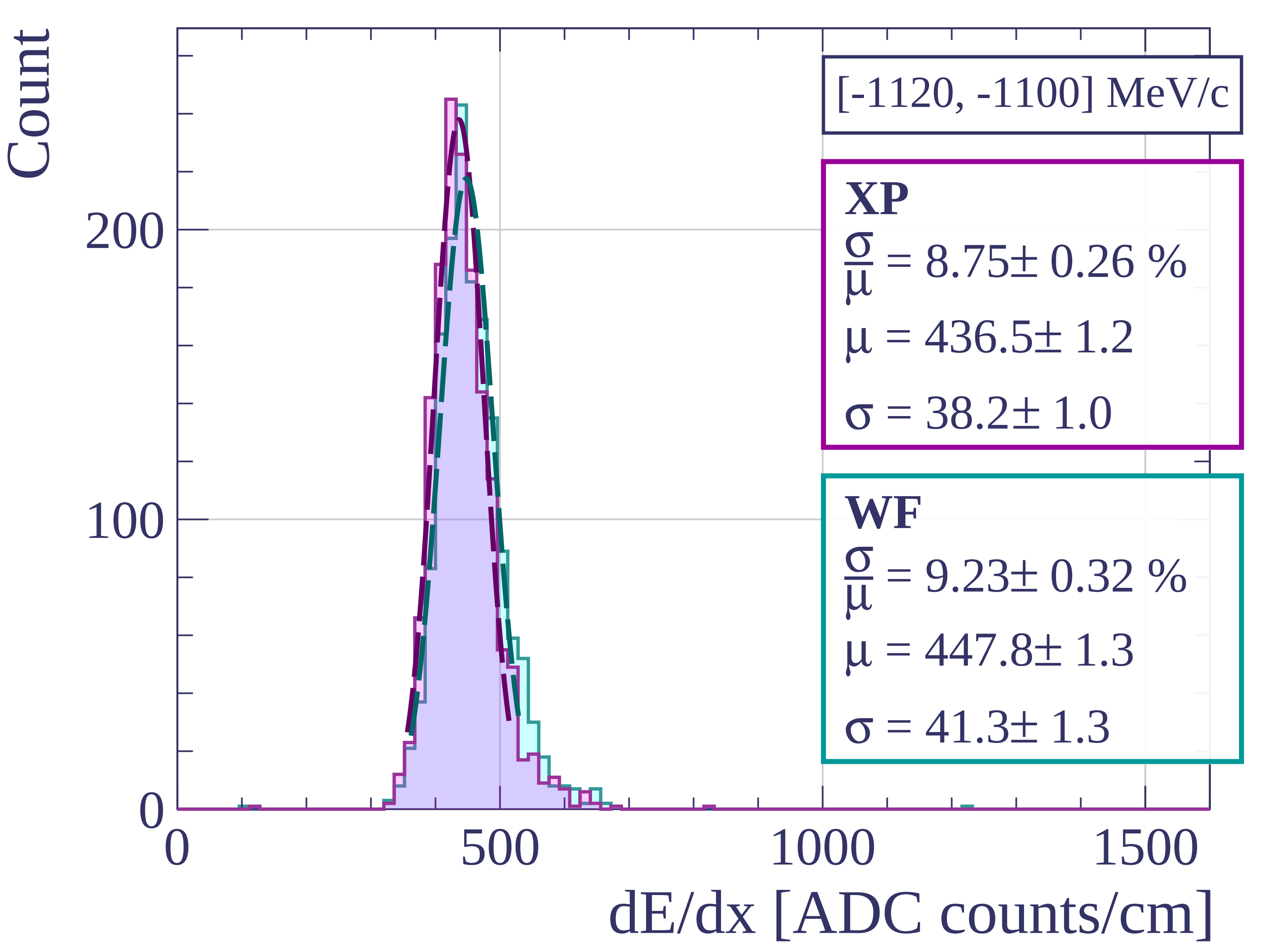

Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1180, -1160] MeV/c
XP
σ/μ = 9.18± 0.27 %
μ = 442.3± 1.2
σ = 40.6± 1.1
WF
σ/μ = 8.94± 0.31 %
μ = 452.0± 1.3
σ = 40.4± 1.3



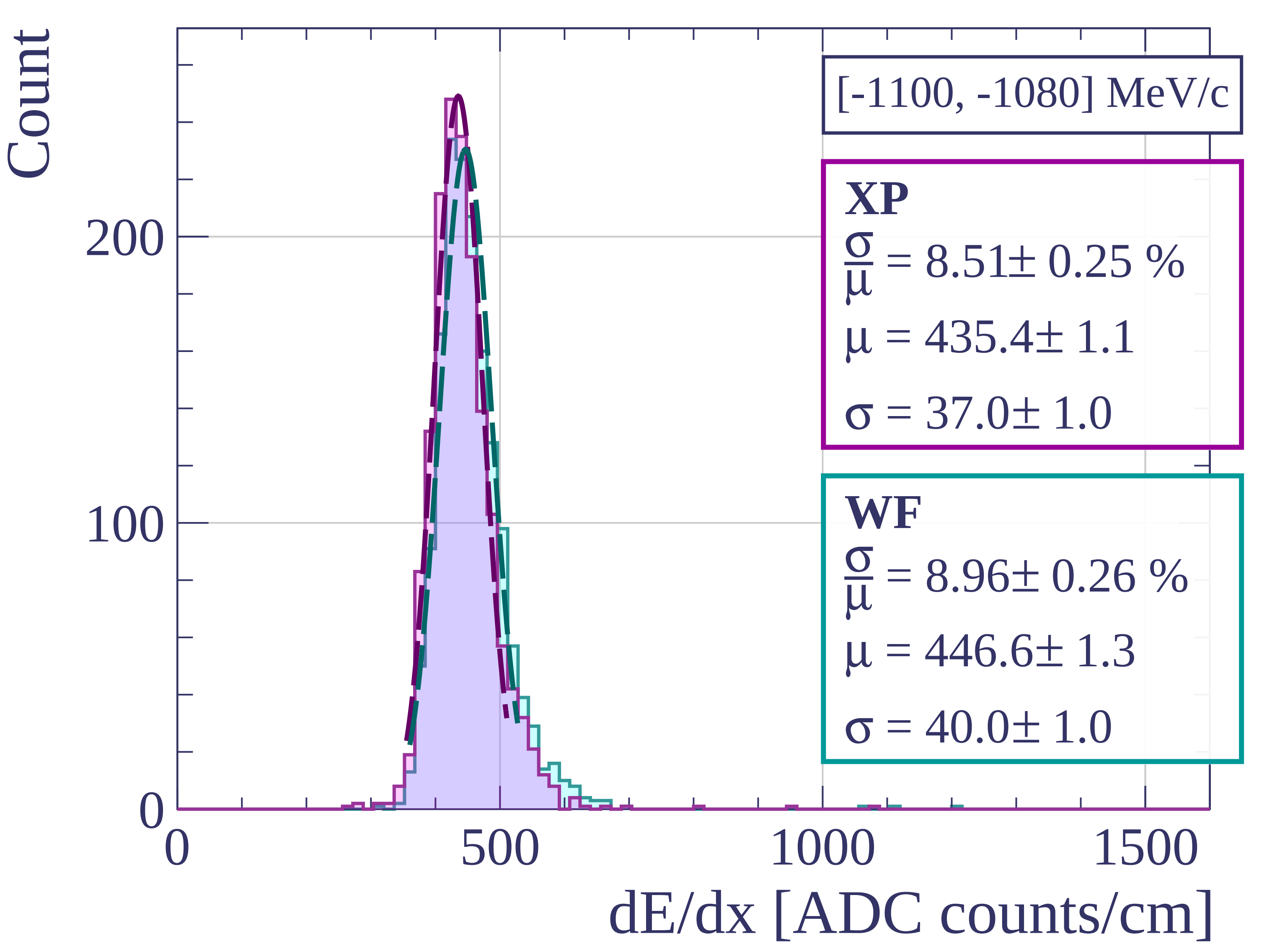
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1160, -1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.85± 0.34 %
μ = 438.0± 1.2
σ = 38.8± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.90± 0.31 %
μ = 450.4± 1.3
σ = 40.1± 1.3

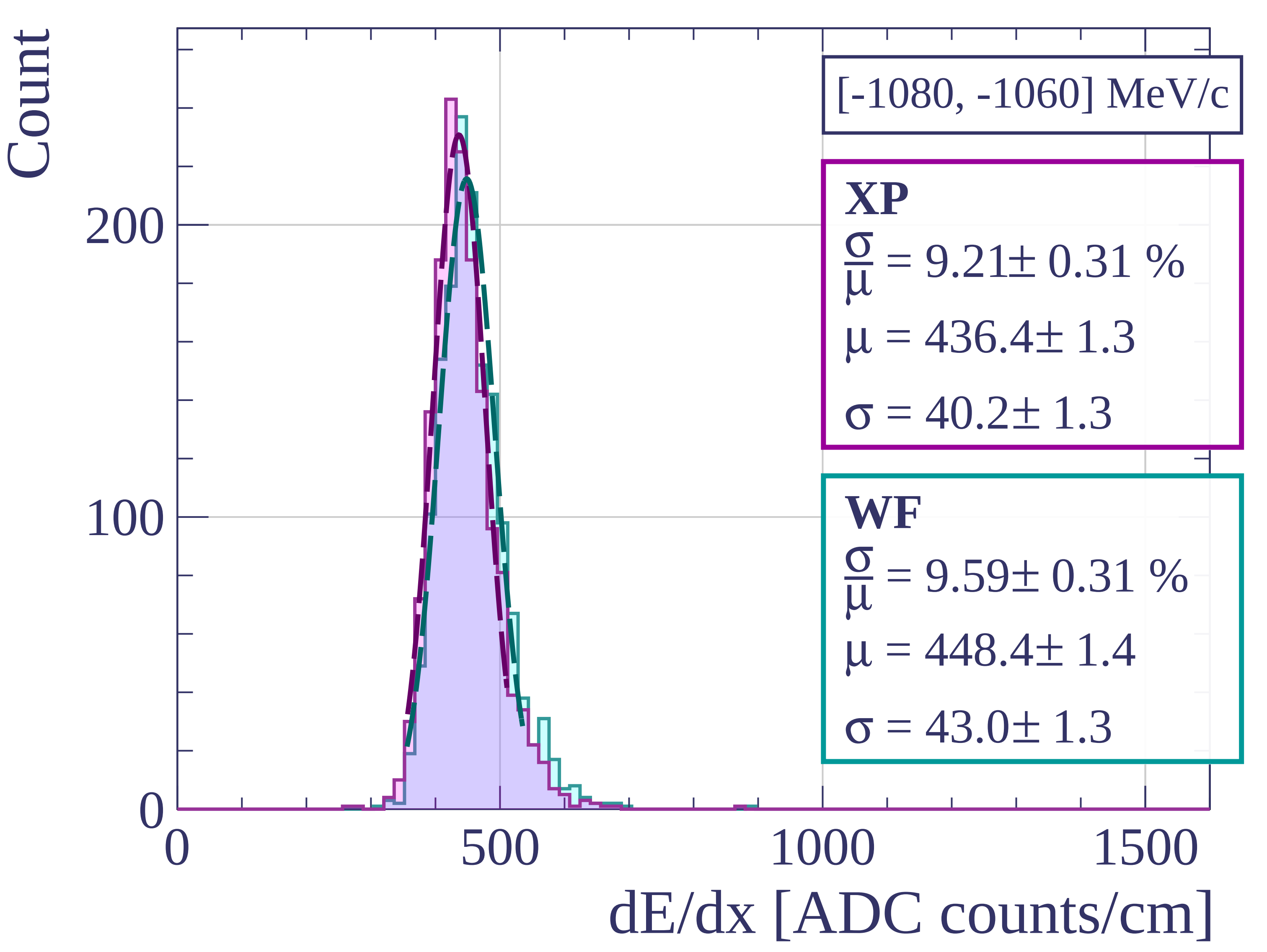


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1140, -1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.02± 0.34 %
μ = 437.6± 1.3
σ = 39.5± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.47± 0.35 %
μ = 449.2± 1.4
σ = 42.6± 1.4

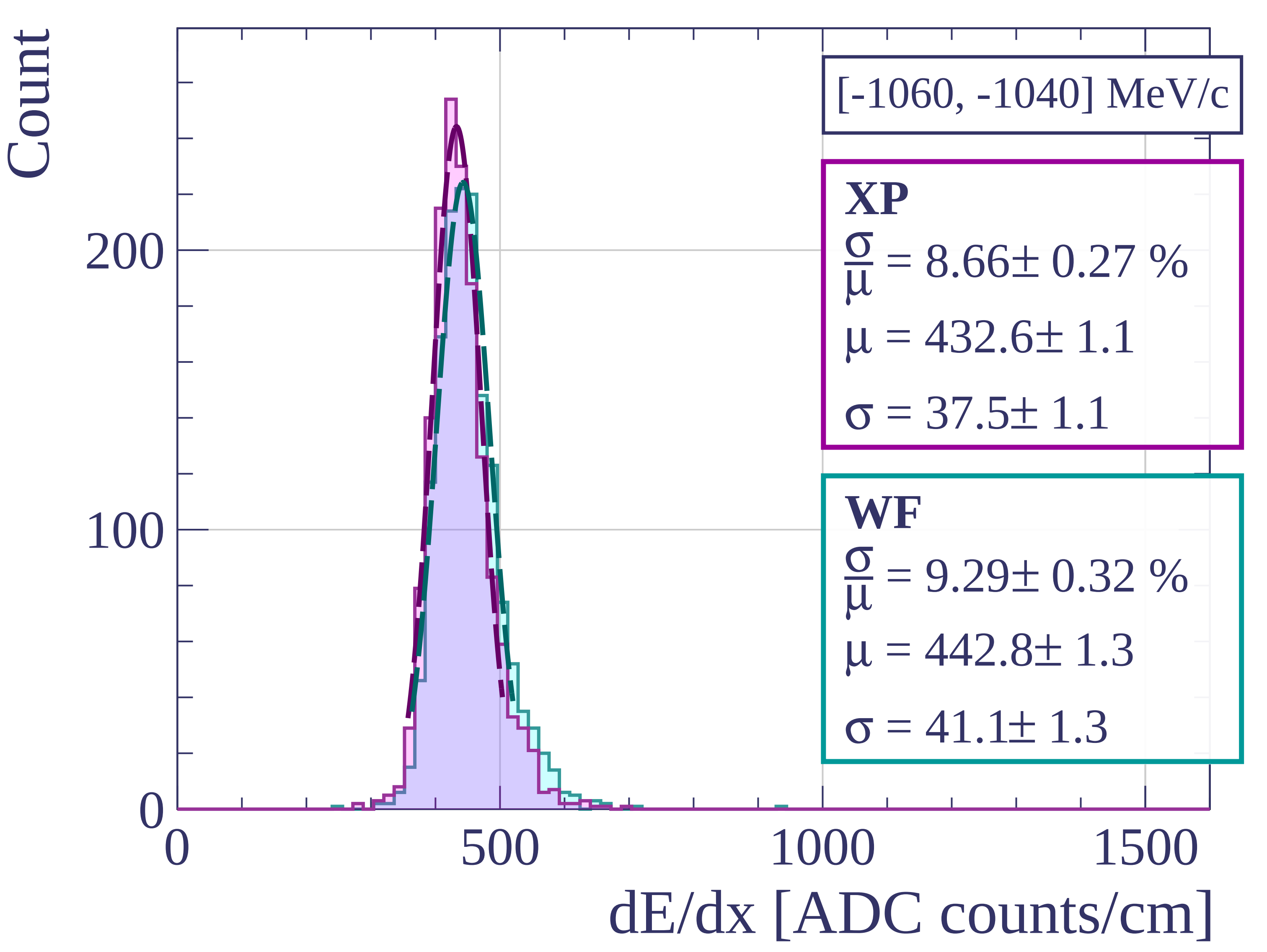


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1120, -1100] MeV/c
XP
σ/μ = 8.75± 0.26 %
μ = 436.5± 1.2
σ = 38.2± 1.0
WF
σ/μ = 9.23± 0.32 %
μ = 447.8± 1.3
σ = 41.3± 1.3

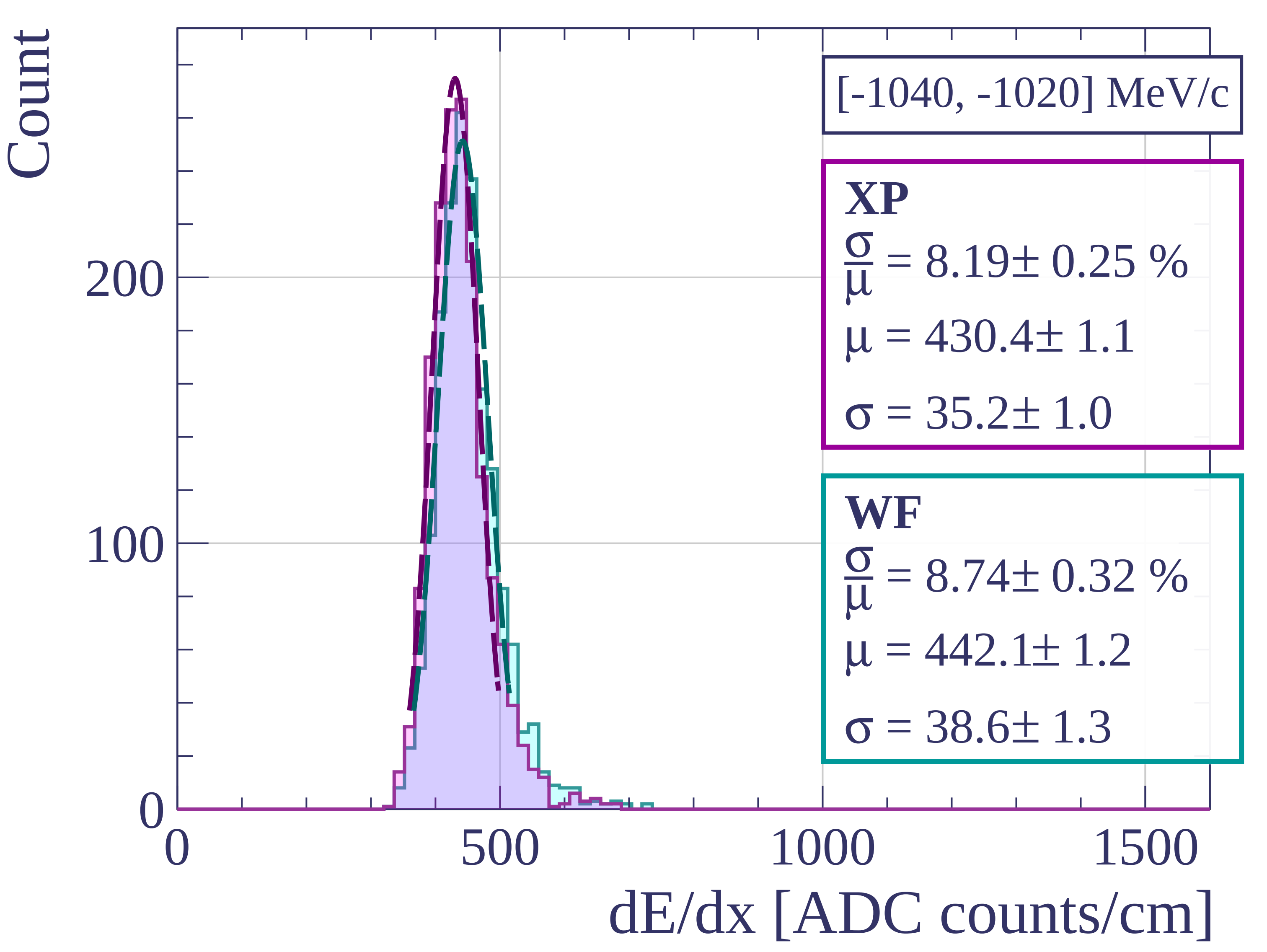

Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1100, -1080] MeV/c
XP
σ/μ = 8.51± 0.25 %
μ = 435.4± 1.1
σ = 37.0± 1.0
WF
σ/μ = 8.96± 0.26 %
μ = 446.6± 1.3
σ = 40.0± 1.0



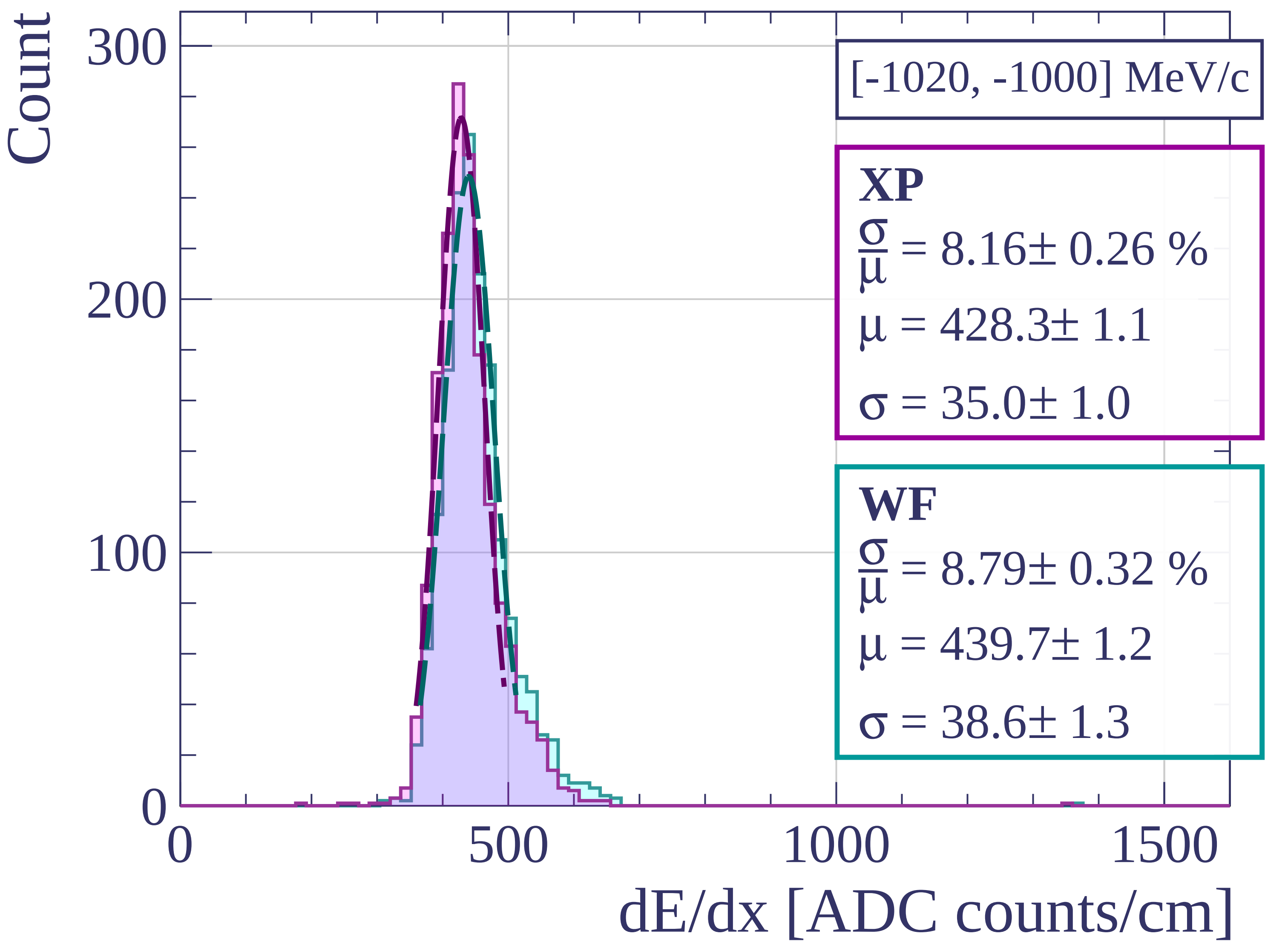
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1080, -1060] MeV/c
XP
σ/μ = 9.21± 0.31 %
μ = 436.4± 1.3
σ = 40.2± 1.3
WF
σ/μ = 9.59± 0.31 %
μ = 448.4± 1.4
σ = 43.0± 1.3



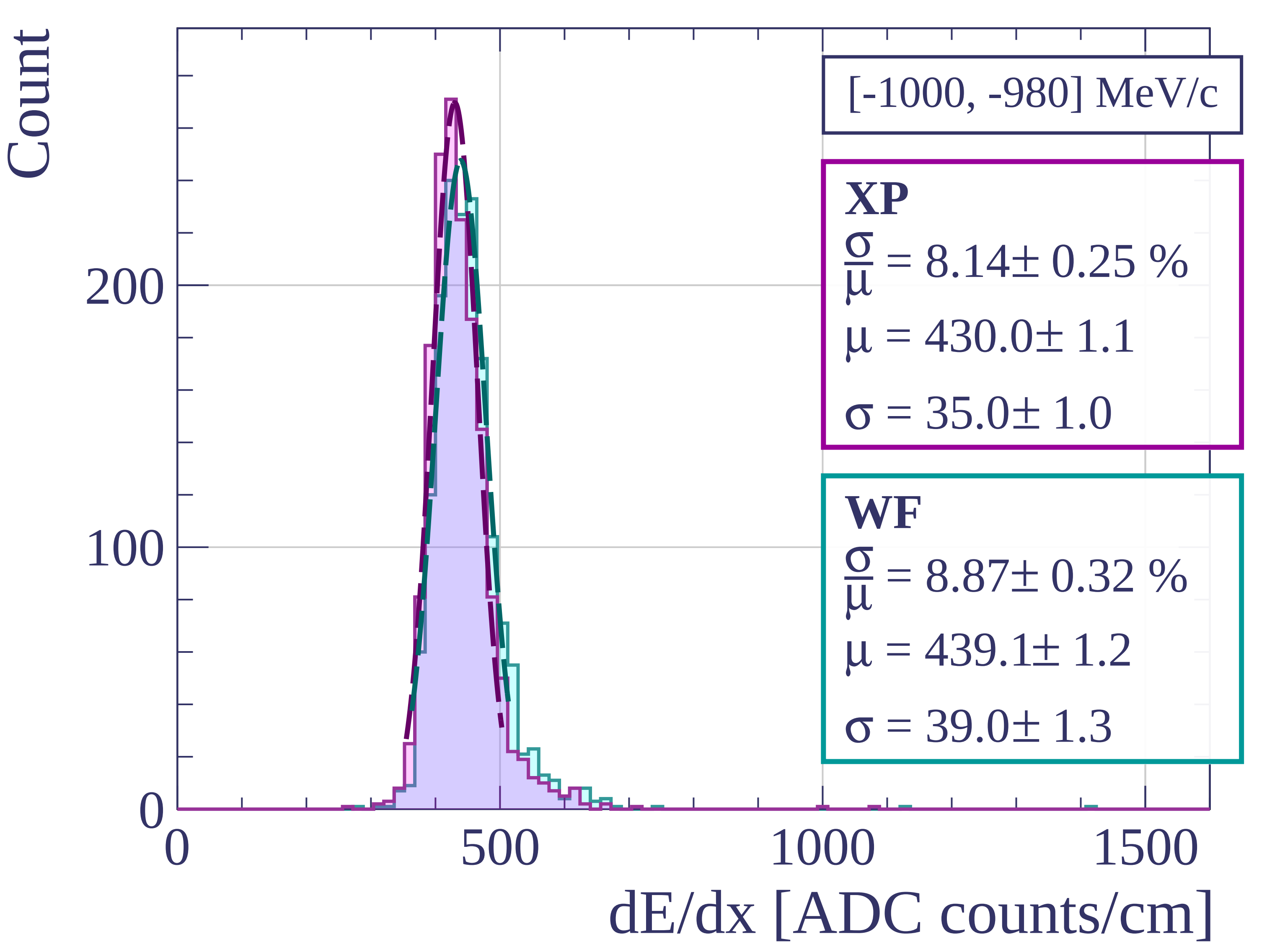
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1060, -1040] MeV/c
XP
σ/μ = 8.66± 0.27 %
μ = 432.6± 1.1
σ = 37.5± 1.1
WF
σ/μ = 9.29± 0.32 %
μ = 442.8± 1.3
σ = 41.1± 1.3



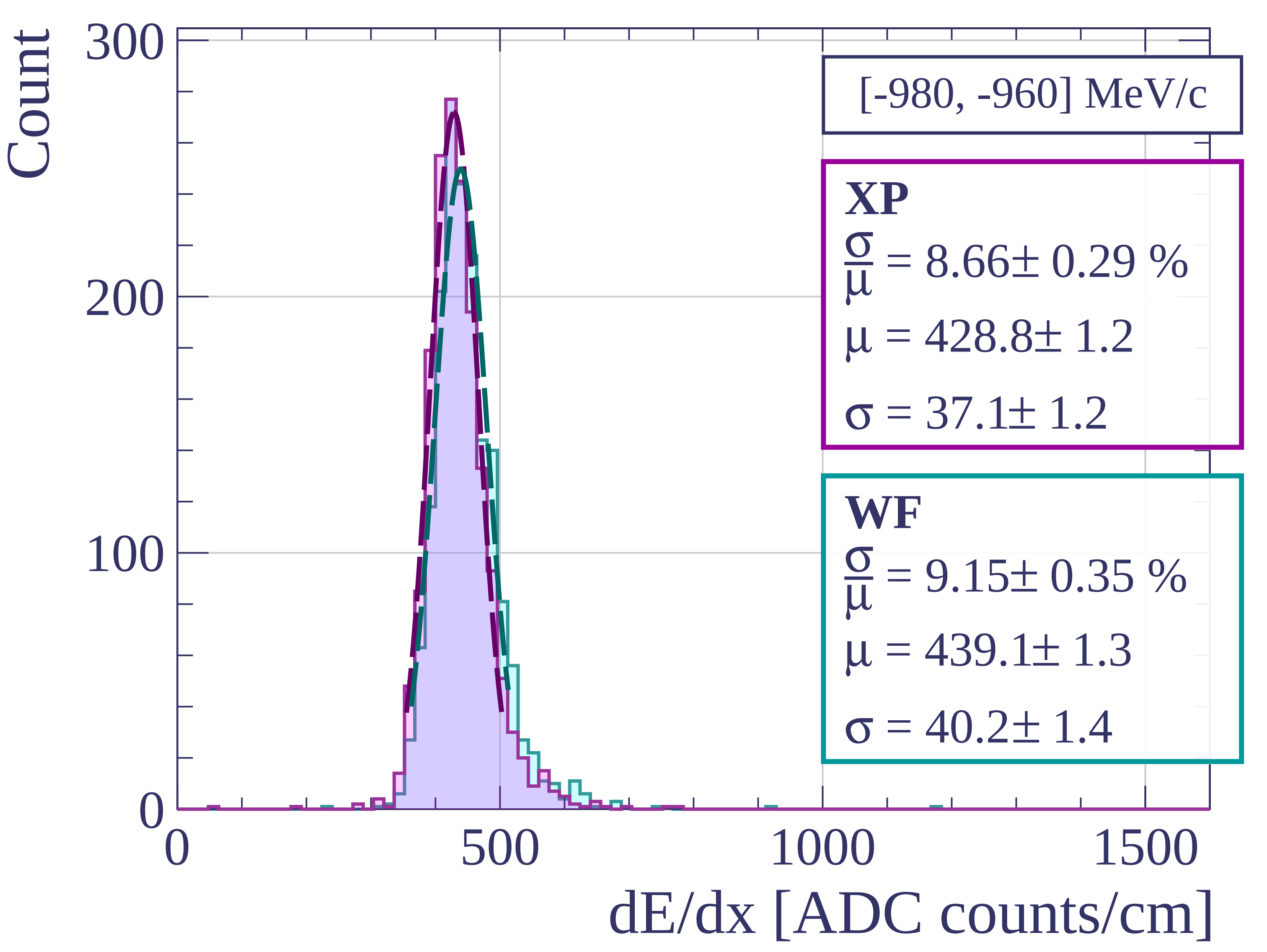
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1040, -1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.25 %
μ = 430.4± 1.1
σ = 35.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.74± 0.32 %
μ = 442.1± 1.2
σ = 38.6± 1.3

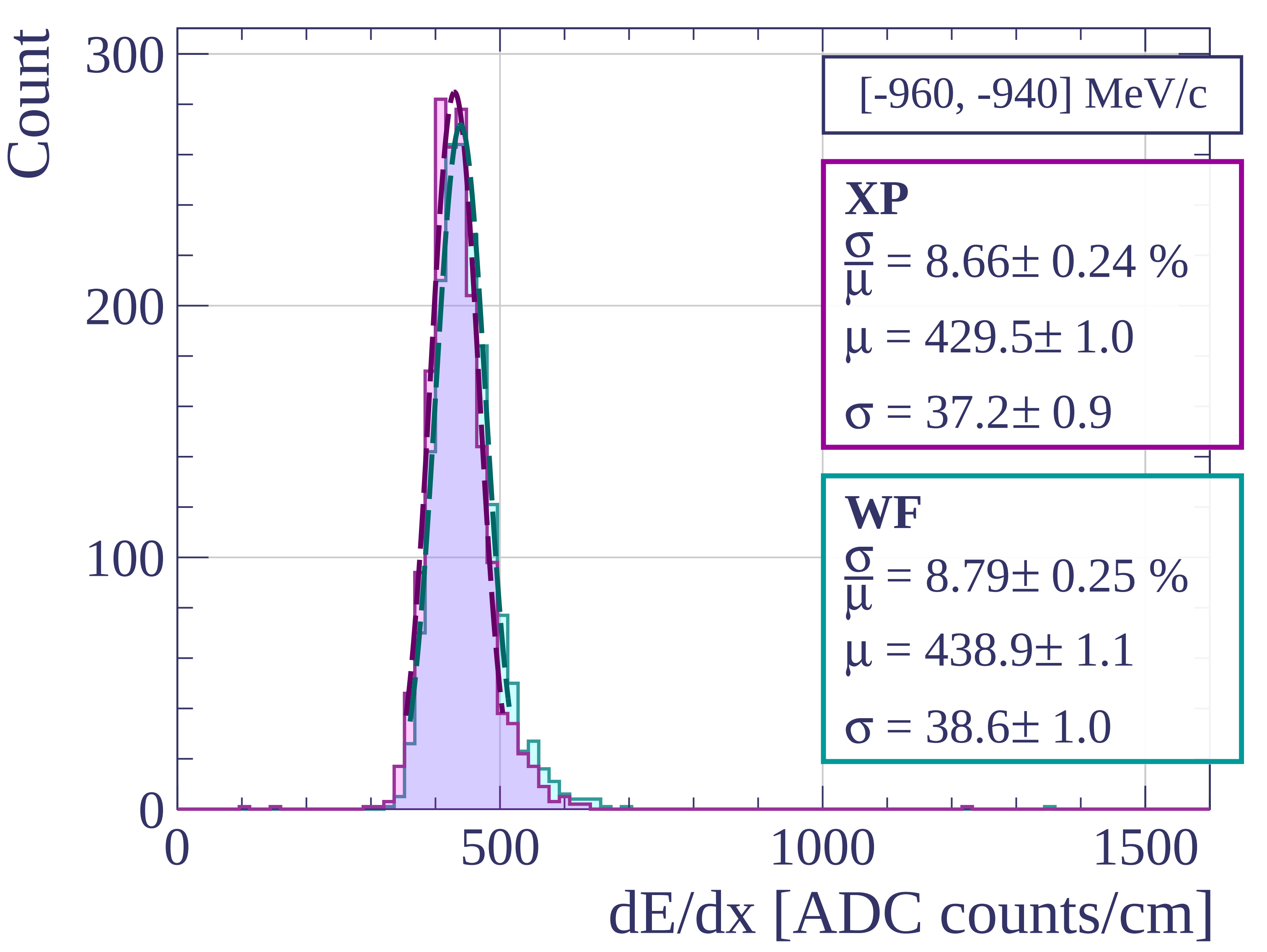


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1020, -1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.16± 0.26 %
μ = 428.3± 1.1
σ = 35.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.79± 0.32 %
μ = 439.7± 1.2
σ = 38.6± 1.3

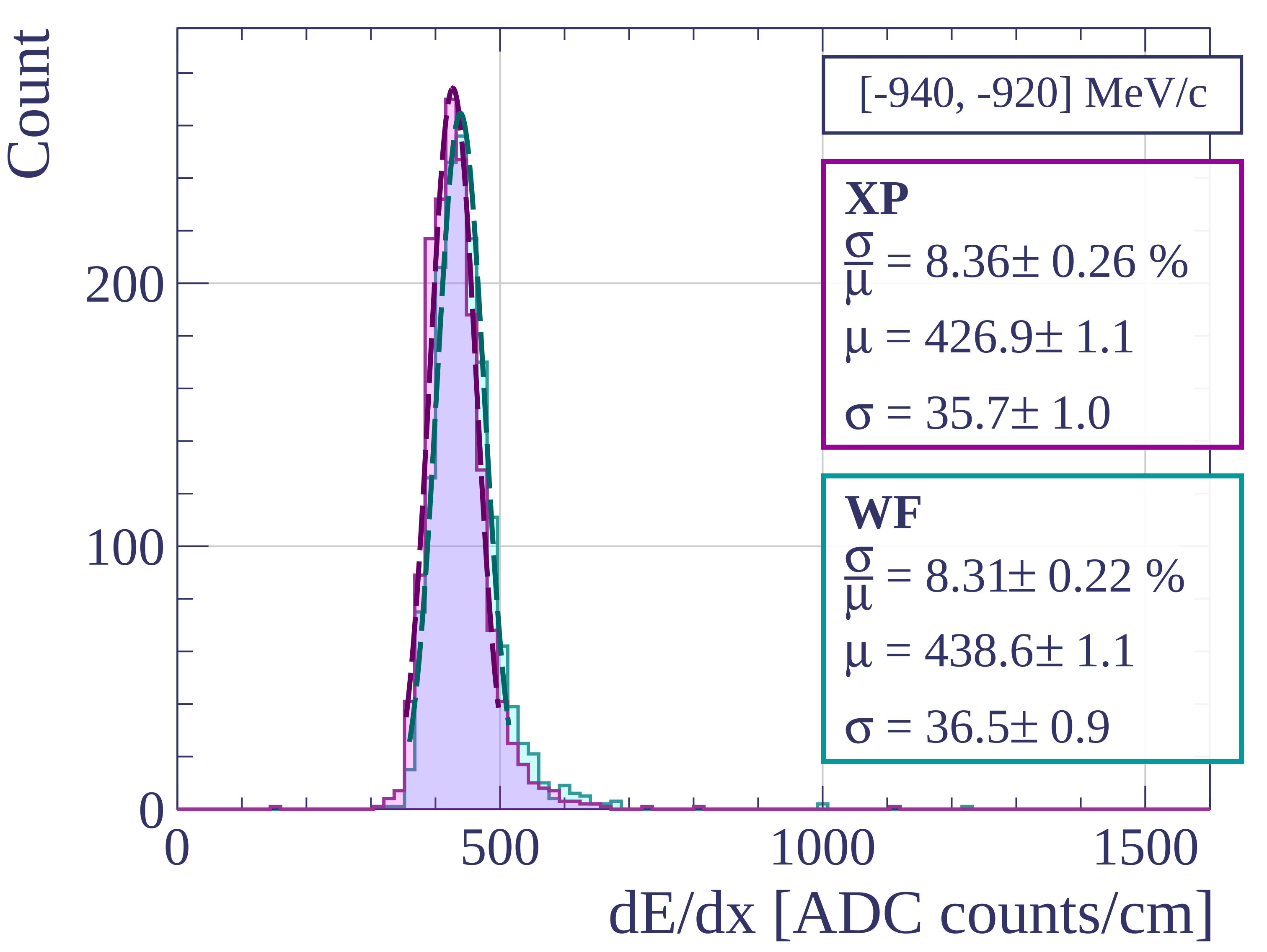


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1000, -980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.14± 0.25 %
μ = 430.0± 1.1
σ = 35.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.87± 0.32 %
μ = 439.1± 1.2
σ = 39.0± 1.3

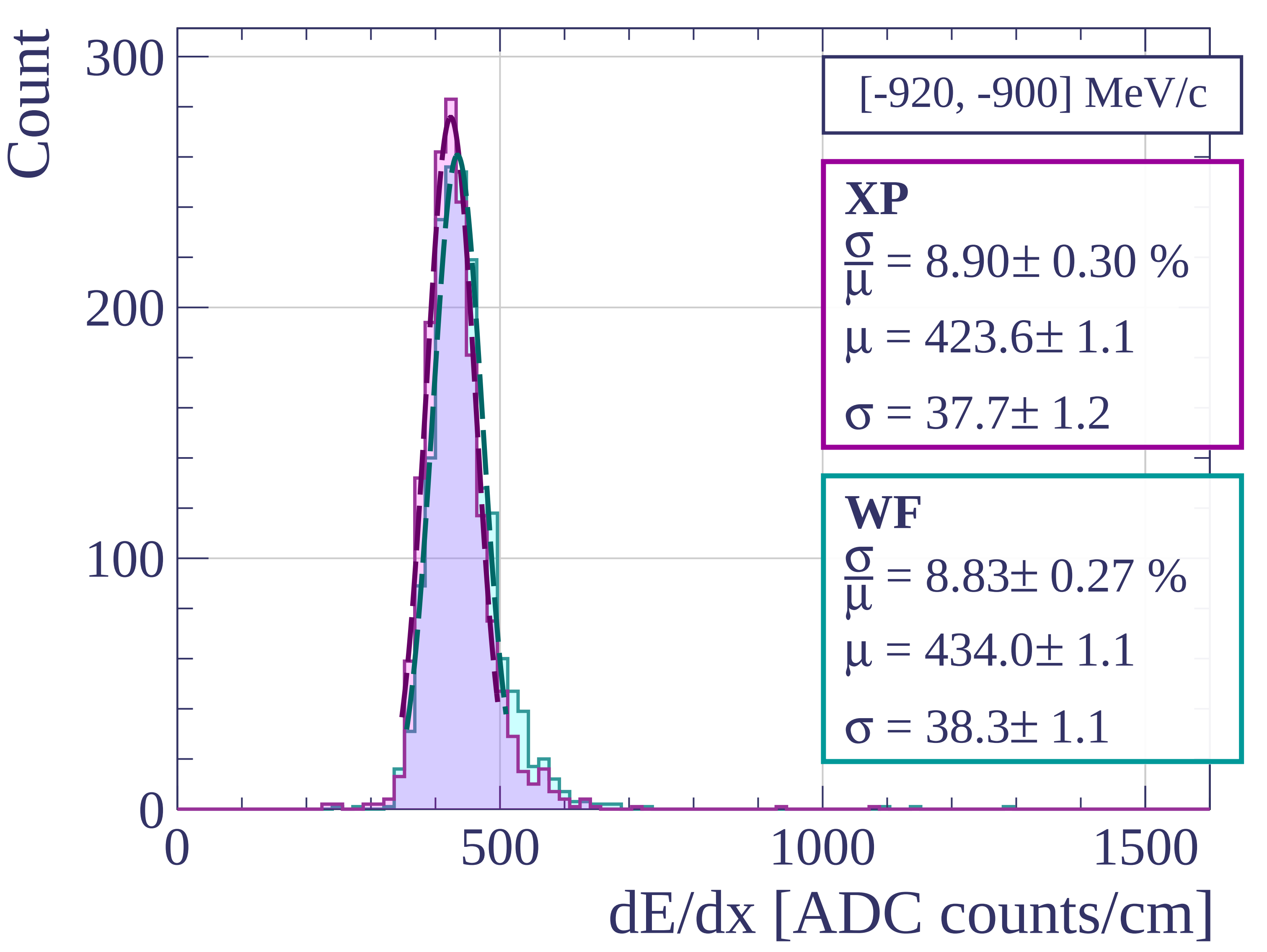

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-980, -960] MeV/c
XP
σ/μ = 8.66± 0.29 %
μ = 428.8± 1.2
σ = 37.1± 1.2
WF
σ/μ = 9.15± 0.35 %
μ = 439.1± 1.3
σ = 40.2± 1.4



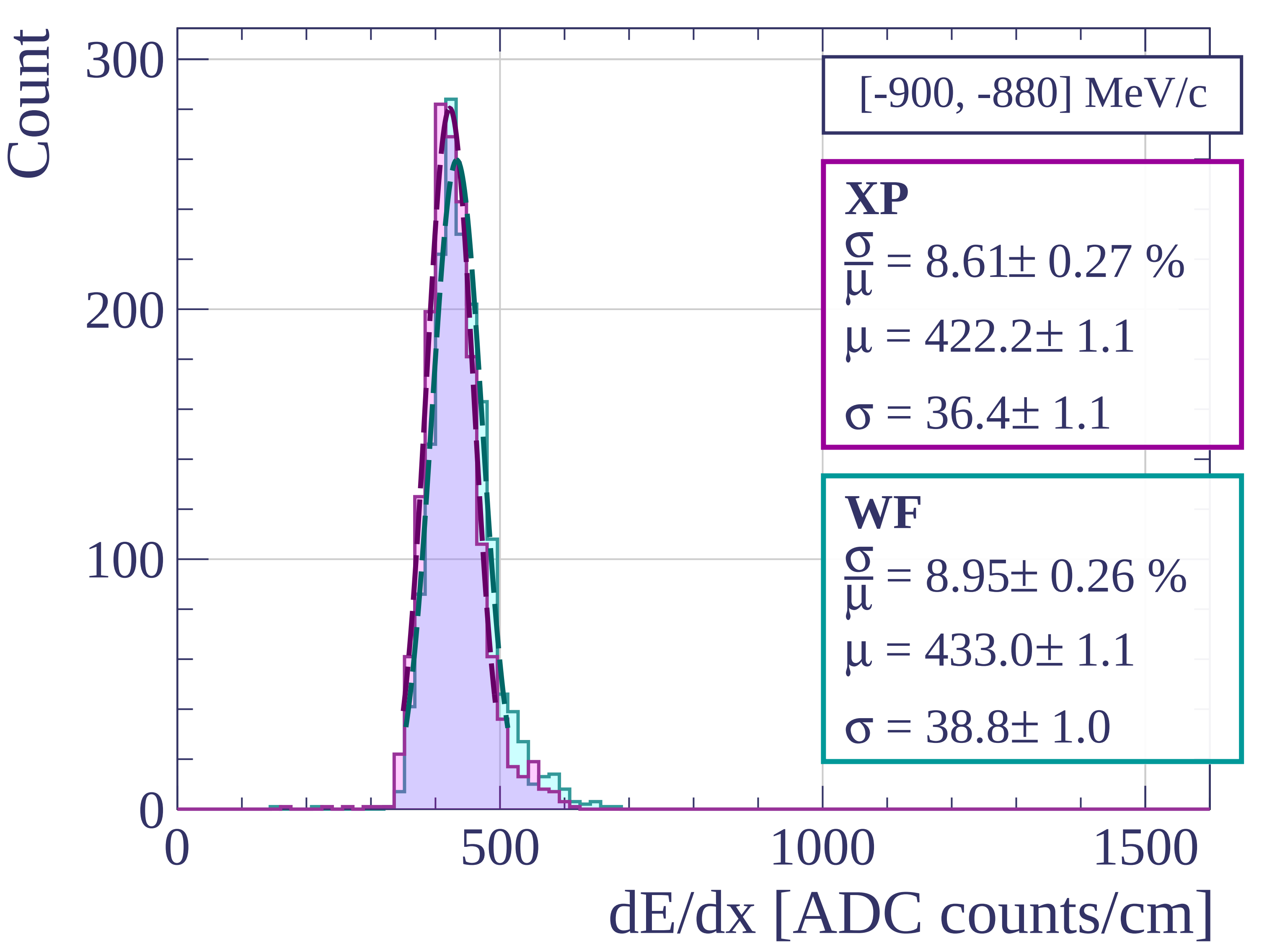
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-960, -940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.66± 0.24 %
μ = 429.5± 1.0
σ = 37.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.79± 0.25 %
μ = 438.9± 1.1
σ = 38.6± 1.0



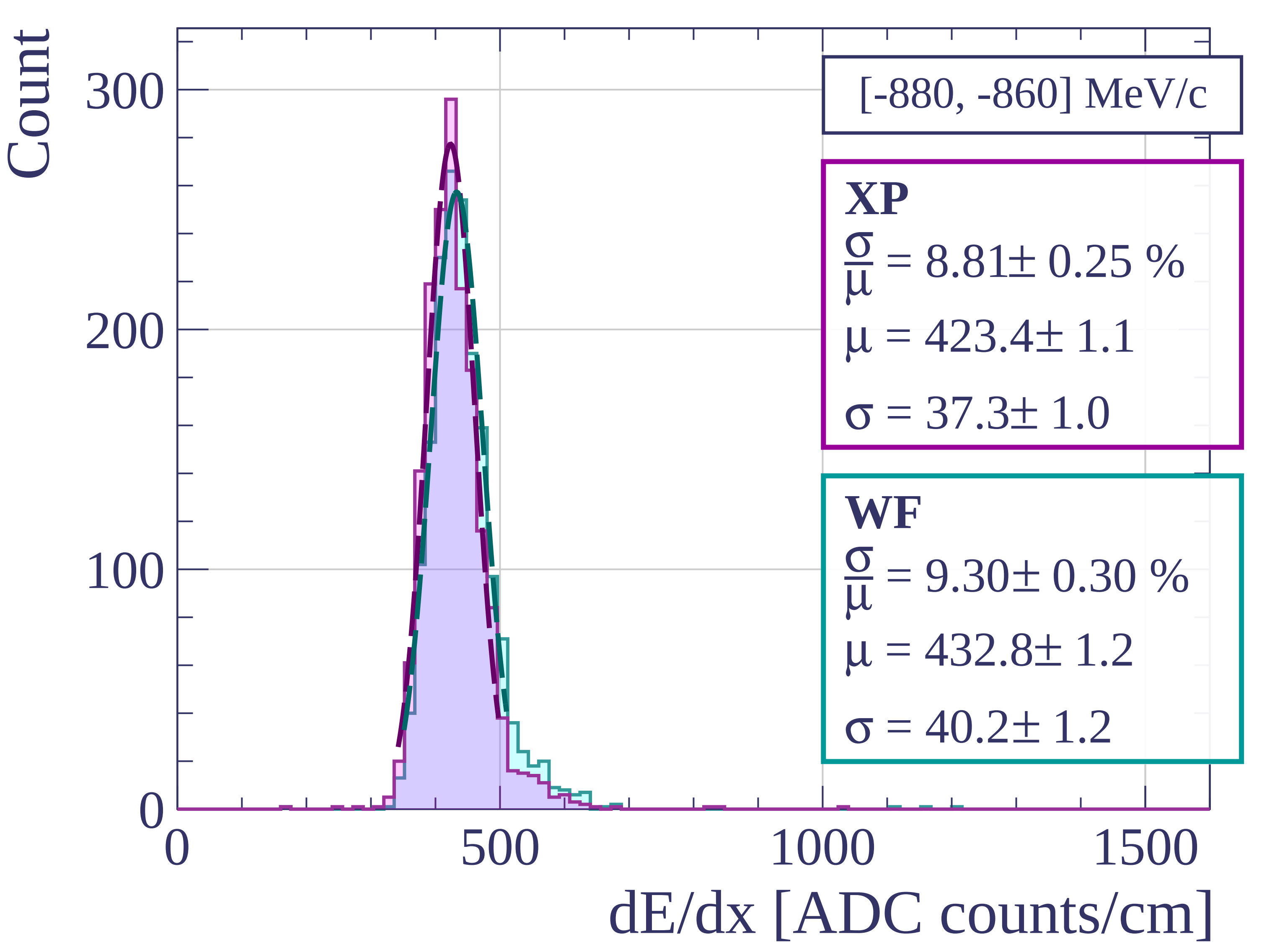
Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
[-940, -920] MeV/c
XP
σ/μ = 8.36± 0.26 %
μ = 426.9± 1.1
σ = 35.7± 1.0
WF
σ/μ = 8.31± 0.22 %
μ = 438.6± 1.1
σ = 36.5± 0.9



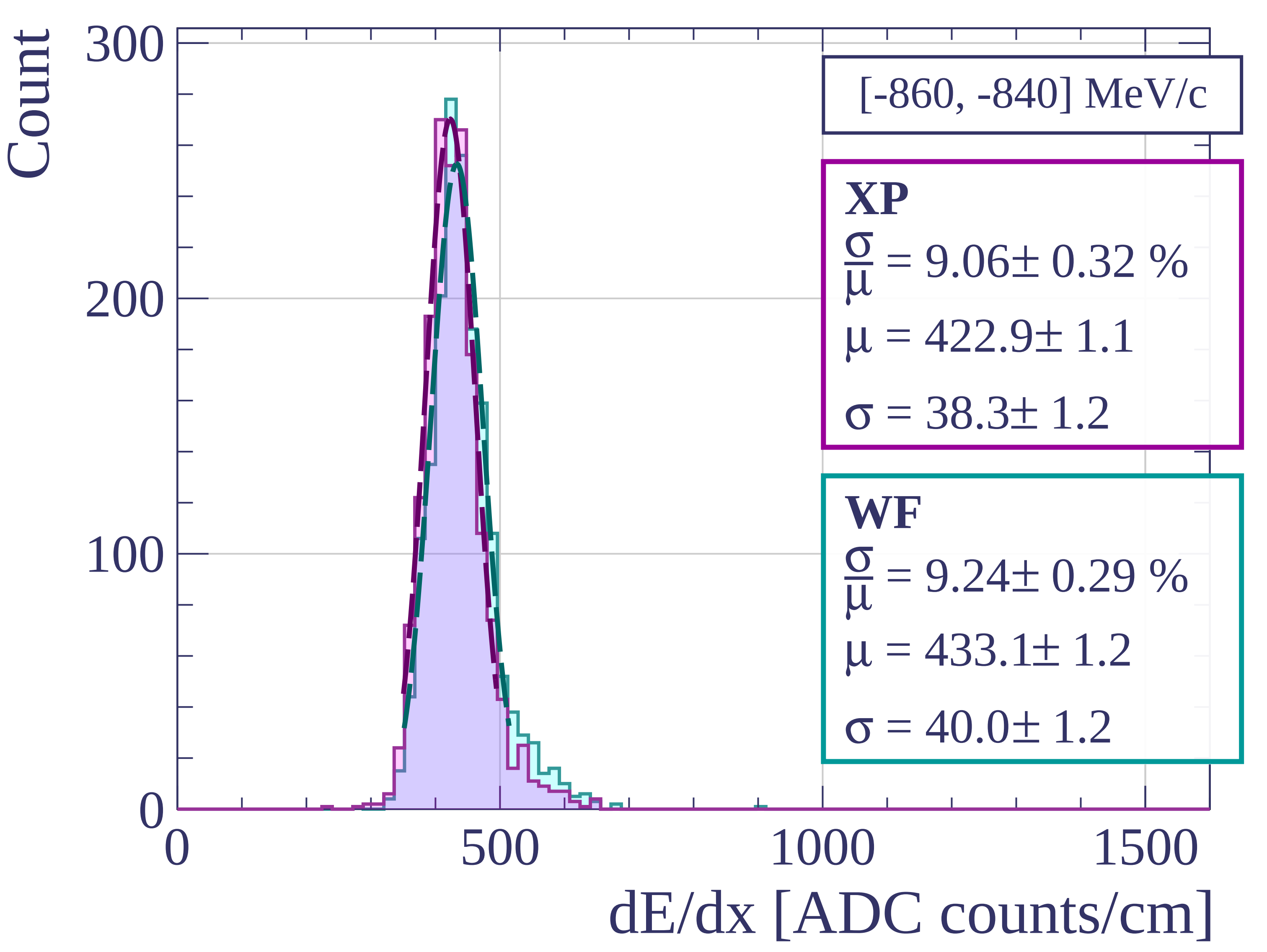
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-920, -900] MeV/c
XP
σ/μ = 8.90± 0.30 %
μ = 423.6± 1.1
σ = 37.7± 1.2
WF
σ/μ = 8.83± 0.27 %
μ = 434.0± 1.1
σ = 38.3± 1.1



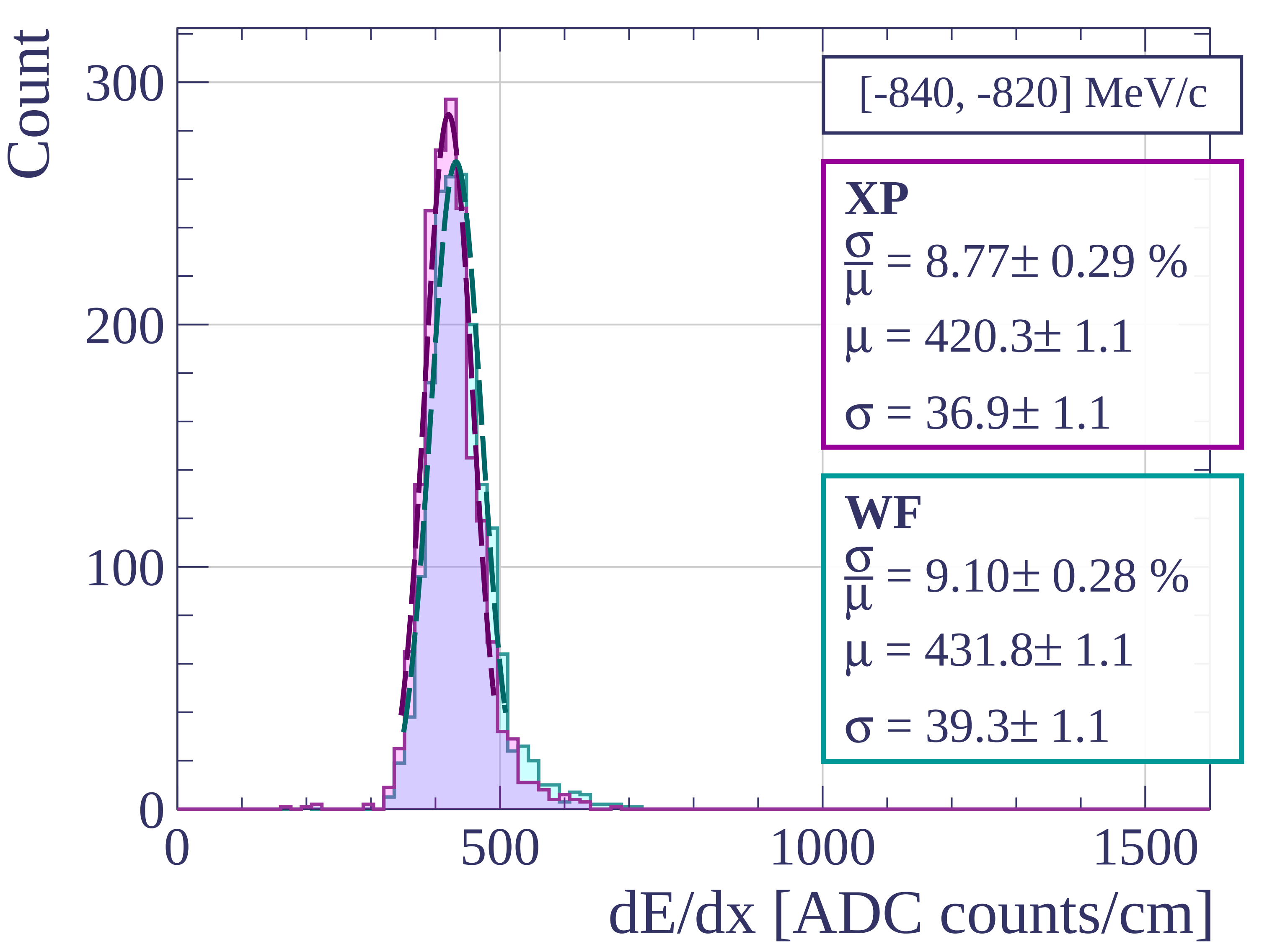
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-900, -880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.61± 0.27 %
μ = 422.2± 1.1
σ = 36.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.95± 0.26 %
μ = 433.0± 1.1
σ = 38.8± 1.0



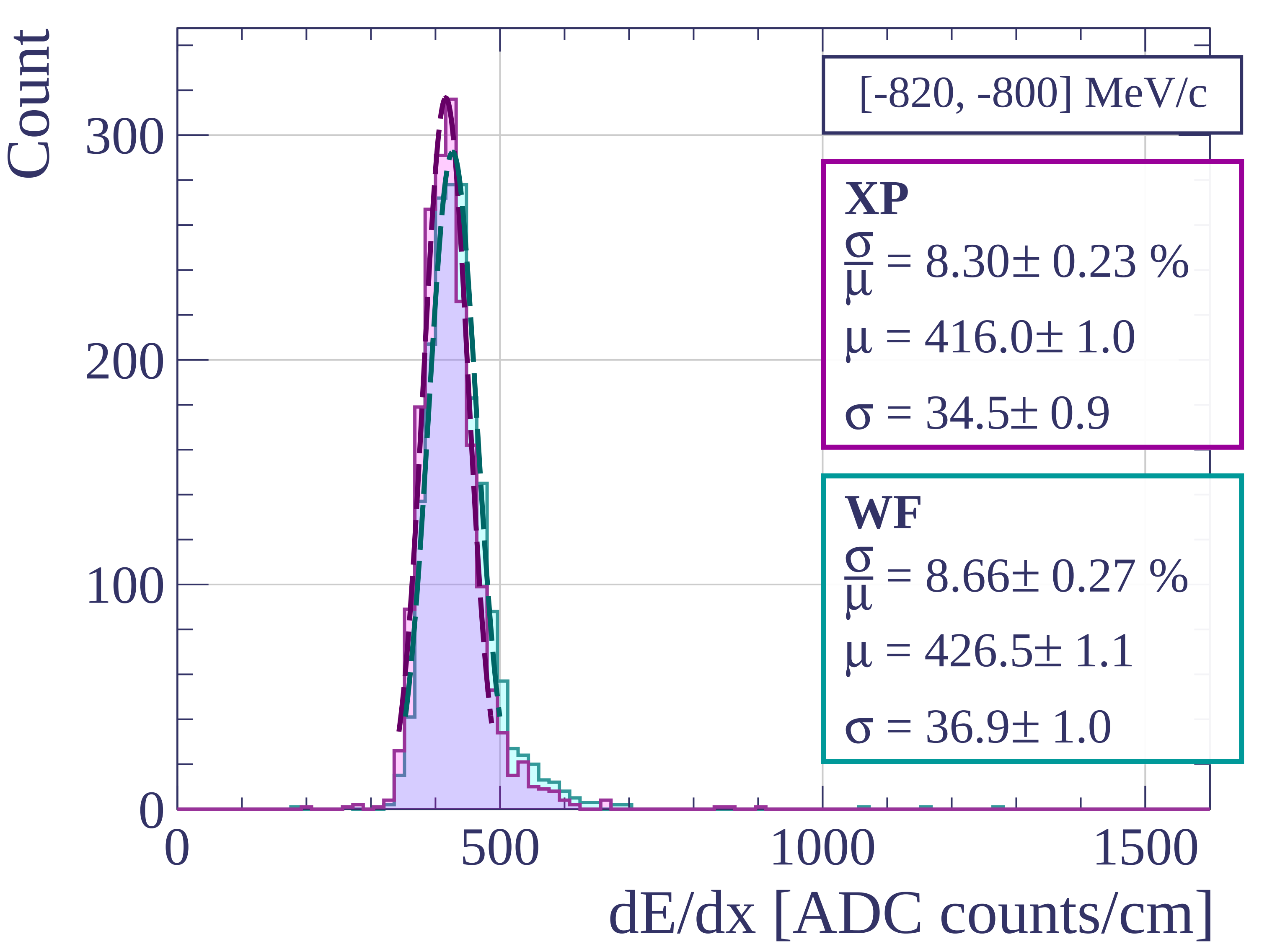
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-880, -860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.81± 0.25 %
μ = 423.4± 1.1
σ = 37.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.30± 0.30 %
μ = 432.8± 1.2
σ = 40.2± 1.2



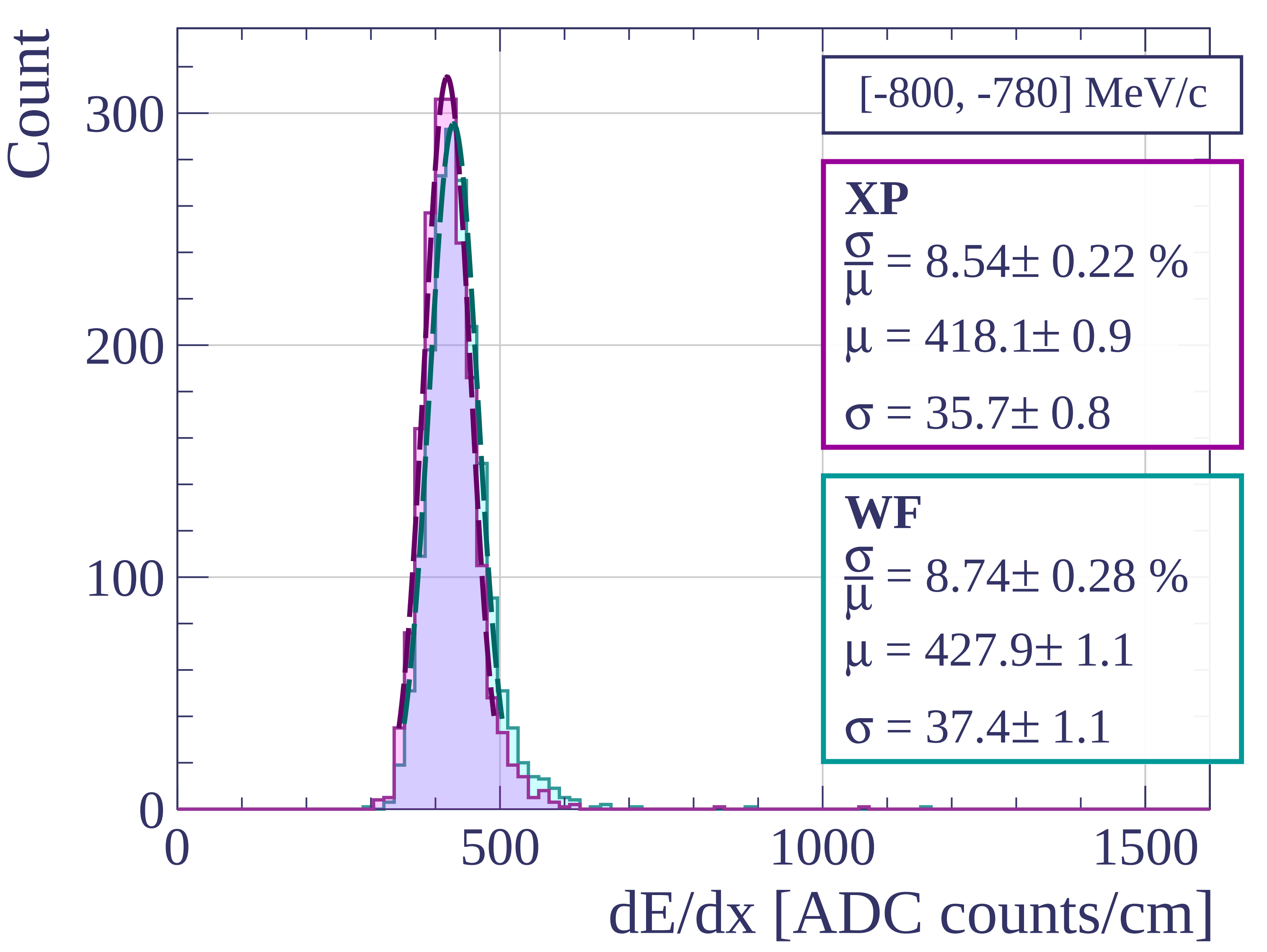
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-860, -840] MeV/c
XP
σ/μ = 9.06± 0.32 %
μ = 422.9± 1.1
σ = 38.3± 1.2
WF
σ/μ = 9.24± 0.29 %
μ = 433.1± 1.2
σ = 40.0± 1.2



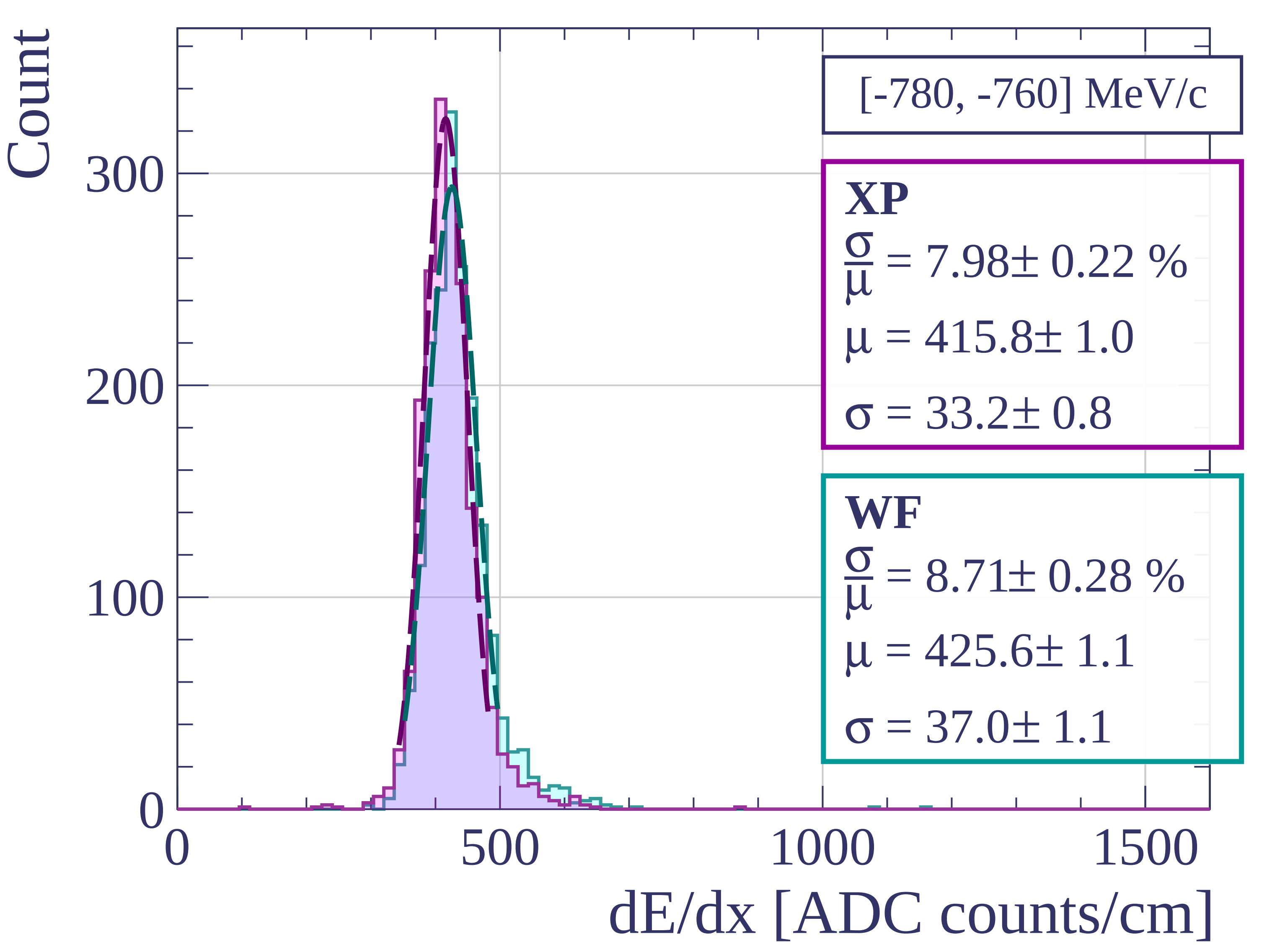
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-840, -820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.77± 0.29 %
μ = 420.3± 1.1
σ = 36.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.10± 0.28 %
μ = 431.8± 1.1
σ = 39.3± 1.1



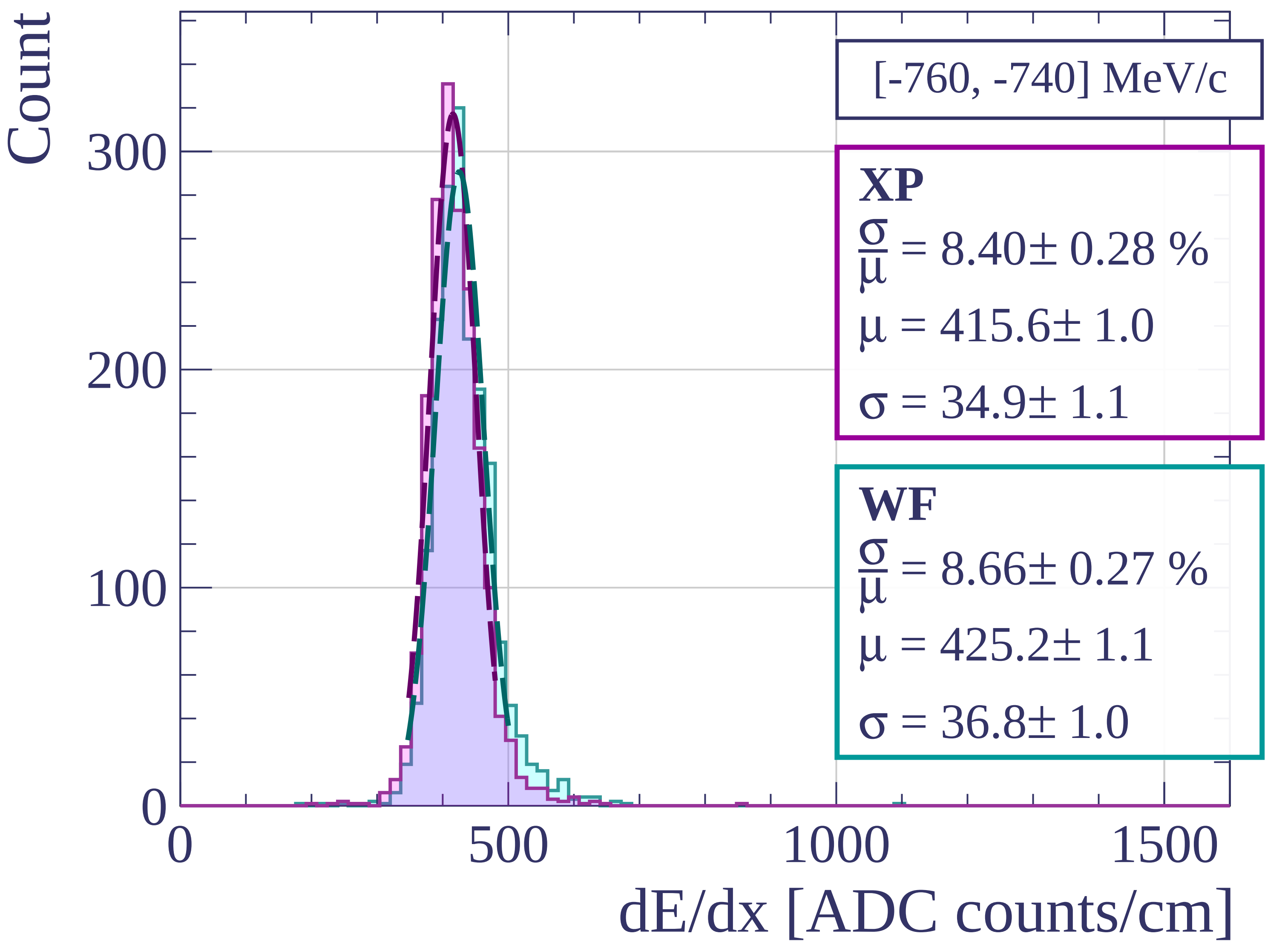
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-820, -800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.30± 0.23 %
μ = 416.0± 1.0
σ = 34.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.66± 0.27 %
μ = 426.5± 1.1
σ = 36.9± 1.0



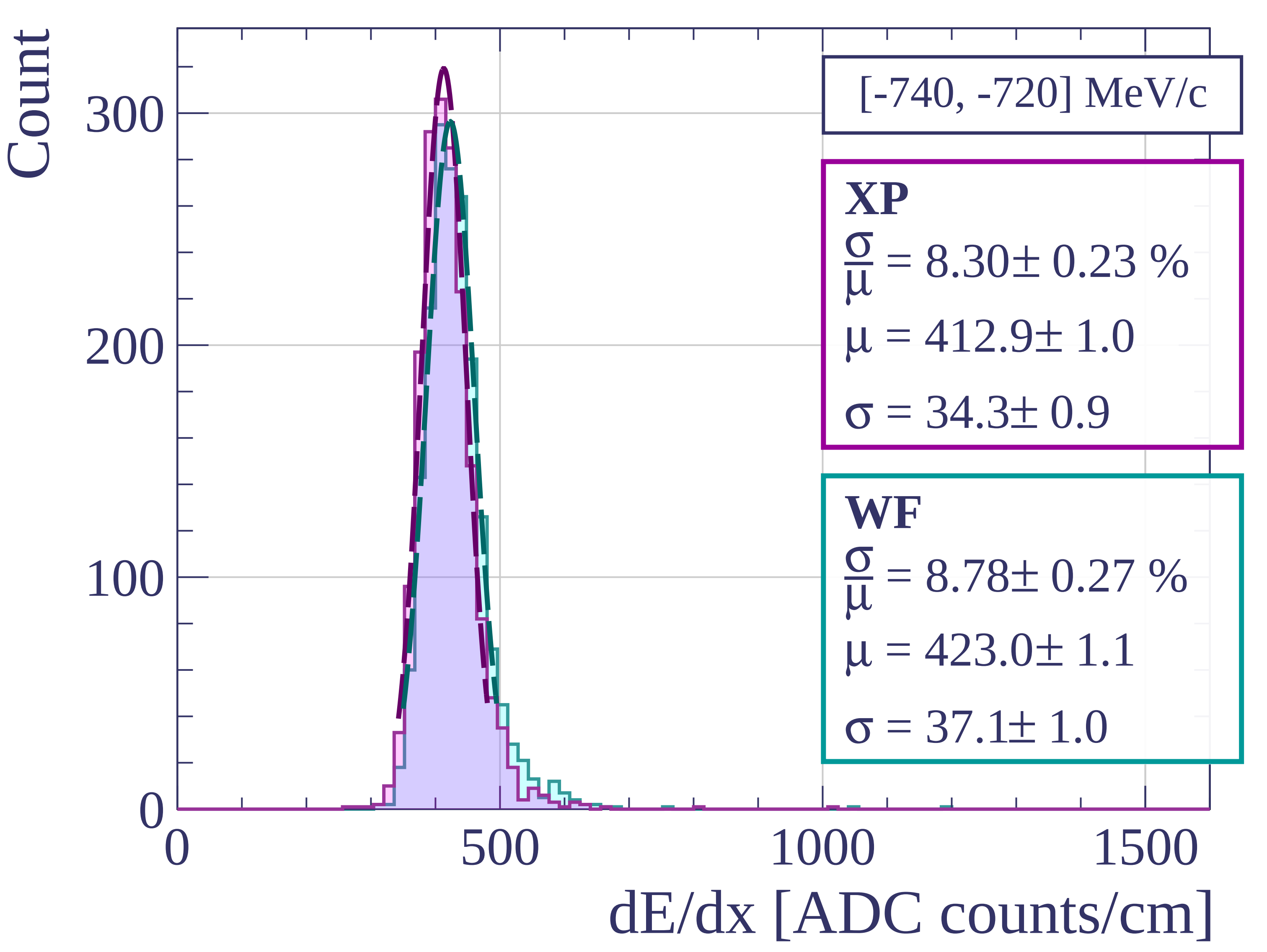
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-800, -780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.54± 0.22 %
μ = 418.1± 0.9
σ = 35.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.74± 0.28 %
μ = 427.9± 1.1
σ = 37.4± 1.1



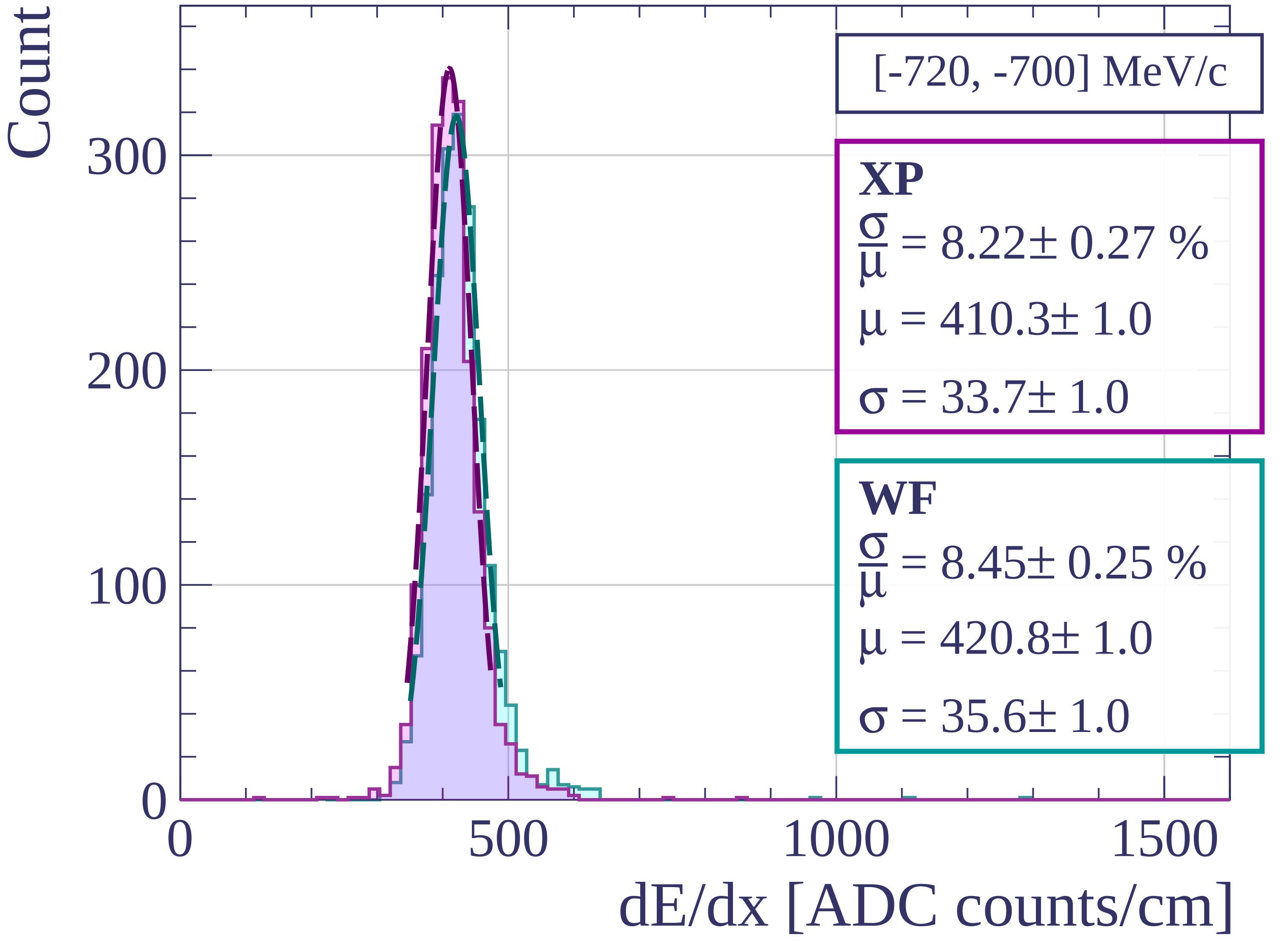
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-780, -760] MeV/c
XP
σ/μ = 7.98± 0.22 %
μ = 415.8± 1.0
σ = 33.2± 0.8
WF
σ/μ = 8.71± 0.28 %
μ = 425.6± 1.1
σ = 37.0± 1.1



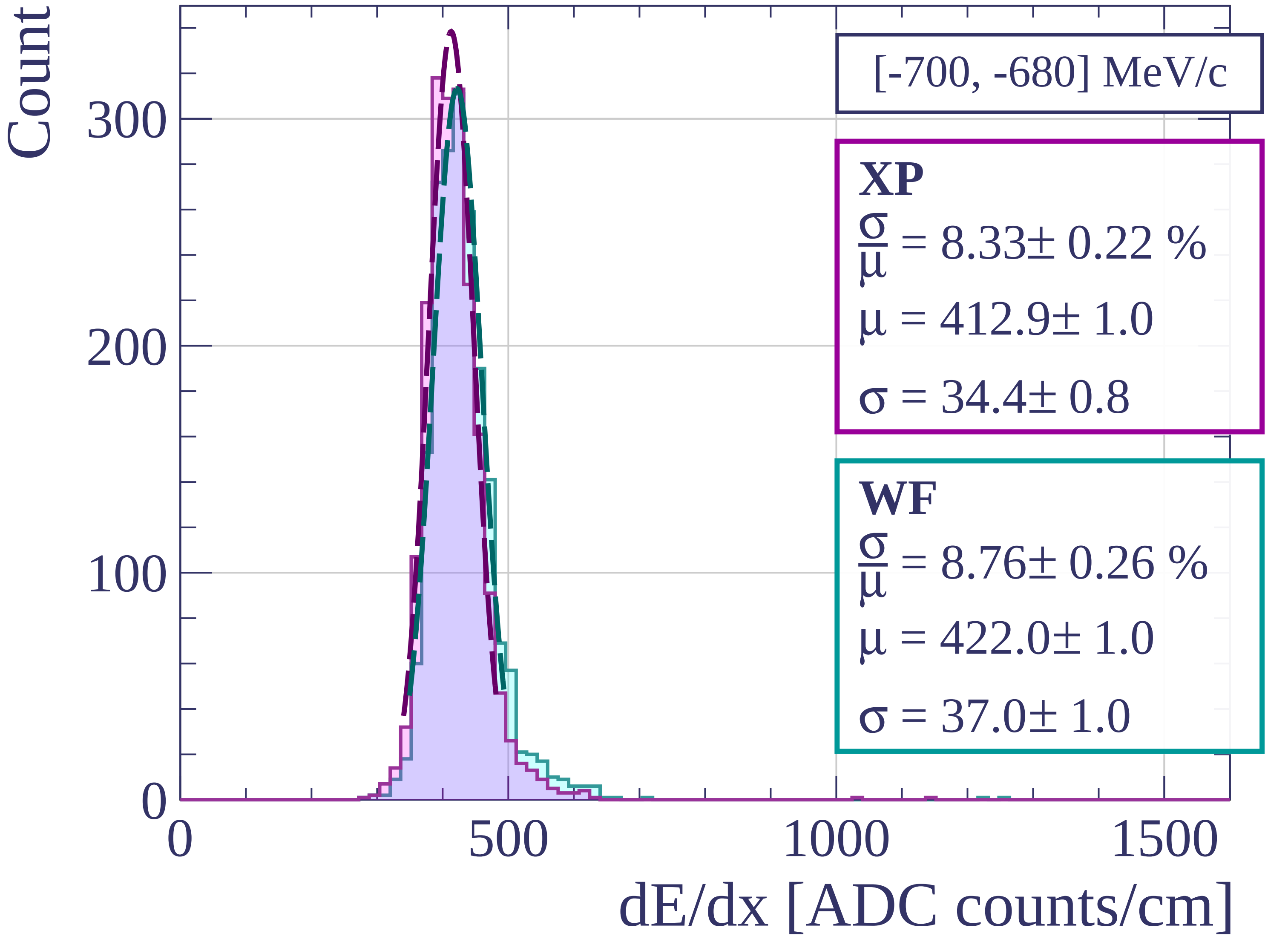
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-760, -740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.40± 0.28 %
μ = 415.6± 1.0
σ = 34.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.66± 0.27 %
μ = 425.2± 1.1
σ = 36.8± 1.0



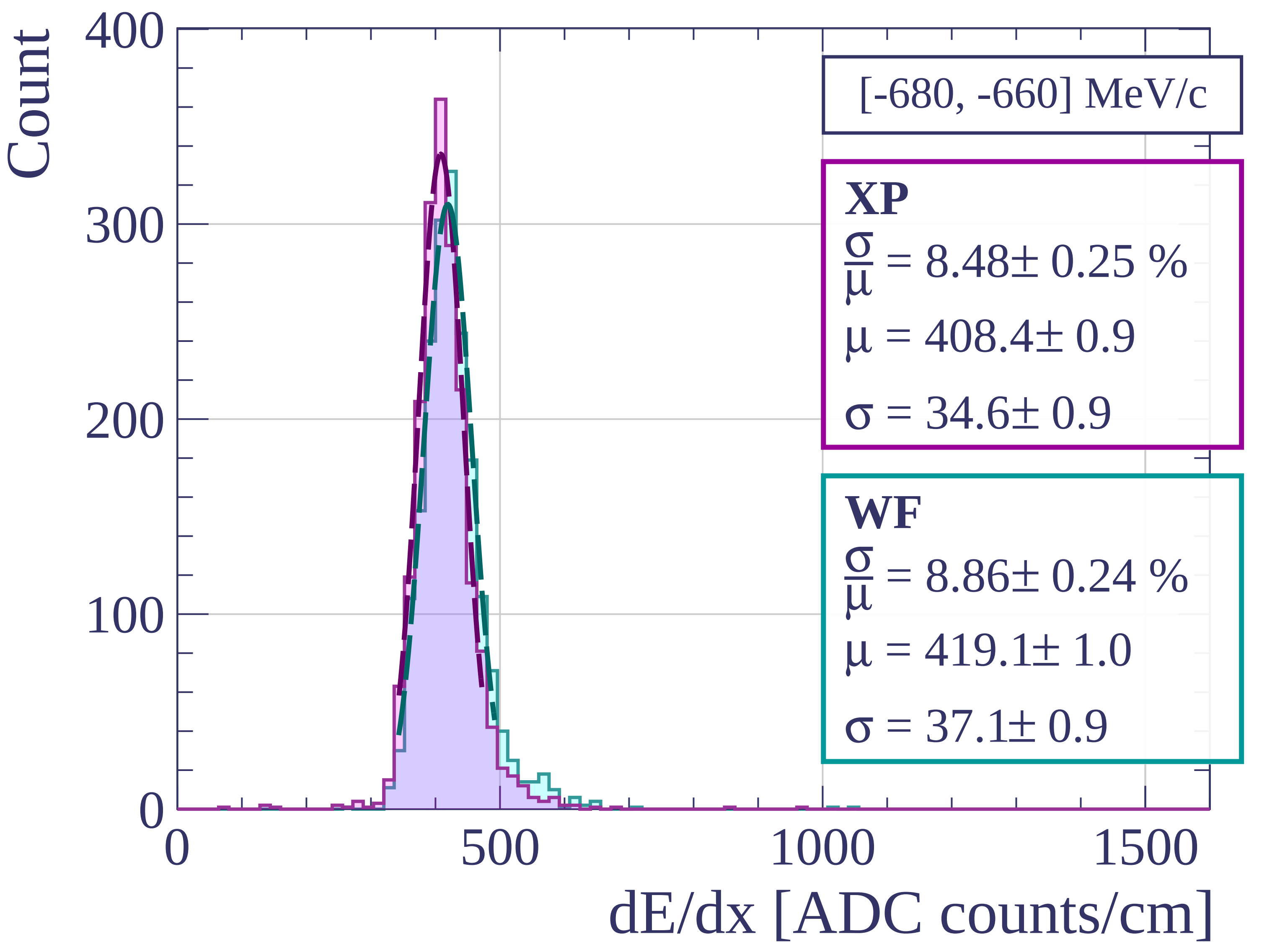
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-740, -720] MeV/c
XP
σ/μ = 8.30± 0.23 %
μ = 412.9± 1.0
σ = 34.3± 0.9
WF
σ/μ = 8.78± 0.27 %
μ = 423.0± 1.1
σ = 37.1± 1.0

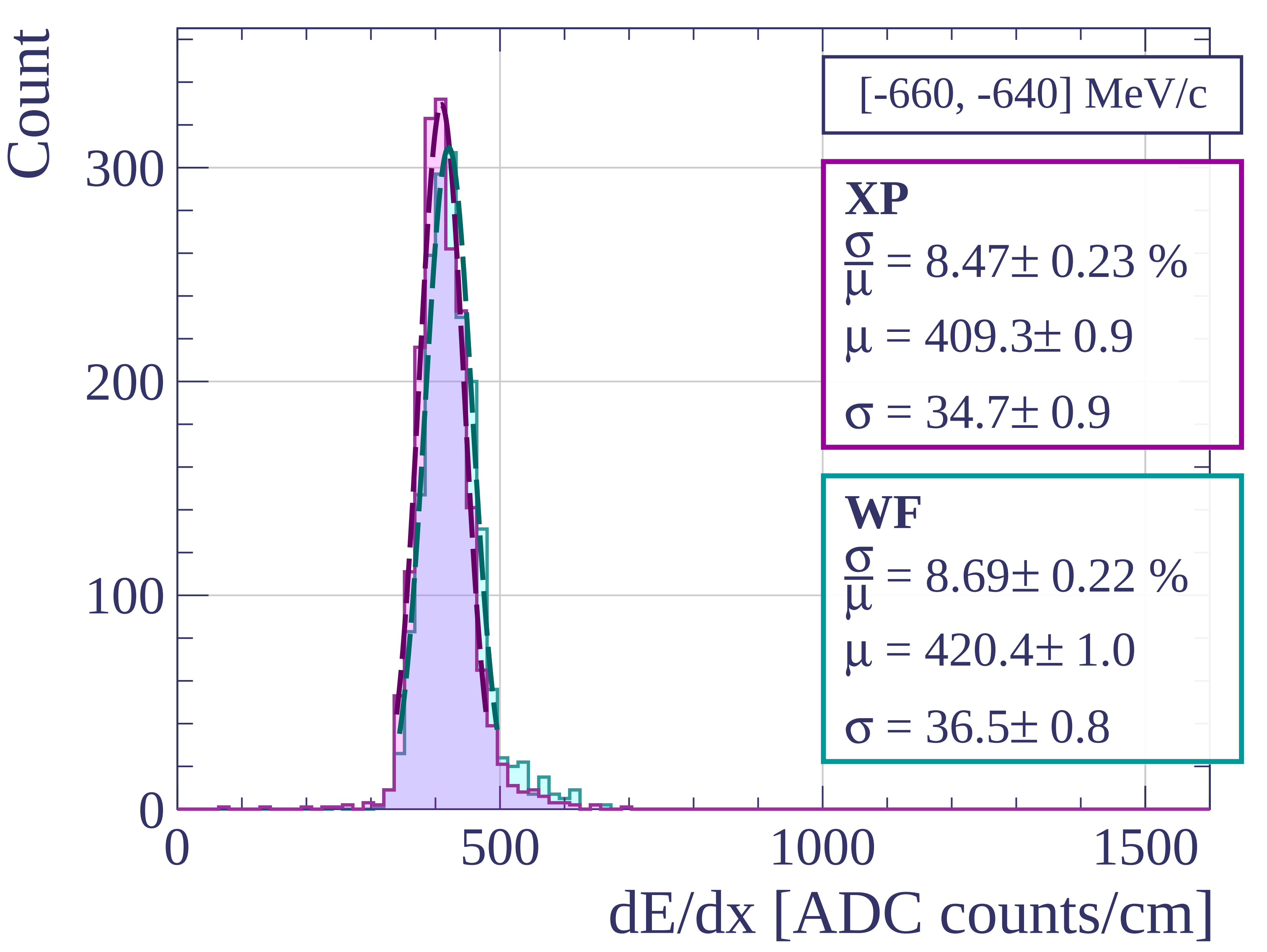


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-720, -700] MeV/c
XP
σ/μ = 8.22± 0.27 %
μ = 410.3± 1.0
σ = 33.7± 1.0
WF
σ/μ = 8.45± 0.25 %
μ = 420.8± 1.0
σ = 35.6± 1.0

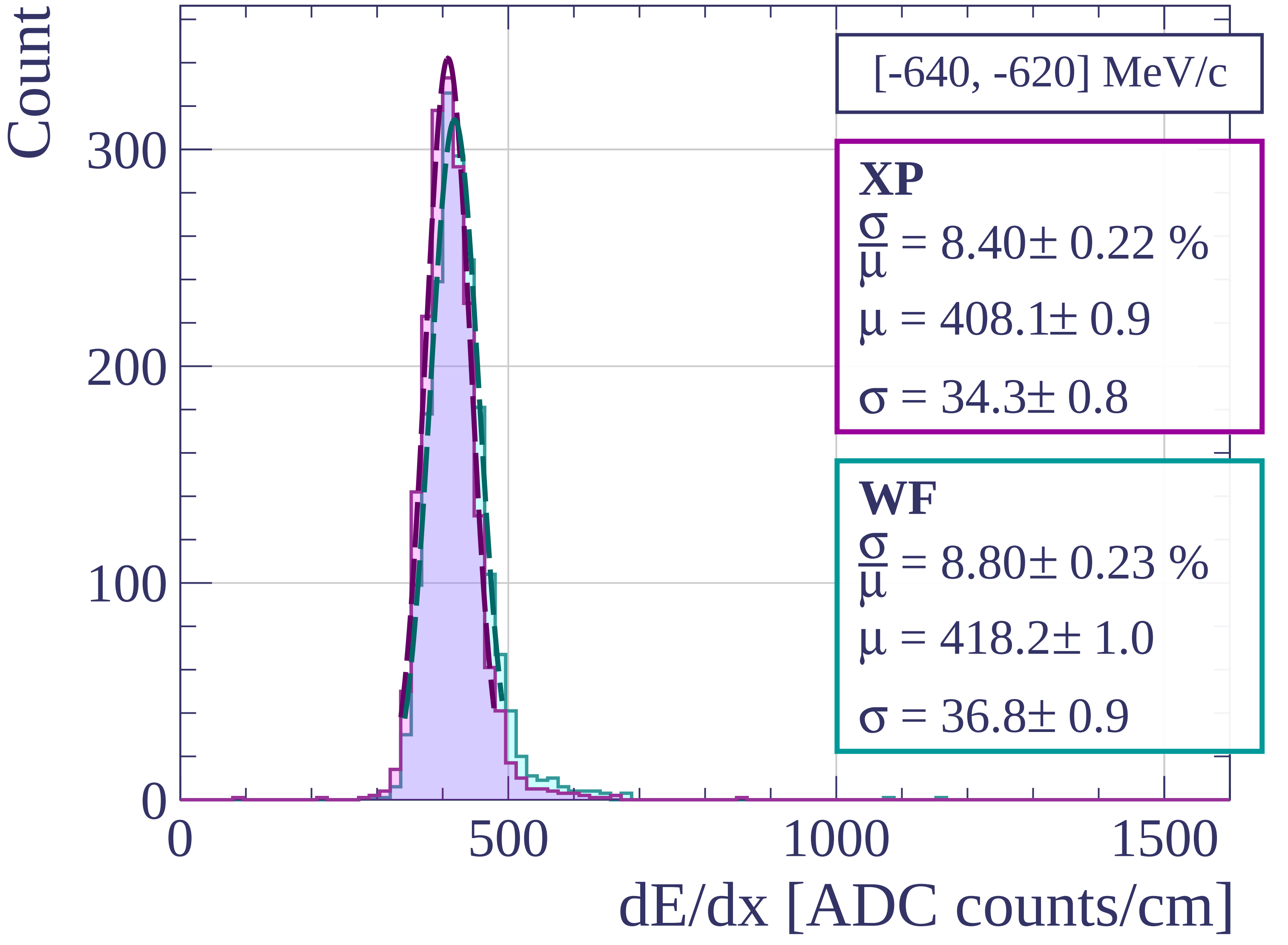


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-700, -680] MeV/c
XP
σ/μ = 8.33± 0.22 %
μ = 412.9± 1.0
σ = 34.4± 0.8
WF
σ/μ = 8.76± 0.26 %
μ = 422.0± 1.0
σ = 37.0± 1.0

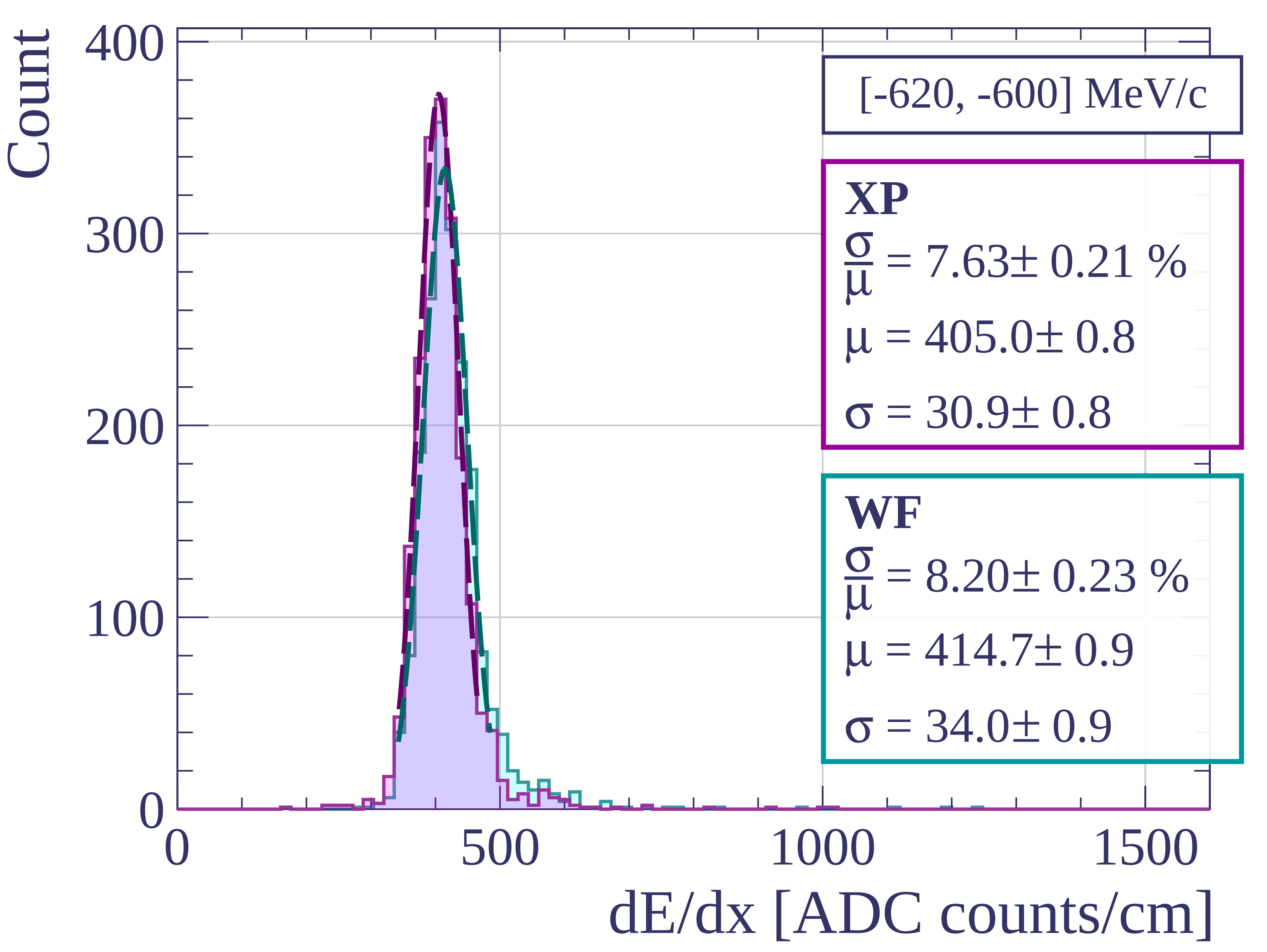

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-680, -660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.48± 0.25 %
μ = 408.4± 0.9
σ = 34.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.86± 0.24 %
μ = 419.1± 1.0
σ = 37.1± 0.9



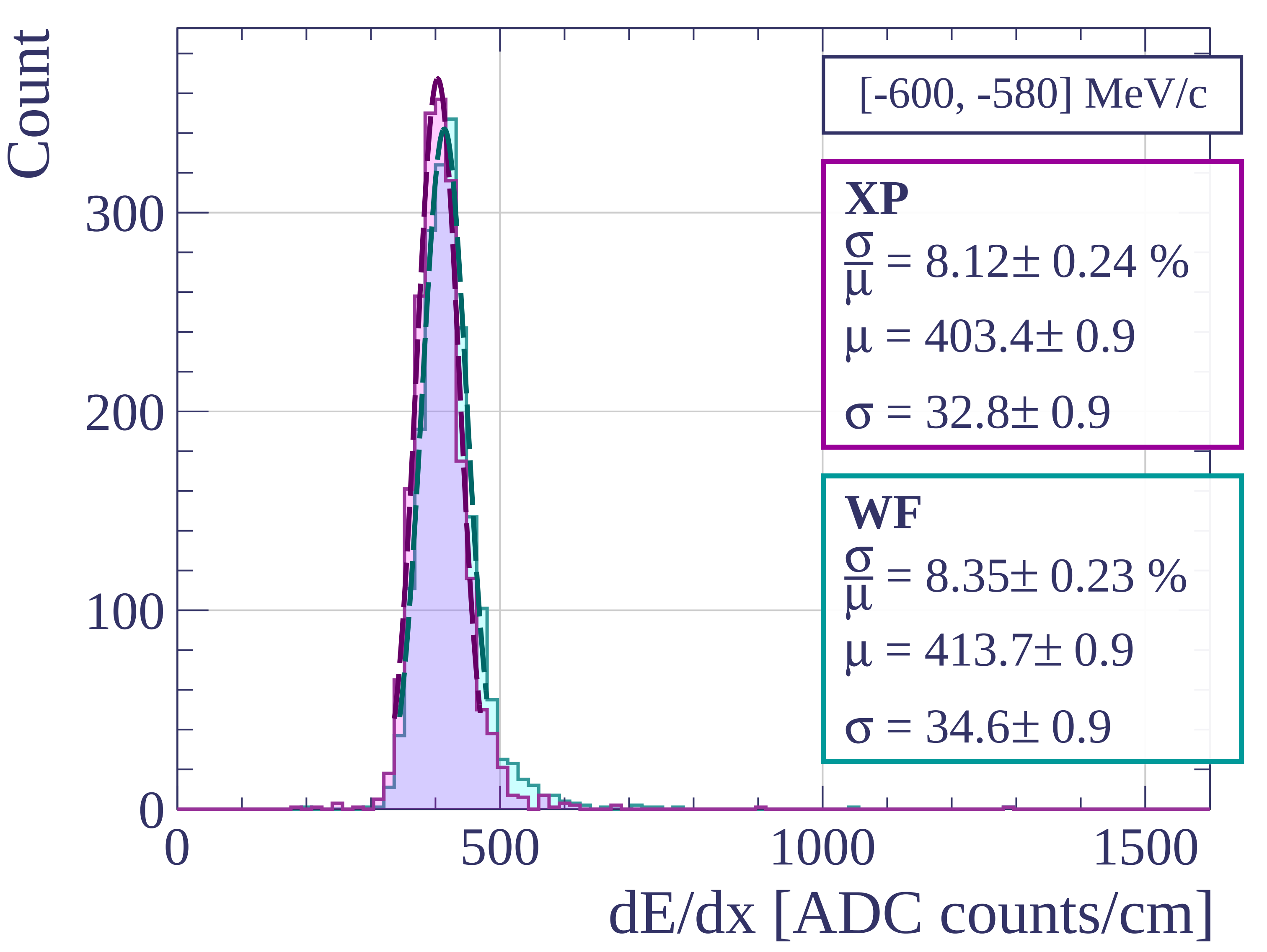
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-660, -640] MeV/c
XP
σ/μ = 8.47± 0.23 %
μ = 409.3± 0.9
σ = 34.7± 0.9
WF
σ/μ = 8.69± 0.22 %
μ = 420.4± 1.0
σ = 36.5± 0.8



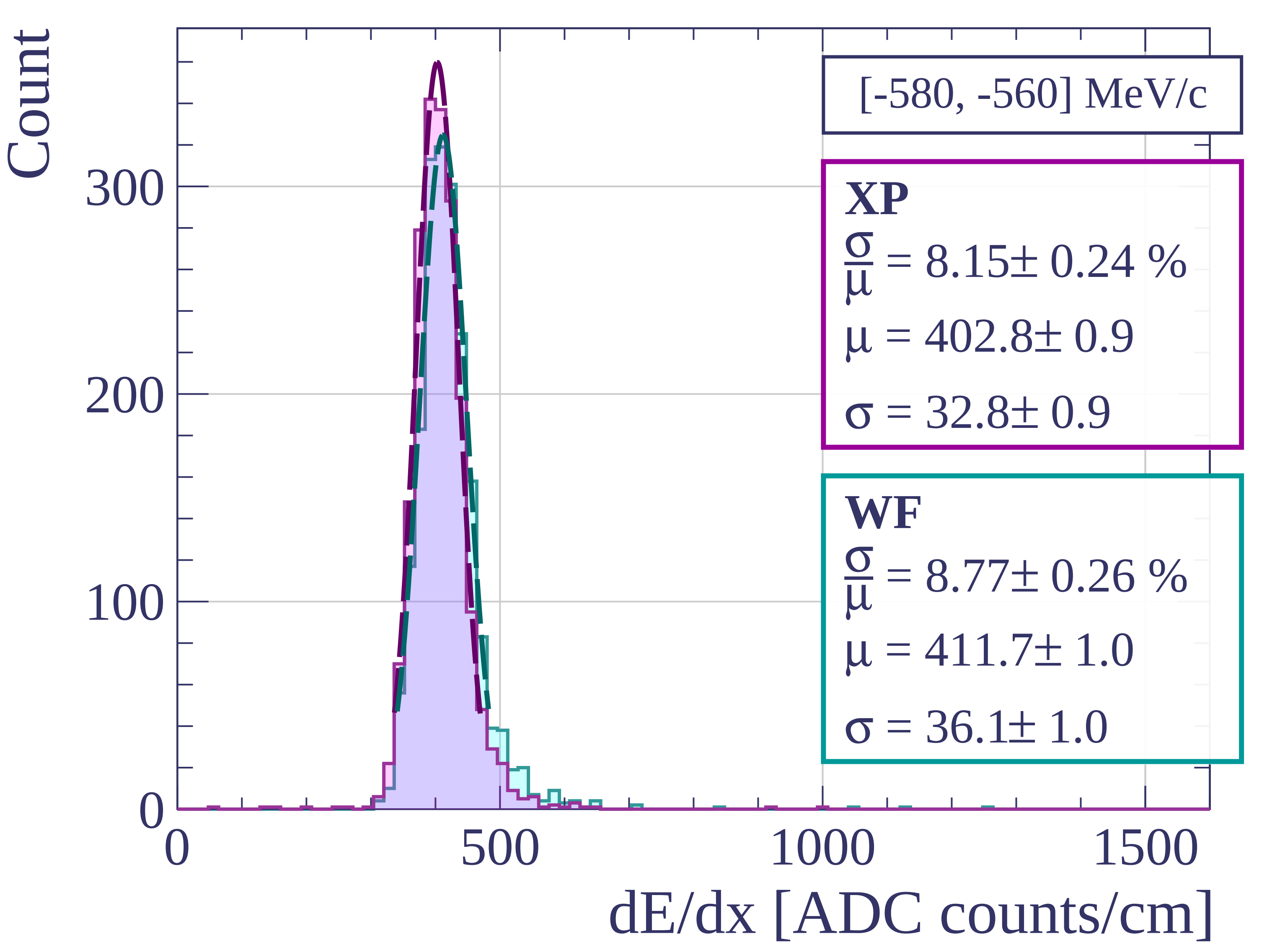
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-640, -620] MeV/c
XP
σ/μ = 8.40± 0.22 %
μ = 408.1± 0.9
σ = 34.3± 0.8
WF
σ/μ = 8.80± 0.23 %
μ = 418.2± 1.0
σ = 36.8± 0.9

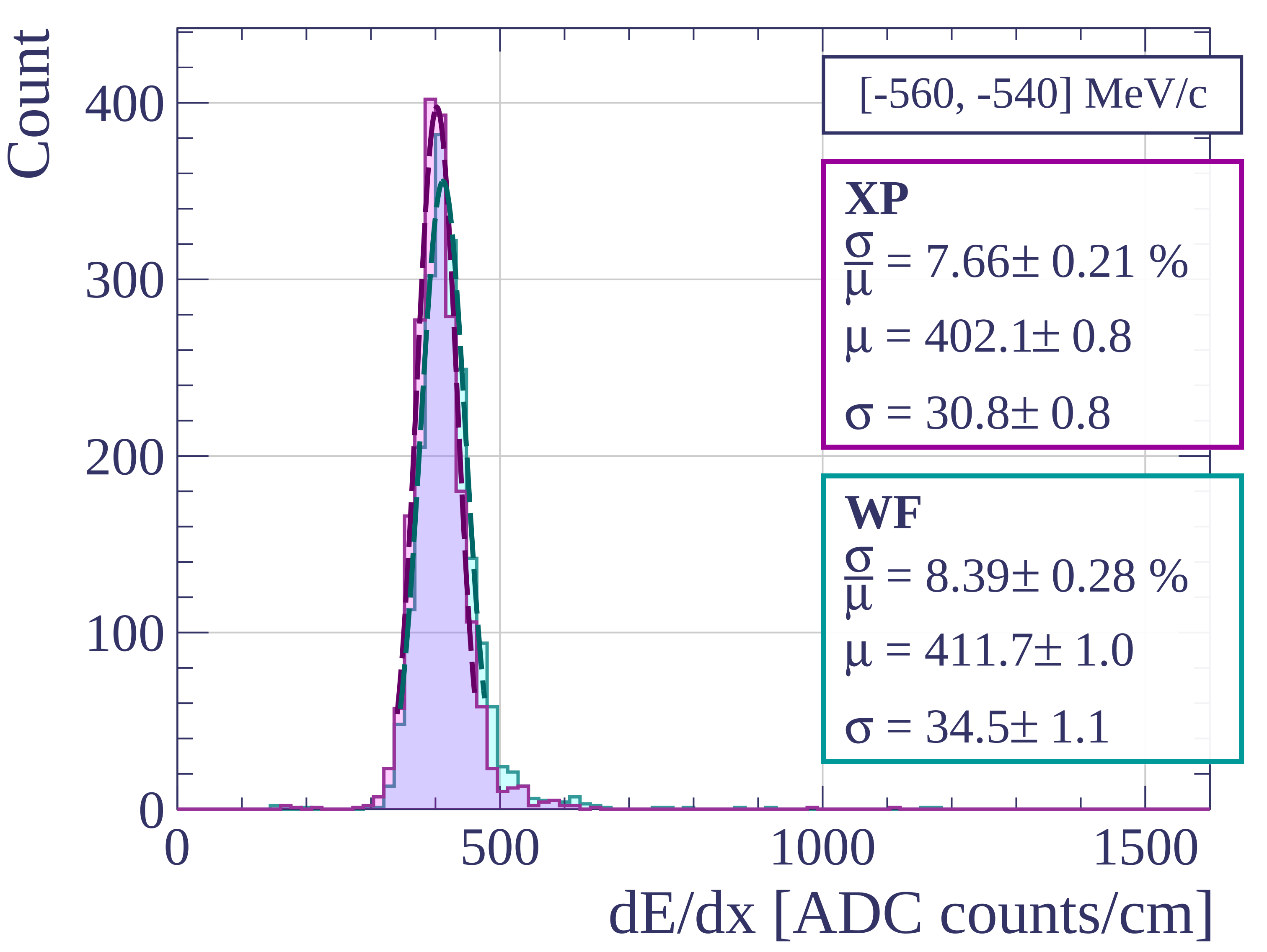


Count
dE/dx [ADC counts/cm]
[-620, -600] MeV/c
XP
σ/μ = 7.63± 0.21 %
μ = 405.0± 0.8
σ = 30.9± 0.8
WF
σ/μ = 8.20± 0.23 %
μ = 414.7± 0.9
σ = 34.0± 0.9
0
100
200
300
400
500
1000
1500

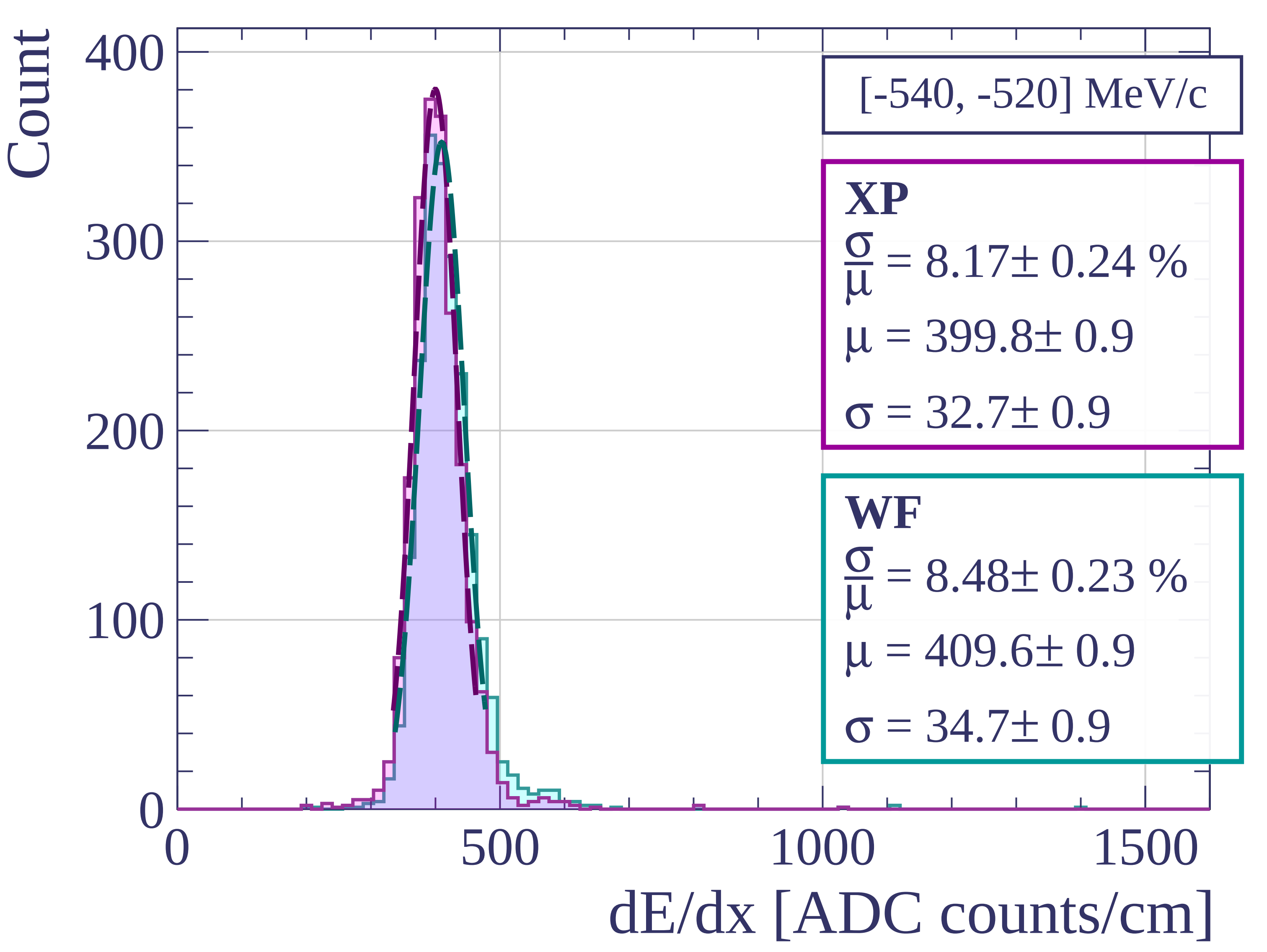


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-600, -580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.12± 0.24 %
μ = 403.4± 0.9
σ = 32.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.35± 0.23 %
μ = 413.7± 0.9
σ = 34.6± 0.9

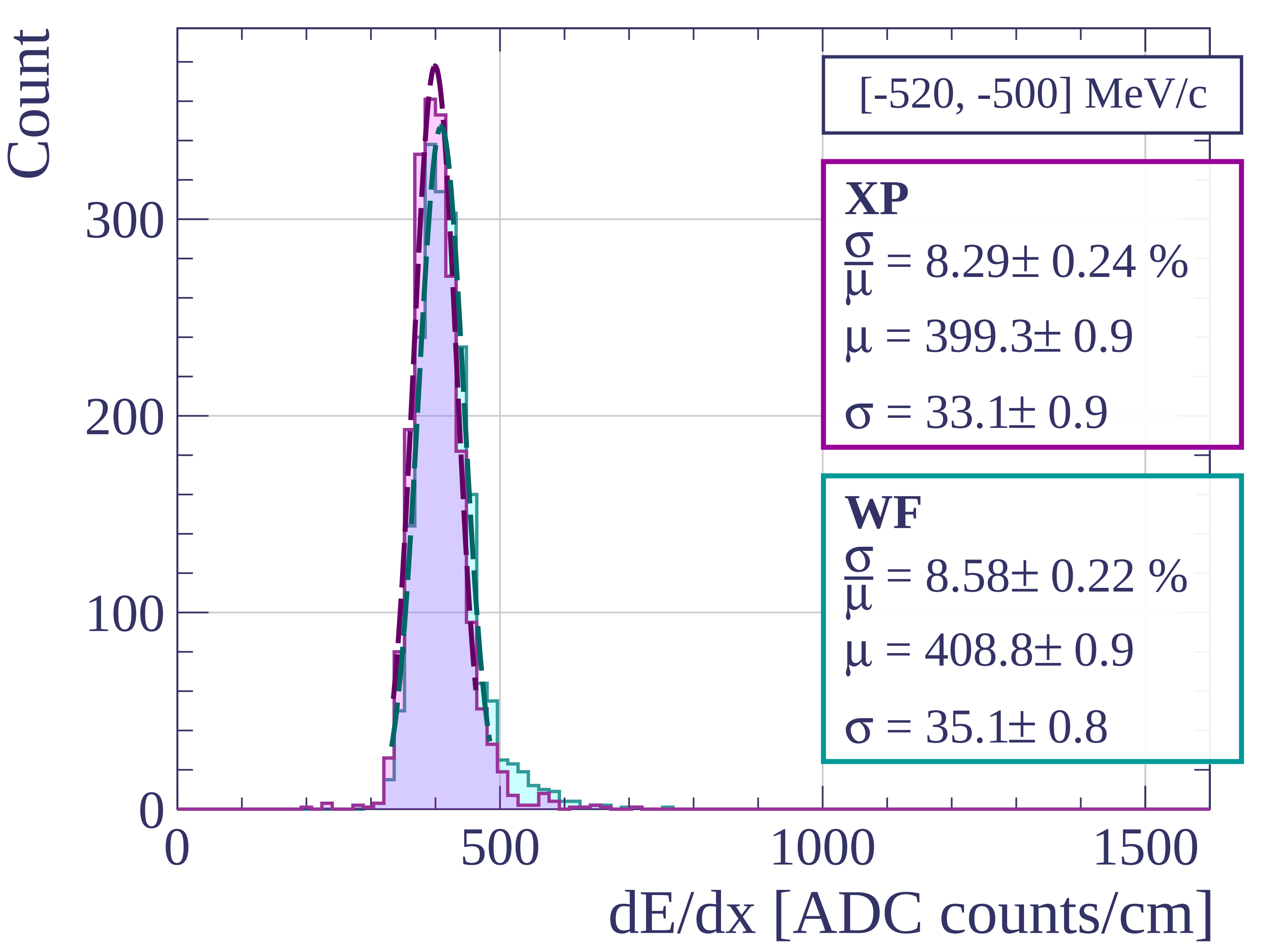

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-580, -560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.15± 0.24 %
μ = 402.8± 0.9
σ = 32.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.77± 0.26 %
μ = 411.7± 1.0
σ = 36.1± 1.0

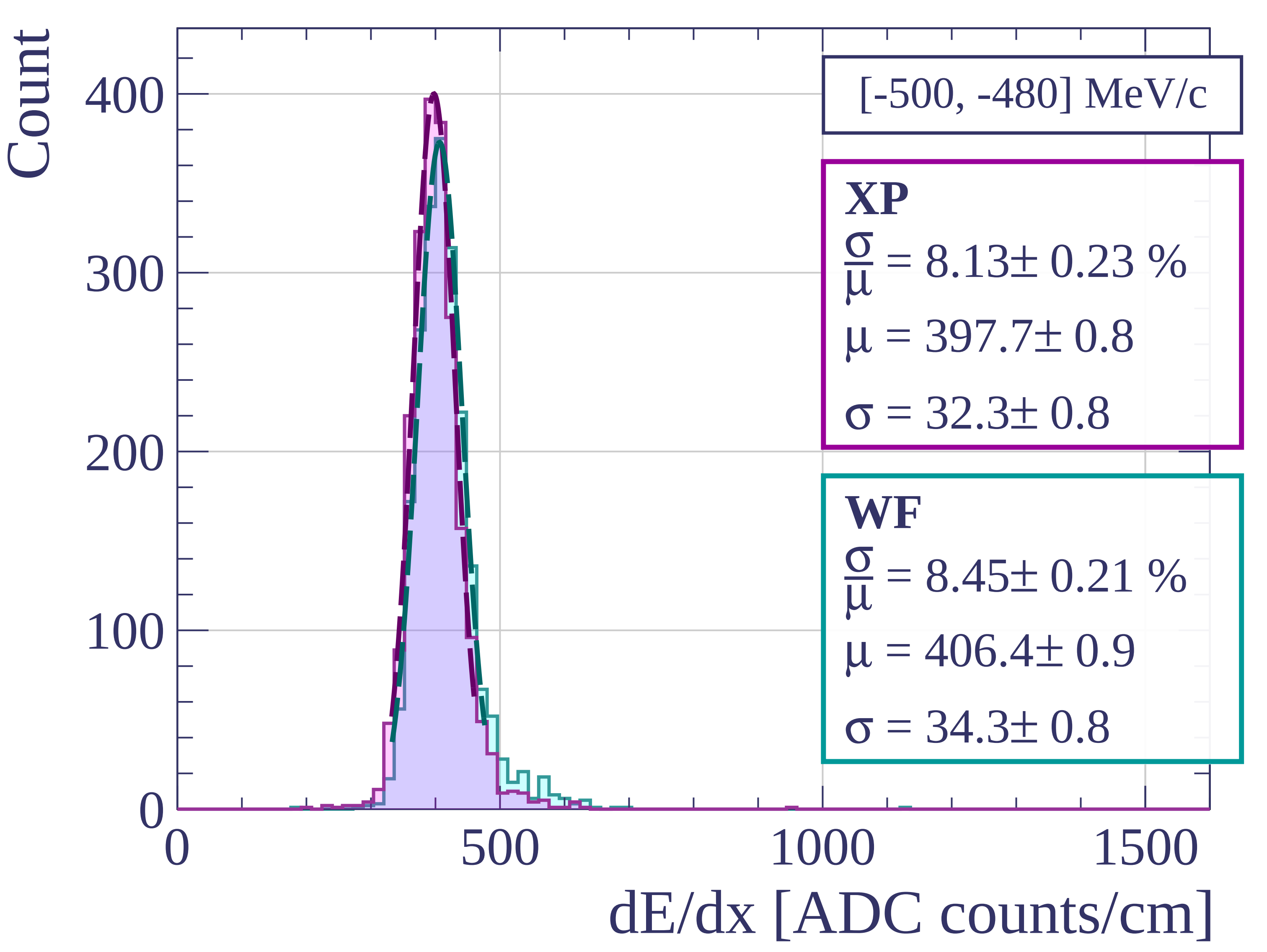


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-560, -540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.66± 0.21 %
μ = 402.1± 0.8
σ = 30.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.39± 0.28 %
μ = 411.7± 1.0
σ = 34.5± 1.1

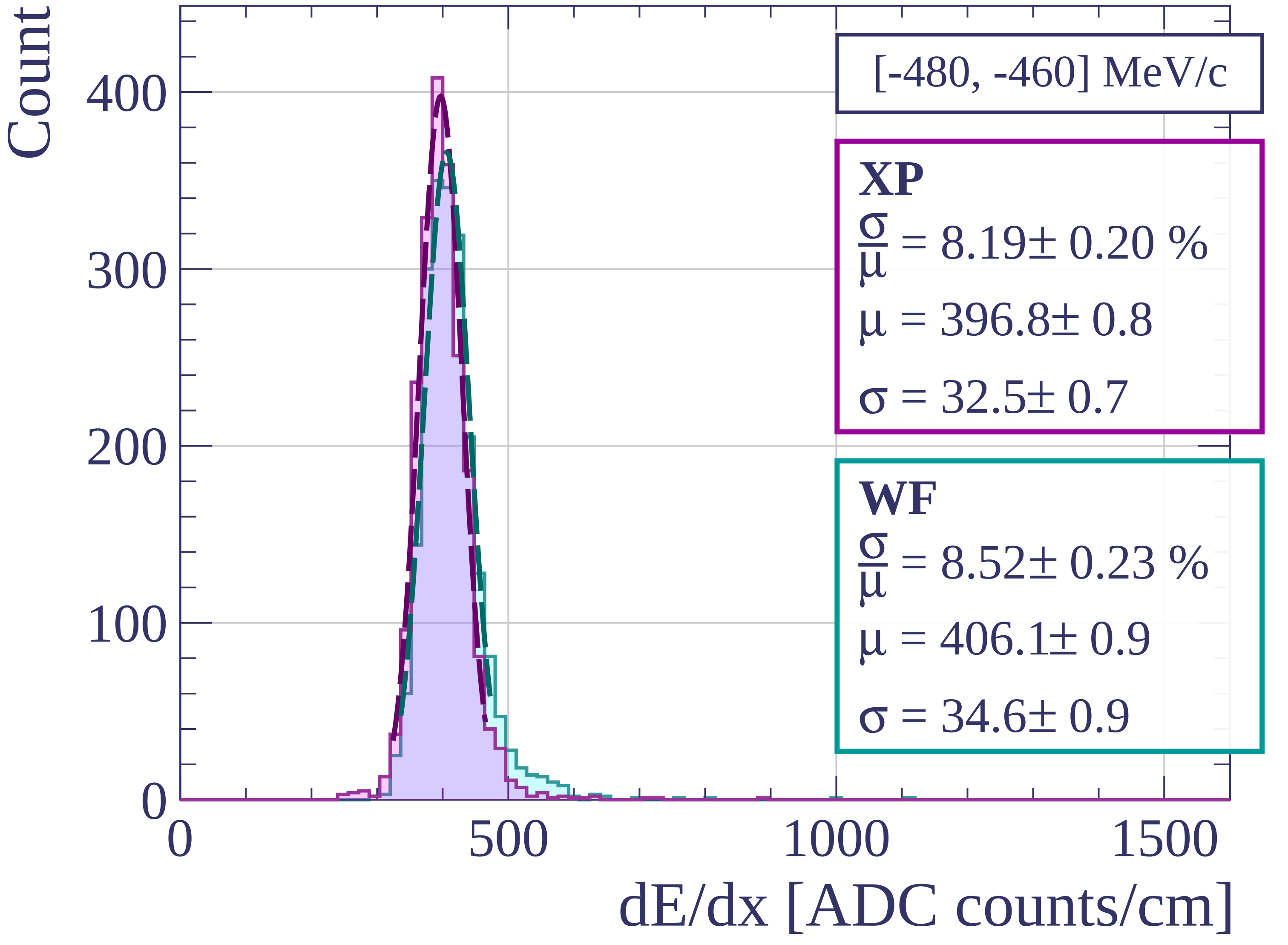


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-540, -520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.17± 0.24 %
μ = 399.8± 0.9
σ = 32.7± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.48± 0.23 %
μ = 409.6± 0.9
σ = 34.7± 0.9

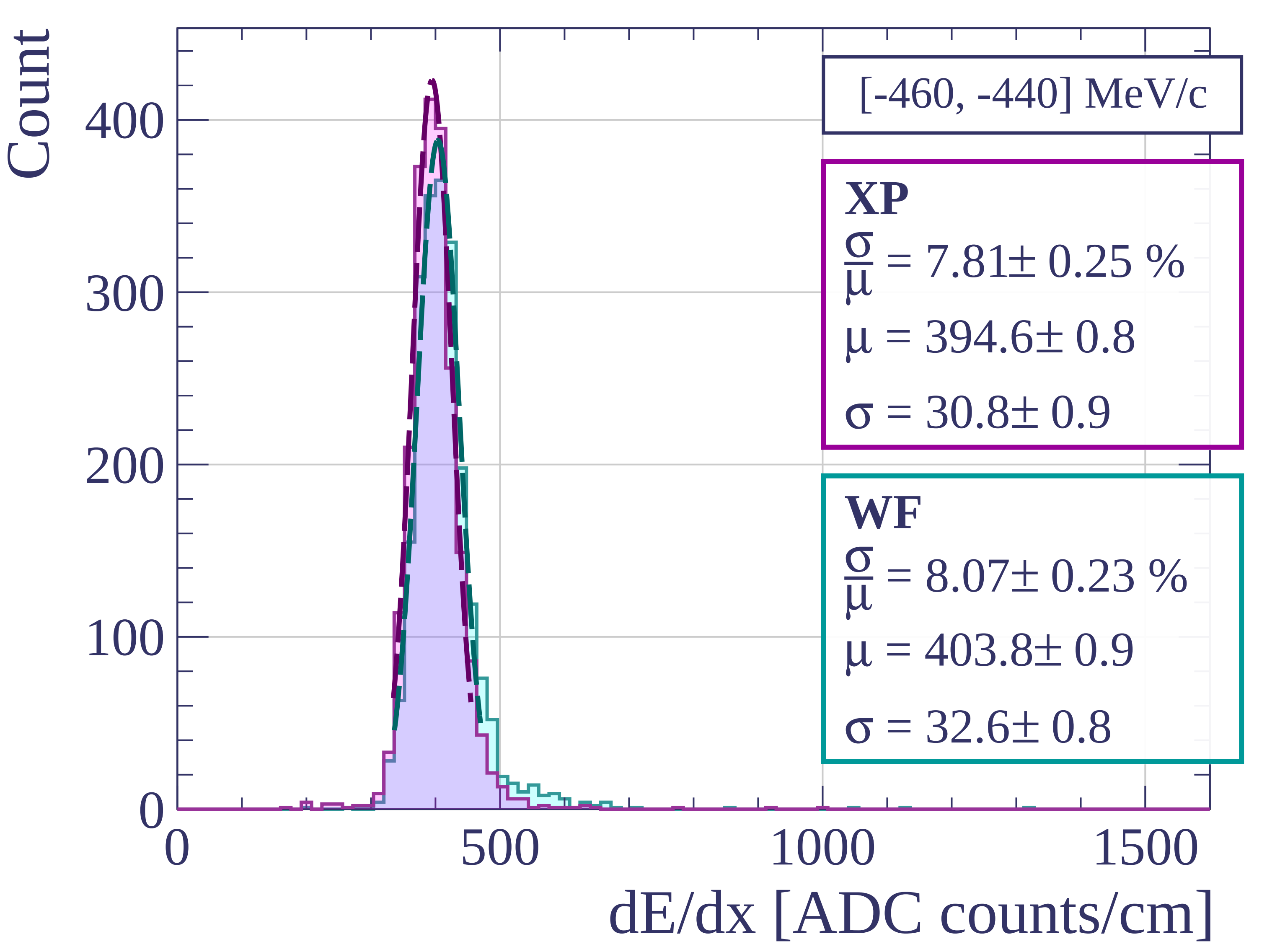

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-520, -500] MeV/c
XP
σ/μ = 8.29± 0.24 %
μ = 399.3± 0.9
σ = 33.1± 0.9
WF
σ/μ = 8.58± 0.22 %
μ = 408.8± 0.9
σ = 35.1± 0.8

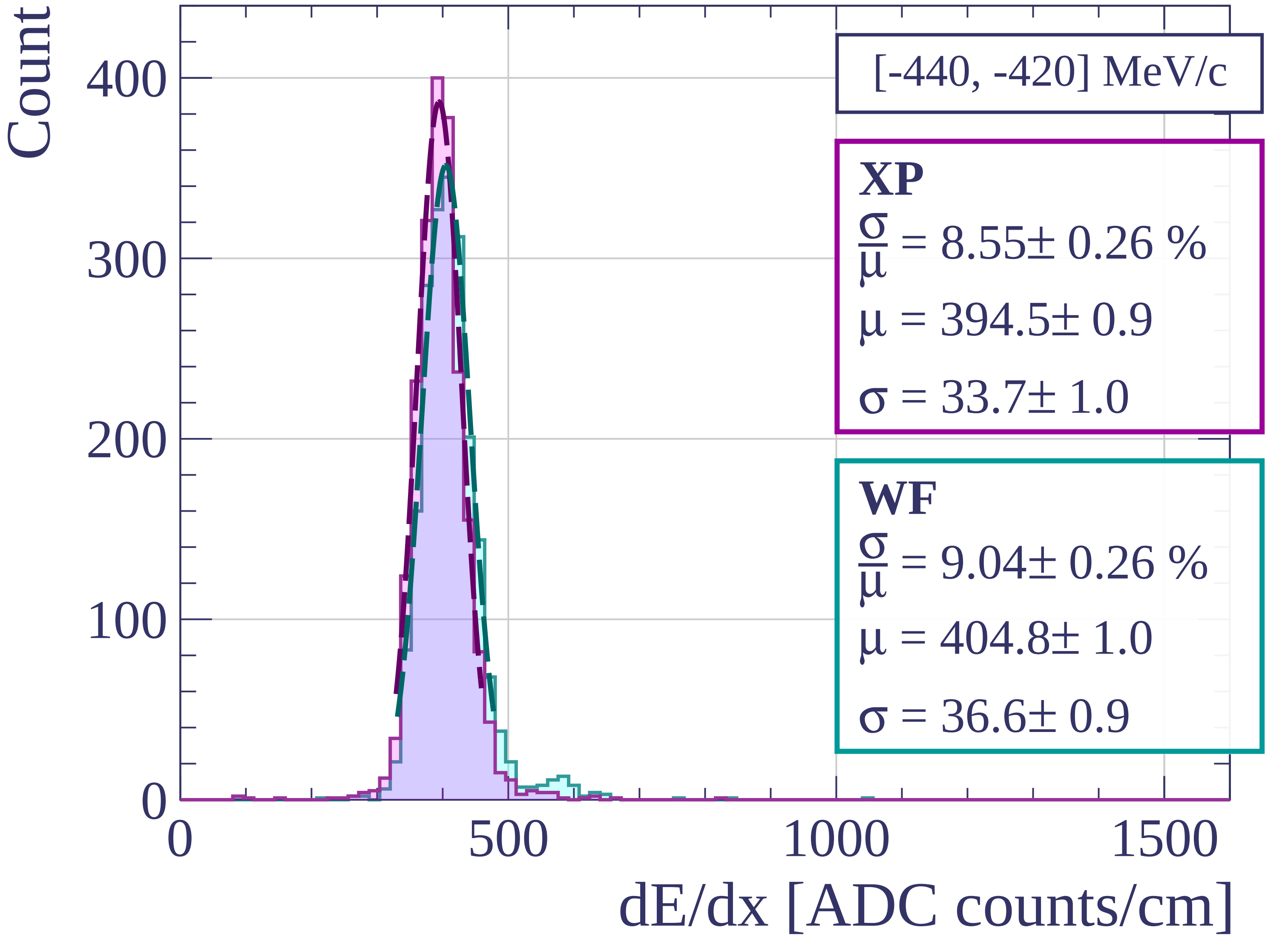


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-500, -480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.13± 0.23 %
μ = 397.7± 0.8
σ = 32.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.45± 0.21 %
μ = 406.4± 0.9
σ = 34.3± 0.8

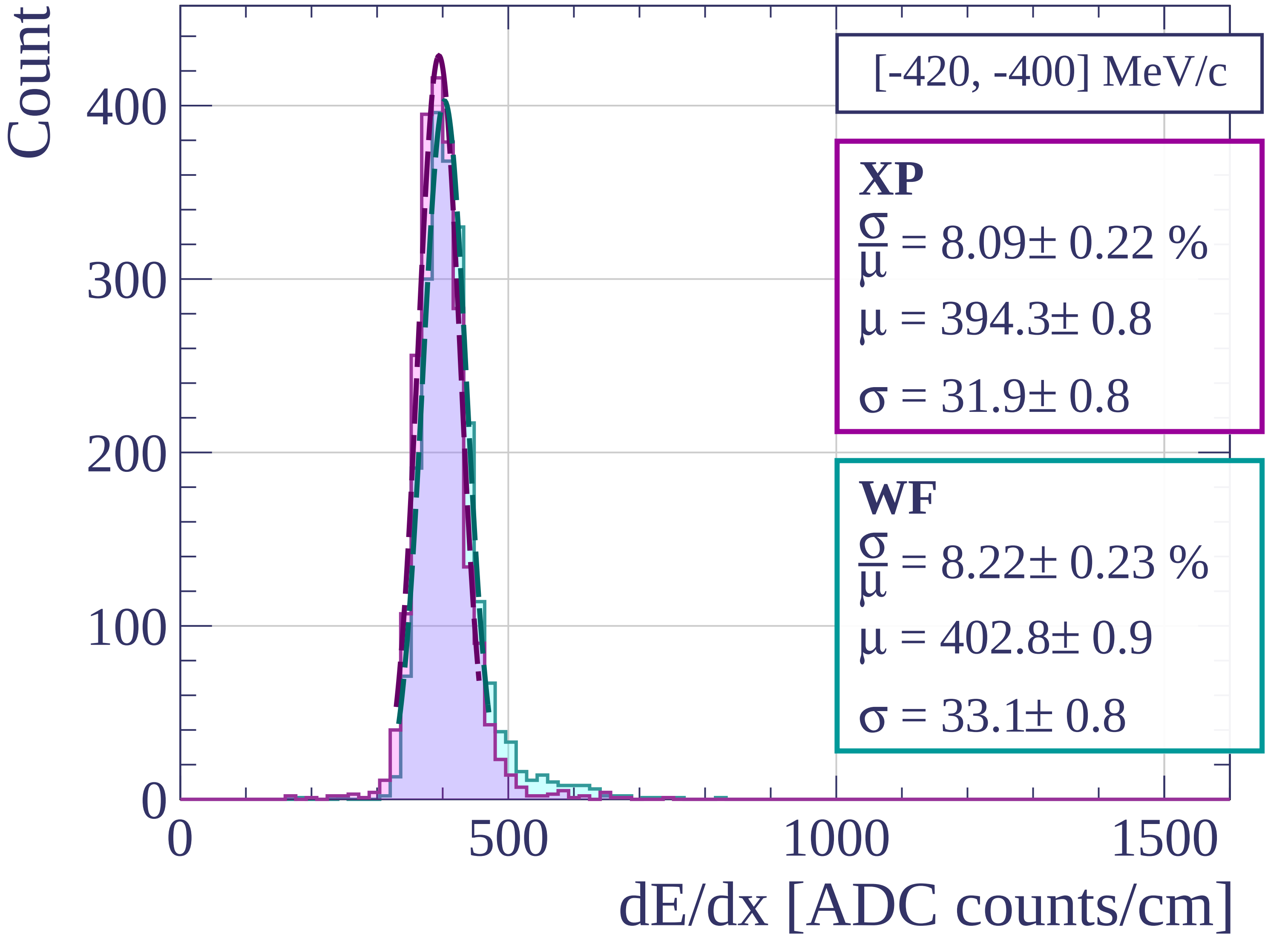


Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
[-480, -460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.20 %
μ = 396.8± 0.8
σ = 32.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.52± 0.23 %
μ = 406.1± 0.9
σ = 34.6± 0.9

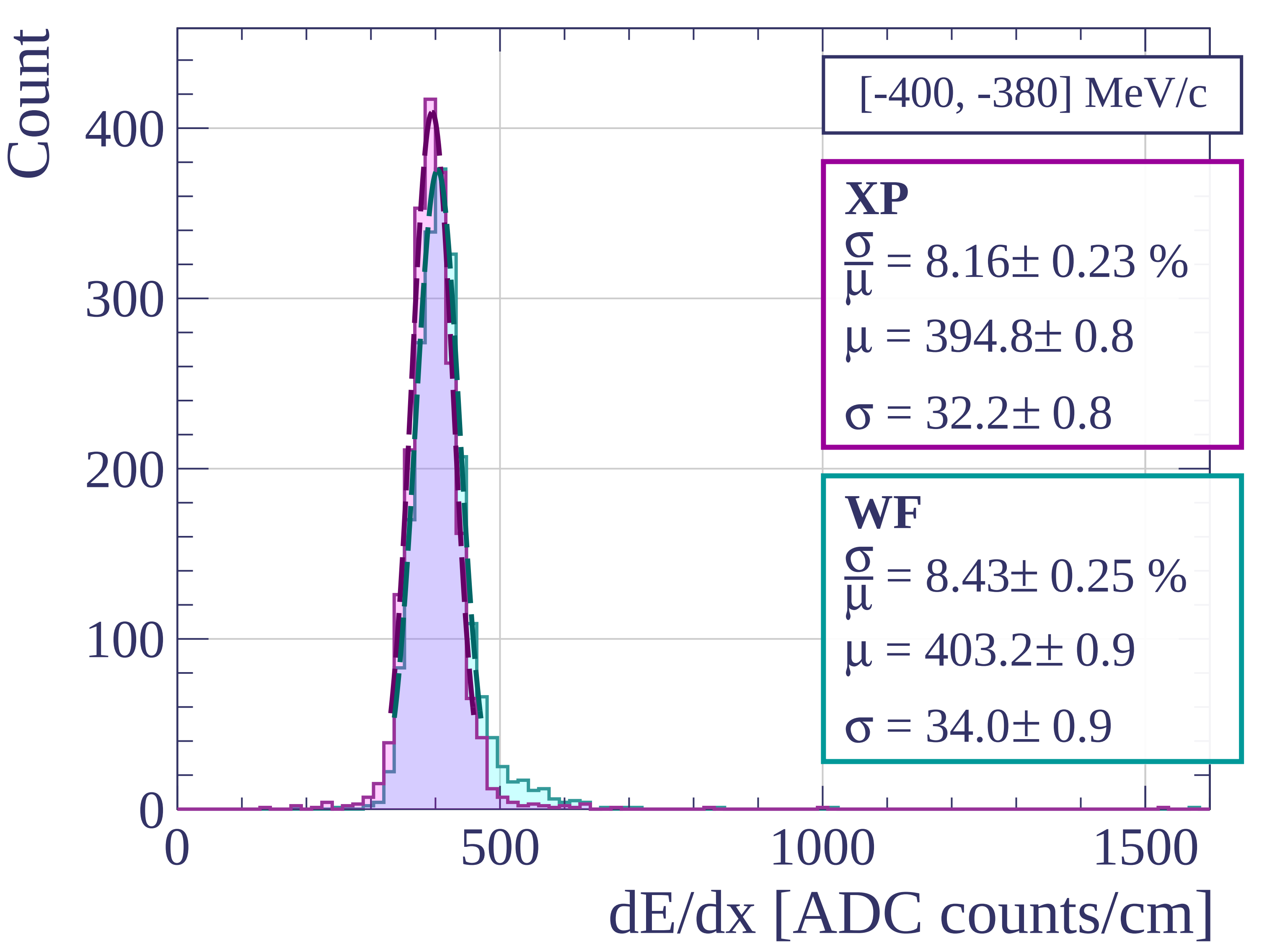

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-460, -440] MeV/c
XP
σ/μ = 7.81± 0.25 %
μ = 394.6± 0.8
σ = 30.8± 0.9
WF
σ/μ = 8.07± 0.23 %
μ = 403.8± 0.9
σ = 32.6± 0.8

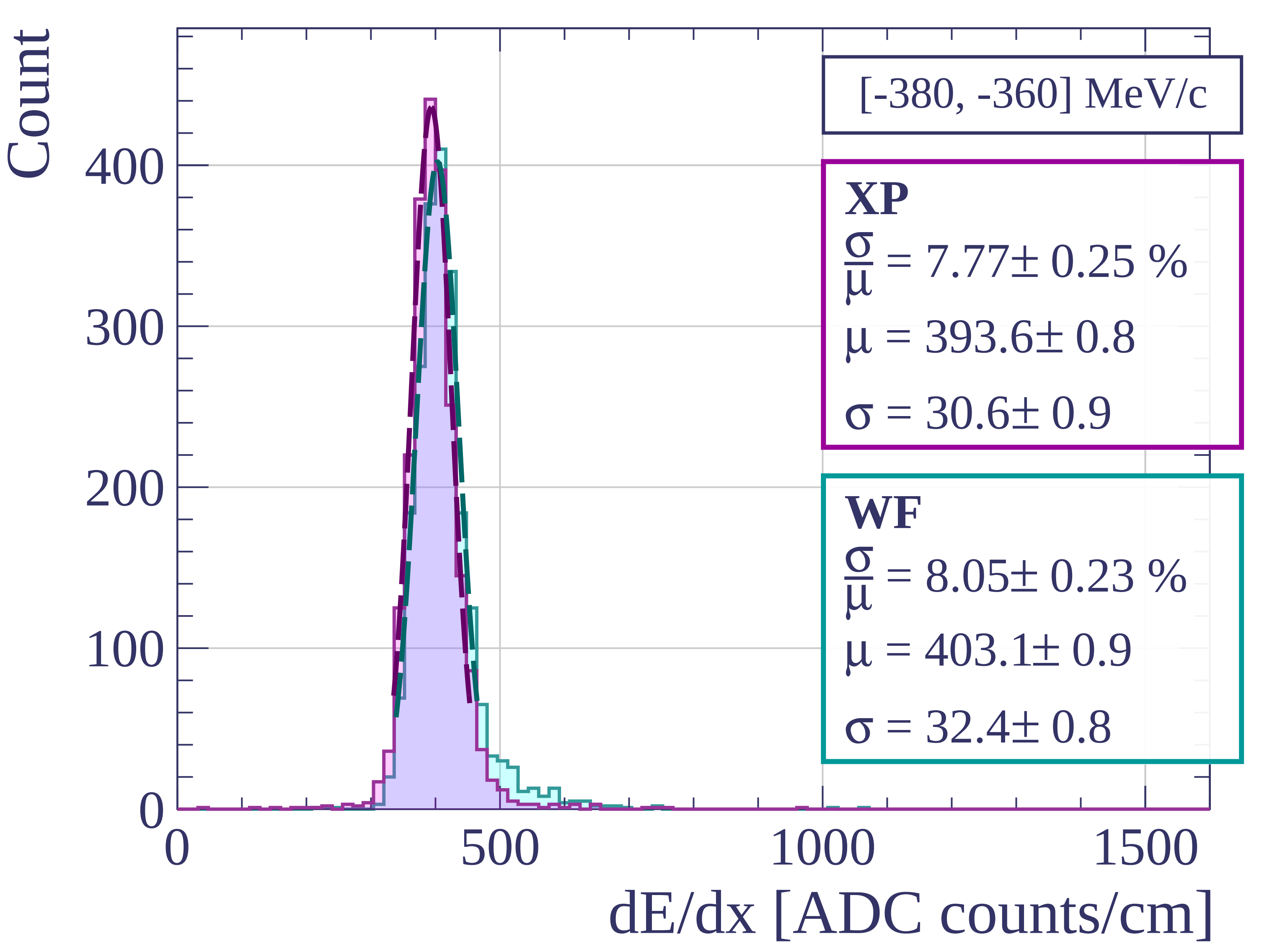


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-440, -420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.55± 0.26 %
μ = 394.5± 0.9
σ = 33.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.04± 0.26 %
μ = 404.8± 1.0
σ = 36.6± 0.9

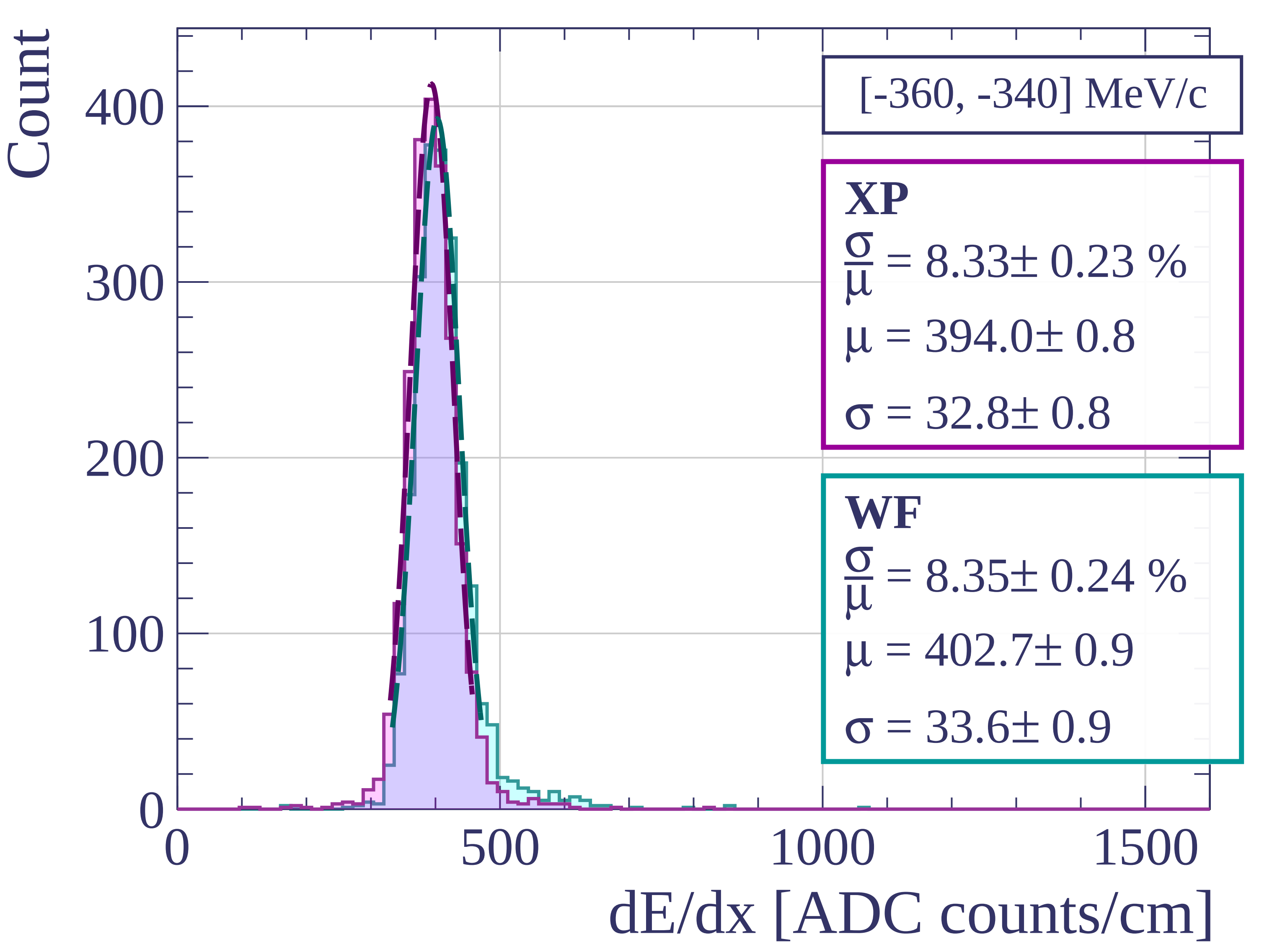


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-420, -400] MeV/c
XP
σ/μ = 8.09± 0.22 %
μ = 394.3± 0.8
σ = 31.9± 0.8
WF
σ/μ = 8.22± 0.23 %
μ = 402.8± 0.9
σ = 33.1± 0.8

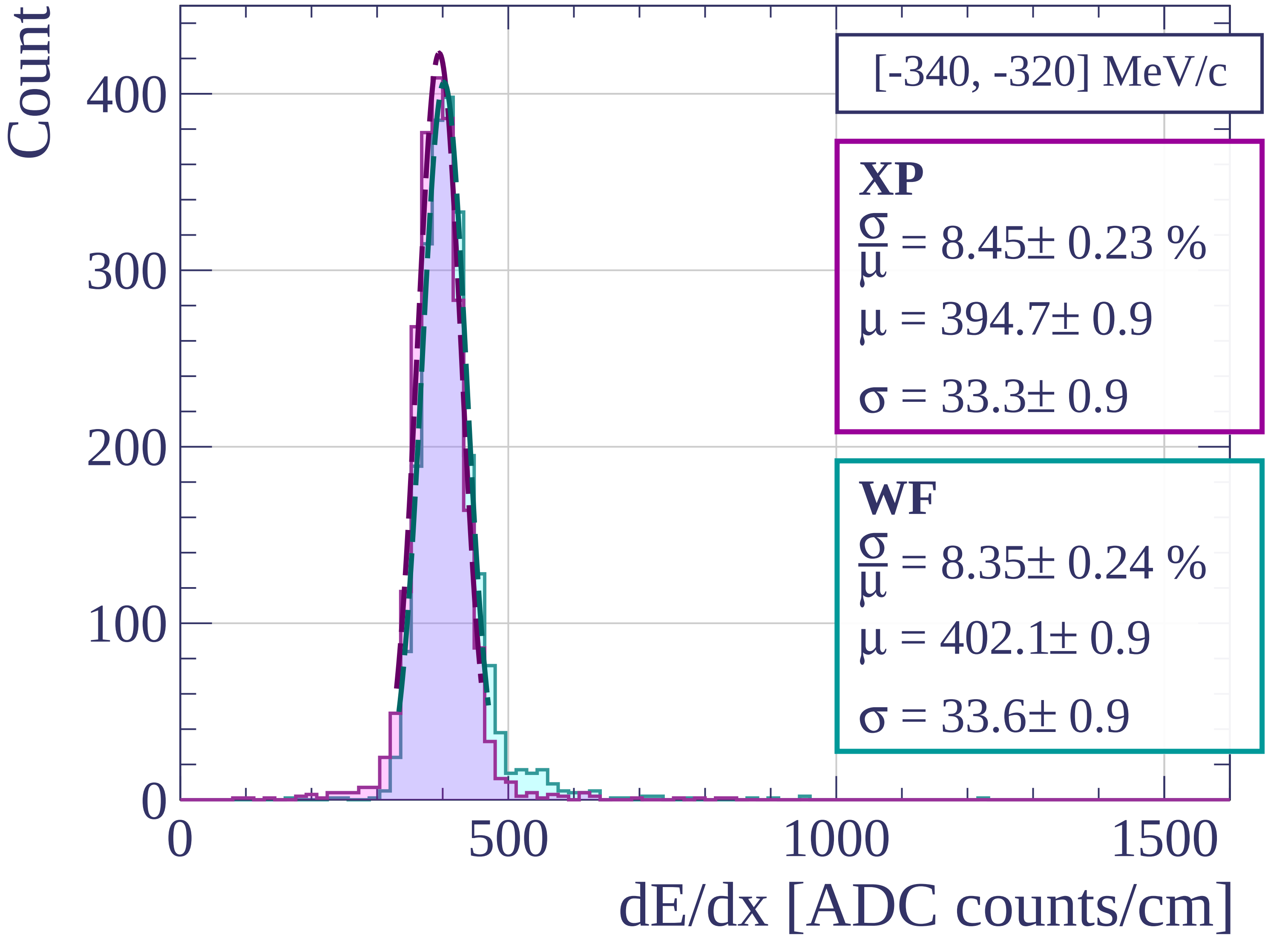

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-400, -380] MeV/c
XP
σ/μ = 8.16± 0.23 %
μ = 394.8± 0.8
σ = 32.2± 0.8
WF
σ/μ = 8.43± 0.25 %
μ = 403.2± 0.9
σ = 34.0± 0.9



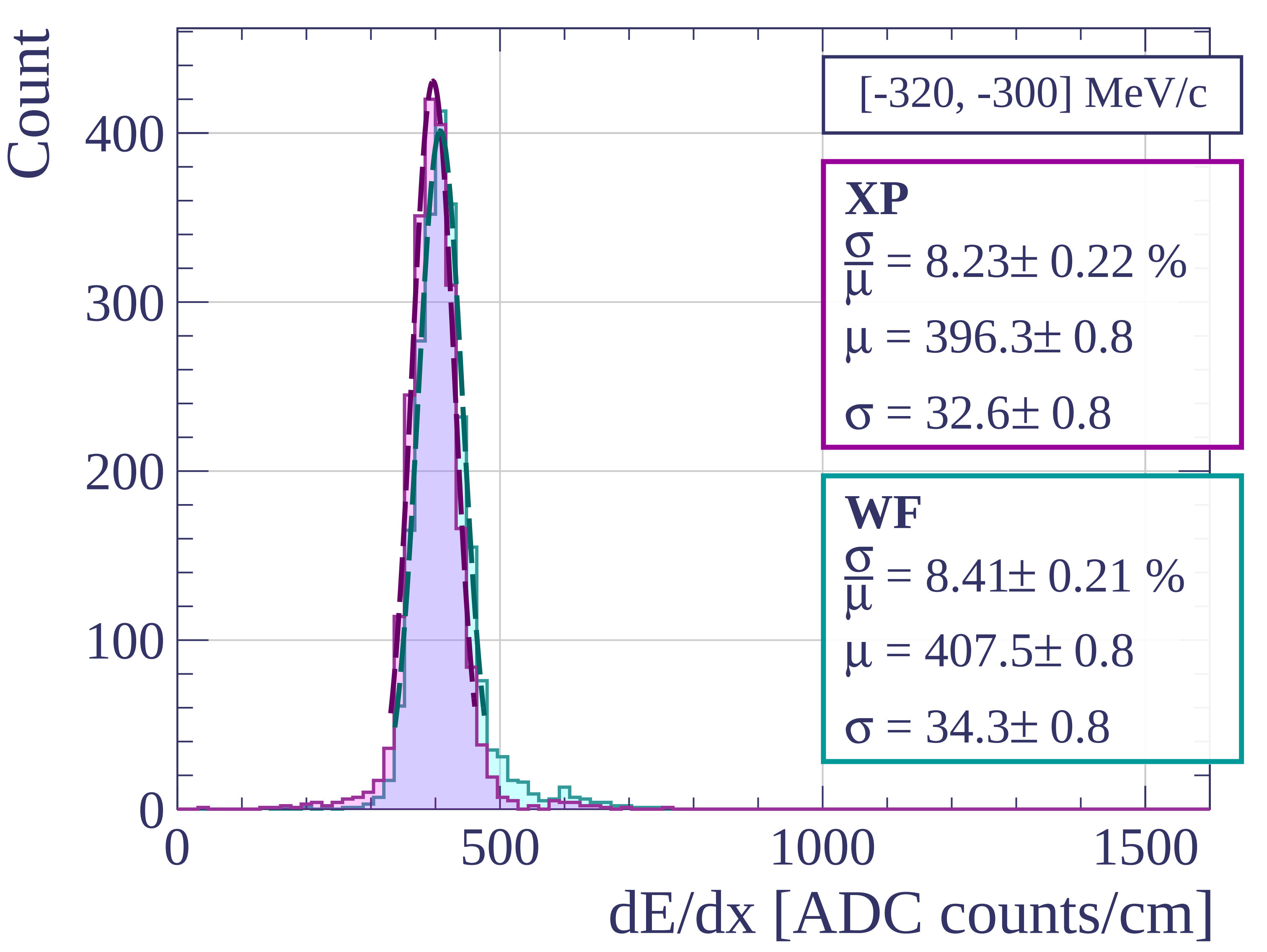
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-380, -360] MeV/c
XP
σ/μ = 7.77± 0.25 %
μ = 393.6± 0.8
σ = 30.6± 0.9
WF
σ/μ = 8.05± 0.23 %
μ = 403.1± 0.9
σ = 32.4± 0.8

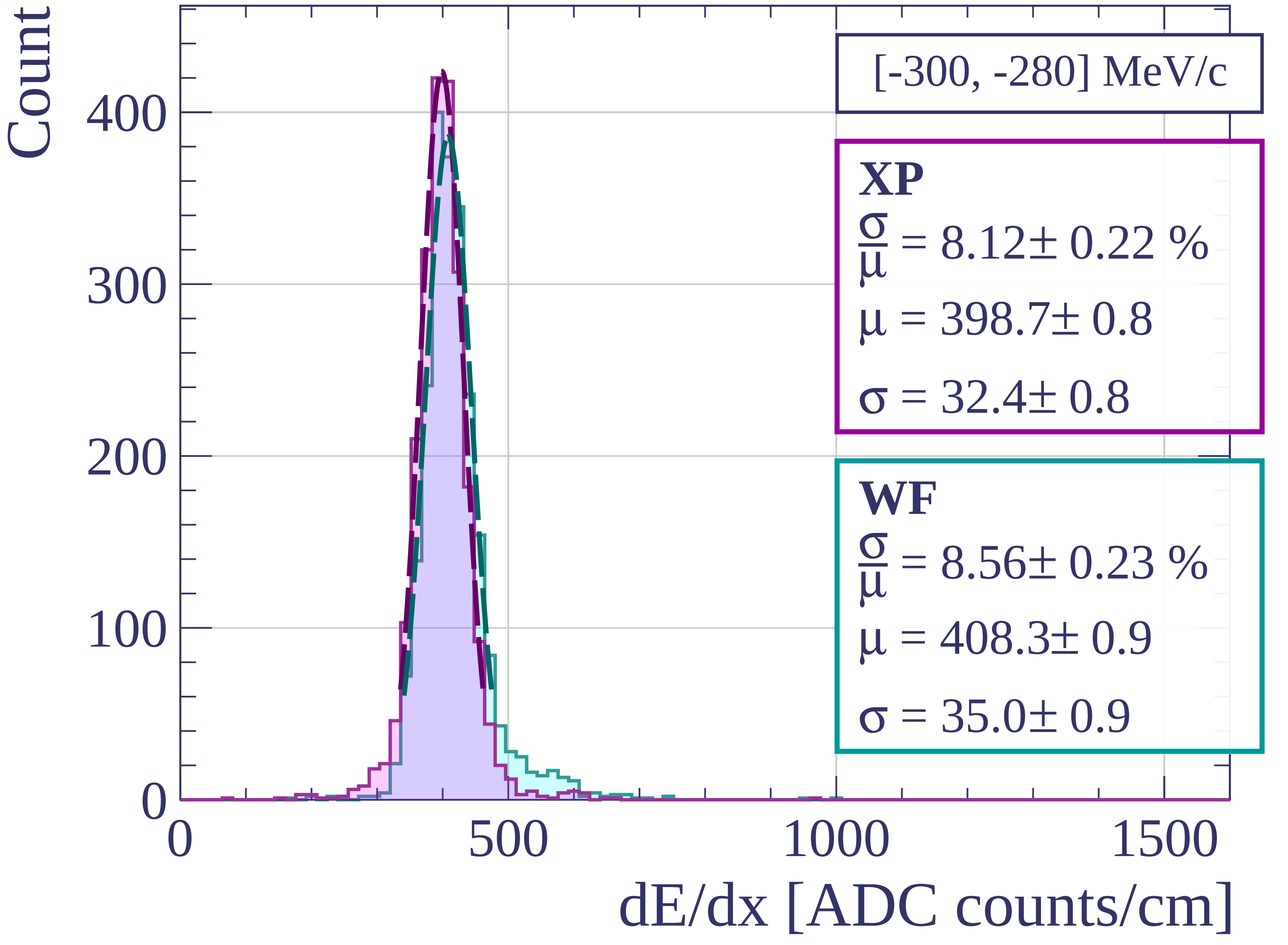


Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
[-360, -340] MeV/c
XP
σ/μ = 8.33± 0.23 %
μ = 394.0± 0.8
σ = 32.8± 0.8
WF
σ/μ = 8.35± 0.24 %
μ = 402.7± 0.9
σ = 33.6± 0.9

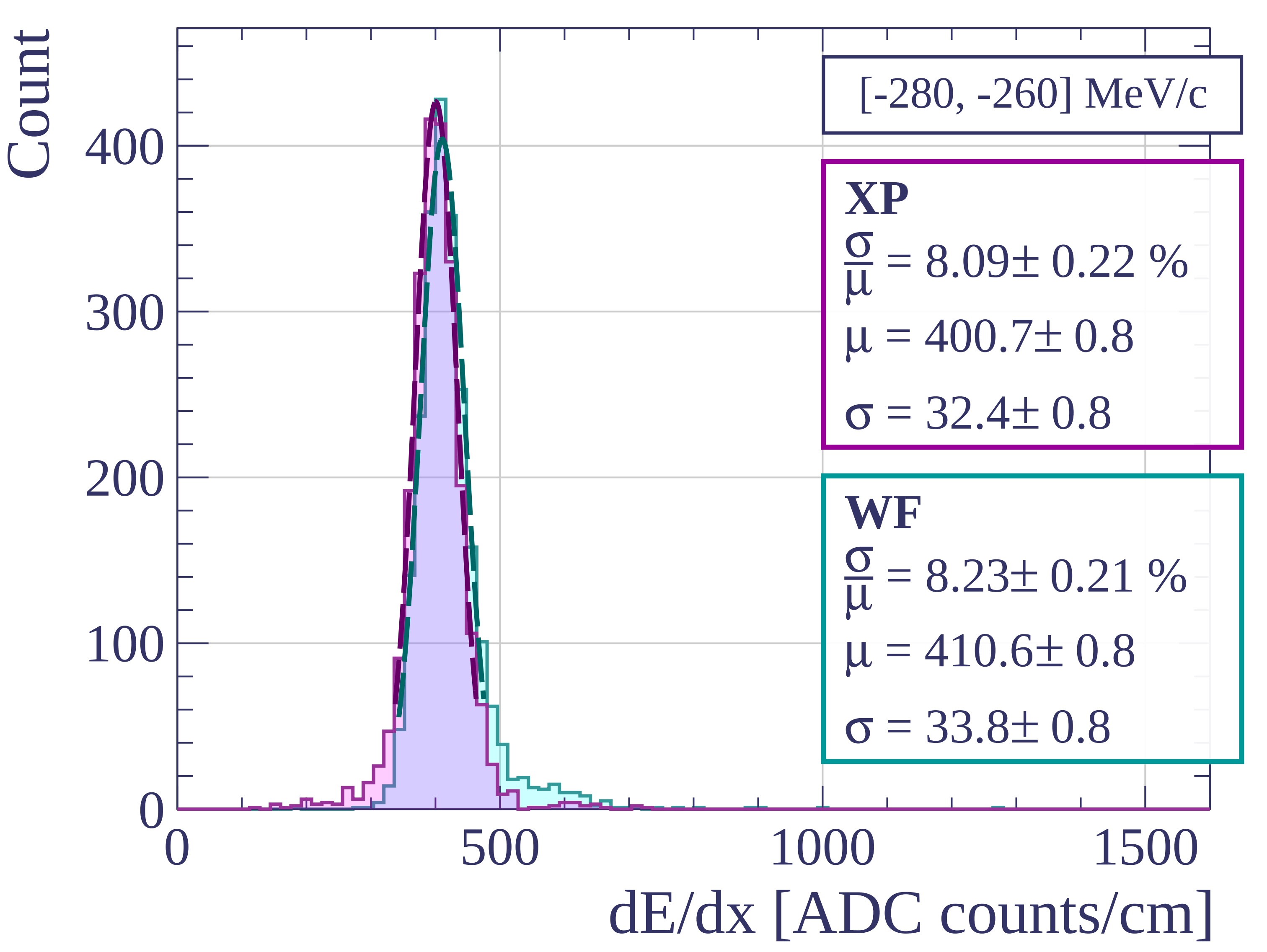


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-340, -320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.45± 0.23 %
μ = 394.7± 0.9
σ = 33.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.35± 0.24 %
μ = 402.1± 0.9
σ = 33.6± 0.9

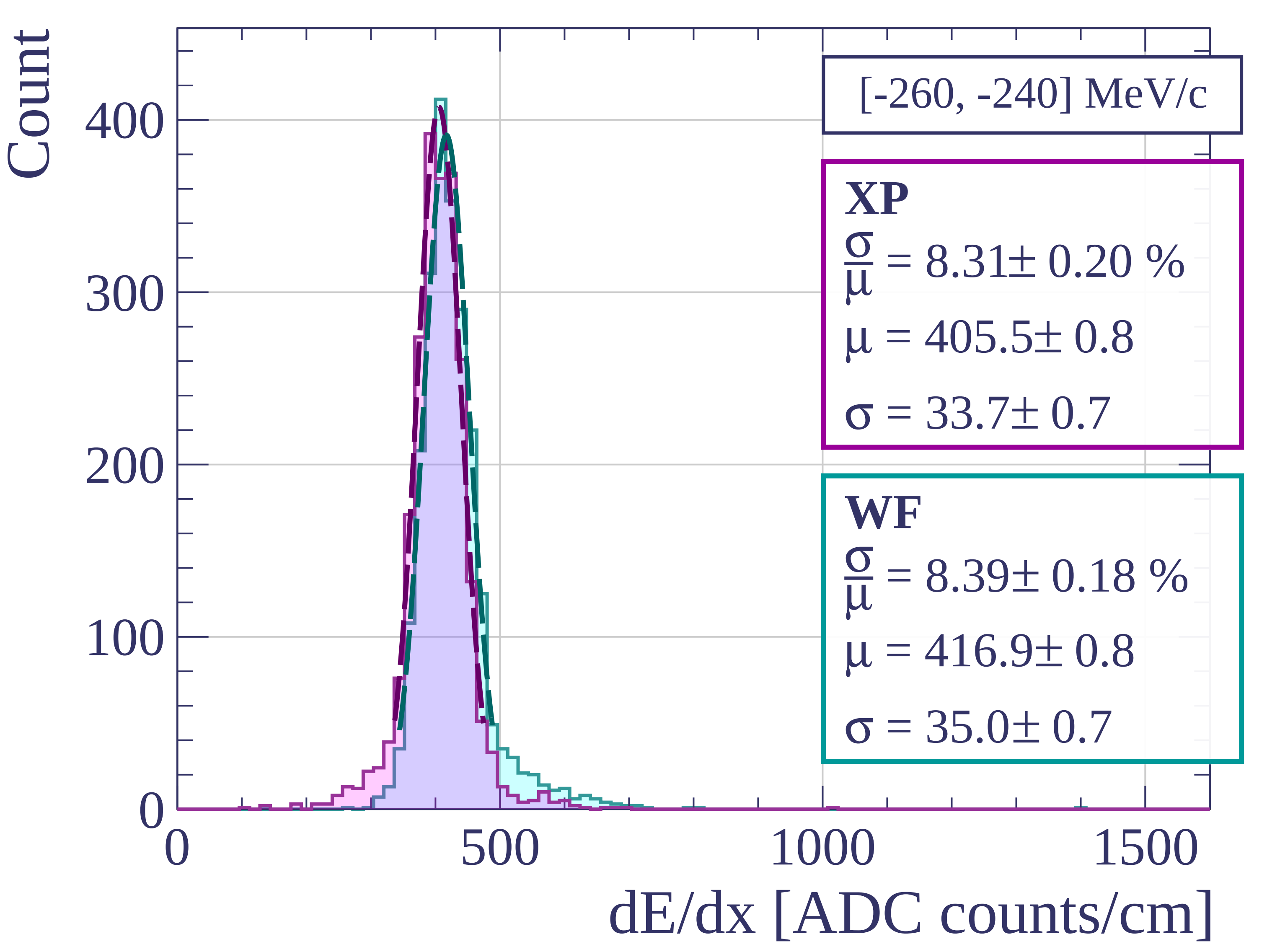

Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
[-320, -300] MeV/c
XP
σ/μ = 8.23± 0.22 %
μ = 396.3± 0.8
σ = 32.6± 0.8
WF
σ/μ = 8.41± 0.21 %
μ = 407.5± 0.8
σ = 34.3± 0.8



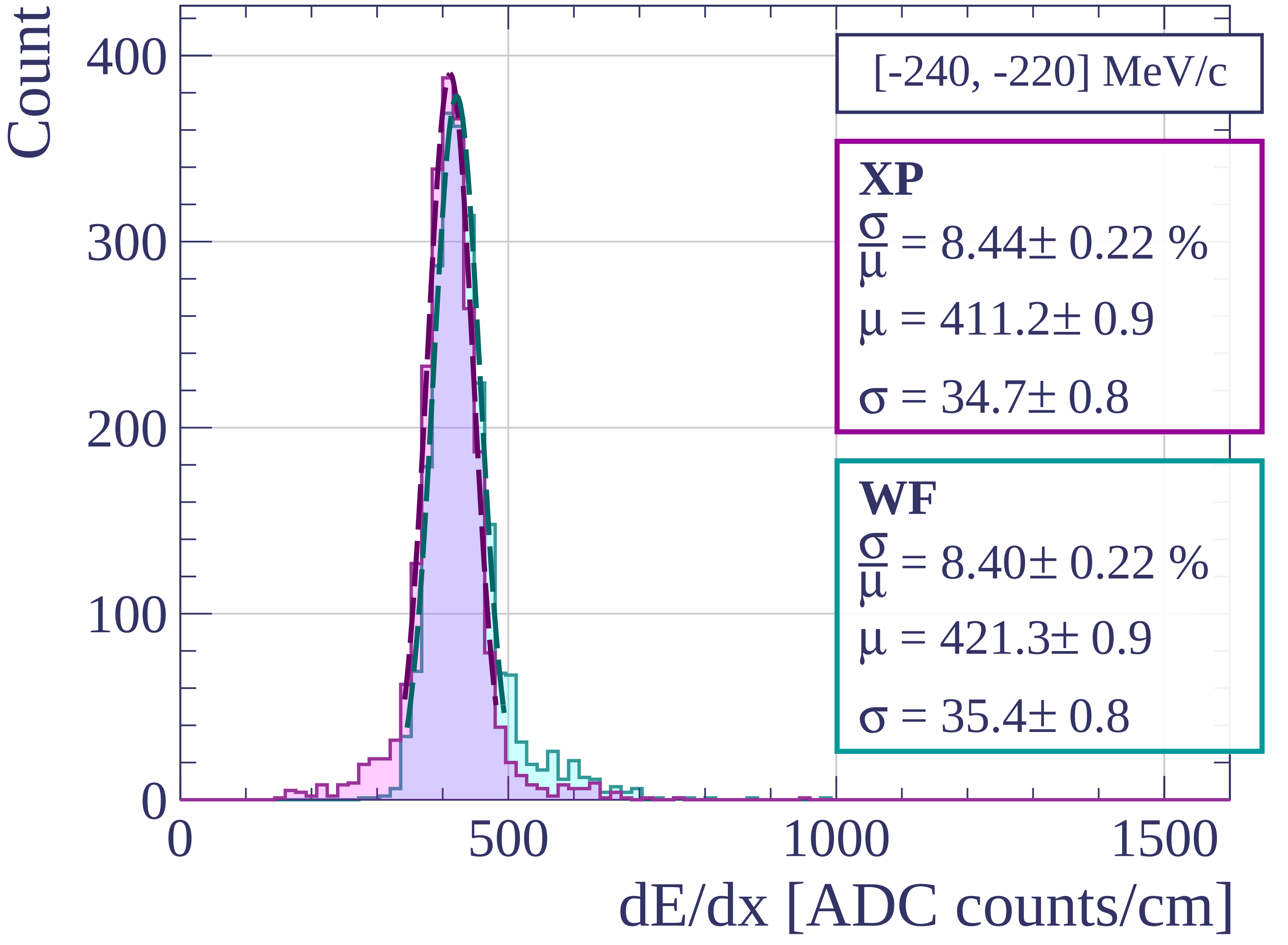
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-300, -280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.12± 0.22 %
μ = 398.7± 0.8
σ = 32.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.56± 0.23 %
μ = 408.3± 0.9
σ = 35.0± 0.9



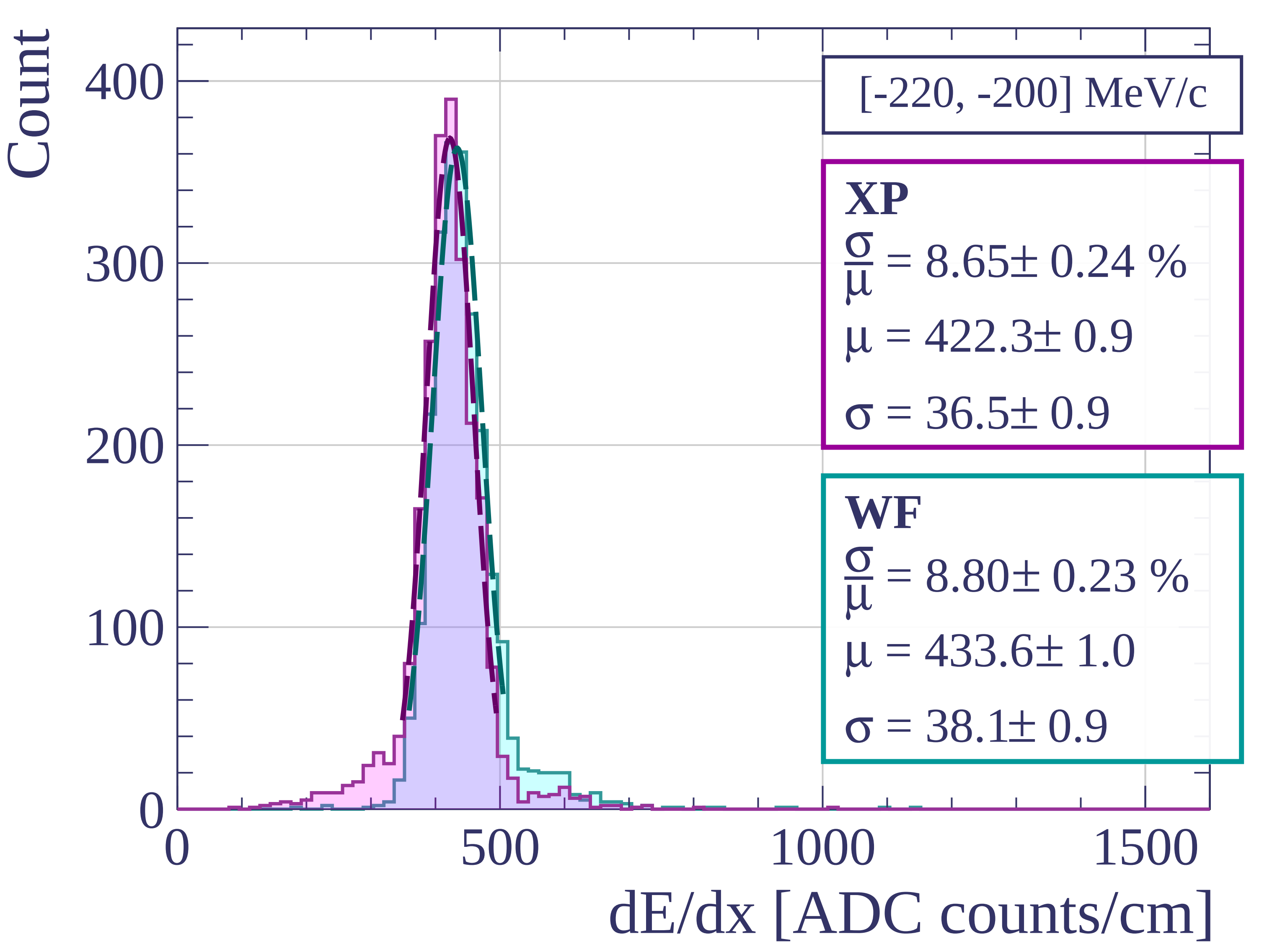
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
100
200
300
400
[-280, -260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.22 %
μ = 400.7± 0.8
σ = 32.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.23± 0.21 %
μ = 410.6± 0.8
σ = 33.8± 0.8

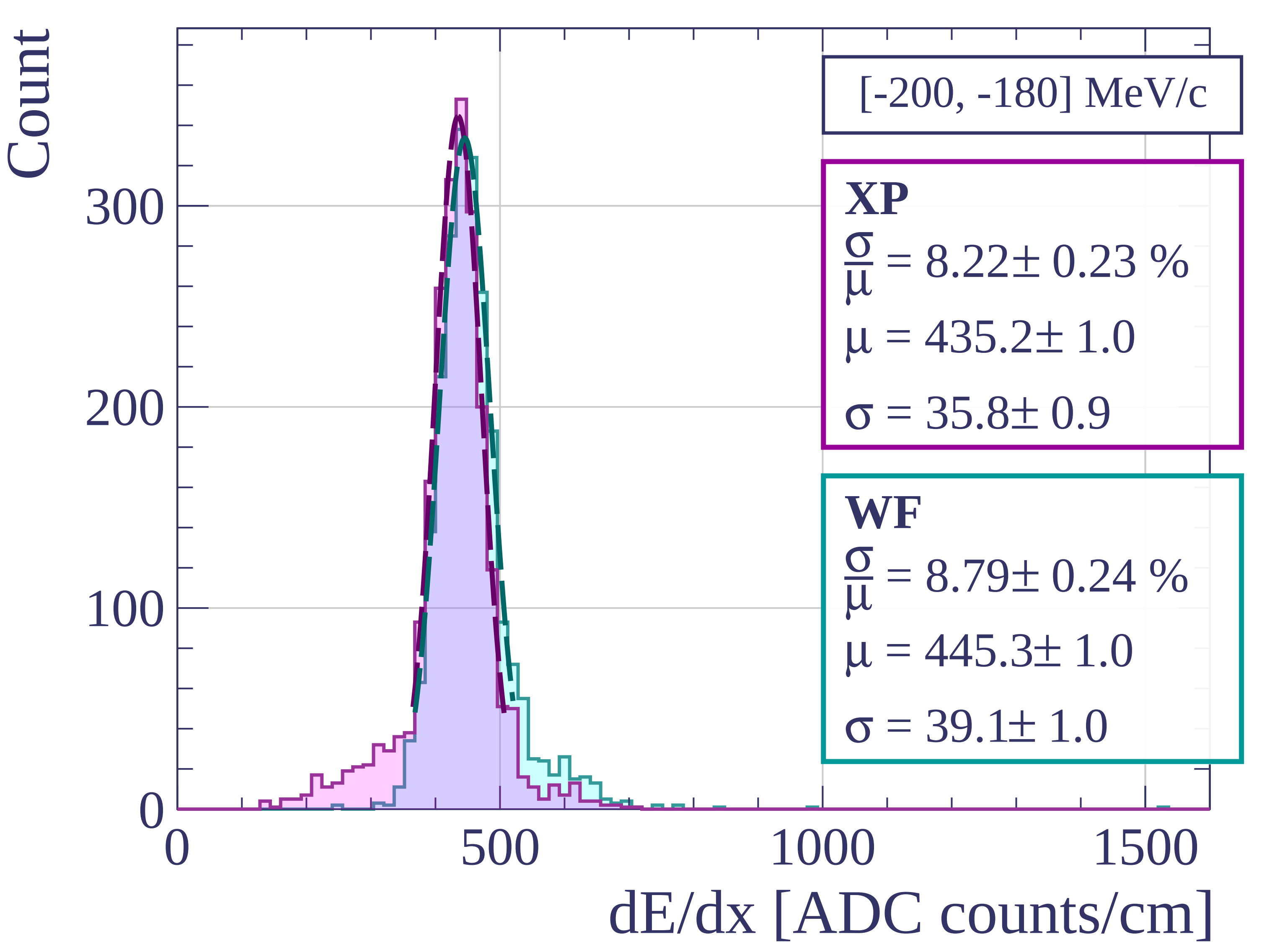


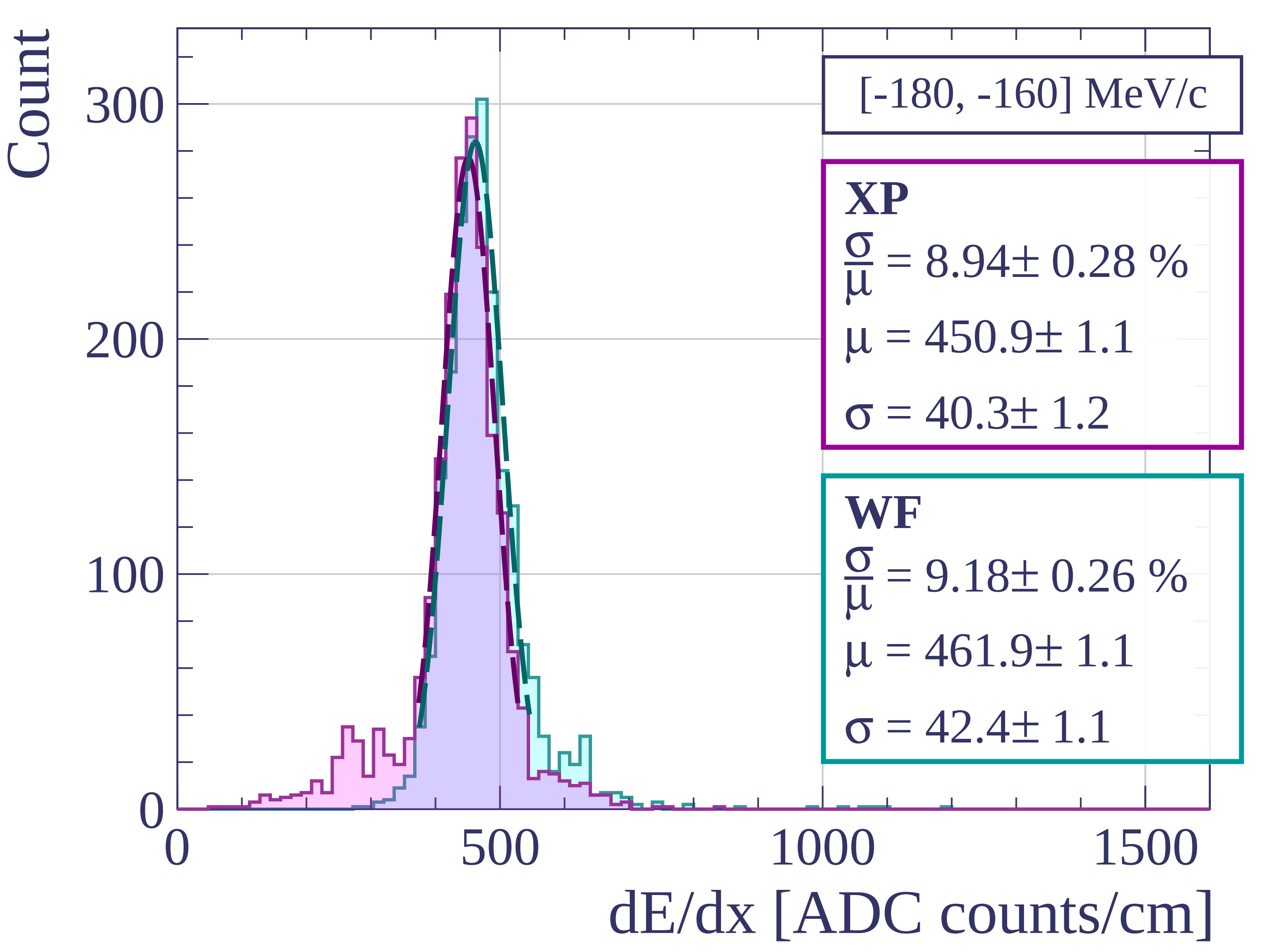
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-260, -240] MeV/c
XP
σ/μ = 8.31± 0.20 %
μ = 405.5± 0.8
σ = 33.7± 0.7
WF
σ/μ = 8.39± 0.18 %
μ = 416.9± 0.8
σ = 35.0± 0.7

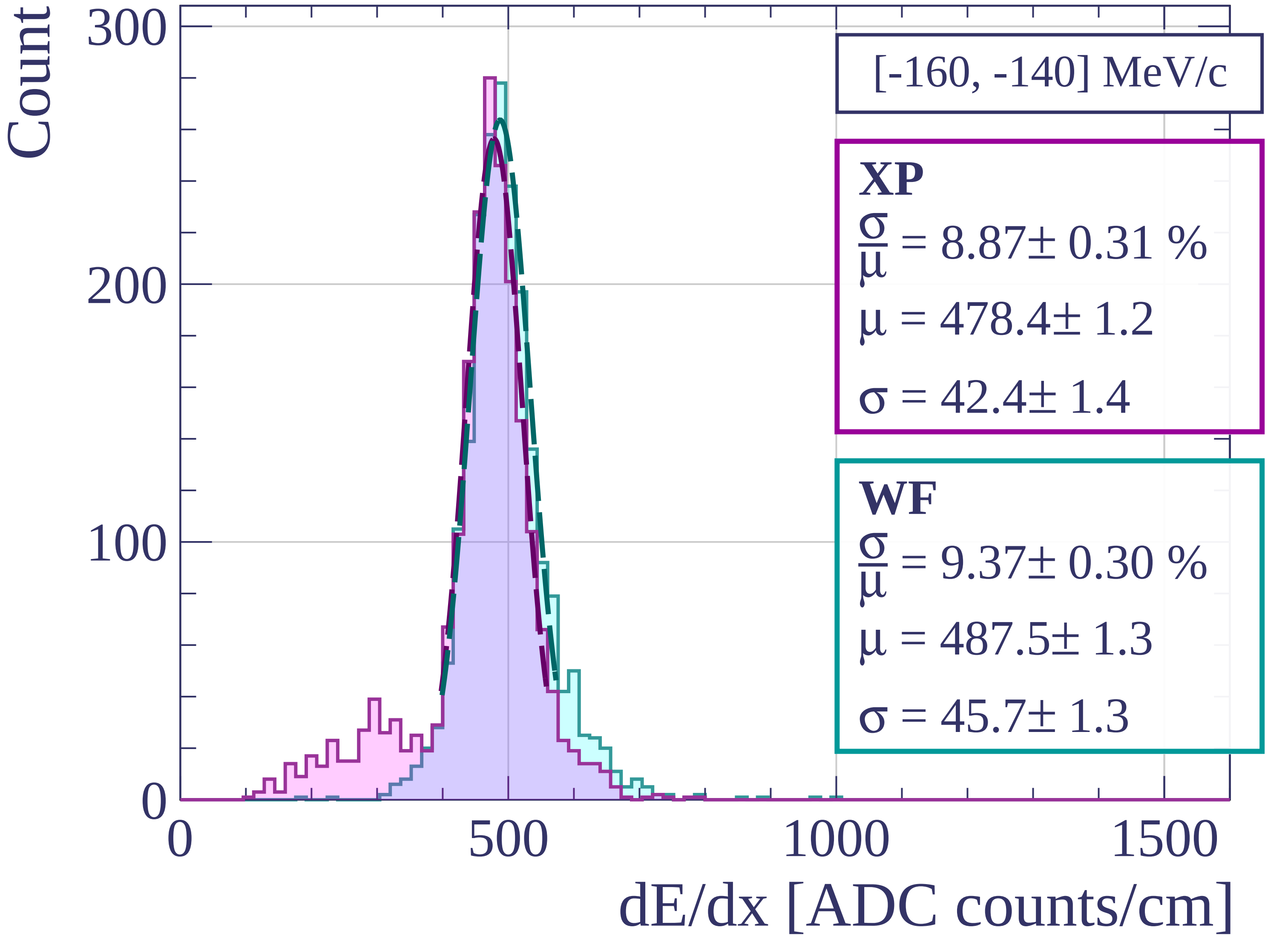


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-240, -220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.44± 0.22 %
μ = 411.2± 0.9
σ = 34.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.40± 0.22 %
μ = 421.3± 0.9
σ = 35.4± 0.8

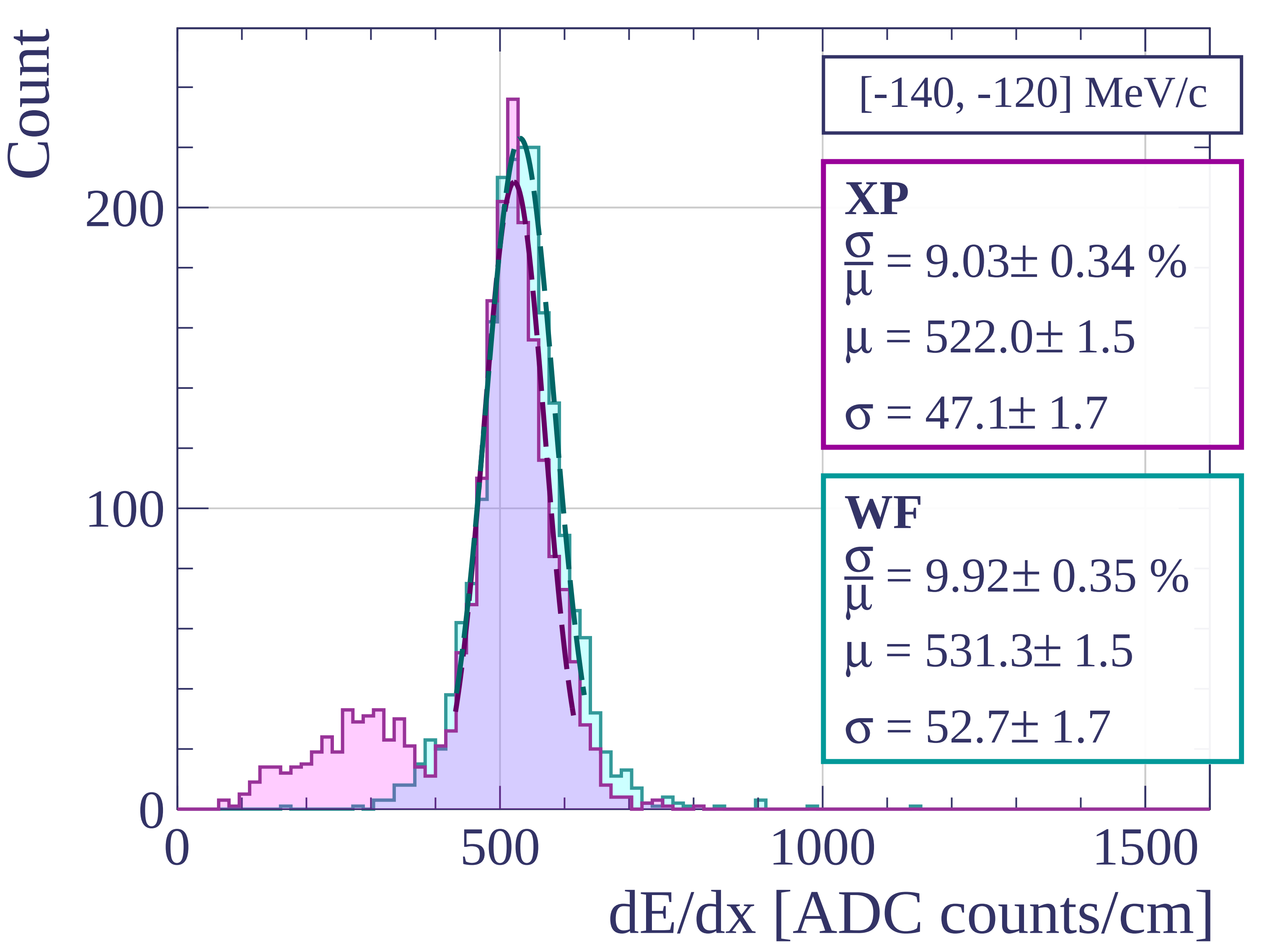

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-220, -200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.65± 0.24 %
μ = 422.3± 0.9
σ = 36.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.80± 0.23 %
μ = 433.6± 1.0
σ = 38.1± 0.9

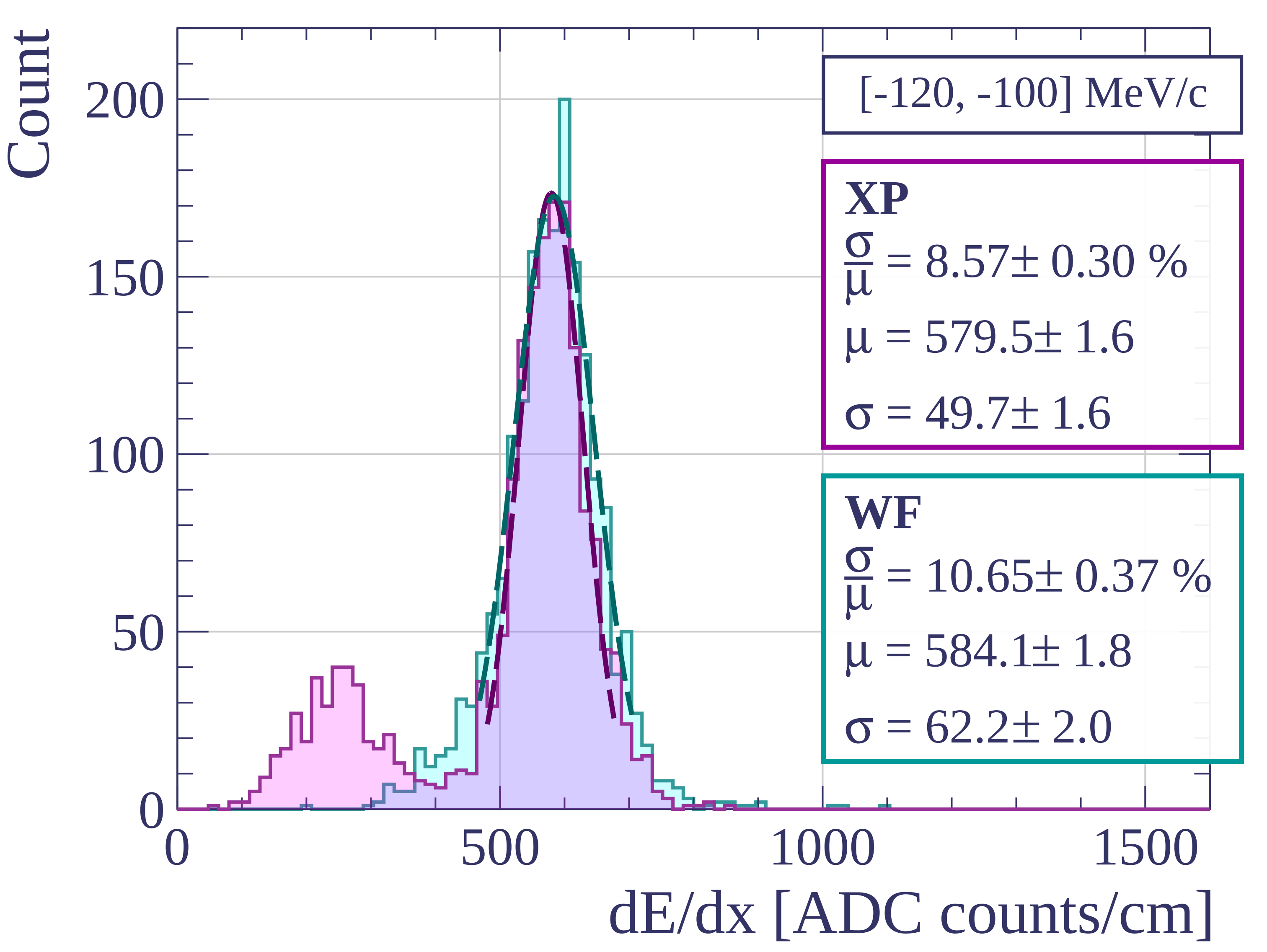

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-200, -180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.22± 0.23 %
μ = 435.2± 1.0
σ = 35.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.79± 0.24 %
μ = 445.3± 1.0
σ = 39.1± 1.0


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-180, -160] MeV/c
XP
σ/μ = 8.94± 0.28 %
μ = 450.9± 1.1
σ = 40.3± 1.2
WF
σ/μ = 9.18± 0.26 %
μ = 461.9± 1.1
σ = 42.4± 1.1

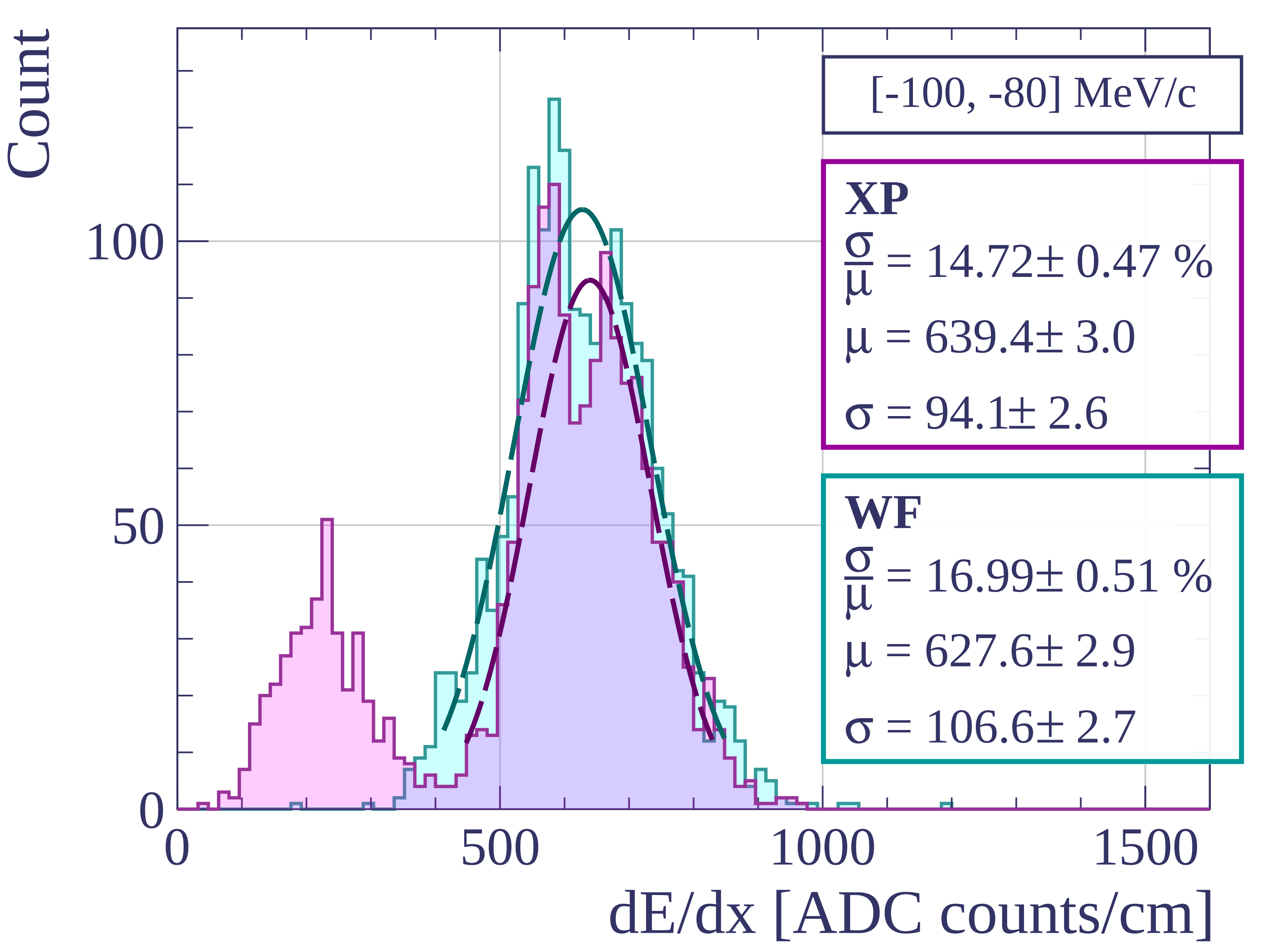


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-160, -140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.87± 0.31 %
μ = 478.4± 1.2
σ = 42.4± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.37± 0.30 %
μ = 487.5± 1.3
σ = 45.7± 1.3

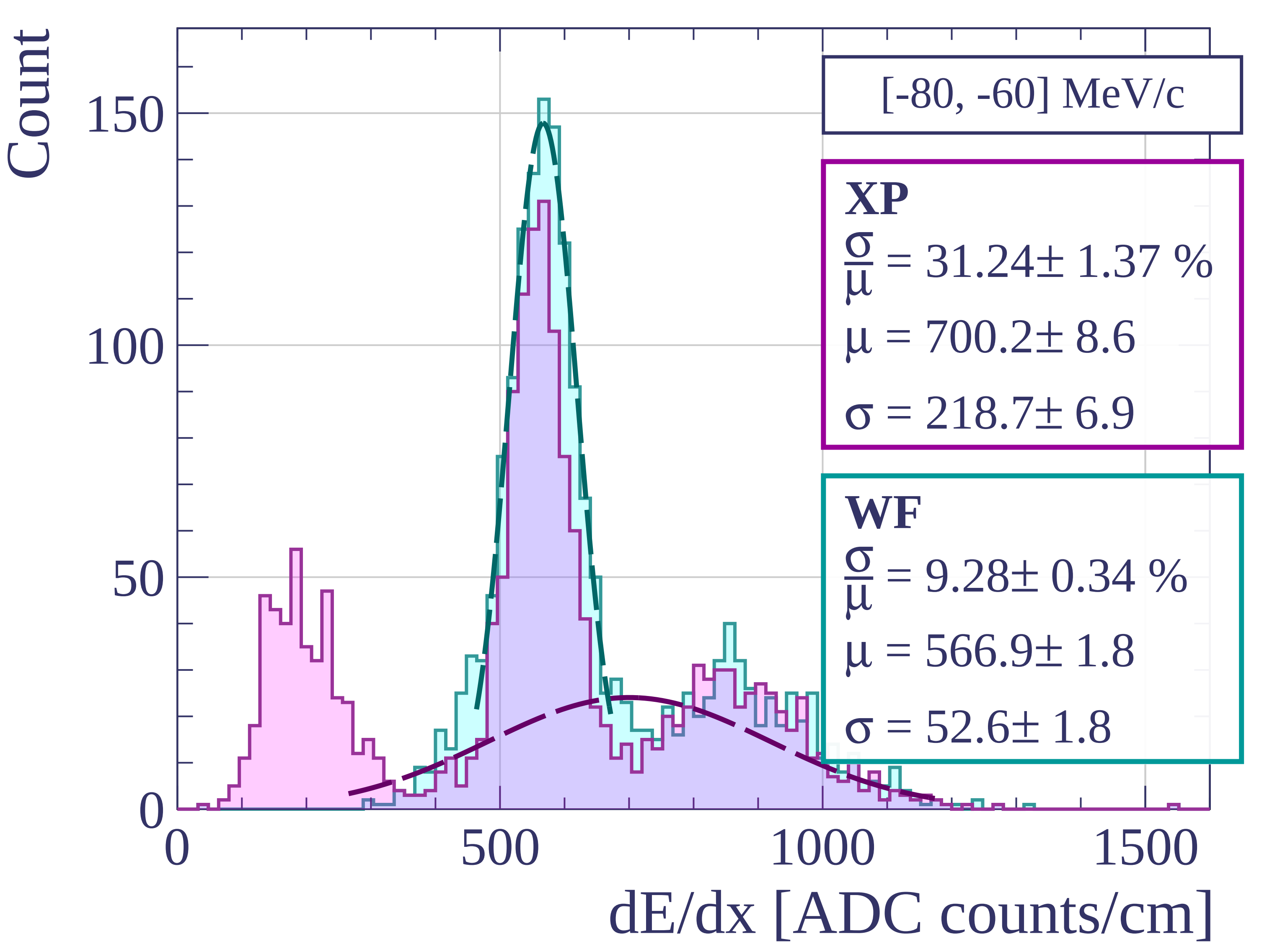

Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-140, -120] MeV/c
XP
σ/μ = 9.03± 0.34 %
μ = 522.0± 1.5
σ = 47.1± 1.7
WF
σ/μ = 9.92± 0.35 %
μ = 531.3± 1.5
σ = 52.7± 1.7



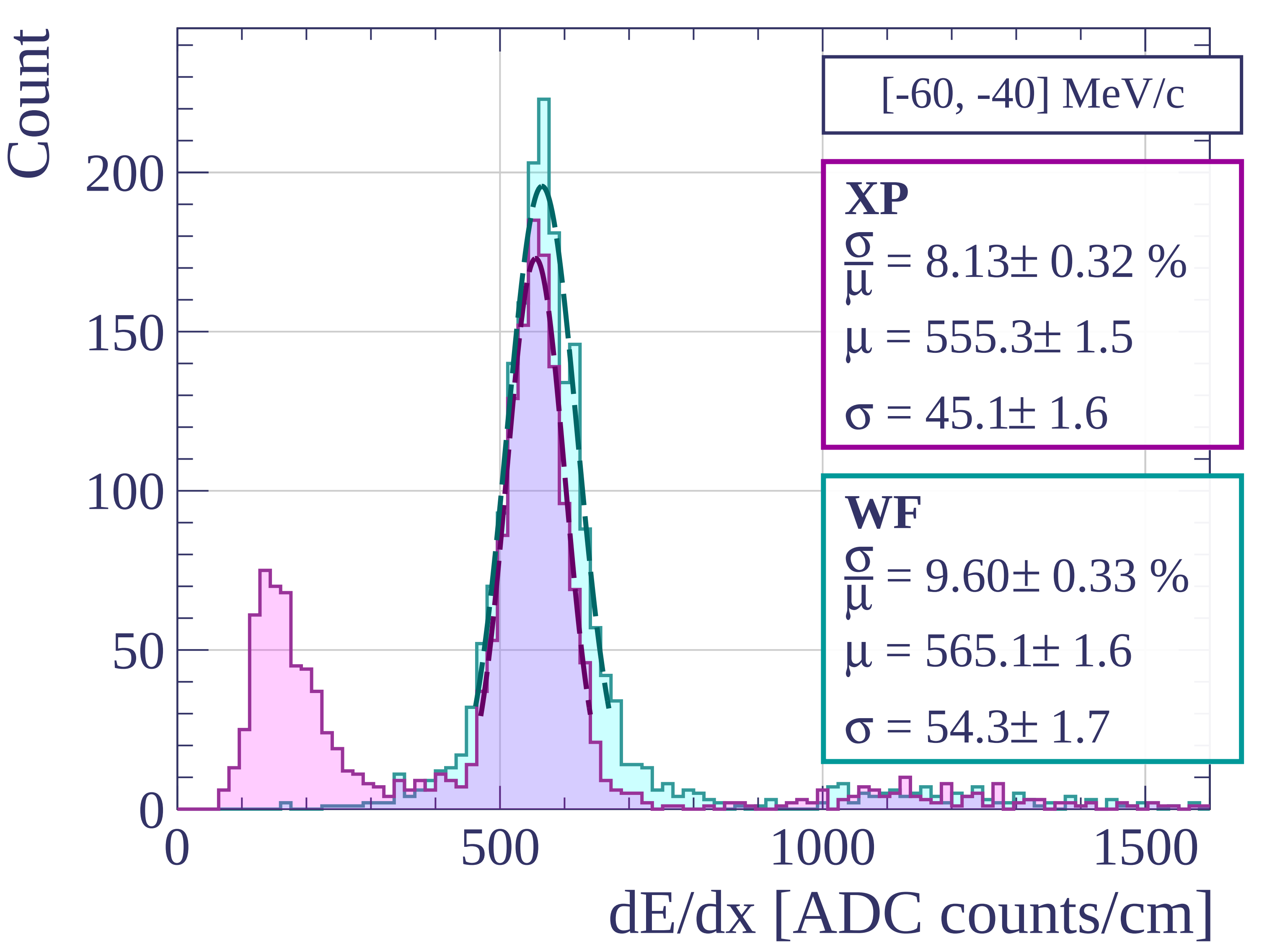
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-120, -100] MeV/c
XP
σ/μ = 8.57± 0.30 %
μ = 579.5± 1.6
σ = 49.7± 1.6
WF
σ/μ = 10.65± 0.37 %
μ = 584.1± 1.8
σ = 62.2± 2.0



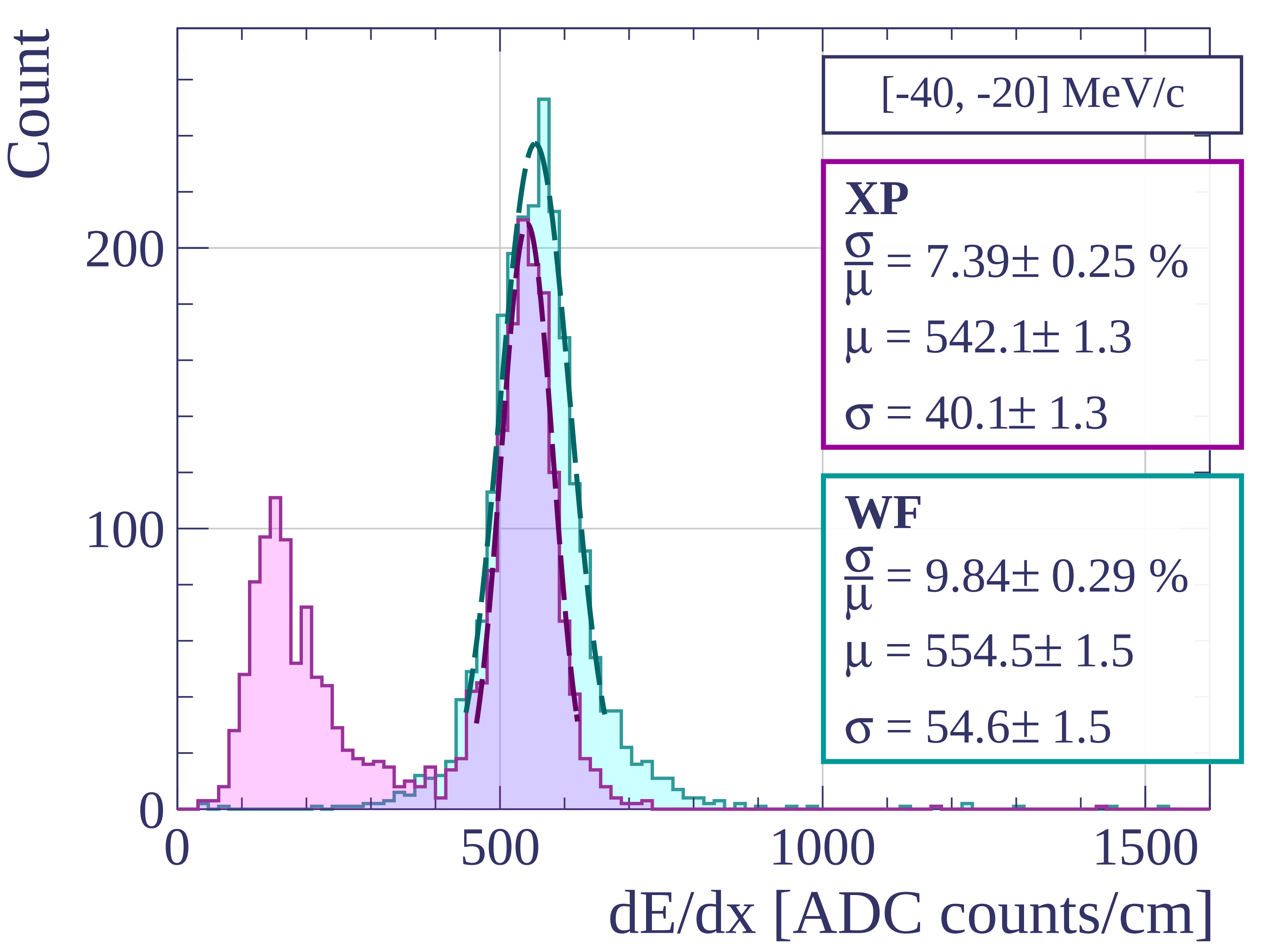
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[-100, -80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.72± 0.47 %
μ = 639.4± 3.0
σ = 94.1± 2.6
WF
$\frac{\sigma}{\mu}$ = 16.99± 0.51 %
μ = 627.6± 2.9
σ = 106.6± 2.7



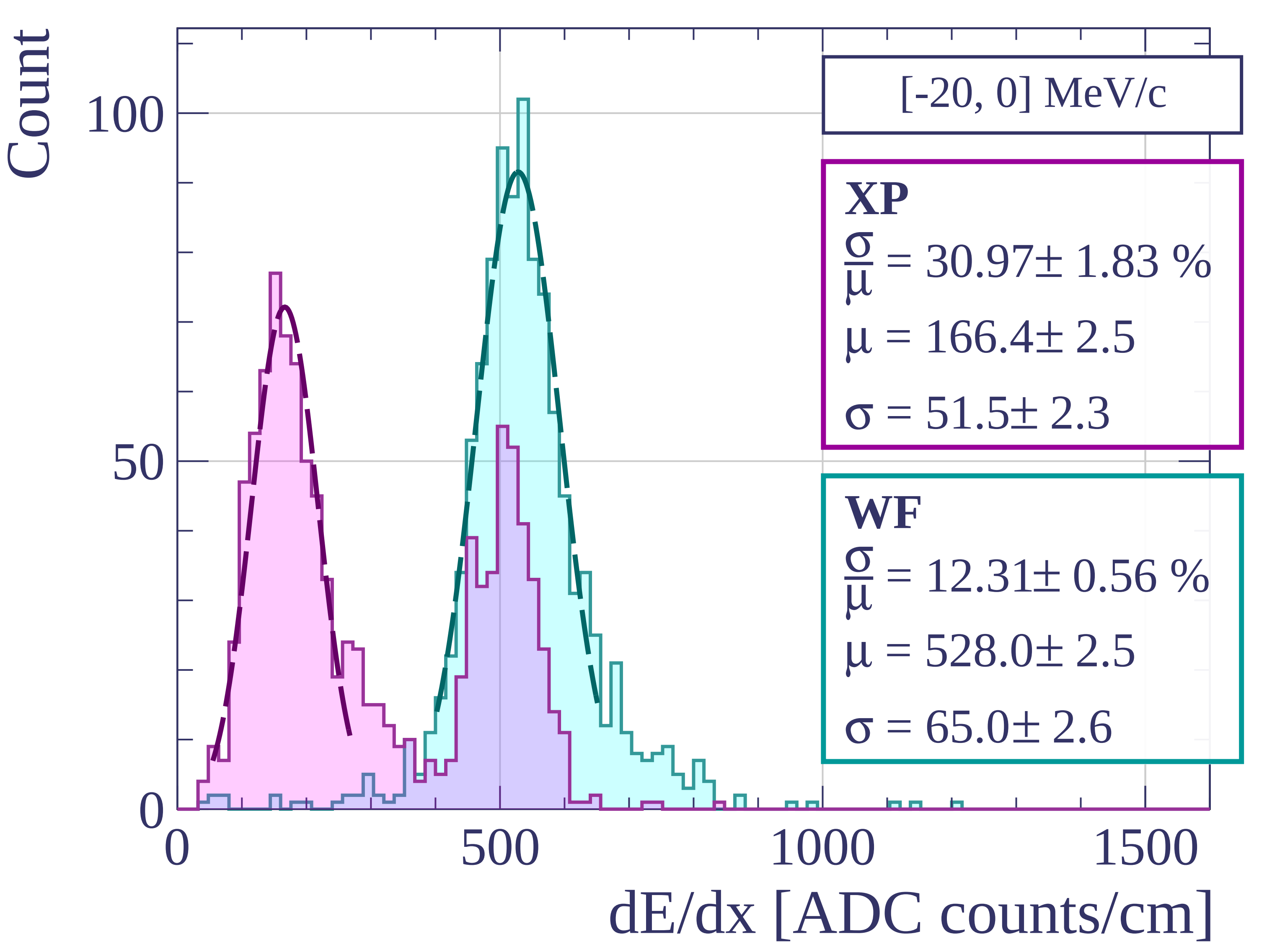
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-80, -60] MeV/c
XP
σ/μ = 31.24± 1.37 %
μ = 700.2± 8.6
σ = 218.7± 6.9
WF
σ/μ = 9.28± 0.34 %
μ = 566.9± 1.8
σ = 52.6± 1.8



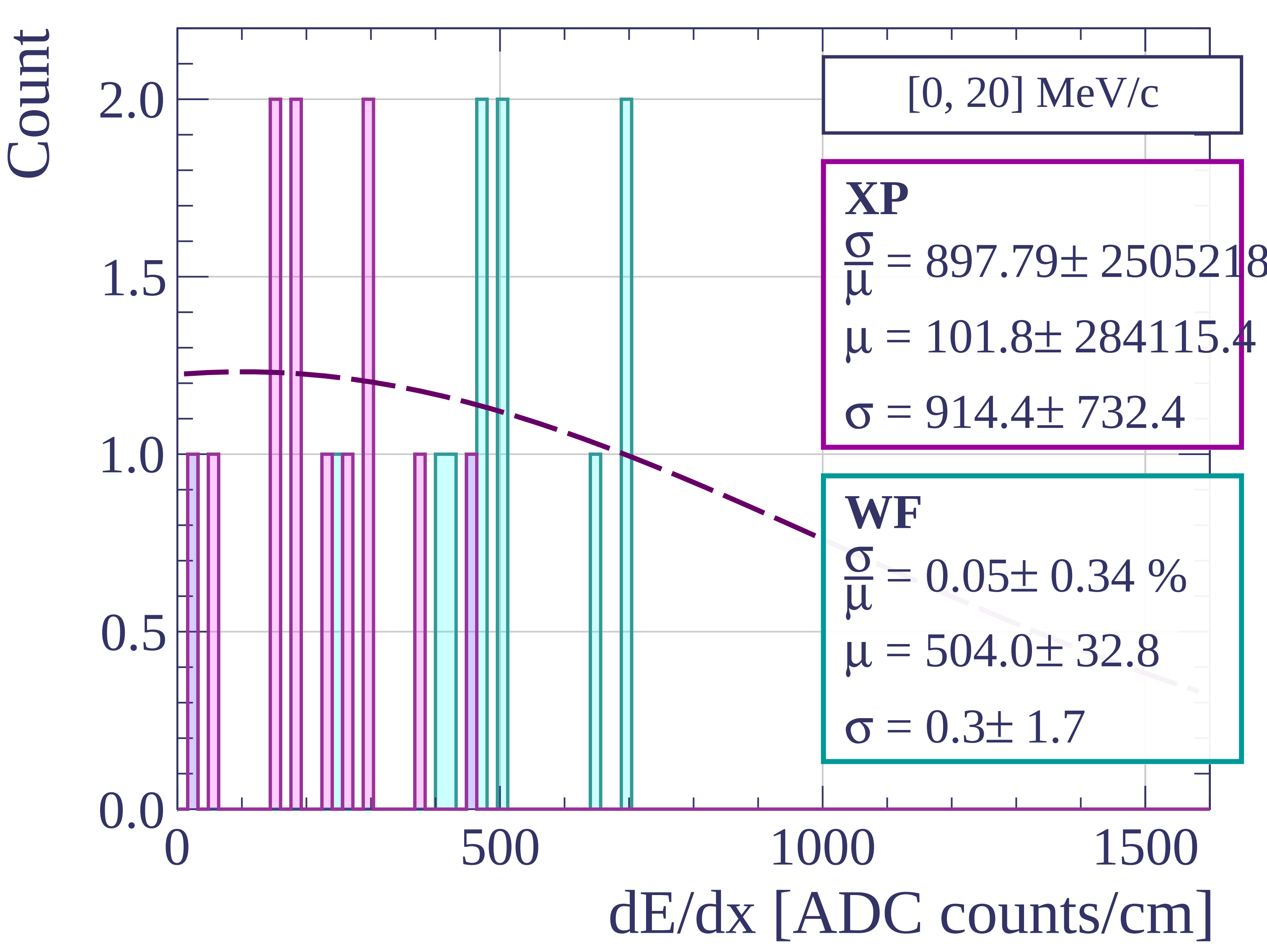
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-60, -40] MeV/c
XP
σ/μ = 8.13± 0.32 %
μ = 555.3± 1.5
σ = 45.1± 1.6
WF
σ/μ = 9.60± 0.33 %
μ = 565.1± 1.6
σ = 54.3± 1.7



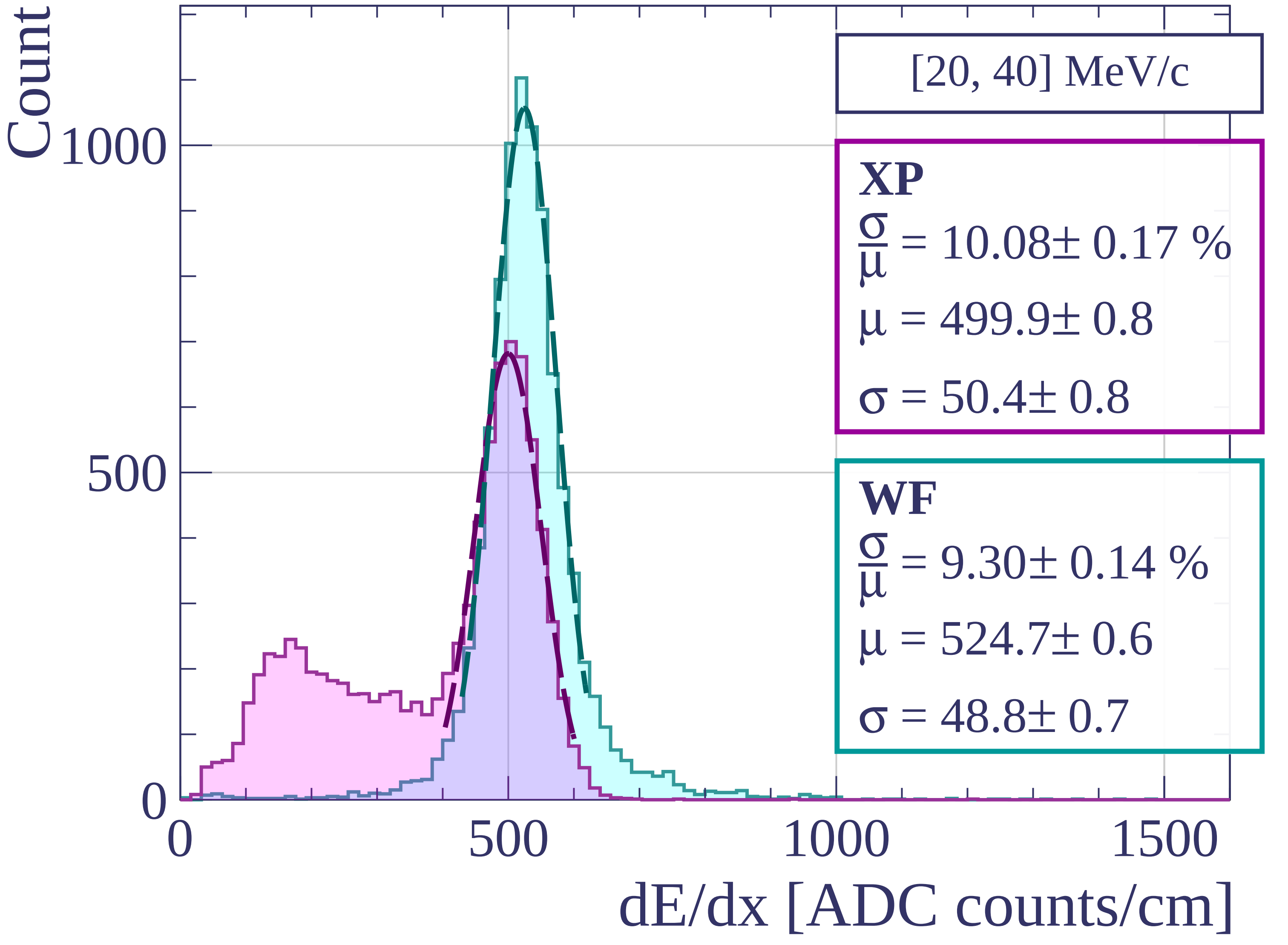
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
[-40, -20] MeV/c
XP
σ/μ = 7.39± 0.25 %
μ = 542.1± 1.3
σ = 40.1± 1.3
WF
σ/μ = 9.84± 0.29 %
μ = 554.5± 1.5
σ = 54.6± 1.5



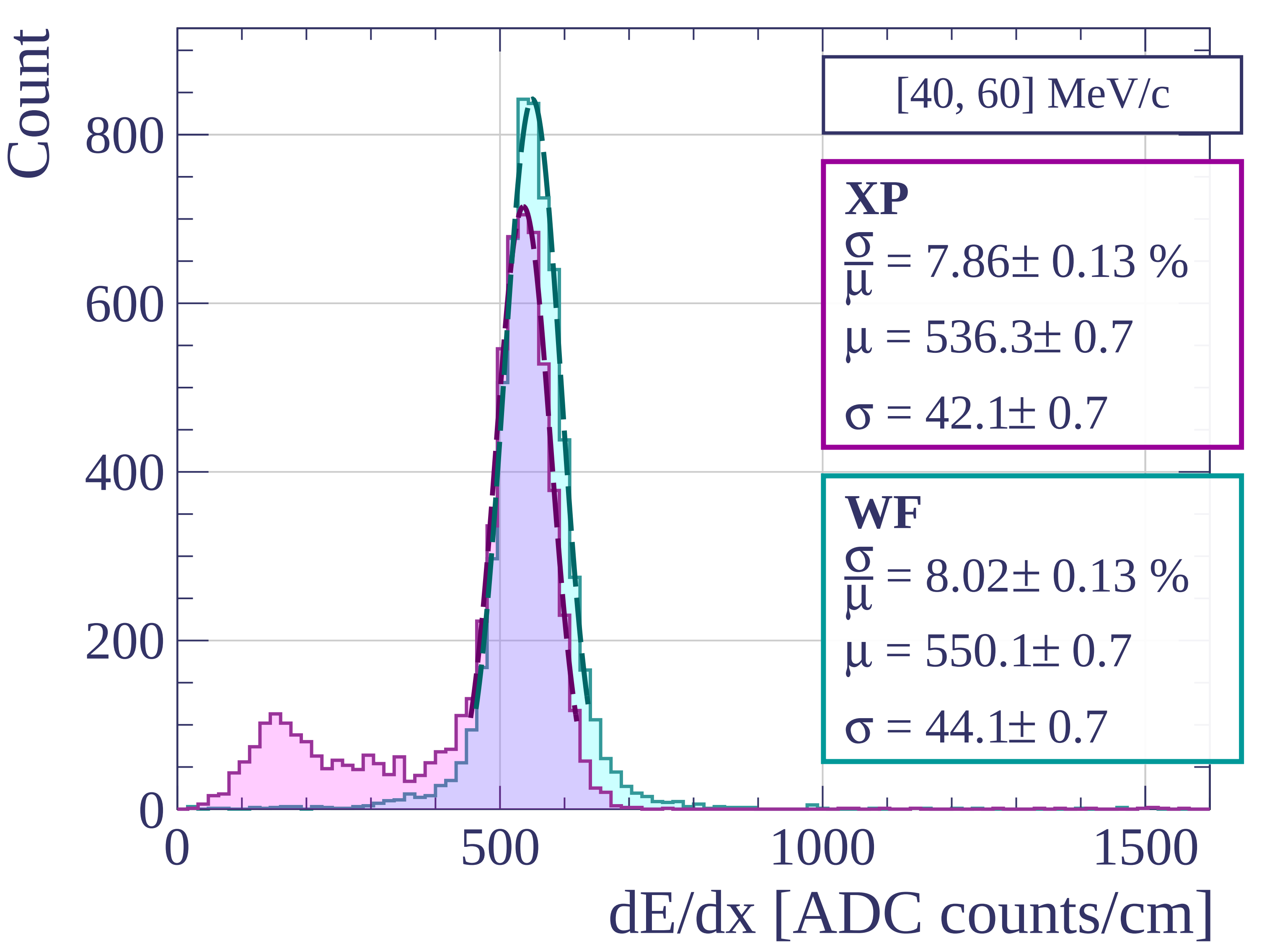
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-20, 0] MeV/c
XP
σ/μ = 30.97± 1.83 %
μ = 166.4± 2.5
σ = 51.5± 2.3
WF
σ/μ = 12.31± 0.56 %
μ = 528.0± 2.5
σ = 65.0± 2.6



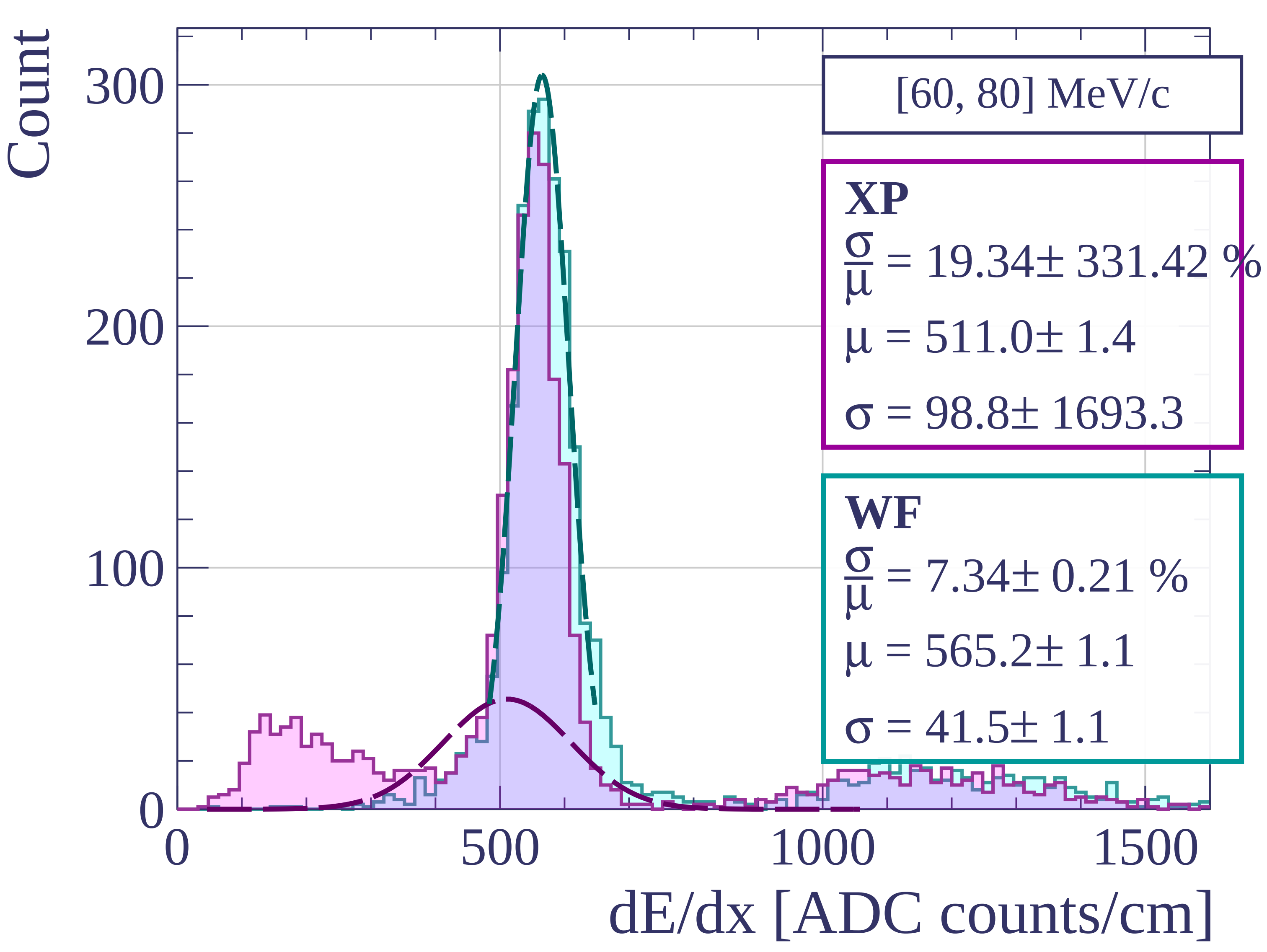
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 20] MeV/c
XP
σ/μ = 897.79± 2505218
μ = 101.8± 284115.4
σ = 914.4± 732.4
WF
σ/μ = 0.05± 0.34 %
μ = 504.0± 32.8
σ = 0.3± 1.7

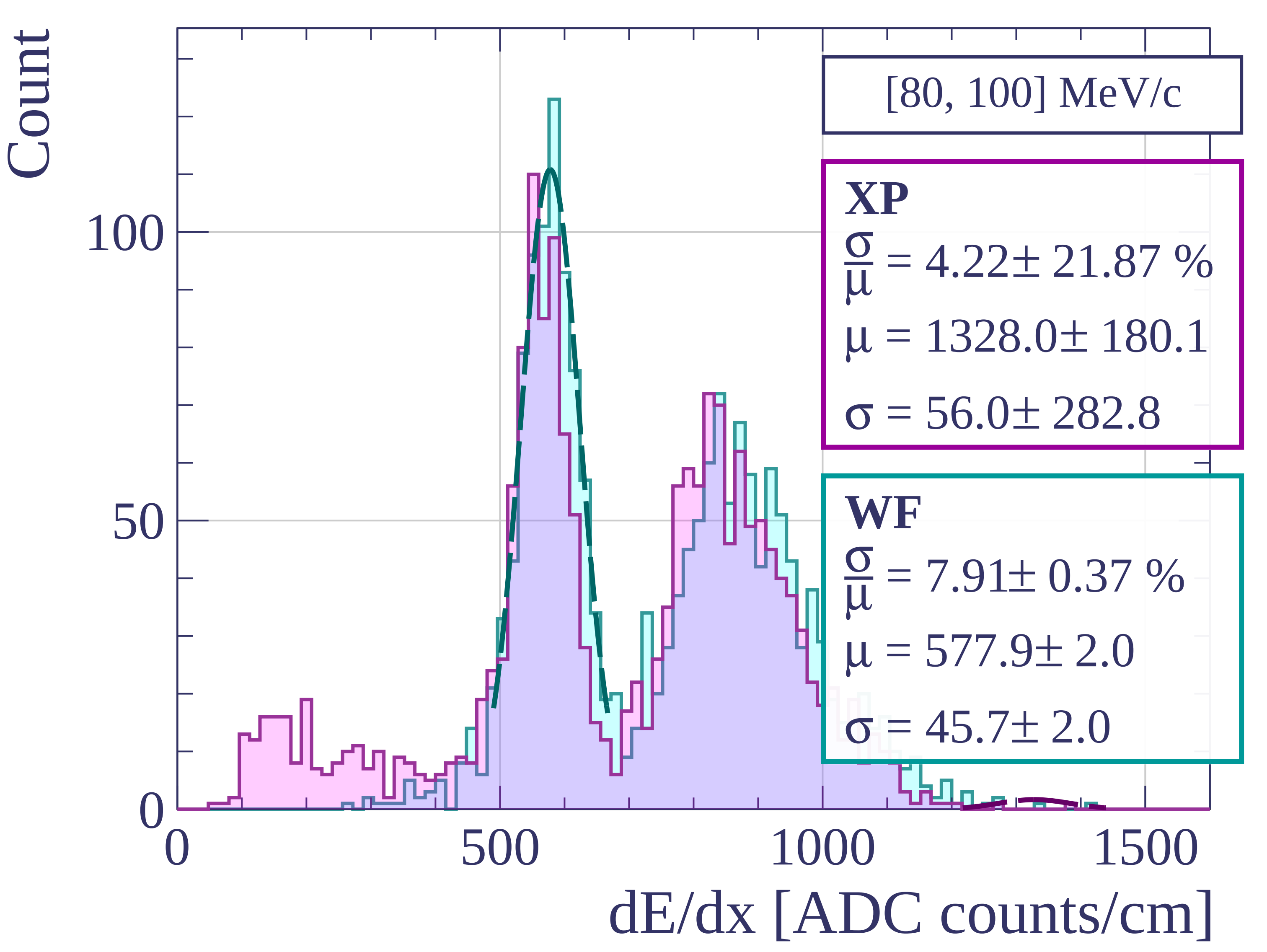


Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[20, 40] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.08± 0.17 %
μ = 499.9± 0.8
σ = 50.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.30± 0.14 %
μ = 524.7± 0.6
σ = 48.8± 0.7

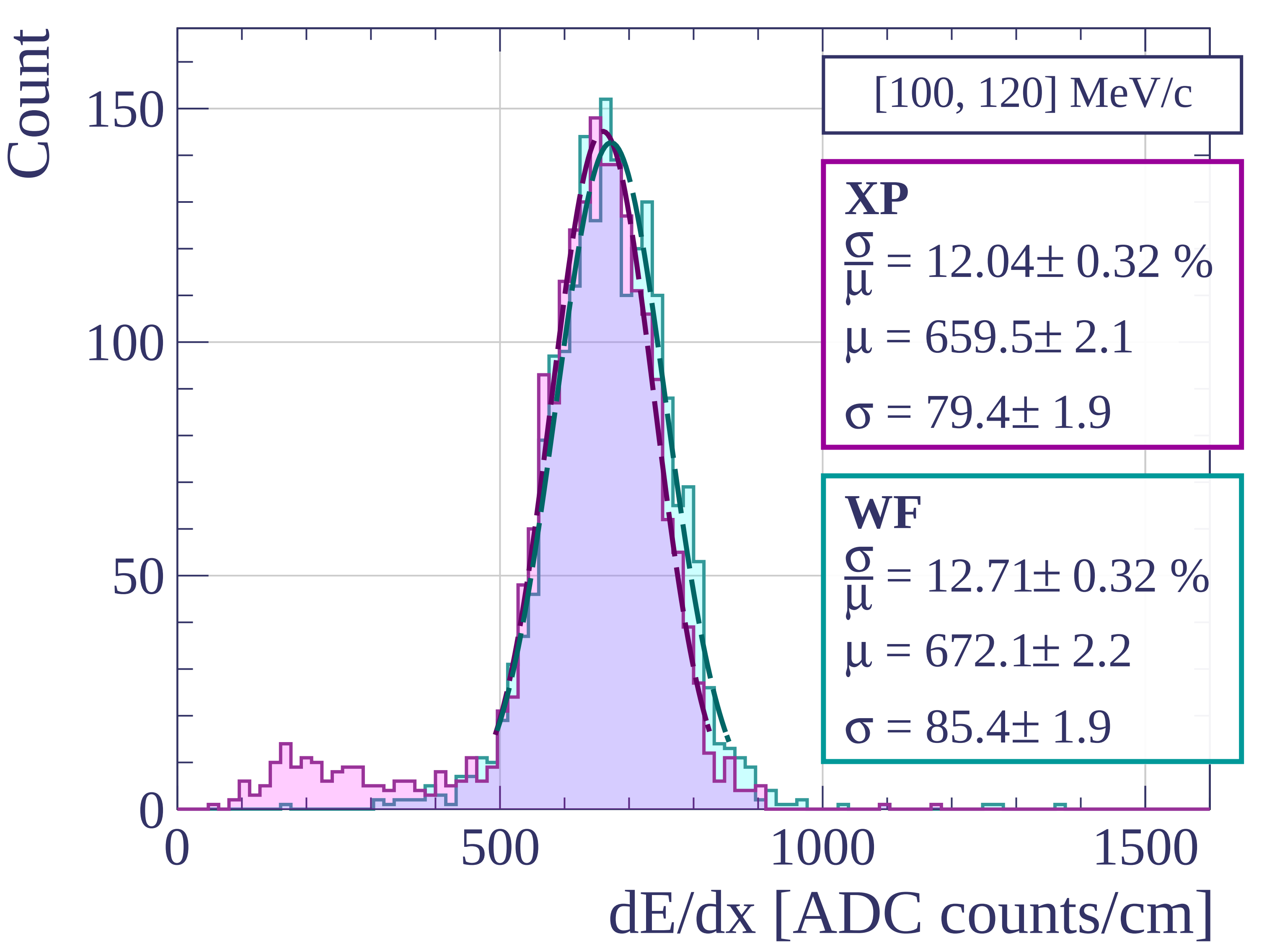


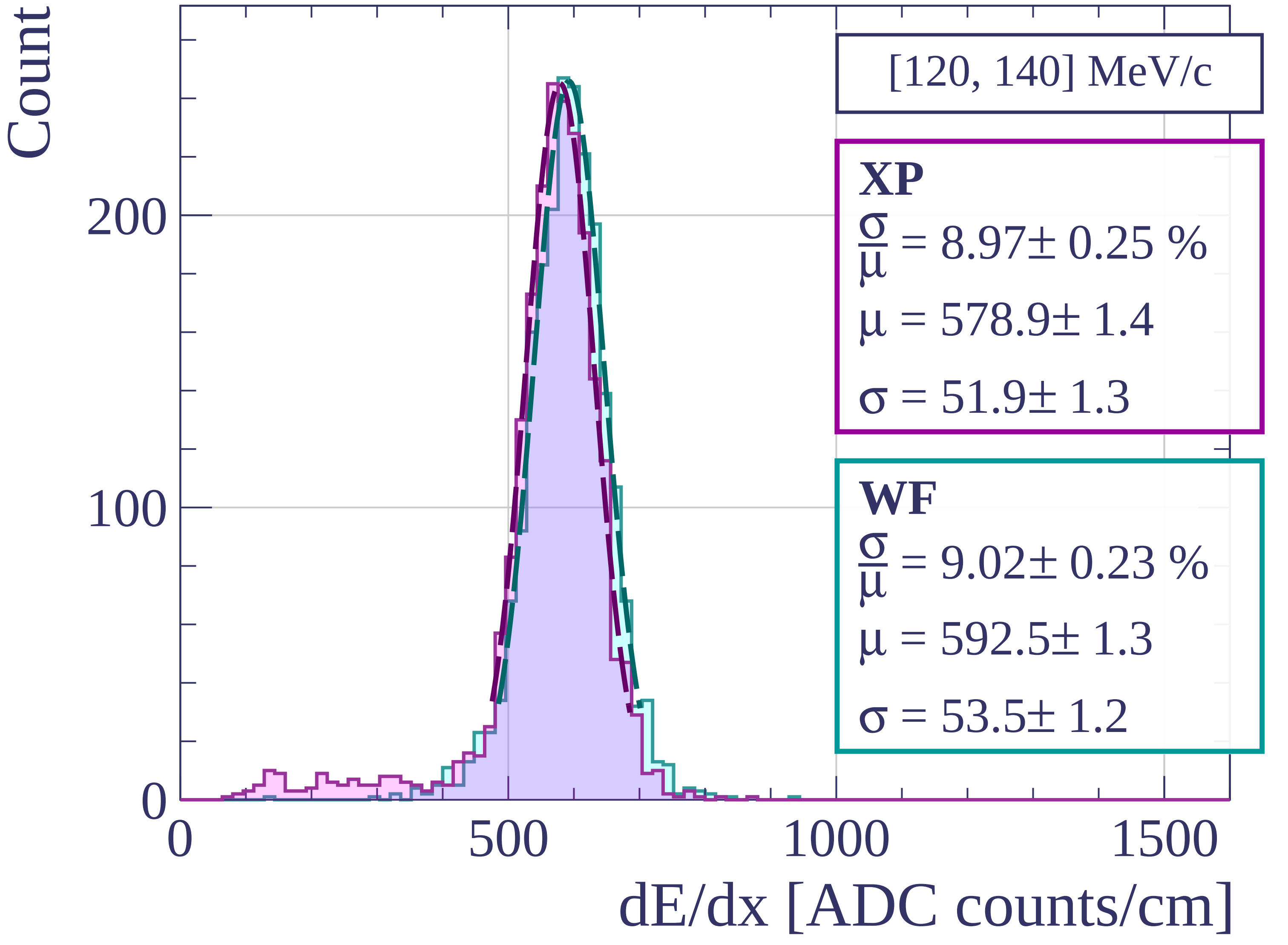
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[40, 60] MeV/c
XP
σ/μ = 7.86± 0.13 %
μ = 536.3± 0.7
σ = 42.1± 0.7
WF
σ/μ = 8.02± 0.13 %
μ = 550.1± 0.7
σ = 44.1± 0.7

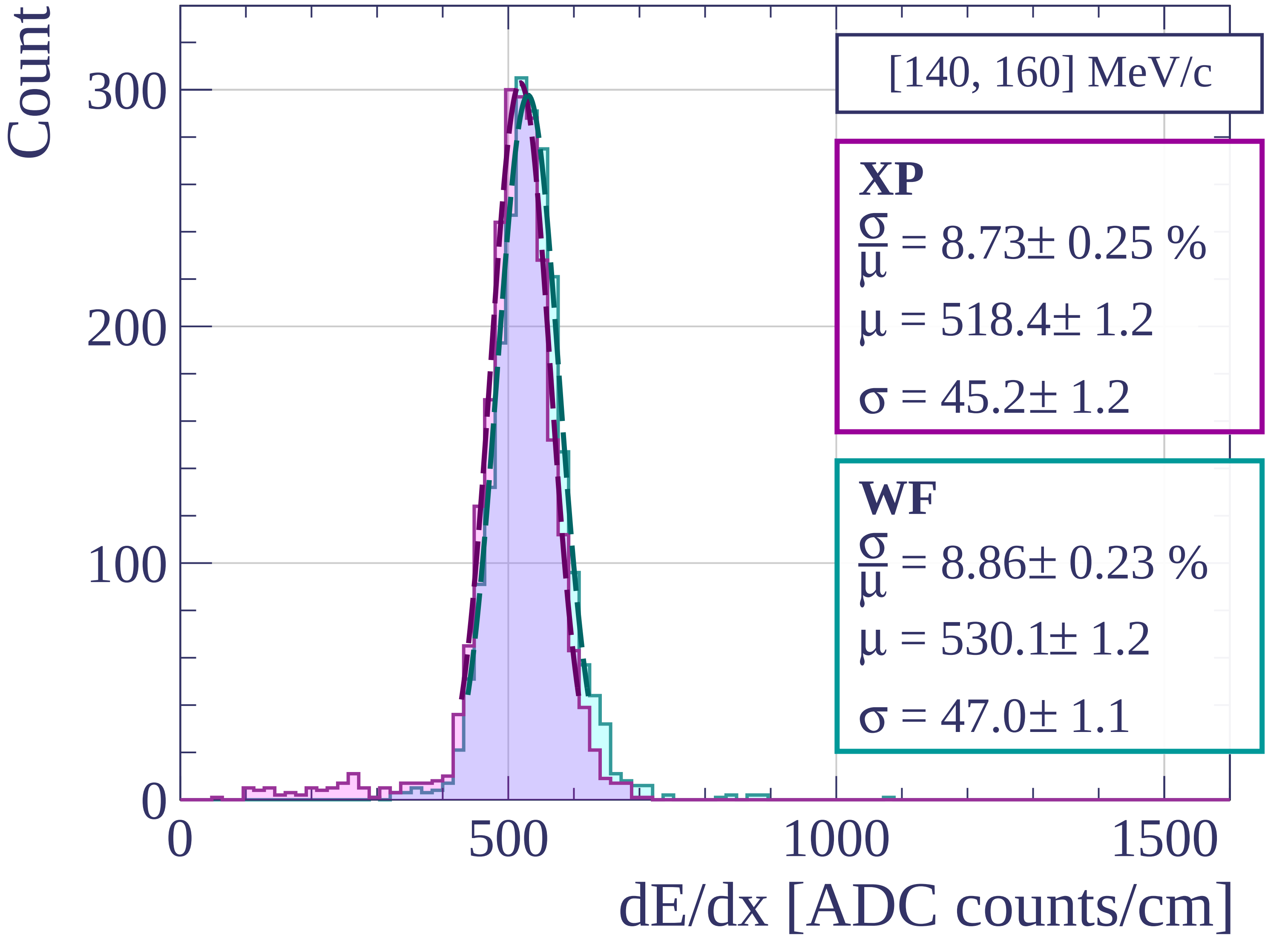

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[60, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 19.34± 331.42 %
μ = 511.0± 1.4
σ = 98.8± 1693.3
WF
$\frac{\sigma}{\mu}$ = 7.34± 0.21 %
μ = 565.2± 1.1
σ = 41.5± 1.1

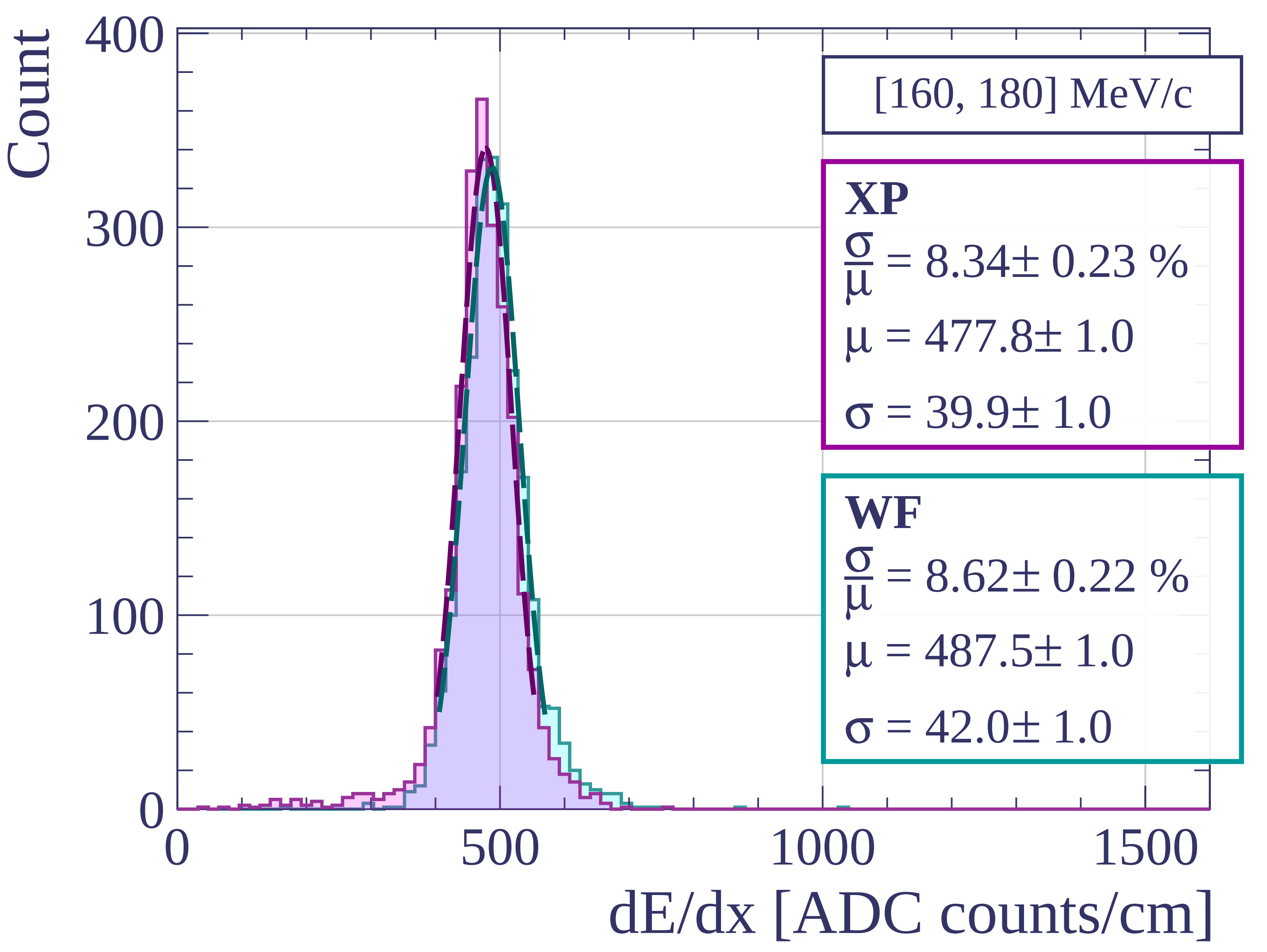


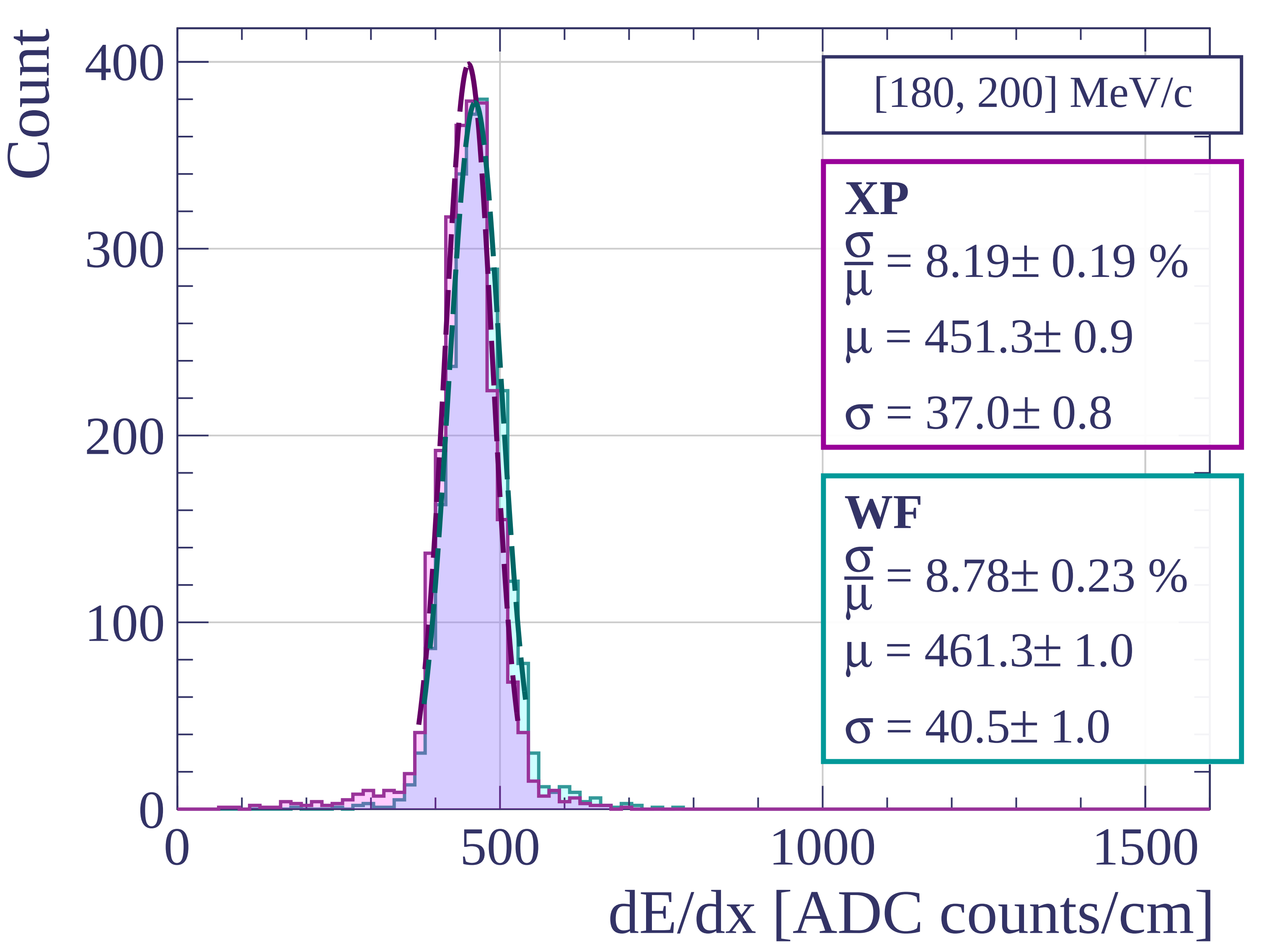
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.22± 21.87 %
μ = 1328.0± 180.1
σ = 56.0± 282.8
WF
$\frac{\sigma}{\mu}$ = 7.91± 0.37 %
μ = 577.9± 2.0
σ = 45.7± 2.0

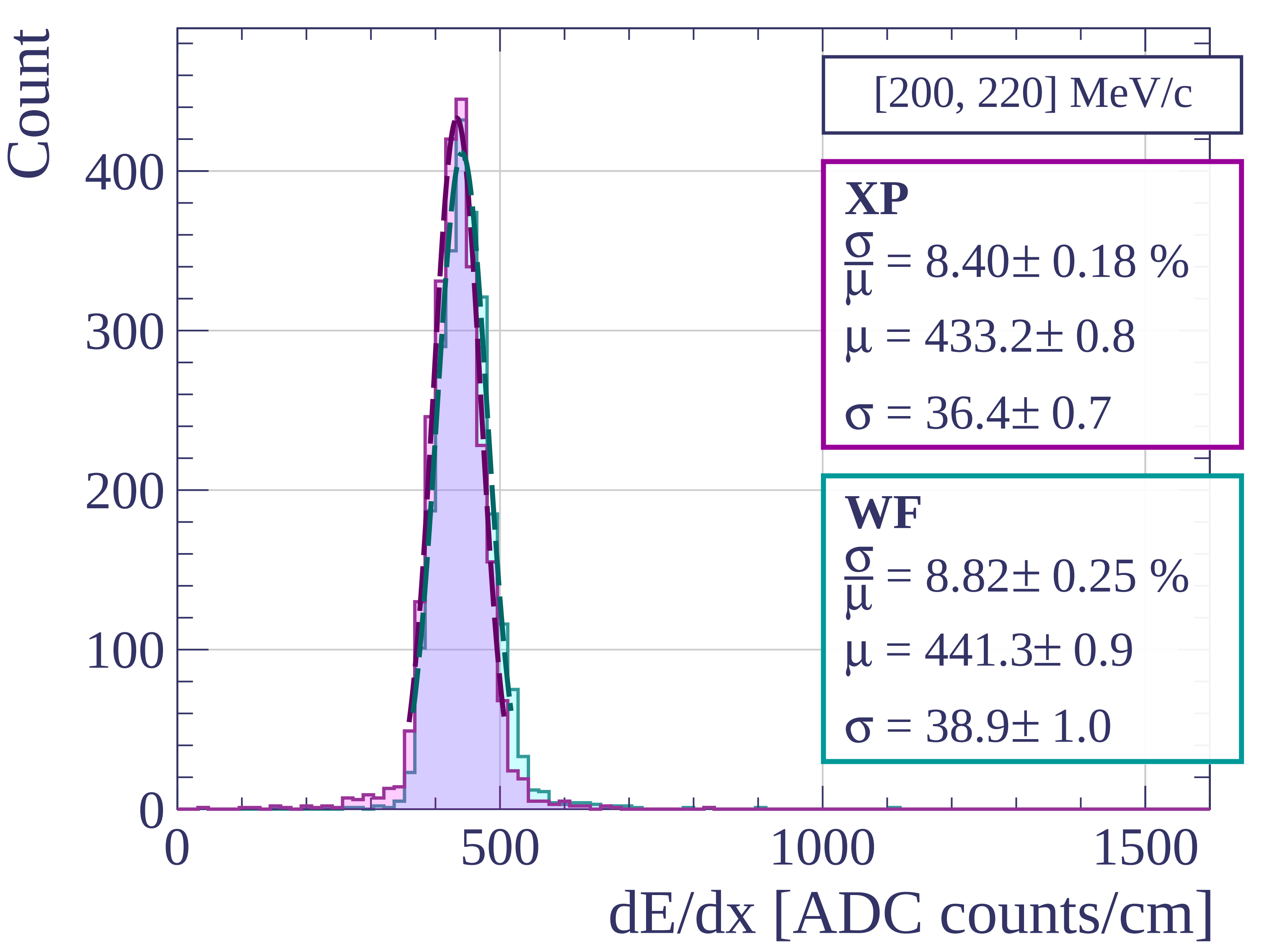

Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[100, 120] MeV/c
XP
σ/μ = 12.04± 0.32 %
μ = 659.5± 2.1
σ = 79.4± 1.9
WF
σ/μ = 12.71± 0.32 %
μ = 672.1± 2.2
σ = 85.4± 1.9


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[120, 140] MeV/c
XP
σ/μ = 8.97± 0.25 %
μ = 578.9± 1.4
σ = 51.9± 1.3
WF
σ/μ = 9.02± 0.23 %
μ = 592.5± 1.3
σ = 53.5± 1.2

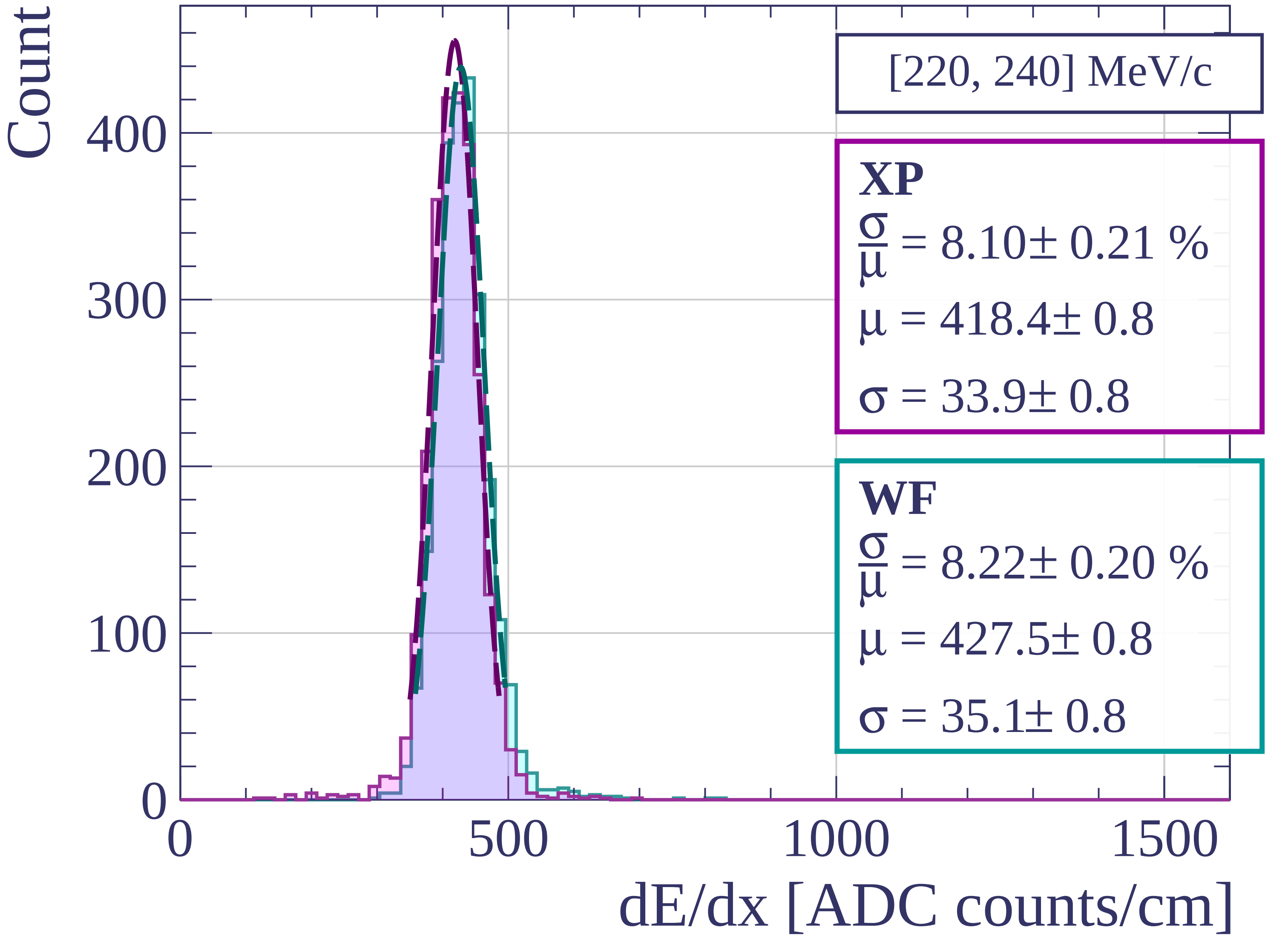

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[140, 160] MeV/c
XP
σ/μ = 8.73± 0.25 %
μ = 518.4± 1.2
σ = 45.2± 1.2
WF
σ/μ = 8.86± 0.23 %
μ = 530.1± 1.2
σ = 47.0± 1.1


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 180] MeV/c
XP
σ/μ = 8.34± 0.23 %
μ = 477.8± 1.0
σ = 39.9± 1.0
WF
σ/μ = 8.62± 0.22 %
μ = 487.5± 1.0
σ = 42.0± 1.0

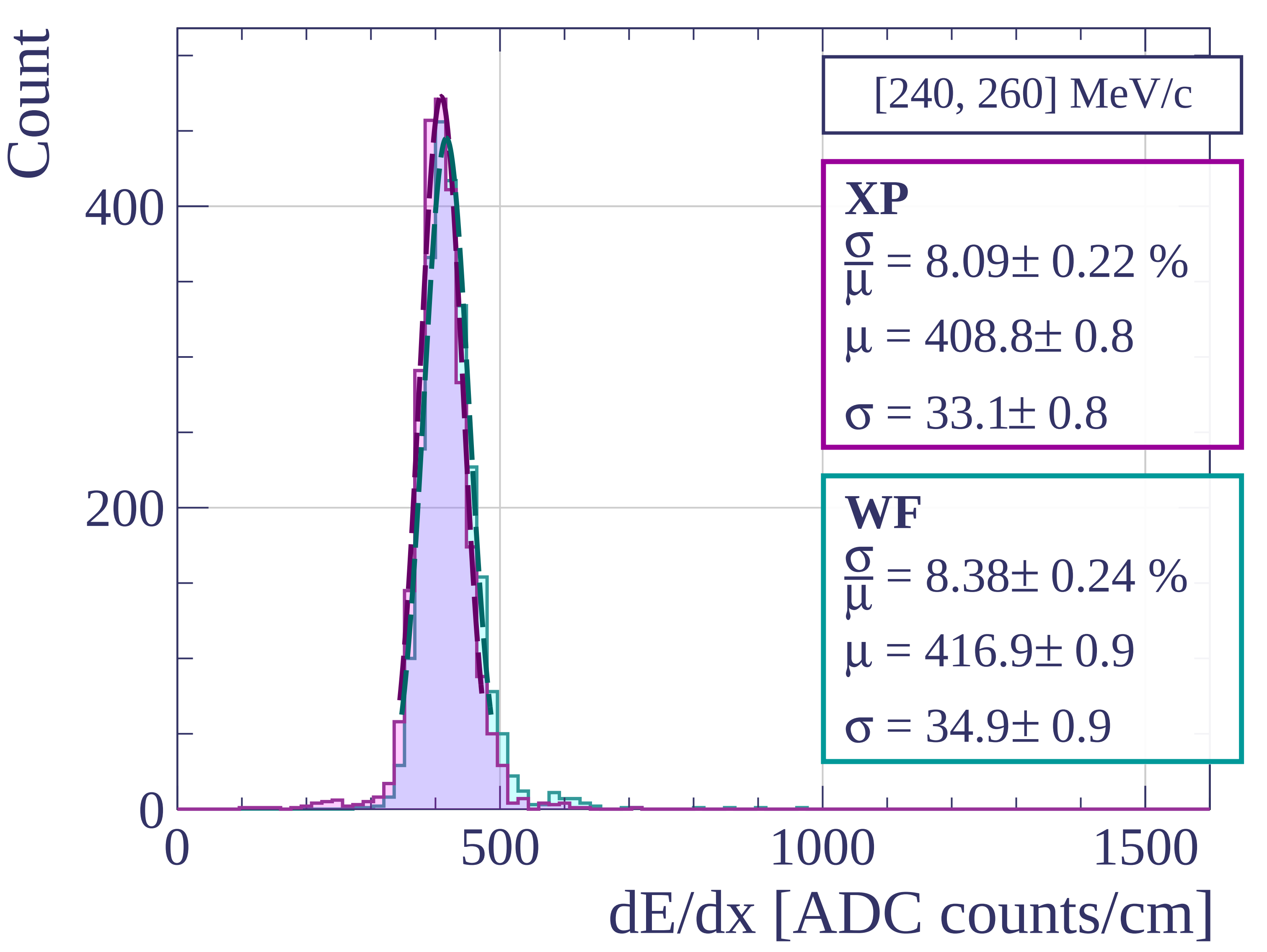

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[180, 200] MeV/c
XP
σ/μ = 8.19± 0.19 %
μ = 451.3± 0.9
σ = 37.0± 0.8
WF
σ/μ = 8.78± 0.23 %
μ = 461.3± 1.0
σ = 40.5± 1.0



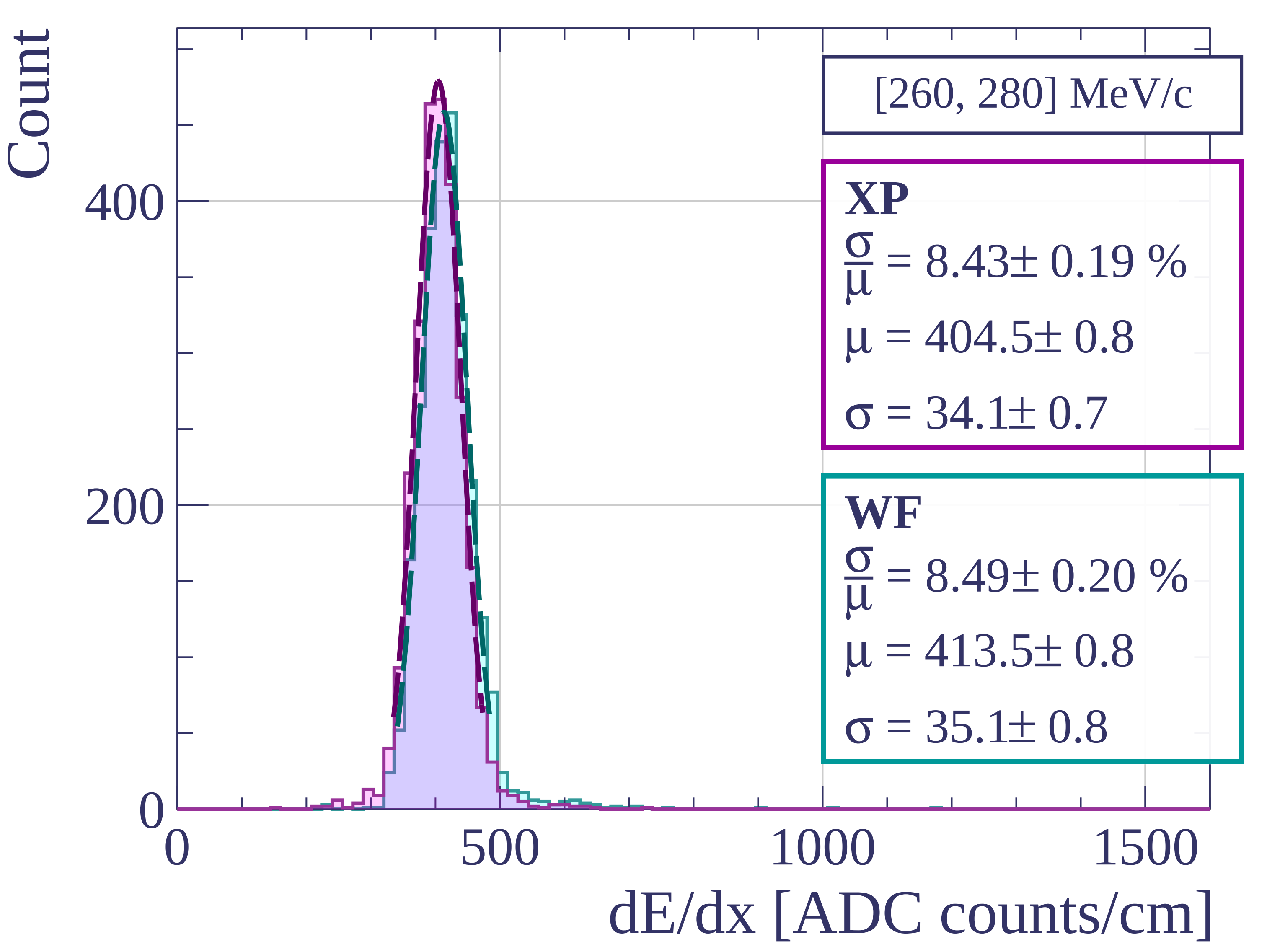
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[200, 220] MeV/c
XP
σ/μ = 8.40± 0.18 %
μ = 433.2± 0.8
σ = 36.4± 0.7
WF
σ/μ = 8.82± 0.25 %
μ = 441.3± 0.9
σ = 38.9± 1.0



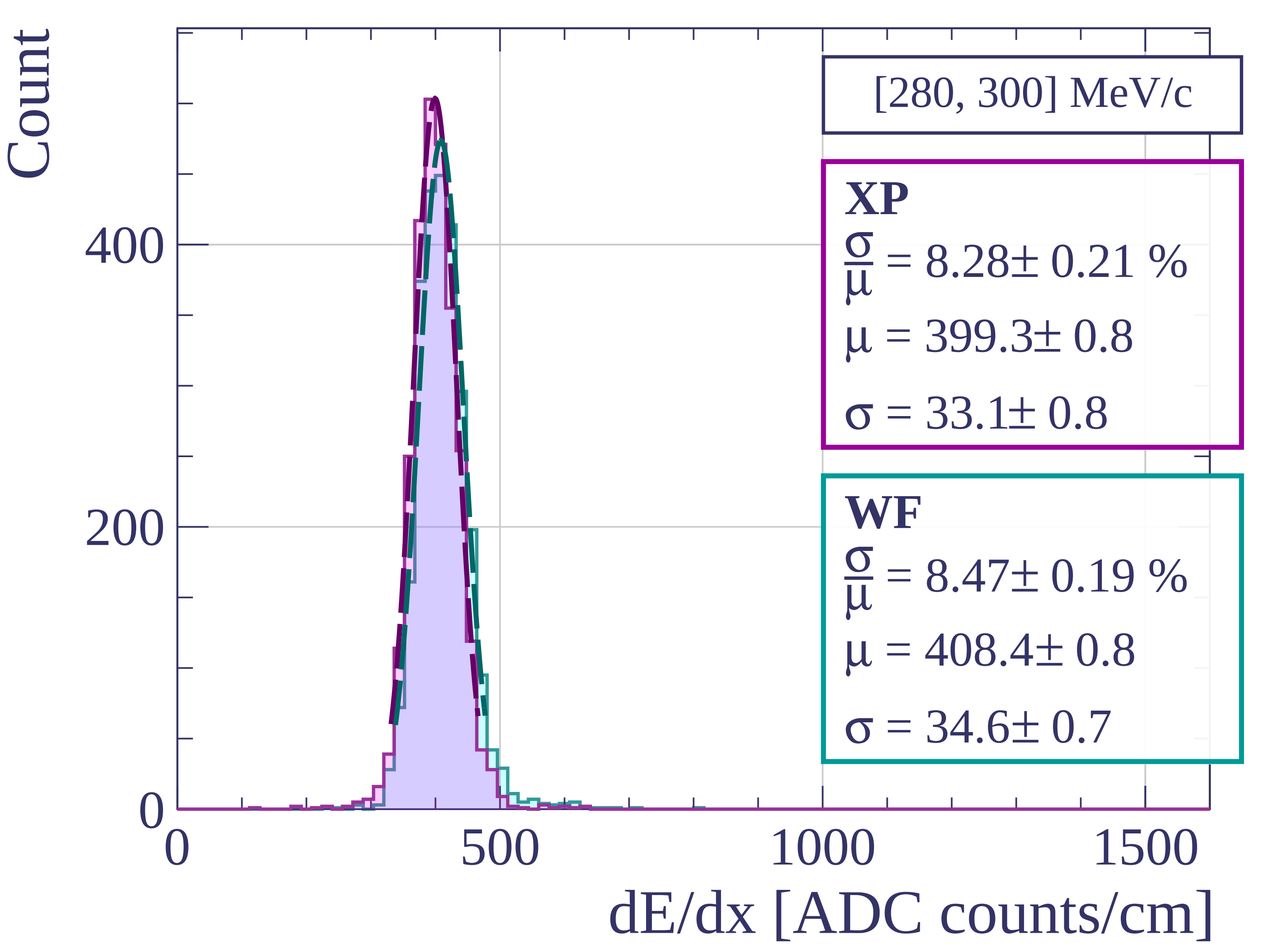
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[220, 240] MeV/c
XP
σ/μ = 8.10± 0.21 %
μ = 418.4± 0.8
σ = 33.9± 0.8
WF
σ/μ = 8.22± 0.20 %
μ = 427.5± 0.8
σ = 35.1± 0.8

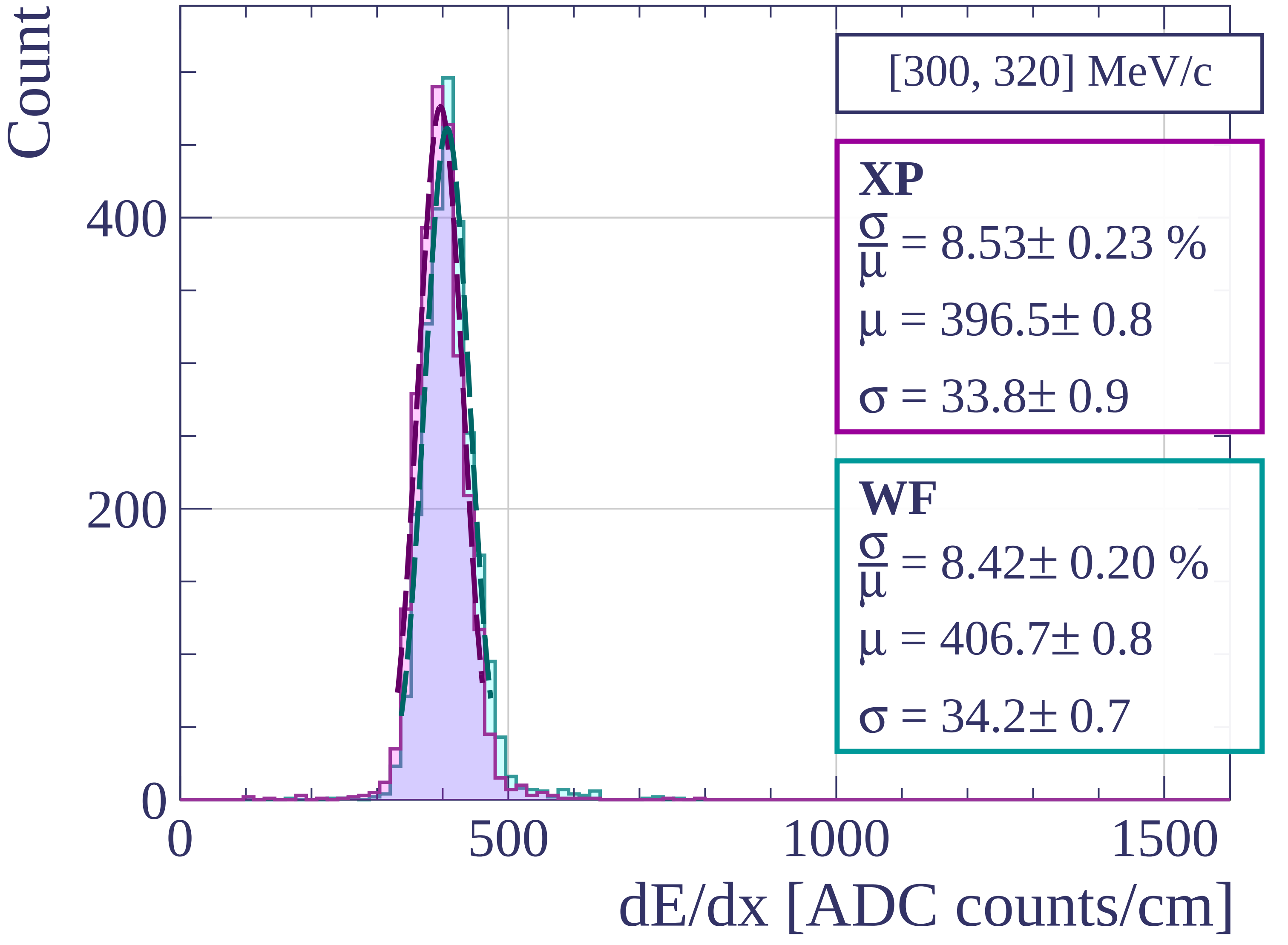


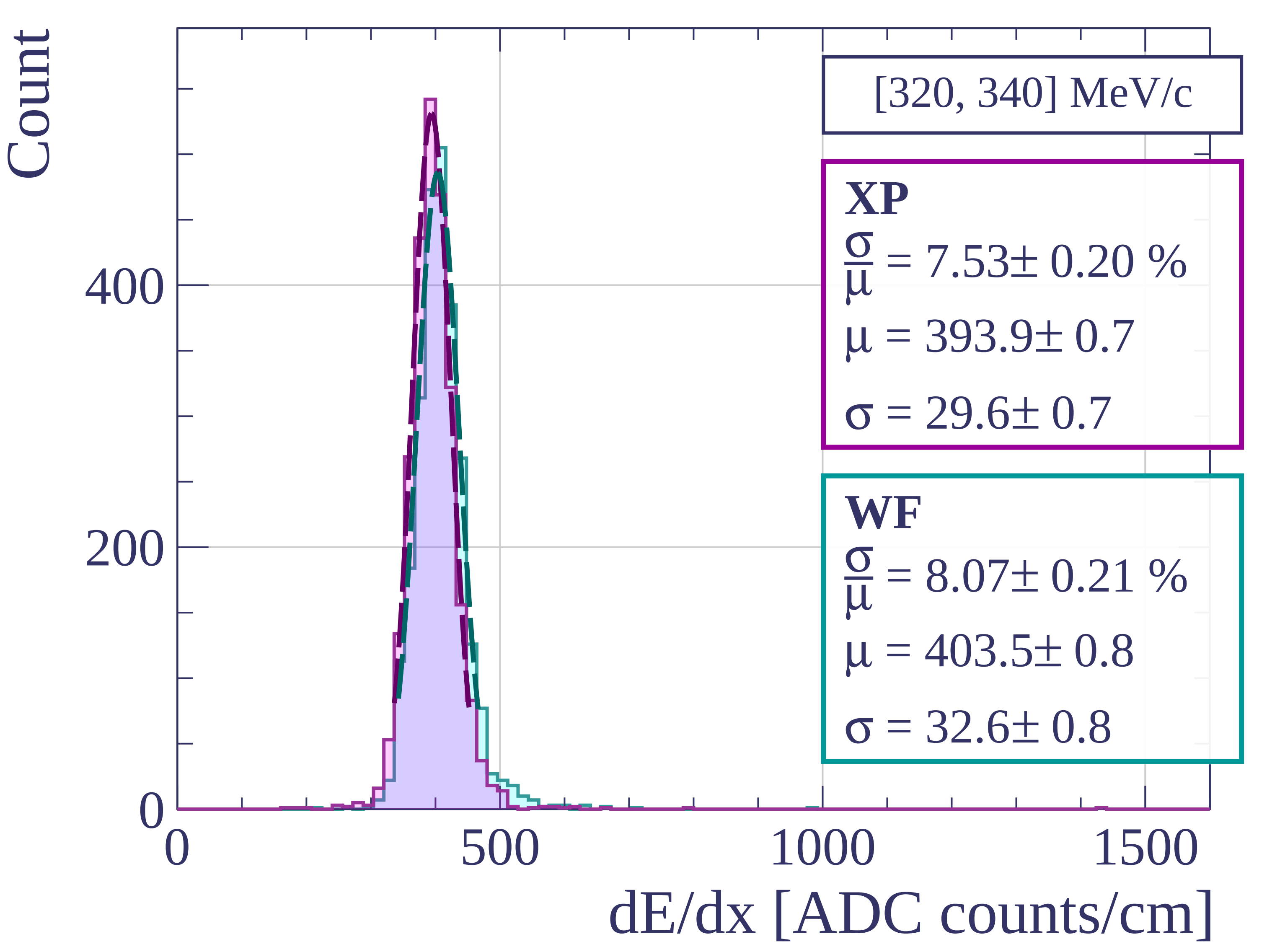
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.22 %
μ = 408.8± 0.8
σ = 33.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.38± 0.24 %
μ = 416.9± 0.9
σ = 34.9± 0.9

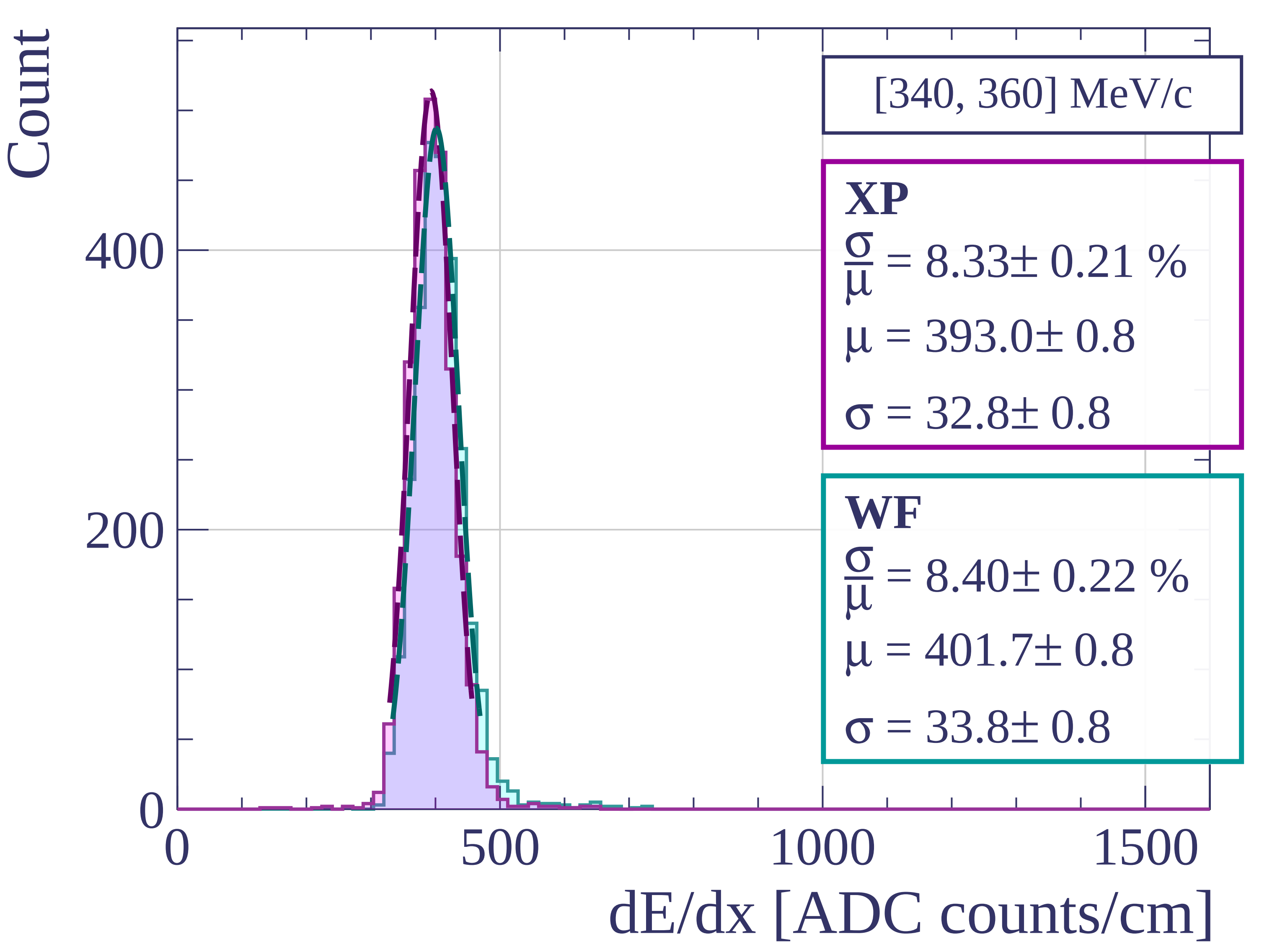


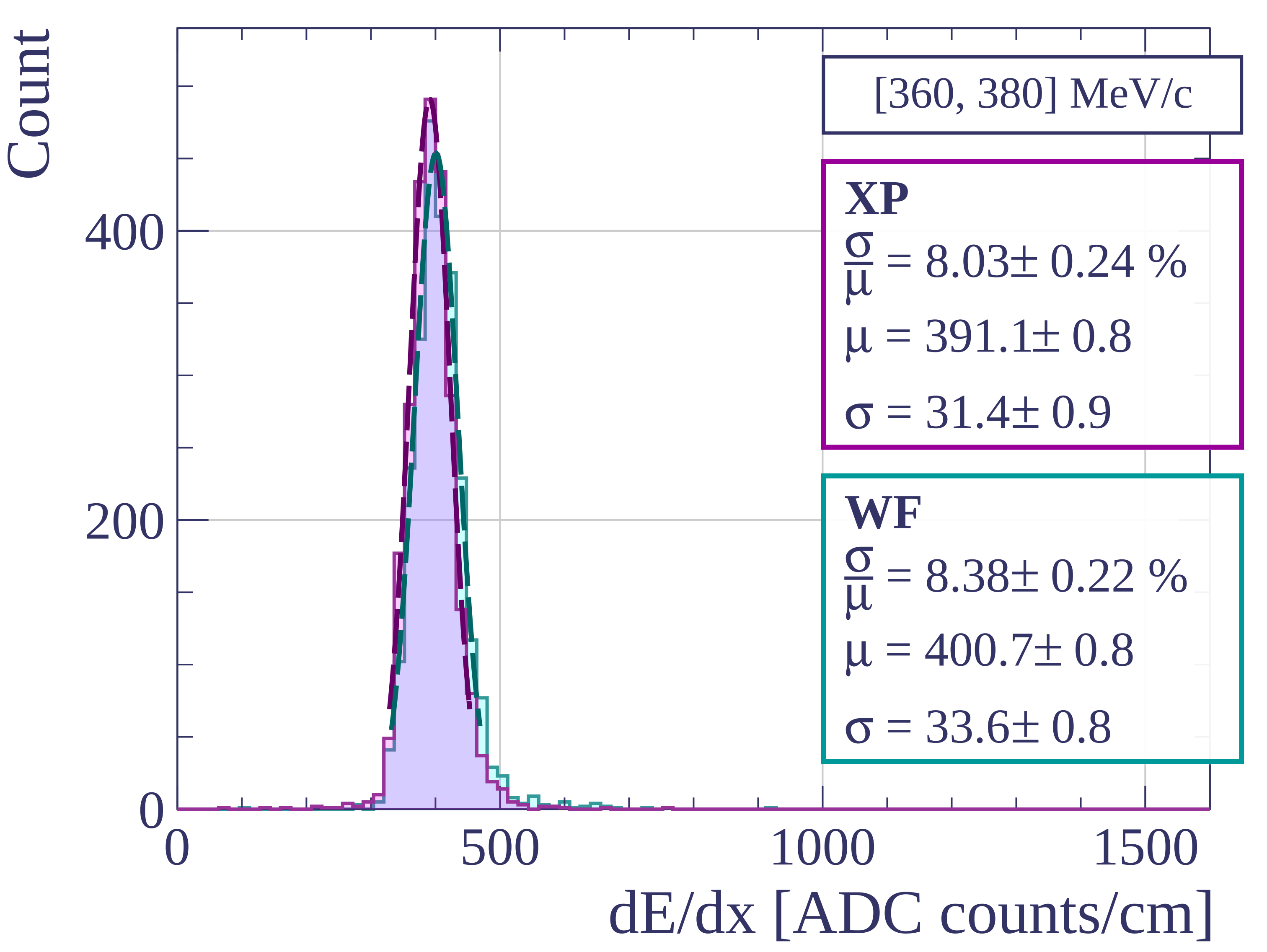
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[260, 280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.43± 0.19 %
μ = 404.5± 0.8
σ = 34.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.49± 0.20 %
μ = 413.5± 0.8
σ = 35.1± 0.8


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[280, 300] MeV/c
XP
σ/μ = 8.28± 0.21 %
μ = 399.3± 0.8
σ = 33.1± 0.8
WF
σ/μ = 8.47± 0.19 %
μ = 408.4± 0.8
σ = 34.6± 0.7

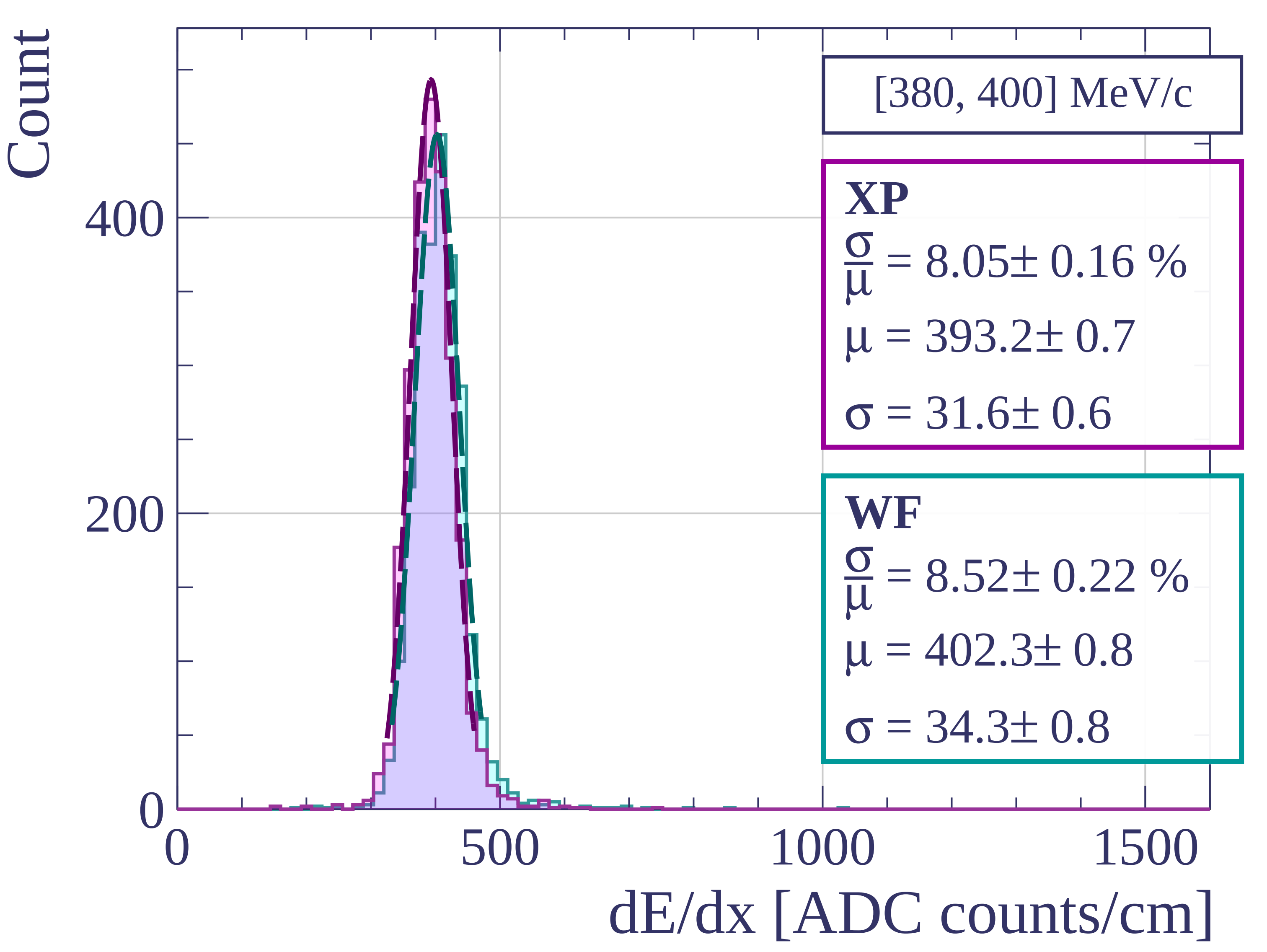

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[300, 320] MeV/c
XP
σ/μ = 8.53± 0.23 %
μ = 396.5± 0.8
σ = 33.8± 0.9
WF
σ/μ = 8.42± 0.20 %
μ = 406.7± 0.8
σ = 34.2± 0.7


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 340] MeV/c
XP
σ/μ = 7.53± 0.20 %
μ = 393.9± 0.7
σ = 29.6± 0.7
WF
σ/μ = 8.07± 0.21 %
μ = 403.5± 0.8
σ = 32.6± 0.8

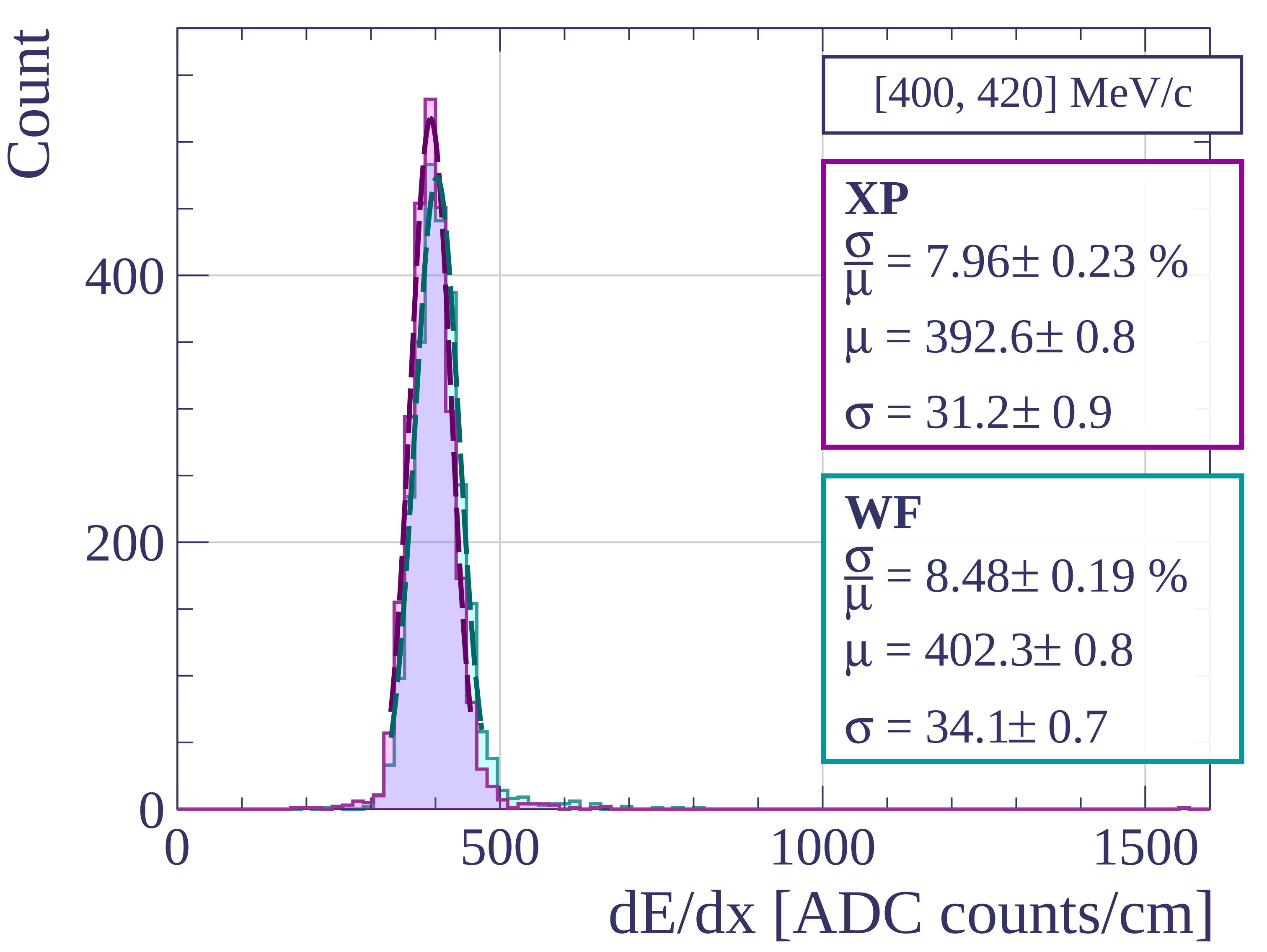

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[340, 360] MeV/c
XP
σ/μ = 8.33± 0.21 %
μ = 393.0± 0.8
σ = 32.8± 0.8
WF
σ/μ = 8.40± 0.22 %
μ = 401.7± 0.8
σ = 33.8± 0.8

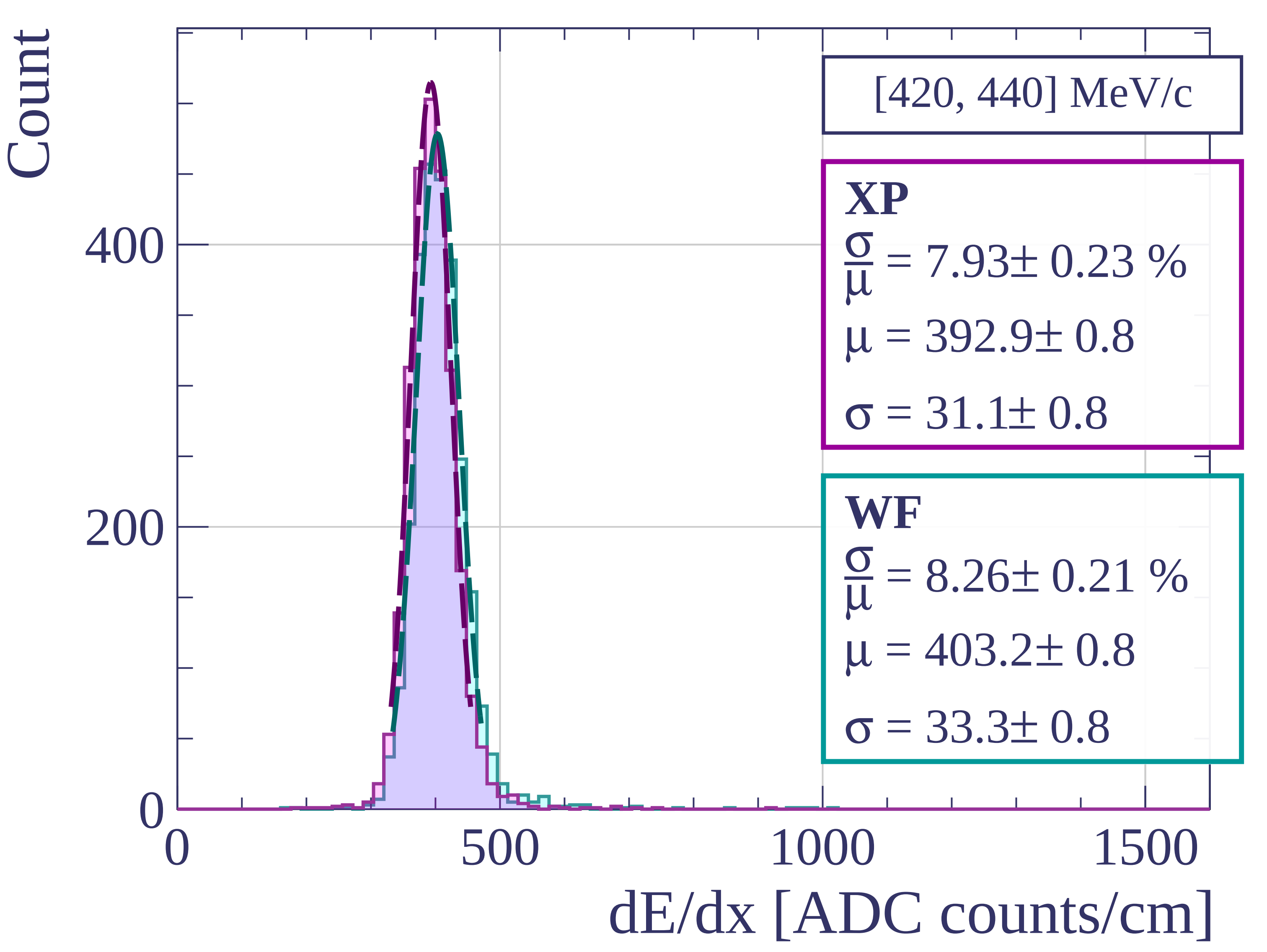


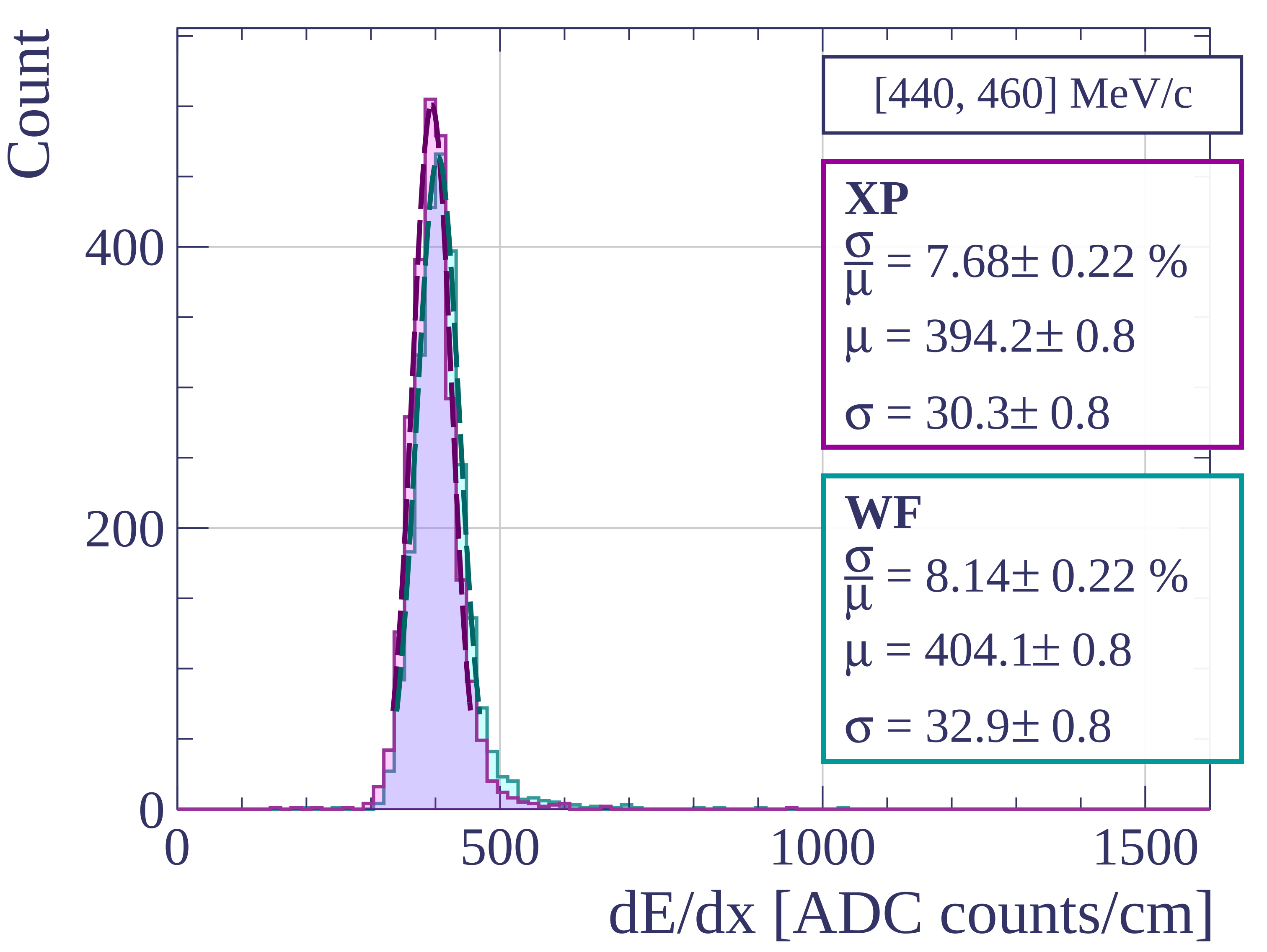
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[360, 380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.03± 0.24 %
μ = 391.1± 0.8
σ = 31.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.38± 0.22 %
μ = 400.7± 0.8
σ = 33.6± 0.8

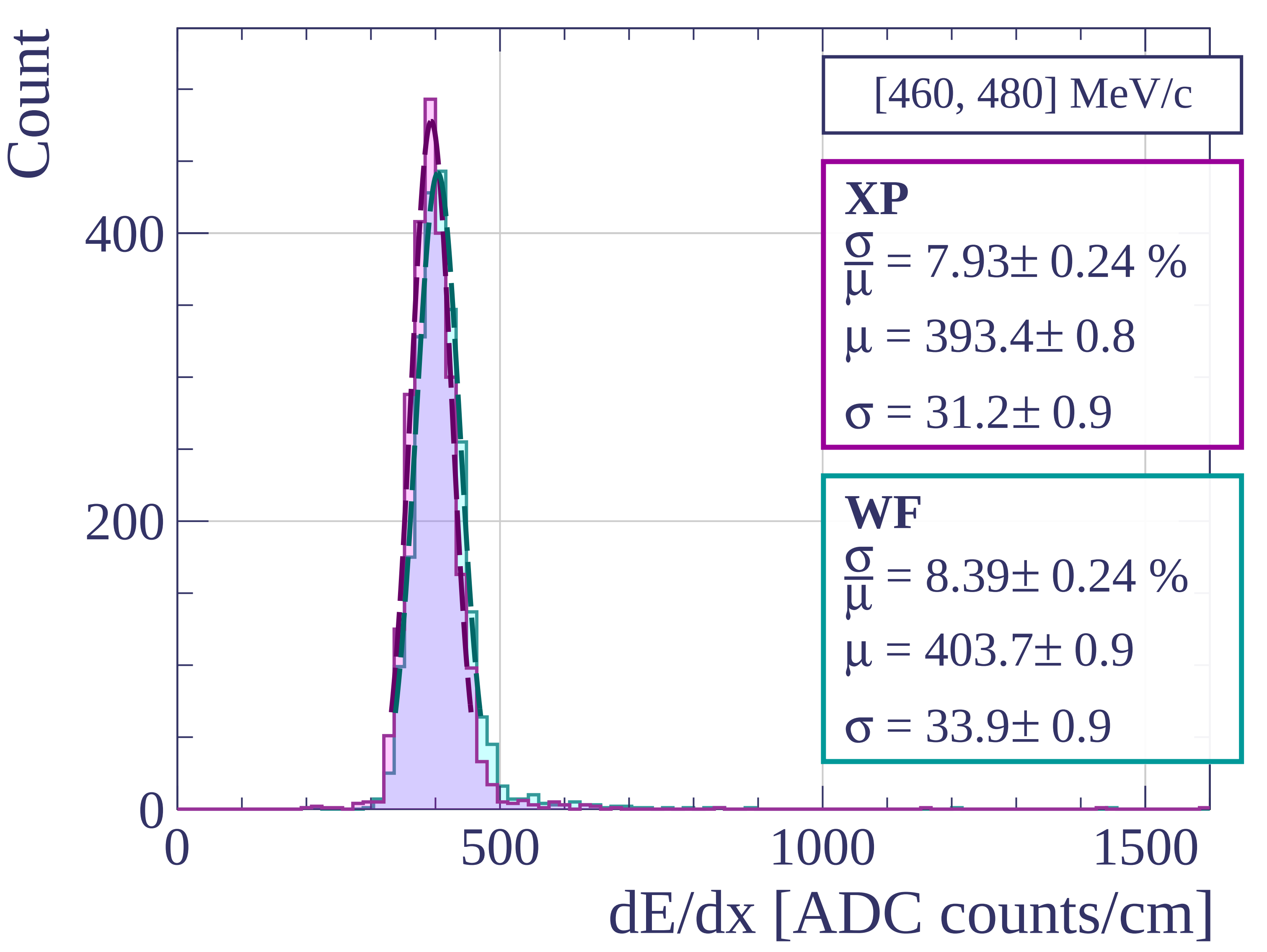


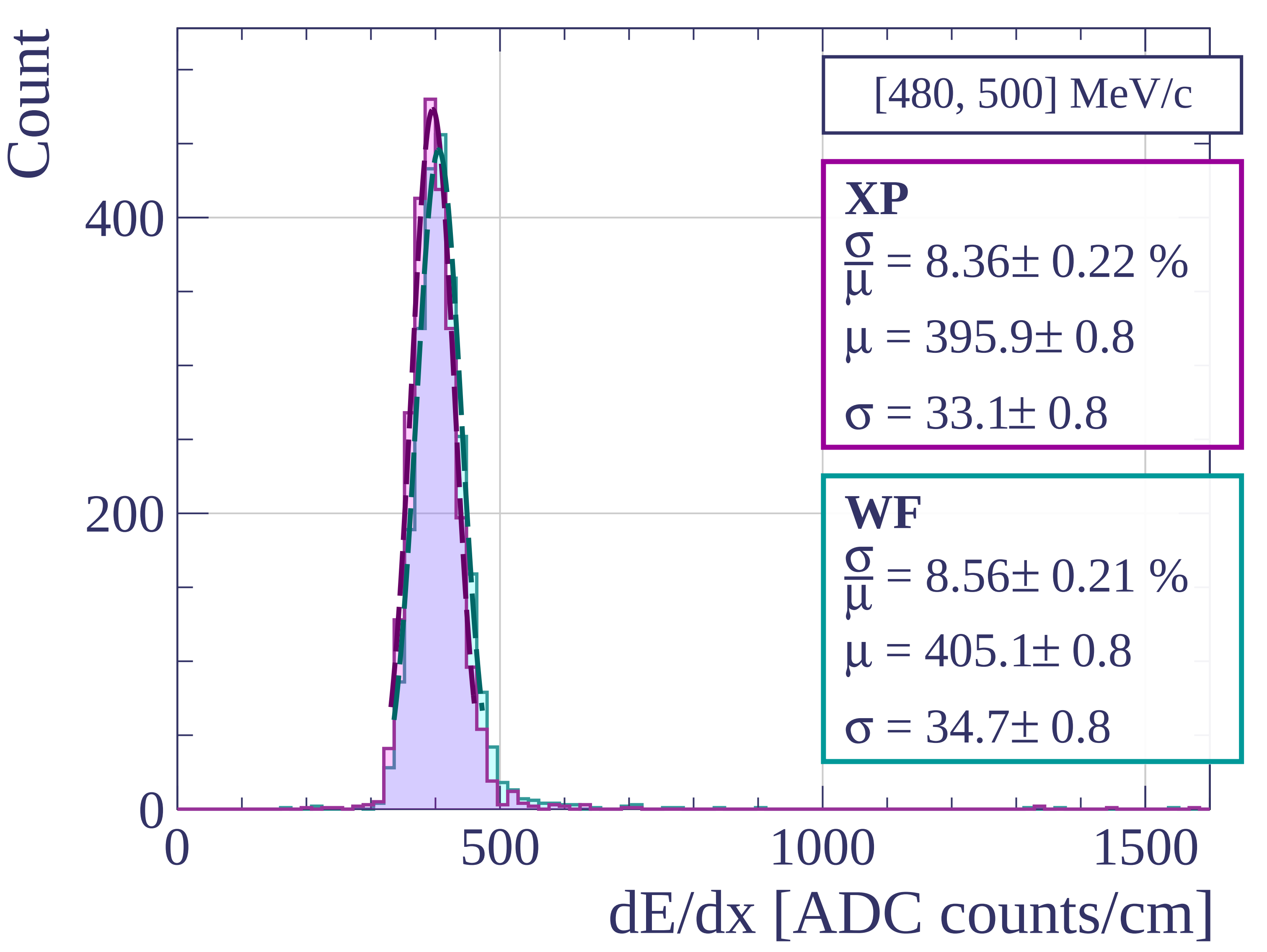
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[380, 400] MeV/c
XP
σ/μ = 8.05± 0.16 %
μ = 393.2± 0.7
σ = 31.6± 0.6
WF
σ/μ = 8.52± 0.22 %
μ = 402.3± 0.8
σ = 34.3± 0.8

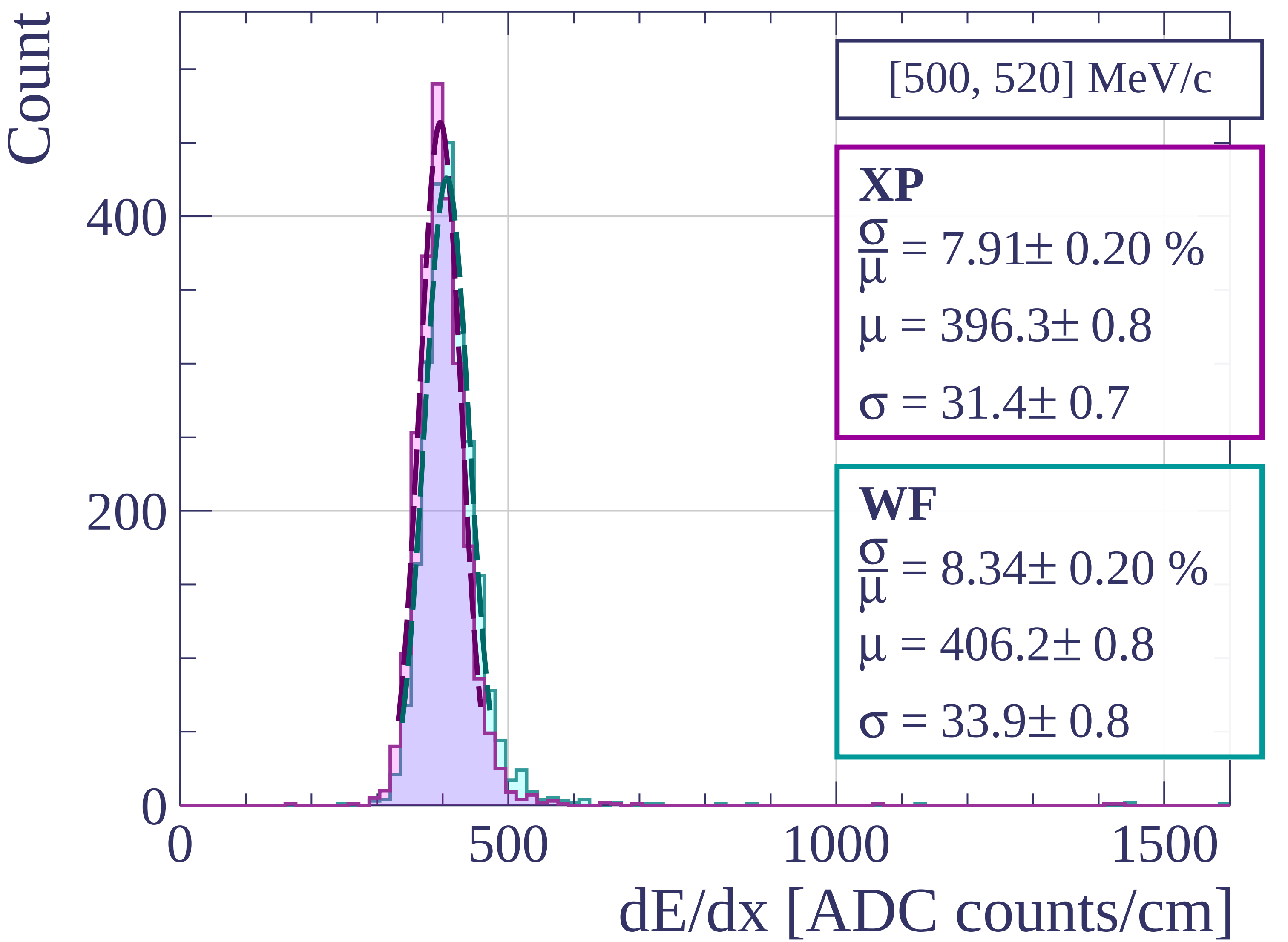

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 420] MeV/c
XP
σ/μ = 7.96± 0.23 %
μ = 392.6± 0.8
σ = 31.2± 0.9
WF
σ/μ = 8.48± 0.19 %
μ = 402.3± 0.8
σ = 34.1± 0.7


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[420, 440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.93± 0.23 %
μ = 392.9± 0.8
σ = 31.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.26± 0.21 %
μ = 403.2± 0.8
σ = 33.3± 0.8

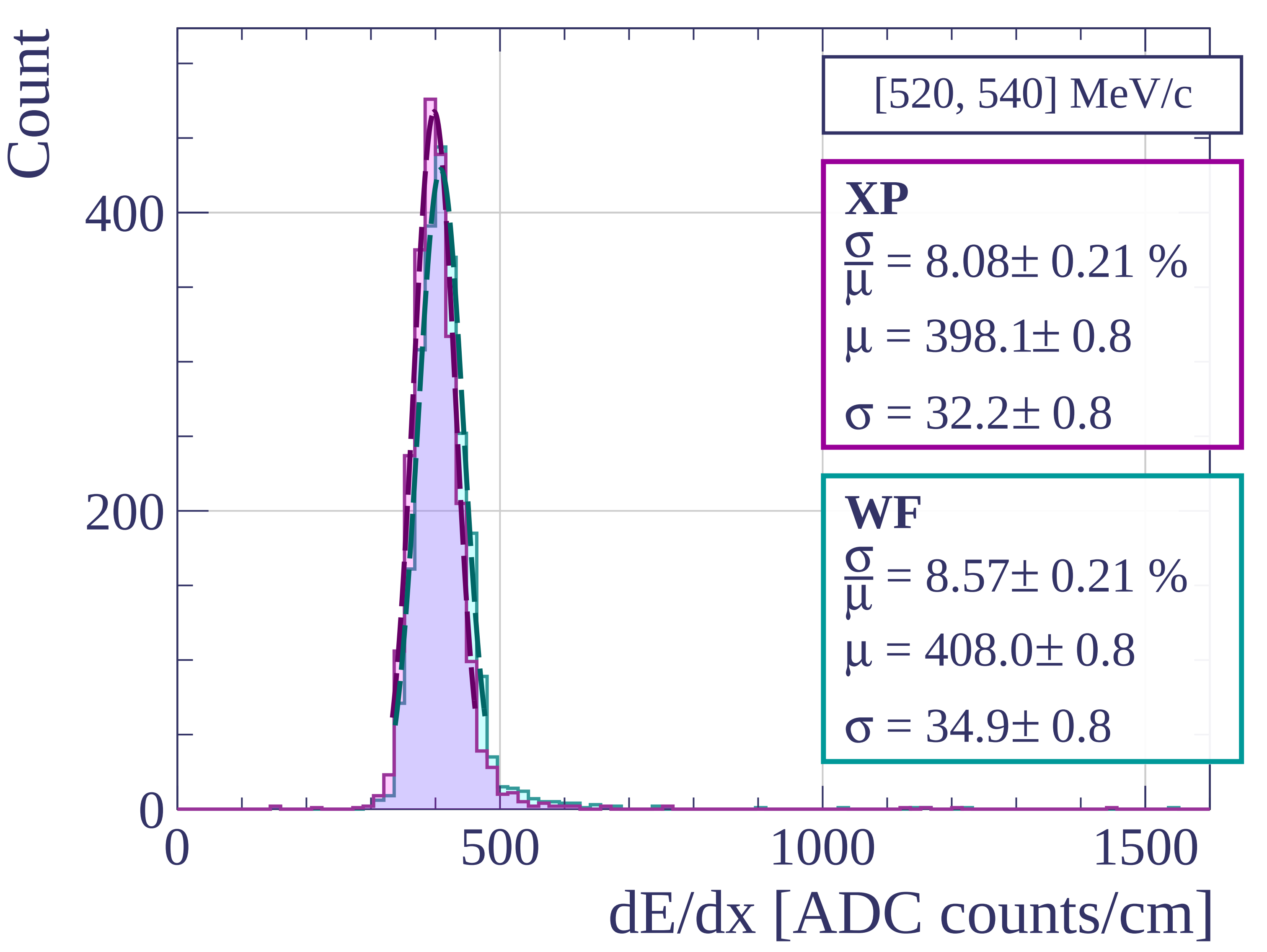

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[440, 460] MeV/c
XP
σ/μ = 7.68± 0.22 %
μ = 394.2± 0.8
σ = 30.3± 0.8
WF
σ/μ = 8.14± 0.22 %
μ = 404.1± 0.8
σ = 32.9± 0.8

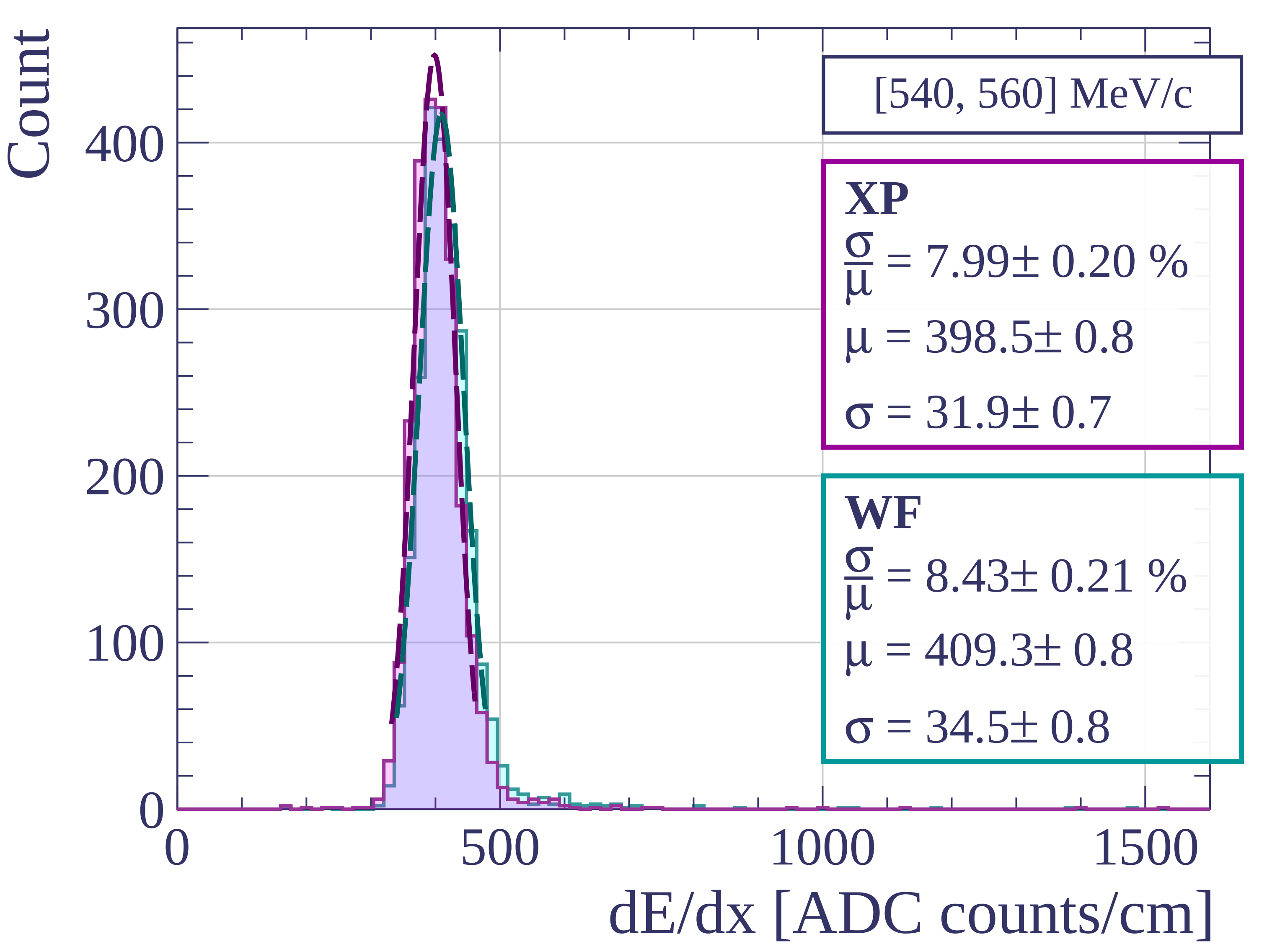

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[460, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.93± 0.24 %
μ = 393.4± 0.8
σ = 31.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.39± 0.24 %
μ = 403.7± 0.9
σ = 33.9± 0.9

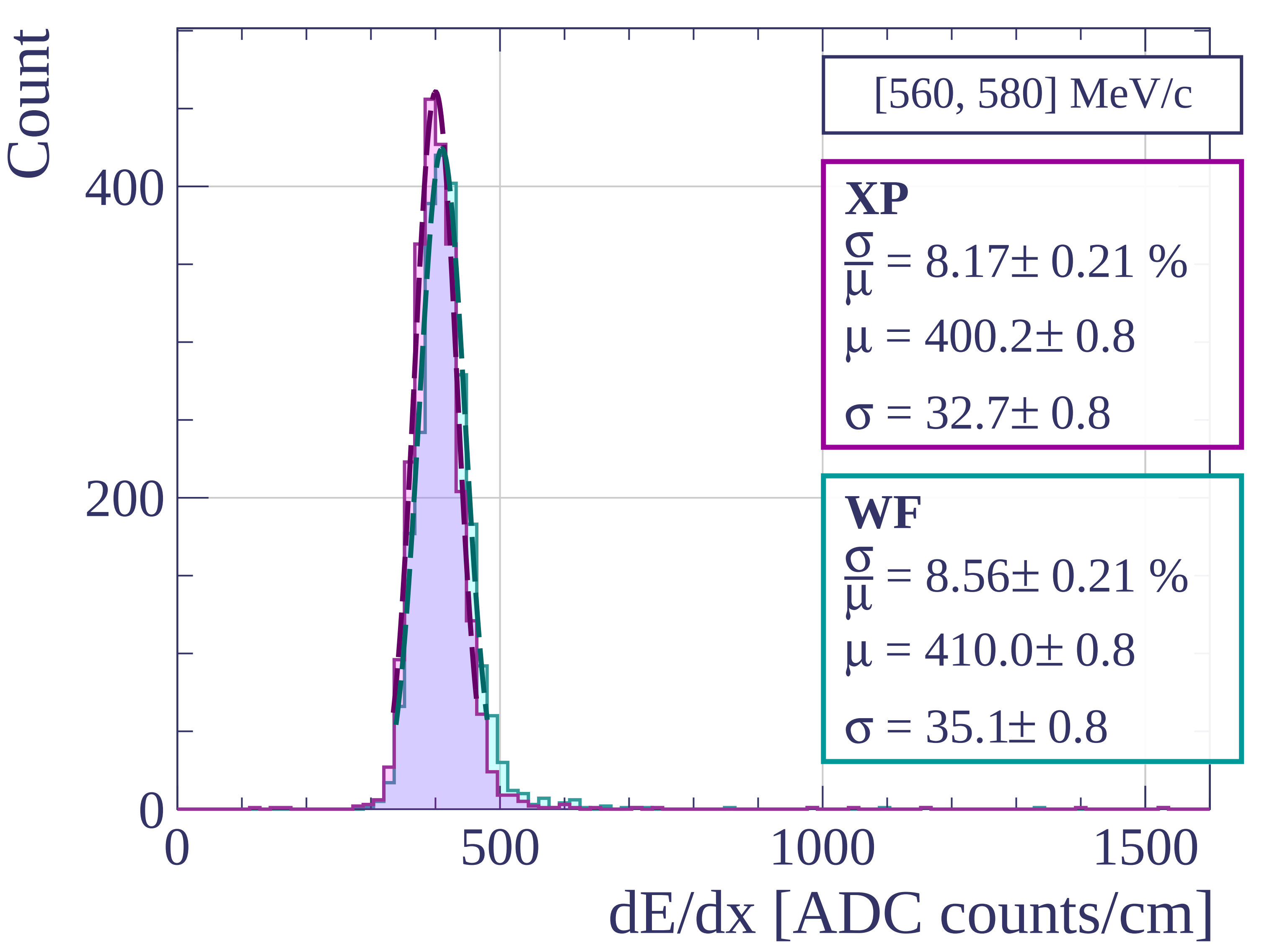

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.36± 0.22 %
μ = 395.9± 0.8
σ = 33.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.56± 0.21 %
μ = 405.1± 0.8
σ = 34.7± 0.8

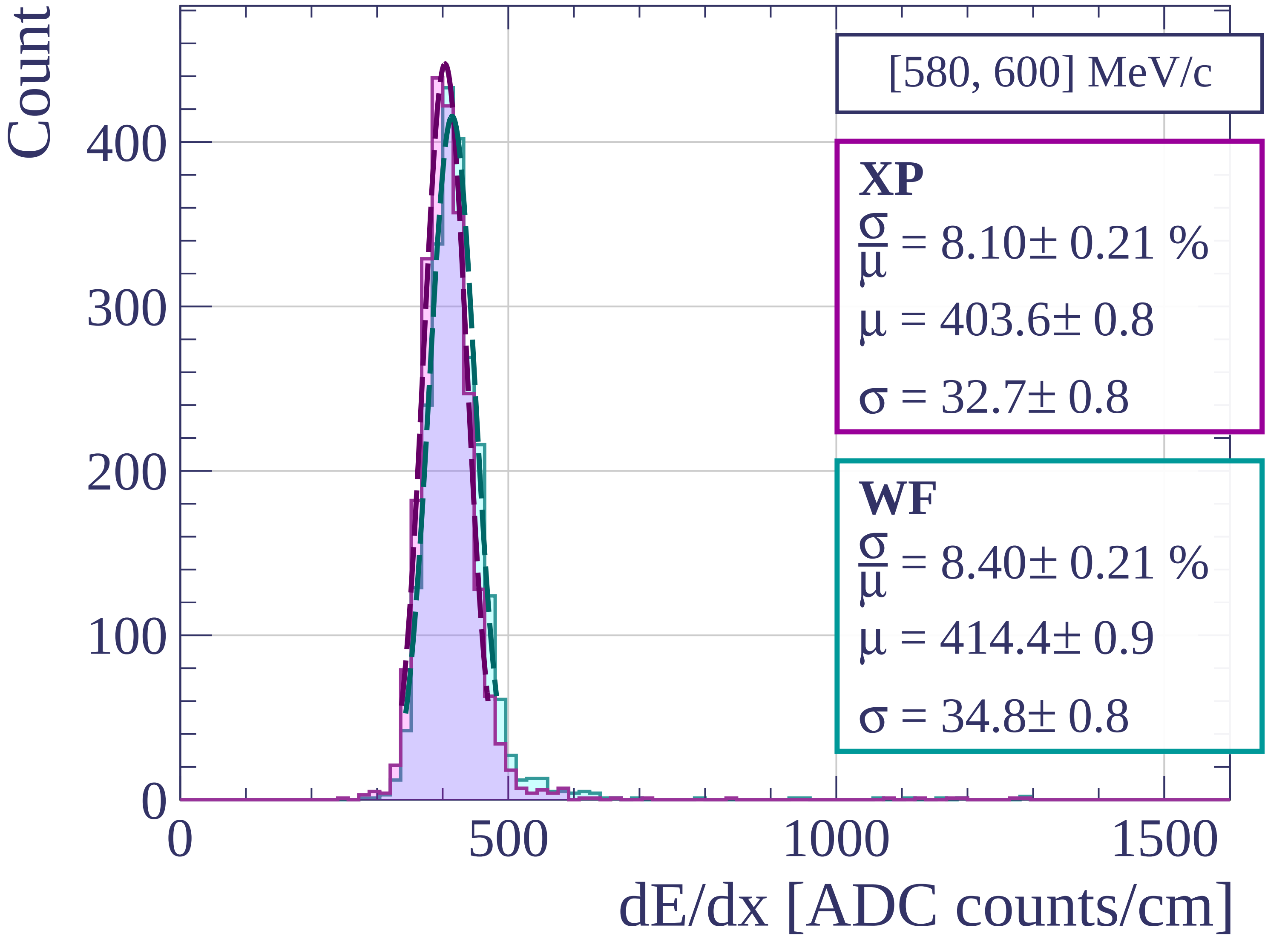


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[500, 520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.91± 0.20 %
μ = 396.3± 0.8
σ = 31.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.34± 0.20 %
μ = 406.2± 0.8
σ = 33.9± 0.8

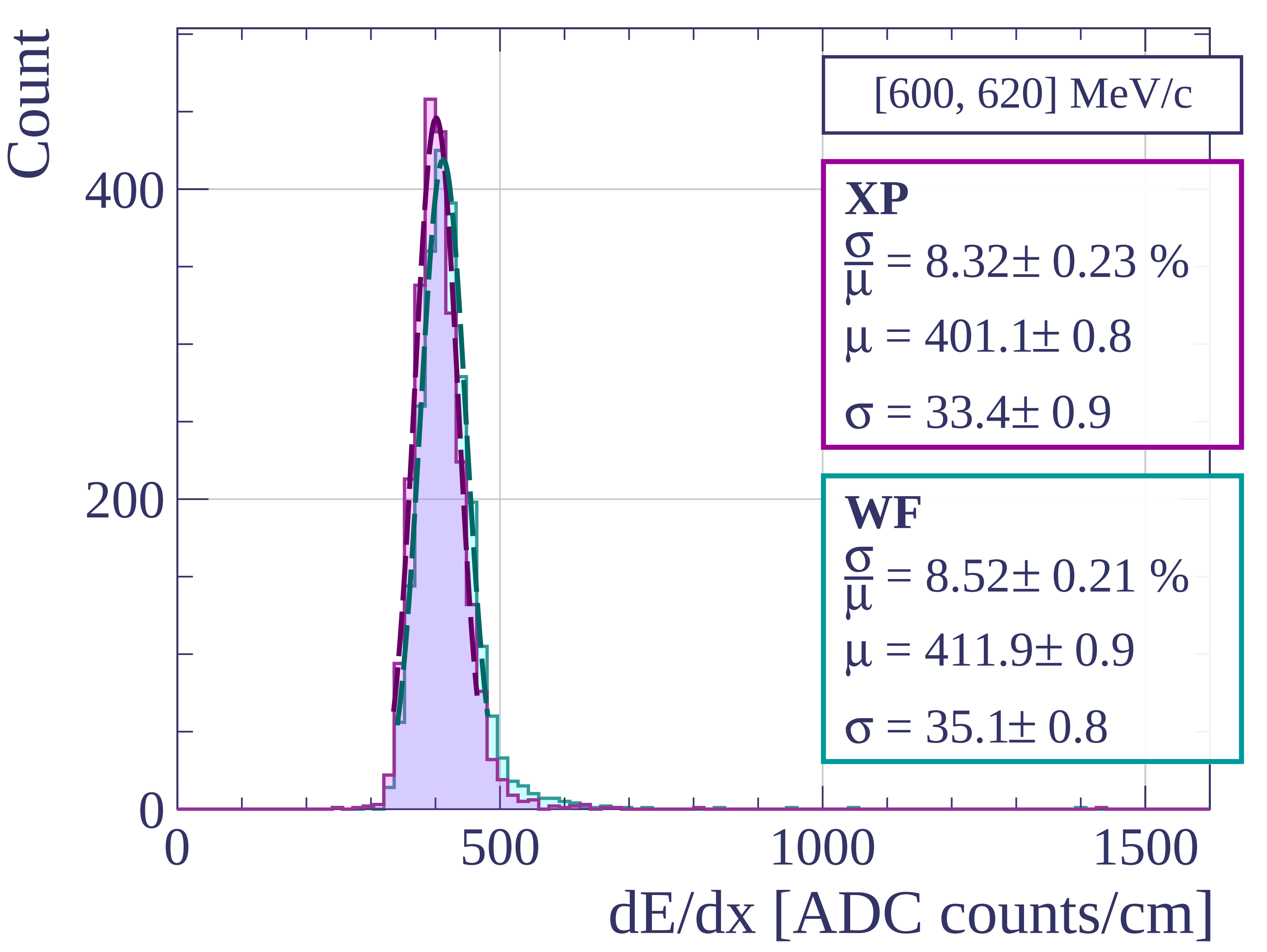

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[520, 540] MeV/c
XP
σ/μ = 8.08± 0.21 %
μ = 398.1± 0.8
σ = 32.2± 0.8
WF
σ/μ = 8.57± 0.21 %
μ = 408.0± 0.8
σ = 34.9± 0.8

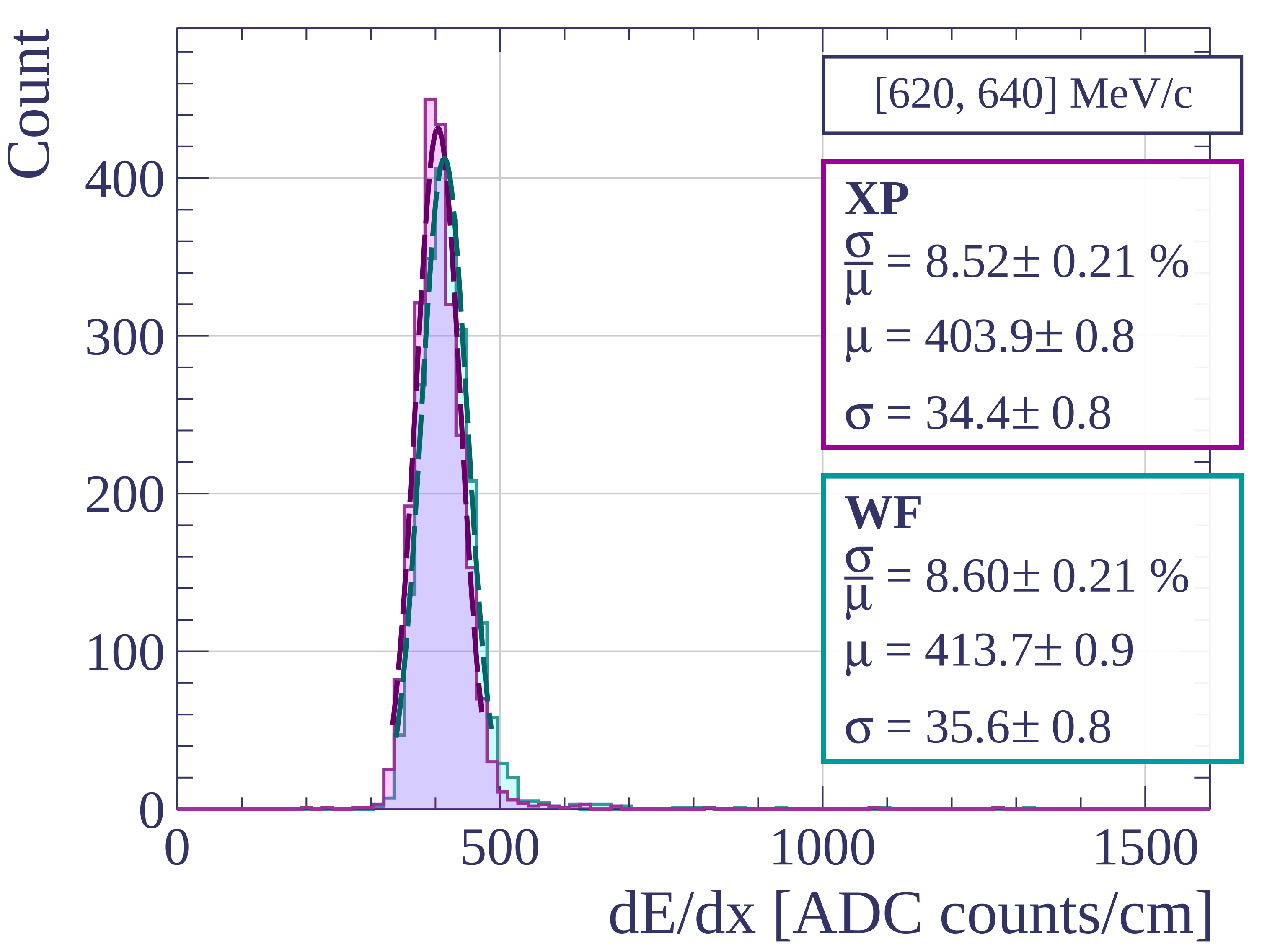

[540, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.99± 0.20 %
μ = 398.5± 0.8
σ = 31.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.43± 0.21 %
μ = 409.3± 0.8
σ = 34.5± 0.8
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.17± 0.21 %
μ = 400.2± 0.8
σ = 32.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.56± 0.21 %
μ = 410.0± 0.8
σ = 35.1± 0.8

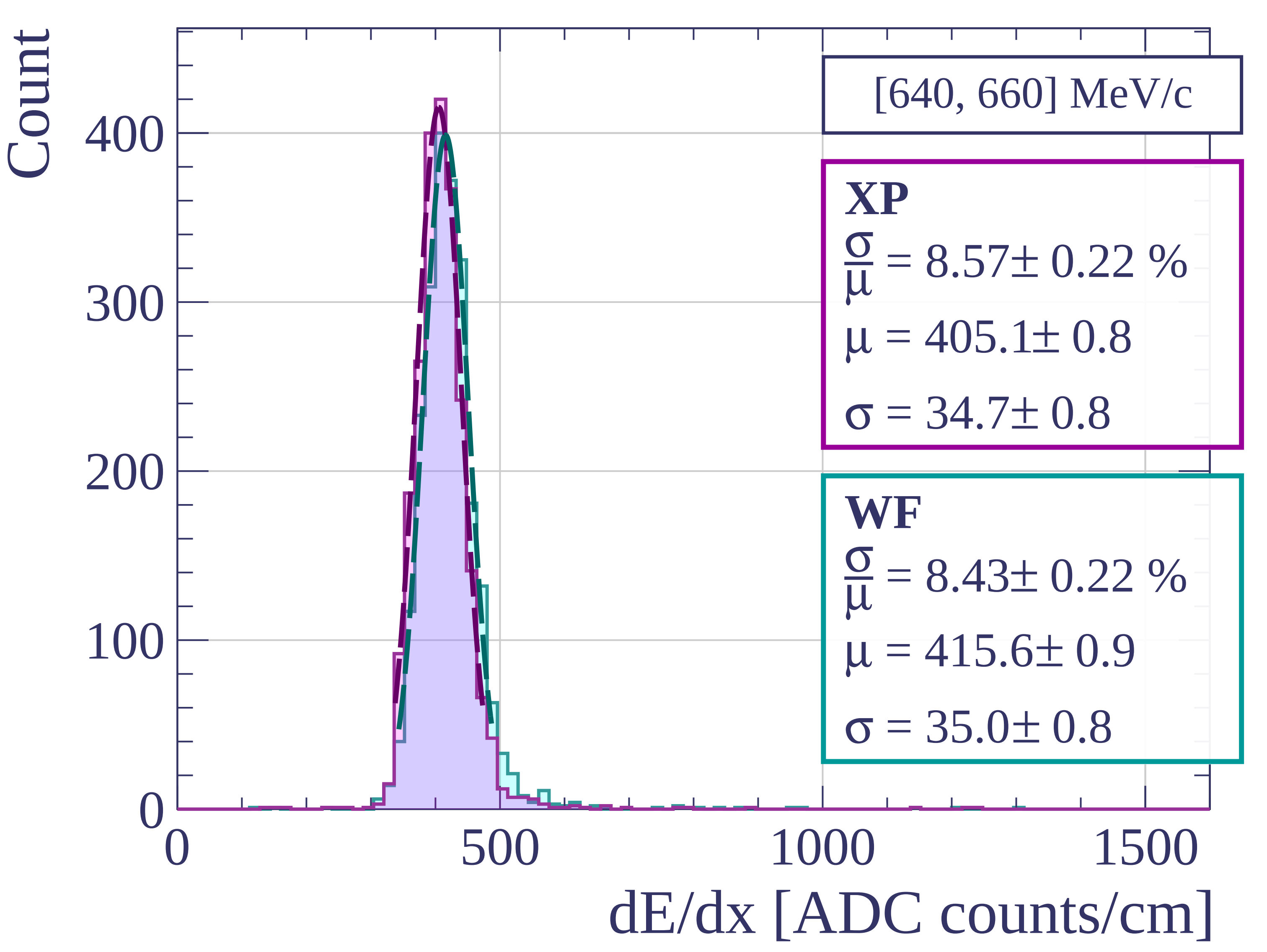


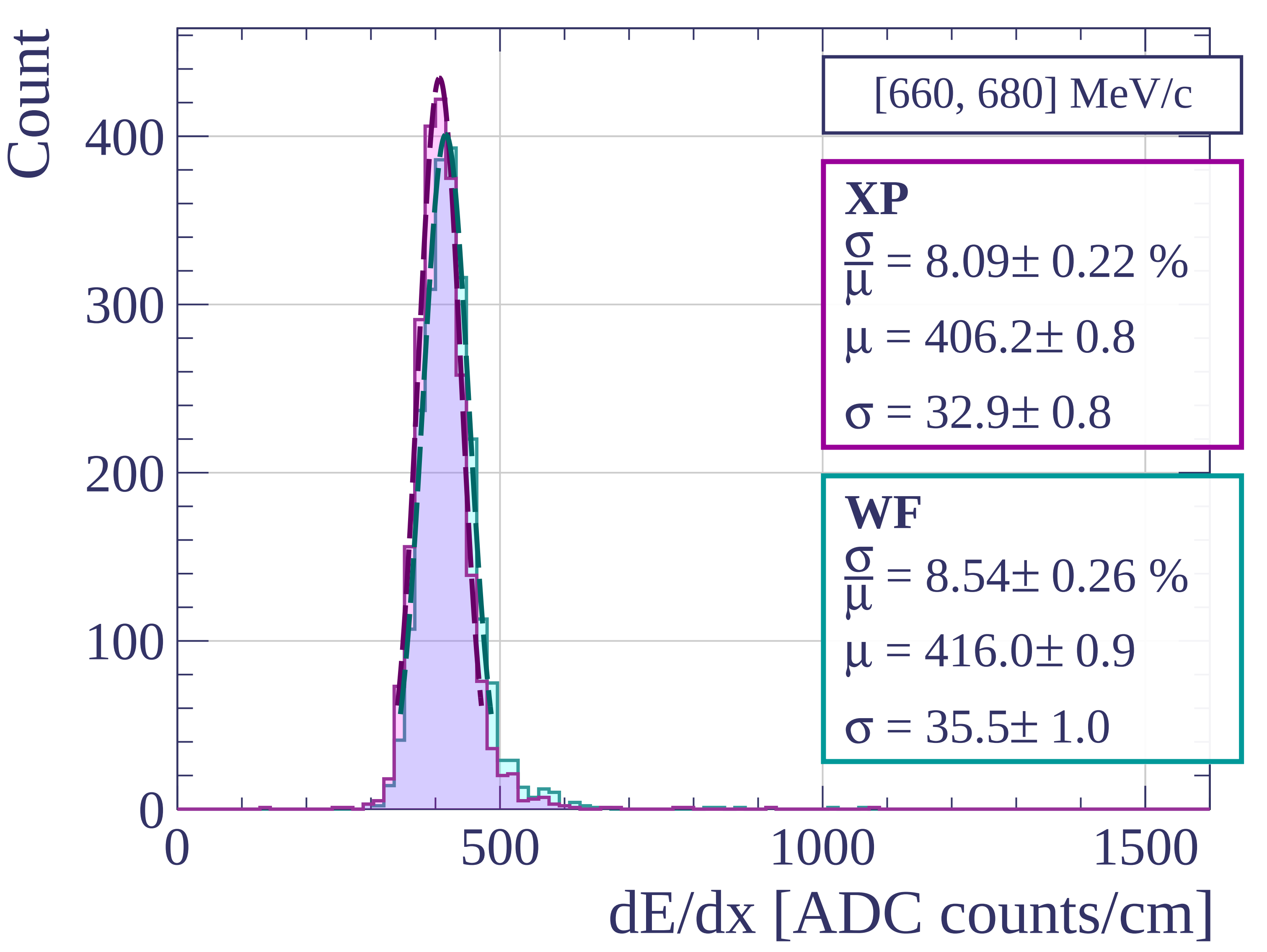
Count
dE/dx [ADC counts/cm]
[580, 600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.10± 0.21 %
μ = 403.6± 0.8
σ = 32.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.40± 0.21 %
μ = 414.4± 0.9
σ = 34.8± 0.8


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[600, 620] MeV/c
XP
σ/μ = 8.32± 0.23 %
μ = 401.1± 0.8
σ = 33.4± 0.9
WF
σ/μ = 8.52± 0.21 %
μ = 411.9± 0.9
σ = 35.1± 0.8

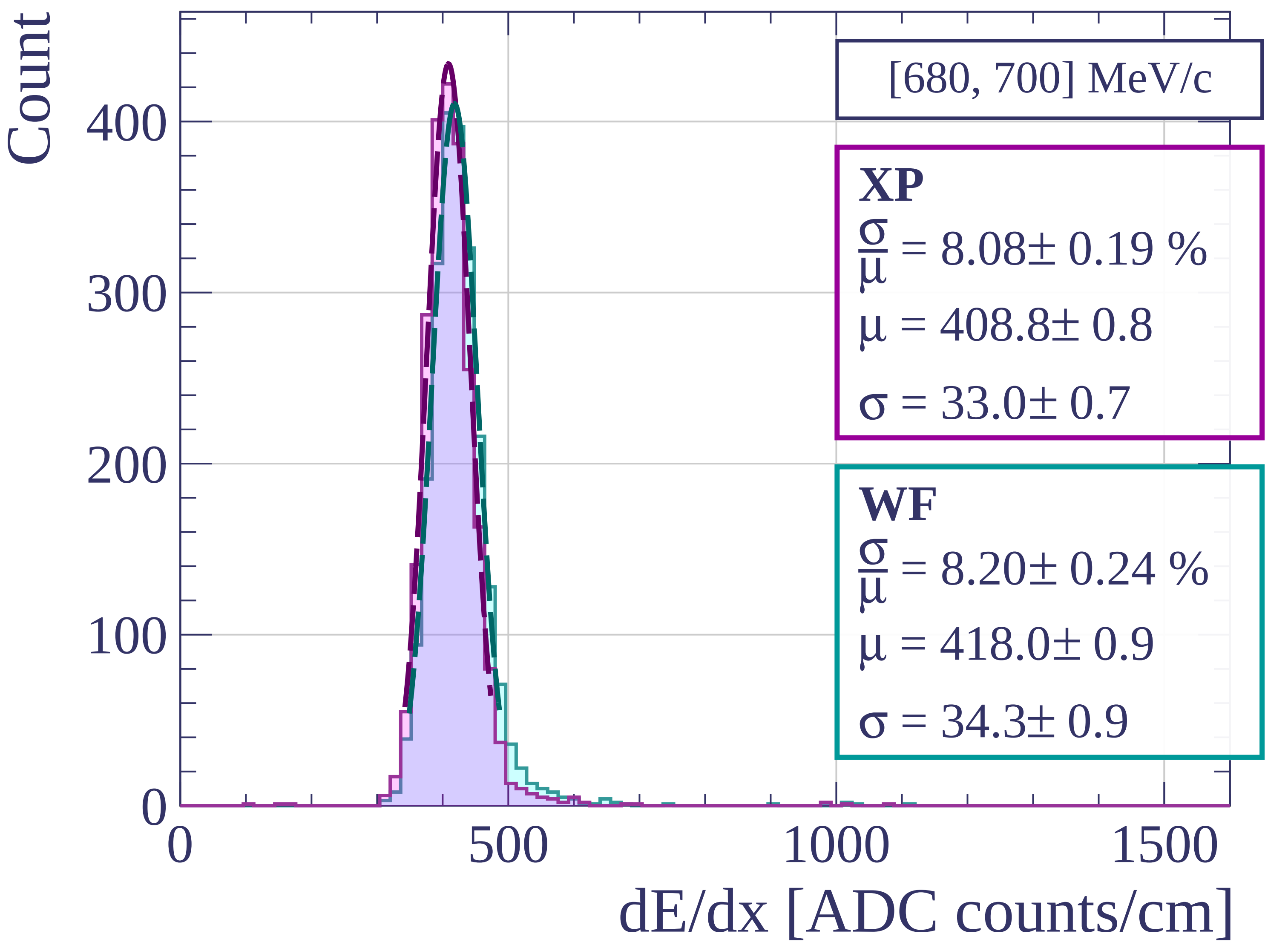


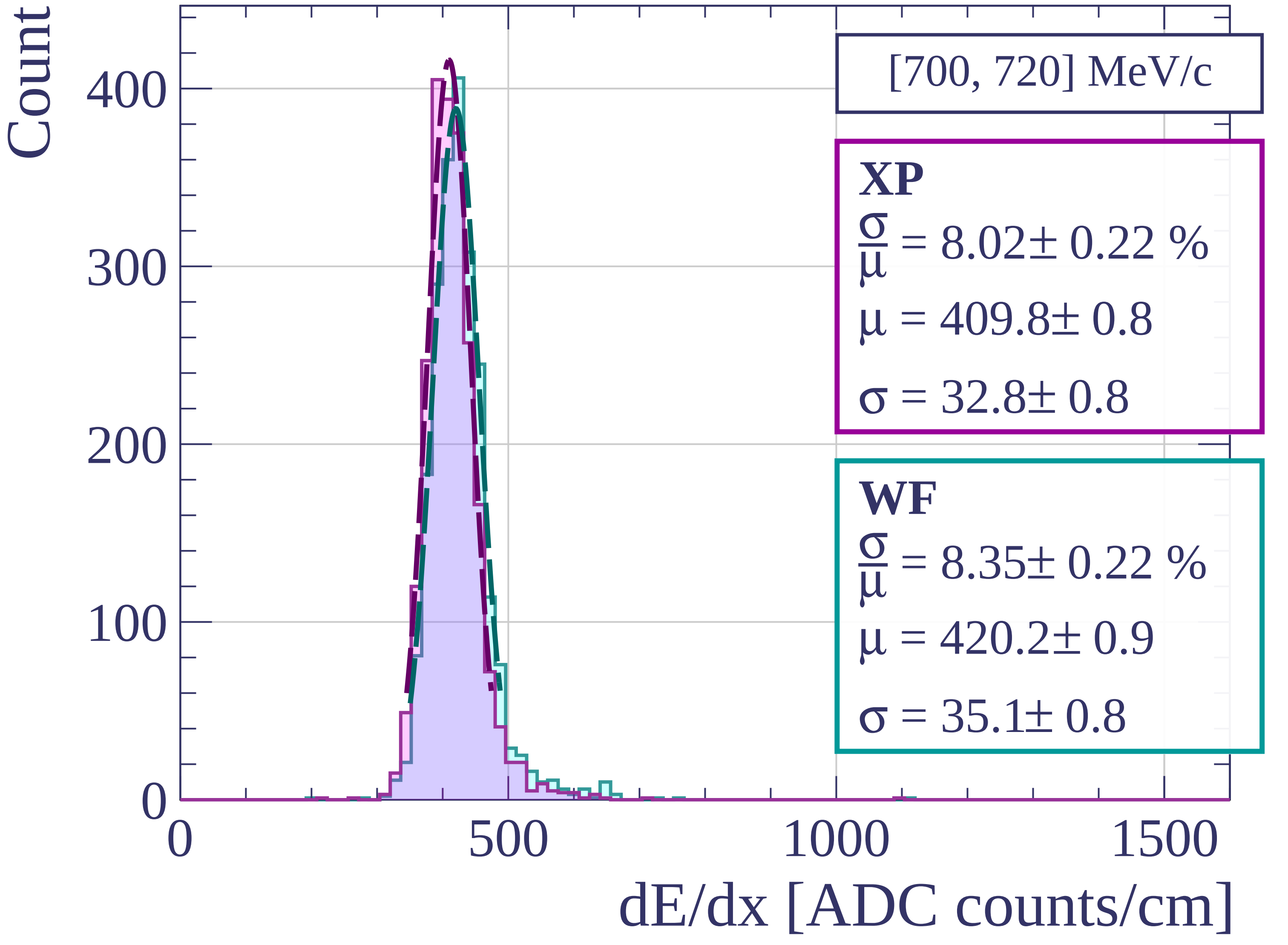
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[620, 640] MeV/c
XP
σ/μ = 8.52± 0.21 %
μ = 403.9± 0.8
σ = 34.4± 0.8
WF
σ/μ = 8.60± 0.21 %
μ = 413.7± 0.9
σ = 35.6± 0.8

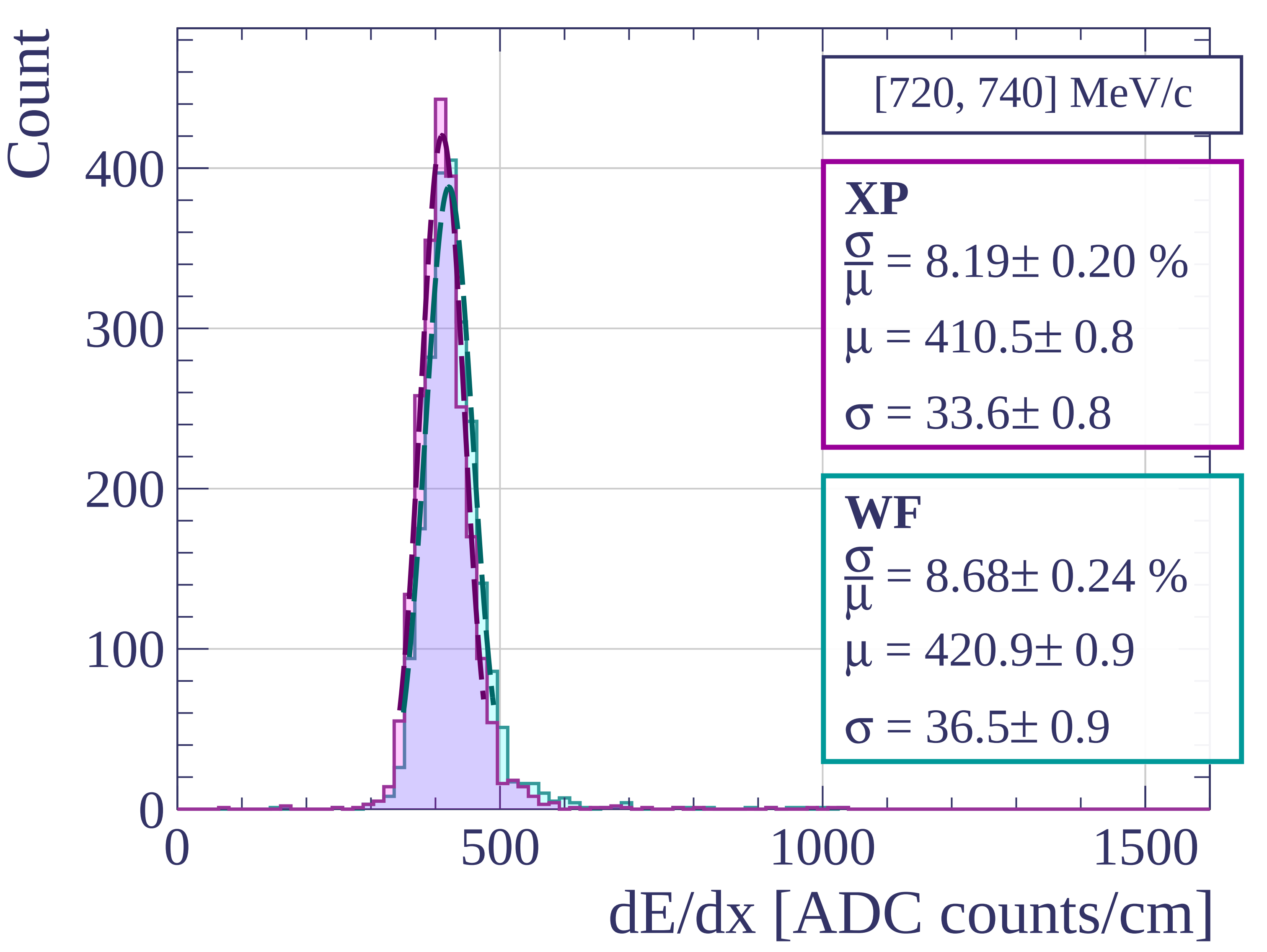

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.57± 0.22 %
μ = 405.1± 0.8
σ = 34.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.43± 0.22 %
μ = 415.6± 0.9
σ = 35.0± 0.8



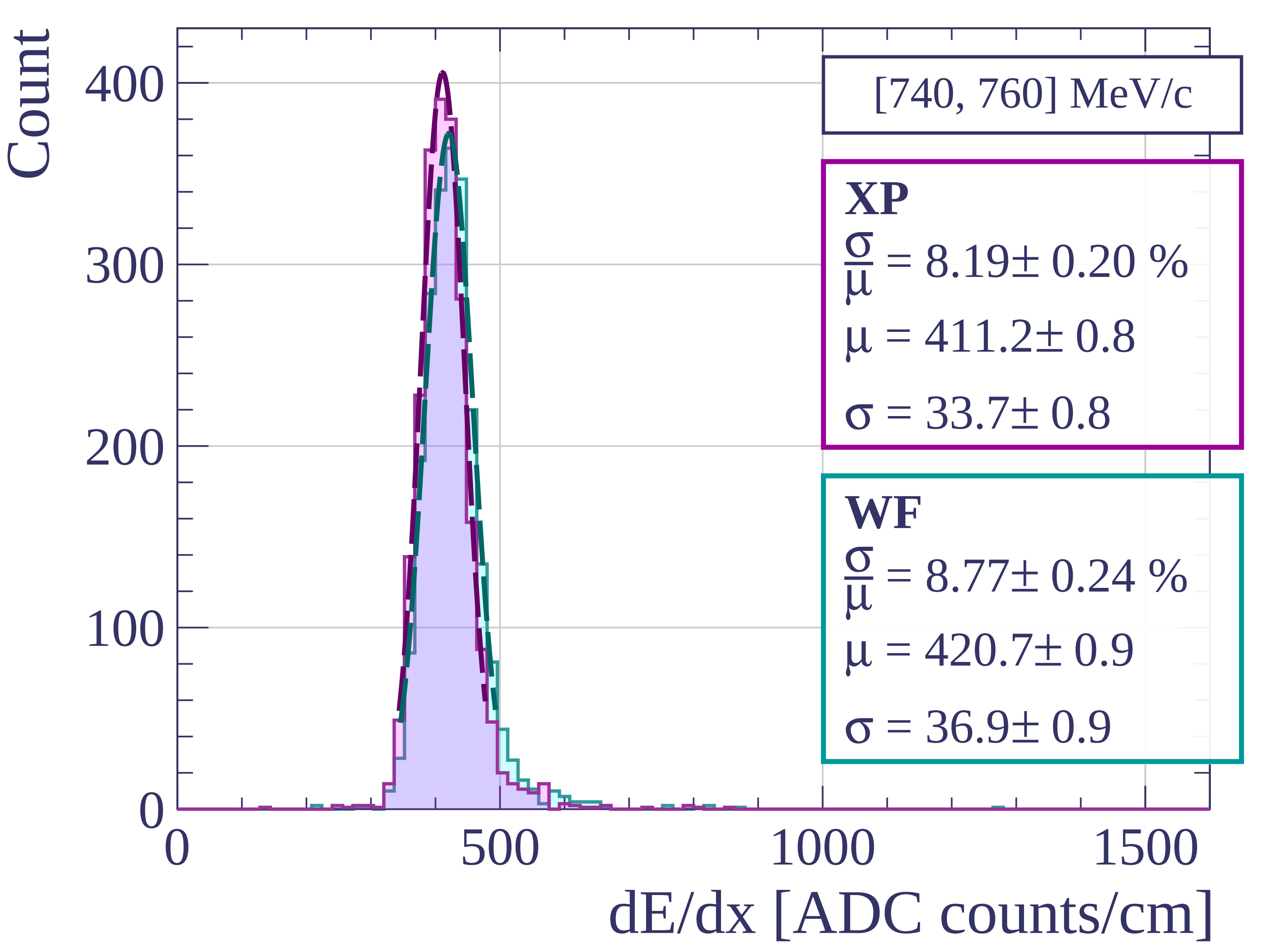
Count
dE/dx [ADC counts/cm]
[660, 680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.22 %
μ = 406.2± 0.8
σ = 32.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.54± 0.26 %
μ = 416.0± 0.9
σ = 35.5± 1.0
0
100
200
300
400
0
500
1000
1500


Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
[680, 700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.08± 0.19 %
μ = 408.8± 0.8
σ = 33.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.20± 0.24 %
μ = 418.0± 0.9
σ = 34.3± 0.9

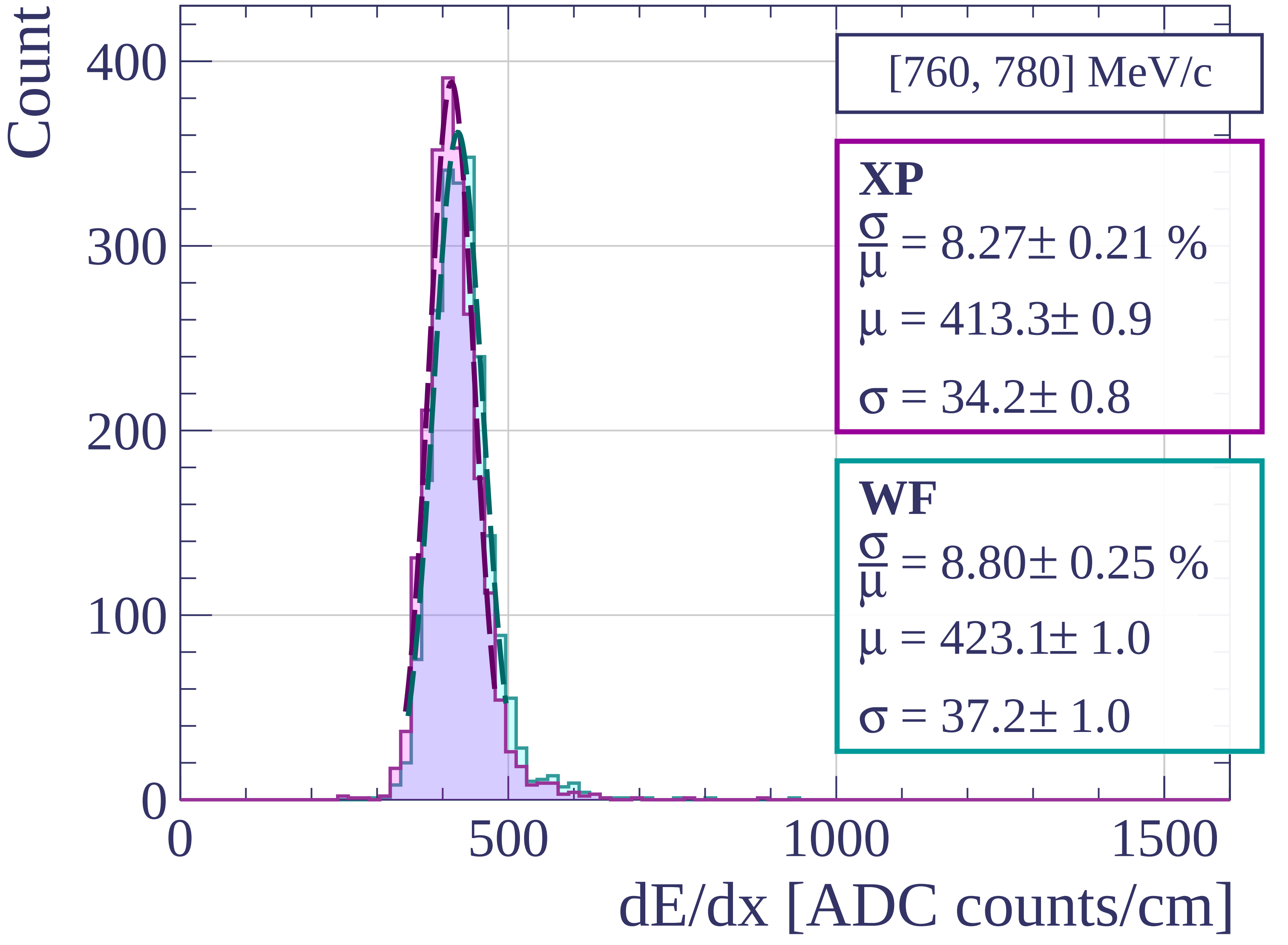

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[700, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.02± 0.22 %
μ = 409.8± 0.8
σ = 32.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.35± 0.22 %
μ = 420.2± 0.9
σ = 35.1± 0.8

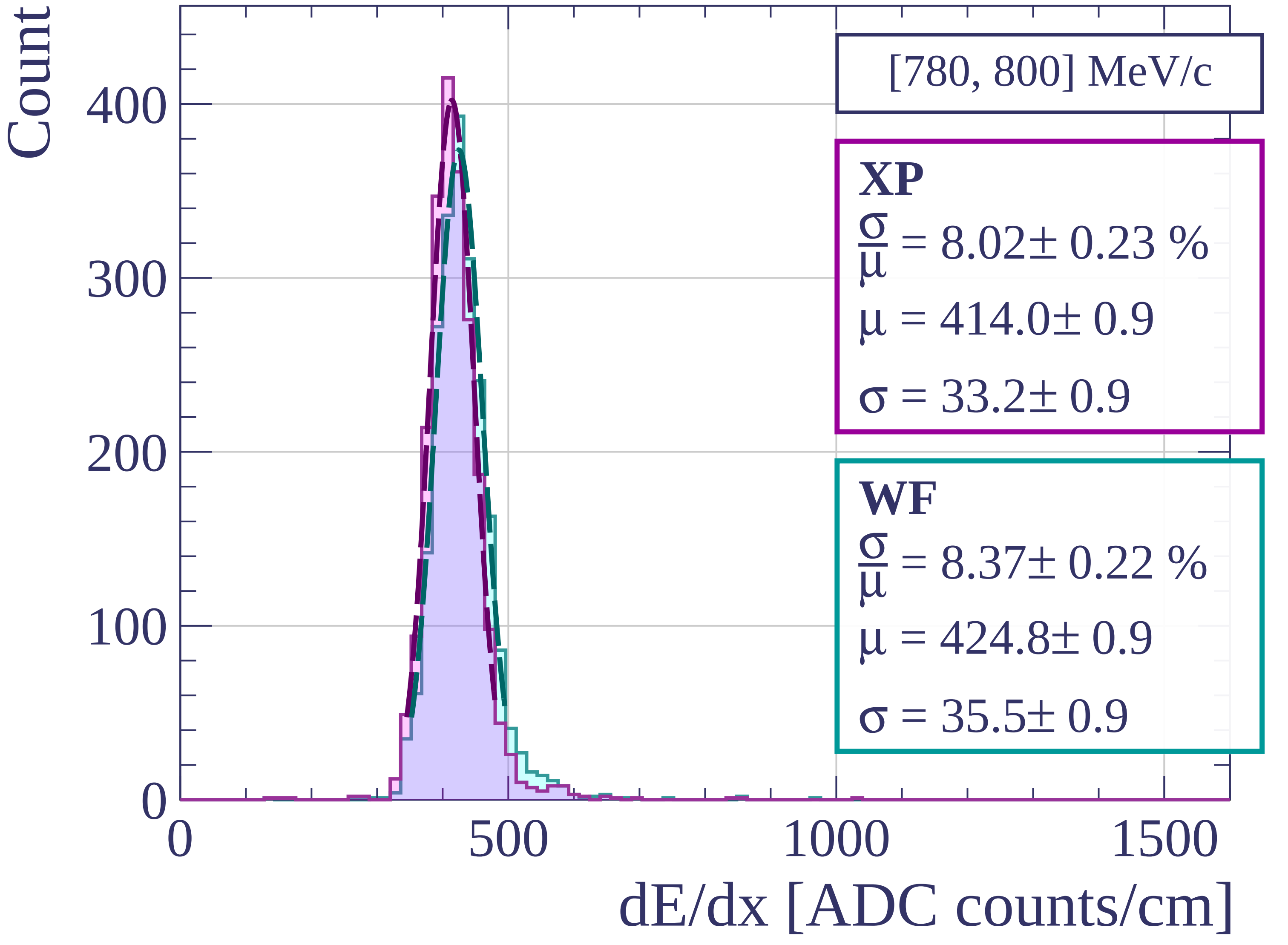


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 740] MeV/c
XP
σ/μ = 8.19± 0.20 %
μ = 410.5± 0.8
σ = 33.6± 0.8
WF
σ/μ = 8.68± 0.24 %
μ = 420.9± 0.9
σ = 36.5± 0.9

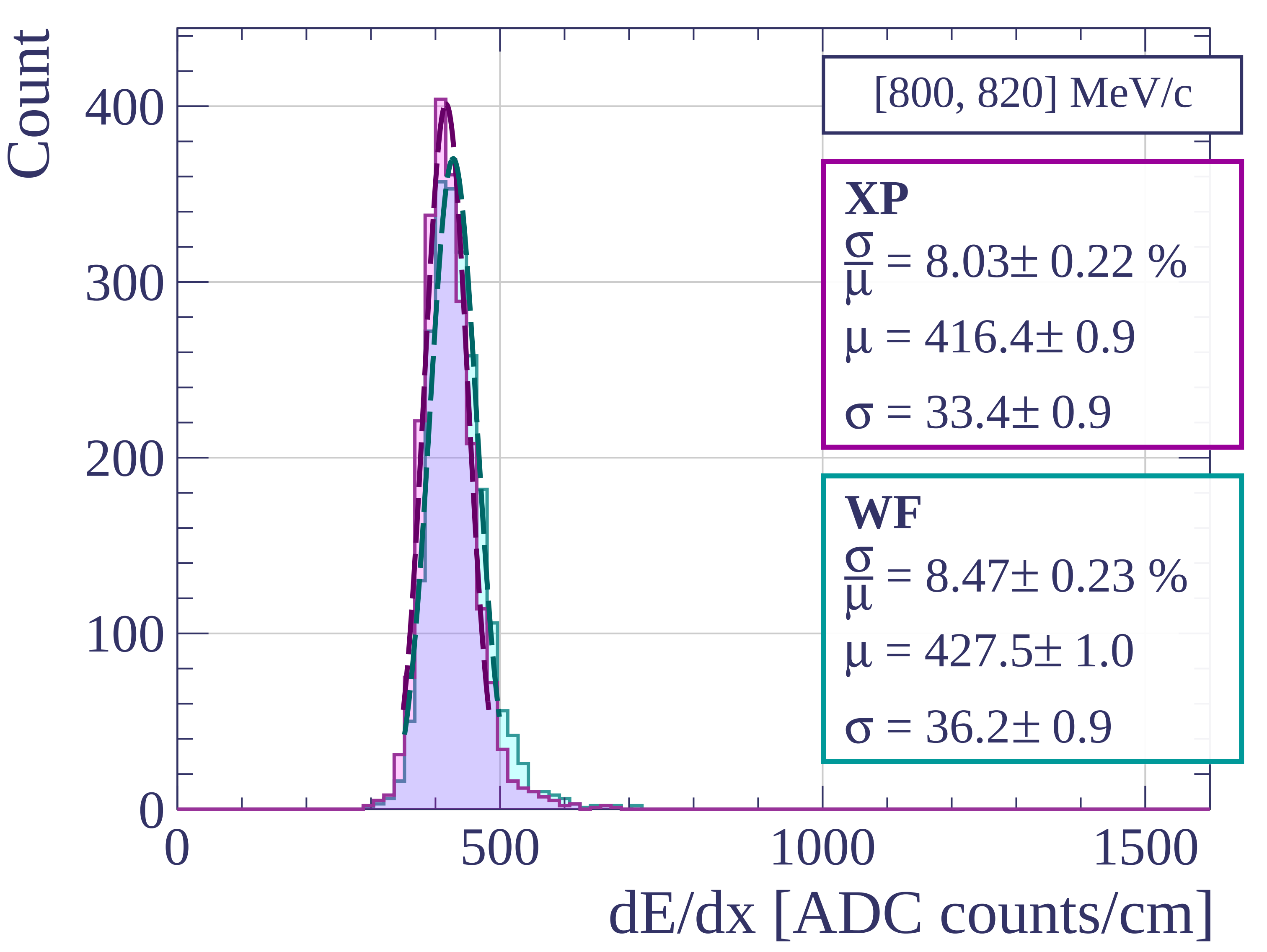


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[740, 760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.20 %
μ = 411.2± 0.8
σ = 33.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.77± 0.24 %
μ = 420.7± 0.9
σ = 36.9± 0.9

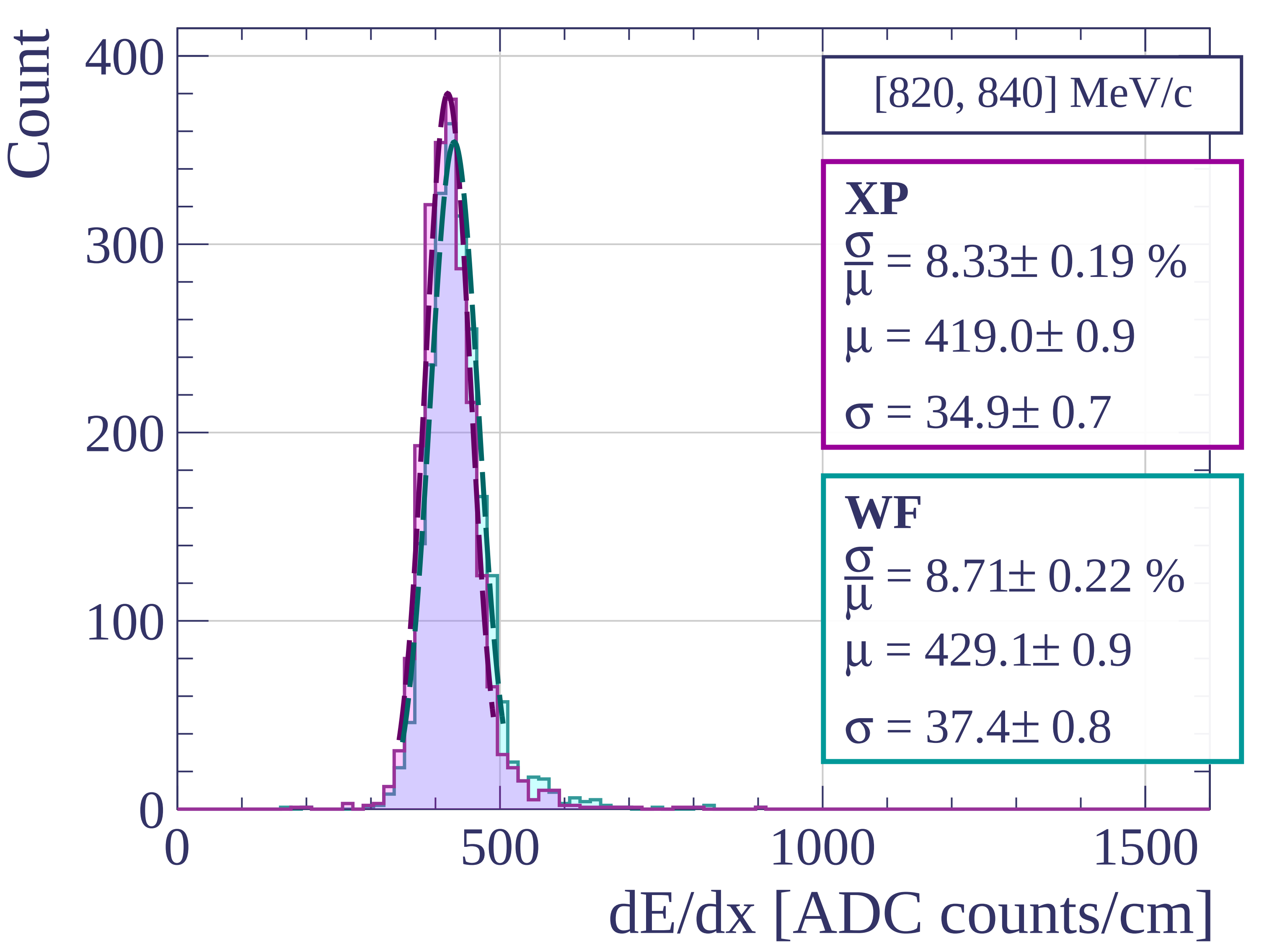

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[760, 780] MeV/c
XP
σ/μ = 8.27± 0.21 %
μ = 413.3± 0.9
σ = 34.2± 0.8
WF
σ/μ = 8.80± 0.25 %
μ = 423.1± 1.0
σ = 37.2± 1.0



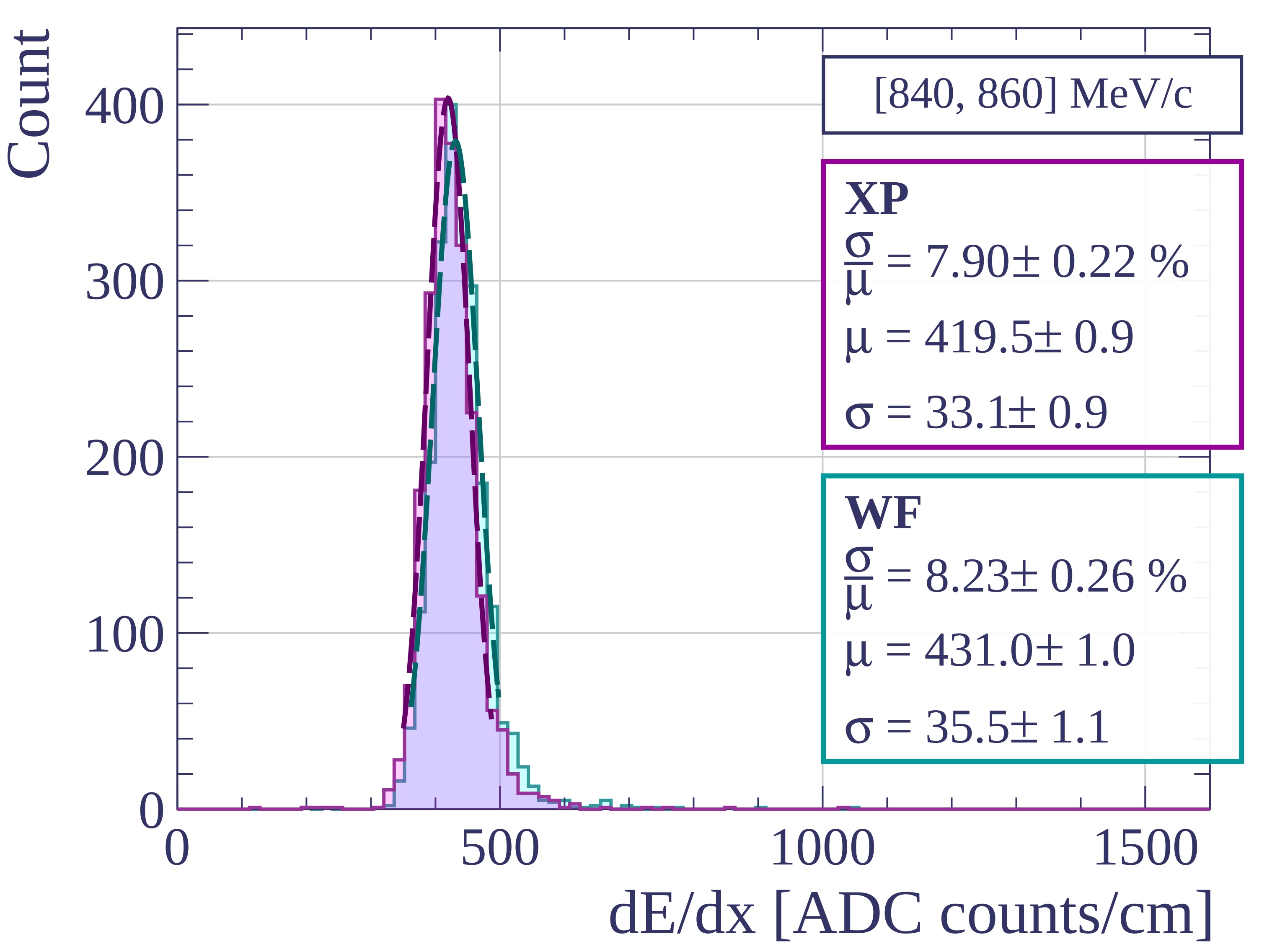
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[780, 800] MeV/c
XP
σ/μ = 8.02± 0.23 %
μ = 414.0± 0.9
σ = 33.2± 0.9
WF
σ/μ = 8.37± 0.22 %
μ = 424.8± 0.9
σ = 35.5± 0.9



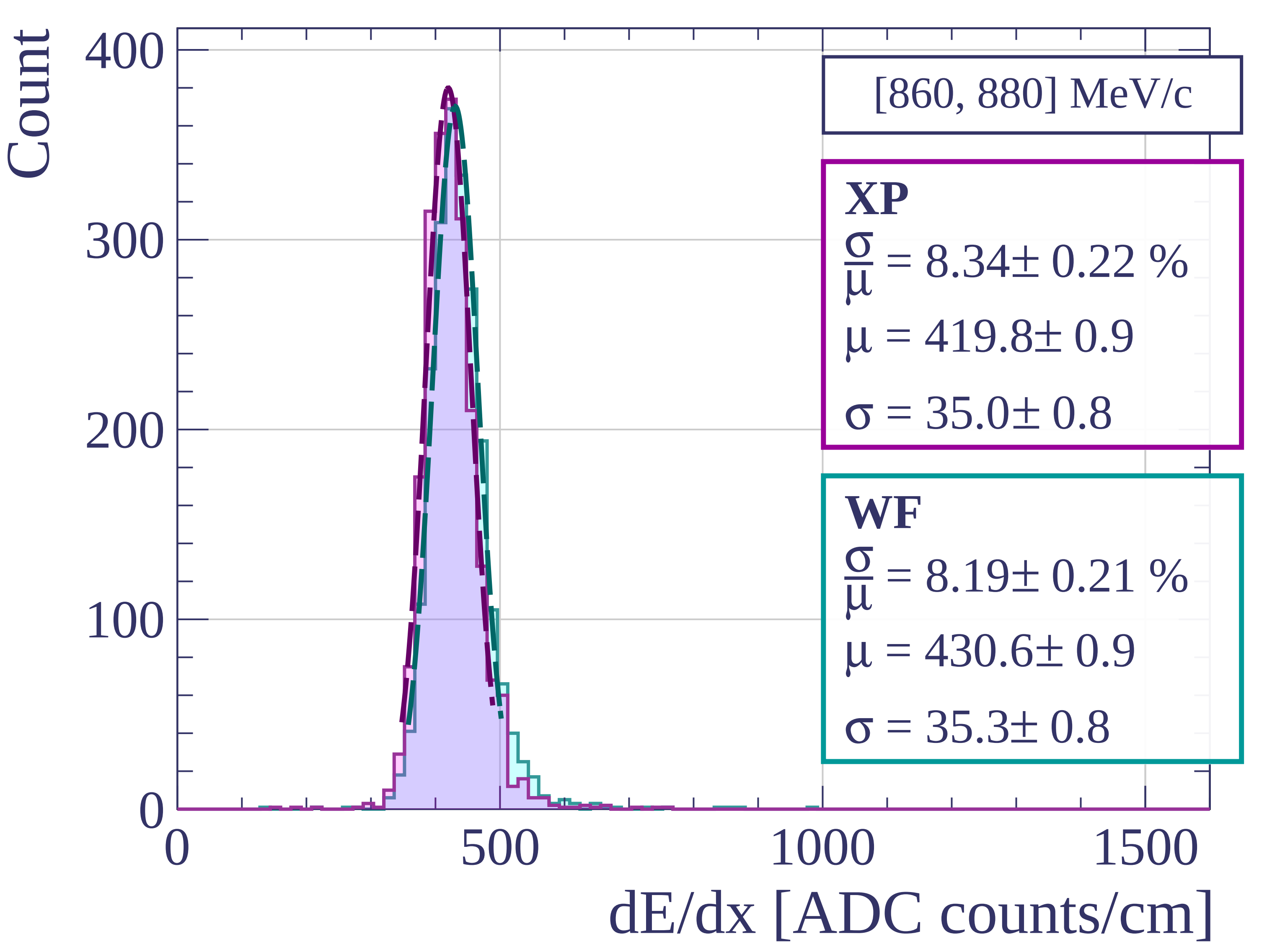
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.03± 0.22 %
μ = 416.4± 0.9
σ = 33.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.47± 0.23 %
μ = 427.5± 1.0
σ = 36.2± 0.9



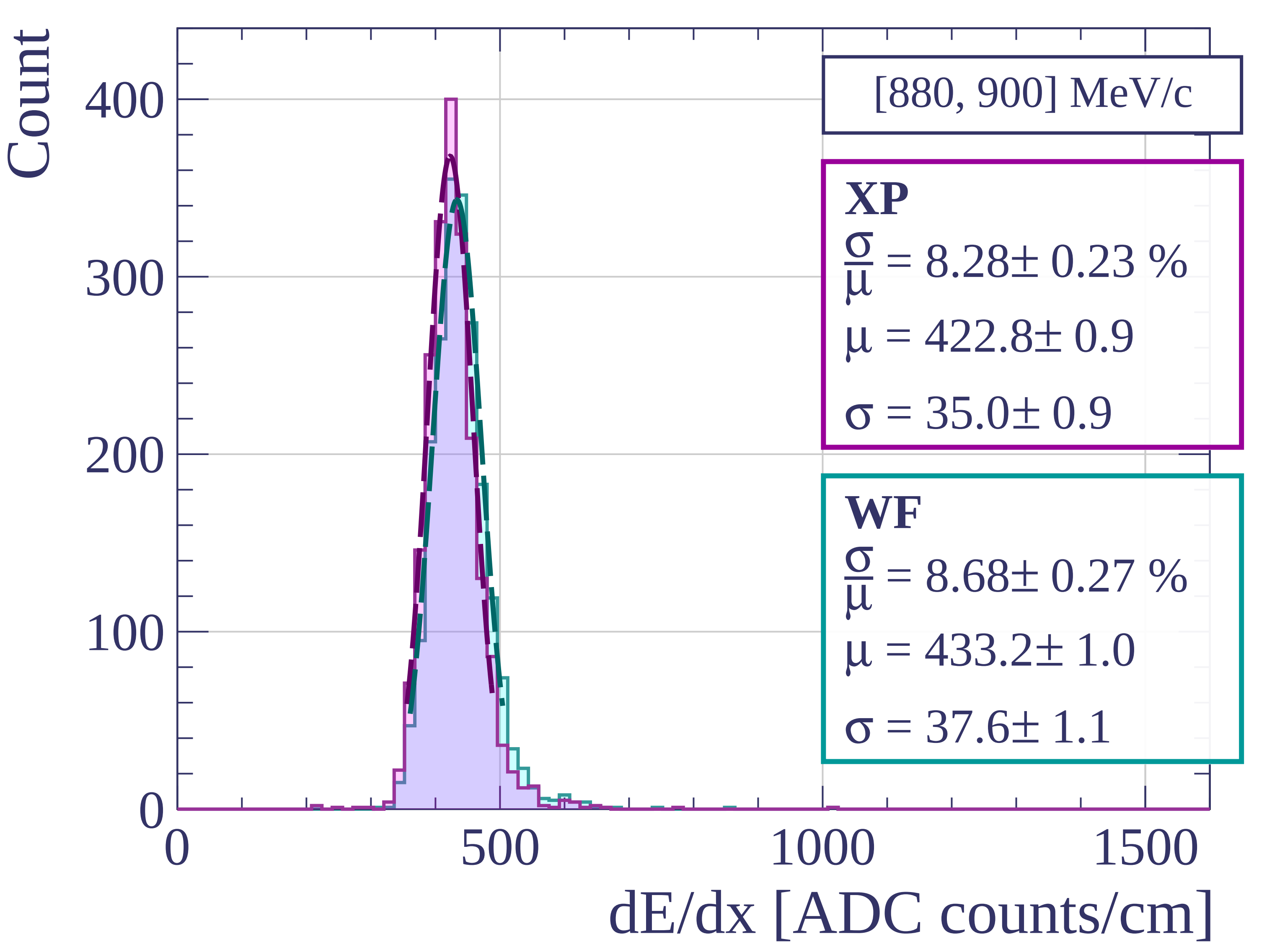
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[820, 840] MeV/c
XP
σ/μ = 8.33± 0.19 %
μ = 419.0± 0.9
σ = 34.9± 0.7
WF
σ/μ = 8.71± 0.22 %
μ = 429.1± 0.9
σ = 37.4± 0.8



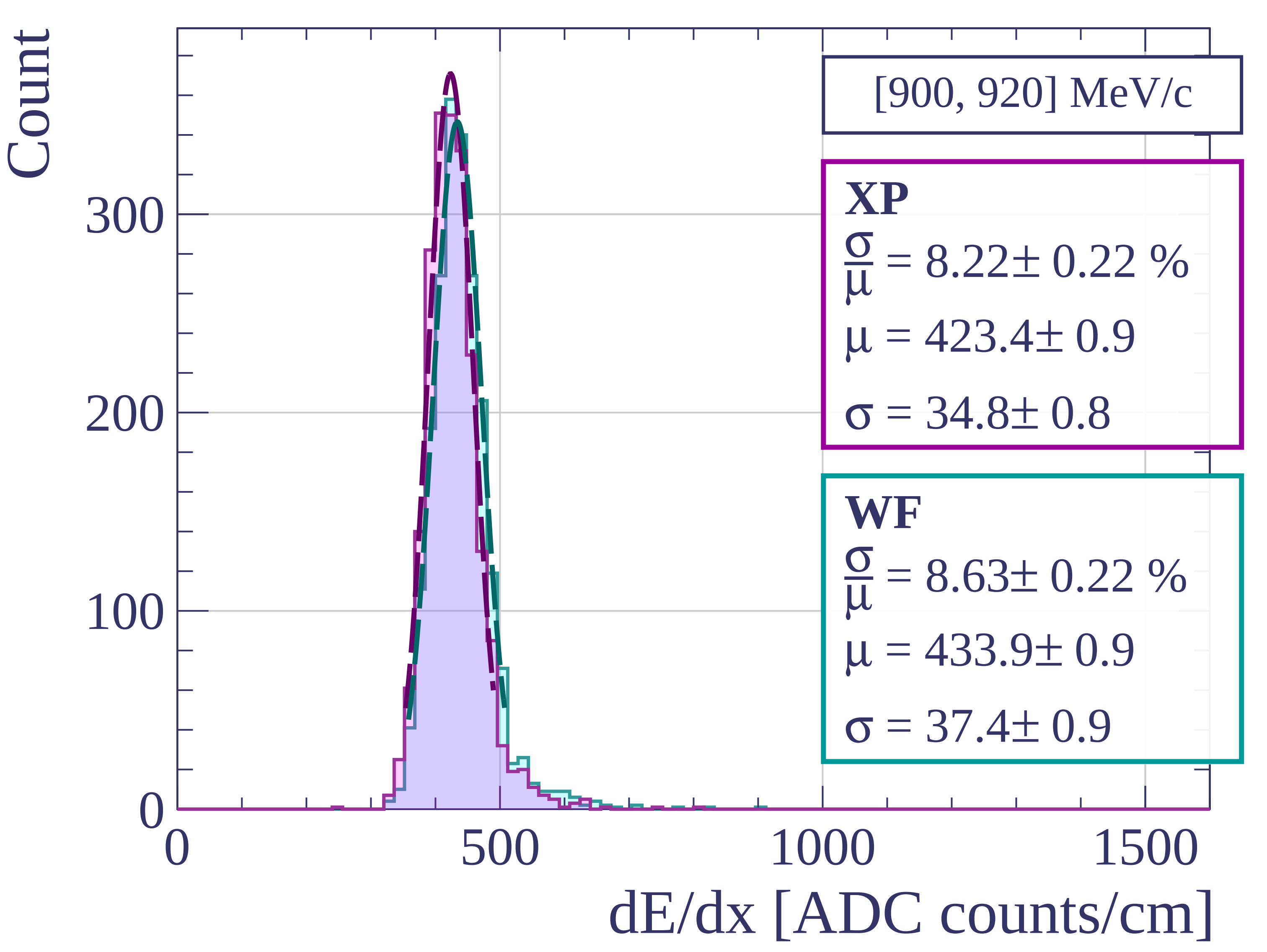
Count
dE/dx [ADC counts/cm]
[840, 860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.90± 0.22 %
μ = 419.5± 0.9
σ = 33.1± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.23± 0.26 %
μ = 431.0± 1.0
σ = 35.5± 1.1
0
100
200
300
400
0
500
1000
1500



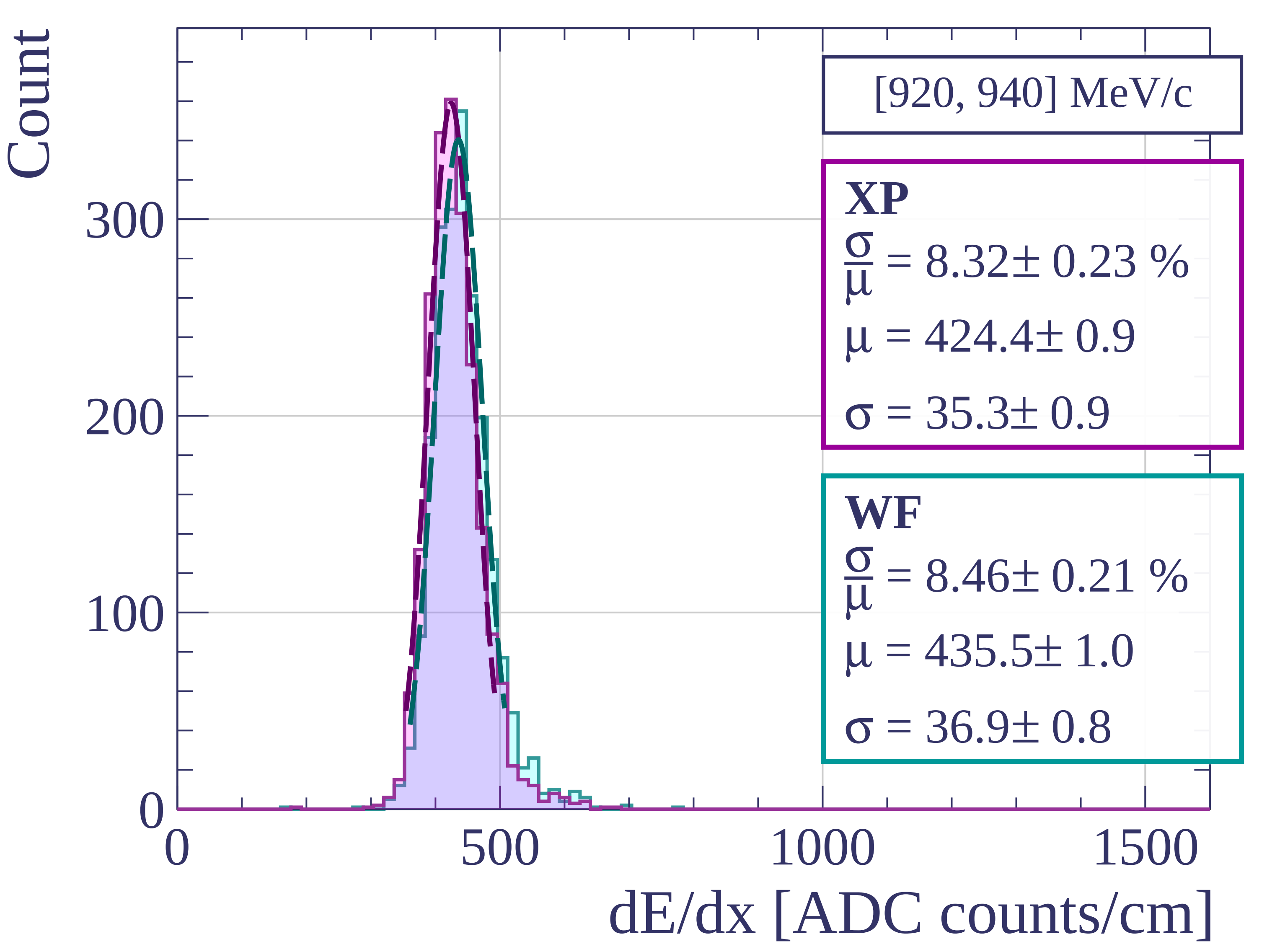
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[860, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.34± 0.22 %
μ = 419.8± 0.9
σ = 35.0± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.19± 0.21 %
μ = 430.6± 0.9
σ = 35.3± 0.8



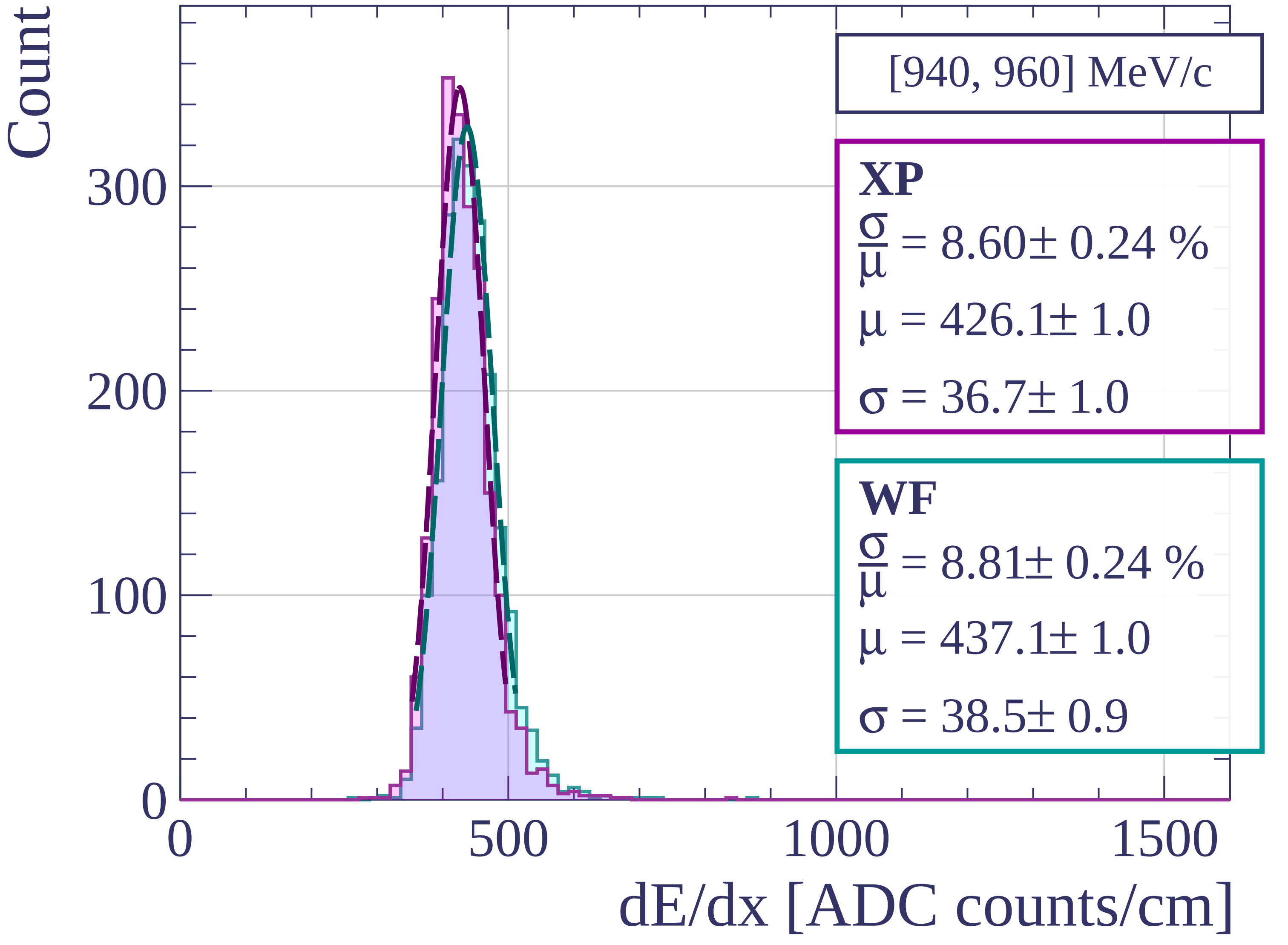
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 900] MeV/c
XP
σ/μ = 8.28± 0.23 %
μ = 422.8± 0.9
σ = 35.0± 0.9
WF
σ/μ = 8.68± 0.27 %
μ = 433.2± 1.0
σ = 37.6± 1.1



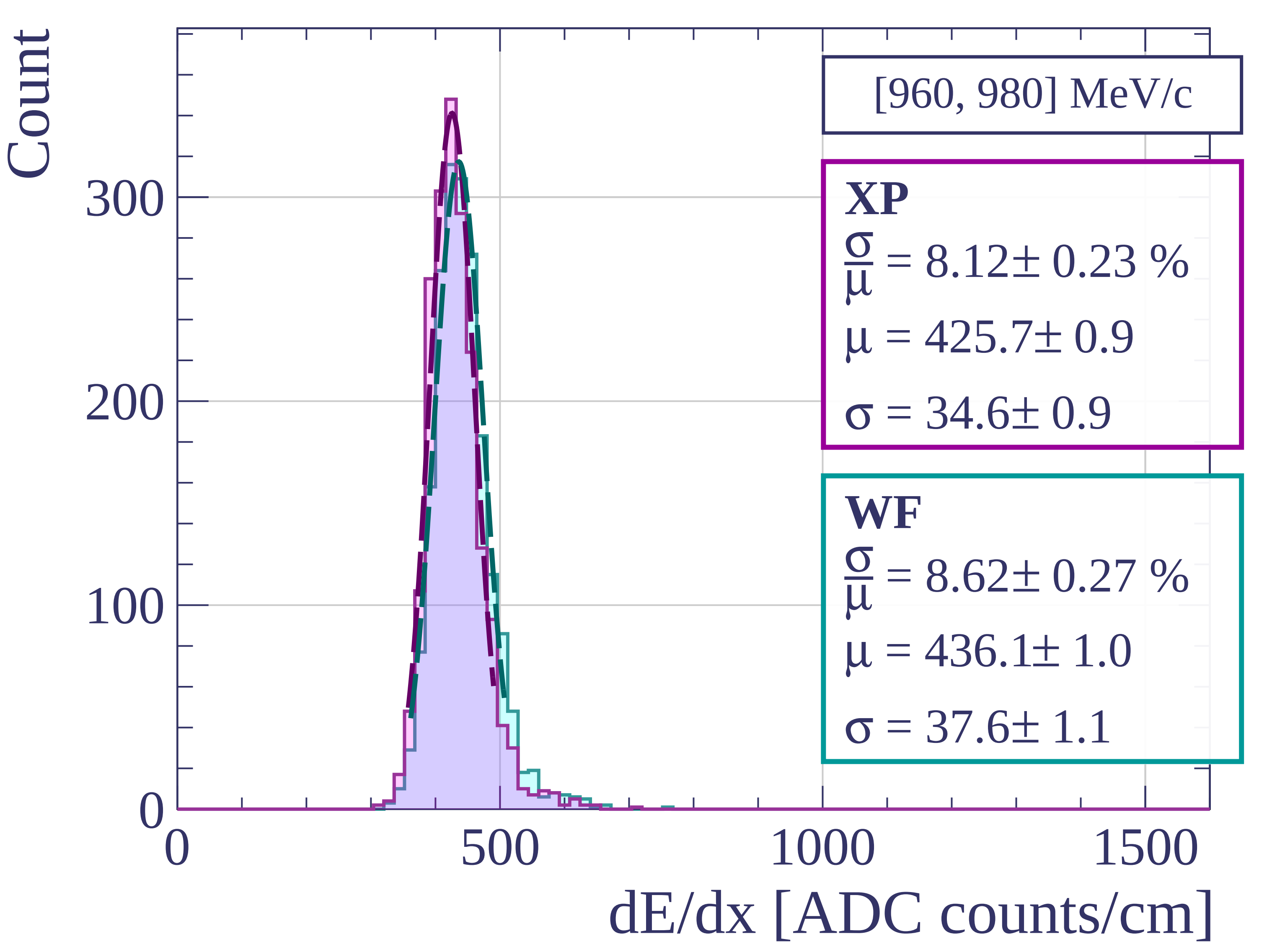
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[900, 920] MeV/c
XP
σ/μ = 8.22± 0.22 %
μ = 423.4± 0.9
σ = 34.8± 0.8
WF
σ/μ = 8.63± 0.22 %
μ = 433.9± 0.9
σ = 37.4± 0.9



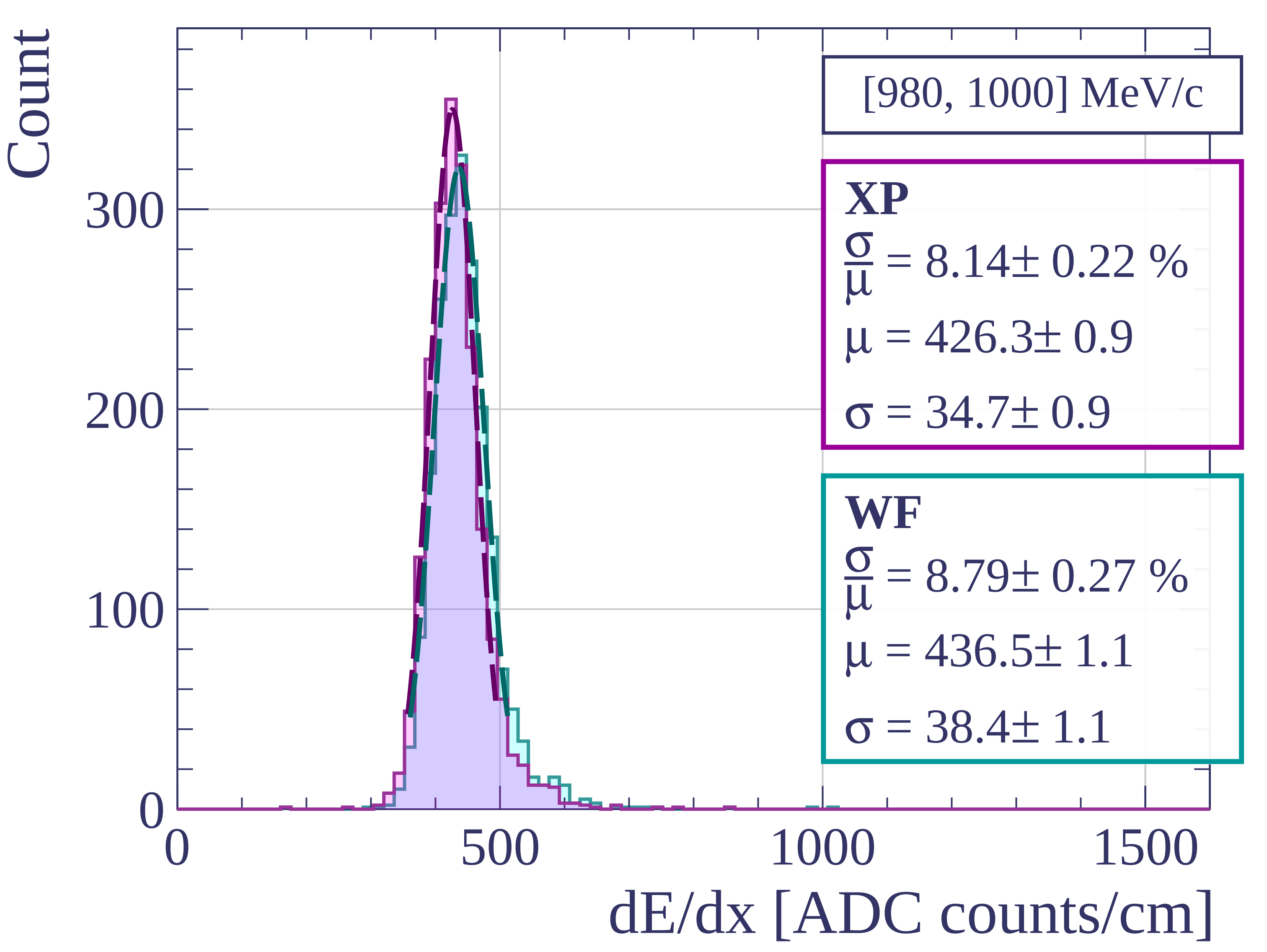
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[920, 940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.32± 0.23 %
μ = 424.4± 0.9
σ = 35.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.21 %
μ = 435.5± 1.0
σ = 36.9± 0.8



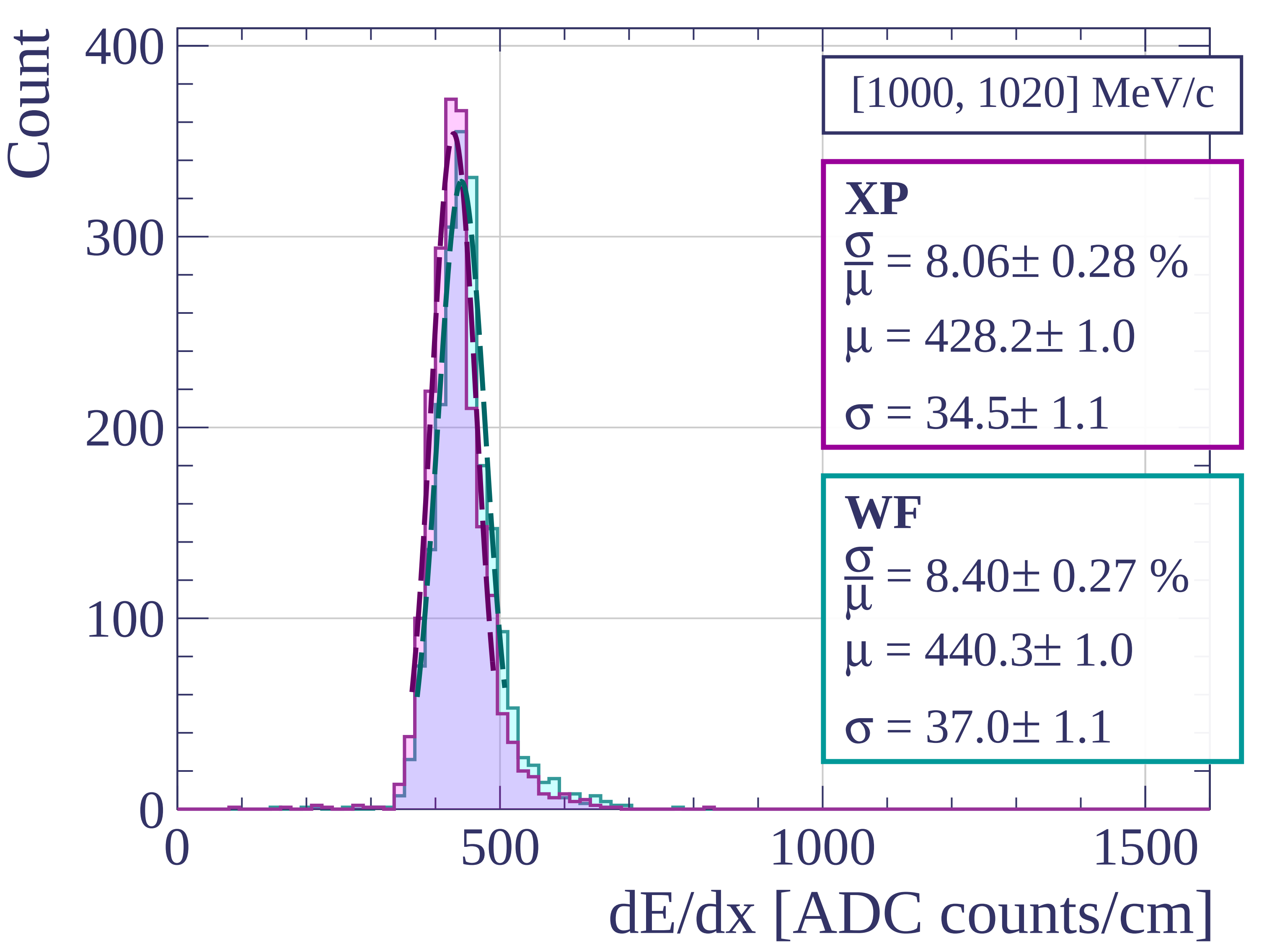
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[940, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.60± 0.24 %
μ = 426.1± 1.0
σ = 36.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.81± 0.24 %
μ = 437.1± 1.0
σ = 38.5± 0.9



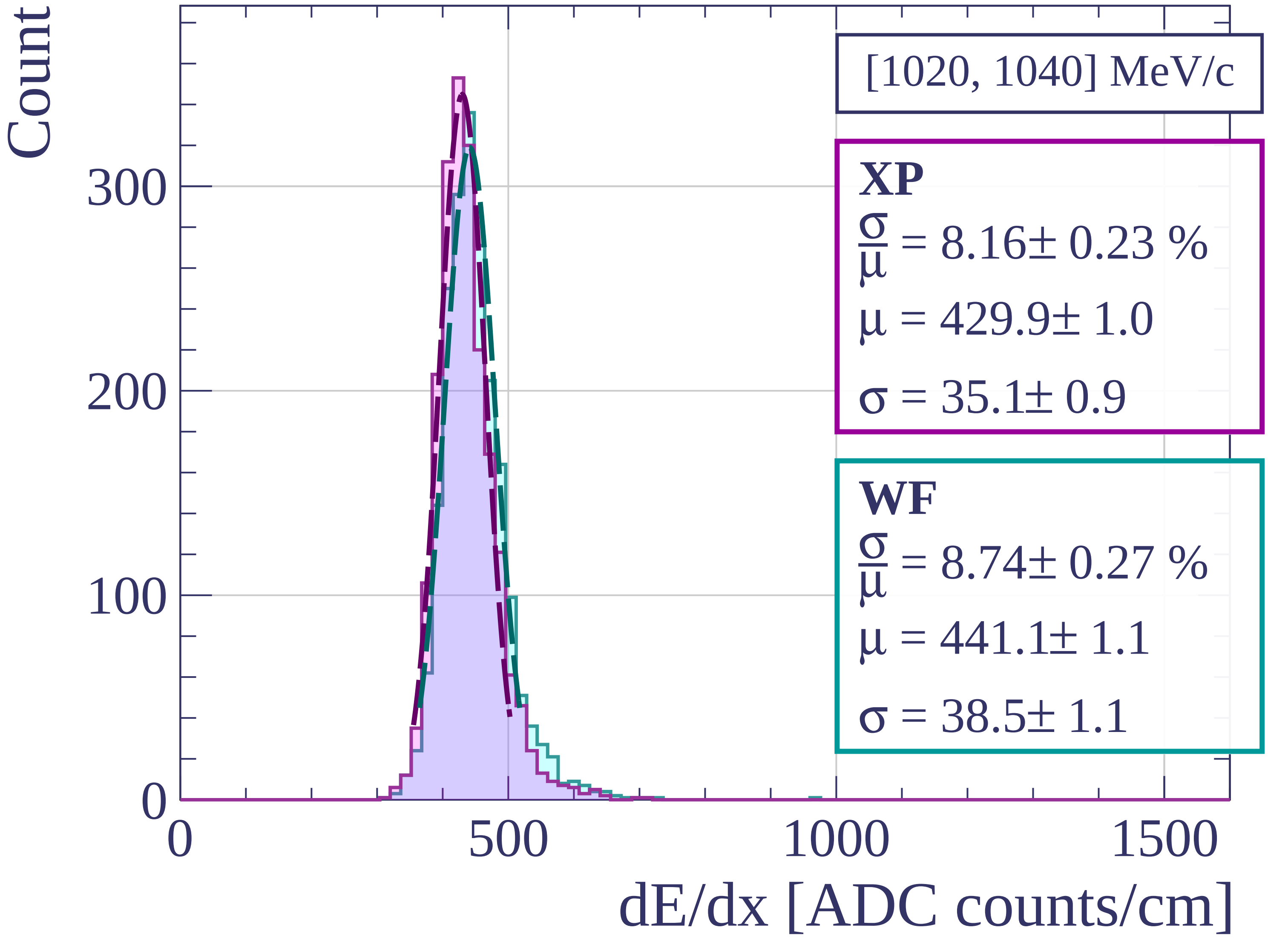
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.12± 0.23 %
μ = 425.7± 0.9
σ = 34.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.27 %
μ = 436.1± 1.0
σ = 37.6± 1.1



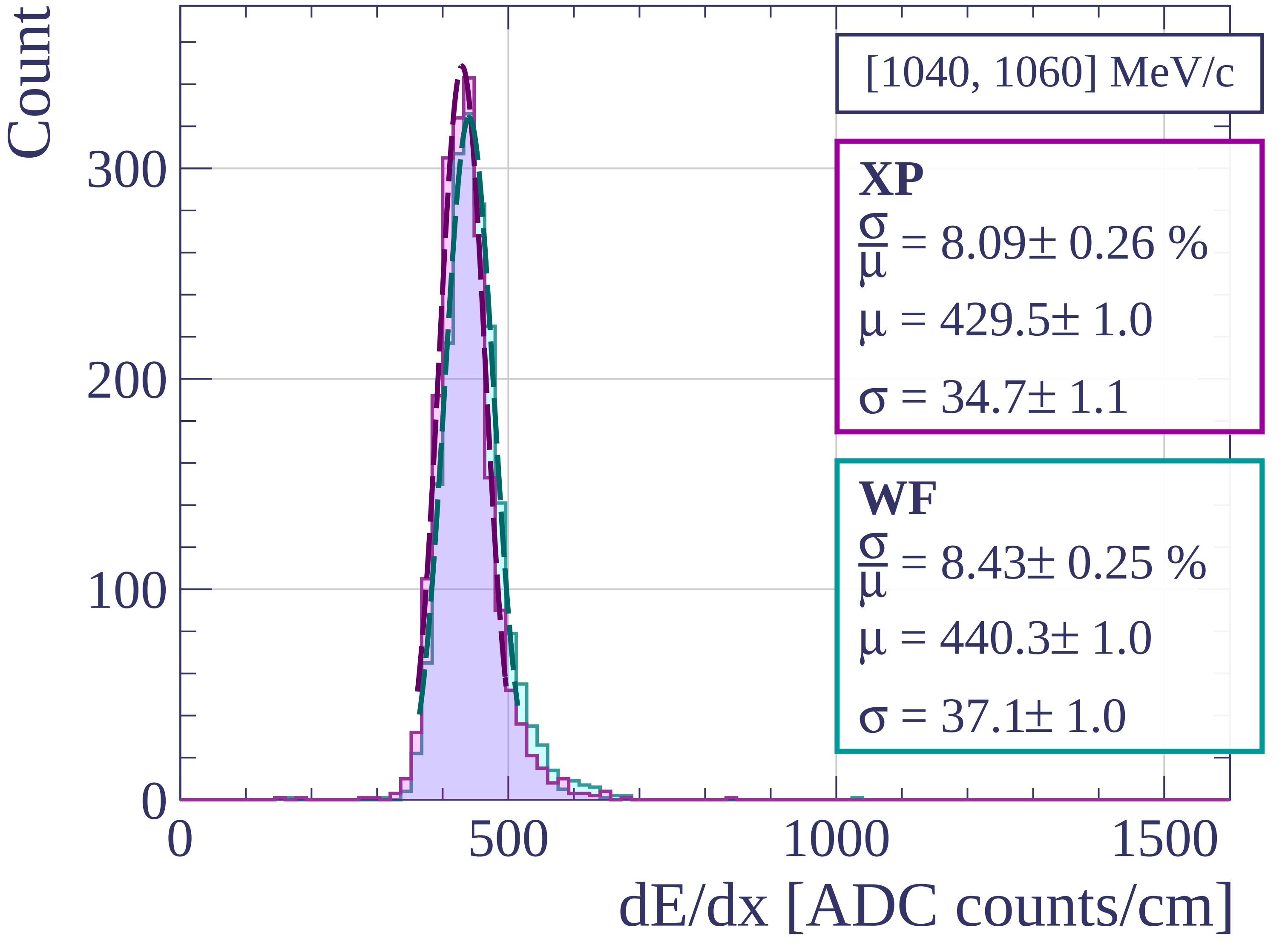
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[980, 1000] MeV/c
XP
σ/μ = 8.14± 0.22 %
μ = 426.3± 0.9
σ = 34.7± 0.9
WF
σ/μ = 8.79± 0.27 %
μ = 436.5± 1.1
σ = 38.4± 1.1



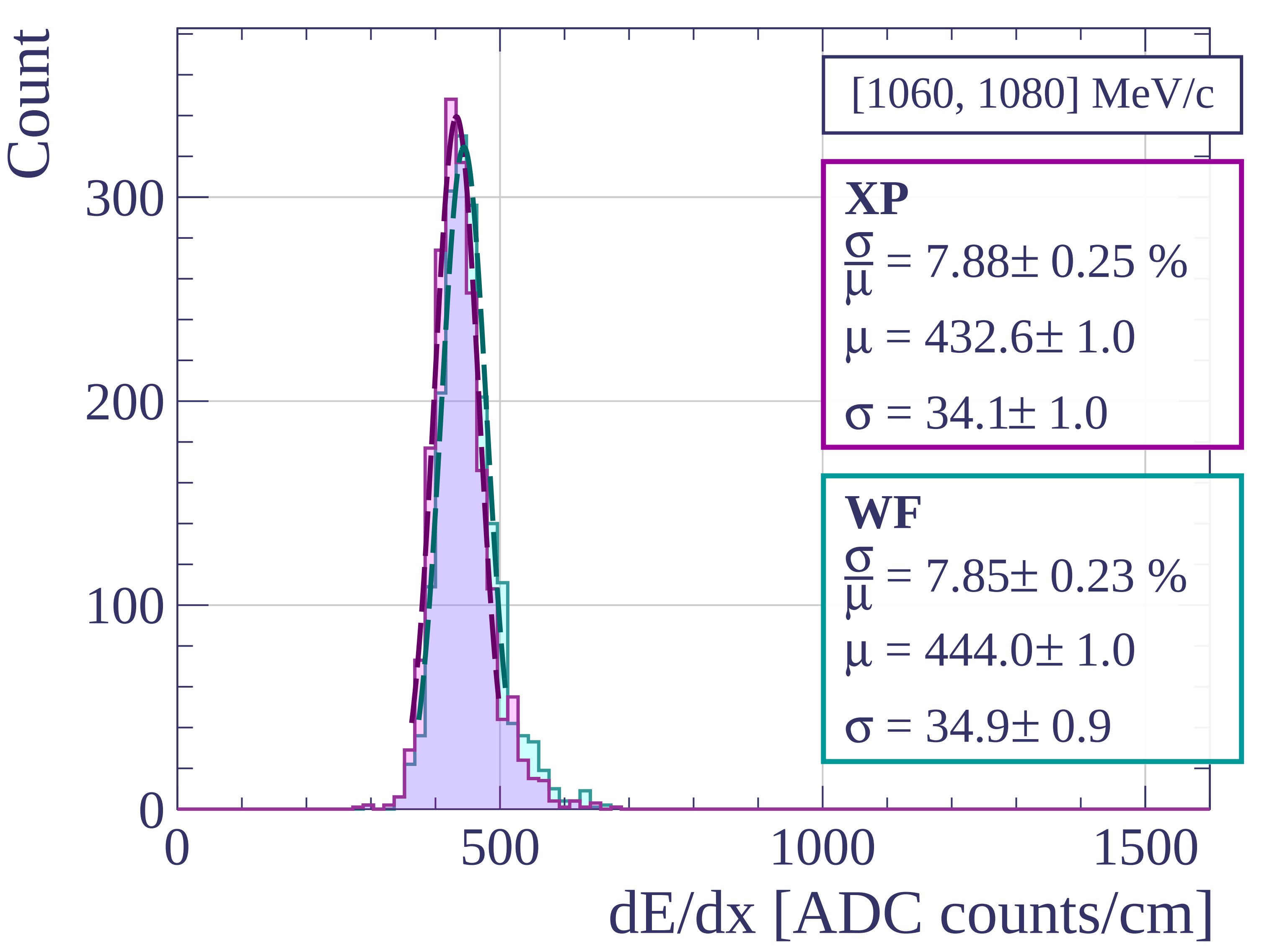
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1000, 1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.06± 0.28 %
μ = 428.2± 1.0
σ = 34.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.40± 0.27 %
μ = 440.3± 1.0
σ = 37.0± 1.1



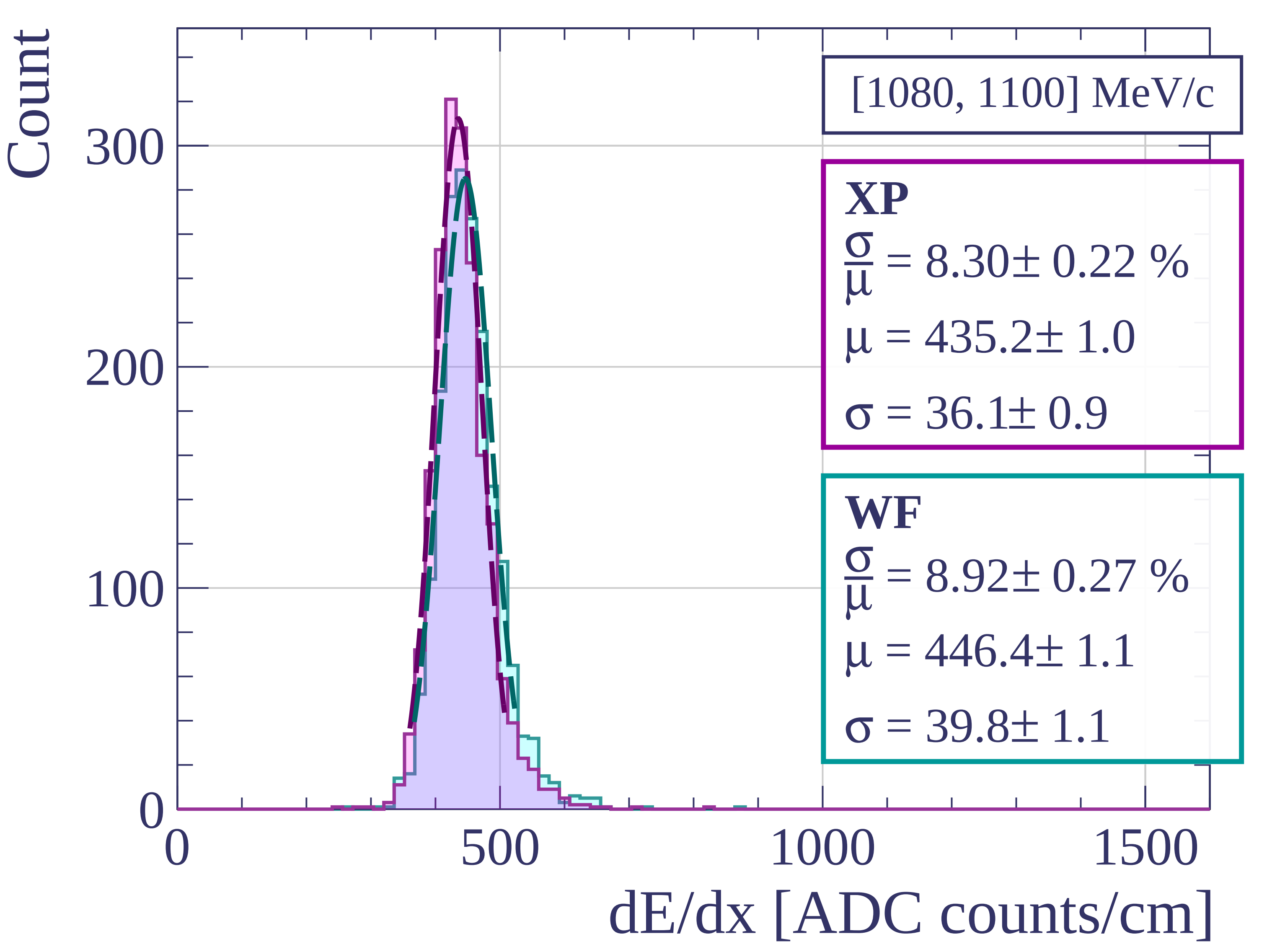
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1020, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.16± 0.23 %
μ = 429.9± 1.0
σ = 35.1± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.74± 0.27 %
μ = 441.1± 1.1
σ = 38.5± 1.1



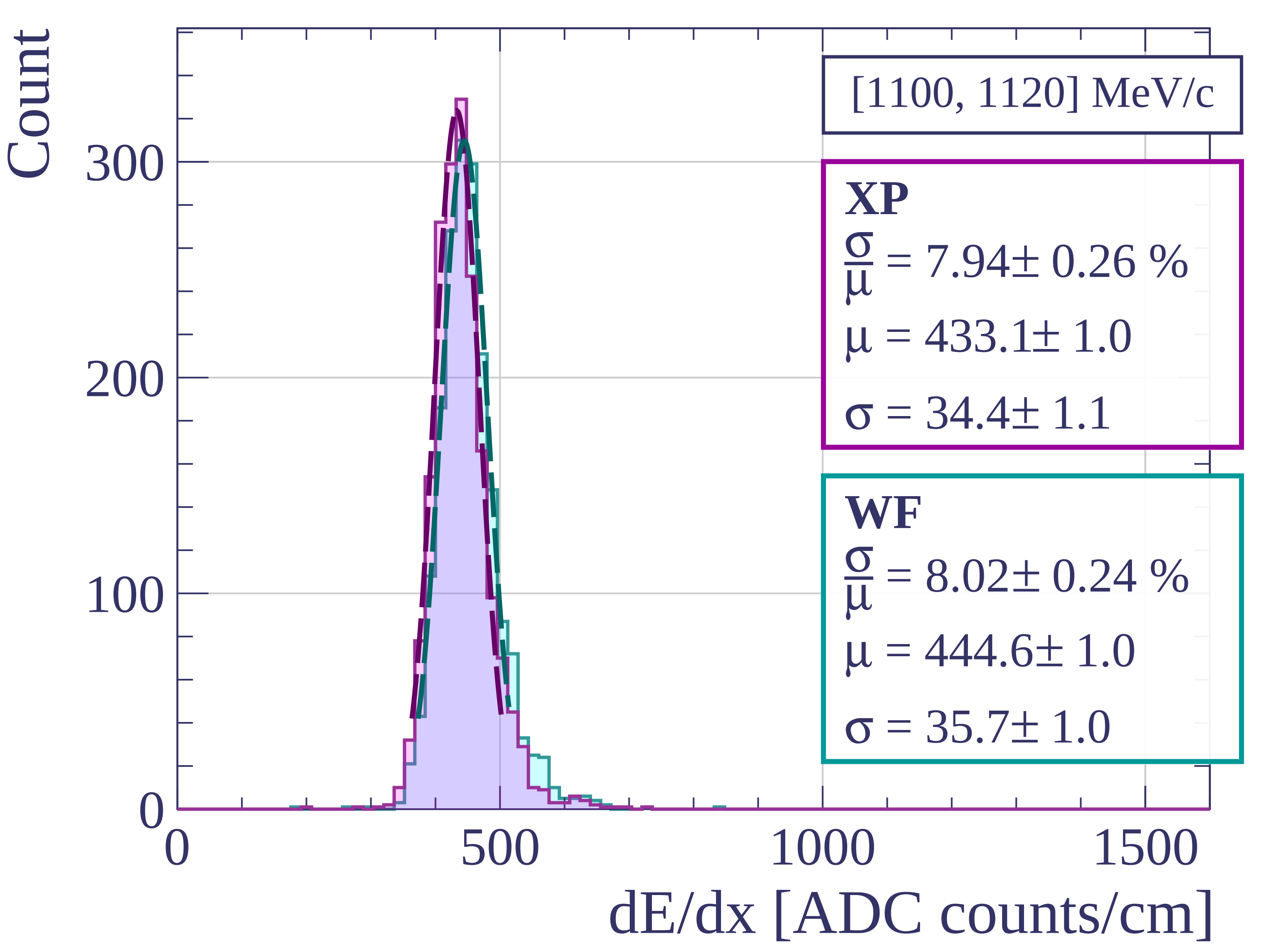
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.26 %
μ = 429.5± 1.0
σ = 34.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.43± 0.25 %
μ = 440.3± 1.0
σ = 37.1± 1.0

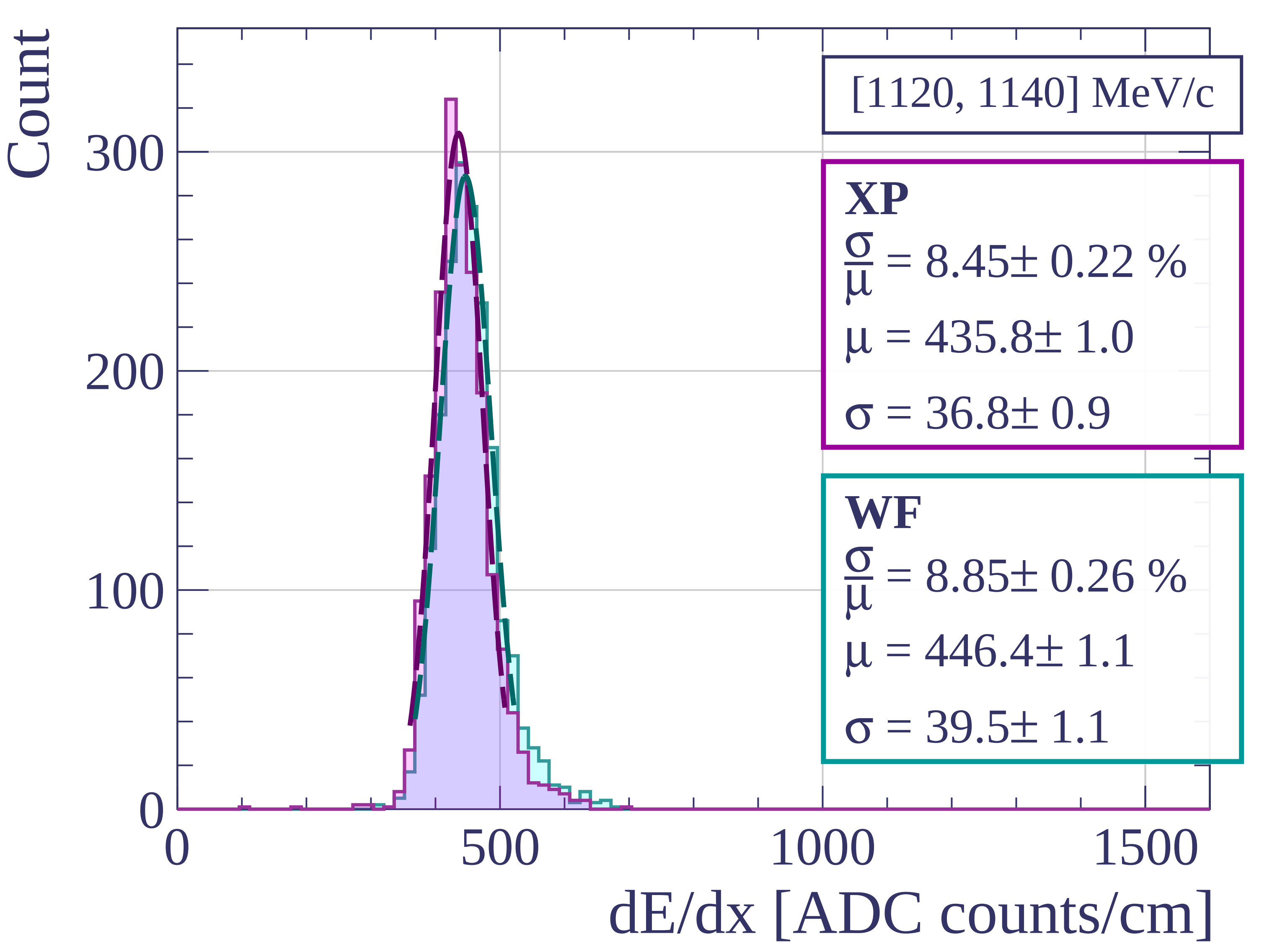


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1060, 1080] MeV/c
XP
σ/μ = 7.88± 0.25 %
μ = 432.6± 1.0
σ = 34.1± 1.0
WF
σ/μ = 7.85± 0.23 %
μ = 444.0± 1.0
σ = 34.9± 0.9

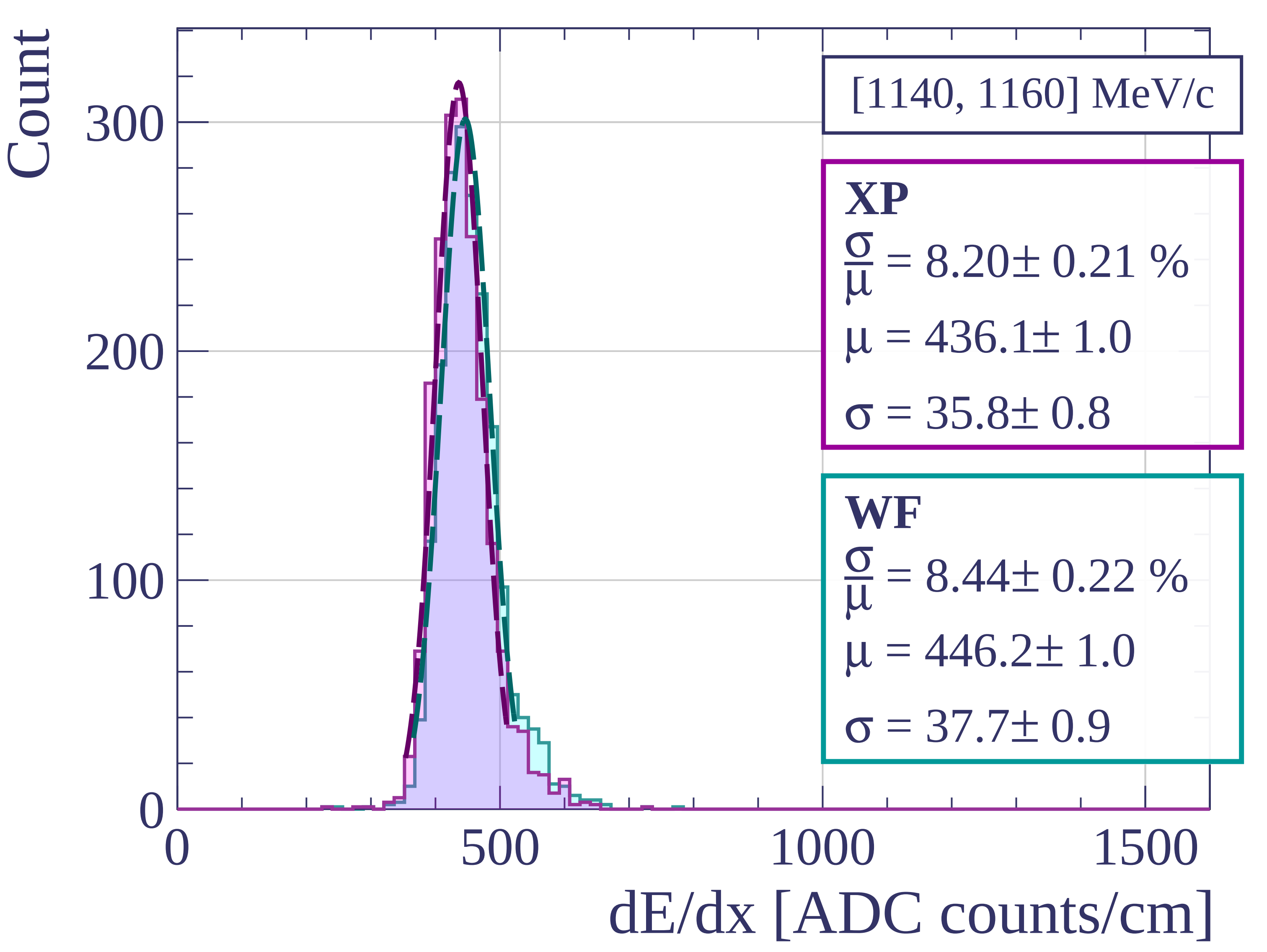


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1080, 1100] MeV/c
XP
σ/μ = 8.30± 0.22 %
μ = 435.2± 1.0
σ = 36.1± 0.9
WF
σ/μ = 8.92± 0.27 %
μ = 446.4± 1.1
σ = 39.8± 1.1

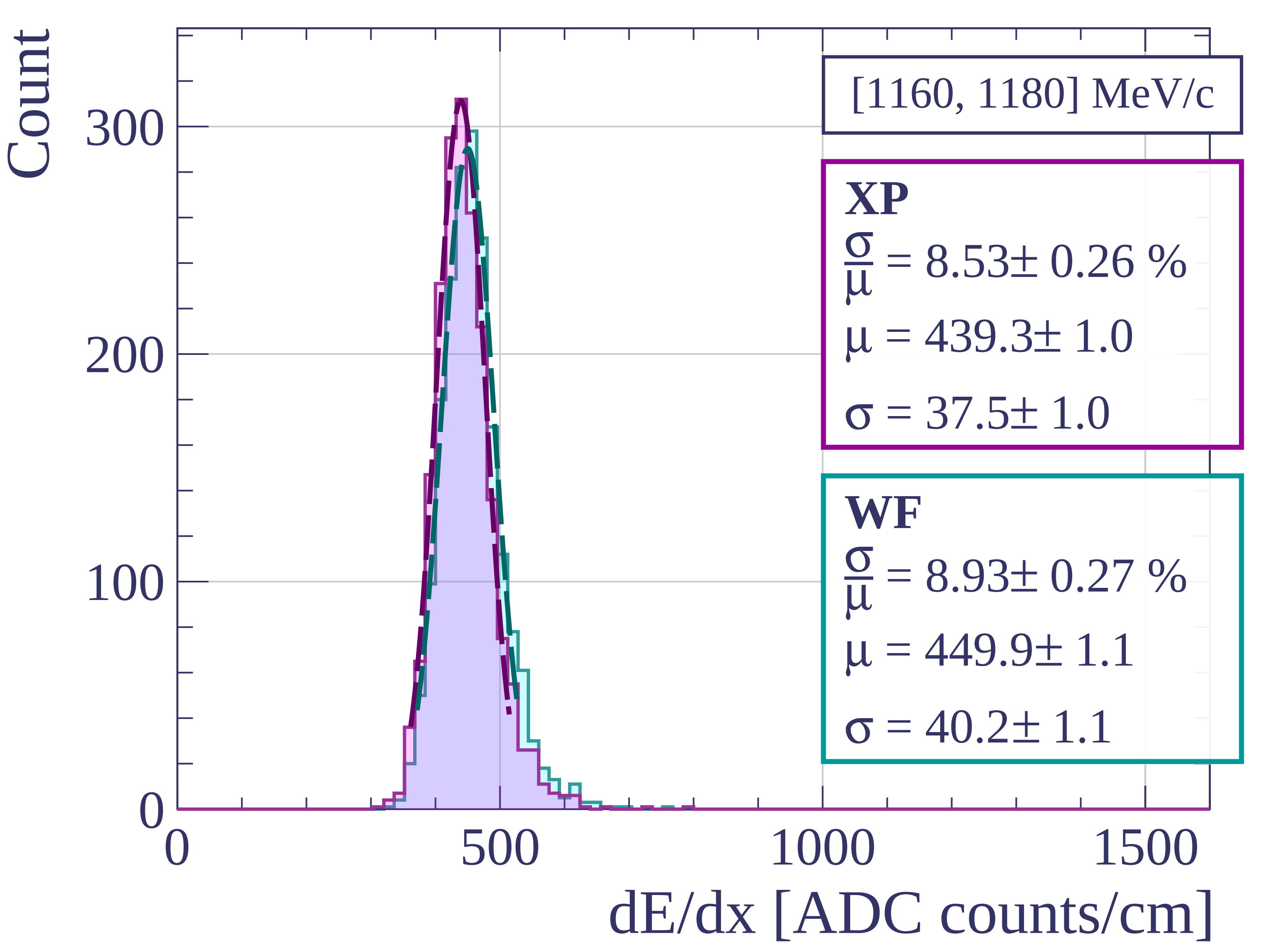

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1100, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.94± 0.26 %
μ = 433.1± 1.0
σ = 34.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.02± 0.24 %
μ = 444.6± 1.0
σ = 35.7± 1.0

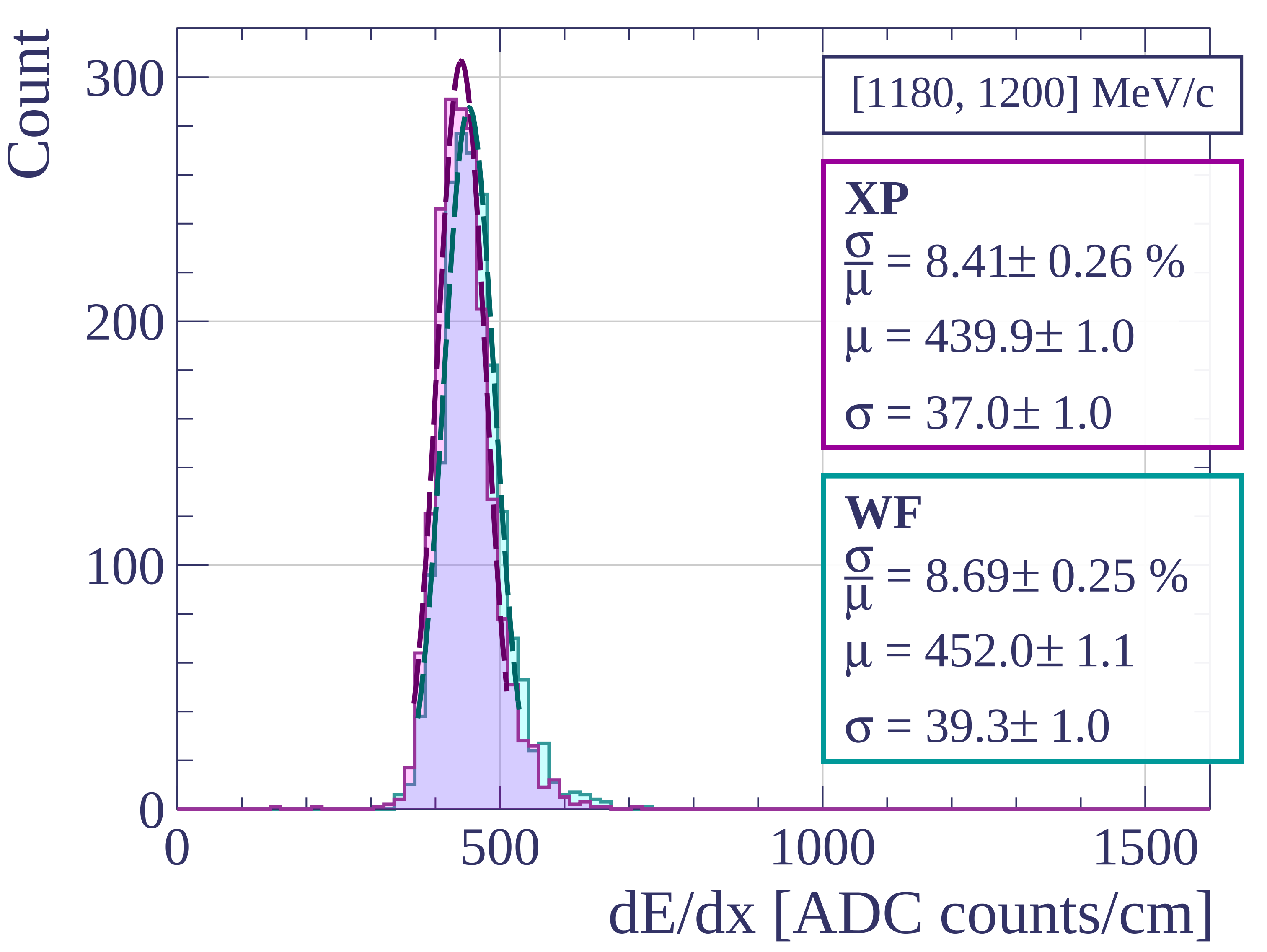


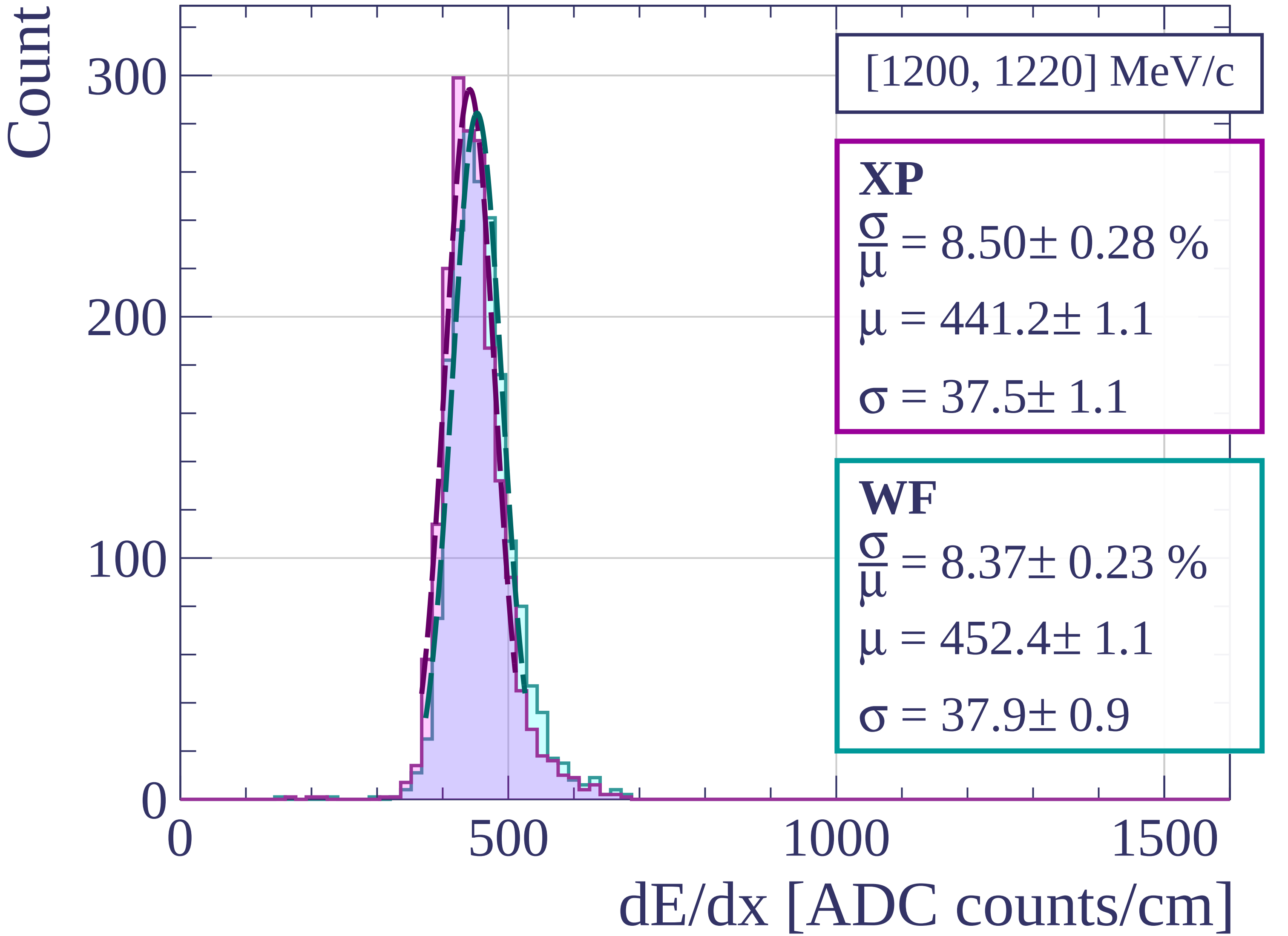
Count
0
100
200
300
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.45± 0.22 %
μ = 435.8± 1.0
σ = 36.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.85± 0.26 %
μ = 446.4± 1.1
σ = 39.5± 1.1



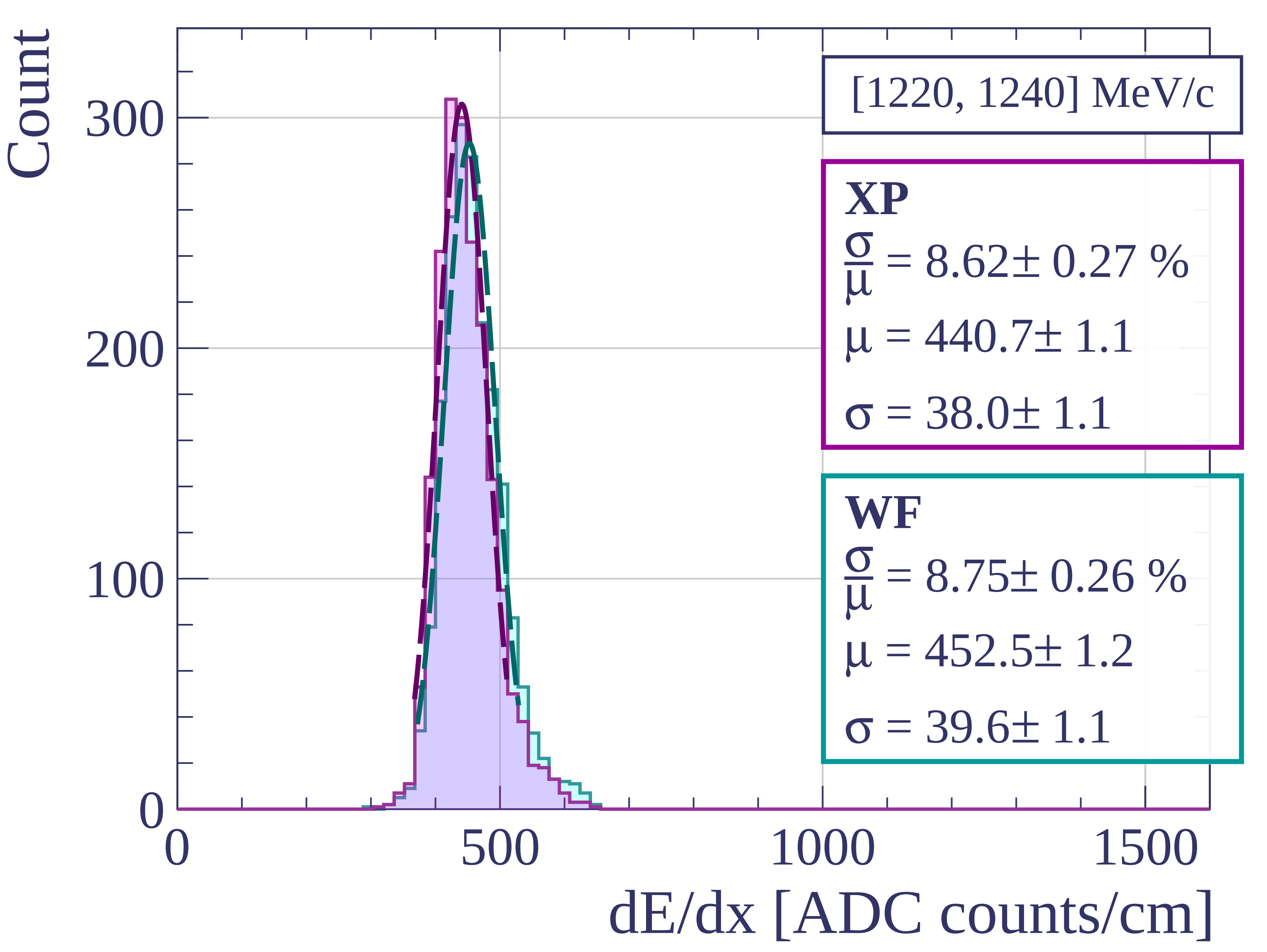
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1140, 1160] MeV/c
XP
σ/μ = 8.20± 0.21 %
μ = 436.1± 1.0
σ = 35.8± 0.8
WF
σ/μ = 8.44± 0.22 %
μ = 446.2± 1.0
σ = 37.7± 0.9


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1160, 1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.53± 0.26 %
μ = 439.3± 1.0
σ = 37.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.93± 0.27 %
μ = 449.9± 1.1
σ = 40.2± 1.1

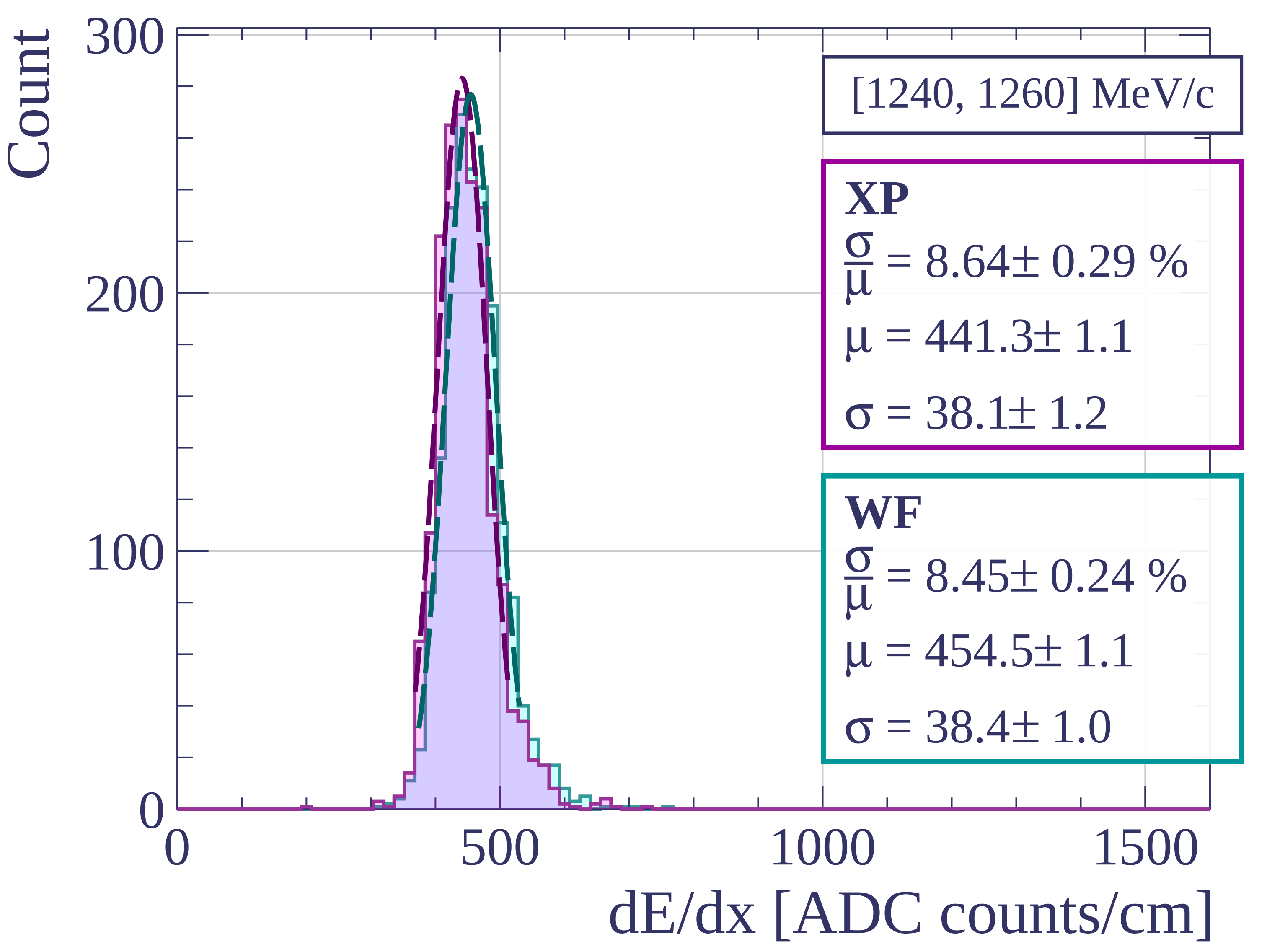

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1180, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.41± 0.26 %
μ = 439.9± 1.0
σ = 37.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.69± 0.25 %
μ = 452.0± 1.1
σ = 39.3± 1.0



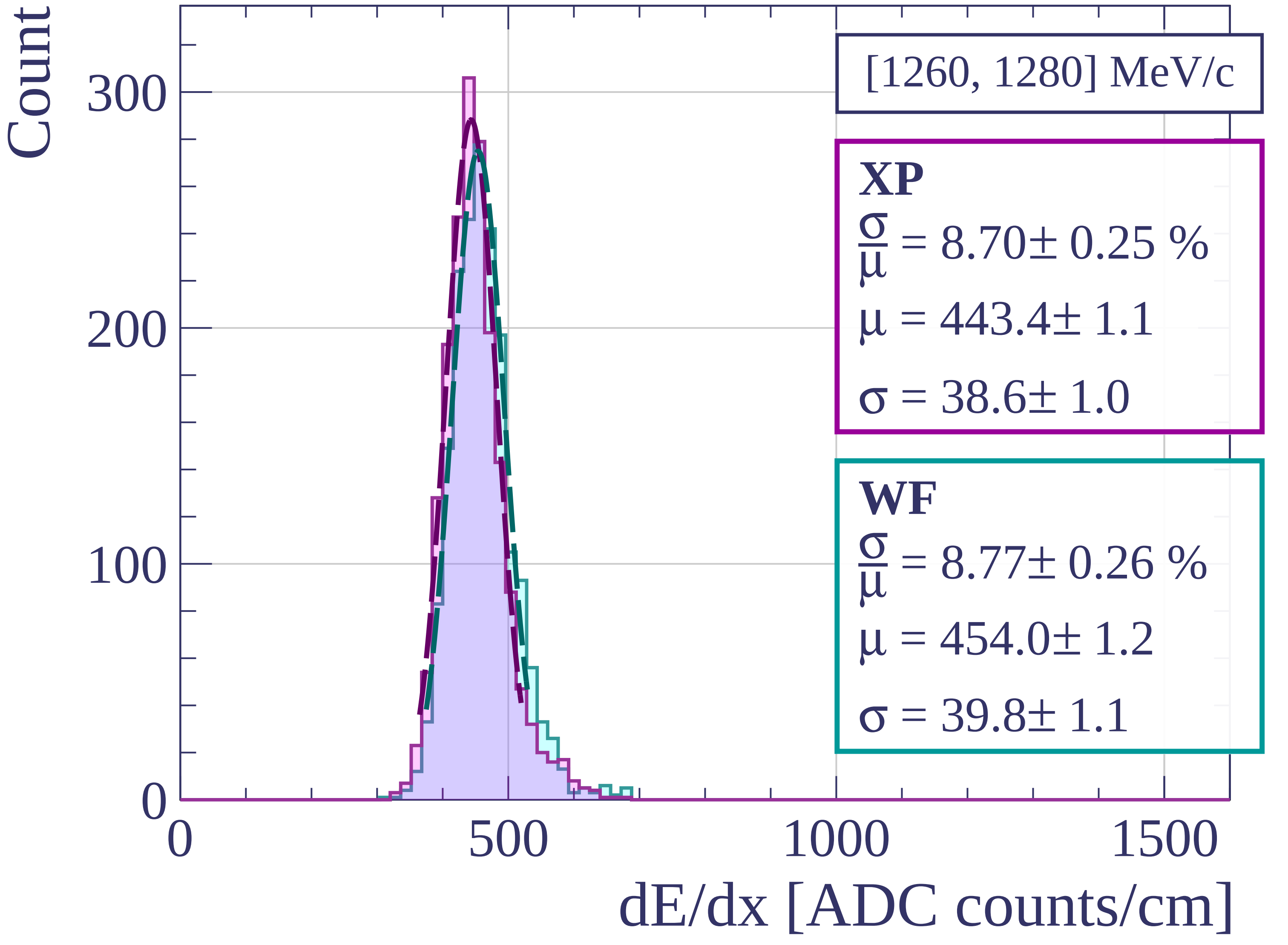
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1220] MeV/c
XP
σ/μ = 8.50± 0.28 %
μ = 441.2± 1.1
σ = 37.5± 1.1
WF
σ/μ = 8.37± 0.23 %
μ = 452.4± 1.1
σ = 37.9± 0.9



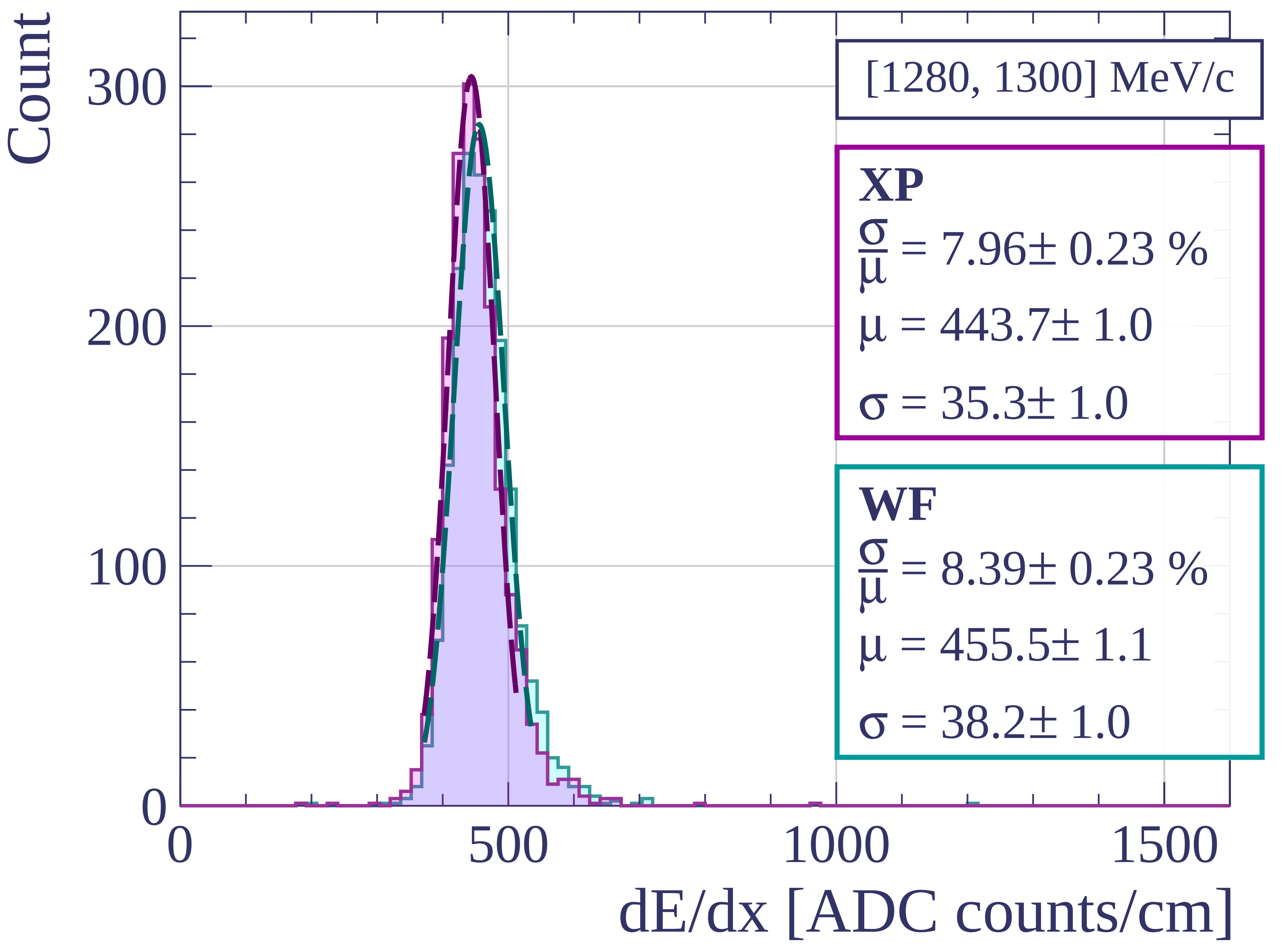
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1220, 1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.62± 0.27 %
μ = 440.7± 1.1
σ = 38.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.75± 0.26 %
μ = 452.5± 1.2
σ = 39.6± 1.1

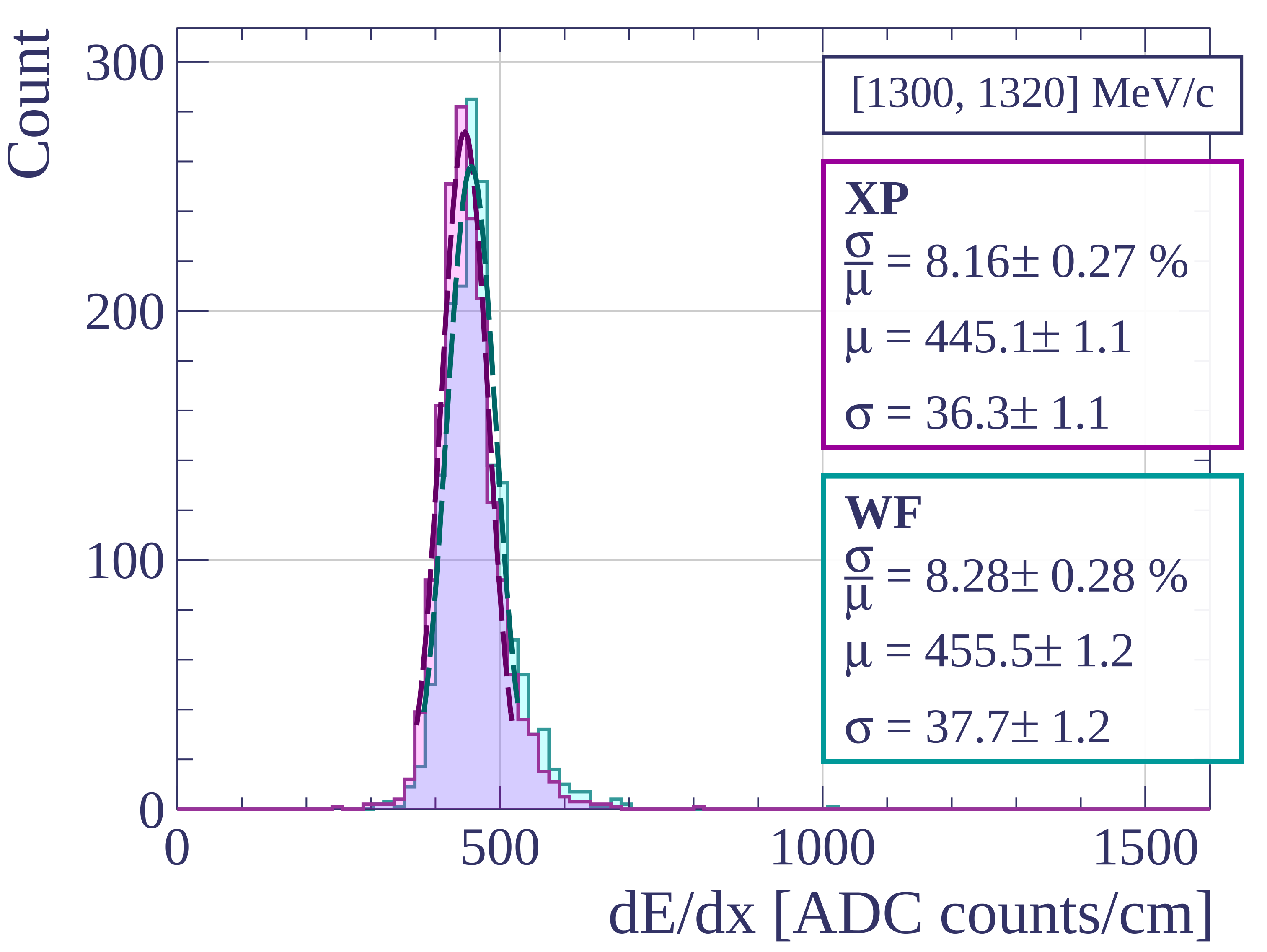


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1240, 1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.64± 0.29 %
μ = 441.3± 1.1
σ = 38.1± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.45± 0.24 %
μ = 454.5± 1.1
σ = 38.4± 1.0

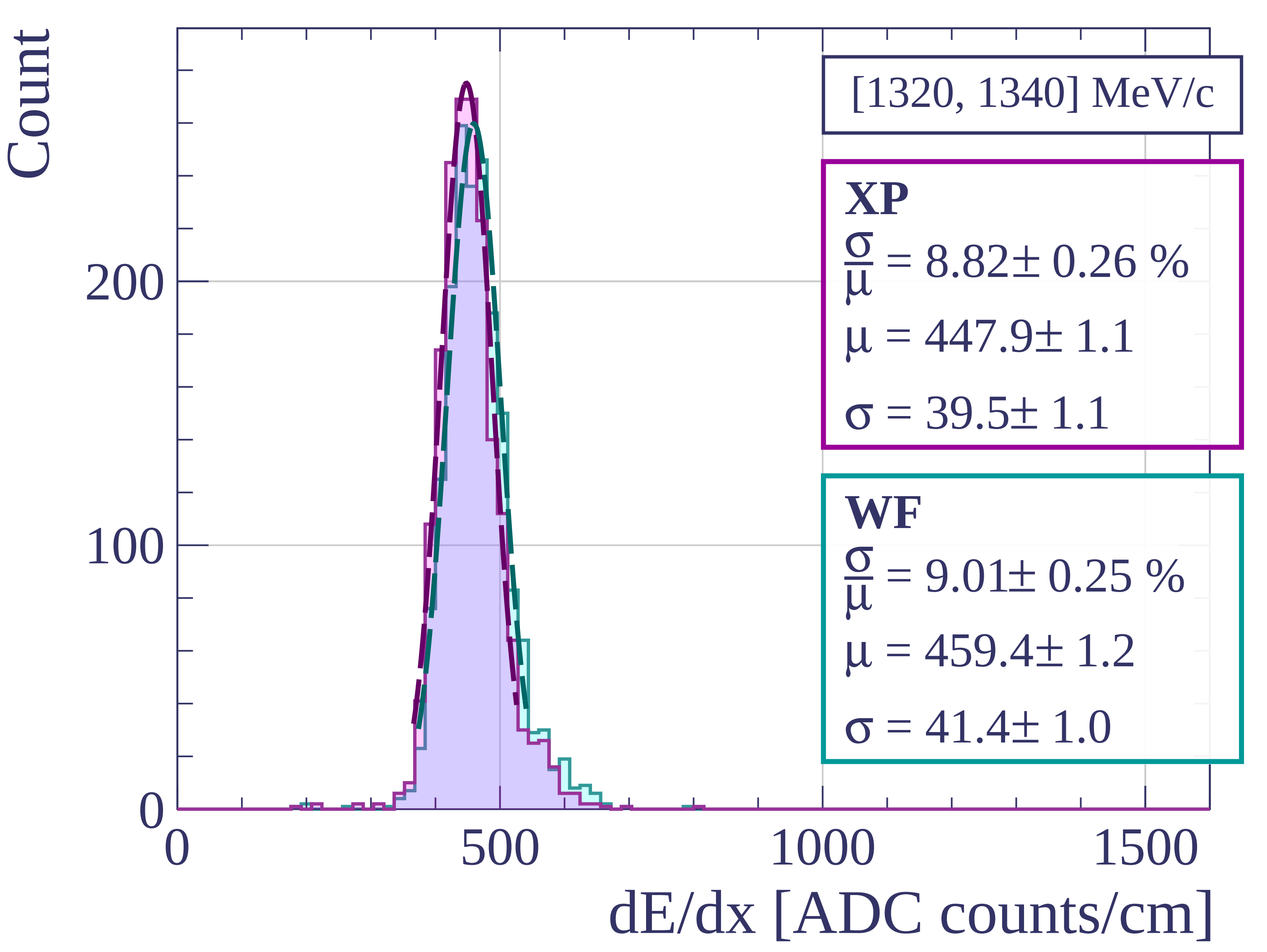


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1260, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.70± 0.25 %
μ = 443.4± 1.1
σ = 38.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.77± 0.26 %
μ = 454.0± 1.2
σ = 39.8± 1.1

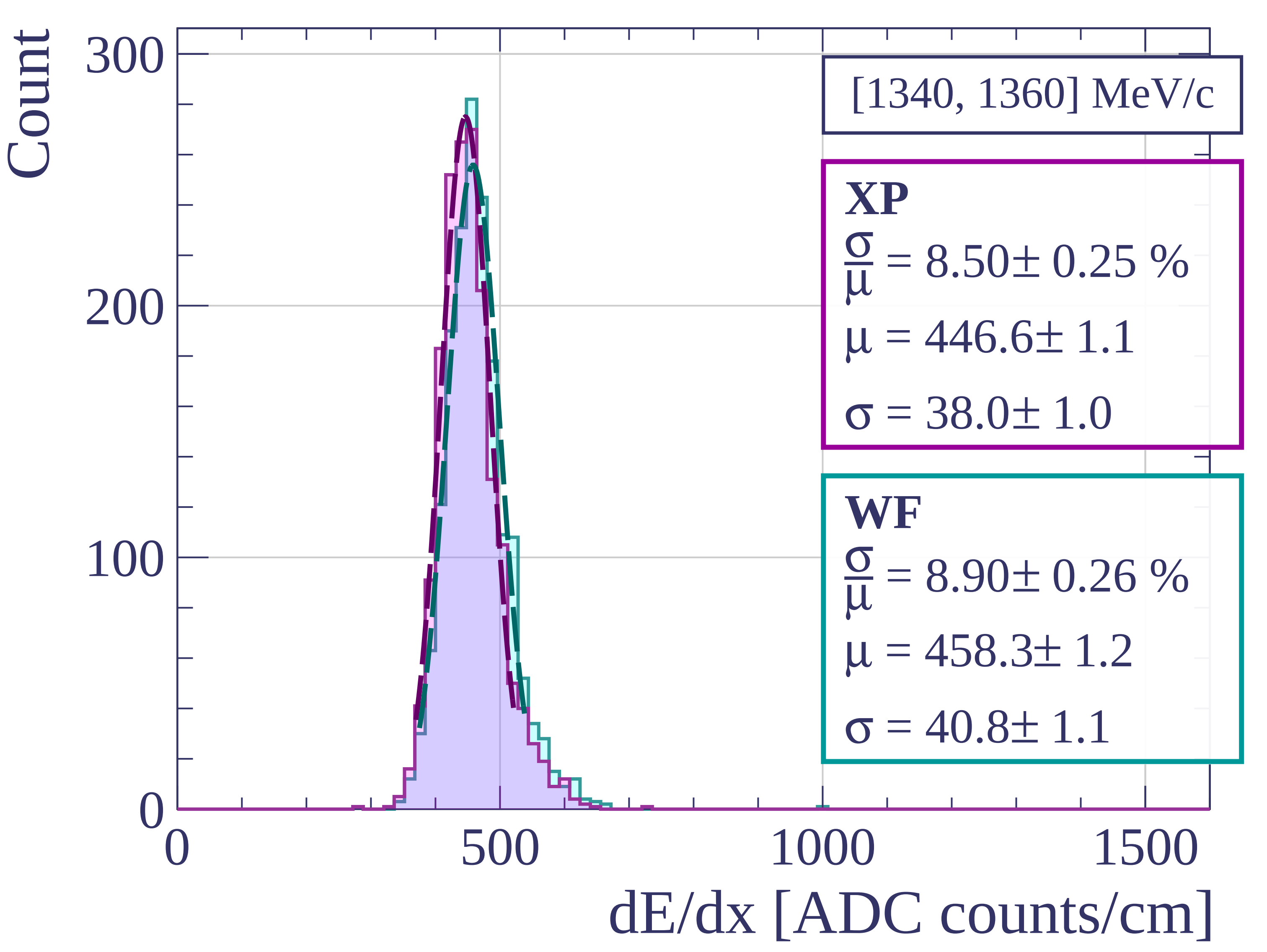

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.96± 0.23 %
μ = 443.7± 1.0
σ = 35.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.39± 0.23 %
μ = 455.5± 1.1
σ = 38.2± 1.0



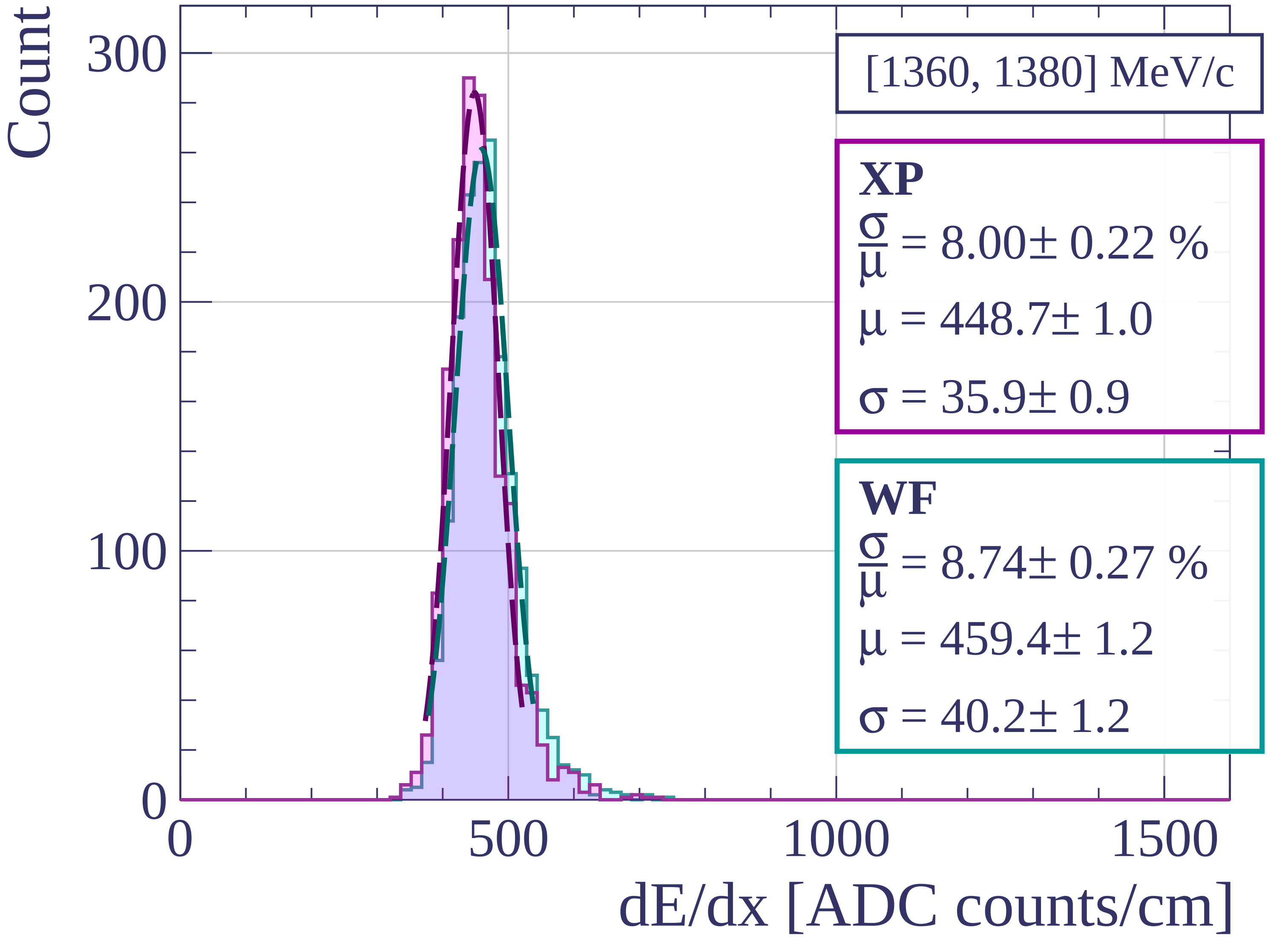
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1300, 1320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.16± 0.27 %
μ = 445.1± 1.1
σ = 36.3± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.28 %
μ = 455.5± 1.2
σ = 37.7± 1.2

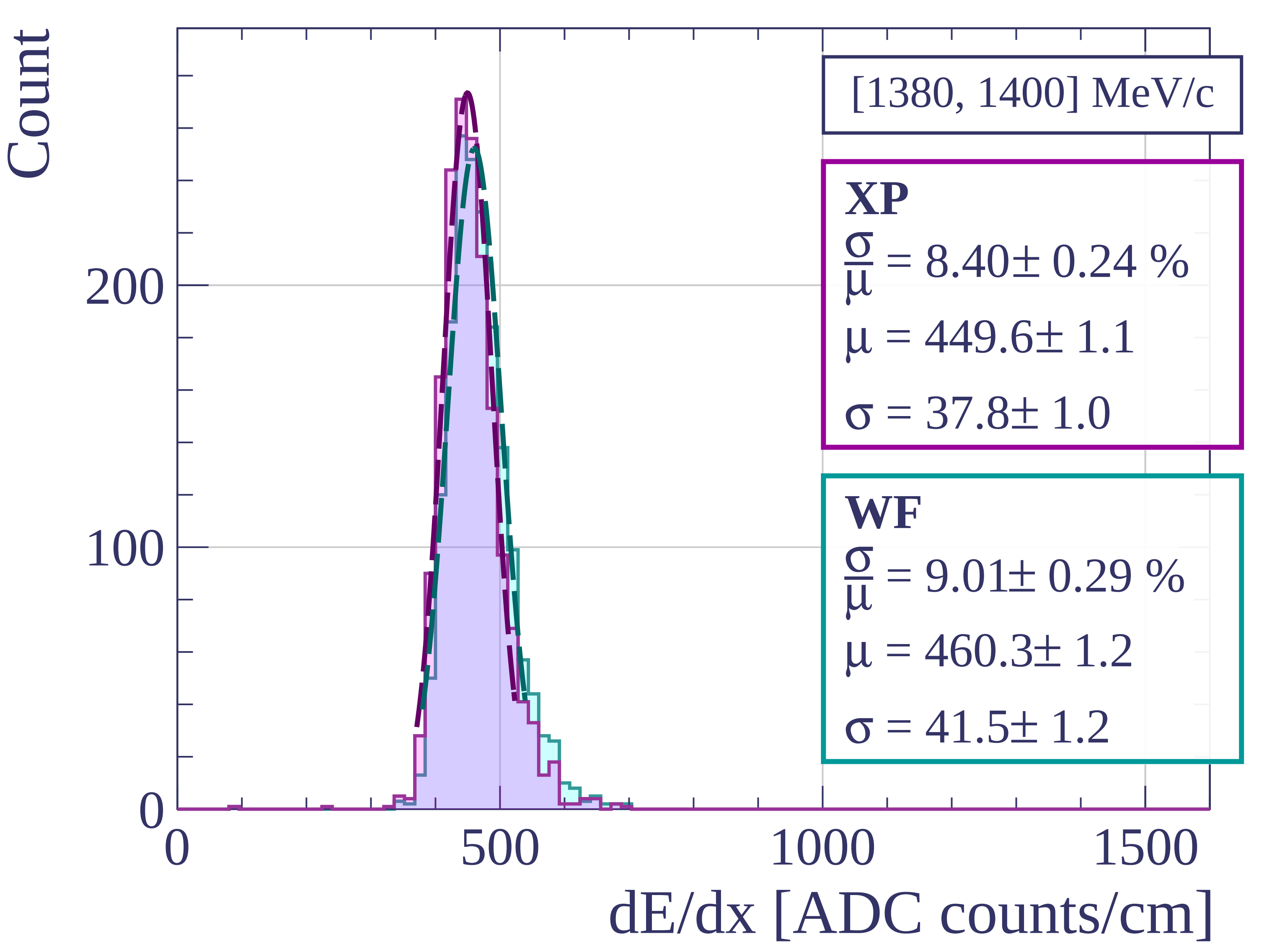


Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
[1320, 1340] MeV/c
XP
σ/μ = 8.82± 0.26 %
μ = 447.9± 1.1
σ = 39.5± 1.1
WF
σ/μ = 9.01± 0.25 %
μ = 459.4± 1.2
σ = 41.4± 1.0

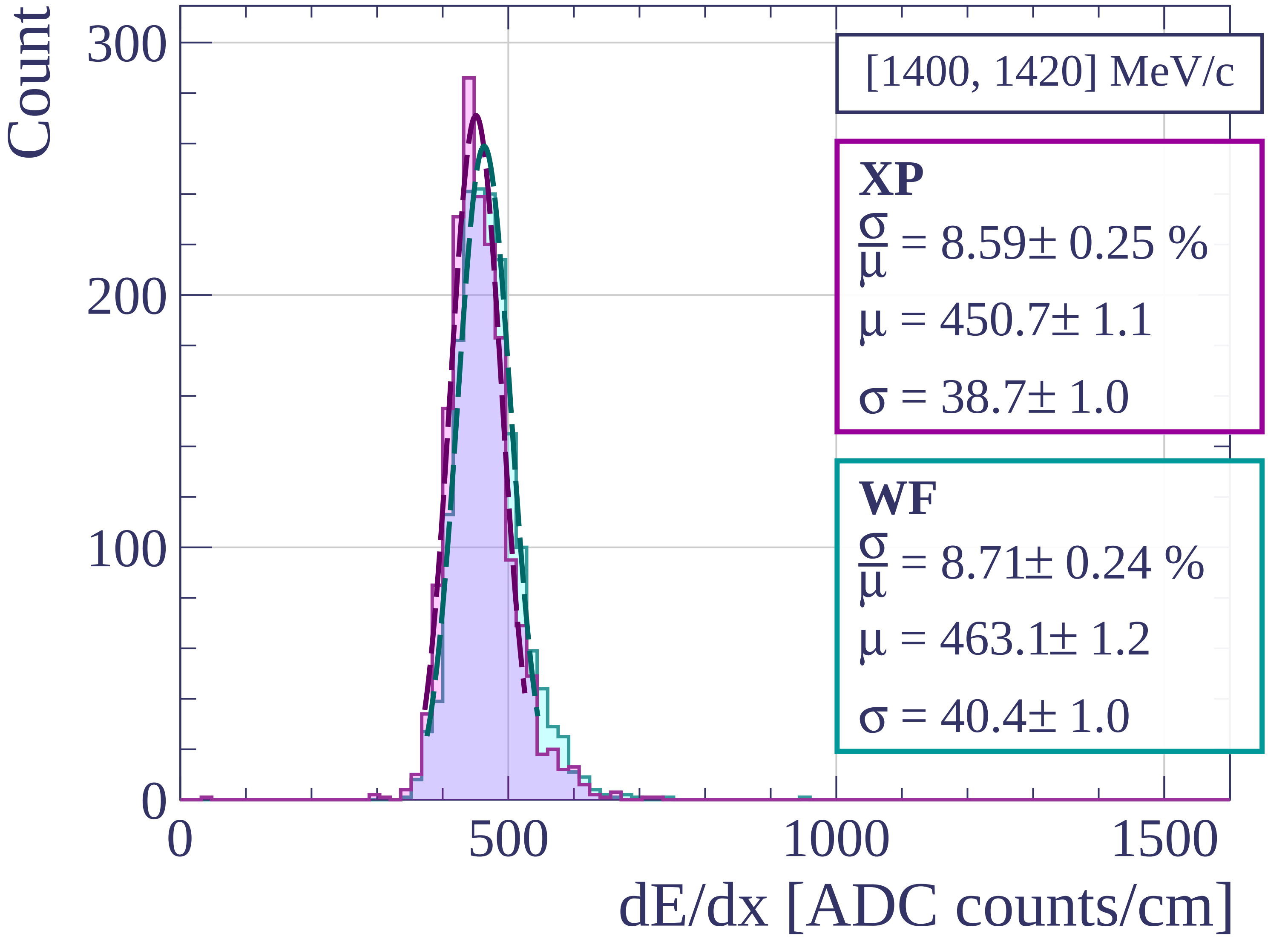


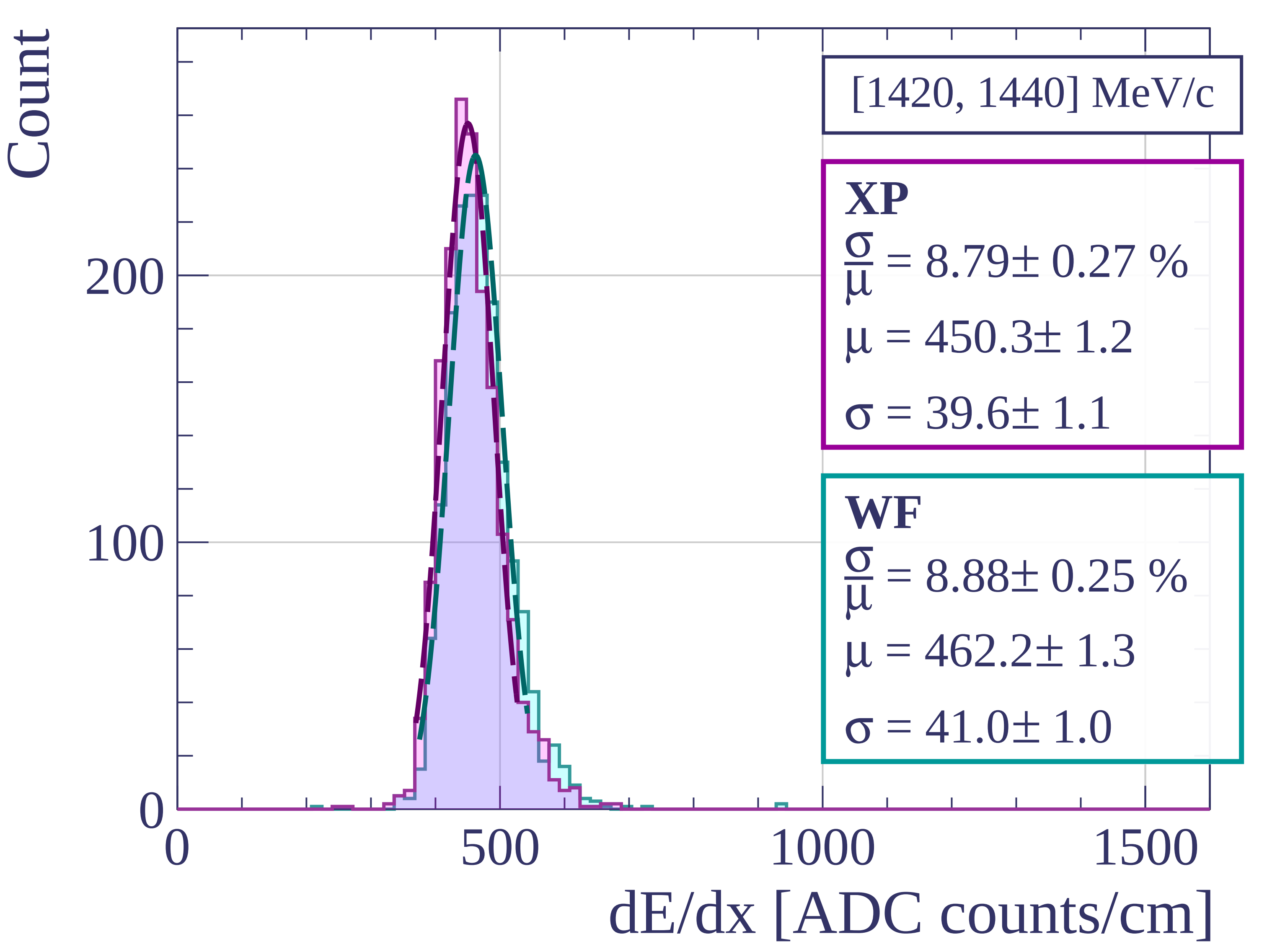
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1340, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.50± 0.25 %
μ = 446.6± 1.1
σ = 38.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.90± 0.26 %
μ = 458.3± 1.2
σ = 40.8± 1.1


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1380] MeV/c
XP
σ/μ = 8.00± 0.22 %
μ = 448.7± 1.0
σ = 35.9± 0.9
WF
σ/μ = 8.74± 0.27 %
μ = 459.4± 1.2
σ = 40.2± 1.2

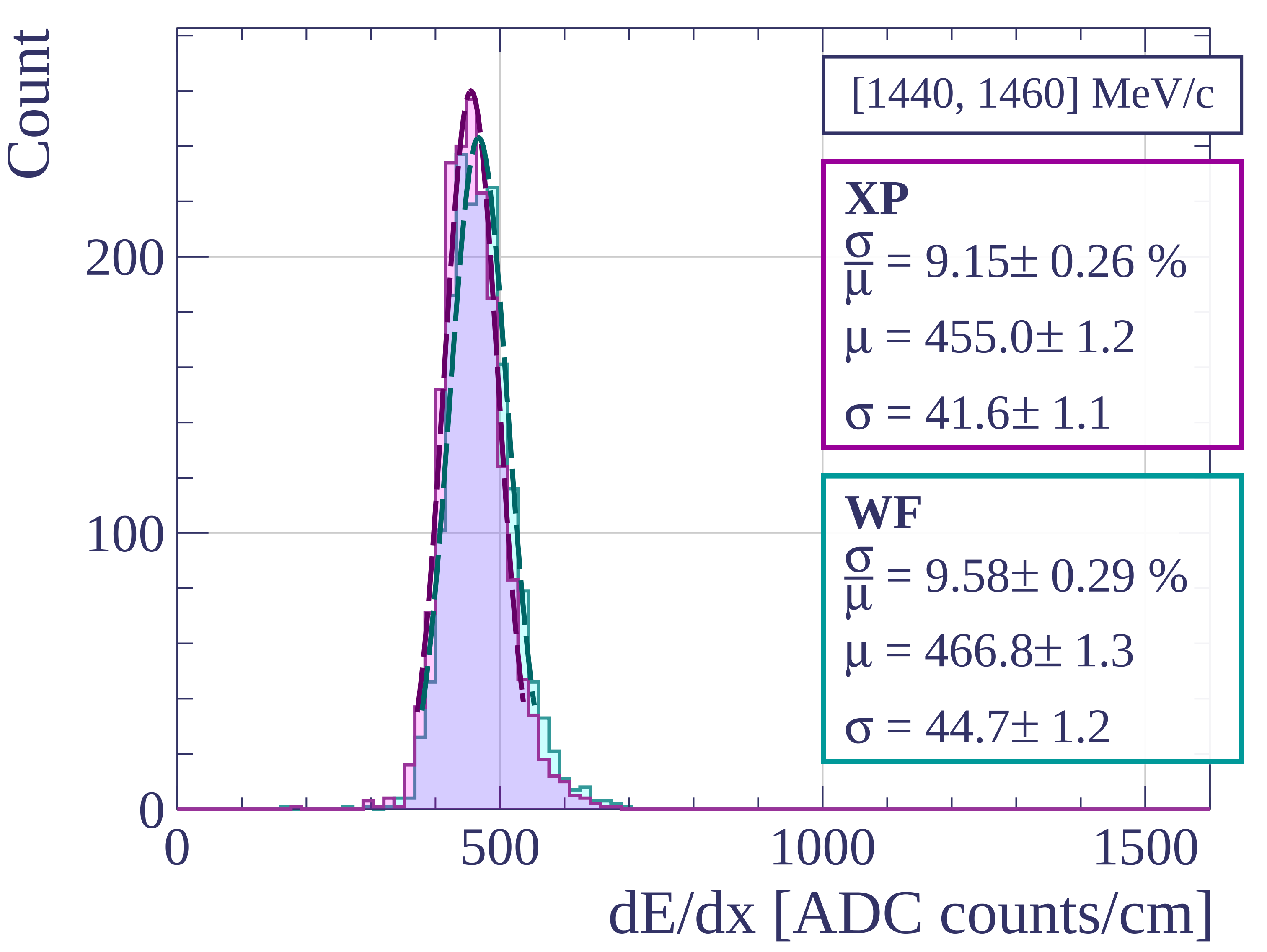


Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
[1380, 1400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.40± 0.24 %
μ = 449.6± 1.1
σ = 37.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.01± 0.29 %
μ = 460.3± 1.2
σ = 41.5± 1.2

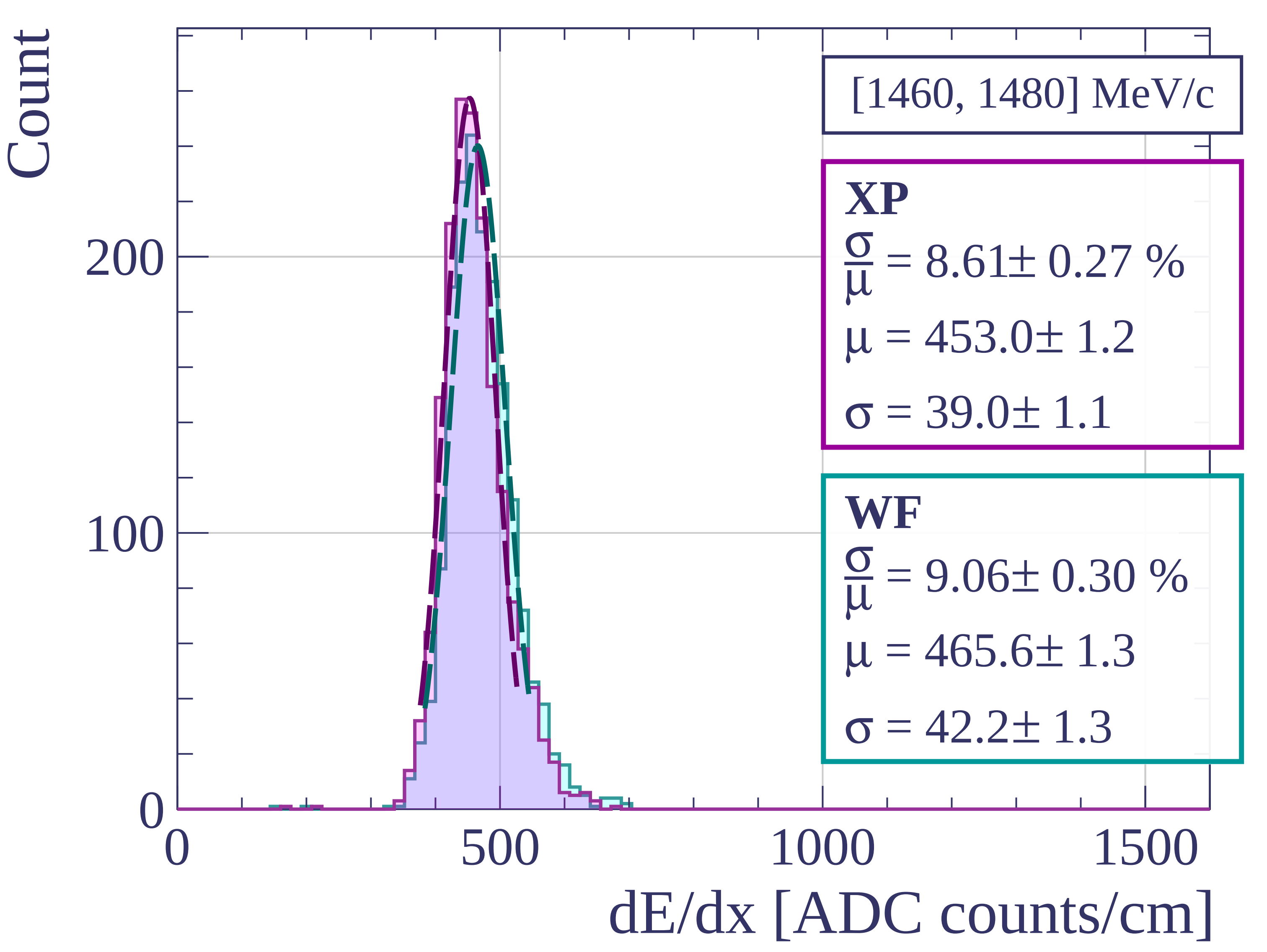

[1400, 1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.59± 0.25 %
μ = 450.7± 1.1
σ = 38.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.71± 0.24 %
μ = 463.1± 1.2
σ = 40.4± 1.0
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]



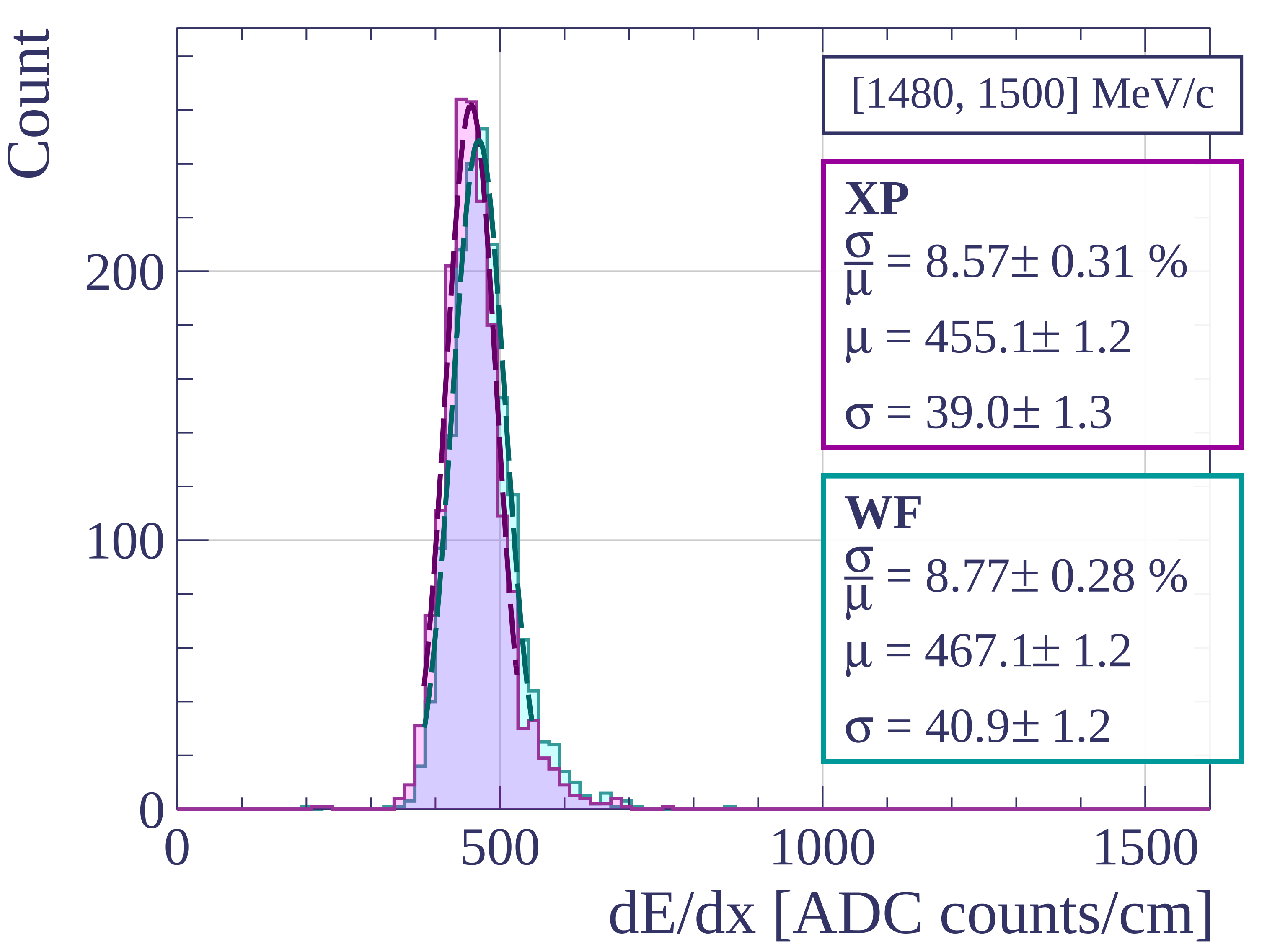
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
[1420, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.79± 0.27 %
μ = 450.3± 1.2
σ = 39.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.88± 0.25 %
μ = 462.2± 1.3
σ = 41.0± 1.0

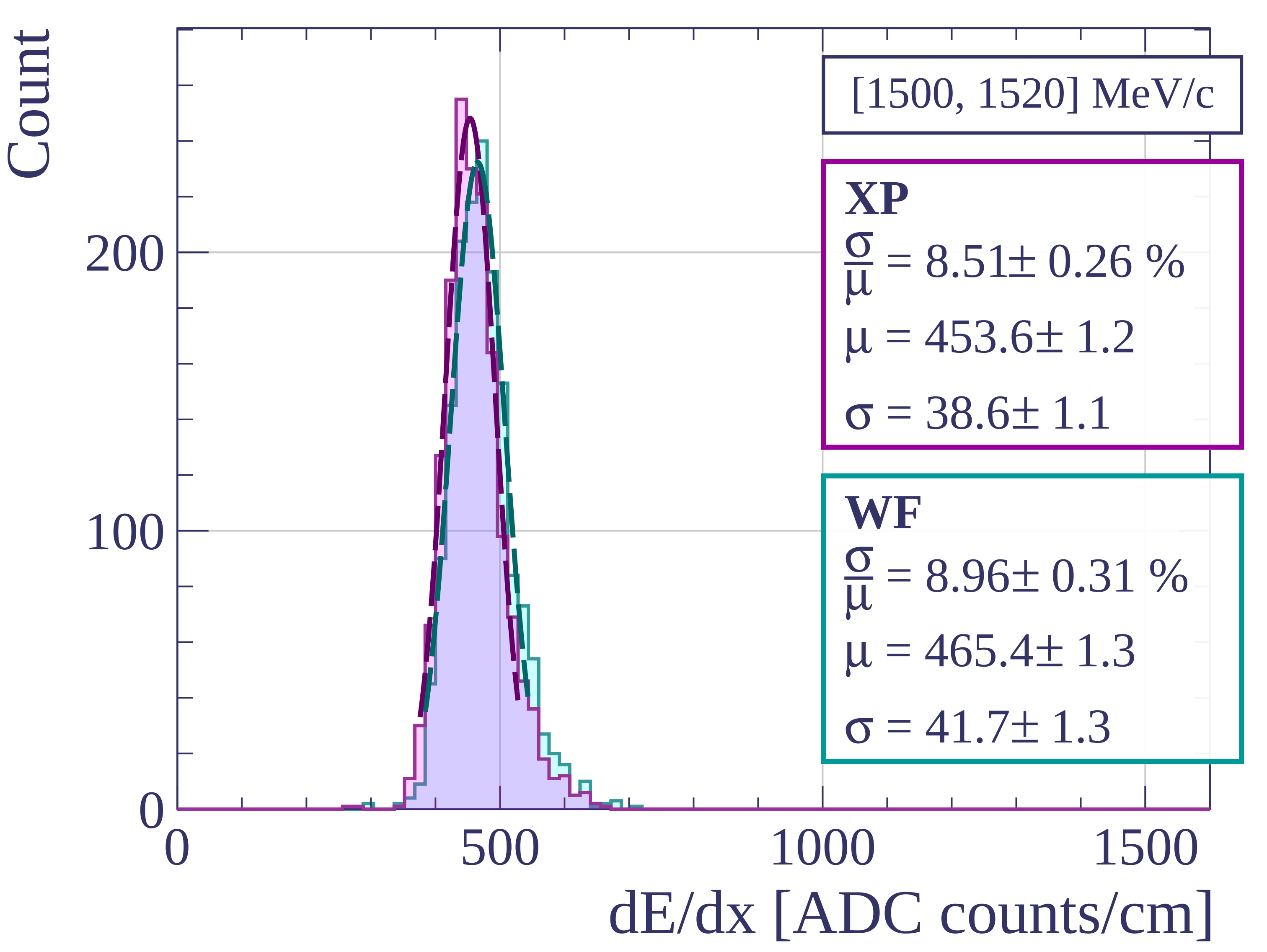


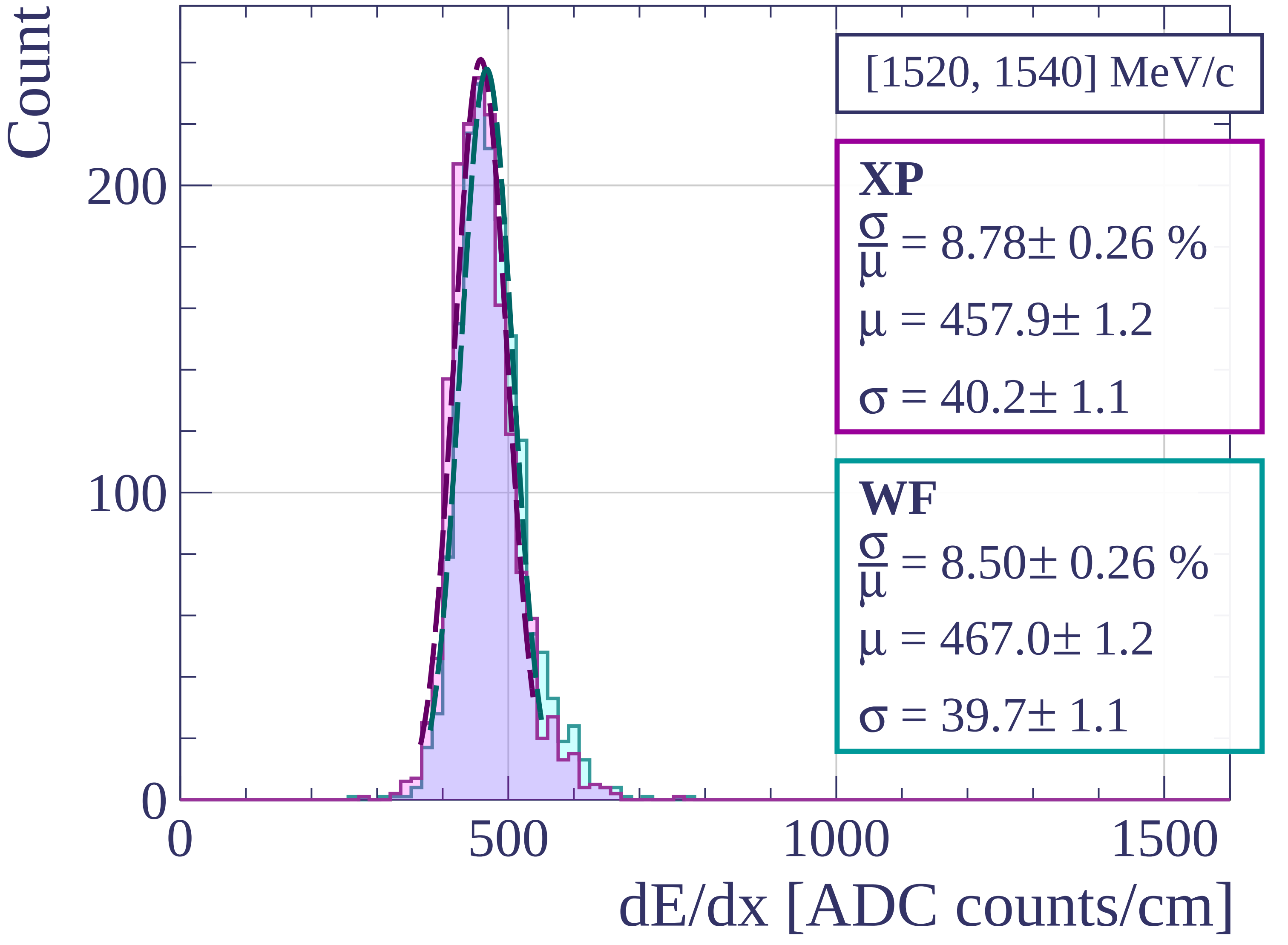
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.15± 0.26 %
μ = 455.0± 1.2
σ = 41.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.58± 0.29 %
μ = 466.8± 1.3
σ = 44.7± 1.2

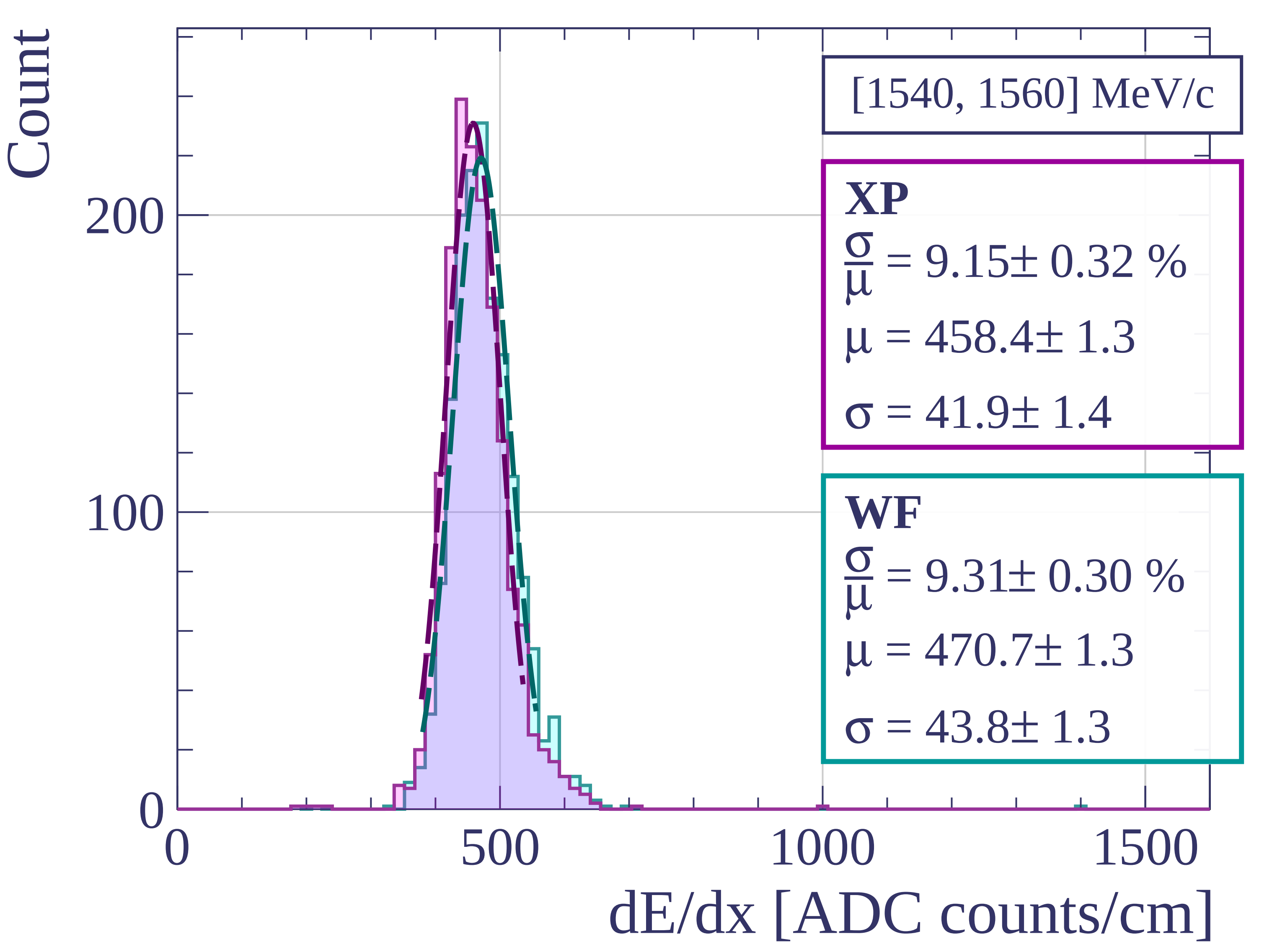


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1460, 1480] MeV/c
XP
σ/μ = 8.61± 0.27 %
μ = 453.0± 1.2
σ = 39.0± 1.1
WF
σ/μ = 9.06± 0.30 %
μ = 465.6± 1.3
σ = 42.2± 1.3

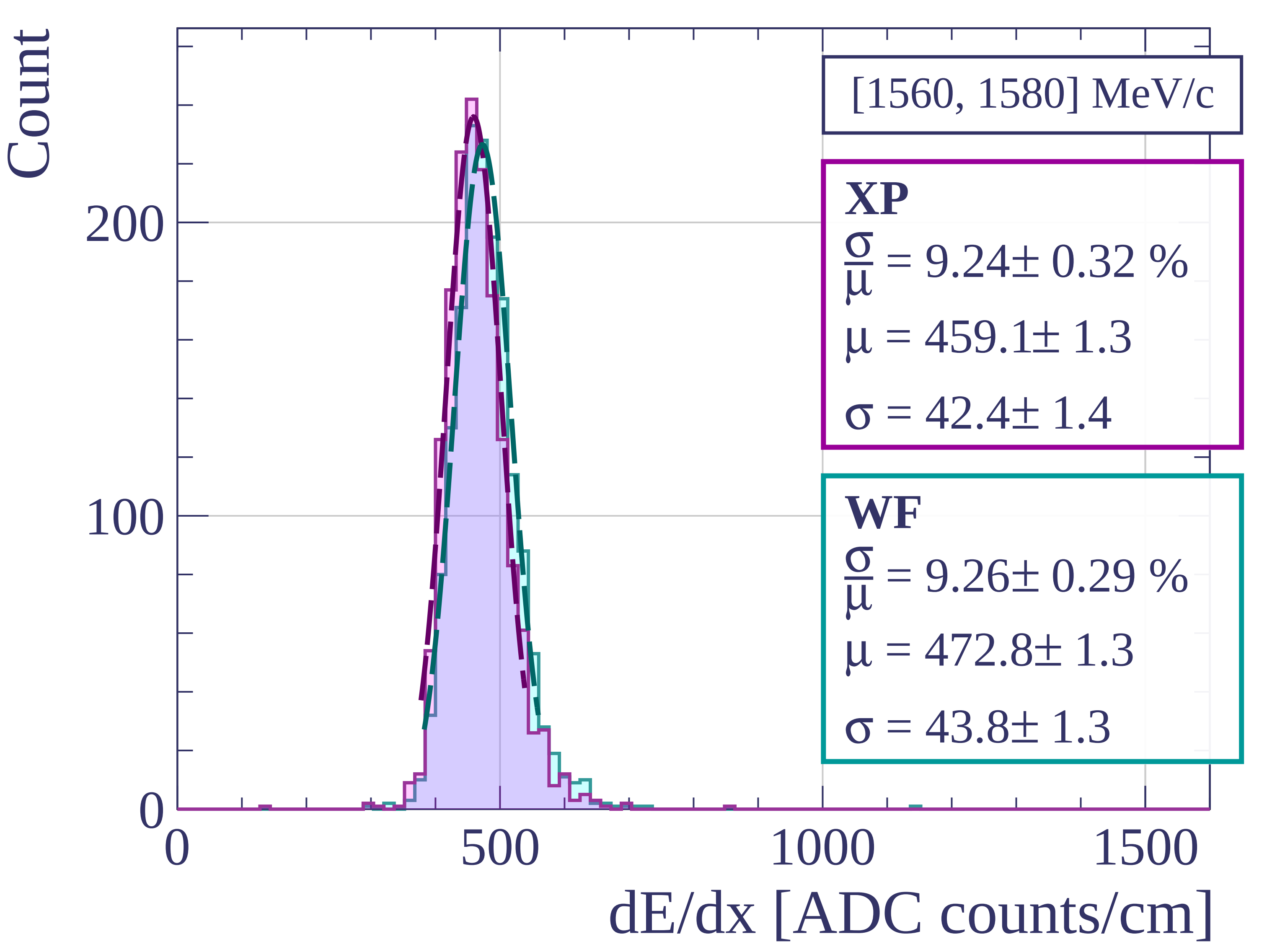

Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1480, 1500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.57± 0.31 %
μ = 455.1± 1.2
σ = 39.0± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.77± 0.28 %
μ = 467.1± 1.2
σ = 40.9± 1.2


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1500, 1520] MeV/c
XP
σ/μ = 8.51± 0.26 %
μ = 453.6± 1.2
σ = 38.6± 1.1
WF
σ/μ = 8.96± 0.31 %
μ = 465.4± 1.3
σ = 41.7± 1.3

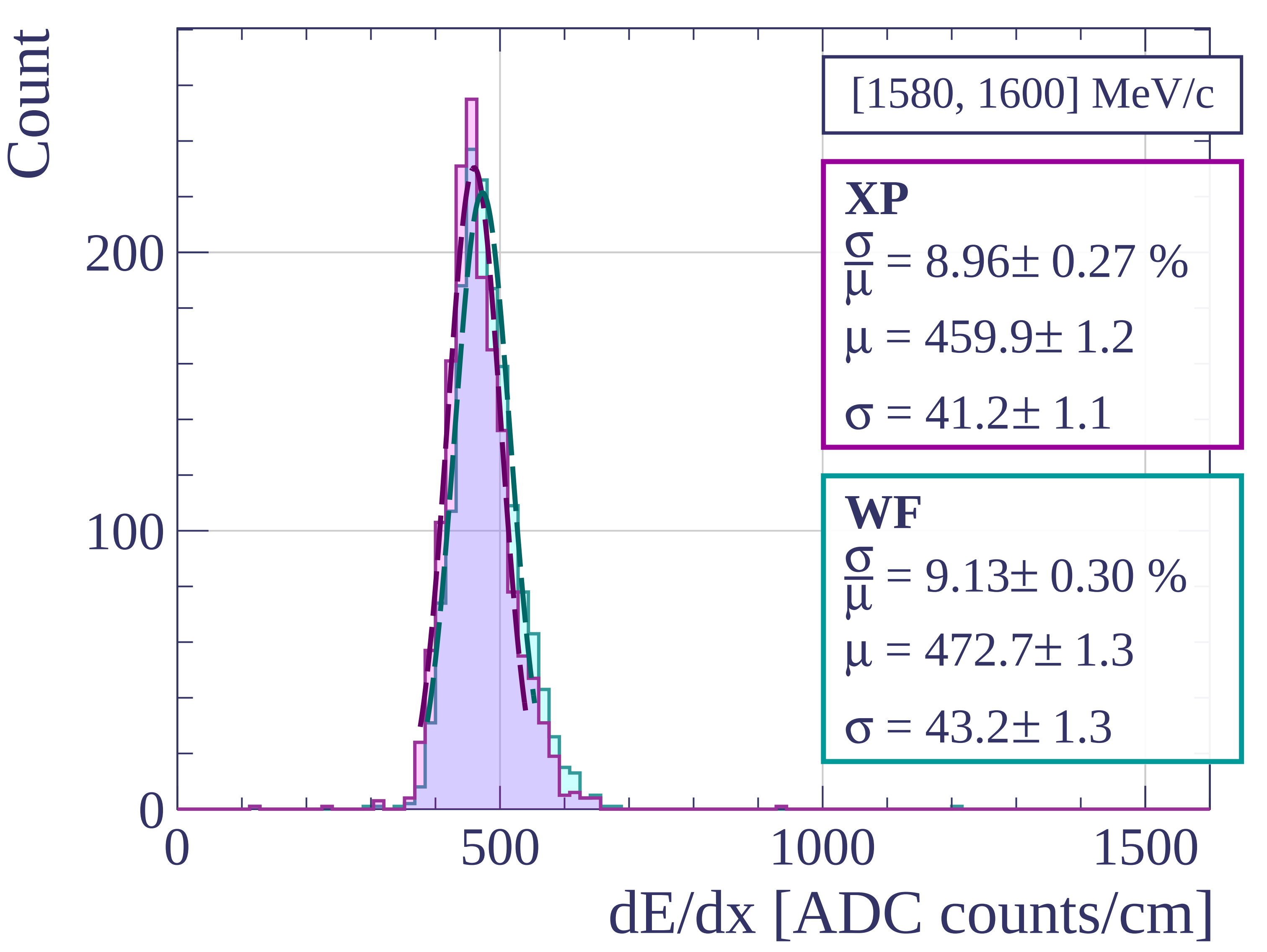

Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
[1520, 1540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.78± 0.26 %
μ = 457.9± 1.2
σ = 40.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.50± 0.26 %
μ = 467.0± 1.2
σ = 39.7± 1.1



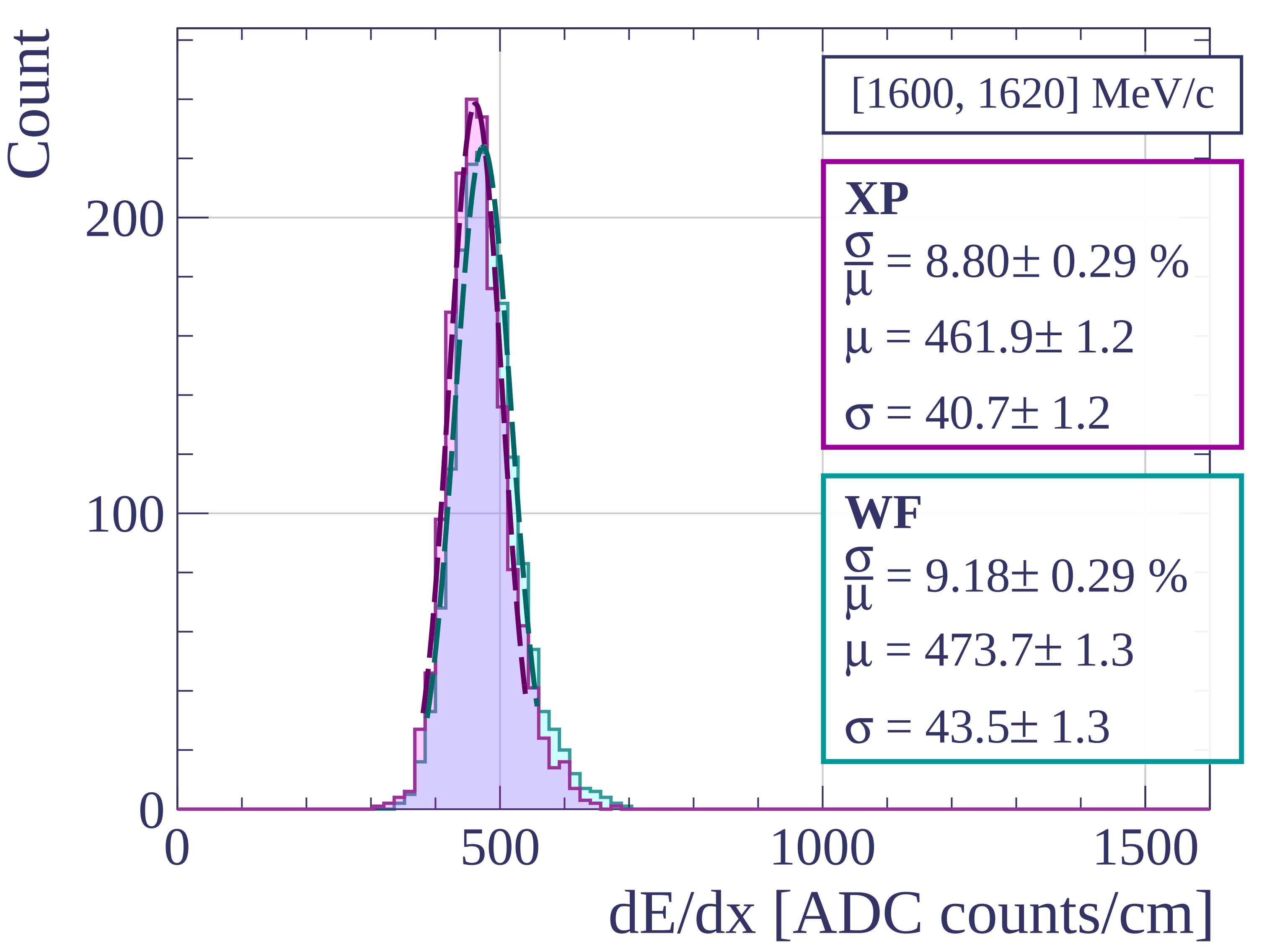
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1540, 1560] MeV/c
XP
σ/μ = 9.15± 0.32 %
μ = 458.4± 1.3
σ = 41.9± 1.4
WF
σ/μ = 9.31± 0.30 %
μ = 470.7± 1.3
σ = 43.8± 1.3



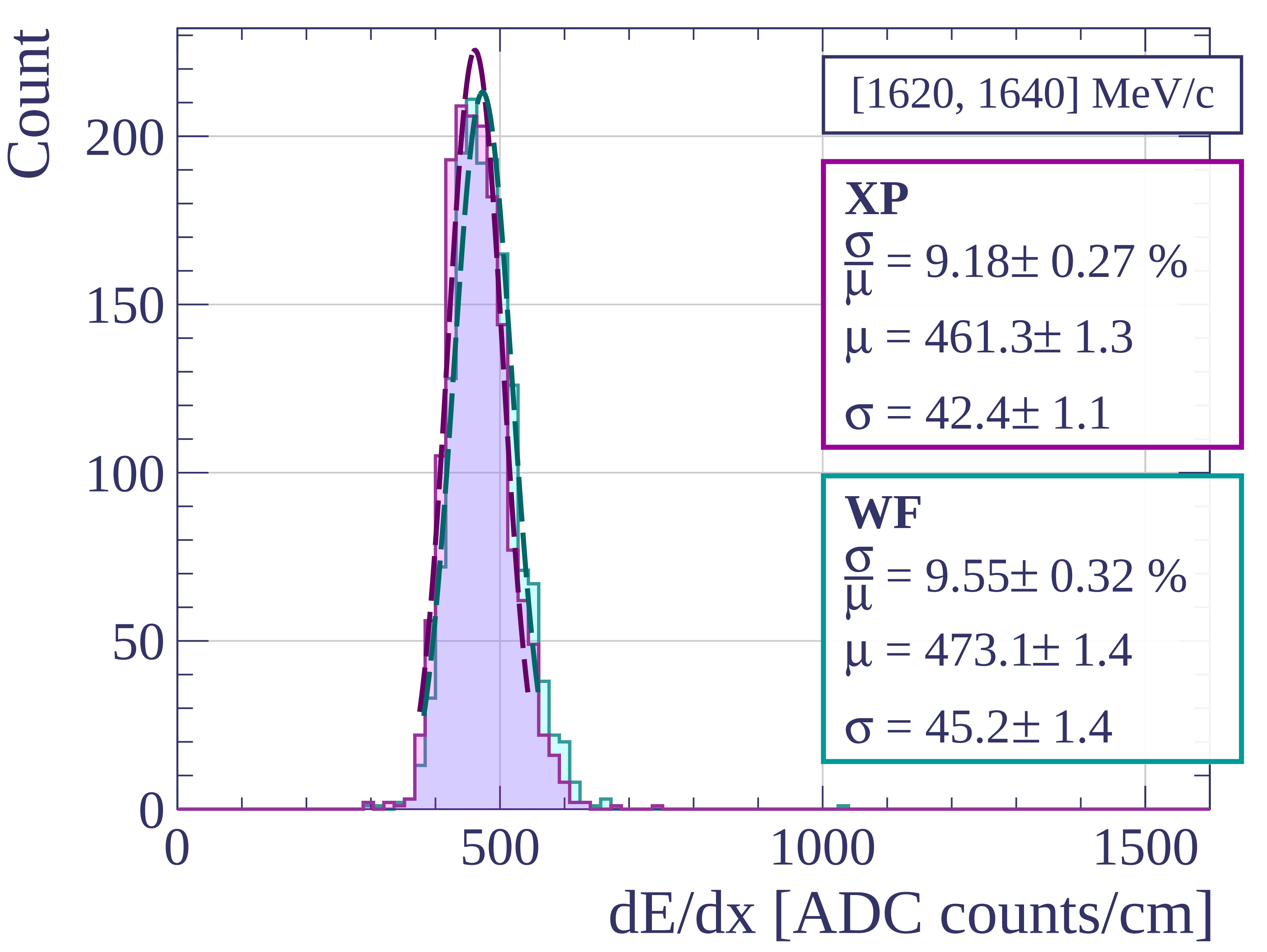
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1560, 1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.24± 0.32 %
μ = 459.1± 1.3
σ = 42.4± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.26± 0.29 %
μ = 472.8± 1.3
σ = 43.8± 1.3

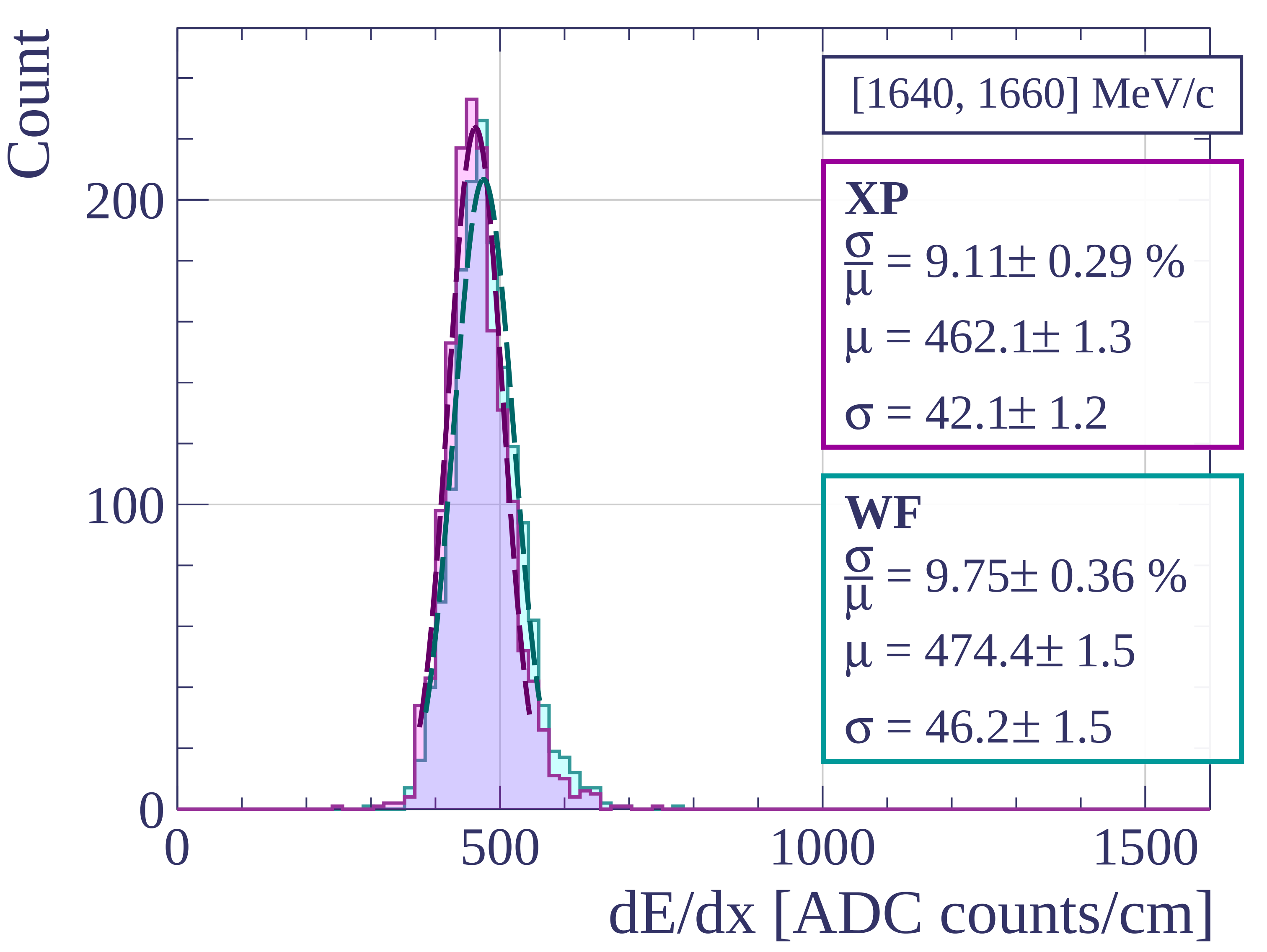


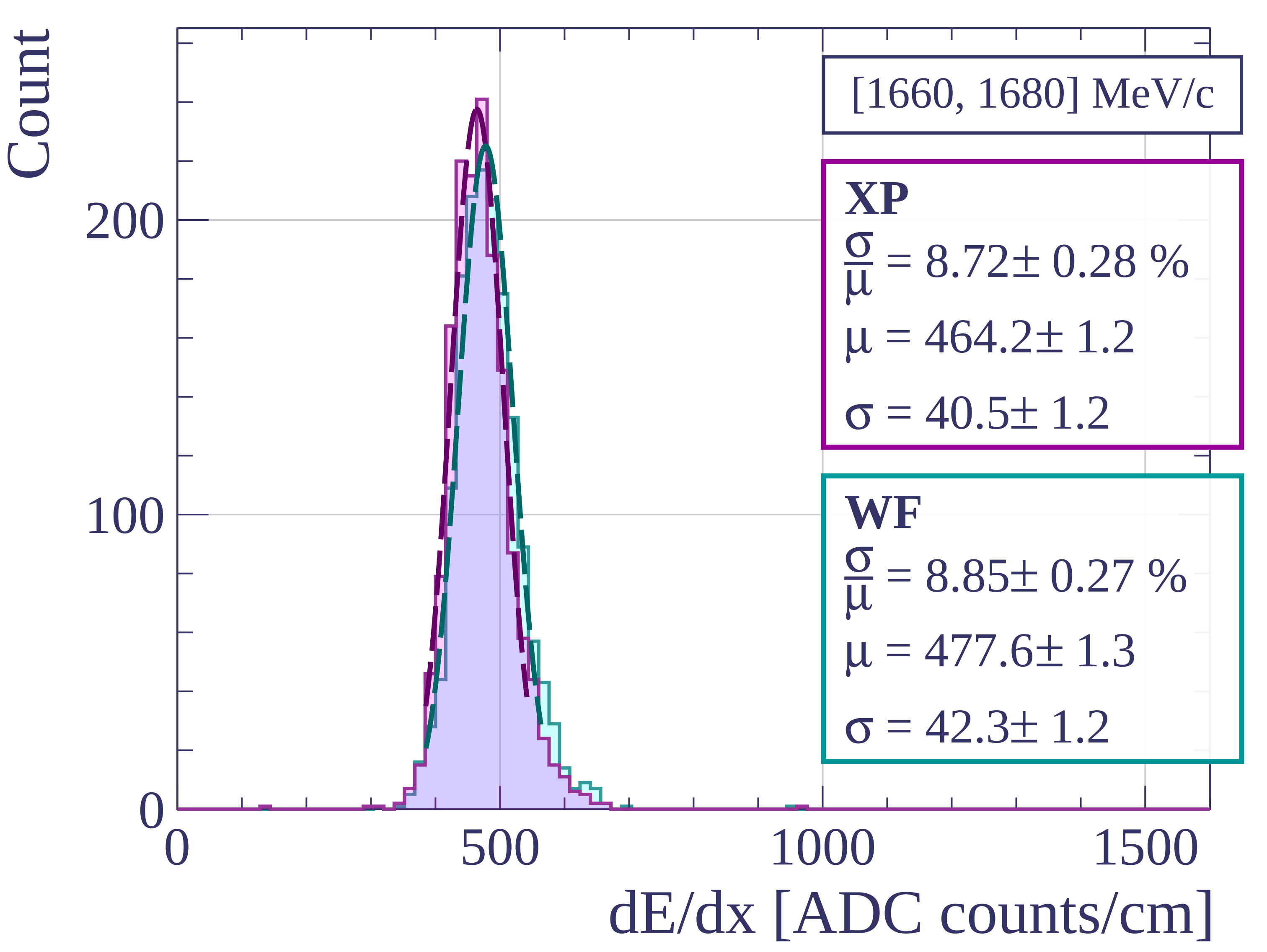
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
[1580, 1600] MeV/c
XP
σ/μ = 8.96± 0.27 %
μ = 459.9± 1.2
σ = 41.2± 1.1
WF
σ/μ = 9.13± 0.30 %
μ = 472.7± 1.3
σ = 43.2± 1.3



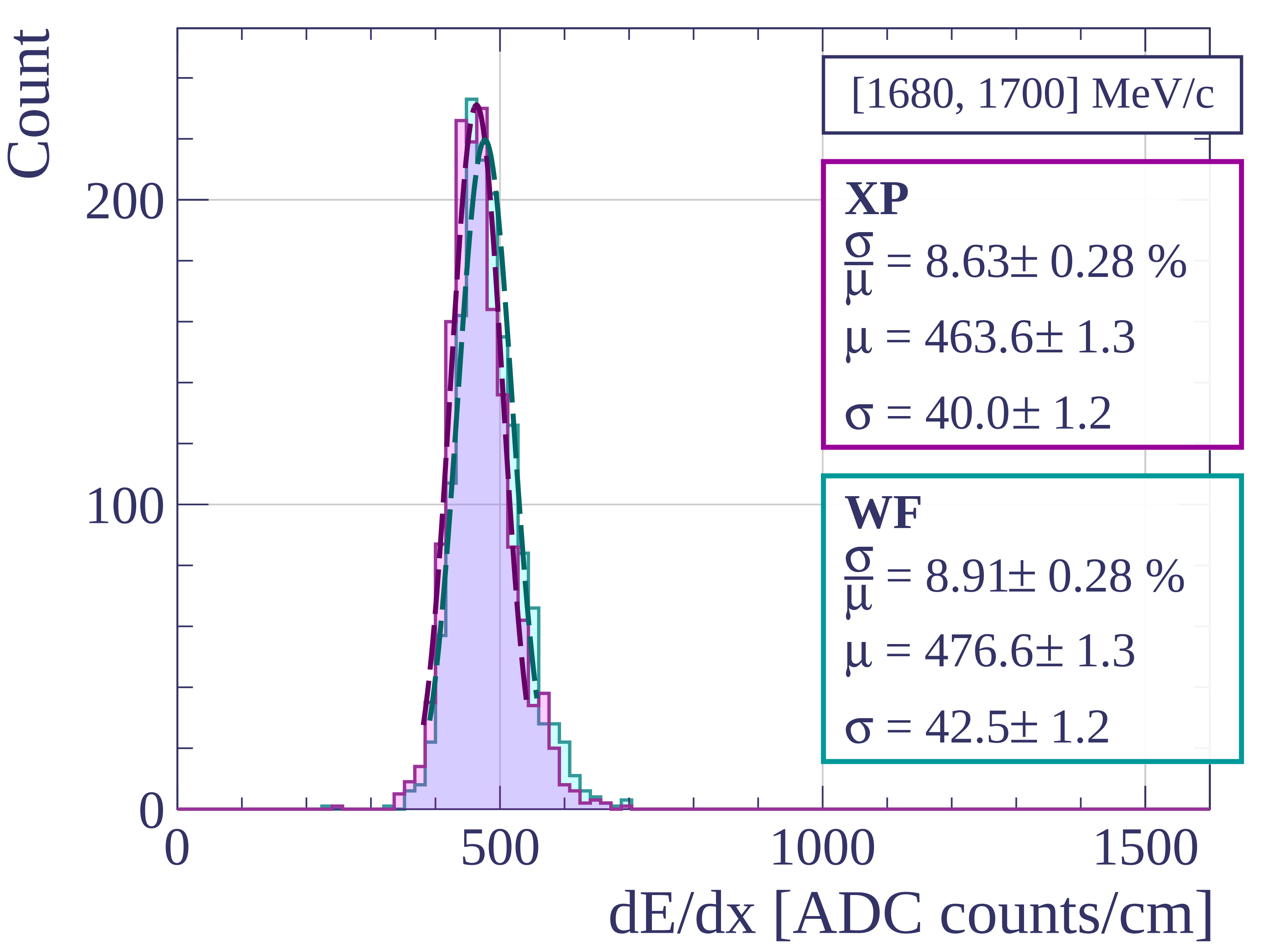
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.80± 0.29 %
μ = 461.9± 1.2
σ = 40.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.18± 0.29 %
μ = 473.7± 1.3
σ = 43.5± 1.3

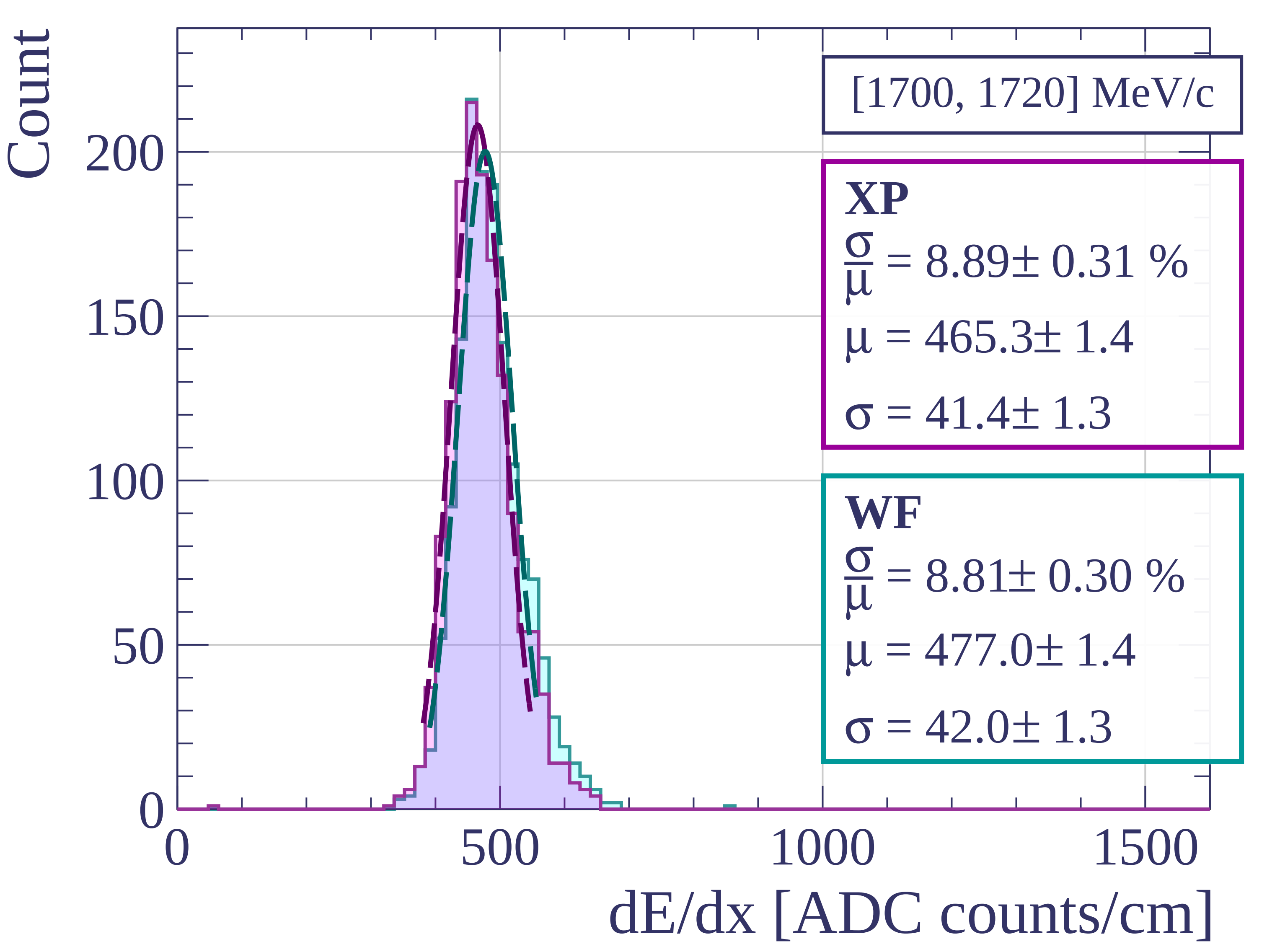

Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
[1620, 1640] MeV/c
XP
σ/μ = 9.18± 0.27 %
μ = 461.3± 1.3
σ = 42.4± 1.1
WF
σ/μ = 9.55± 0.32 %
μ = 473.1± 1.4
σ = 45.2± 1.4


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1640, 1660] MeV/c
XP
σ/μ = 9.11± 0.29 %
μ = 462.1± 1.3
σ = 42.1± 1.2
WF
σ/μ = 9.75± 0.36 %
μ = 474.4± 1.5
σ = 46.2± 1.5

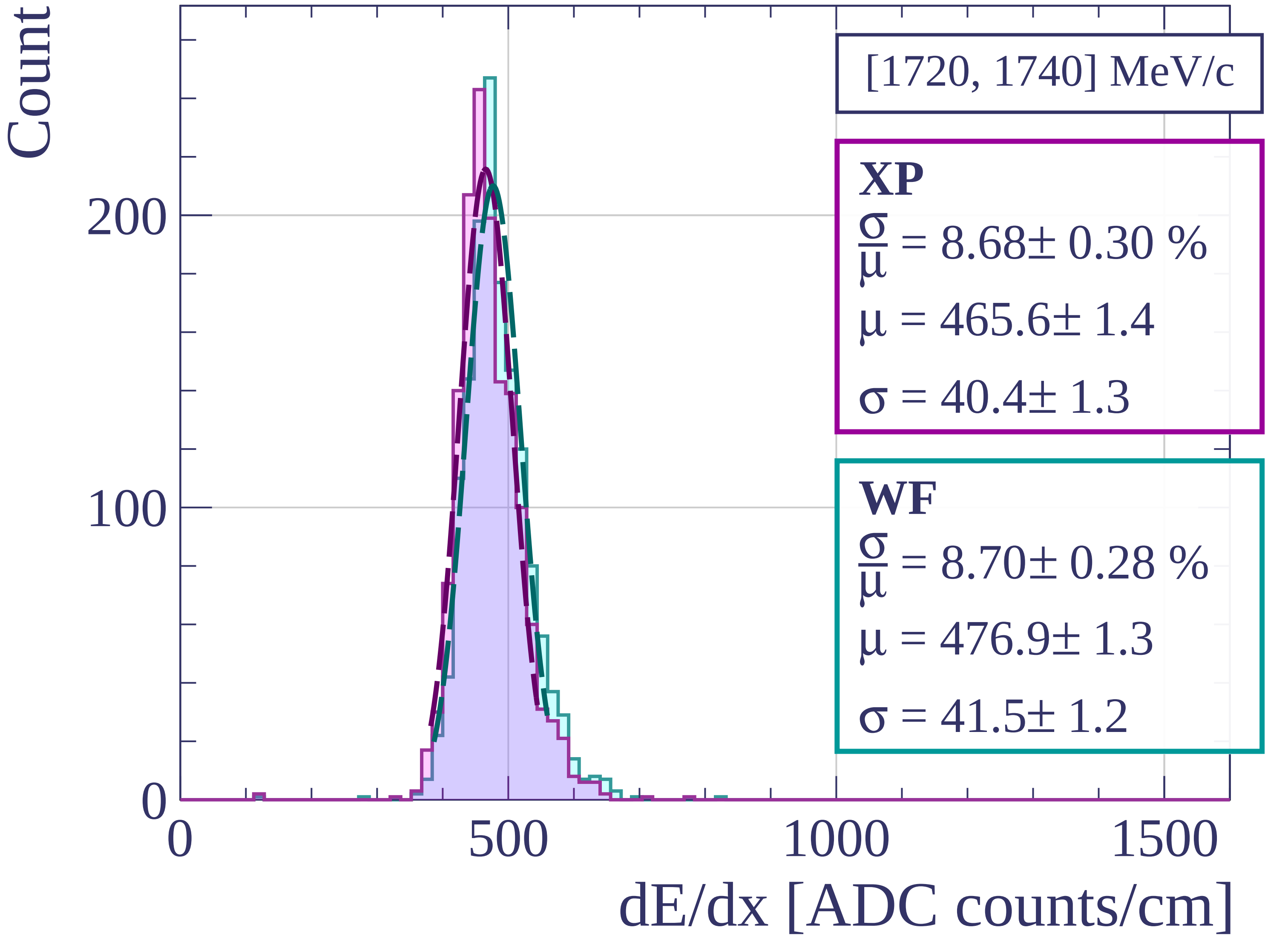


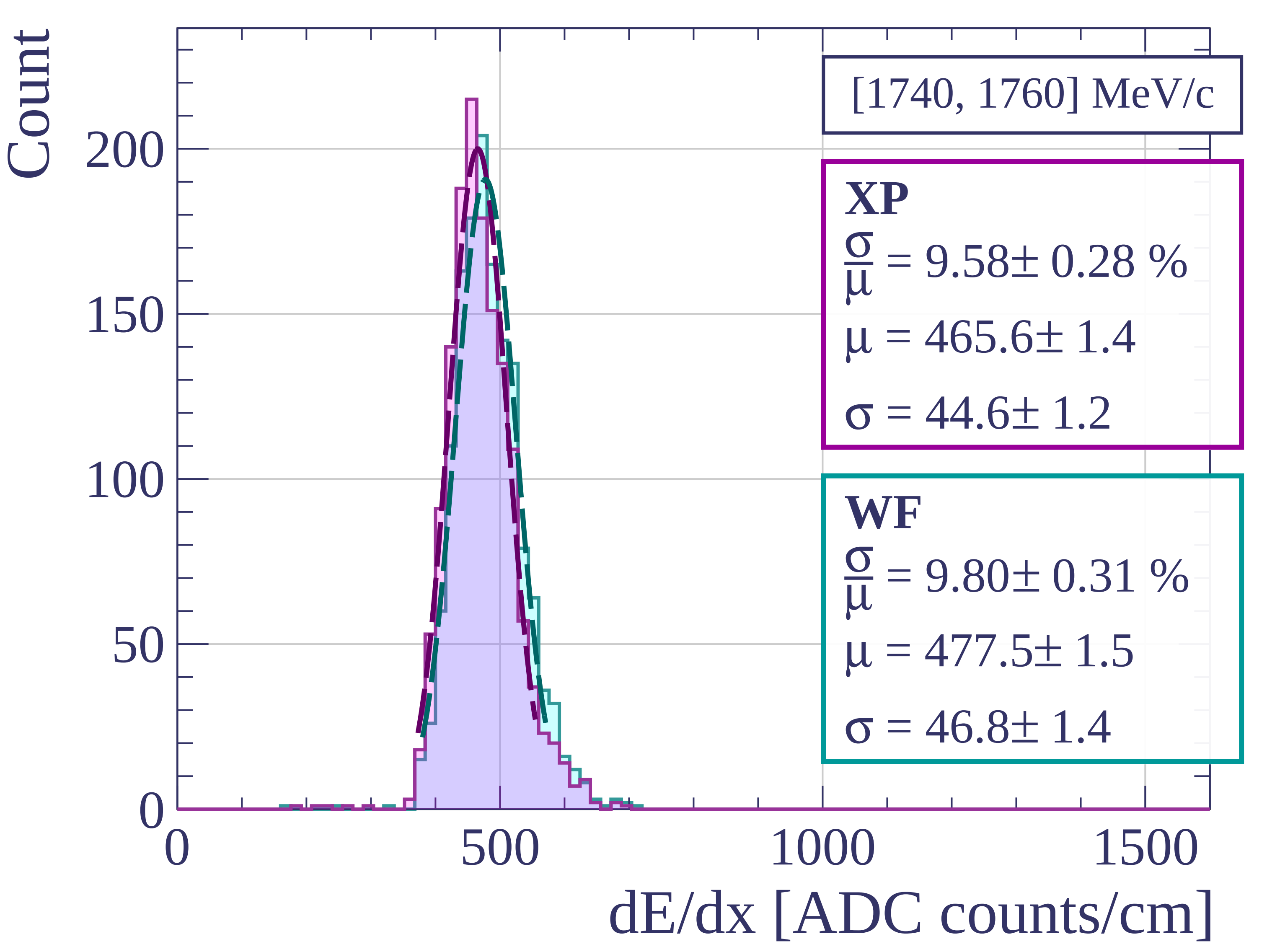
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1660, 1680] MeV/c
XP
σ/μ = 8.72± 0.28 %
μ = 464.2± 1.2
σ = 40.5± 1.2
WF
σ/μ = 8.85± 0.27 %
μ = 477.6± 1.3
σ = 42.3± 1.2

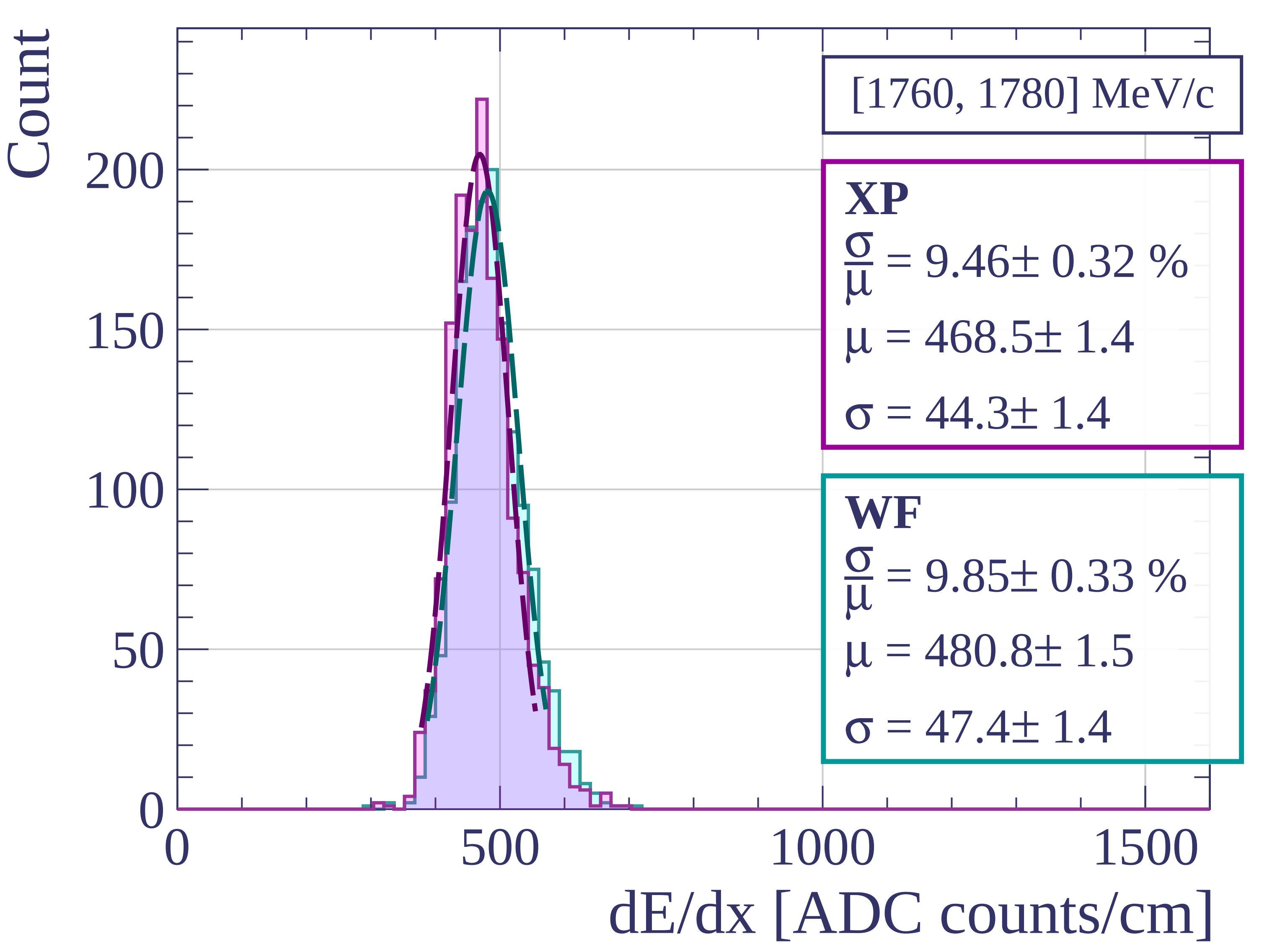

Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1700] MeV/c
XP
σ/μ = 8.63± 0.28 %
μ = 463.6± 1.3
σ = 40.0± 1.2
WF
σ/μ = 8.91± 0.28 %
μ = 476.6± 1.3
σ = 42.5± 1.2



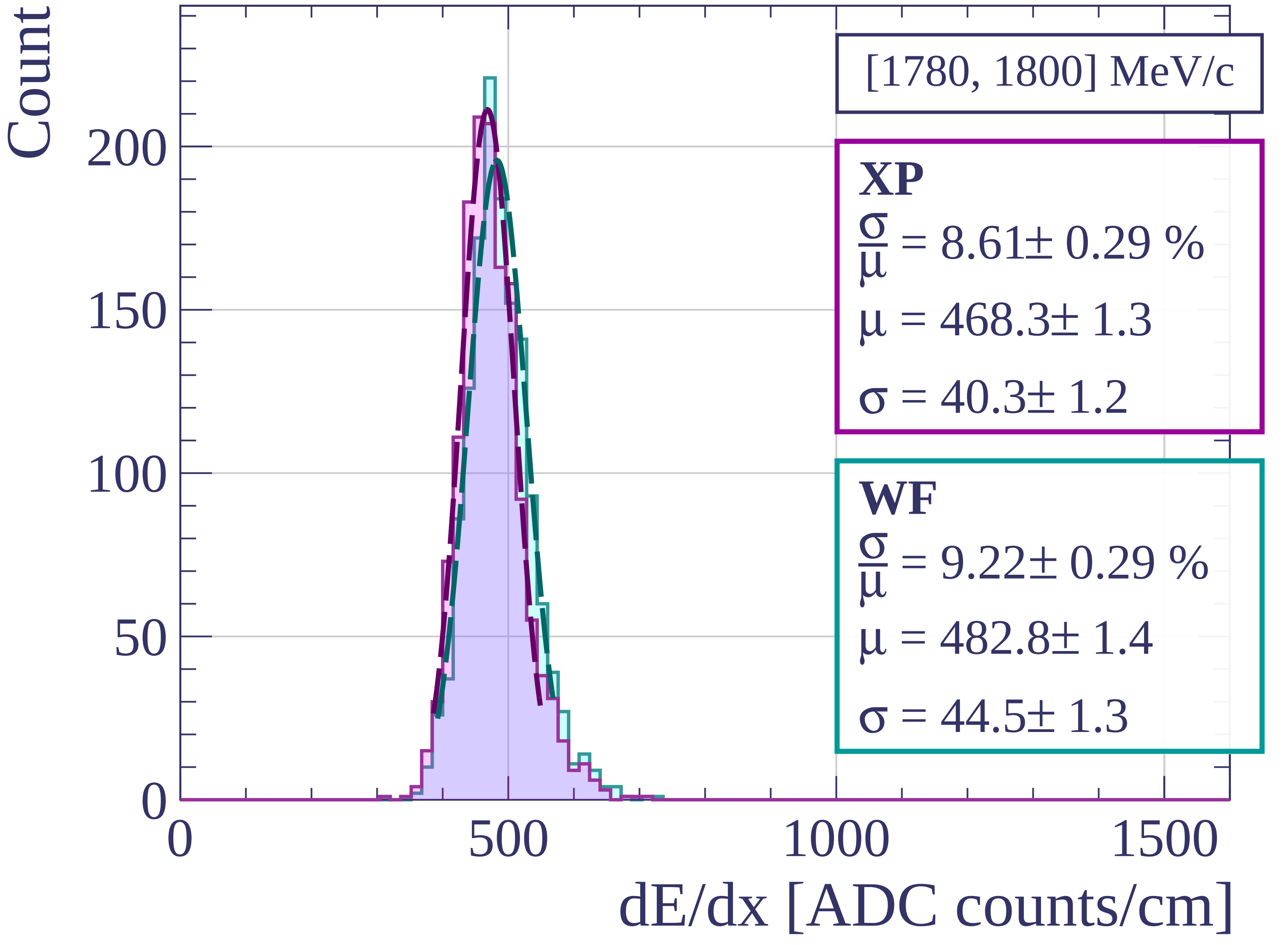
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1700, 1720] MeV/c
XP
σ/μ = 8.89± 0.31 %
μ = 465.3± 1.4
σ = 41.4± 1.3
WF
σ/μ = 8.81± 0.30 %
μ = 477.0± 1.4
σ = 42.0± 1.3

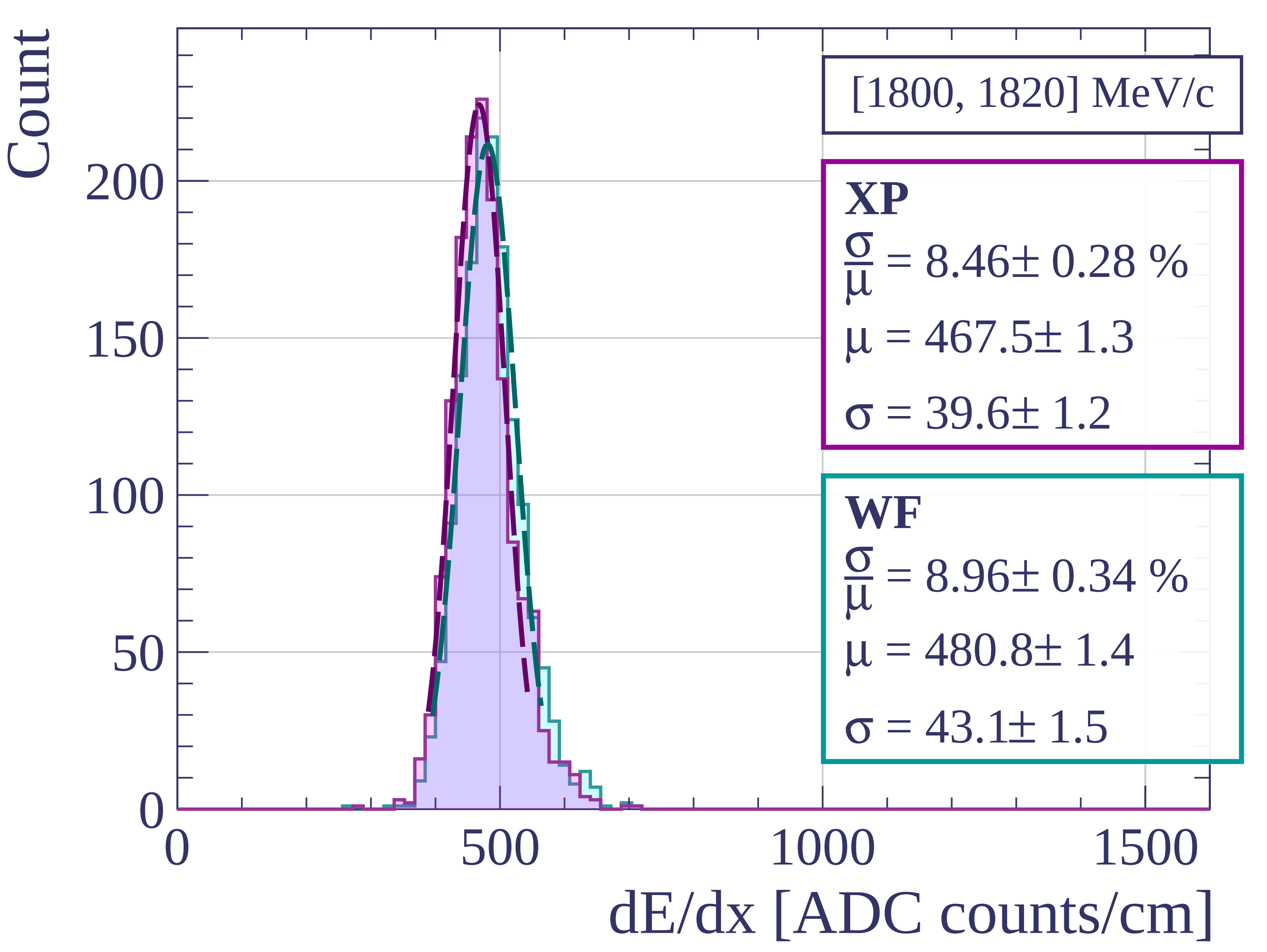

Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1720, 1740] MeV/c
XP
σ/μ = 8.68± 0.30 %
μ = 465.6± 1.4
σ = 40.4± 1.3
WF
σ/μ = 8.70± 0.28 %
μ = 476.9± 1.3
σ = 41.5± 1.2


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1740, 1760] MeV/c
XP
σ/μ = 9.58± 0.28 %
μ = 465.6± 1.4
σ = 44.6± 1.2
WF
σ/μ = 9.80± 0.31 %
μ = 477.5± 1.5
σ = 46.8± 1.4

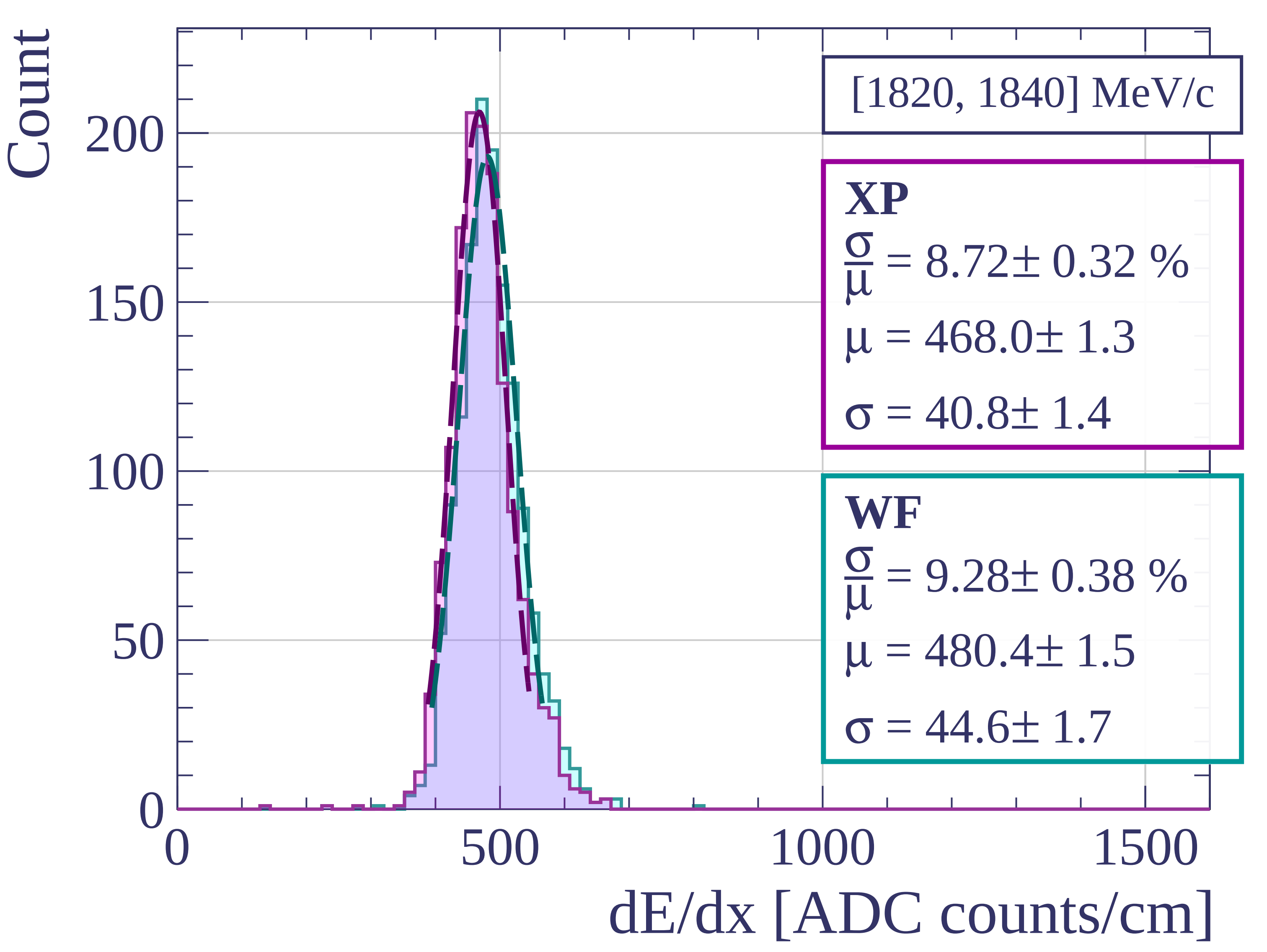


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.32 %
μ = 468.5± 1.4
σ = 44.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.85± 0.33 %
μ = 480.8± 1.5
σ = 47.4± 1.4

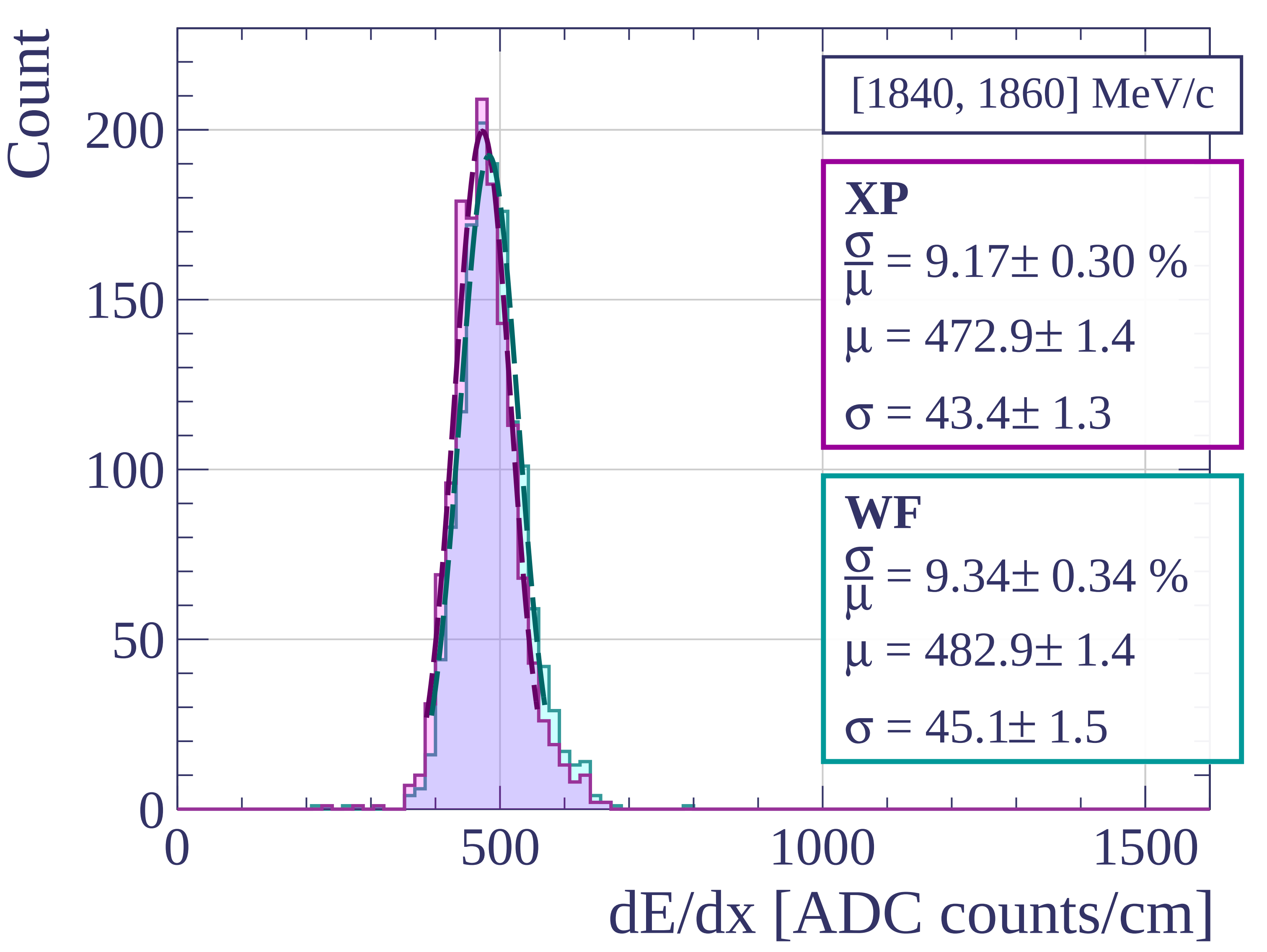

Count
dE/dx [ADC counts/cm]
[1780, 1800] MeV/c
XP
σ/μ = 8.61± 0.29 %
μ = 468.3± 1.3
σ = 40.3± 1.2
WF
σ/μ = 9.22± 0.29 %
μ = 482.8± 1.4
σ = 44.5± 1.3
0
50
100
150
200
0
500
1000
1500

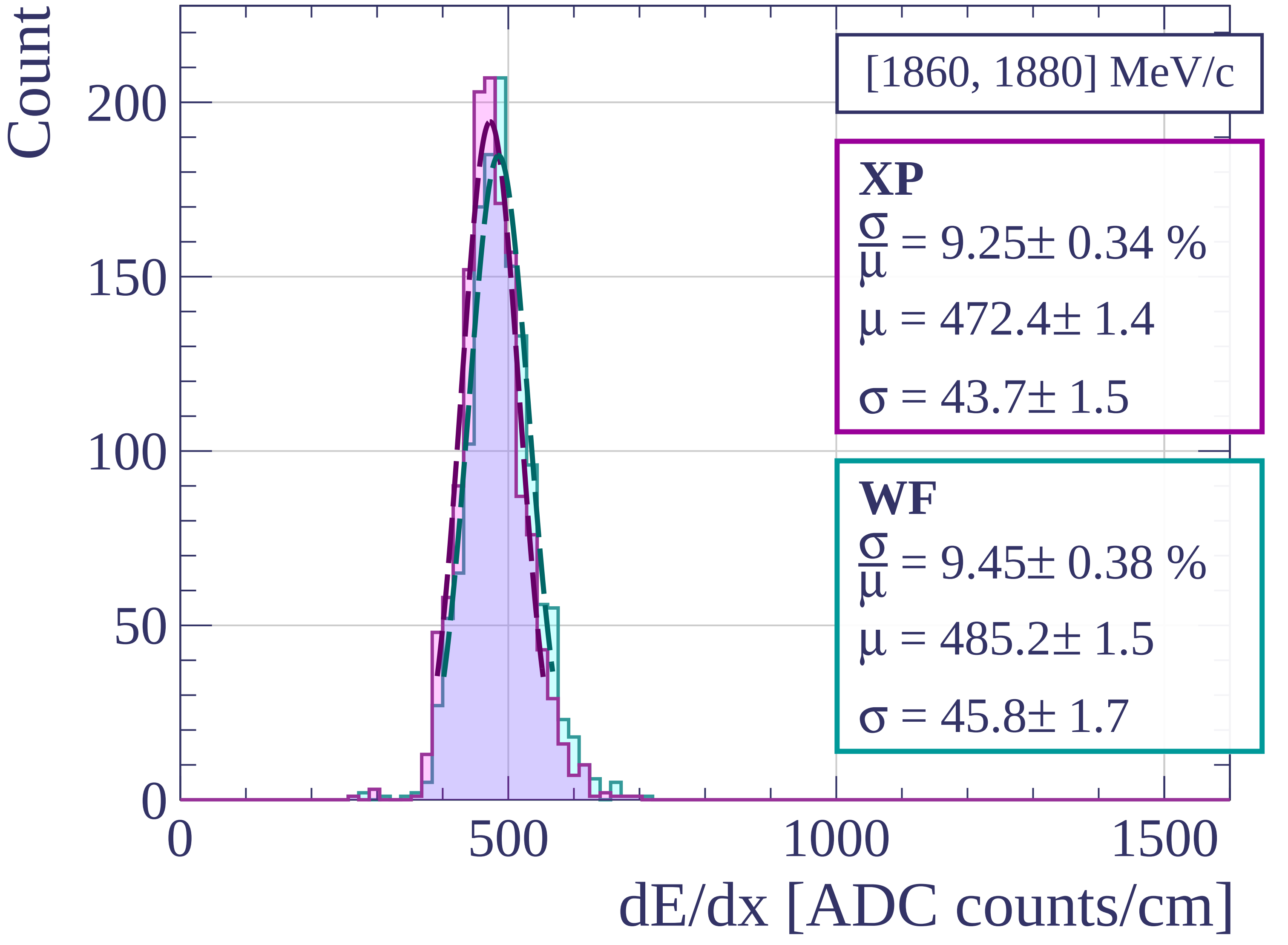


Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
[1800, 1820] MeV/c
XP
σ/μ = 8.46± 0.28 %
μ = 467.5± 1.3
σ = 39.6± 1.2
WF
σ/μ = 8.96± 0.34 %
μ = 480.8± 1.4
σ = 43.1± 1.5

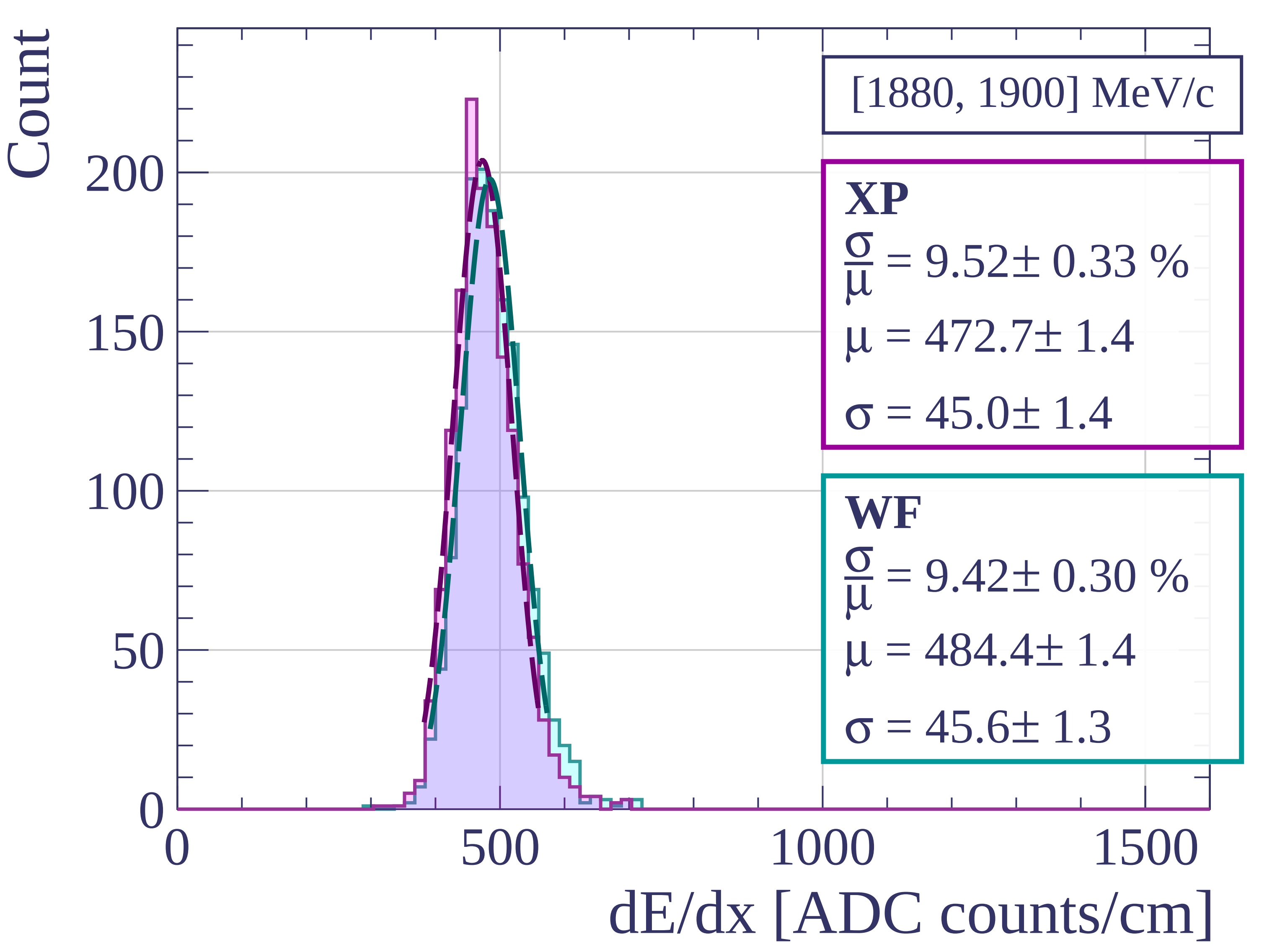


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1820, 1840] MeV/c
XP
σ/μ = 8.72± 0.32 %
μ = 468.0± 1.3
σ = 40.8± 1.4
WF
σ/μ = 9.28± 0.38 %
μ = 480.4± 1.5
σ = 44.6± 1.7

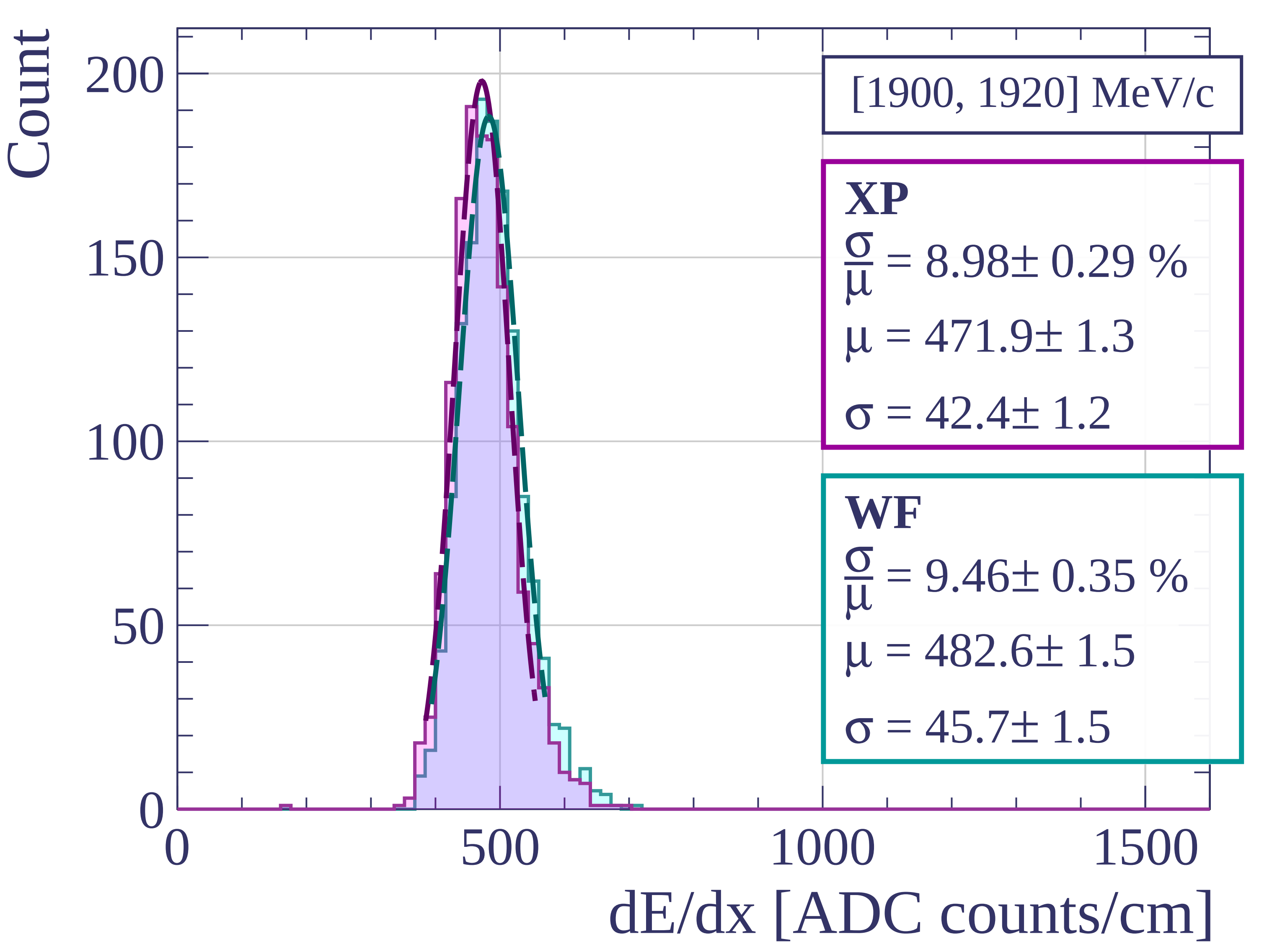

Count
dE/dx [ADC counts/cm]
[1840, 1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.17± 0.30 %
μ = 472.9± 1.4
σ = 43.4± 1.3
WF
$\frac{\sigma}{\mu}$ = 9.34± 0.34 %
μ = 482.9± 1.4
σ = 45.1± 1.5
0
50
100
150
200
0
500
1000
1500



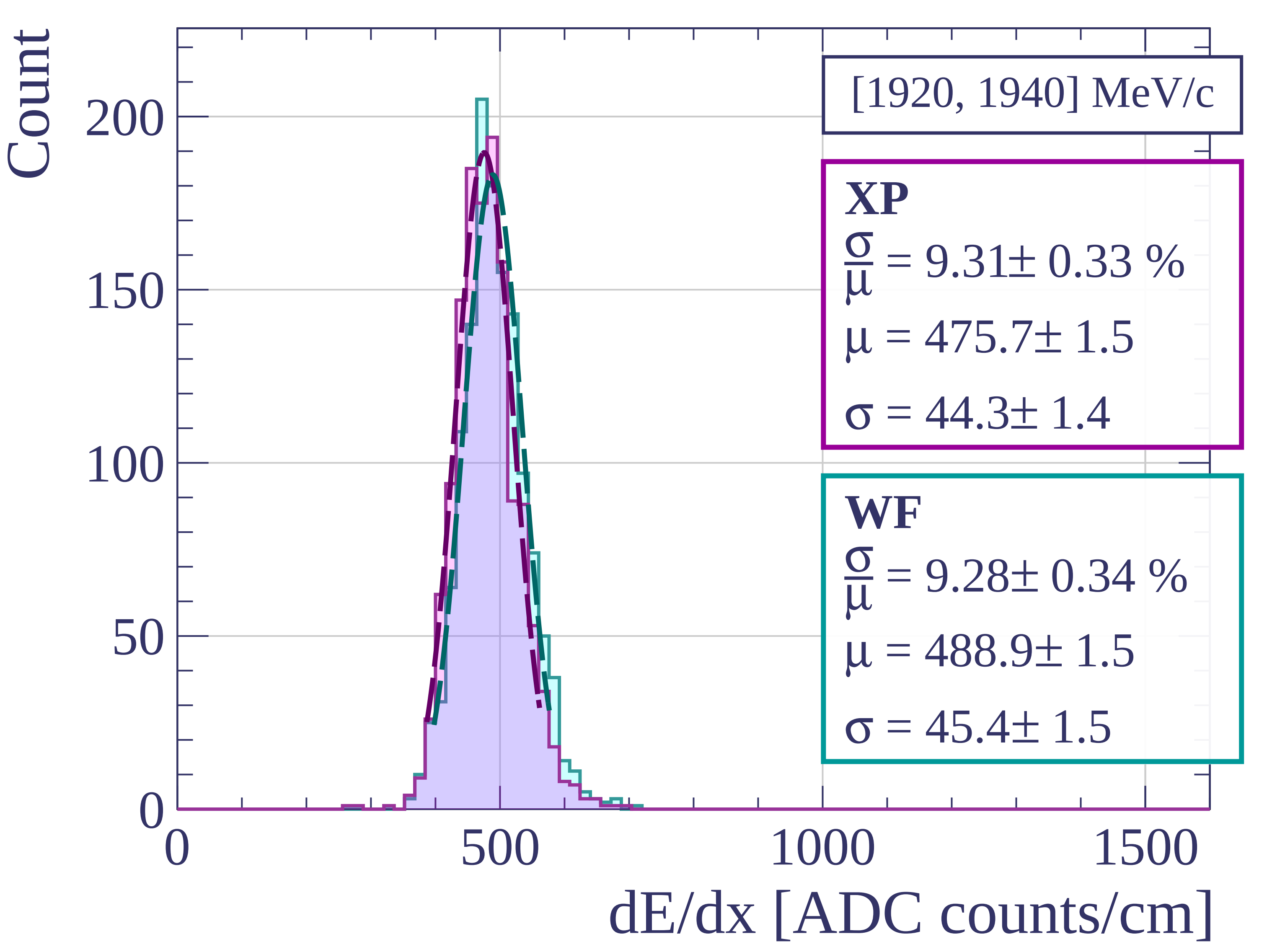
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1860, 1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.25± 0.34 %
μ = 472.4± 1.4
σ = 43.7± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.45± 0.38 %
μ = 485.2± 1.5
σ = 45.8± 1.7

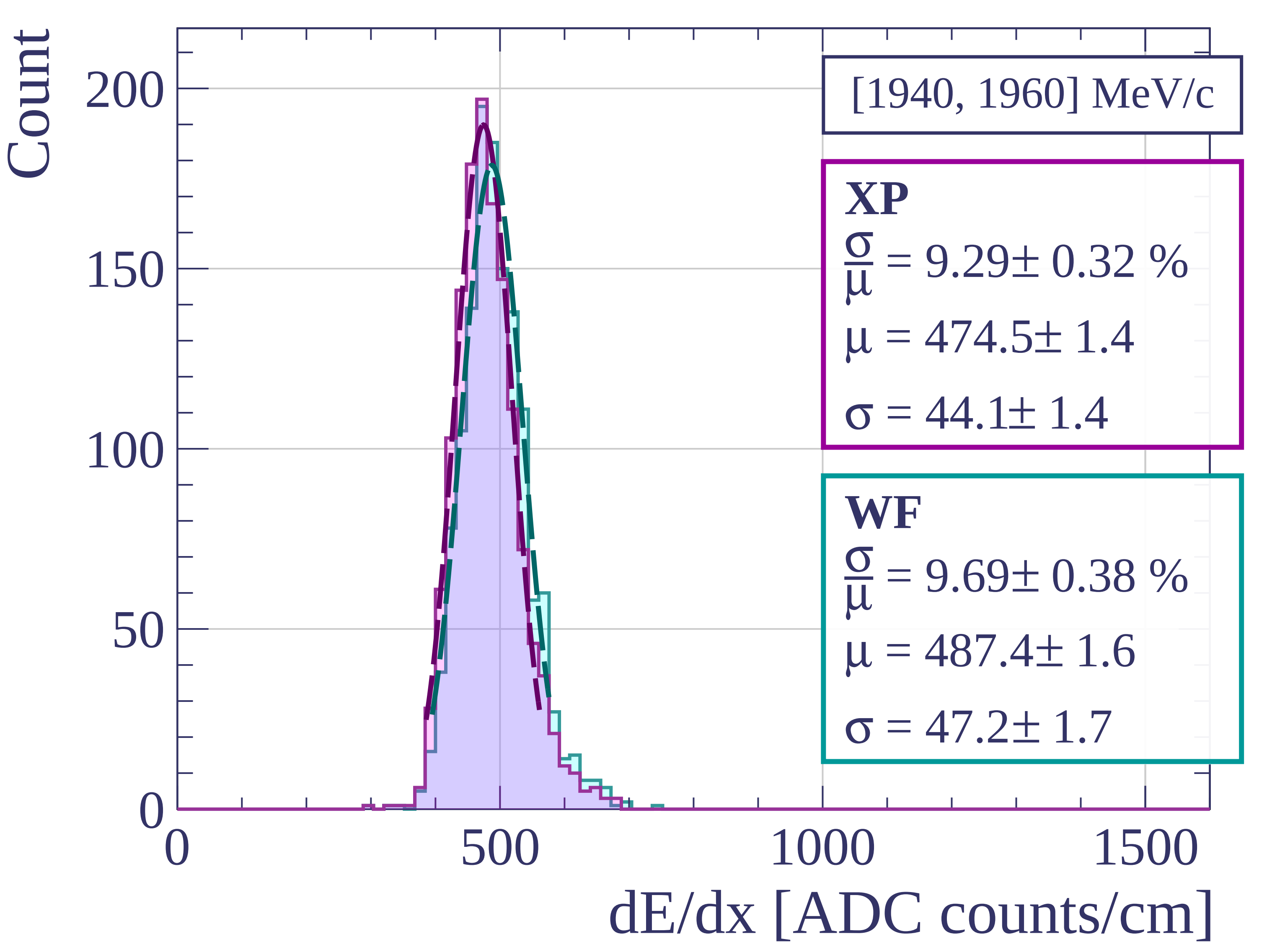


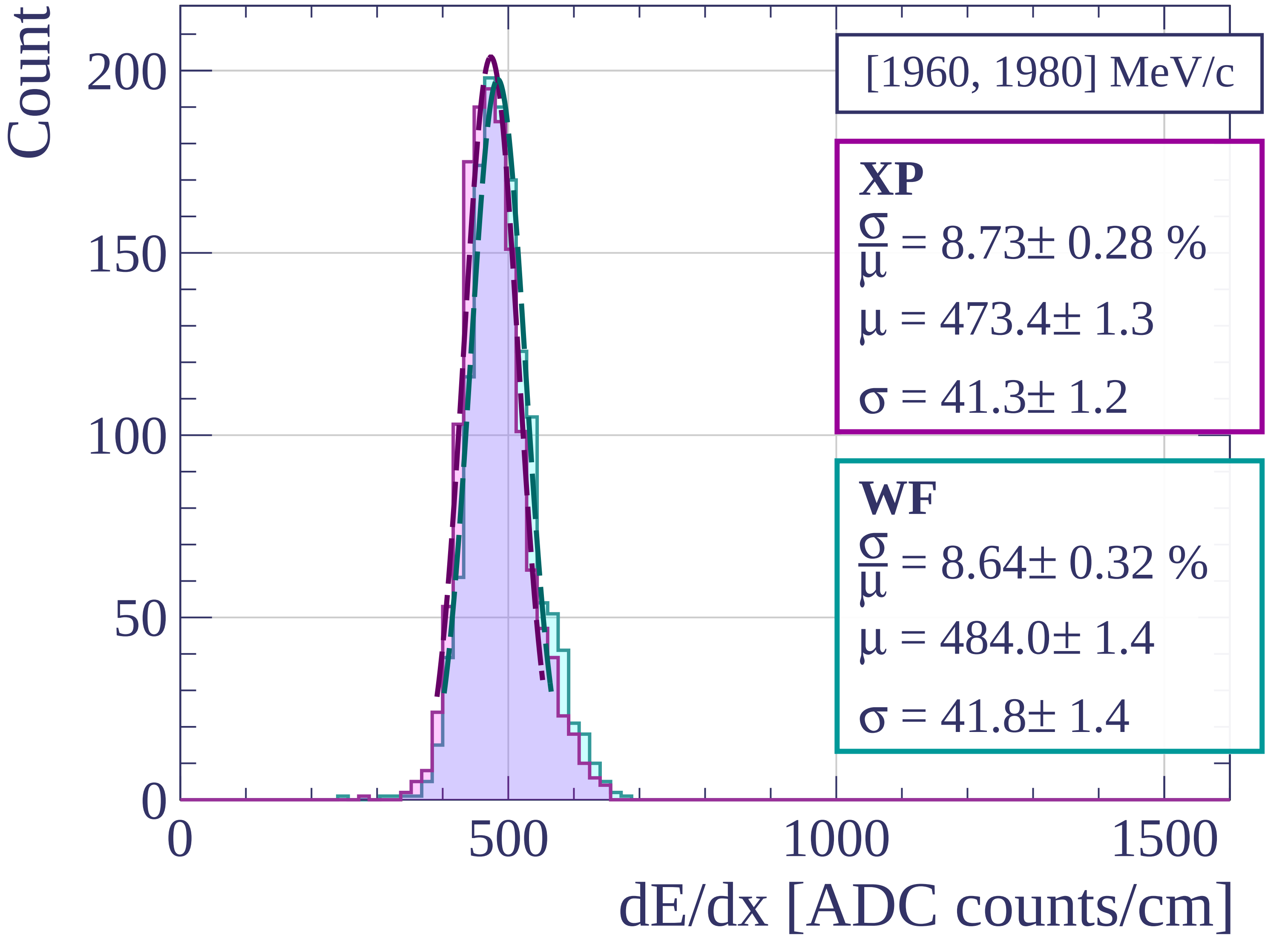
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1880, 1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.52± 0.33 %
μ = 472.7± 1.4
σ = 45.0± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.42± 0.30 %
μ = 484.4± 1.4
σ = 45.6± 1.3



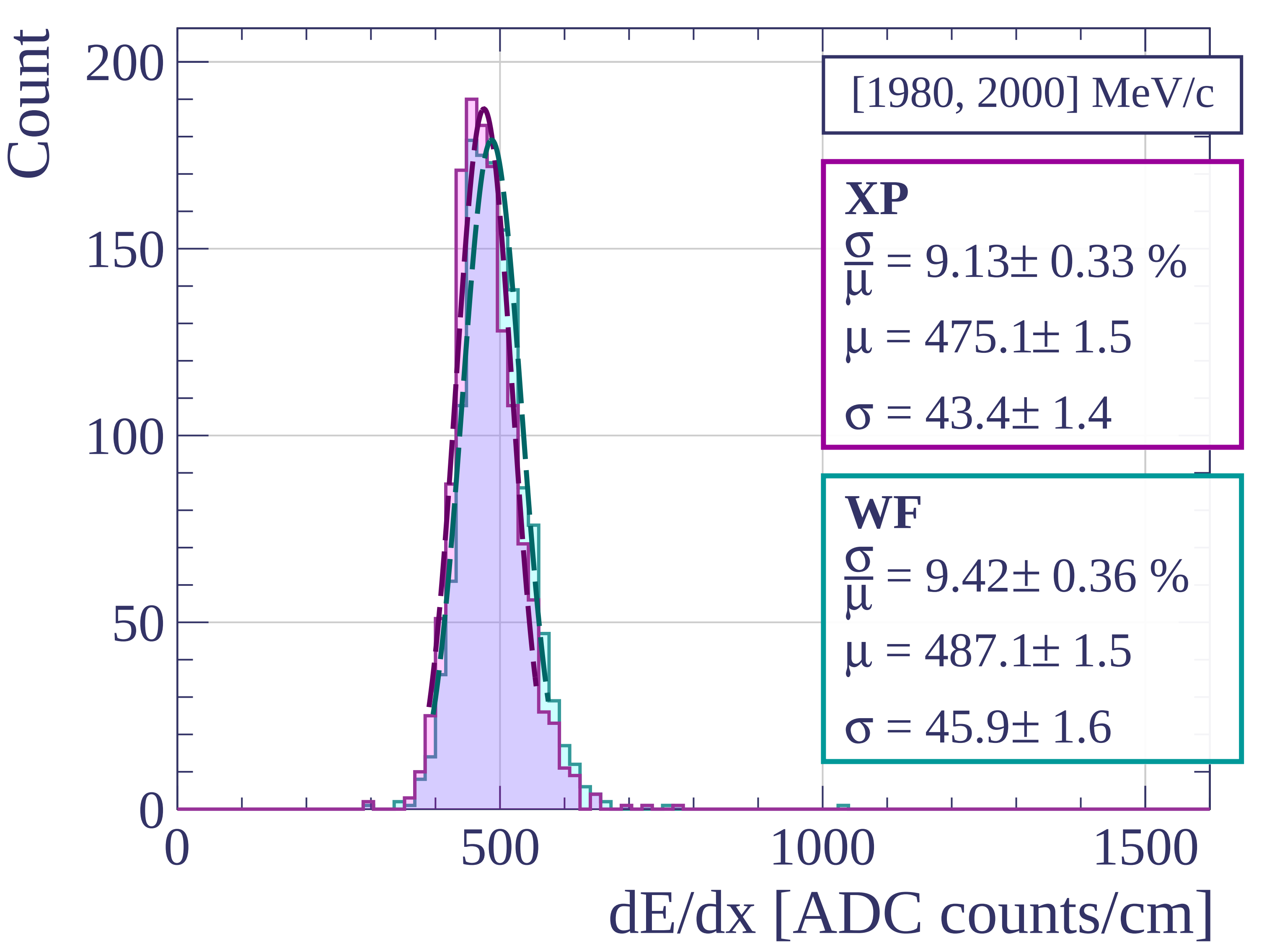
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1900, 1920] MeV/c
XP
σ/μ = 8.98± 0.29 %
μ = 471.9± 1.3
σ = 42.4± 1.2
WF
σ/μ = 9.46± 0.35 %
μ = 482.6± 1.5
σ = 45.7± 1.5


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 1940] MeV/c
XP
σ/μ = 9.31± 0.33 %
μ = 475.7± 1.5
σ = 44.3± 1.4
WF
σ/μ = 9.28± 0.34 %
μ = 488.9± 1.5
σ = 45.4± 1.5

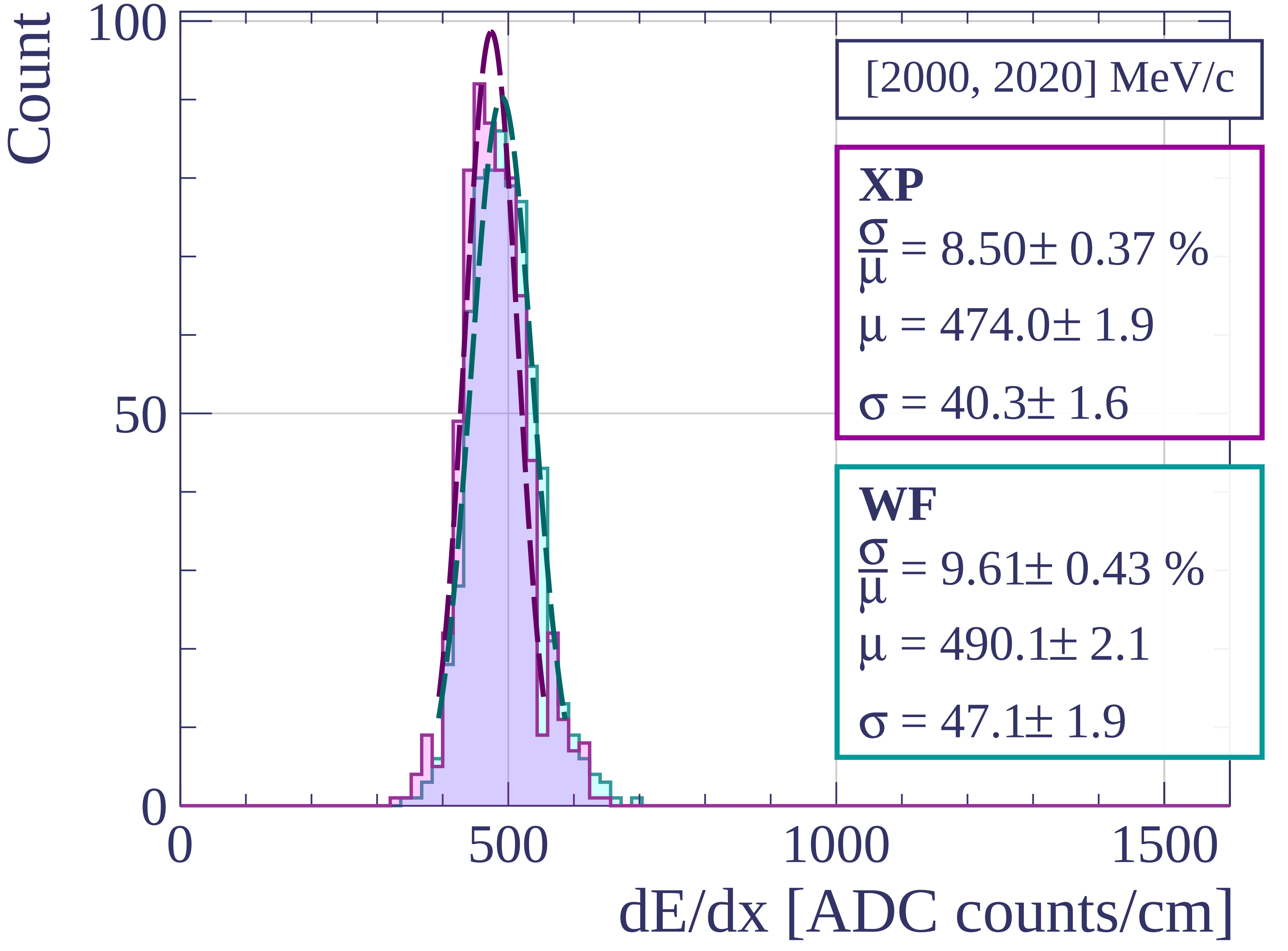

Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1940, 1960] MeV/c
XP
σ/μ = 9.29± 0.32 %
μ = 474.5± 1.4
σ = 44.1± 1.4
WF
σ/μ = 9.69± 0.38 %
μ = 487.4± 1.6
σ = 47.2± 1.7

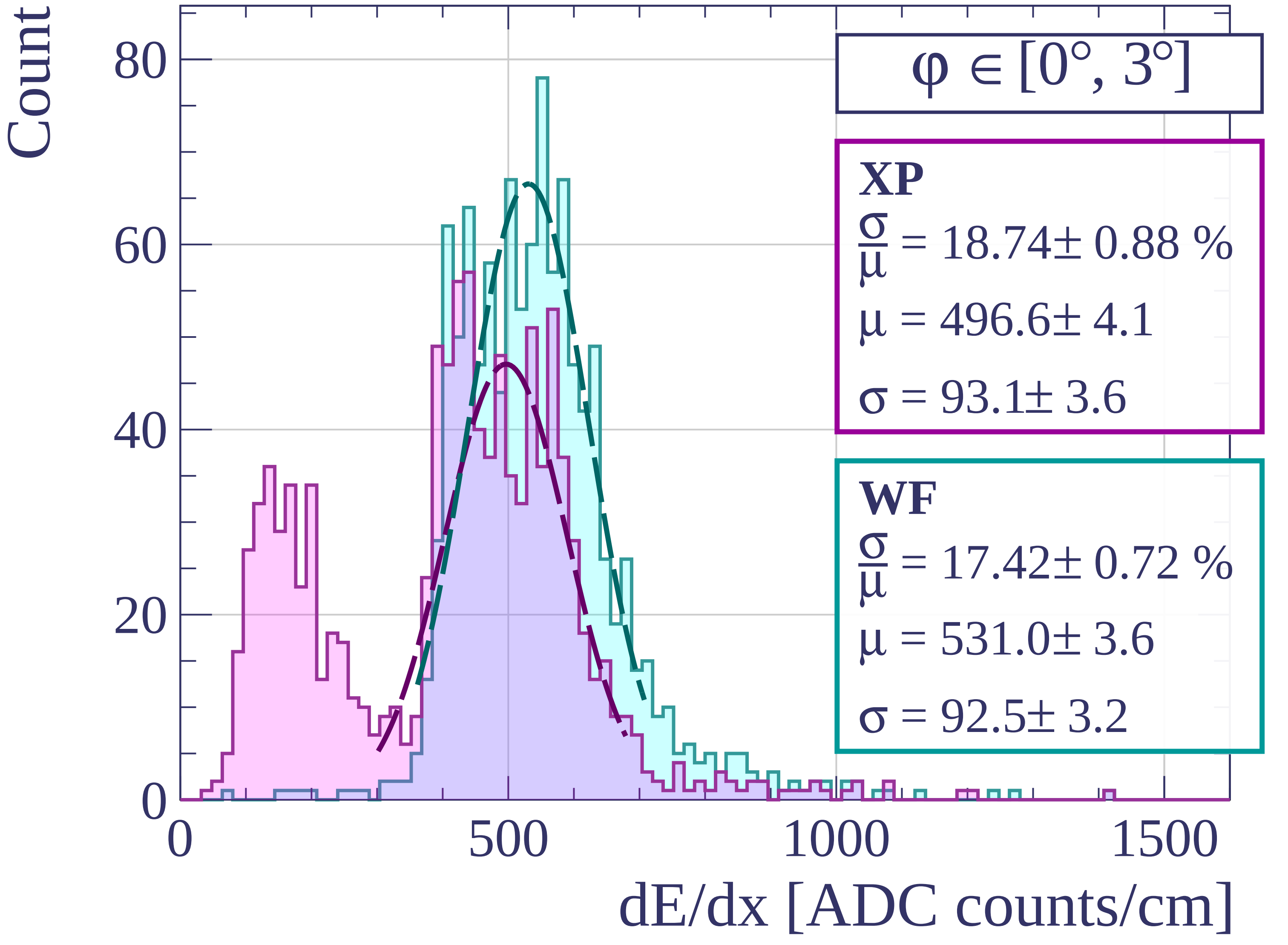


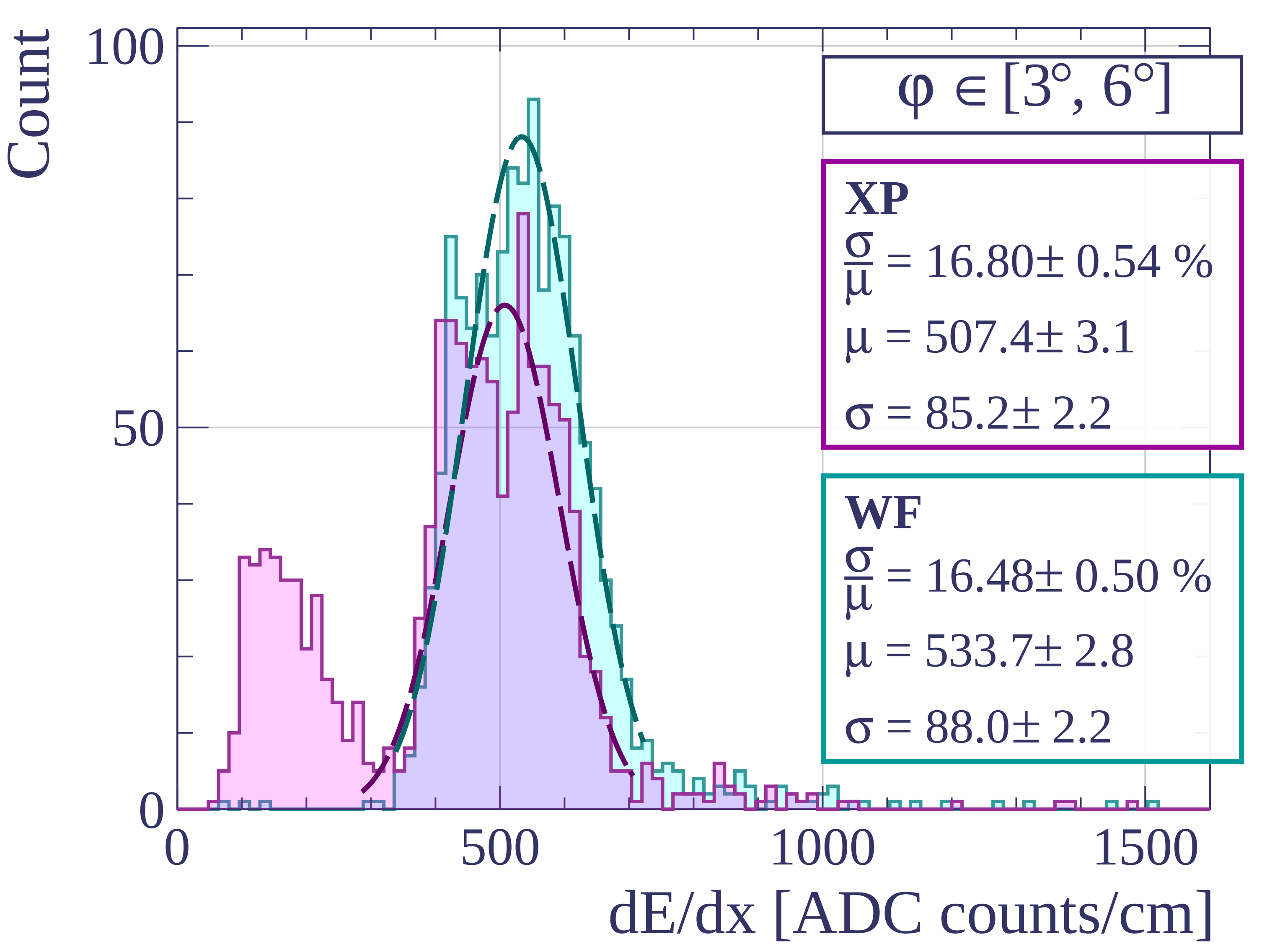
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1960, 1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.73± 0.28 %
μ = 473.4± 1.3
σ = 41.3± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.64± 0.32 %
μ = 484.0± 1.4
σ = 41.8± 1.4

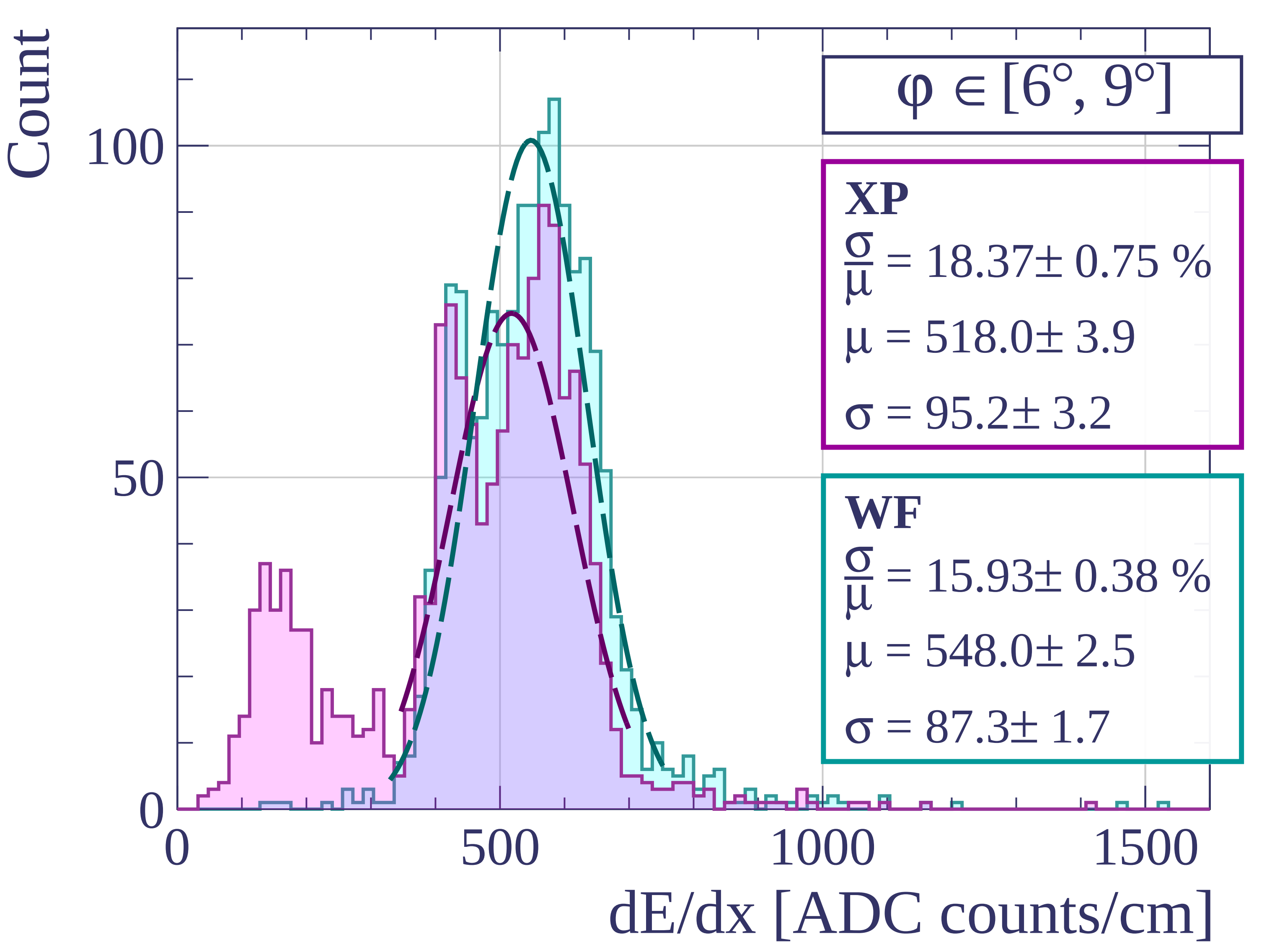


Count
dE/dx [ADC counts/cm]
[1980, 2000] MeV/c
XP
σ/μ = 9.13± 0.33 %
μ = 475.1± 1.5
σ = 43.4± 1.4
WF
σ/μ = 9.42± 0.36 %
μ = 487.1± 1.5
σ = 45.9± 1.6

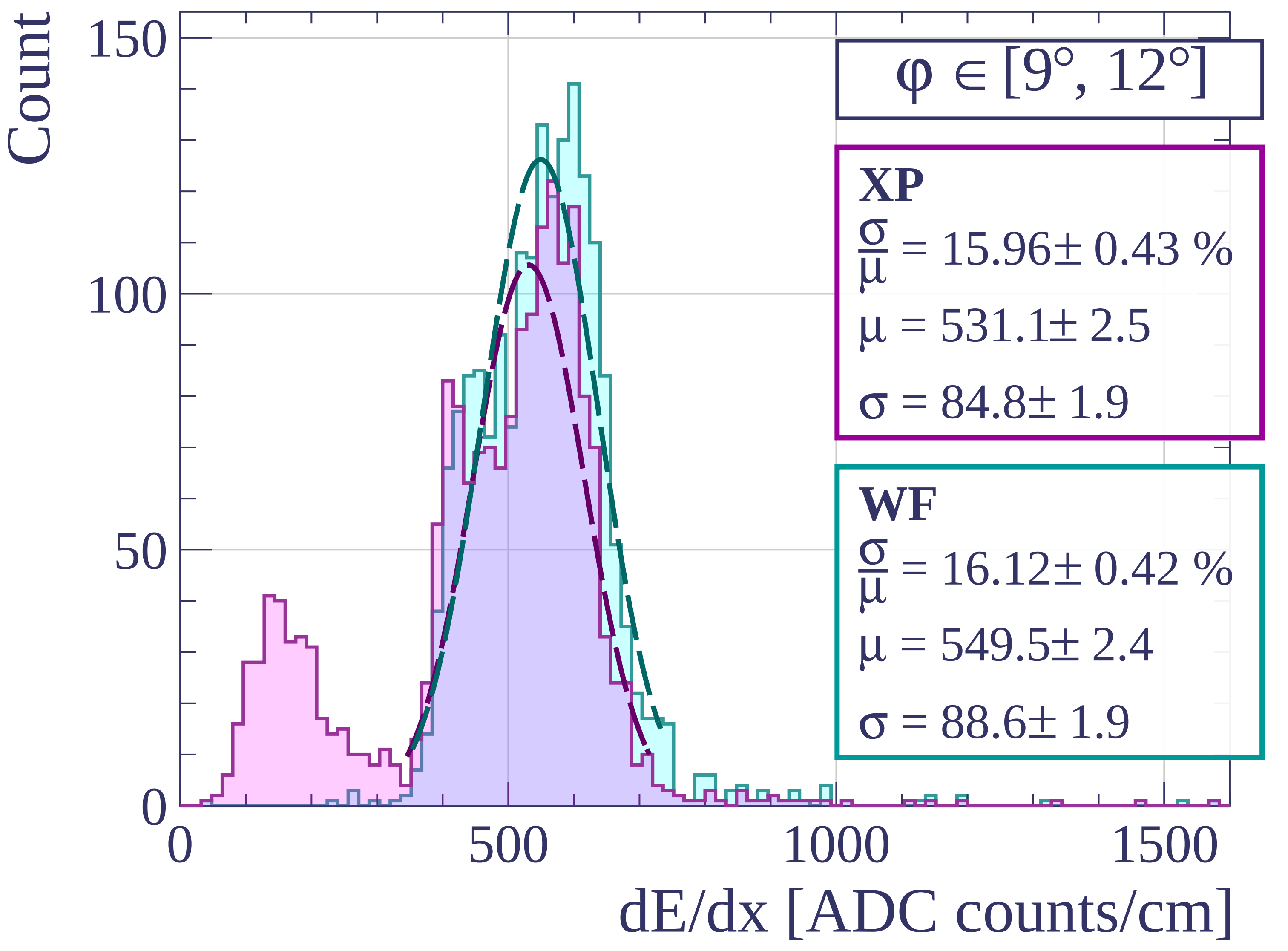

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[2000, 2020] MeV/c
XP
σ/μ = 8.50± 0.37 %
μ = 474.0± 1.9
σ = 40.3± 1.6
WF
σ/μ = 9.61± 0.43 %
μ = 490.1± 2.1
σ = 47.1± 1.9


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [0°, 3°]
XP
σ/μ = 18.74± 0.88 %
μ = 496.6± 4.1
σ = 93.1± 3.6
WF
σ/μ = 17.42± 0.72 %
μ = 531.0± 3.6
σ = 92.5± 3.2

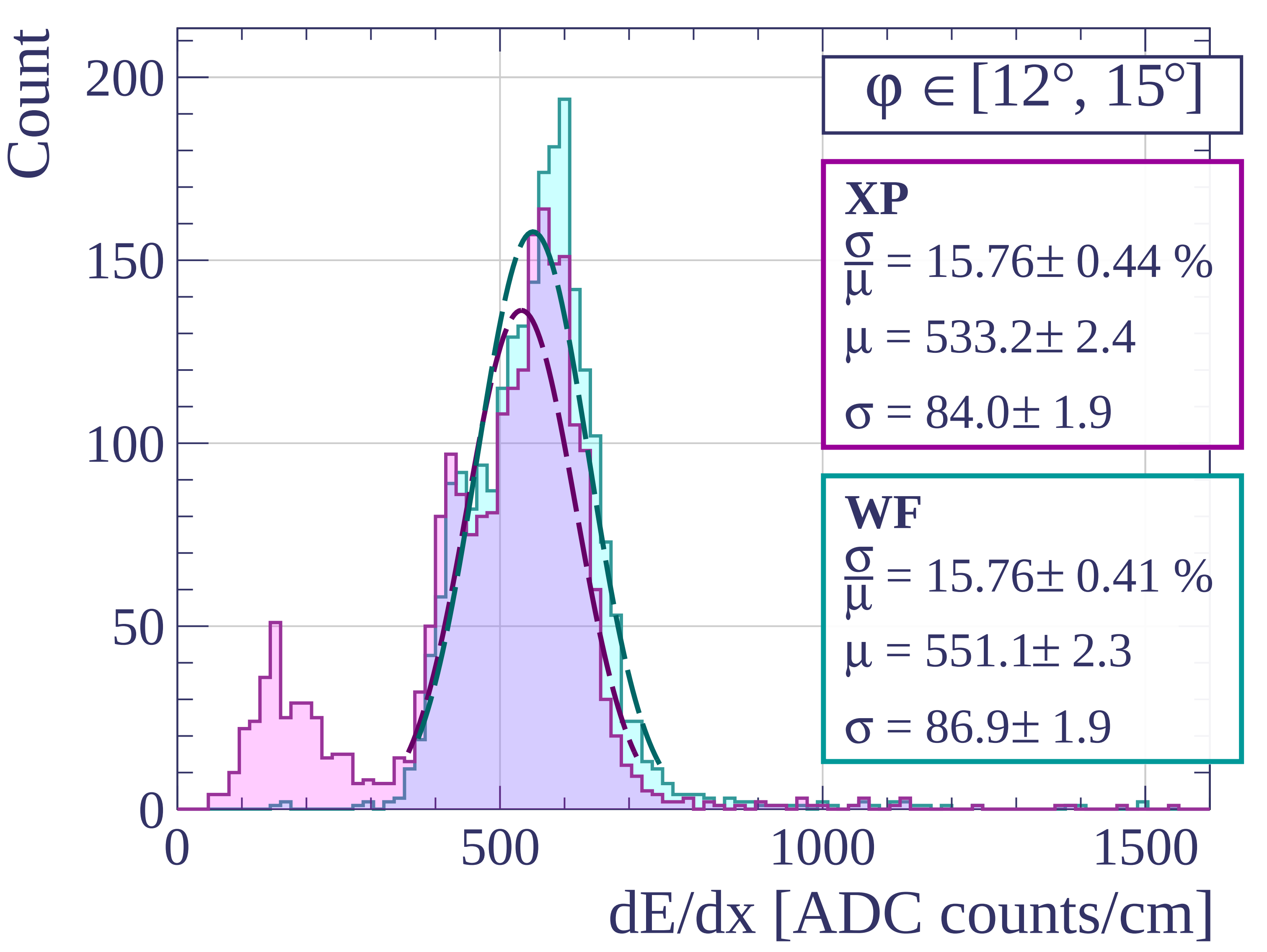

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [3°, 6°]
XP
σ/μ = 16.80± 0.54 %
μ = 507.4± 3.1
σ = 85.2± 2.2
WF
σ/μ = 16.48± 0.50 %
μ = 533.7± 2.8
σ = 88.0± 2.2



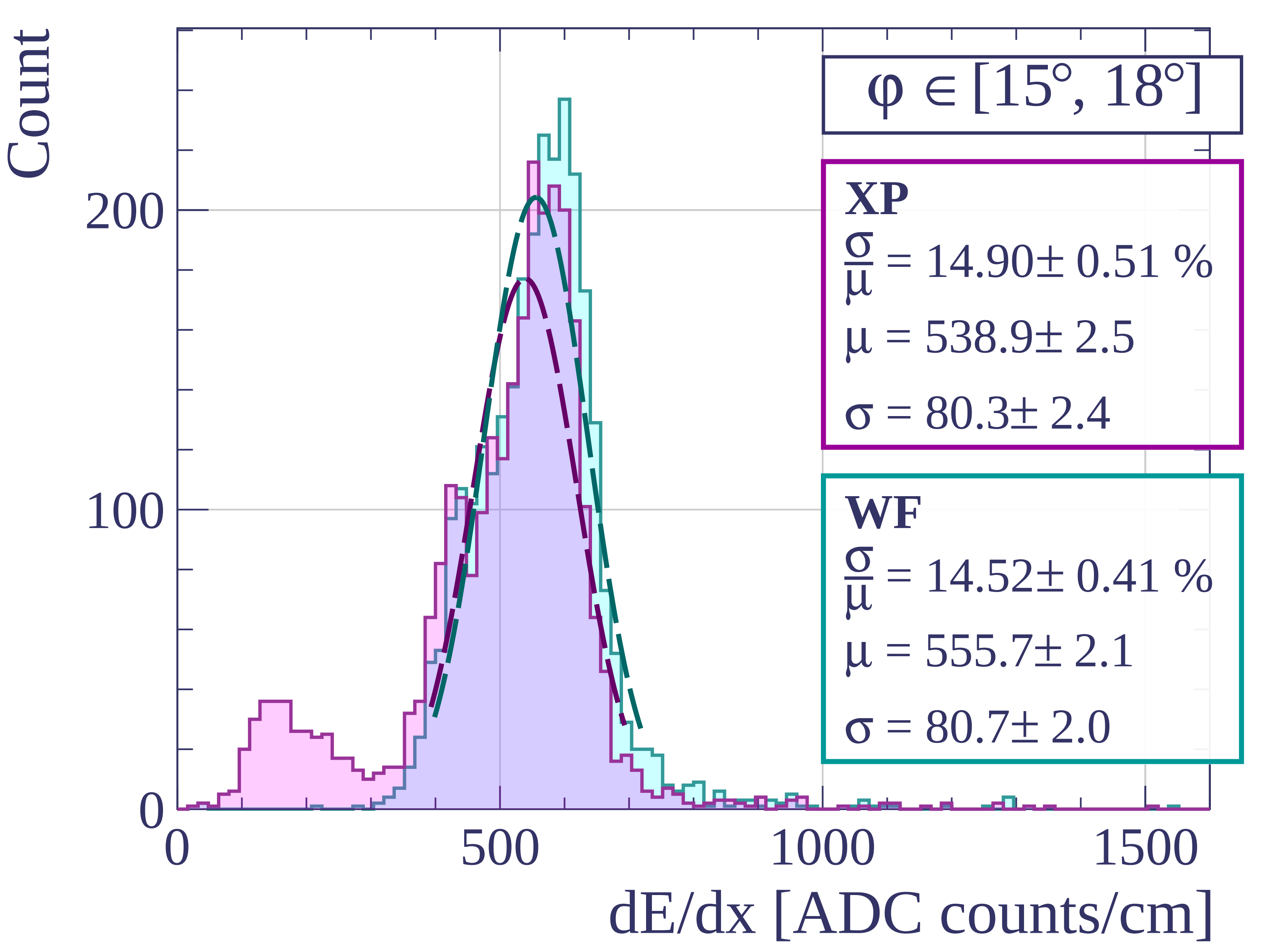
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [6°, 9°]
XP
σ/μ = 18.37± 0.75 %
μ = 518.0± 3.9
σ = 95.2± 3.2
WF
σ/μ = 15.93± 0.38 %
μ = 548.0± 2.5
σ = 87.3± 1.7



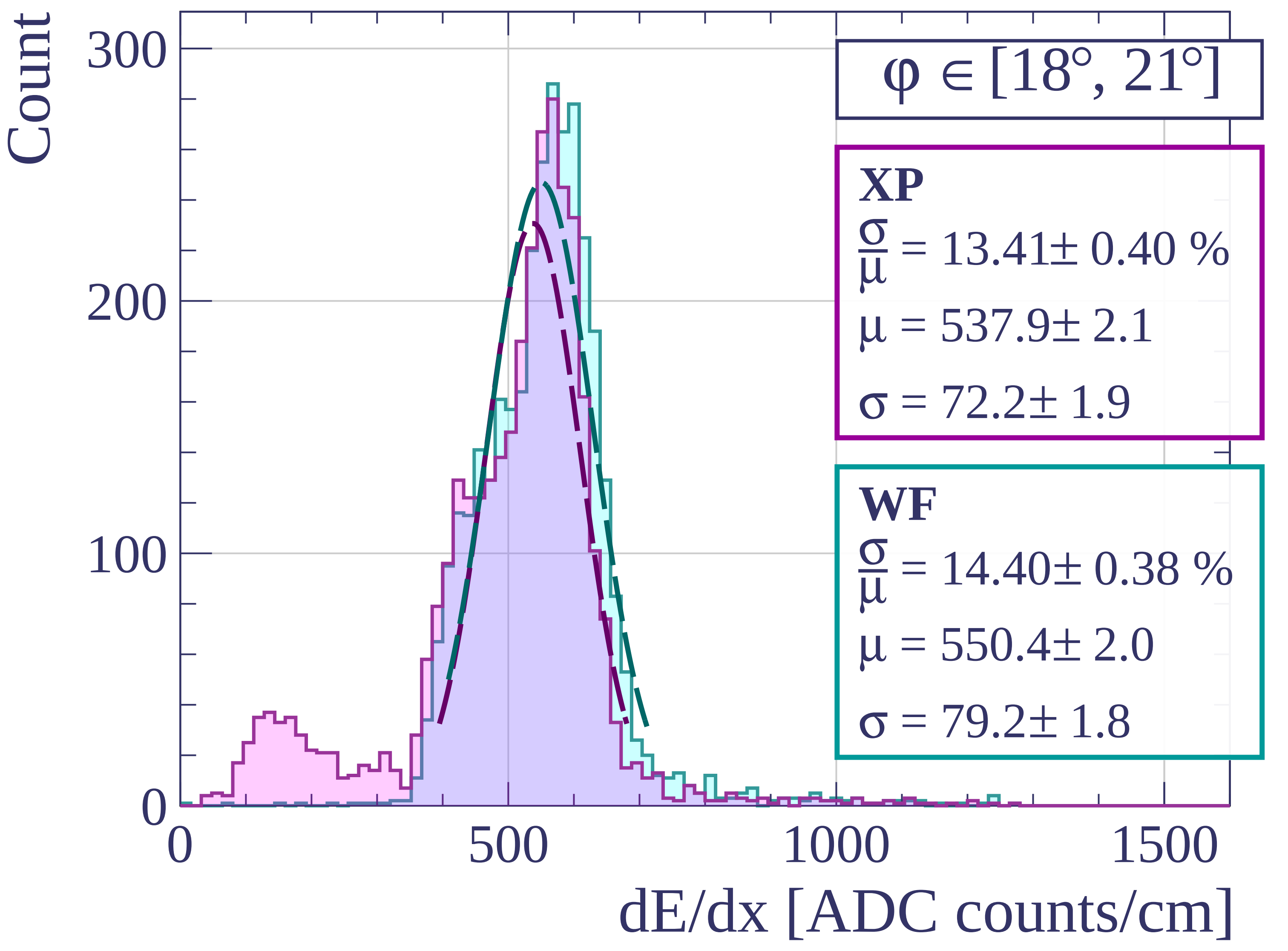
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [9°, 12°]
XP
σ/μ = 15.96± 0.43 %
μ = 531.1± 2.5
σ = 84.8± 1.9
WF
σ/μ = 16.12± 0.42 %
μ = 549.5± 2.4
σ = 88.6± 1.9



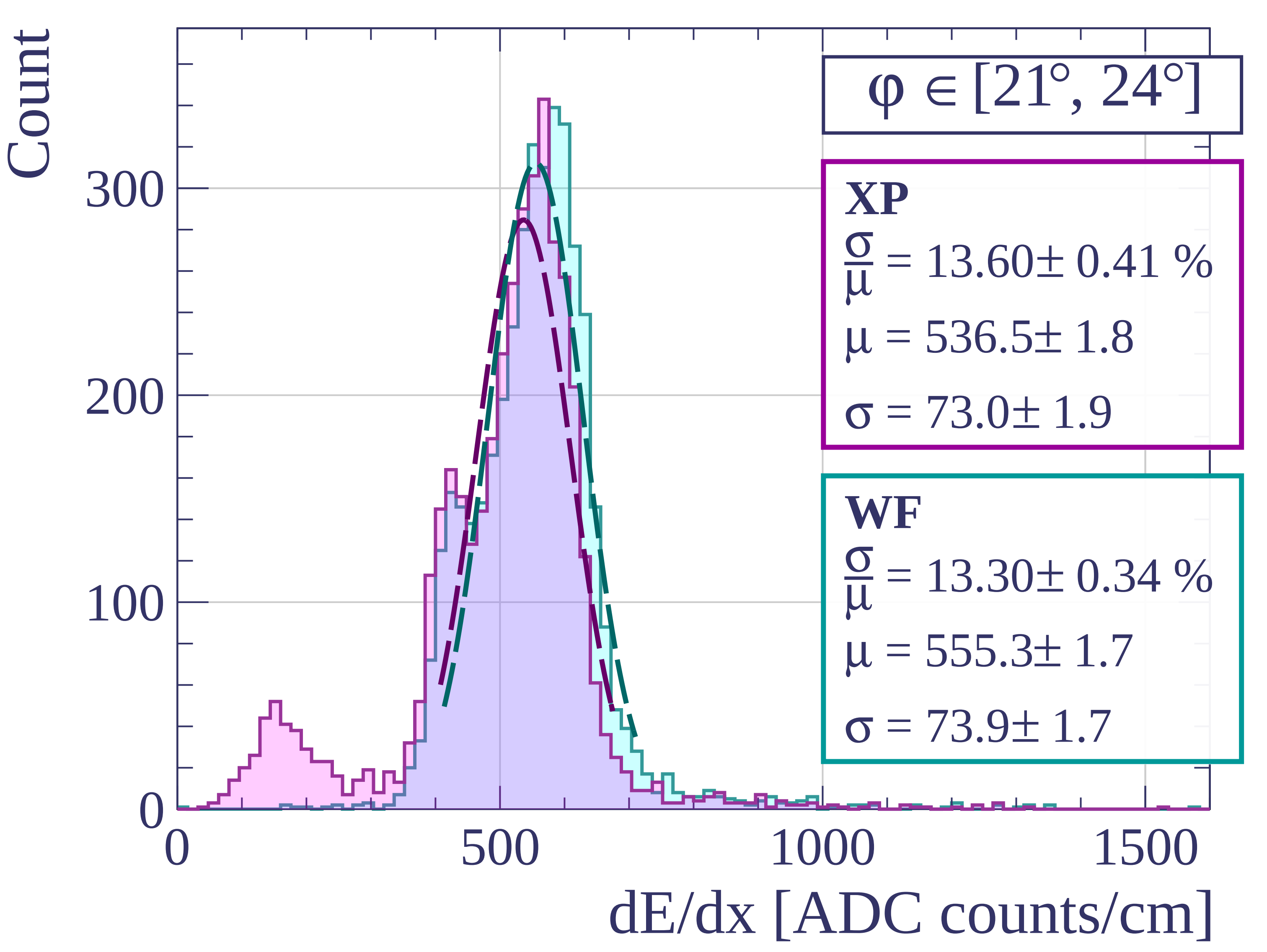
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [12°, 15°]
XP
σ/μ = 15.76± 0.44 %
μ = 533.2± 2.4
σ = 84.0± 1.9
WF
σ/μ = 15.76± 0.41 %
μ = 551.1± 2.3
σ = 86.9± 1.9



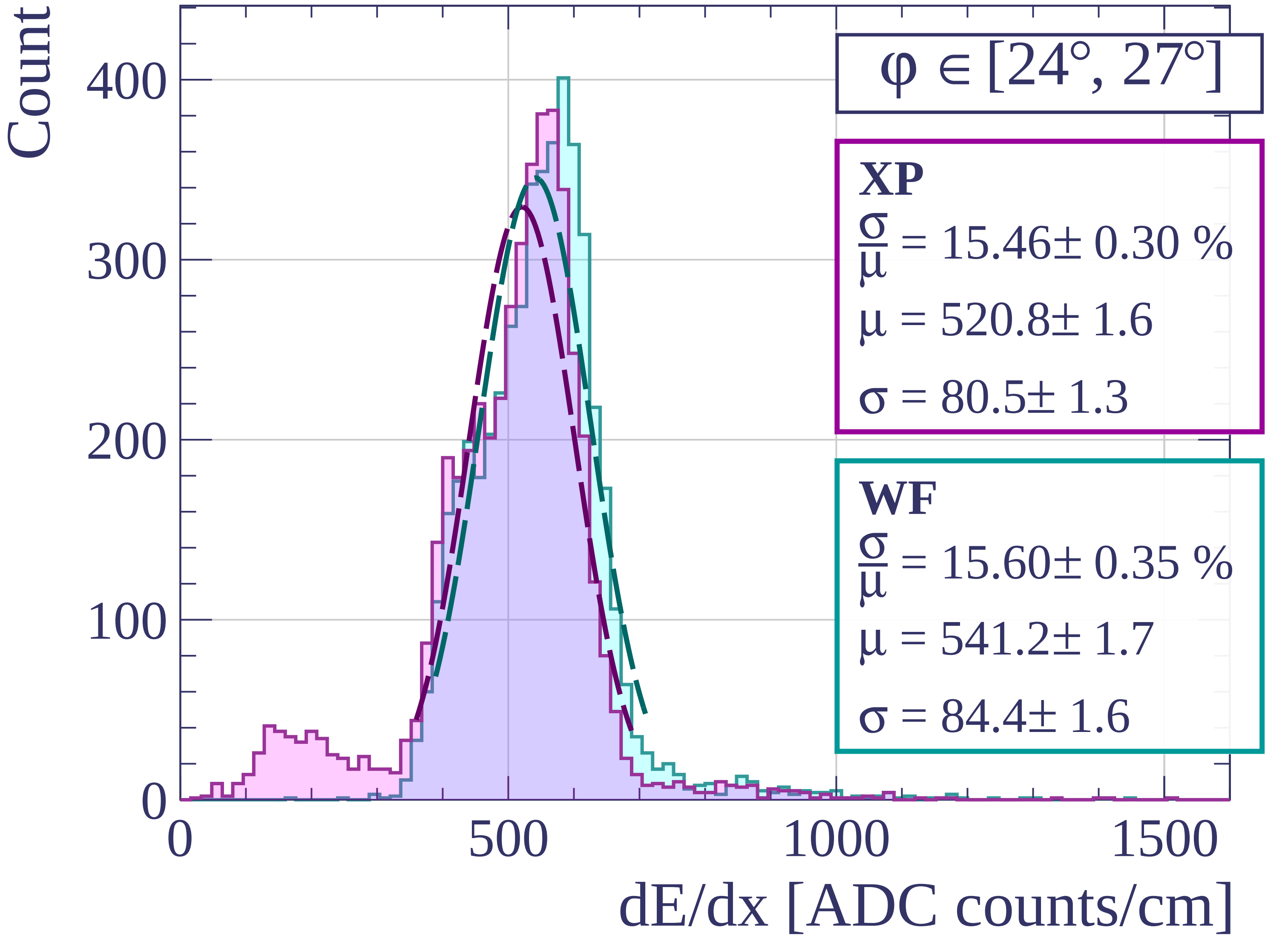
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [15°, 18°]
XP
σ/μ = 14.90± 0.51 %
μ = 538.9± 2.5
σ = 80.3± 2.4
WF
σ/μ = 14.52± 0.41 %
μ = 555.7± 2.1
σ = 80.7± 2.0



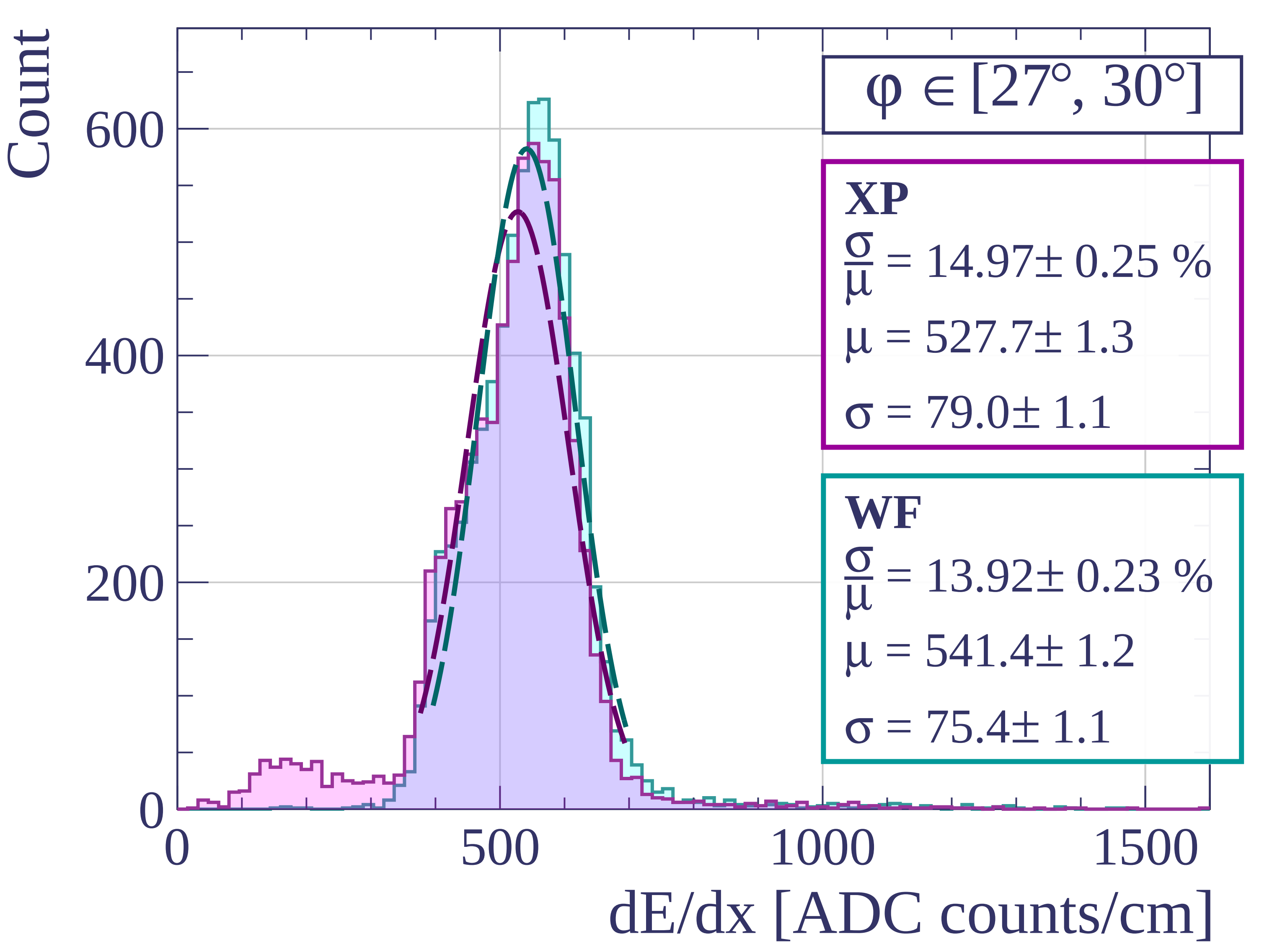
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [18°, 21°]
XP
σ/μ = 13.41± 0.40 %
μ = 537.9± 2.1
σ = 72.2± 1.9
WF
σ/μ = 14.40± 0.38 %
μ = 550.4± 2.0
σ = 79.2± 1.8



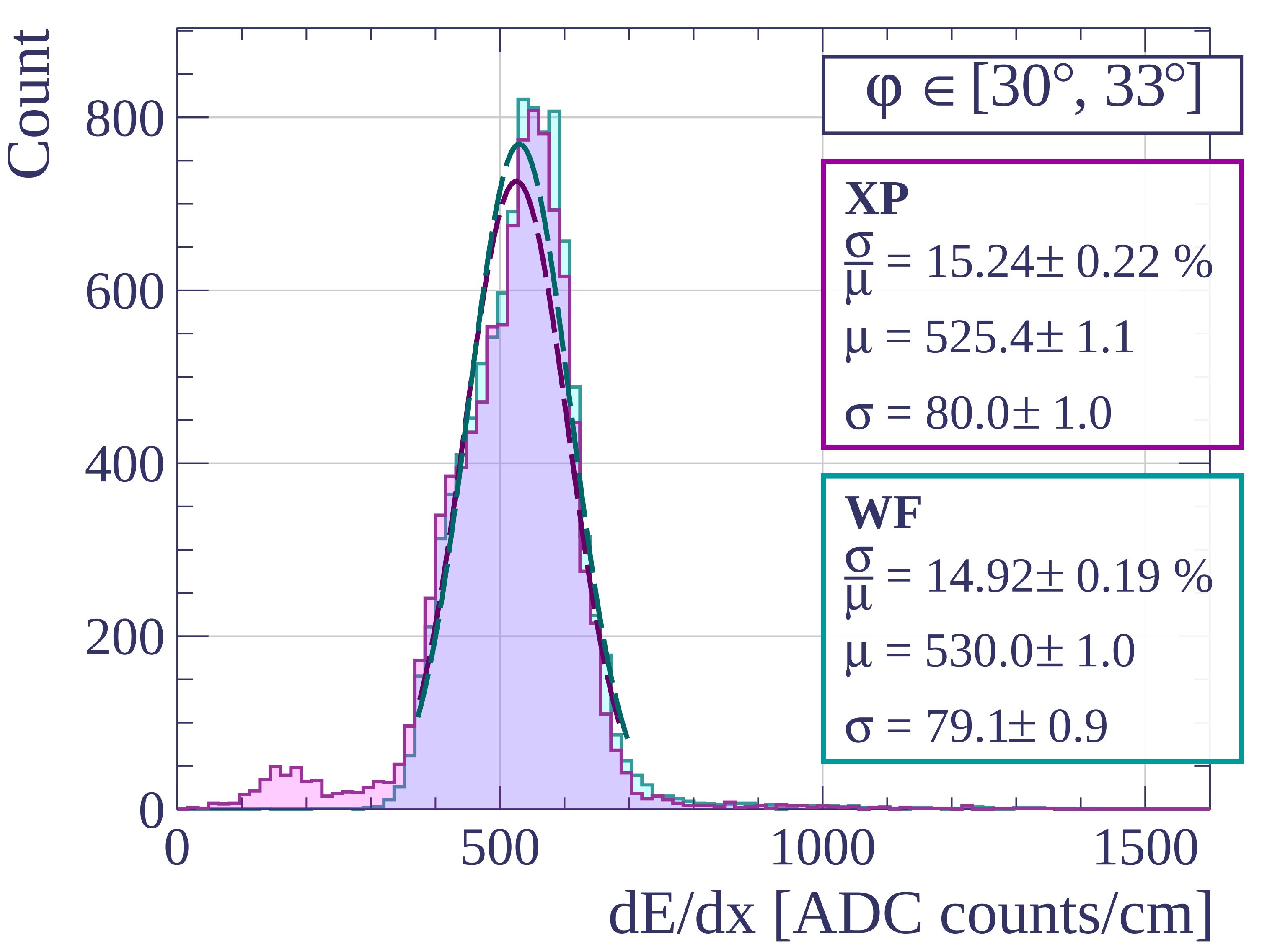
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [21°, 24°]
XP
σ/μ = 13.60± 0.41 %
μ = 536.5± 1.8
σ = 73.0± 1.9
WF
σ/μ = 13.30± 0.34 %
μ = 555.3± 1.7
σ = 73.9± 1.7



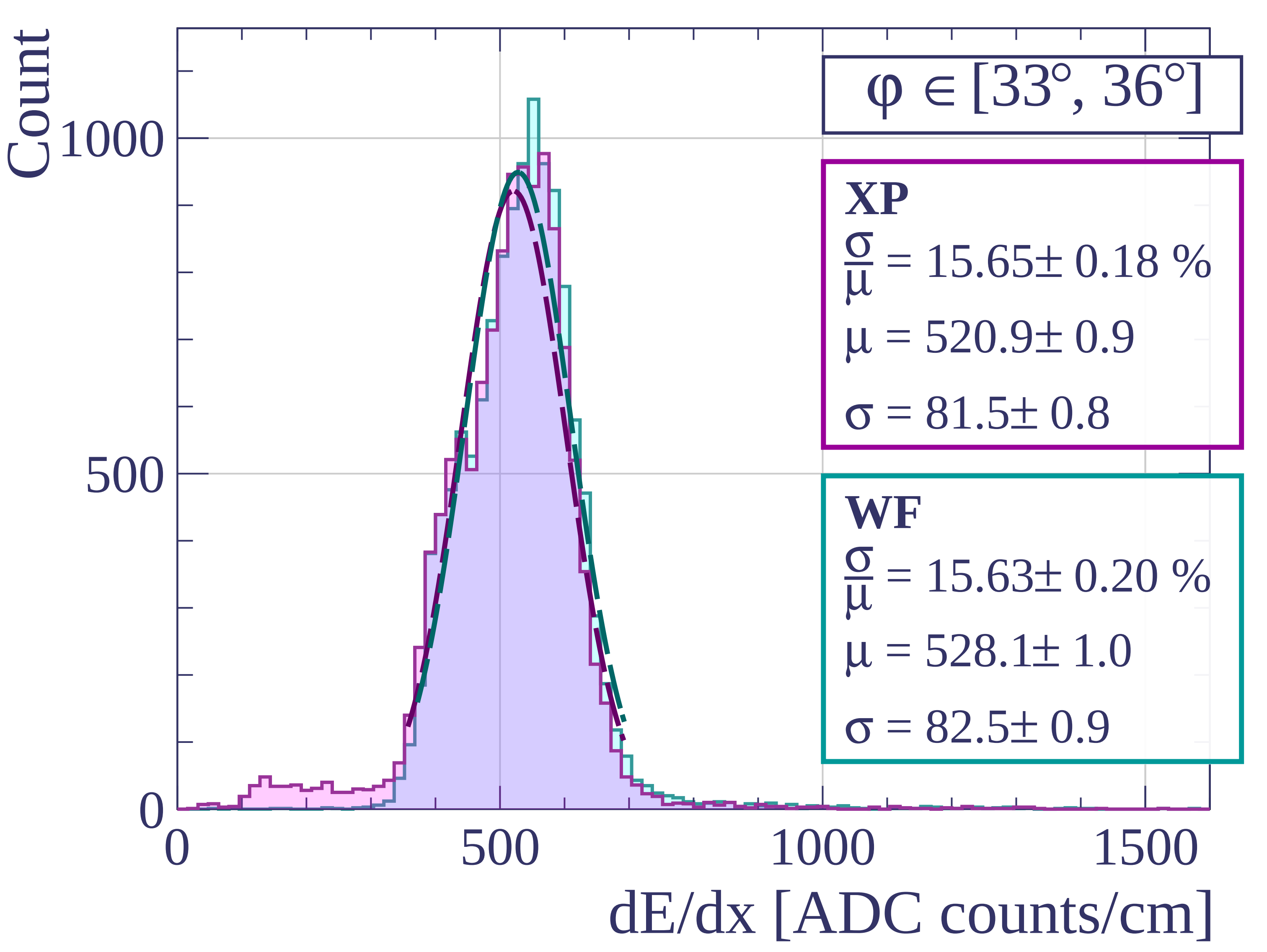
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [24°, 27°]
XP
σ/μ = 15.46± 0.30 %
μ = 520.8± 1.6
σ = 80.5± 1.3
WF
σ/μ = 15.60± 0.35 %
μ = 541.2± 1.7
σ = 84.4± 1.6

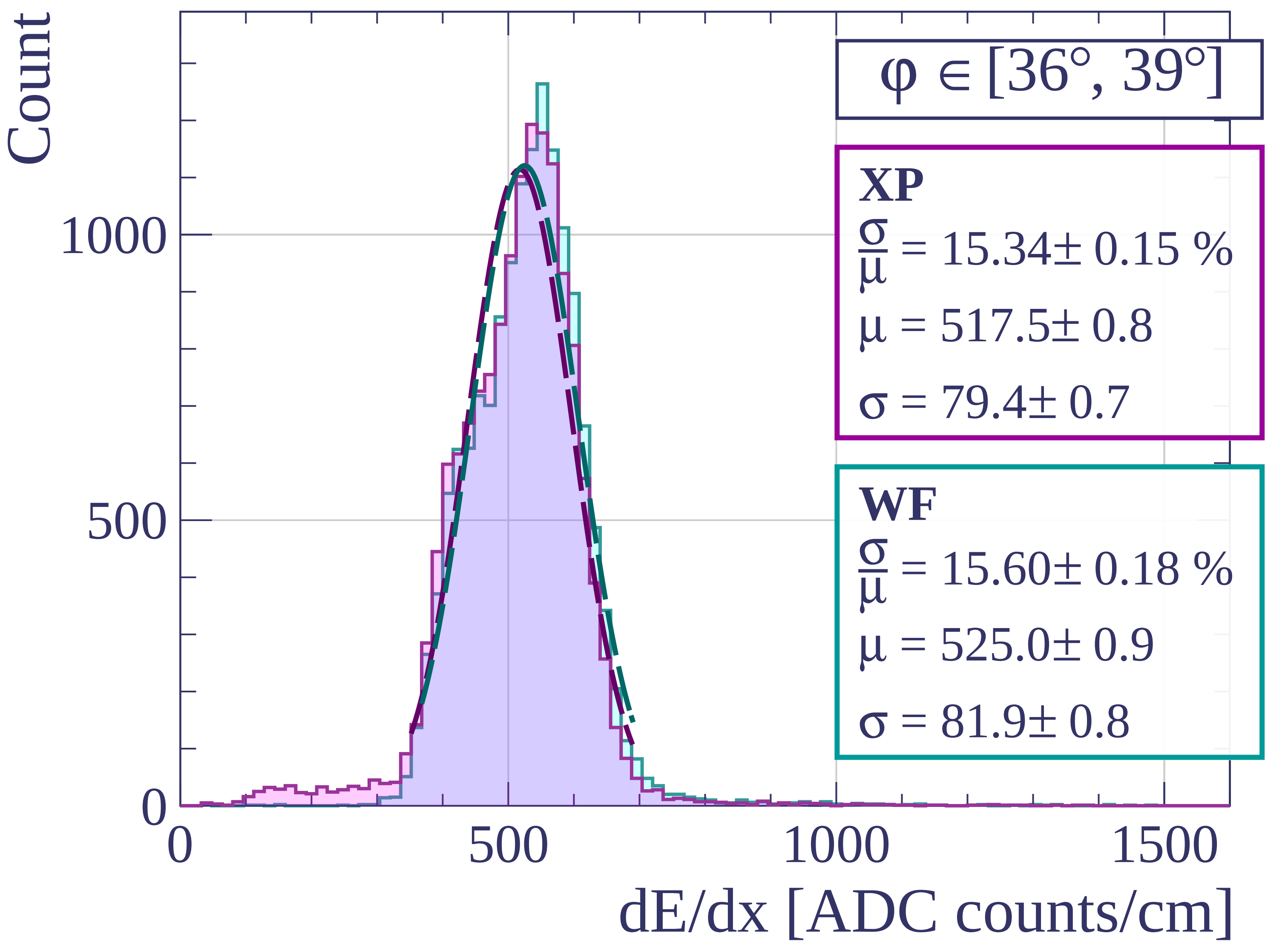


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [27°, 30°]
XP
σ/μ = 14.97± 0.25 %
μ = 527.7± 1.3
σ = 79.0± 1.1
WF
σ/μ = 13.92± 0.23 %
μ = 541.4± 1.2
σ = 75.4± 1.1

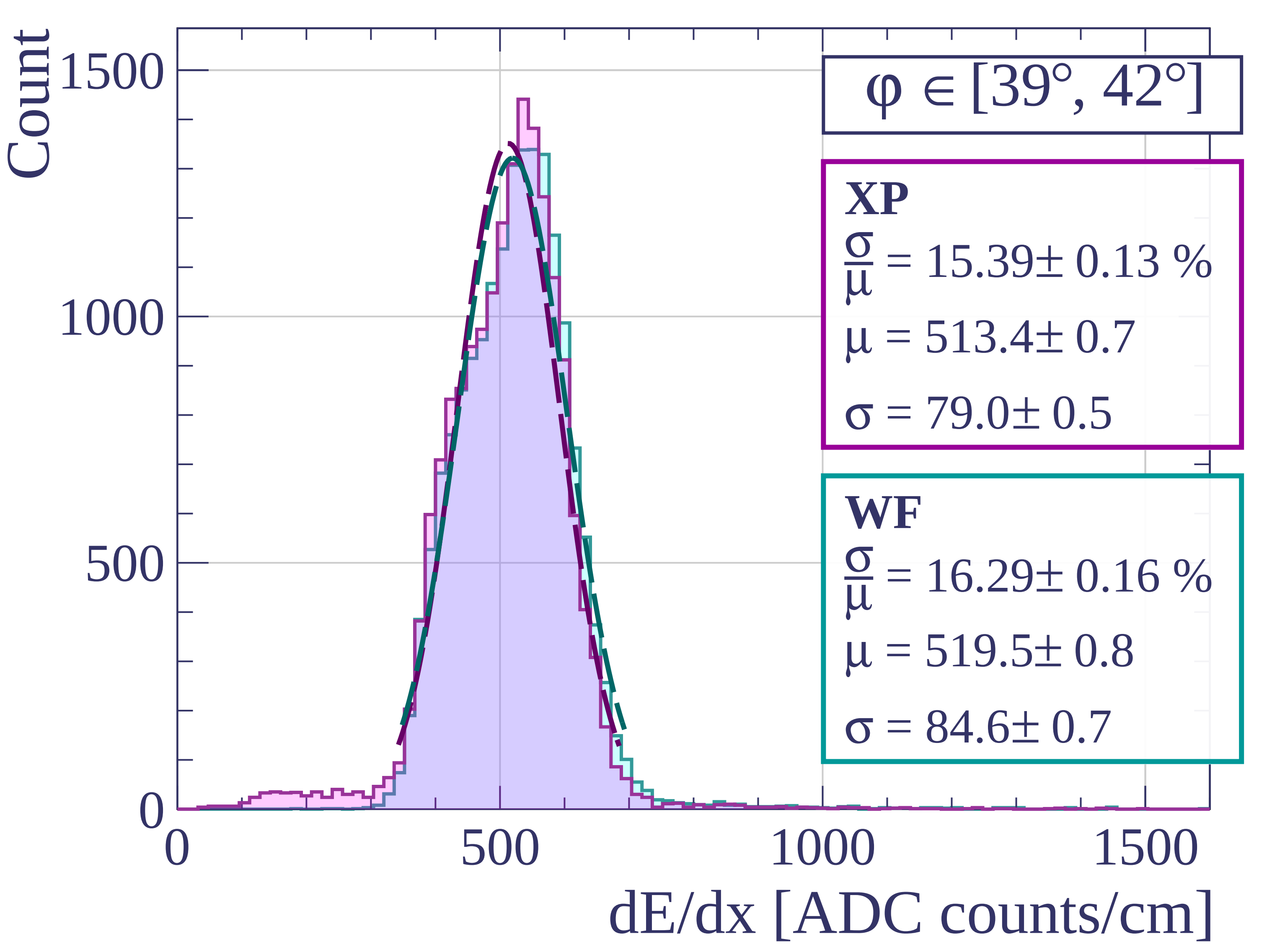


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
200
400
600
800
φ ∈ [30°, 33°]
XP
σ/μ = 15.24± 0.22 %
μ = 525.4± 1.1
σ = 80.0± 1.0
WF
σ/μ = 14.92± 0.19 %
μ = 530.0± 1.0
σ = 79.1± 0.9

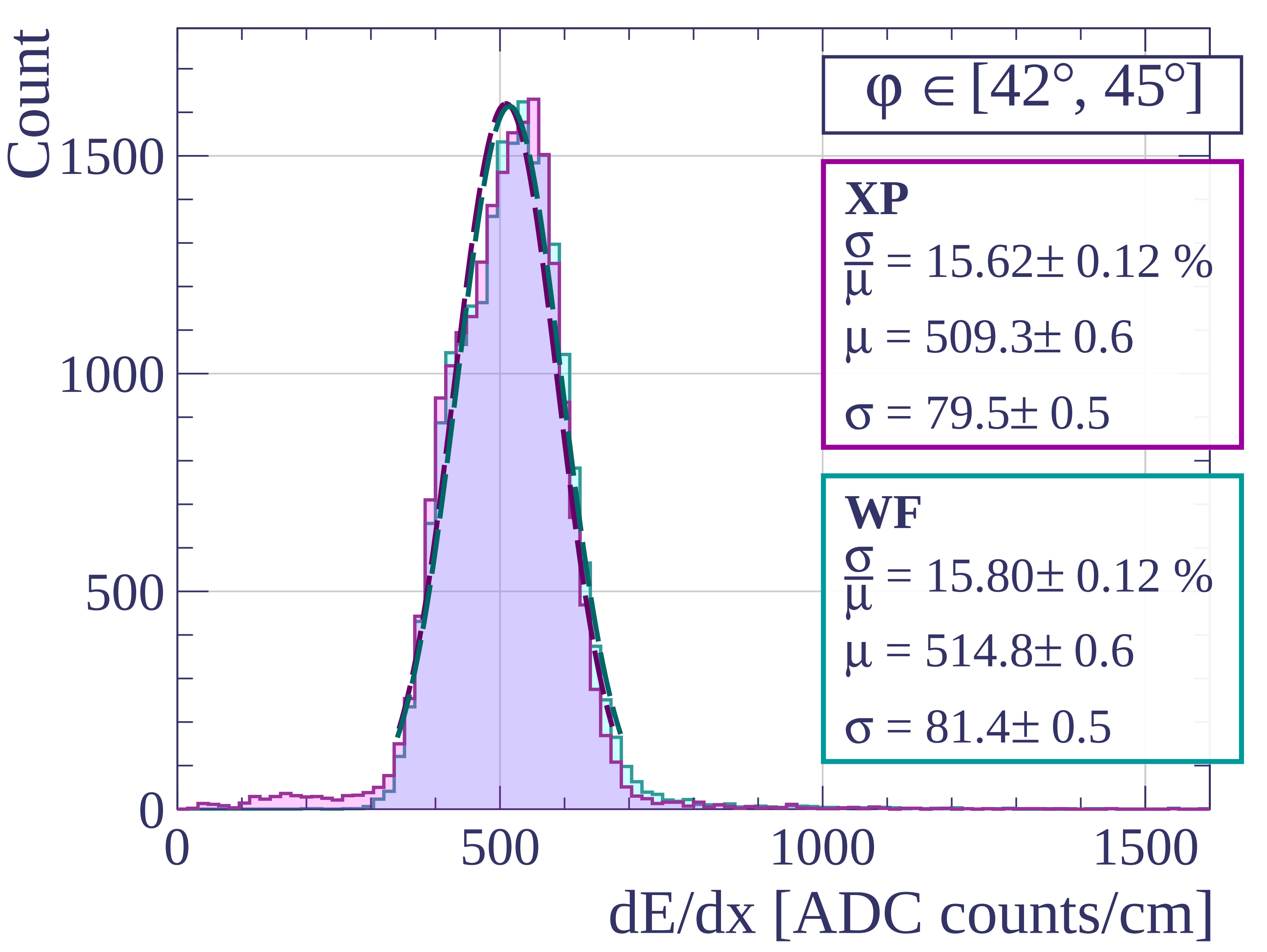

Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [33°, 36°]
XP
σ/μ = 15.65± 0.18 %
μ = 520.9± 0.9
σ = 81.5± 0.8
WF
σ/μ = 15.63± 0.20 %
μ = 528.1± 1.0
σ = 82.5± 0.9



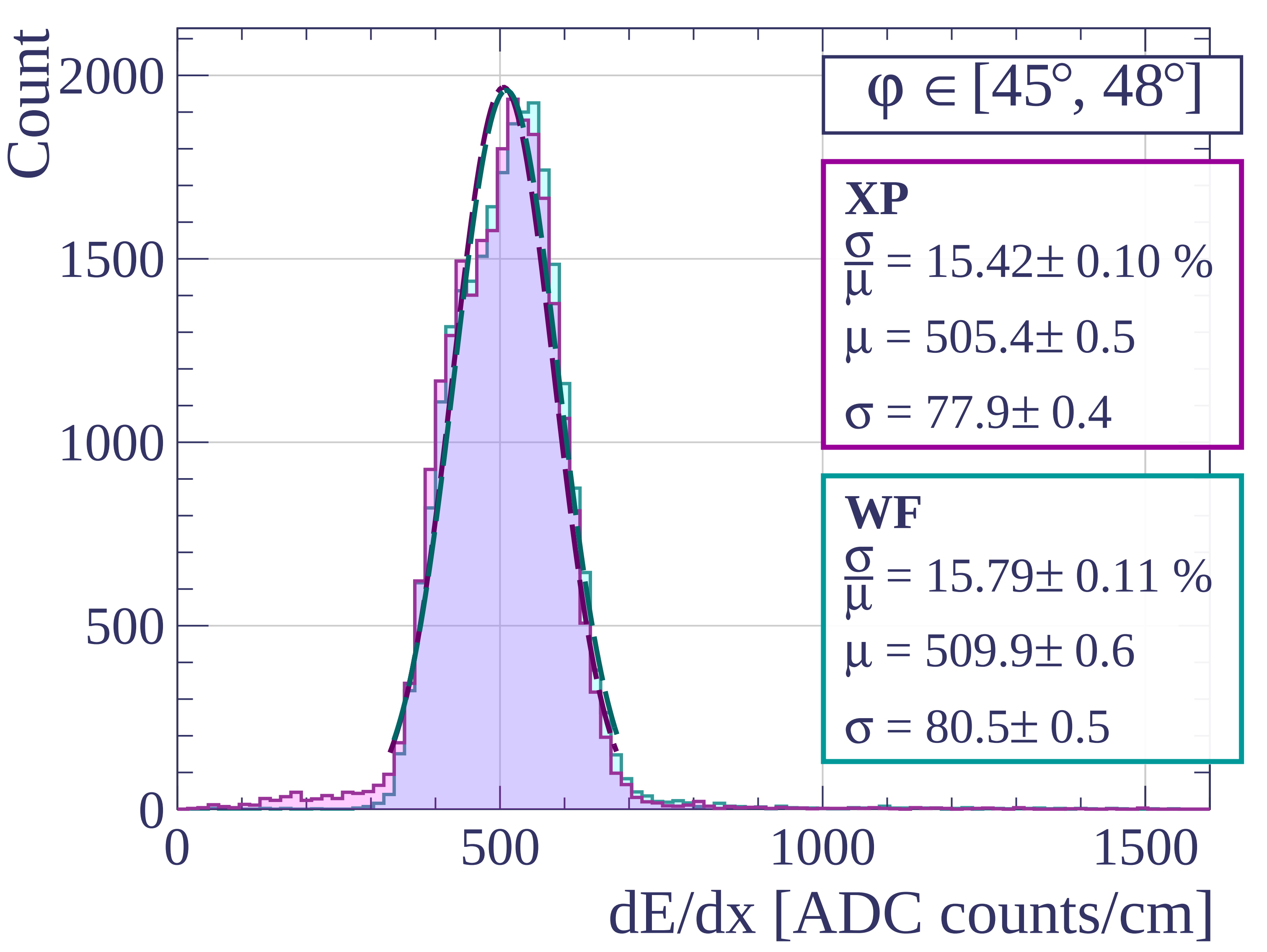
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [36°, 39°]
XP
σ/μ = 15.34± 0.15 %
μ = 517.5± 0.8
σ = 79.4± 0.7
WF
σ/μ = 15.60± 0.18 %
μ = 525.0± 0.9
σ = 81.9± 0.8



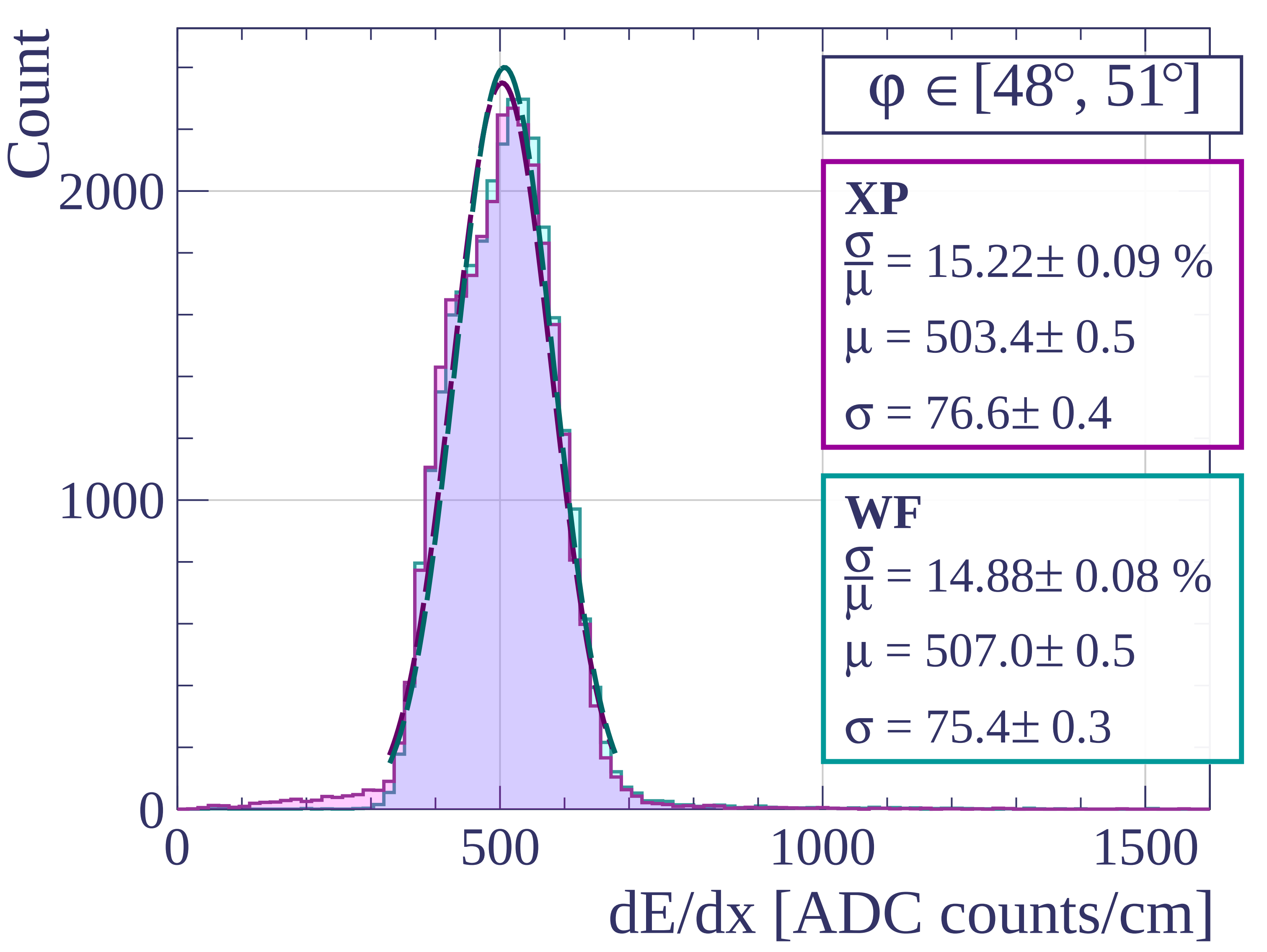
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [39°, 42°]
XP
σ/μ = 15.39± 0.13 %
μ = 513.4± 0.7
σ = 79.0± 0.5
WF
σ/μ = 16.29± 0.16 %
μ = 519.5± 0.8
σ = 84.6± 0.7



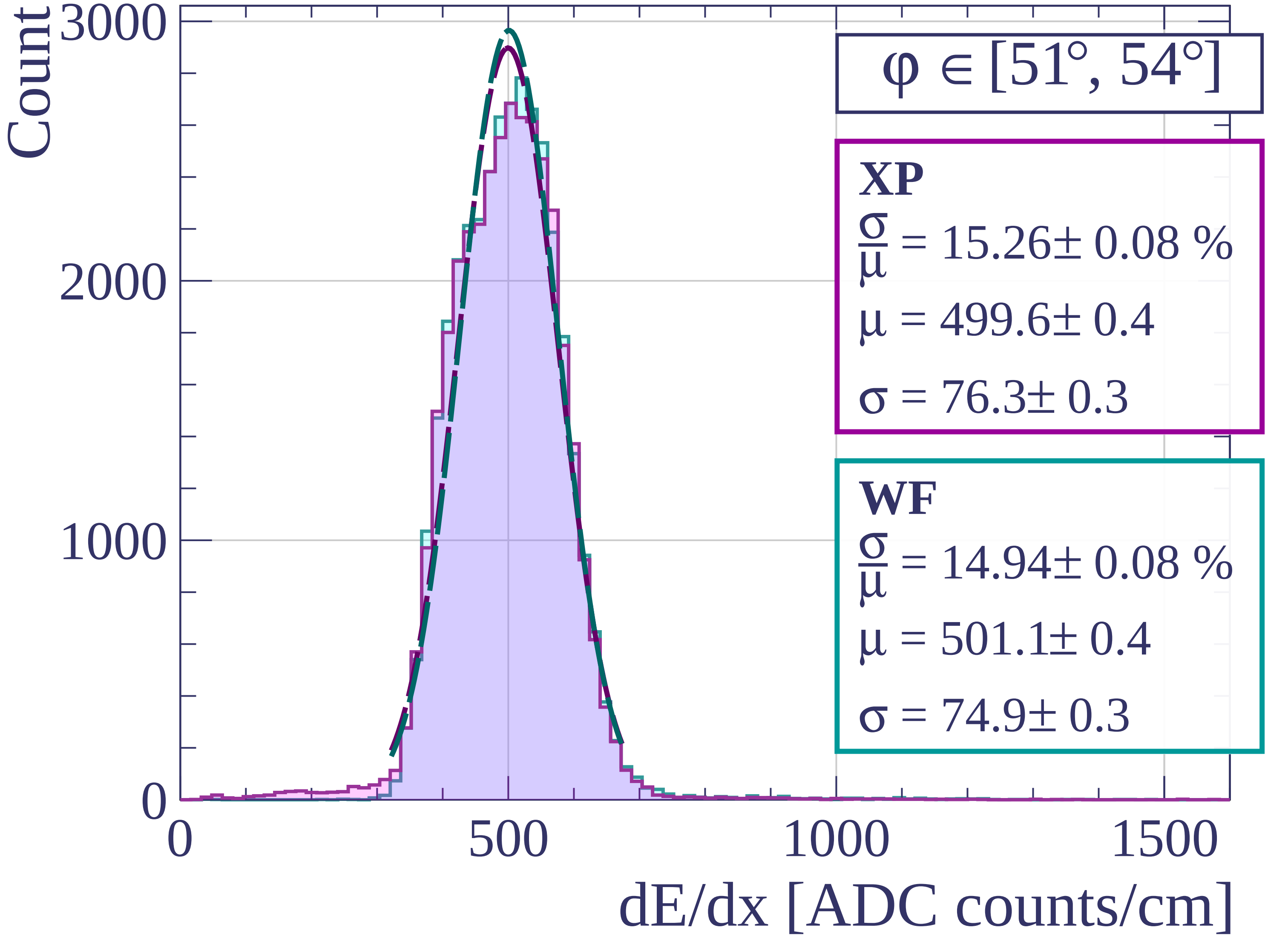
Count
dE/dx [ADC counts/cm]
φ ∈ [42°, 45°]
XP
σ/μ = 15.62± 0.12 %
μ = 509.3± 0.6
σ = 79.5± 0.5
WF
σ/μ = 15.80± 0.12 %
μ = 514.8± 0.6
σ = 81.4± 0.5
0
500
1000
1500



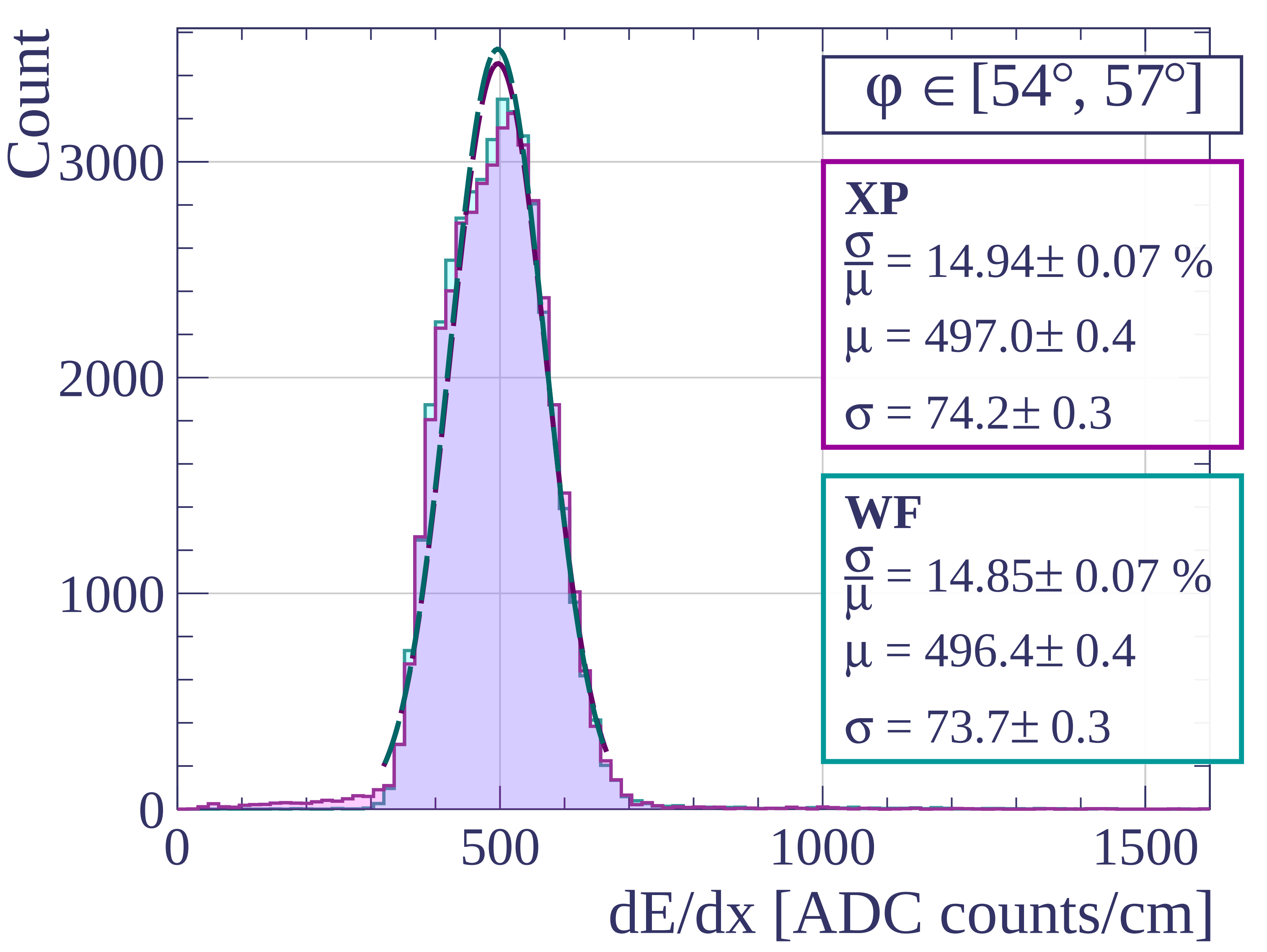
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [45°, 48°]
XP
σ/μ = 15.42± 0.10 %
μ = 505.4± 0.5
σ = 77.9± 0.4
WF
σ/μ = 15.79± 0.11 %
μ = 509.9± 0.6
σ = 80.5± 0.5

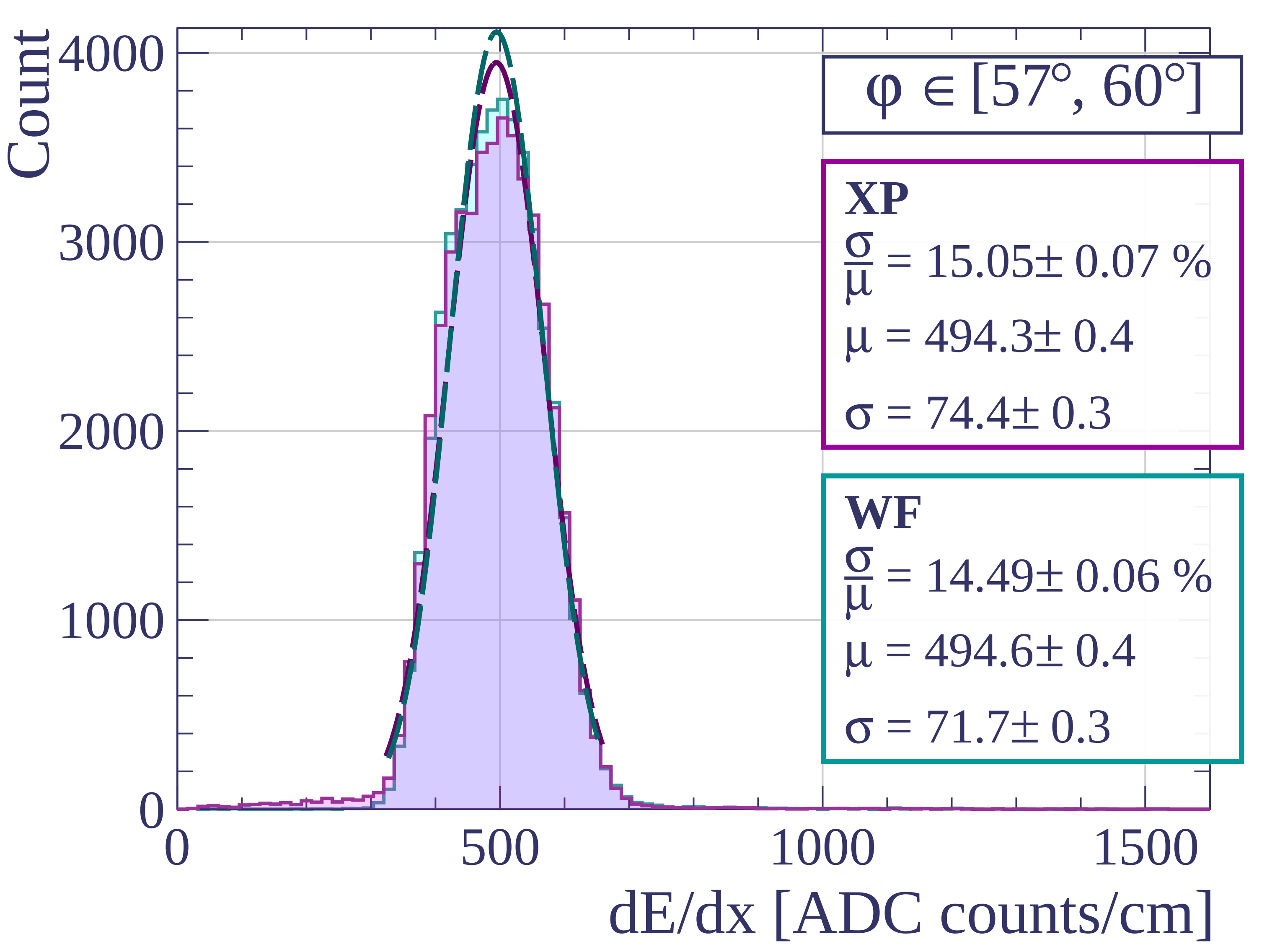


Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [48°, 51°]
XP
σ/μ = 15.22± 0.09 %
μ = 503.4± 0.5
σ = 76.6± 0.4
WF
σ/μ = 14.88± 0.08 %
μ = 507.0± 0.5
σ = 75.4± 0.3

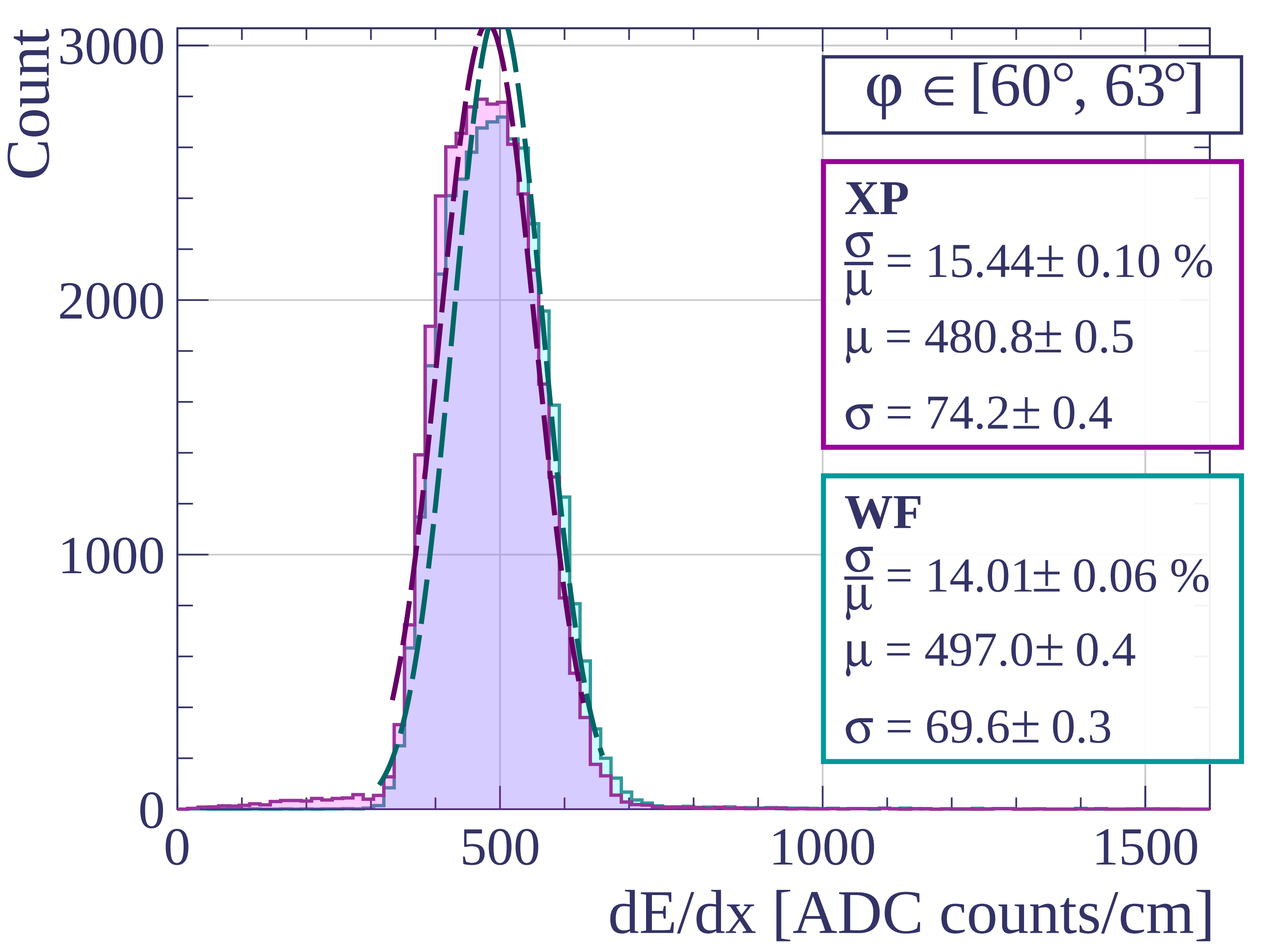


Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [51°, 54°]
XP
σ/μ = 15.26± 0.08 %
μ = 499.6± 0.4
σ = 76.3± 0.3
WF
σ/μ = 14.94± 0.08 %
μ = 501.1± 0.4
σ = 74.9± 0.3

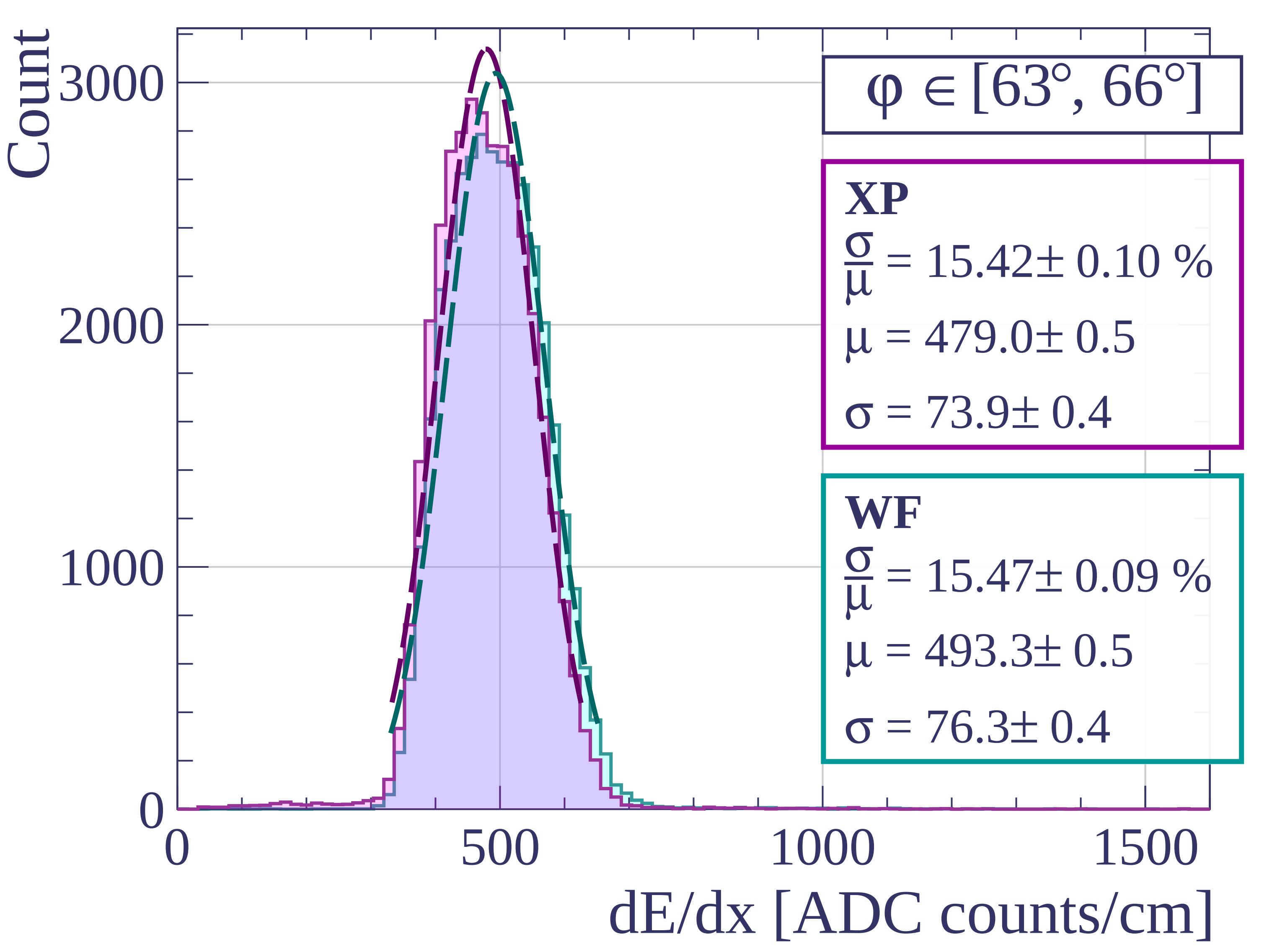

Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [54°, 57°]
XP
σ/μ = 14.94± 0.07 %
μ = 497.0± 0.4
σ = 74.2± 0.3
WF
σ/μ = 14.85± 0.07 %
μ = 496.4± 0.4
σ = 73.7± 0.3



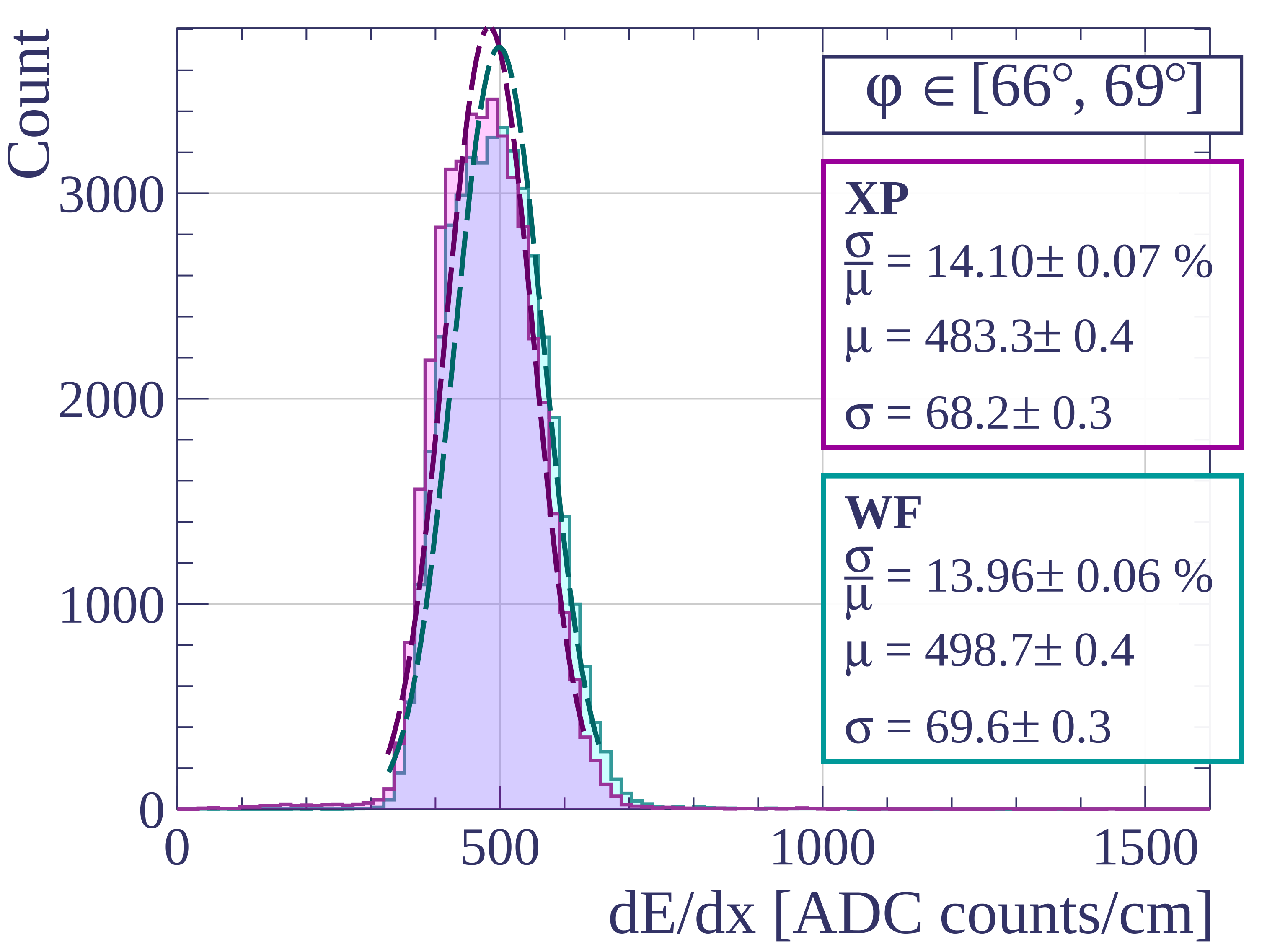
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [57°, 60°]
XP
σ/μ = 15.05± 0.07 %
μ = 494.3± 0.4
σ = 74.4± 0.3
WF
σ/μ = 14.49± 0.06 %
μ = 494.6± 0.4
σ = 71.7± 0.3



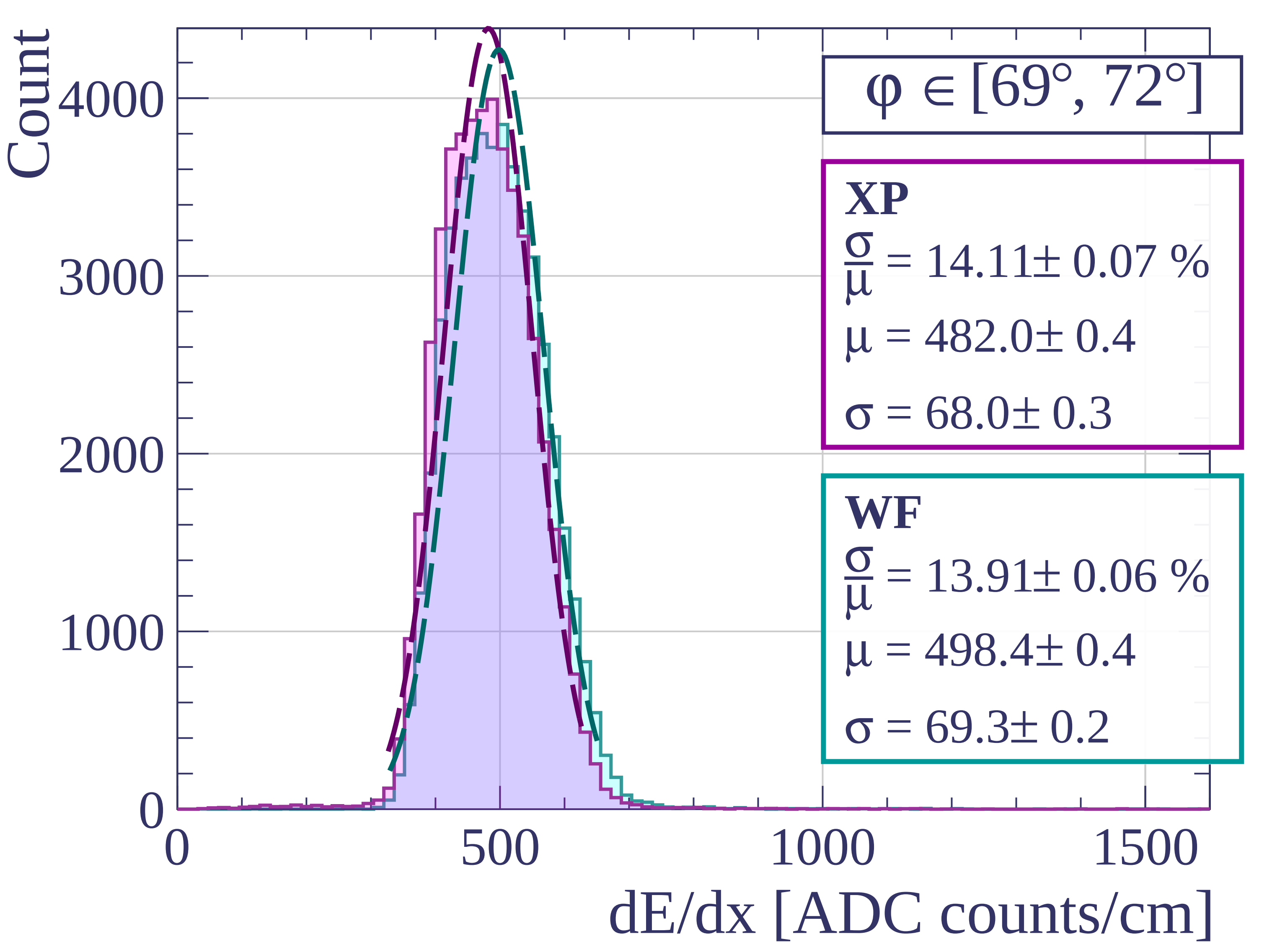
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [60°, 63°]
XP
σ/μ = 15.44± 0.10 %
μ = 480.8± 0.5
σ = 74.2± 0.4
WF
σ/μ = 14.01± 0.06 %
μ = 497.0± 0.4
σ = 69.6± 0.3

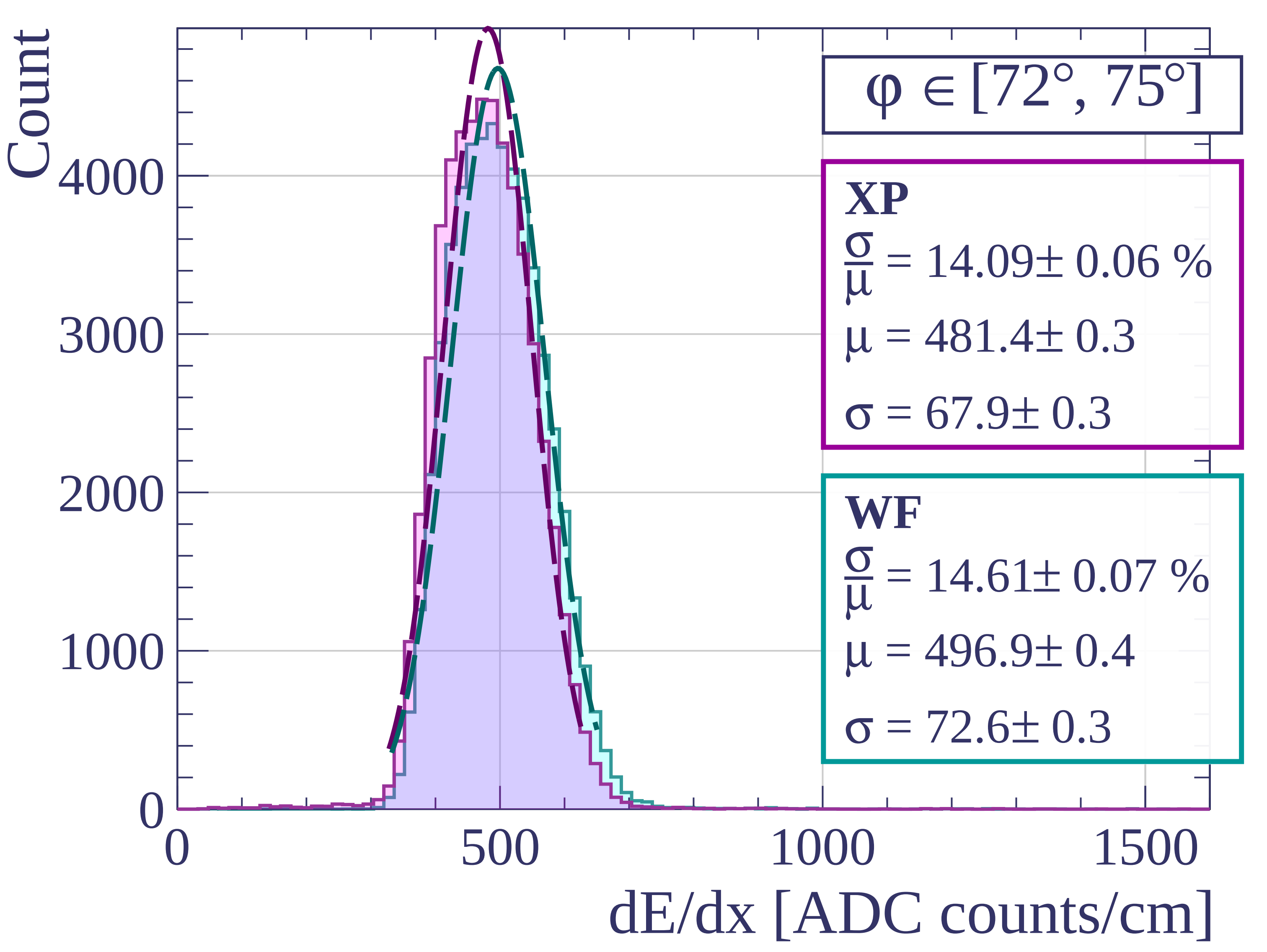


Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [63°, 66°]
XP
$\frac{\sigma}{\mu}$ = 15.42± 0.10 %
μ = 479.0± 0.5
σ = 73.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.47± 0.09 %
μ = 493.3± 0.5
σ = 76.3± 0.4

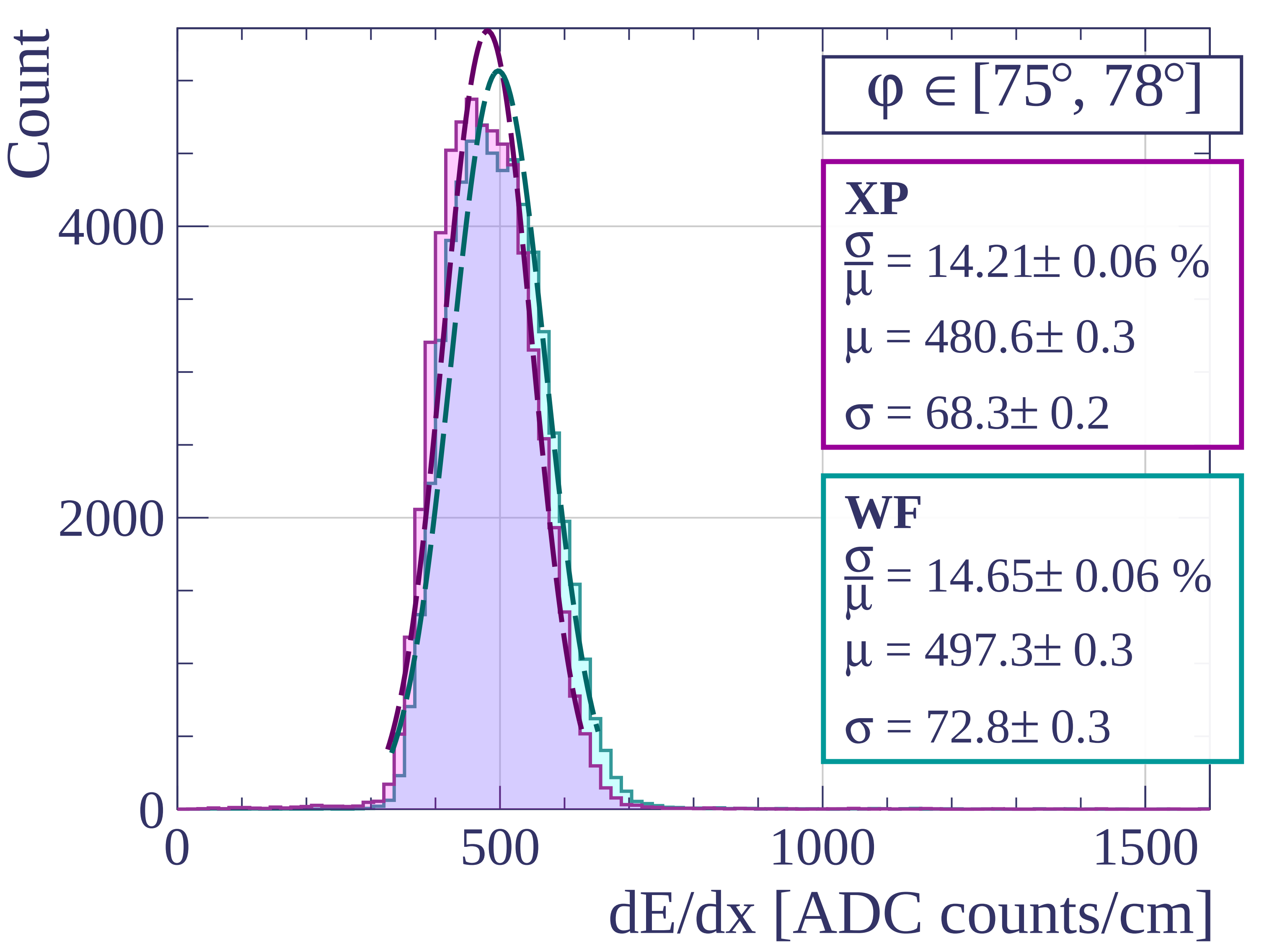


Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [66°, 69°]
XP
σ/μ = 14.10± 0.07 %
μ = 483.3± 0.4
σ = 68.2± 0.3
WF
σ/μ = 13.96± 0.06 %
μ = 498.7± 0.4
σ = 69.6± 0.3

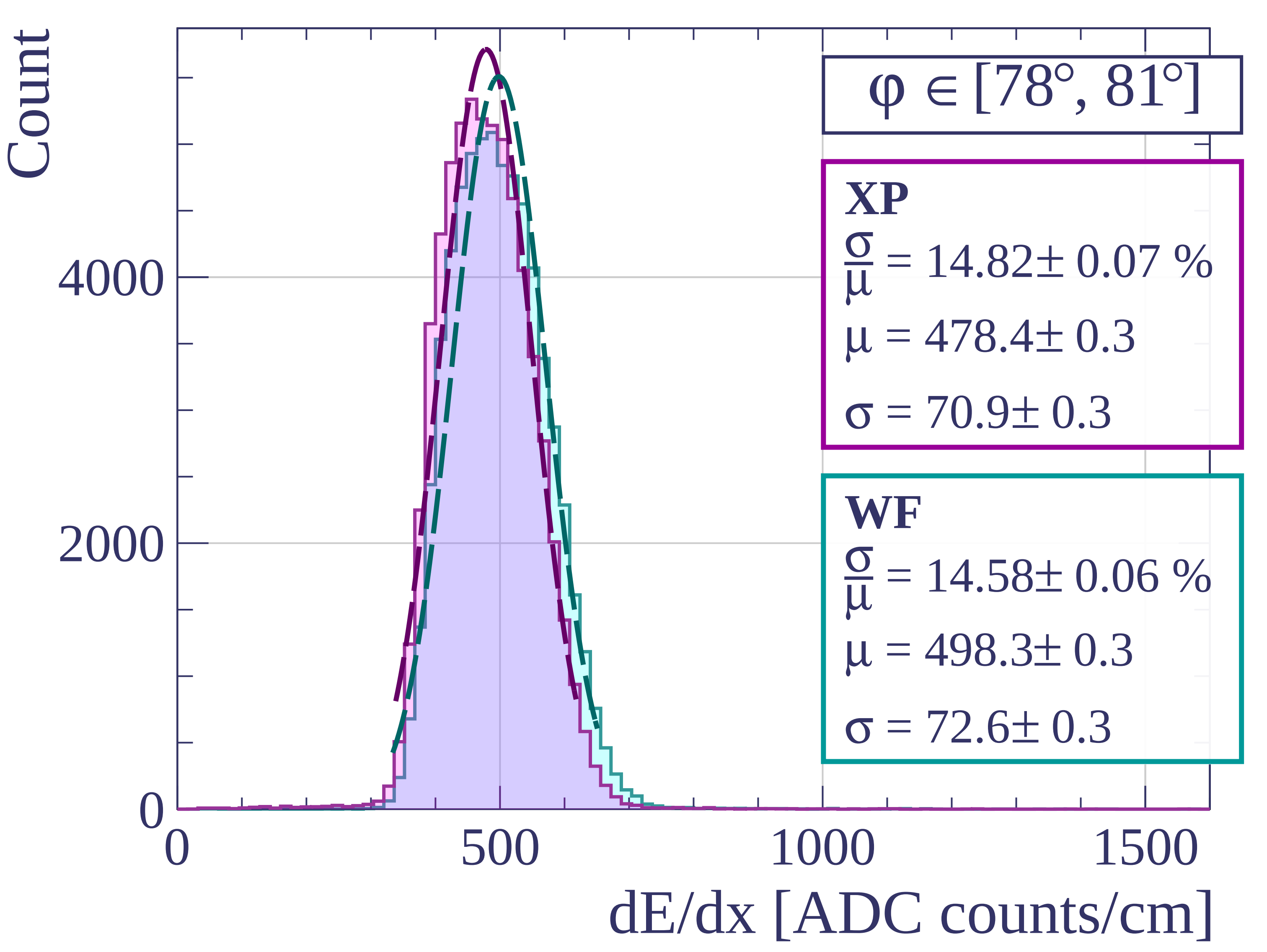

Count
dE/dx [ADC counts/cm]
φ ∈ [69°, 72°]
XP
σ/μ = 14.11± 0.07 %
μ = 482.0± 0.4
σ = 68.0± 0.3
WF
σ/μ = 13.91± 0.06 %
μ = 498.4± 0.4
σ = 69.3± 0.2
0
1000
2000
3000
4000
0
500
1000
1500



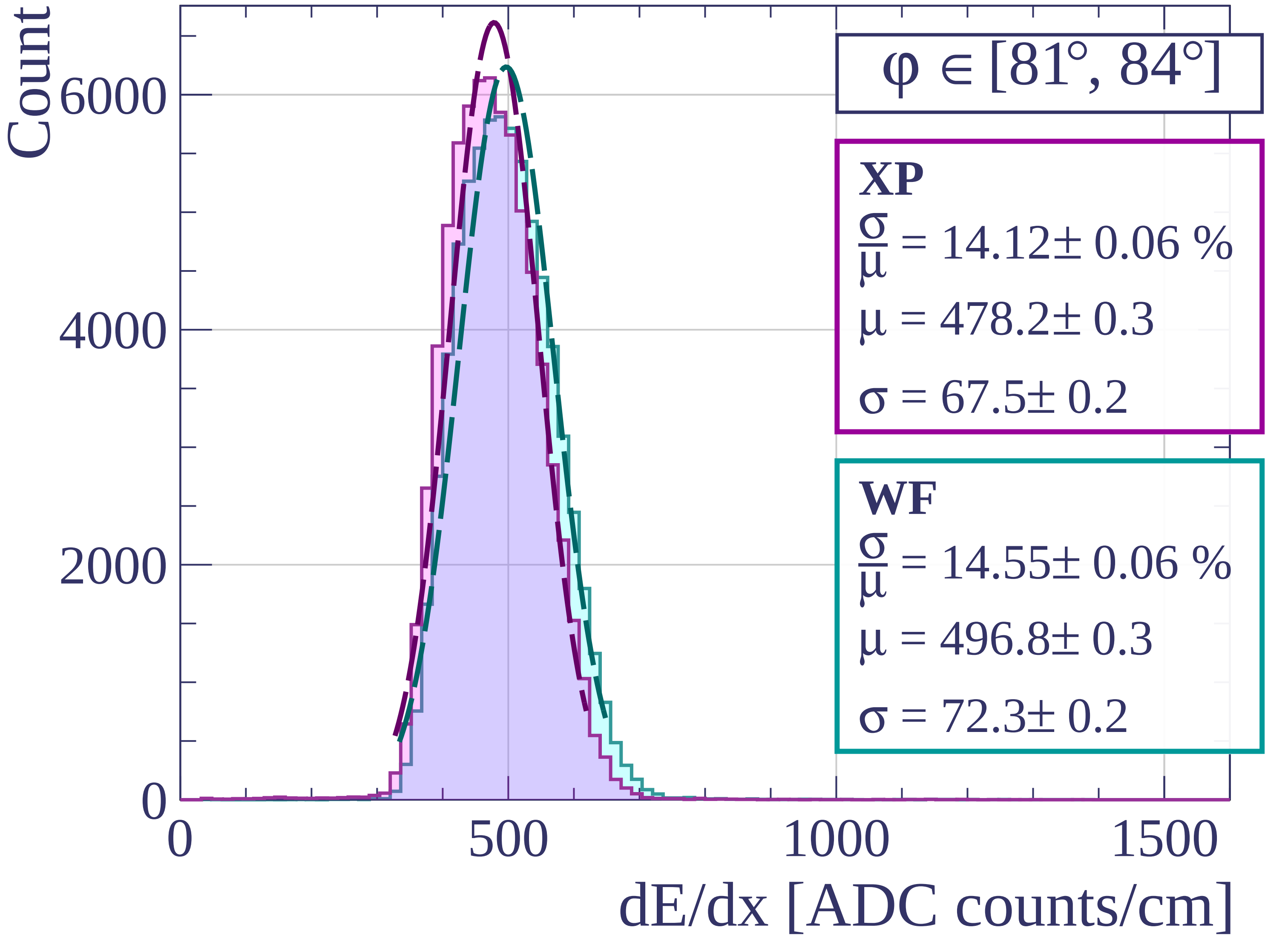
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [72°, 75°]
XP
σ/μ = 14.09± 0.06 %
μ = 481.4± 0.3
σ = 67.9± 0.3
WF
σ/μ = 14.61± 0.07 %
μ = 496.9± 0.4
σ = 72.6± 0.3

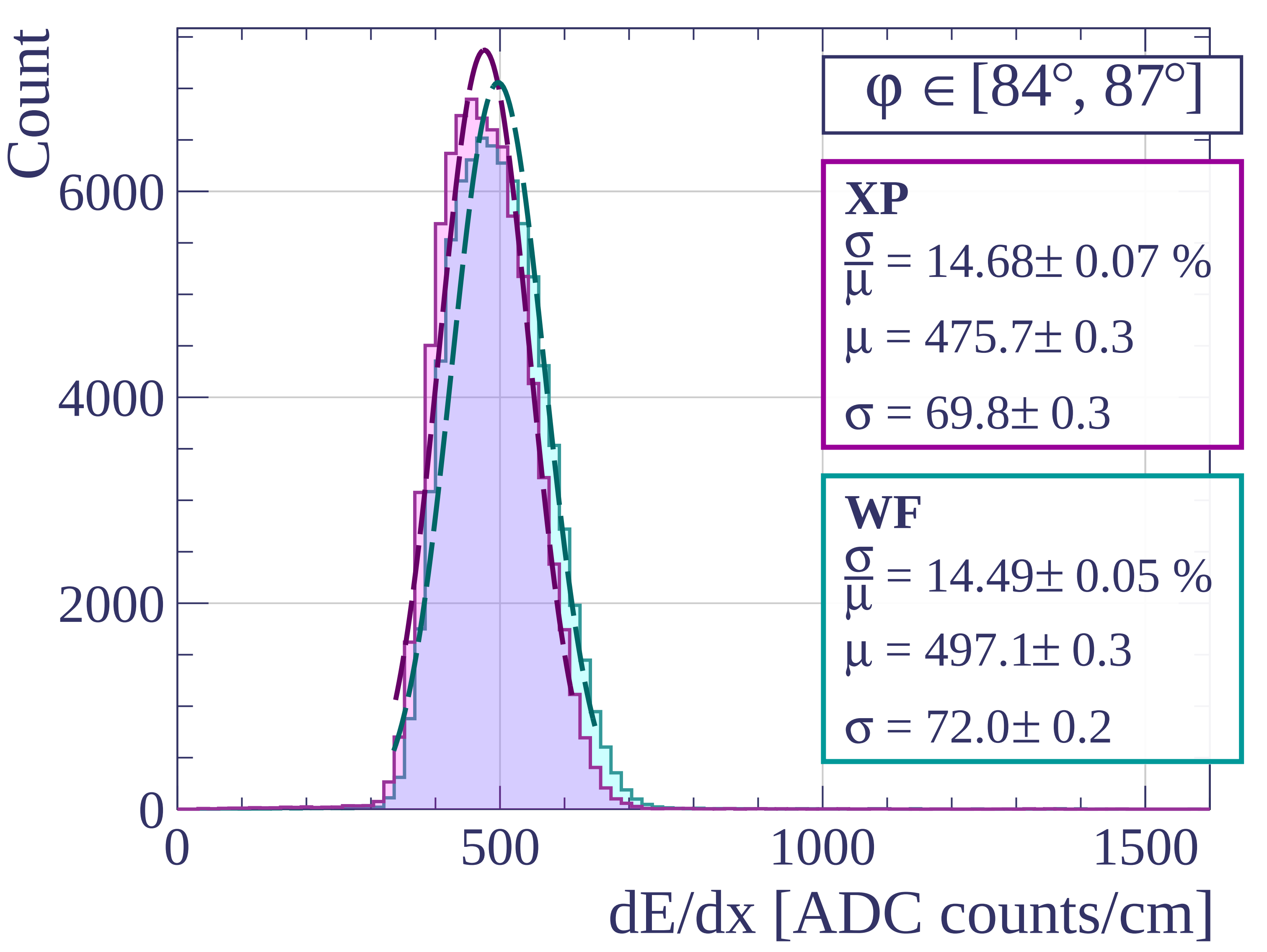


Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [75°, 78°]
XP
σ/μ = 14.21± 0.06 %
μ = 480.6± 0.3
σ = 68.3± 0.2
WF
σ/μ = 14.65± 0.06 %
μ = 497.3± 0.3
σ = 72.8± 0.3

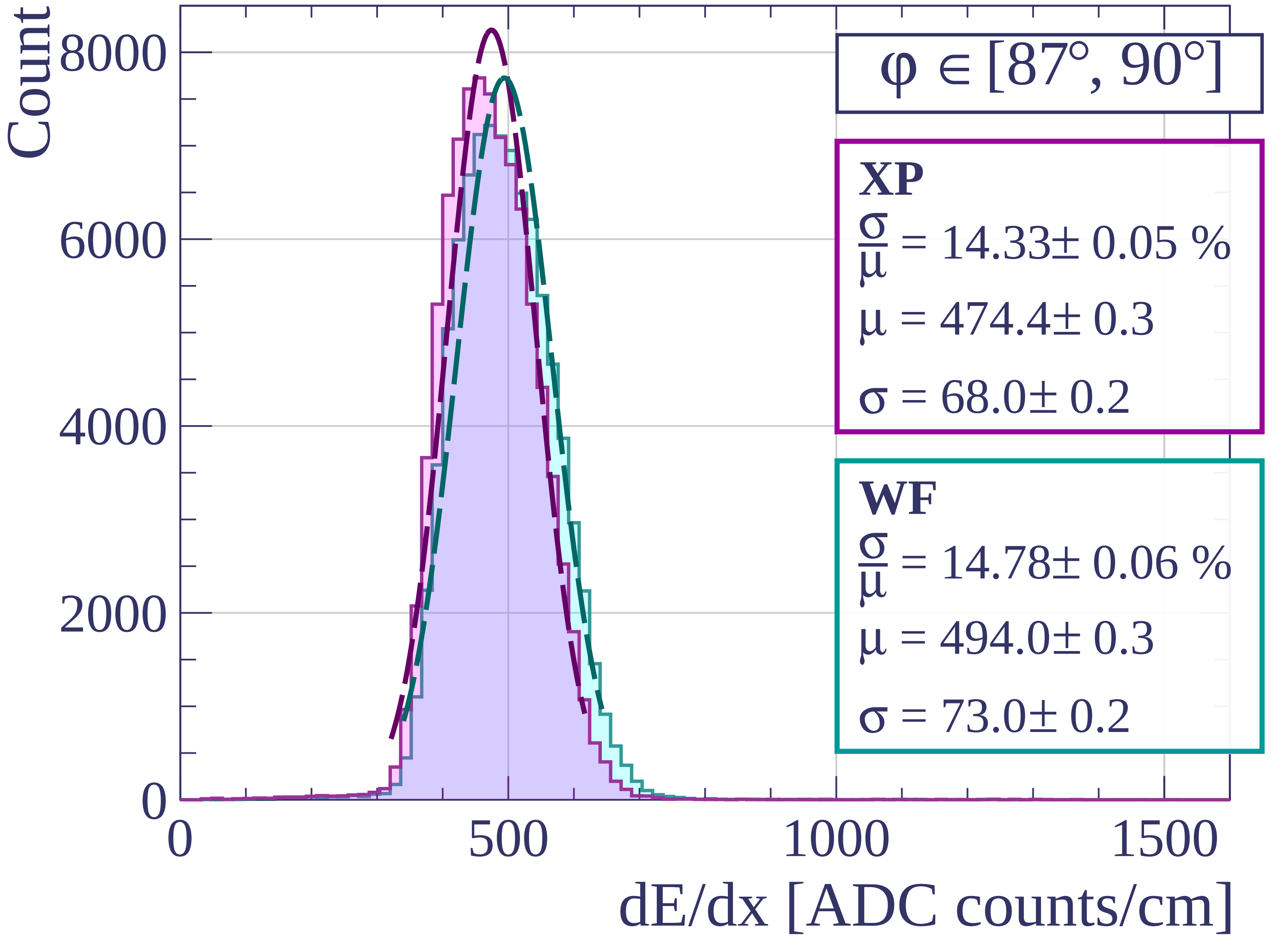


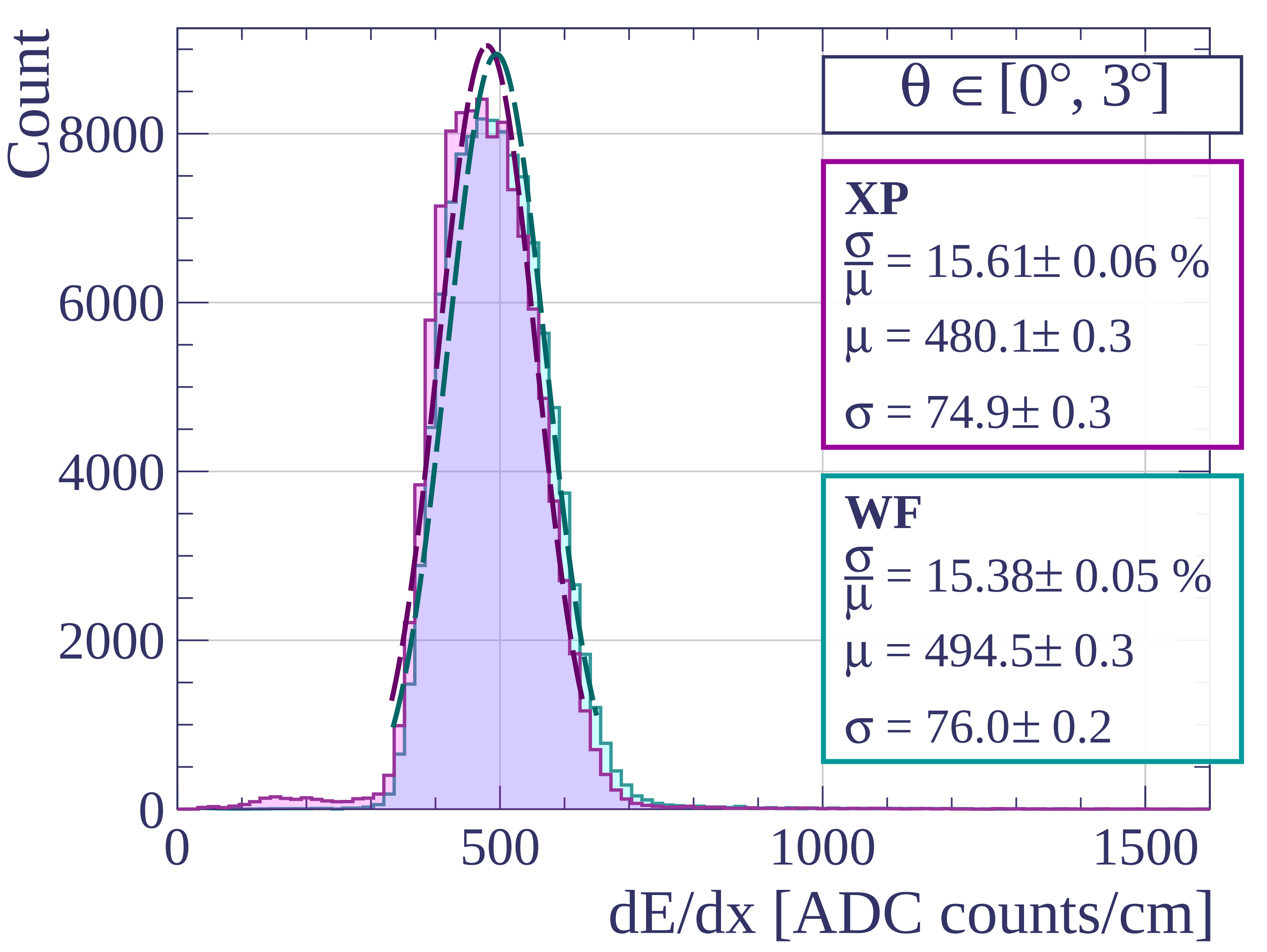
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [78°, 81°]
XP
σ/μ = 14.82± 0.07 %
μ = 478.4± 0.3
σ = 70.9± 0.3
WF
σ/μ = 14.58± 0.06 %
μ = 498.3± 0.3
σ = 72.6± 0.3


Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [81°, 84°]
XP
σ/μ = 14.12± 0.06 %
μ = 478.2± 0.3
σ = 67.5± 0.2
WF
σ/μ = 14.55± 0.06 %
μ = 496.8± 0.3
σ = 72.3± 0.2

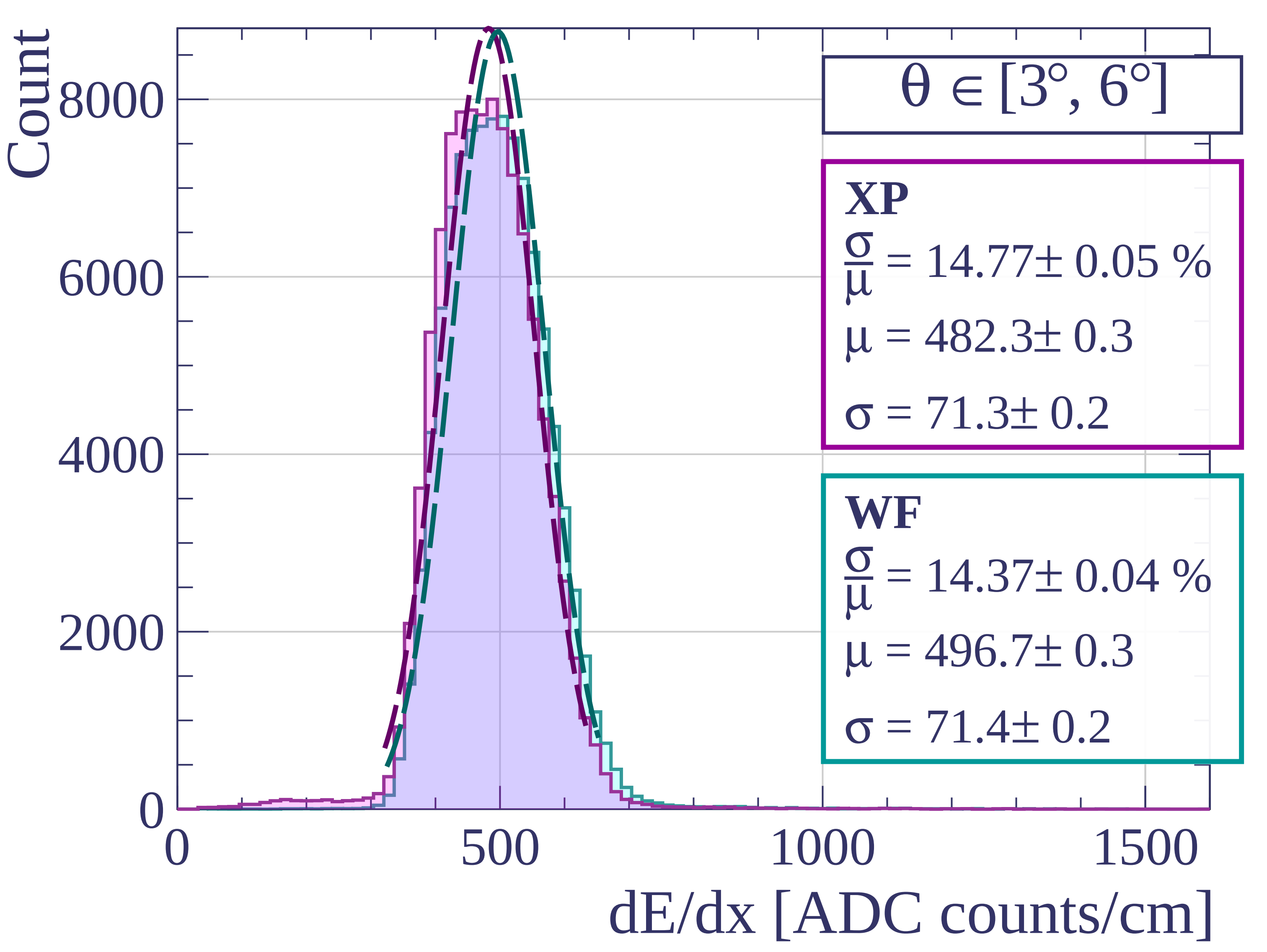


Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [84°, 87°]
XP
σ/μ = 14.68± 0.07 %
μ = 475.7± 0.3
σ = 69.8± 0.3
WF
σ/μ = 14.49± 0.05 %
μ = 497.1± 0.3
σ = 72.0± 0.2

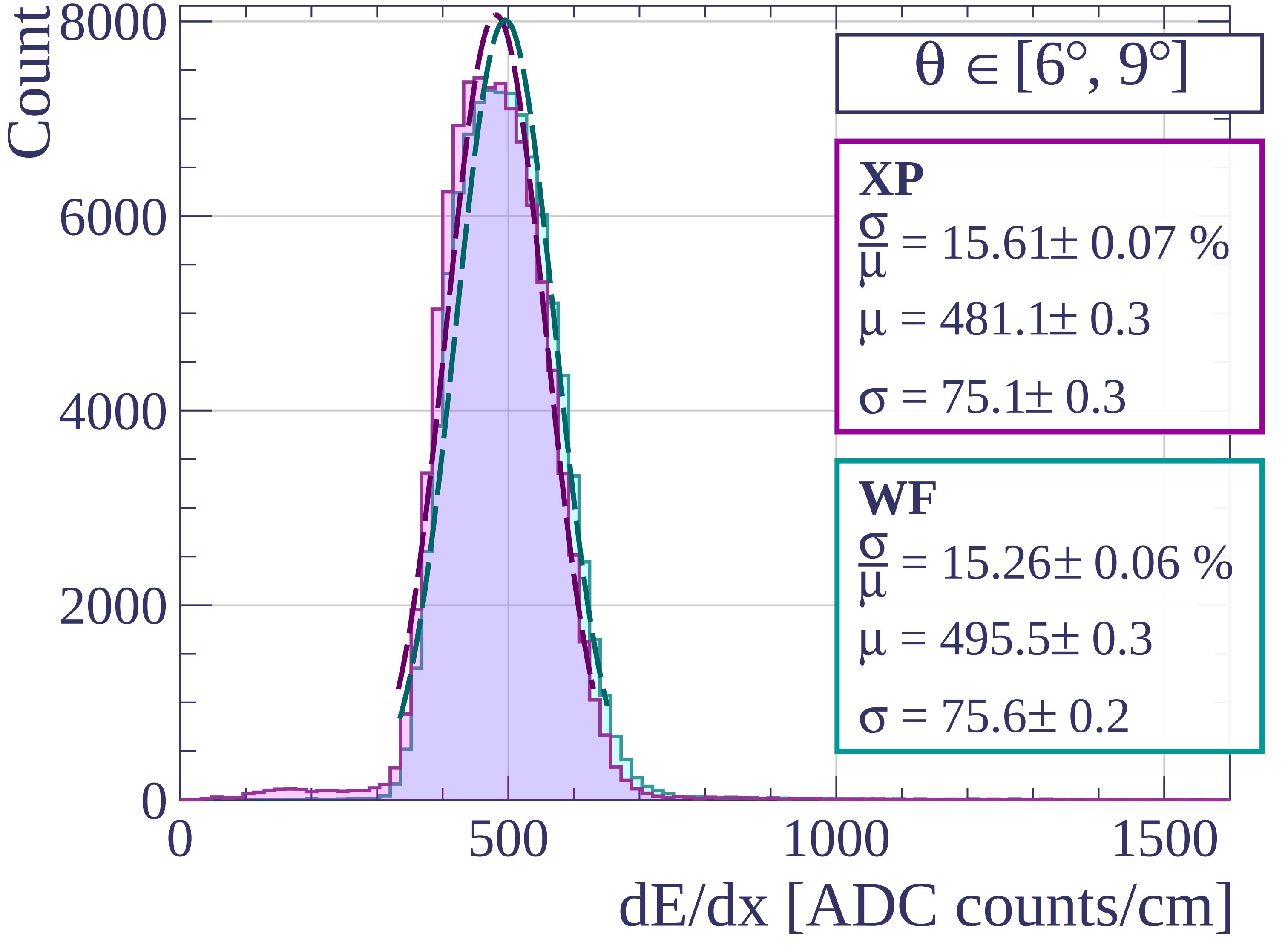

Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [87°, 90°]
XP
σ/μ = 14.33± 0.05 %
μ = 474.4± 0.3
σ = 68.0± 0.2
WF
σ/μ = 14.78± 0.06 %
μ = 494.0± 0.3
σ = 73.0± 0.2



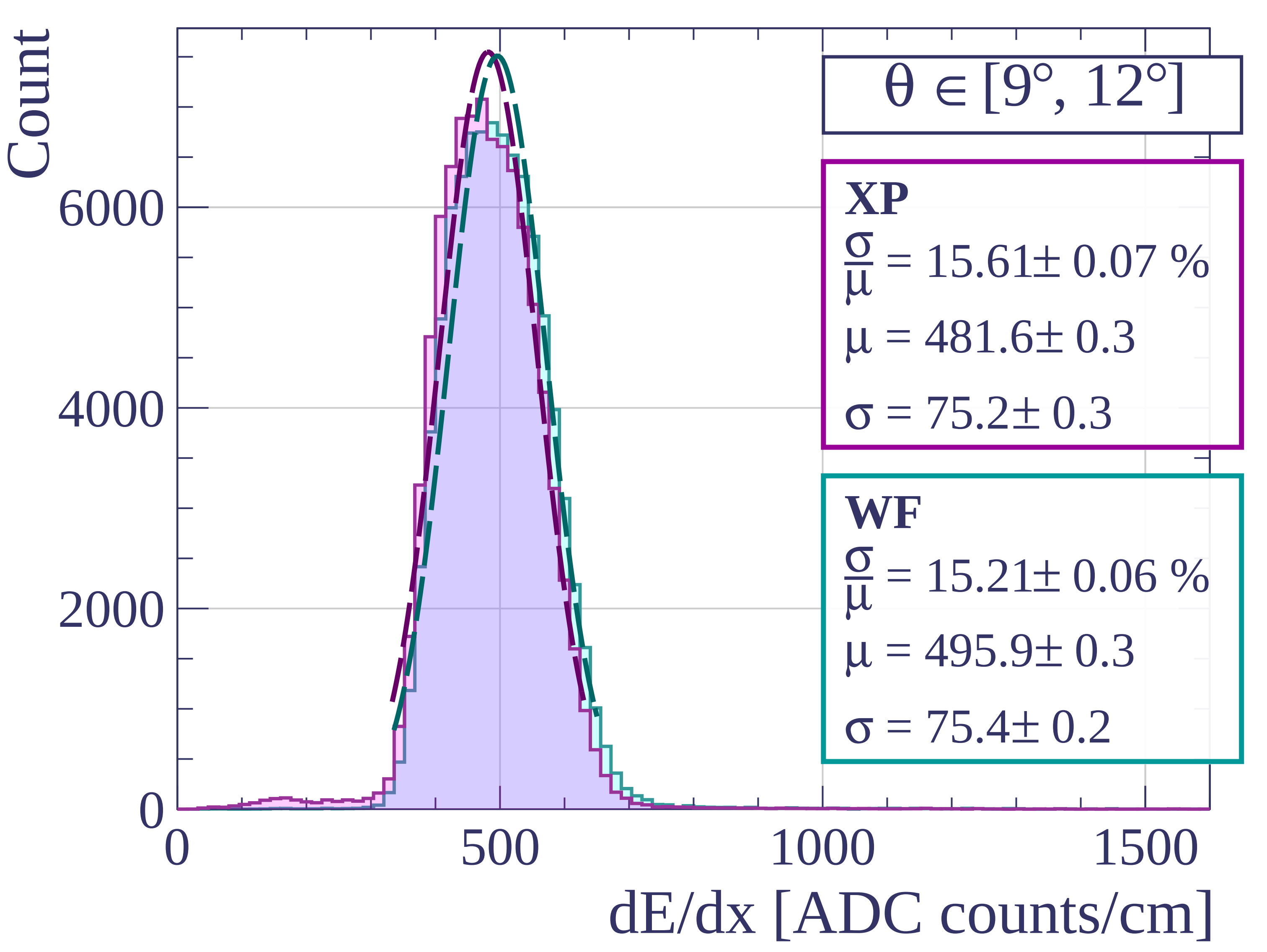
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [0°, 3°]
XP
σ/μ = 15.61± 0.06 %
μ = 480.1± 0.3
σ = 74.9± 0.3
WF
σ/μ = 15.38± 0.05 %
μ = 494.5± 0.3
σ = 76.0± 0.2



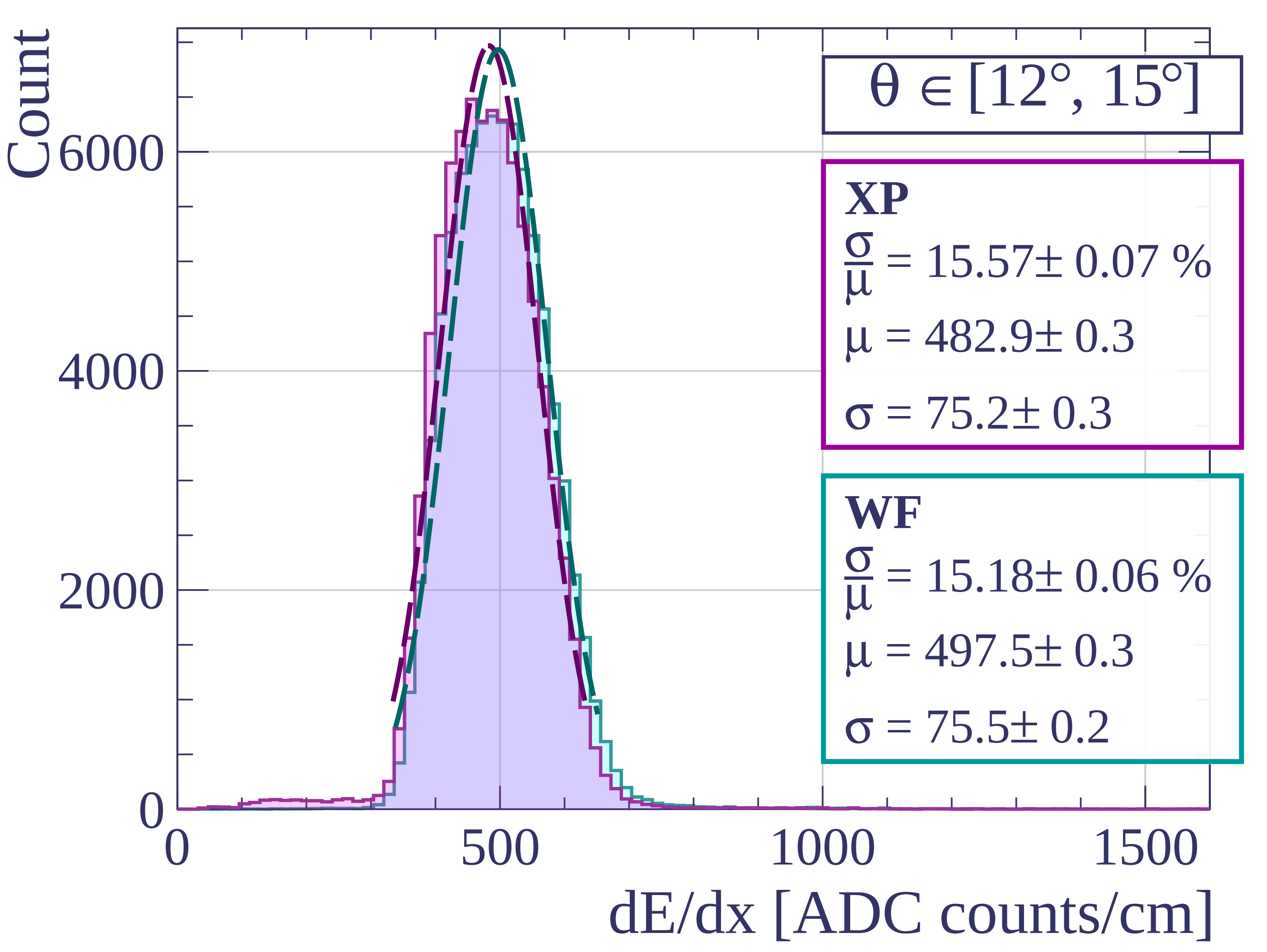
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [3°, 6°]
XP
σ/μ = 14.77± 0.05 %
μ = 482.3± 0.3
σ = 71.3± 0.2
WF
σ/μ = 14.37± 0.04 %
μ = 496.7± 0.3
σ = 71.4± 0.2



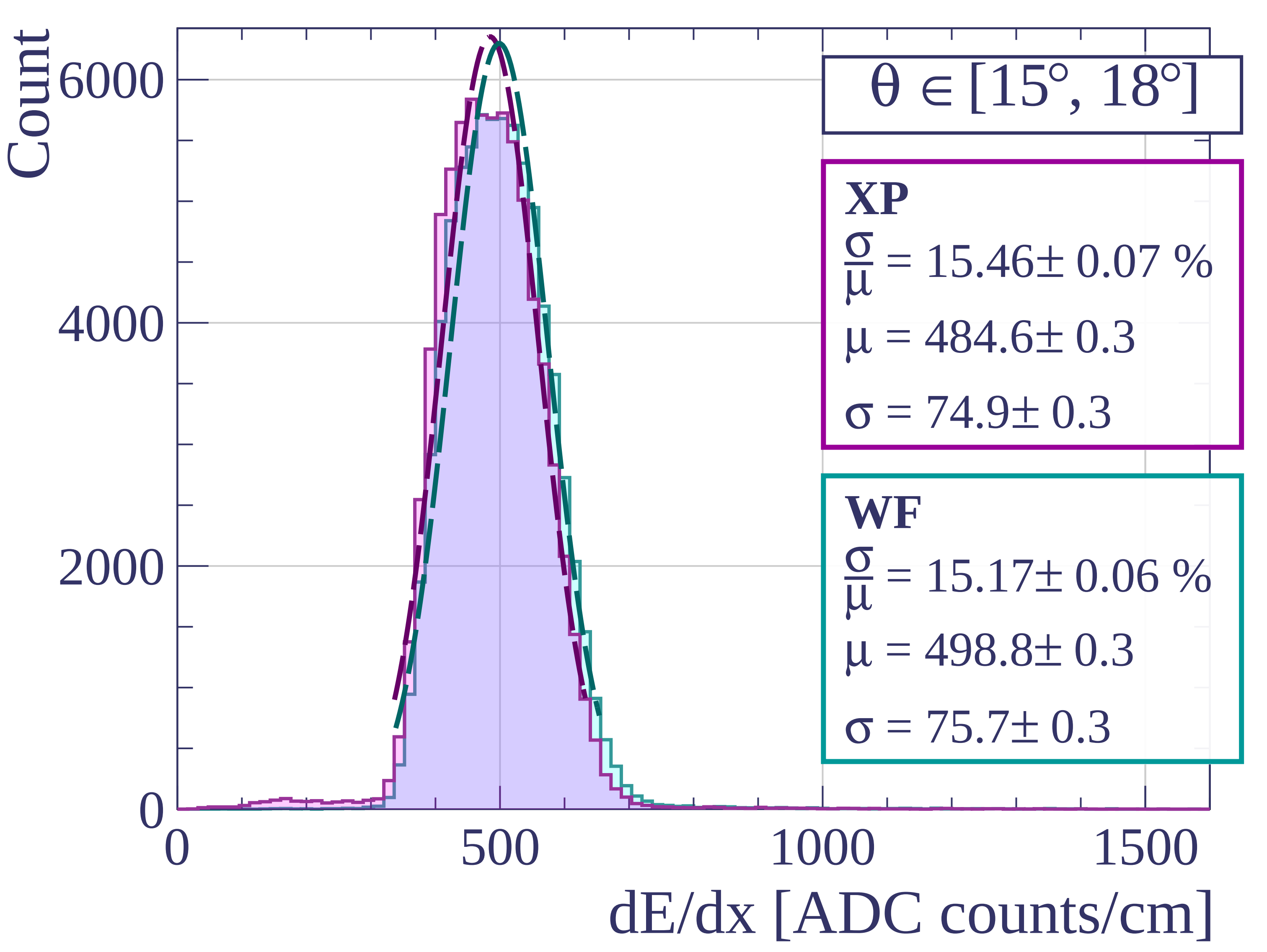
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [6°, 9°]
XP
σ/μ = 15.61± 0.07 %
μ = 481.1± 0.3
σ = 75.1± 0.3
WF
σ/μ = 15.26± 0.06 %
μ = 495.5± 0.3
σ = 75.6± 0.2



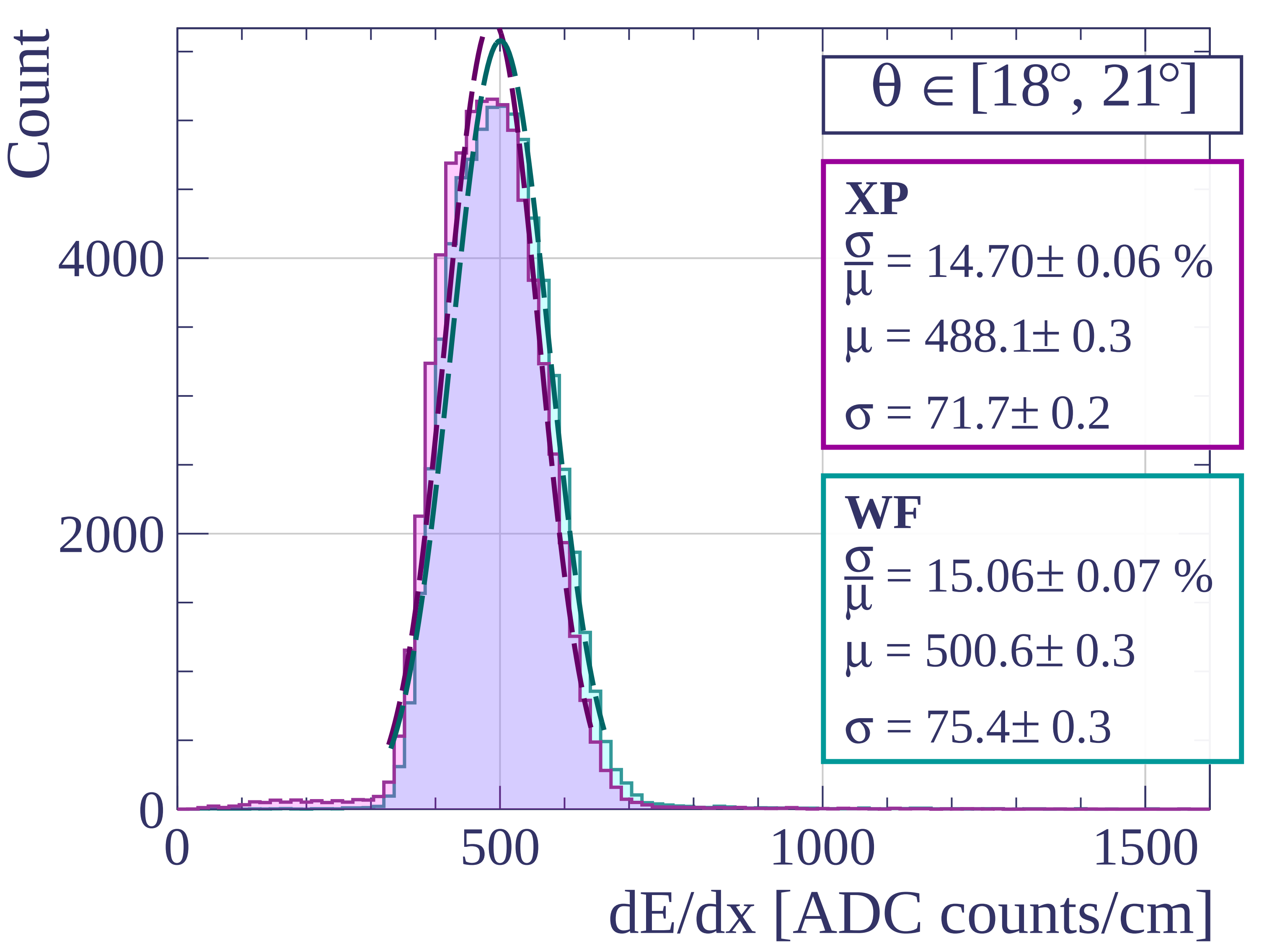
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [9°, 12°]
XP
σ/μ = 15.61± 0.07 %
μ = 481.6± 0.3
σ = 75.2± 0.3
WF
σ/μ = 15.21± 0.06 %
μ = 495.9± 0.3
σ = 75.4± 0.2



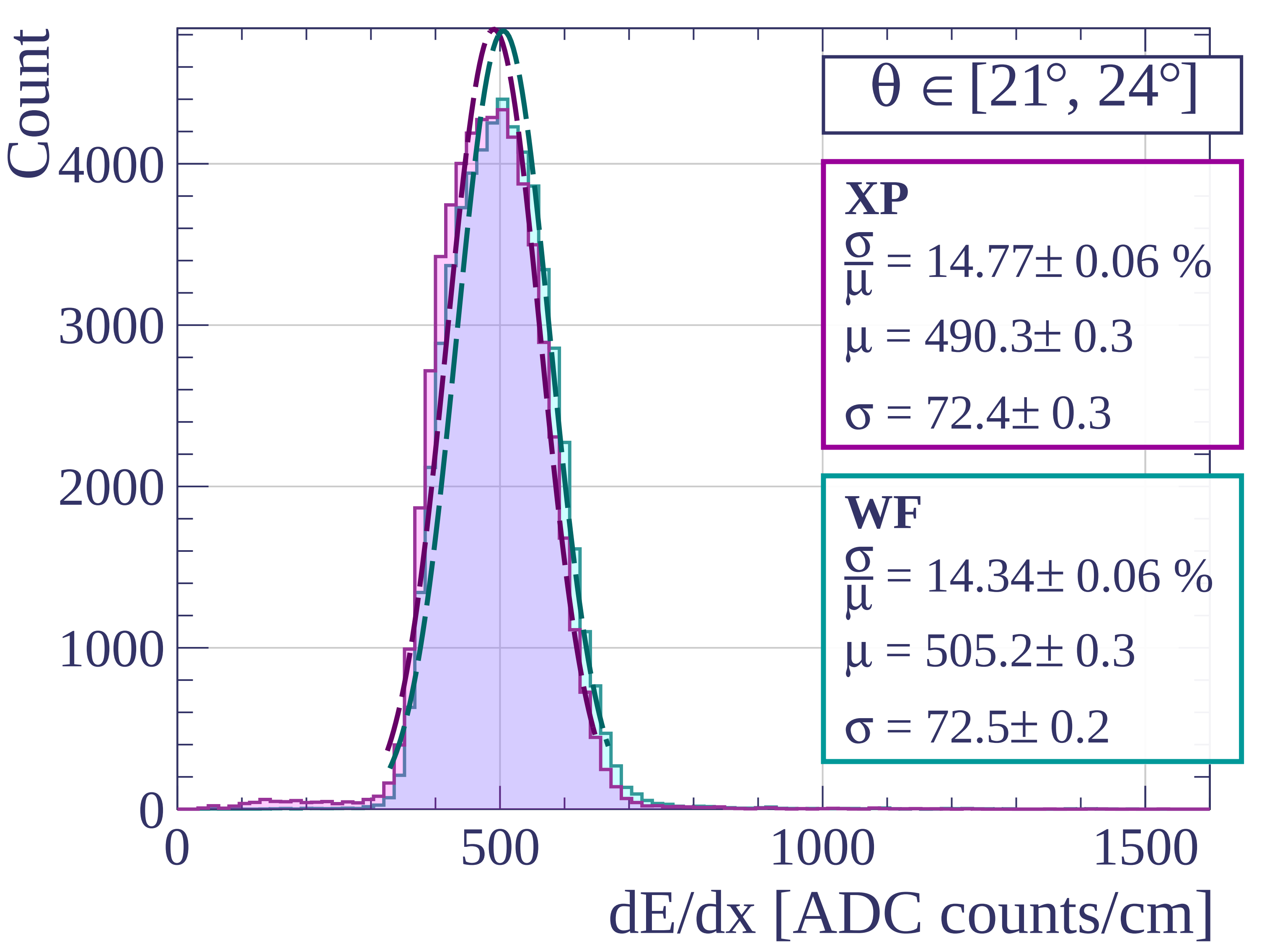
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [12°, 15°]
XP
σ/μ = 15.57± 0.07 %
μ = 482.9± 0.3
σ = 75.2± 0.3
WF
σ/μ = 15.18± 0.06 %
μ = 497.5± 0.3
σ = 75.5± 0.2



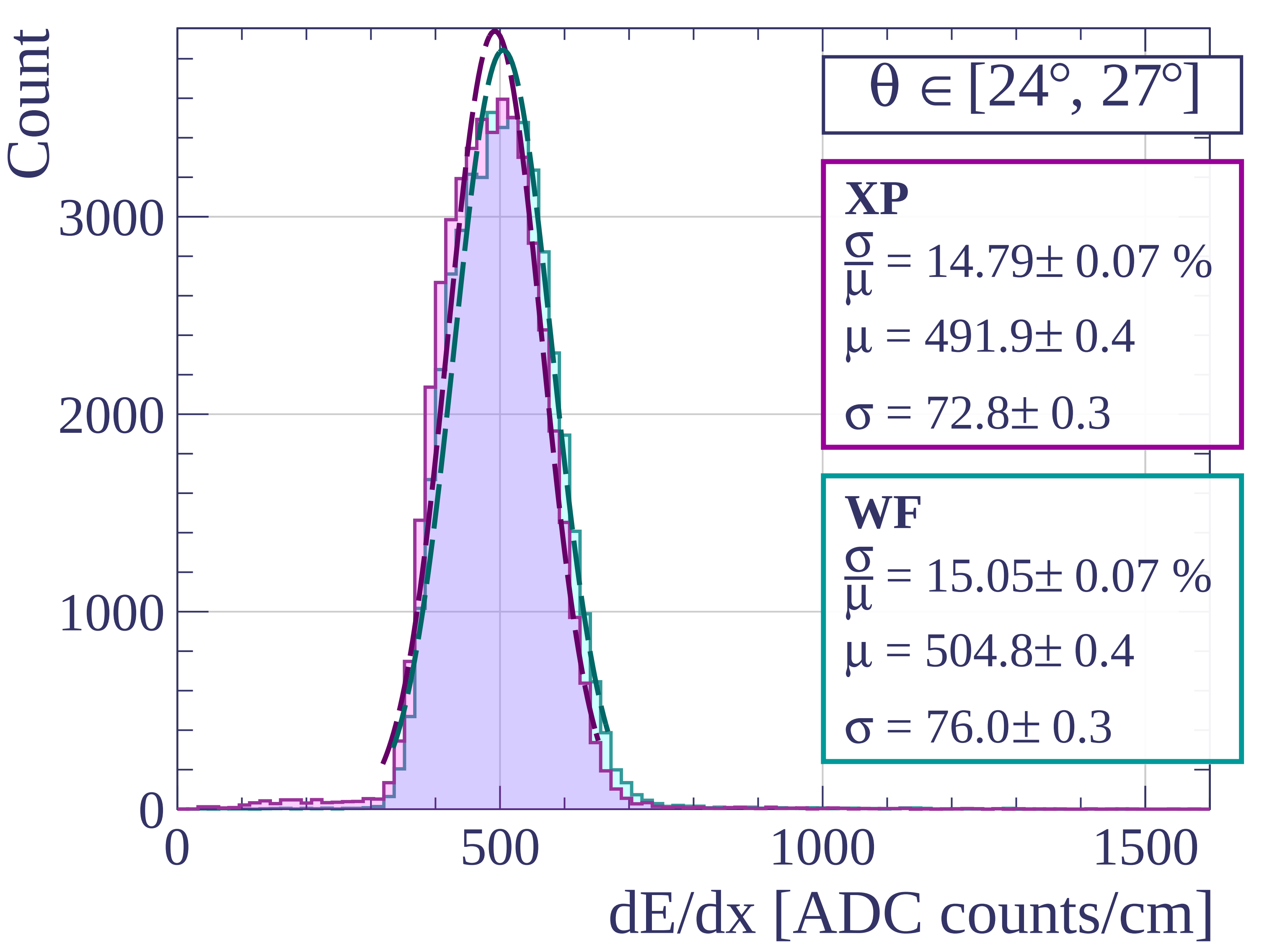
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [15°, 18°]
XP
σ/μ = 15.46± 0.07 %
μ = 484.6± 0.3
σ = 74.9± 0.3
WF
σ/μ = 15.17± 0.06 %
μ = 498.8± 0.3
σ = 75.7± 0.3



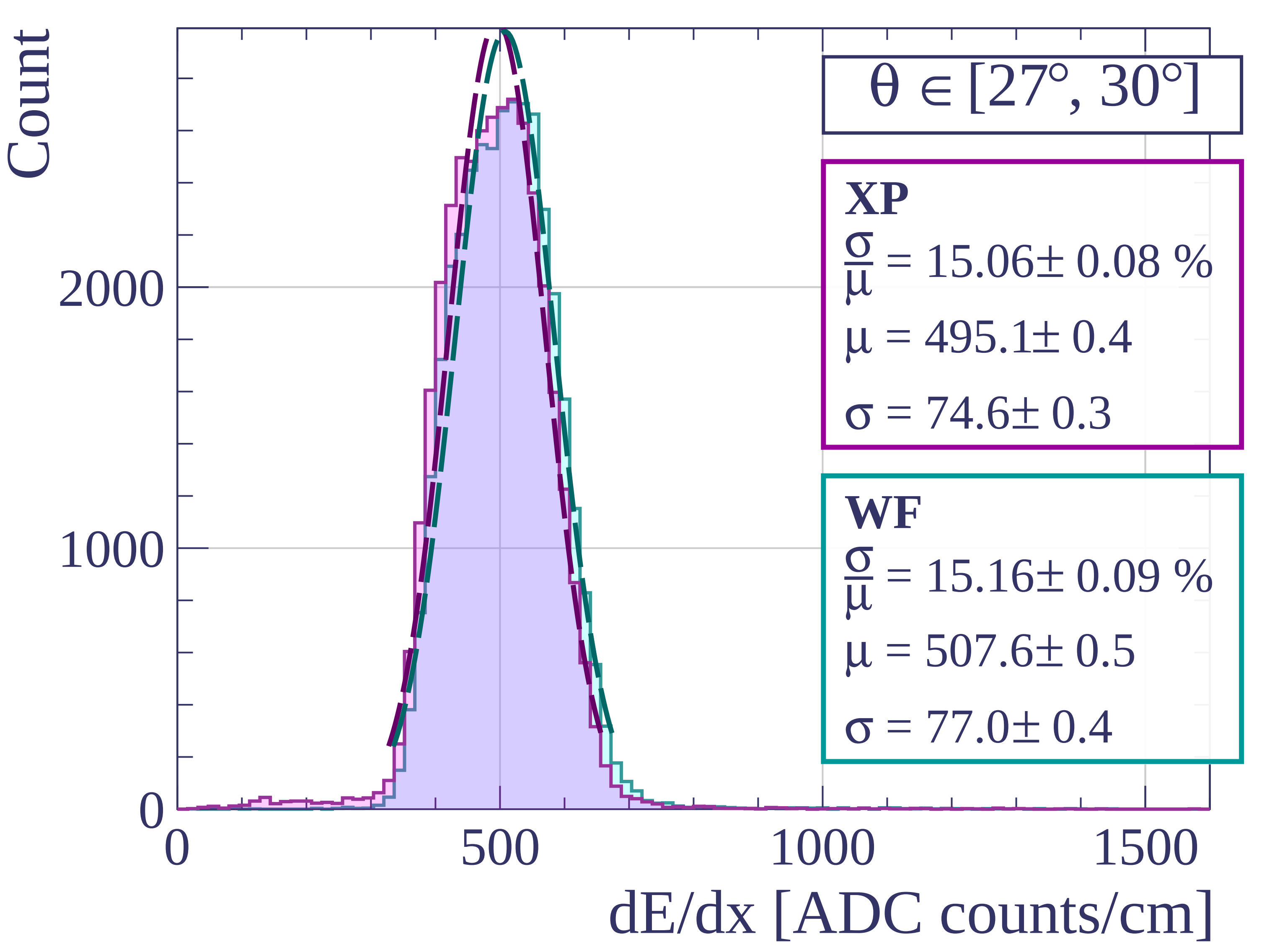
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [18°, 21°]
XP
σ/μ = 14.70± 0.06 %
μ = 488.1± 0.3
σ = 71.7± 0.2
WF
σ/μ = 15.06± 0.07 %
μ = 500.6± 0.3
σ = 75.4± 0.3



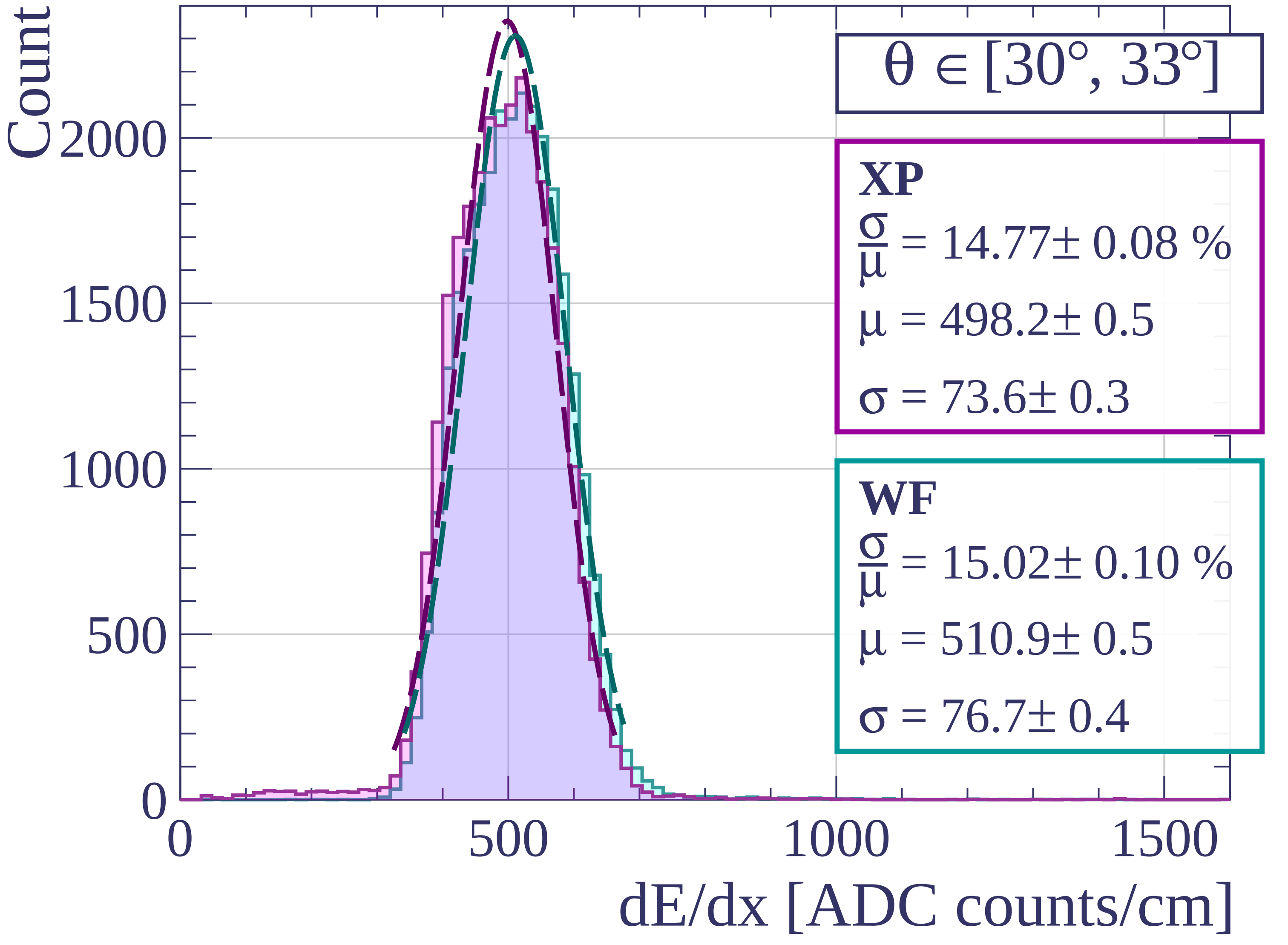
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
4000
0
500
1000
1500
θ ∈ [21°, 24°]
XP
σ/μ = 14.77± 0.06 %
μ = 490.3± 0.3
σ = 72.4± 0.3
WF
σ/μ = 14.34± 0.06 %
μ = 505.2± 0.3
σ = 72.5± 0.2



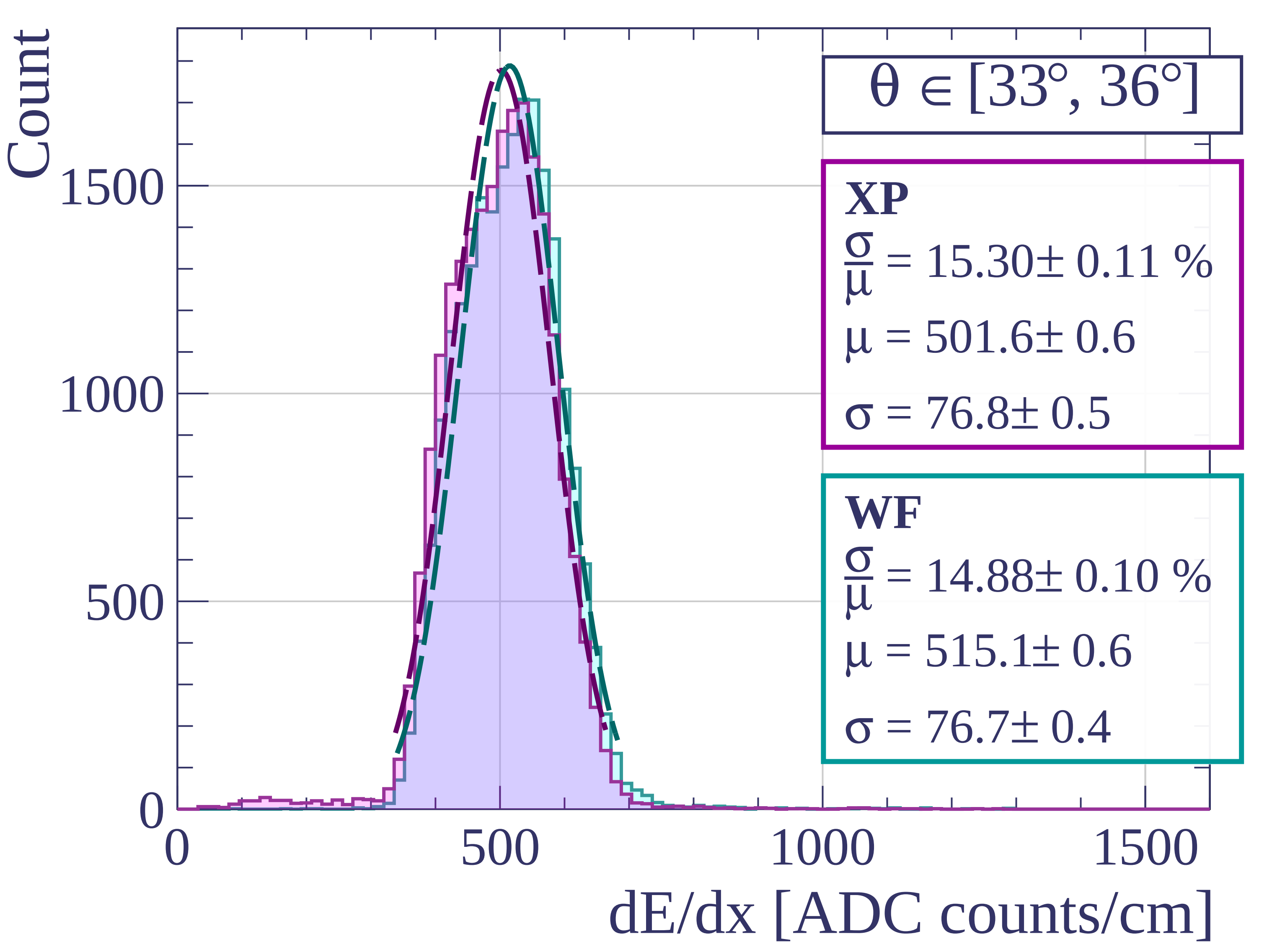
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [24°, 27°]
XP
σ/μ = 14.79± 0.07 %
μ = 491.9± 0.4
σ = 72.8± 0.3
WF
σ/μ = 15.05± 0.07 %
μ = 504.8± 0.4
σ = 76.0± 0.3



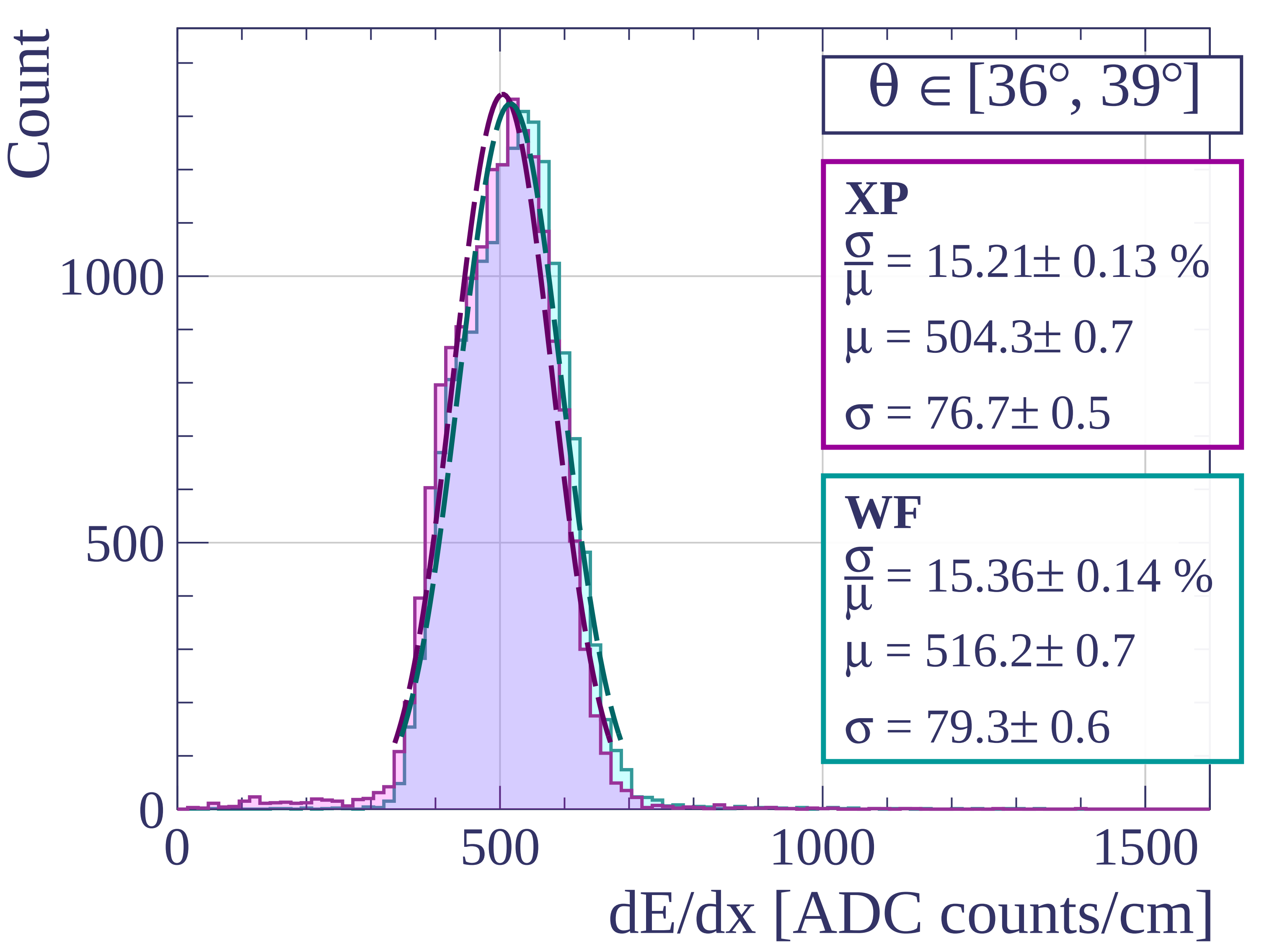
Count
0
1000
2000
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [27°, 30°]
XP
σ/μ = 15.06± 0.08 %
μ = 495.1± 0.4
σ = 74.6± 0.3
WF
σ/μ = 15.16± 0.09 %
μ = 507.6± 0.5
σ = 77.0± 0.4

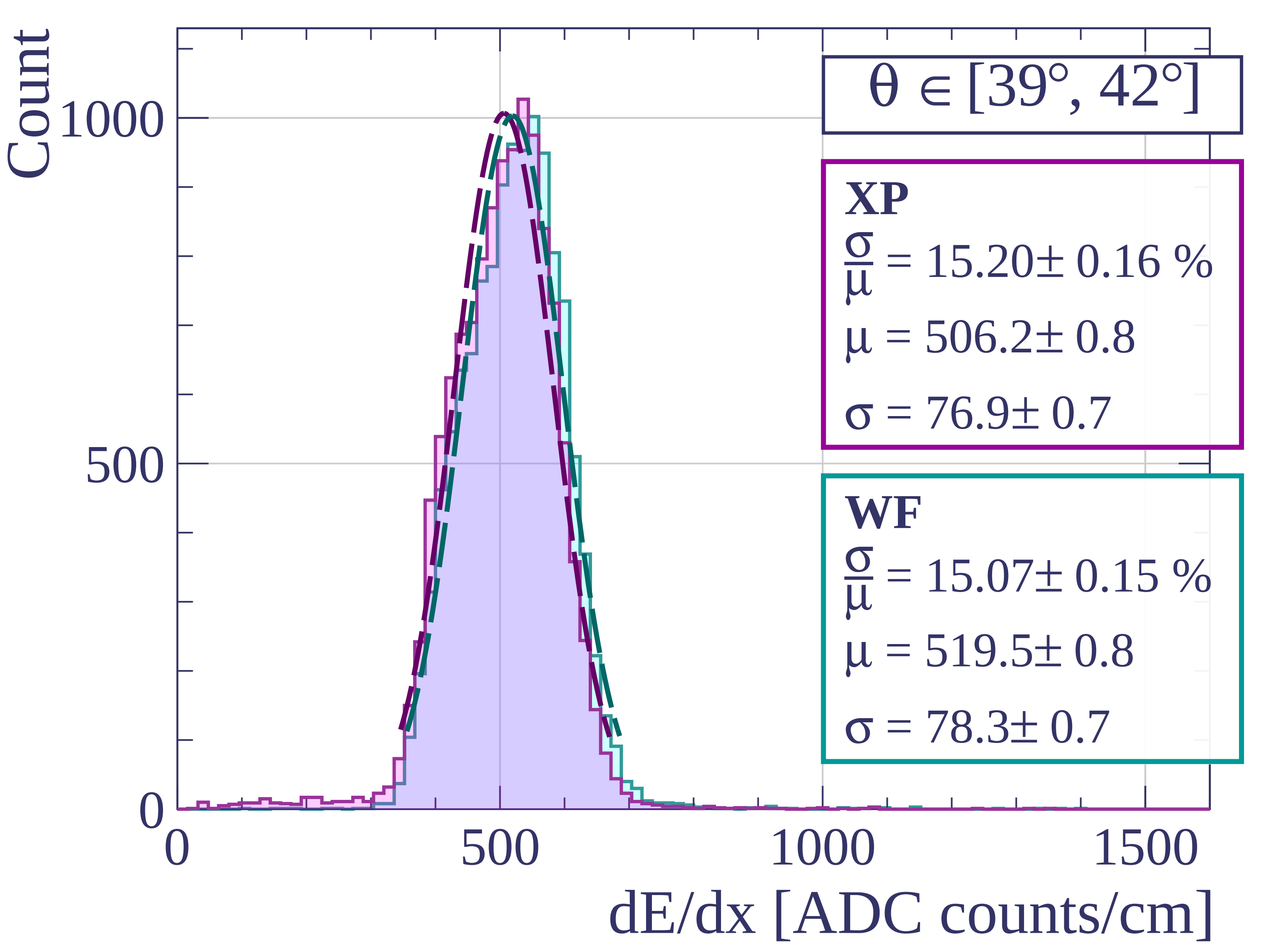


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
θ ∈ [30°, 33°]
XP
σ/μ = 14.77± 0.08 %
μ = 498.2± 0.5
σ = 73.6± 0.3
WF
σ/μ = 15.02± 0.10 %
μ = 510.9± 0.5
σ = 76.7± 0.4

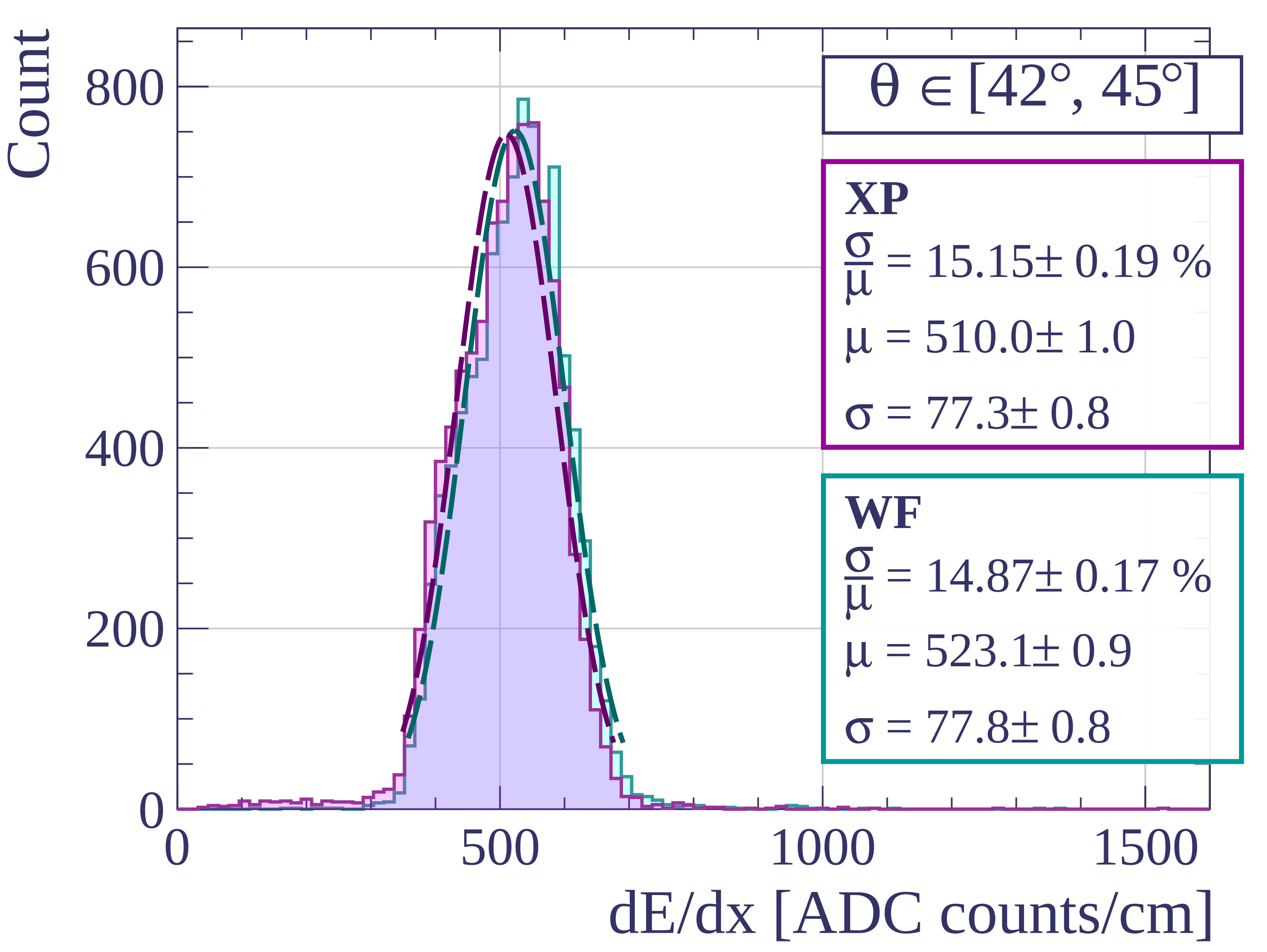


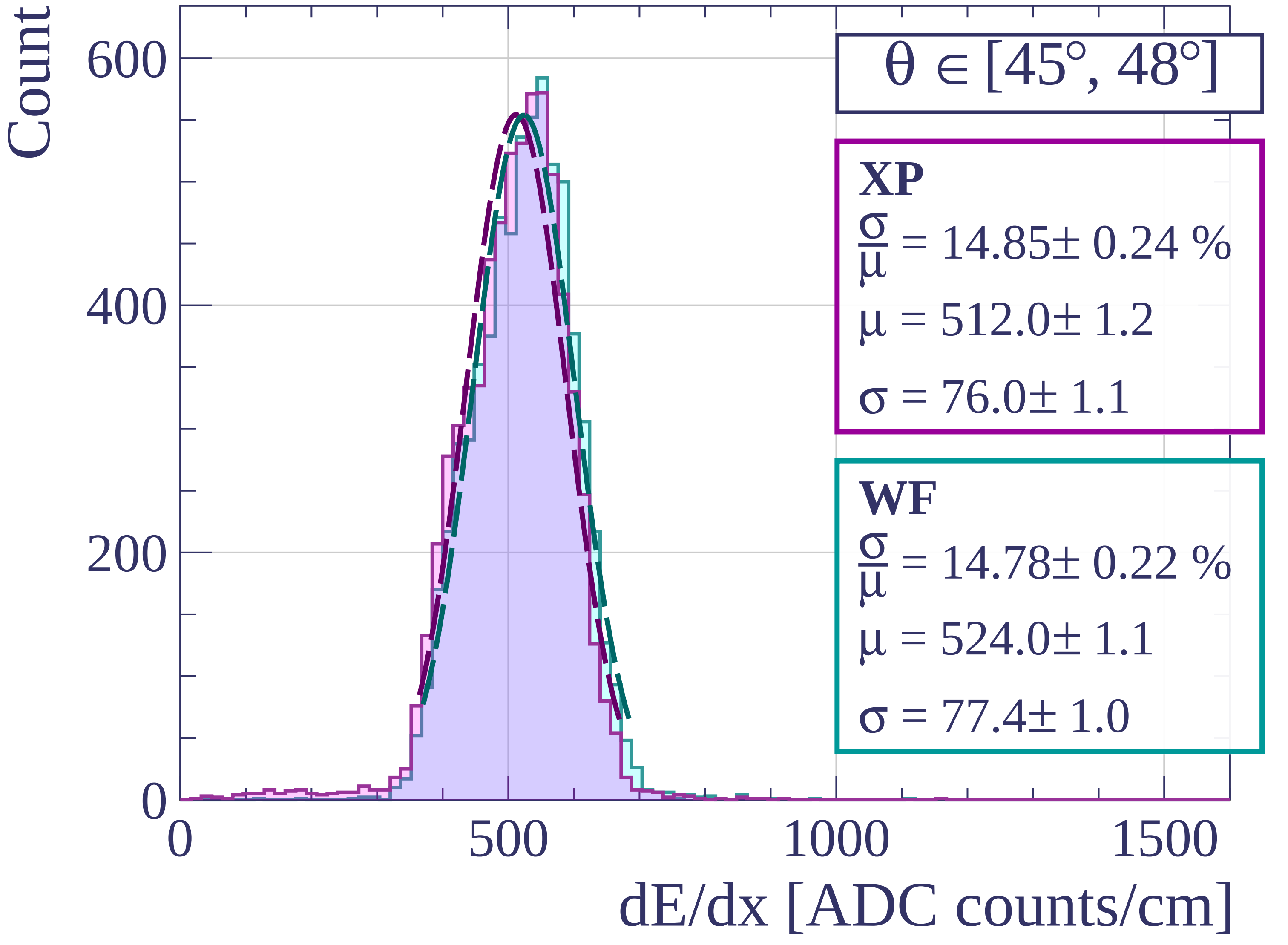
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [33°, 36°]
XP
σ/μ = 15.30± 0.11 %
μ = 501.6± 0.6
σ = 76.8± 0.5
WF
σ/μ = 14.88± 0.10 %
μ = 515.1± 0.6
σ = 76.7± 0.4

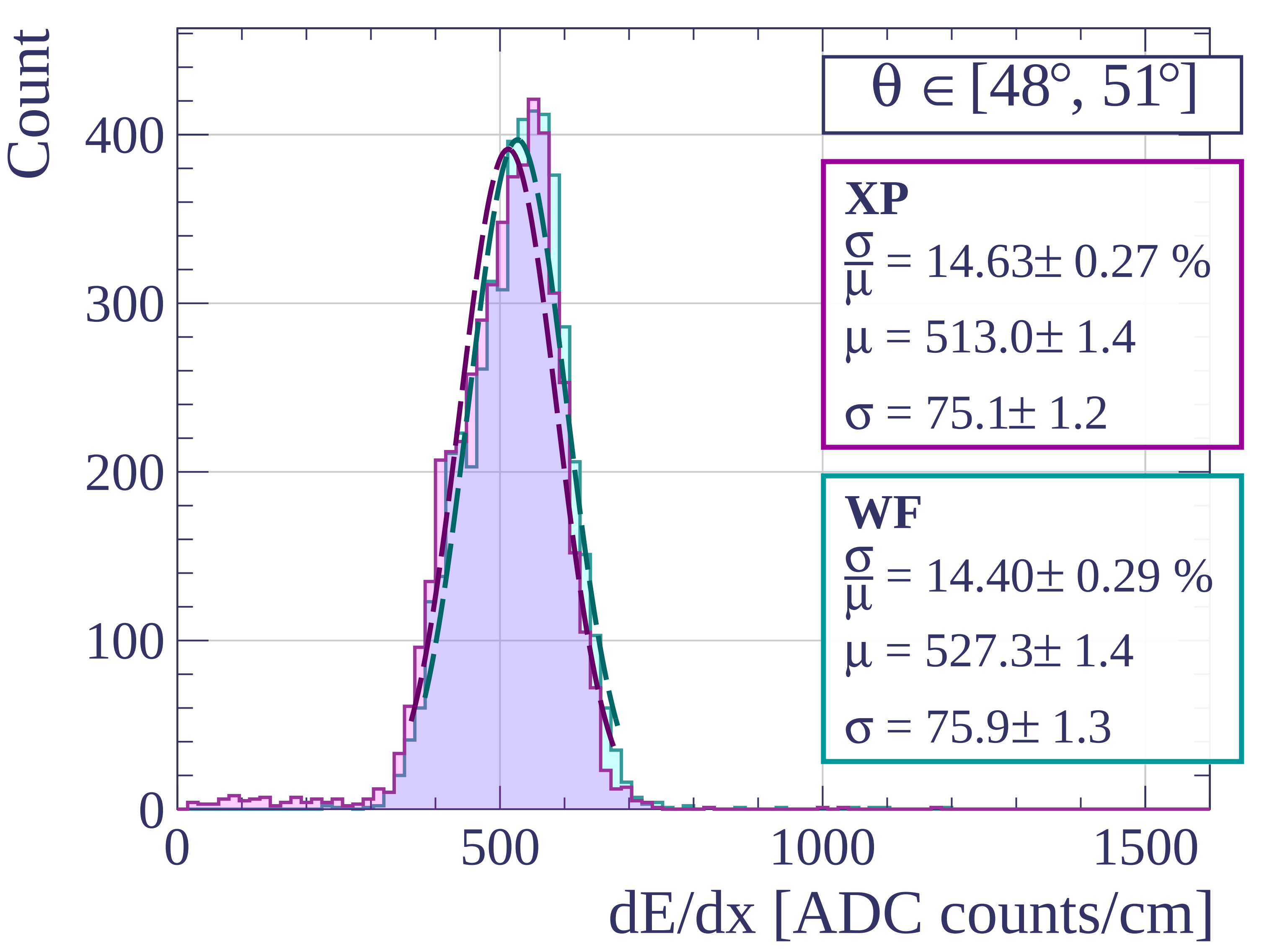

Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [36°, 39°]
XP
σ/μ = 15.21± 0.13 %
μ = 504.3± 0.7
σ = 76.7± 0.5
WF
σ/μ = 15.36± 0.14 %
μ = 516.2± 0.7
σ = 79.3± 0.6



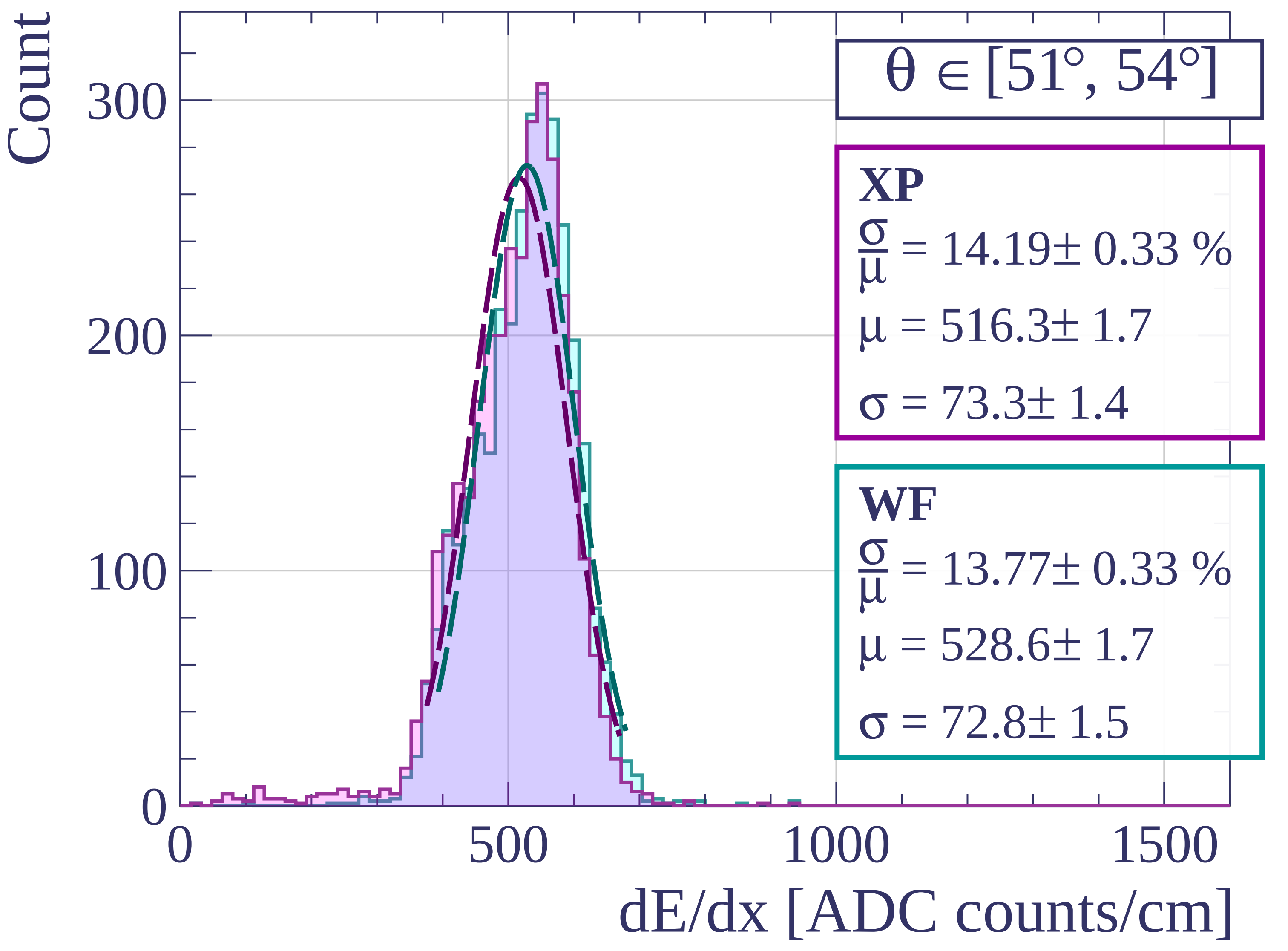
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [39°, 42°]
XP
σ/μ = 15.20± 0.16 %
μ = 506.2± 0.8
σ = 76.9± 0.7
WF
σ/μ = 15.07± 0.15 %
μ = 519.5± 0.8
σ = 78.3± 0.7

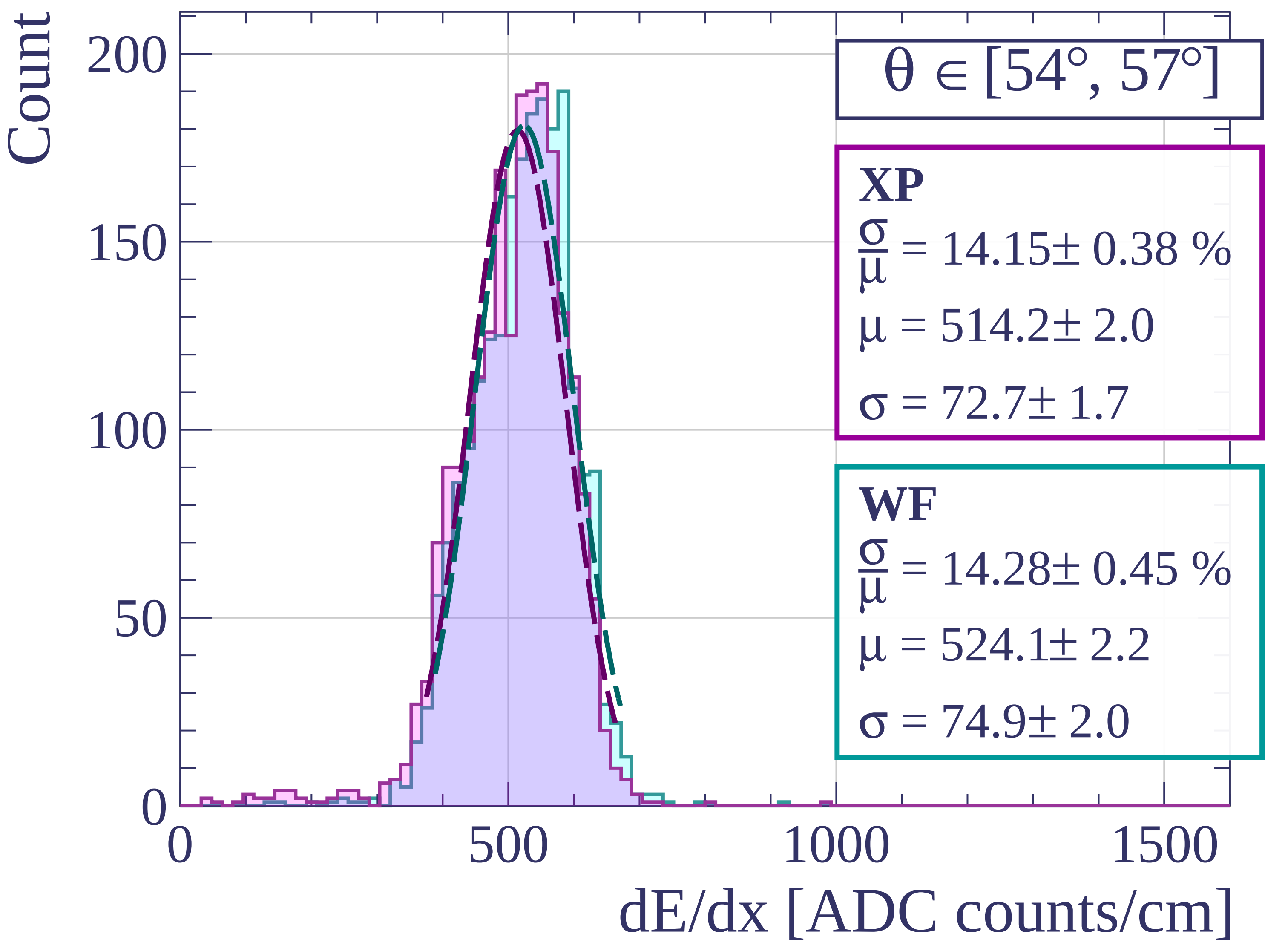

Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [42°, 45°]
XP
σ/μ = 15.15± 0.19 %
μ = 510.0± 1.0
σ = 77.3± 0.8
WF
σ/μ = 14.87± 0.17 %
μ = 523.1± 0.9
σ = 77.8± 0.8


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [45°, 48°]
XP
σ/μ = 14.85± 0.24 %
μ = 512.0± 1.2
σ = 76.0± 1.1
WF
σ/μ = 14.78± 0.22 %
μ = 524.0± 1.1
σ = 77.4± 1.0

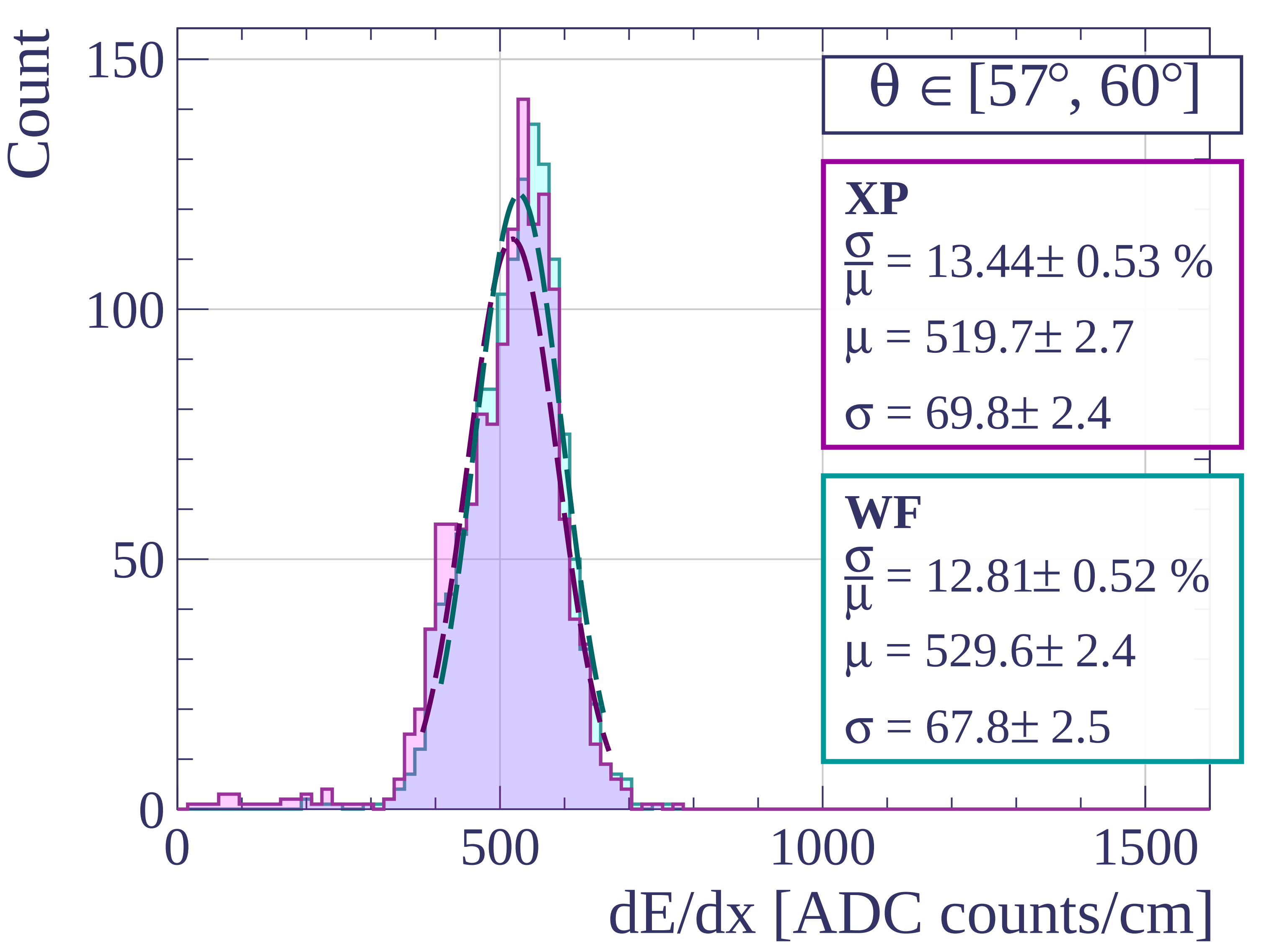


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [48°, 51°]
XP
σ/μ = 14.63± 0.27 %
μ = 513.0± 1.4
σ = 75.1± 1.2
WF
σ/μ = 14.40± 0.29 %
μ = 527.3± 1.4
σ = 75.9± 1.3

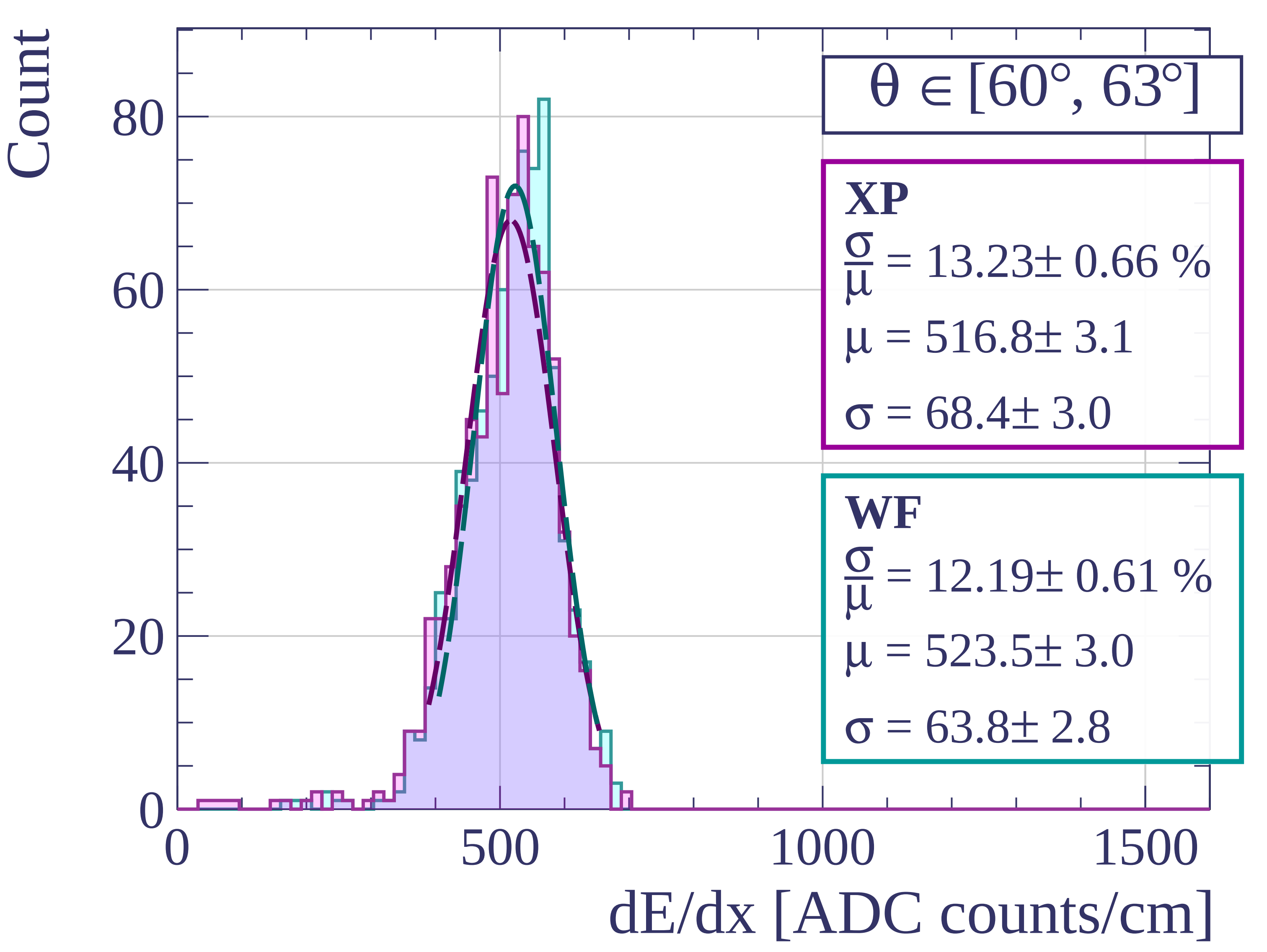

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [51°, 54°]
XP
σ/μ = 14.19± 0.33 %
μ = 516.3± 1.7
σ = 73.3± 1.4
WF
σ/μ = 13.77± 0.33 %
μ = 528.6± 1.7
σ = 72.8± 1.5



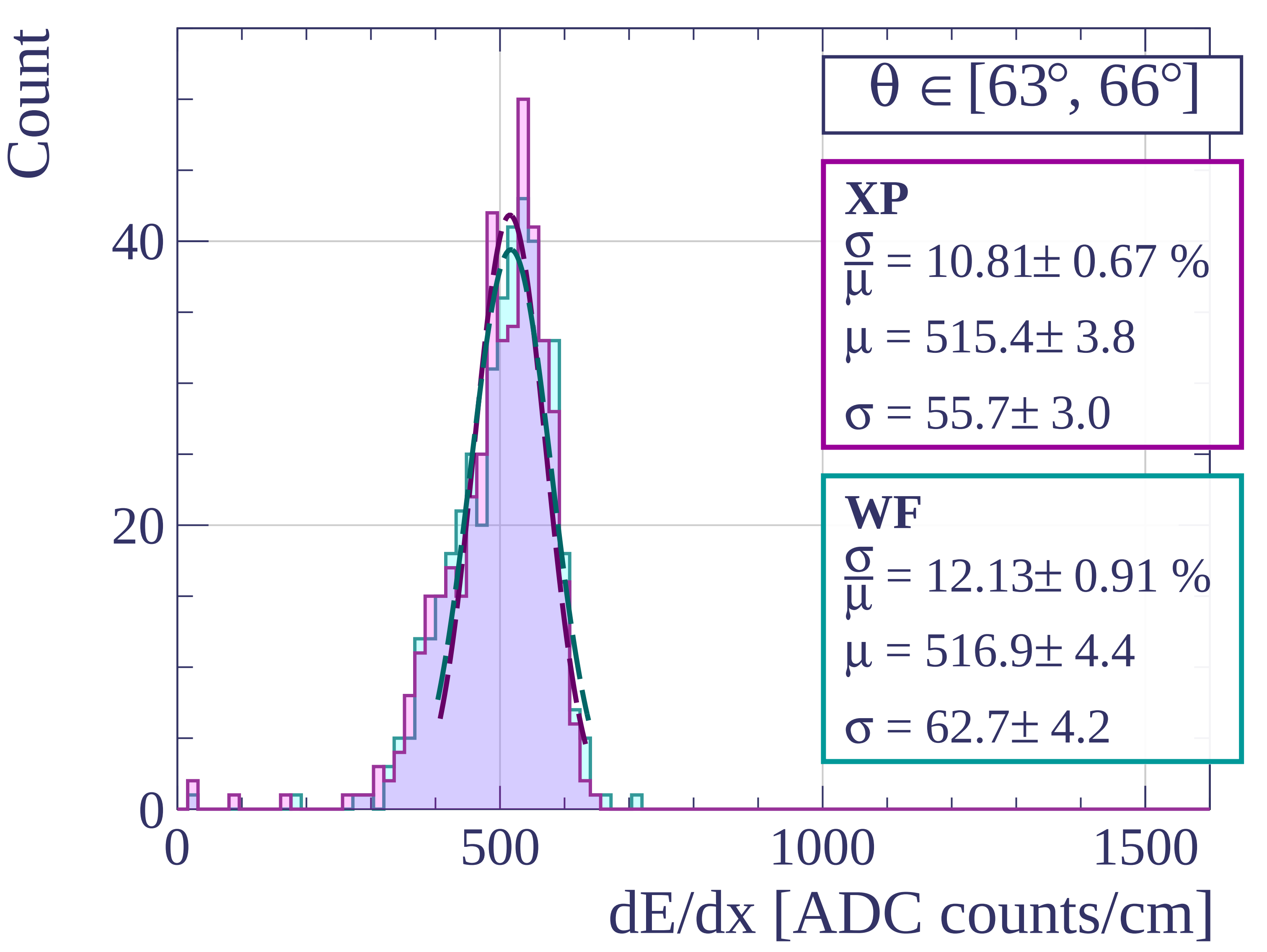
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [54°, 57°]
XP
σ/μ = 14.15± 0.38 %
μ = 514.2± 2.0
σ = 72.7± 1.7
WF
σ/μ = 14.28± 0.45 %
μ = 524.1± 2.2
σ = 74.9± 2.0



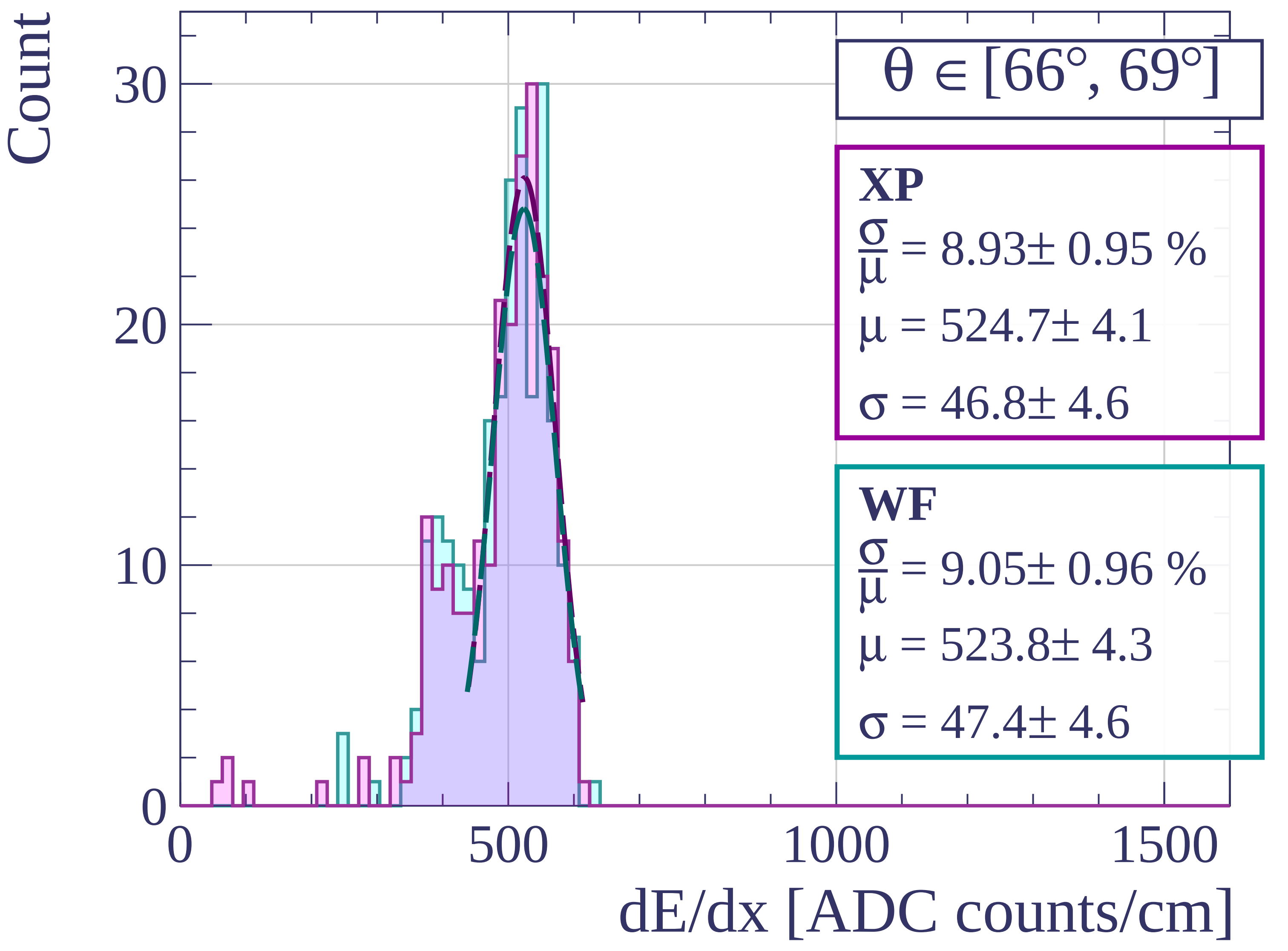
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [57°, 60°]
XP
σ/μ = 13.44± 0.53 %
μ = 519.7± 2.7
σ = 69.8± 2.4
WF
σ/μ = 12.81± 0.52 %
μ = 529.6± 2.4
σ = 67.8± 2.5

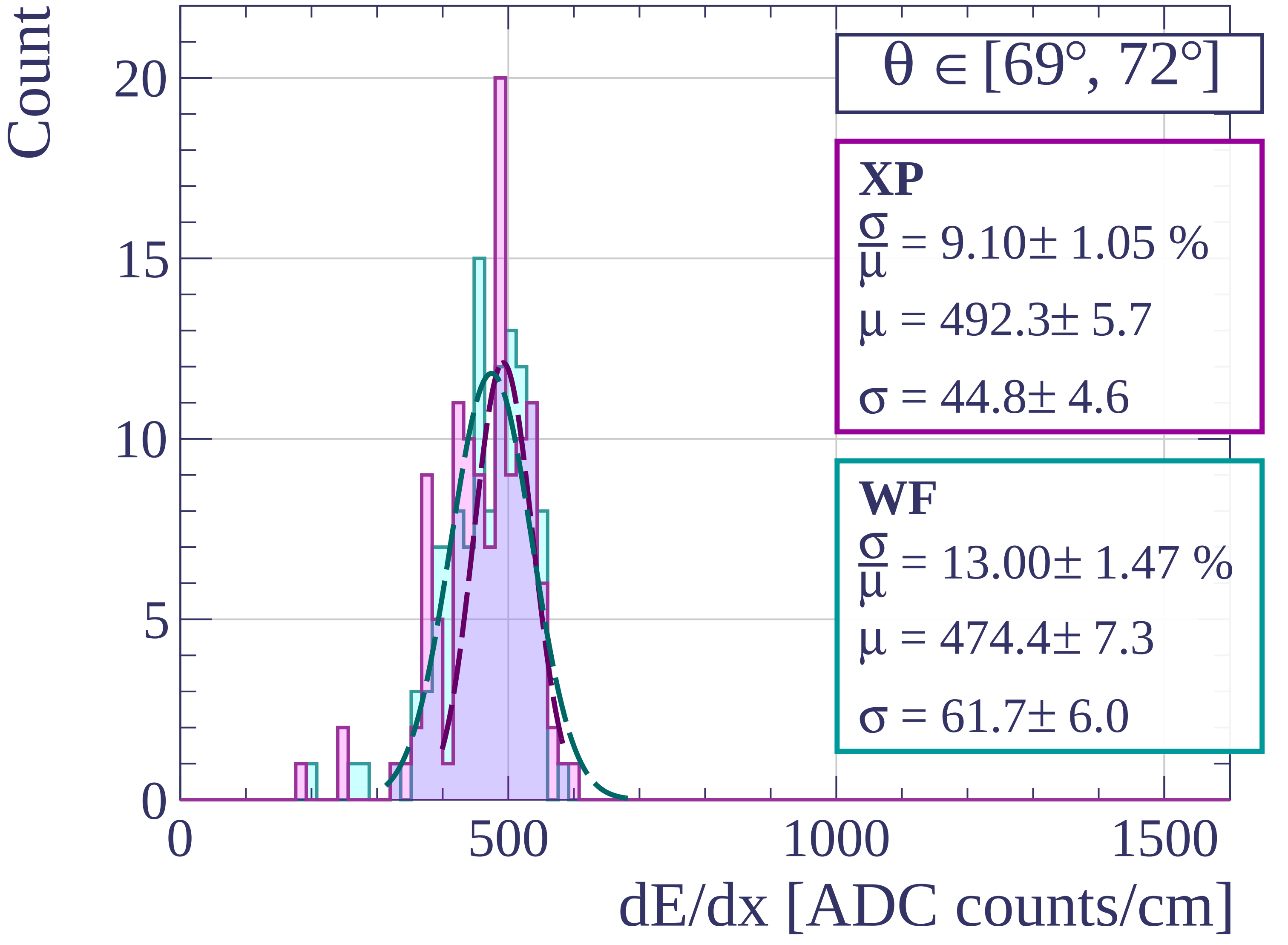


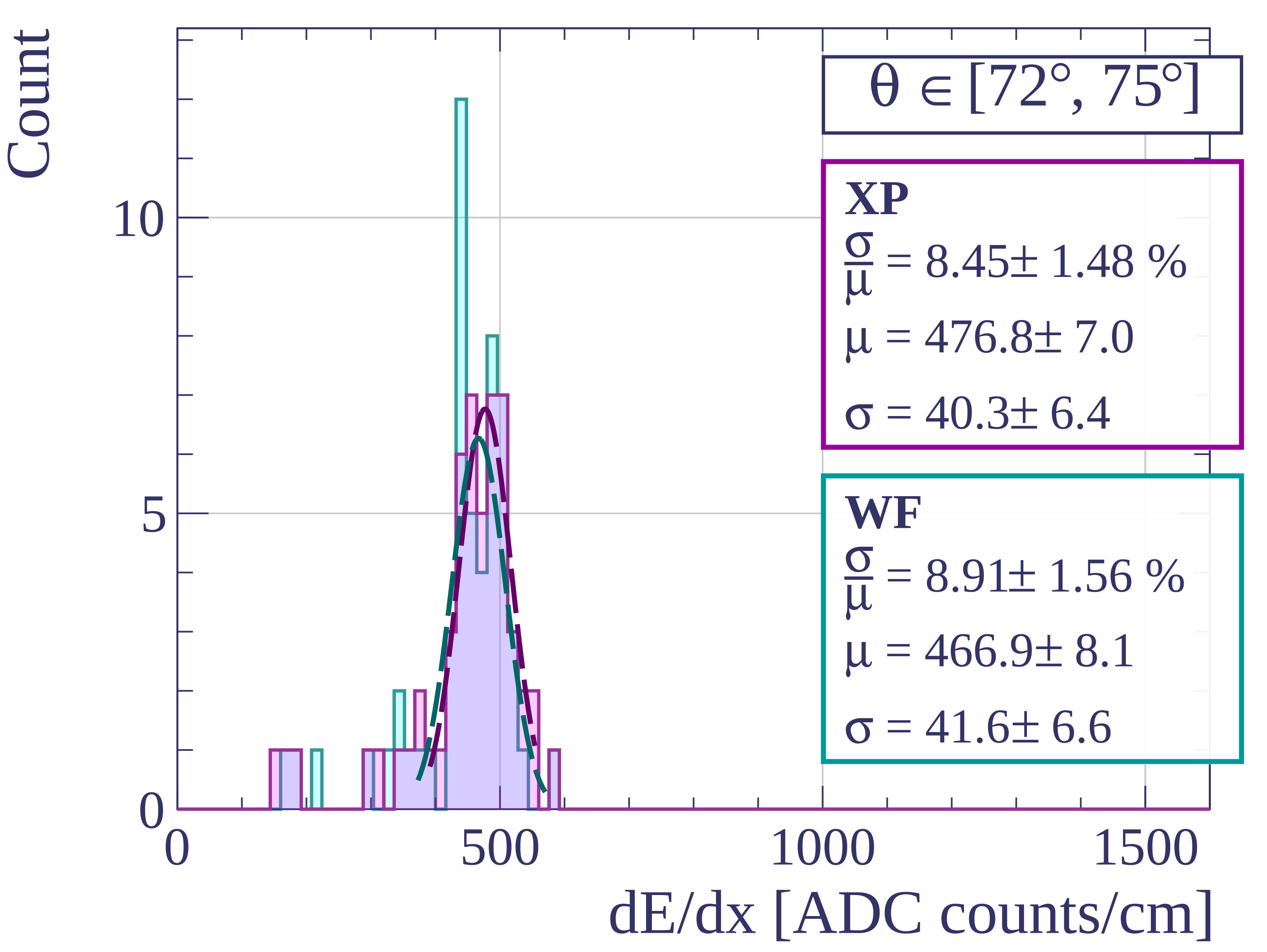
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [60°, 63°]
XP
σ/μ = 13.23± 0.66 %
μ = 516.8± 3.1
σ = 68.4± 3.0
WF
σ/μ = 12.19± 0.61 %
μ = 523.5± 3.0
σ = 63.8± 2.8



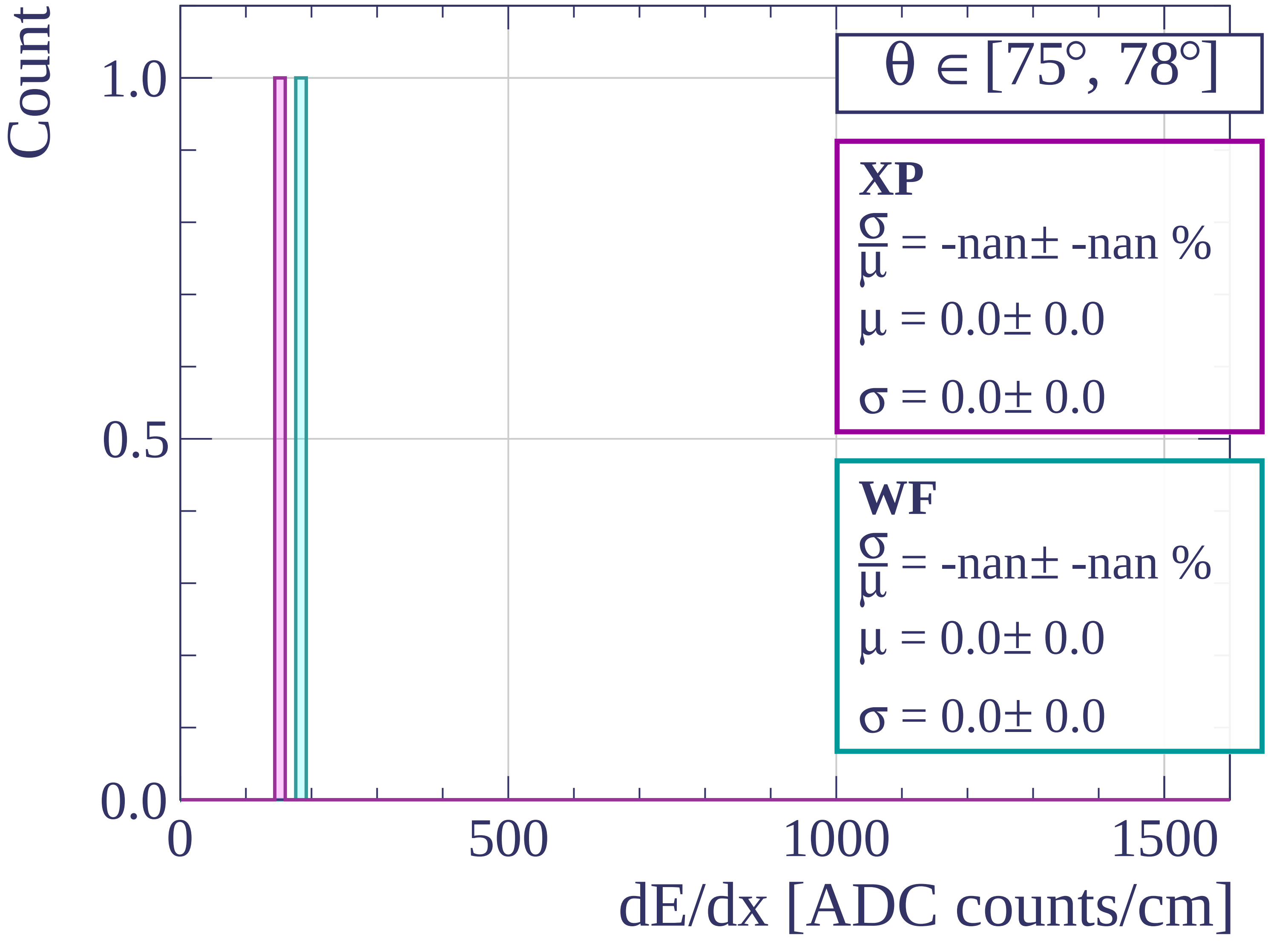
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [63°, 66°]
XP
σ/μ = 10.81± 0.67 %
μ = 515.4± 3.8
σ = 55.7± 3.0
WF
σ/μ = 12.13± 0.91 %
μ = 516.9± 4.4
σ = 62.7± 4.2


θ ∈ [66°, 69°]
XP
σ/μ = 8.93± 0.95 %
μ = 524.7± 4.1
σ = 46.8± 4.6
WF
σ/μ = 9.05± 0.96 %
μ = 523.8± 4.3
σ = 47.4± 4.6
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500

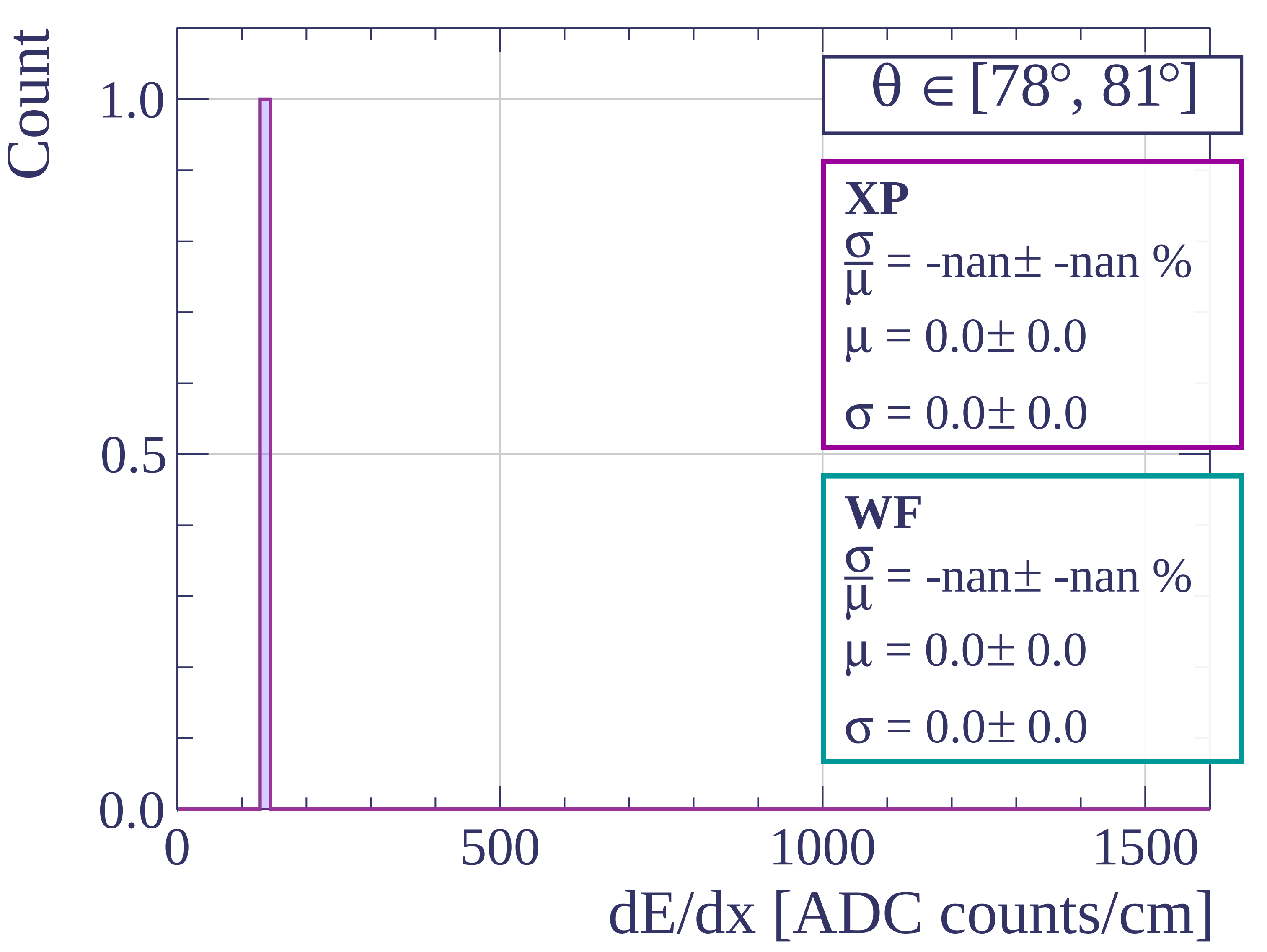

Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
θ ∈ [69°, 72°]
XP
σ/μ = 9.10± 1.05 %
μ = 492.3± 5.7
σ = 44.8± 4.6
WF
σ/μ = 13.00± 1.47 %
μ = 474.4± 7.3
σ = 61.7± 6.0

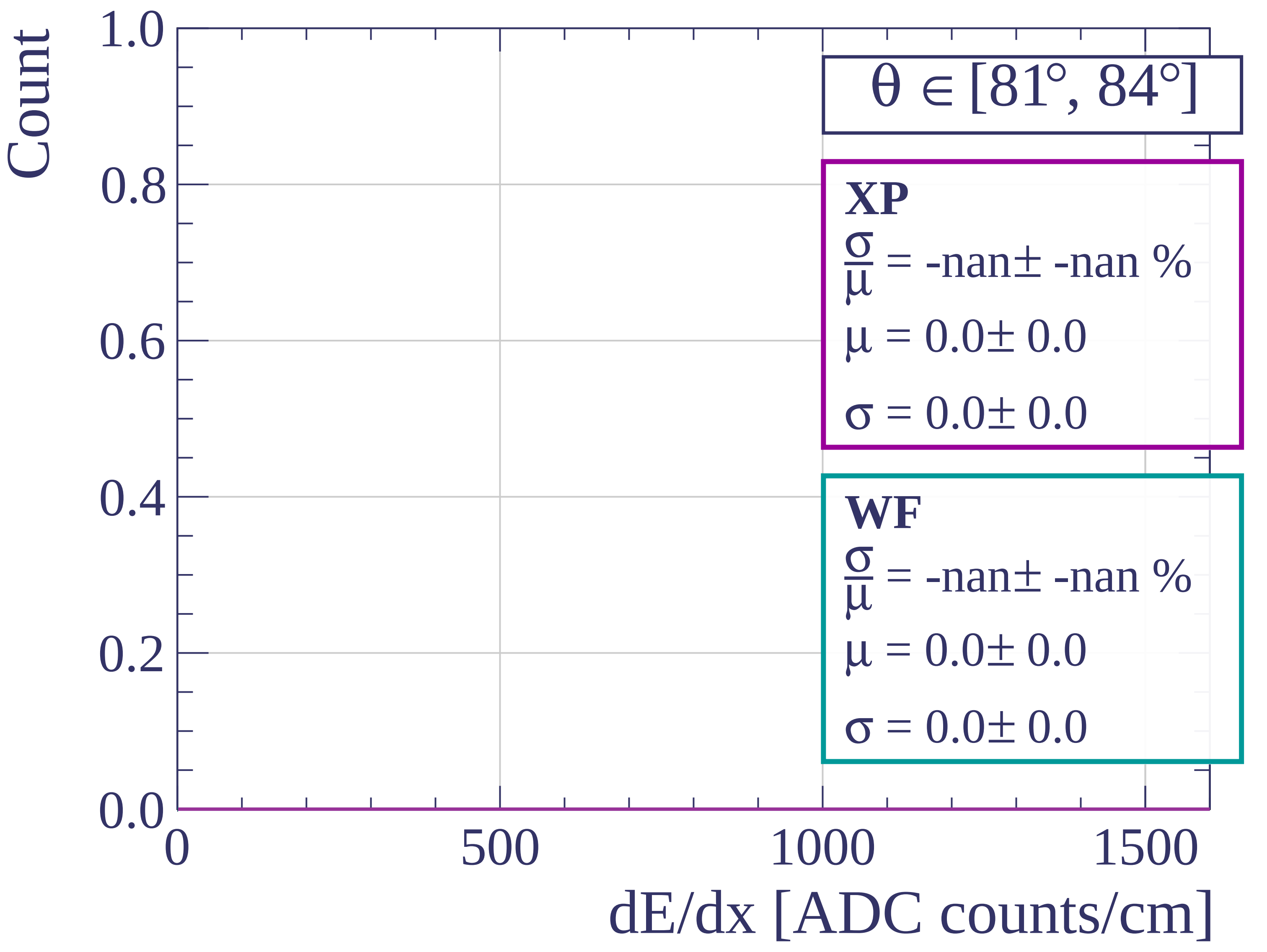


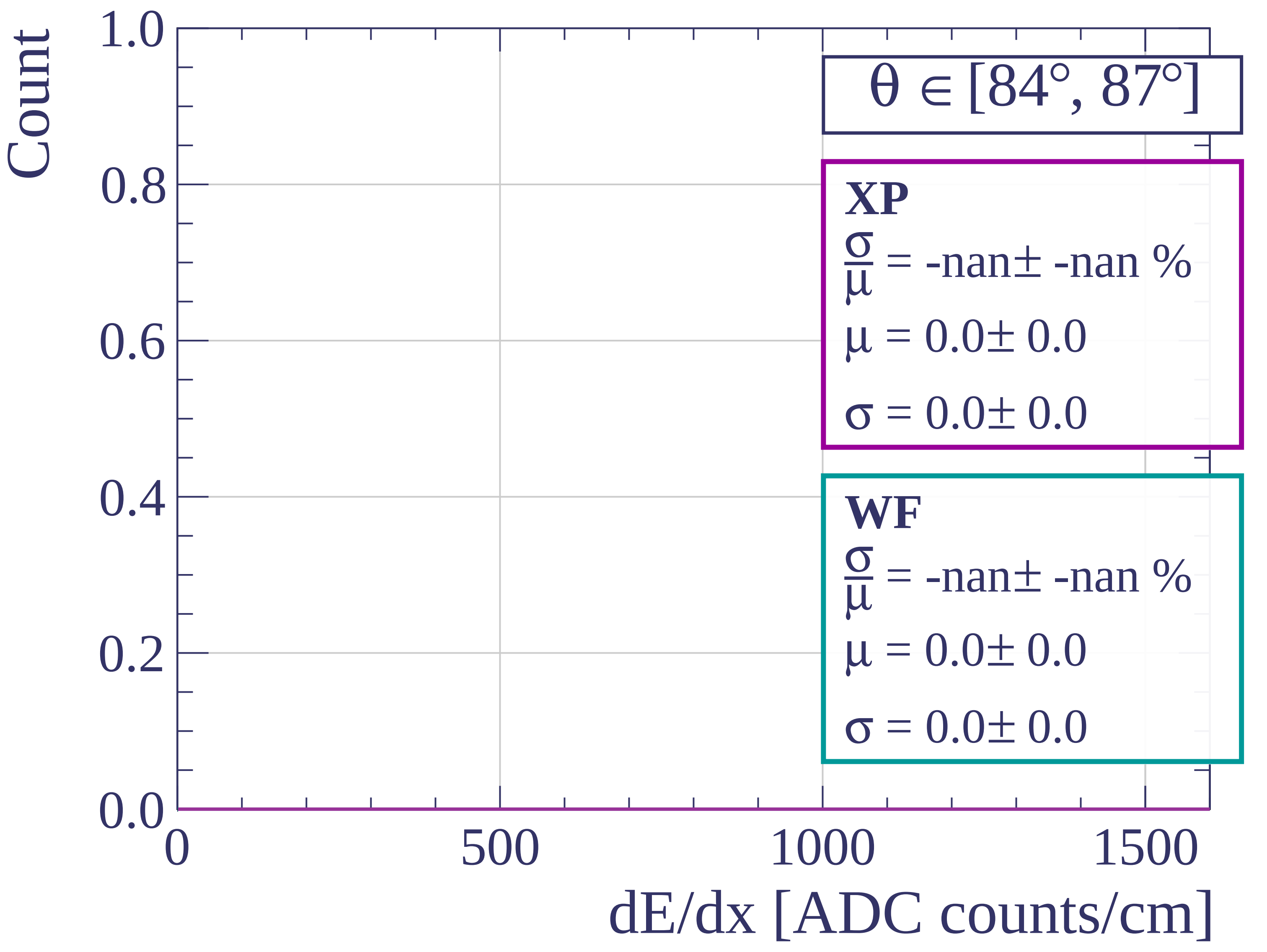
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [72°, 75°]
XP
σ/μ = 8.45± 1.48 %
μ = 476.8± 7.0
σ = 40.3± 6.4
WF
σ/μ = 8.91± 1.56 %
μ = 466.9± 8.1
σ = 41.6± 6.6

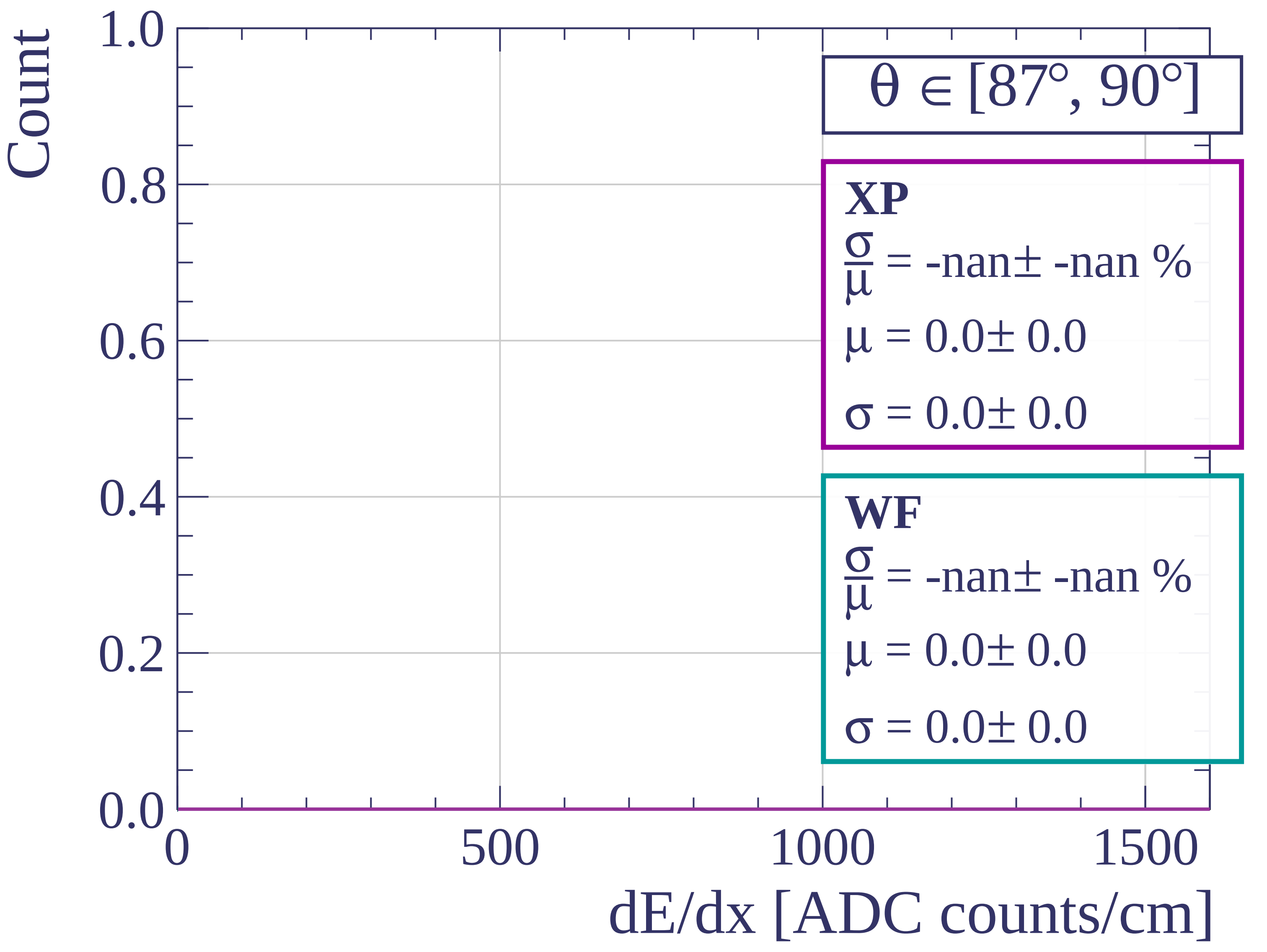


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [75°, 78°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

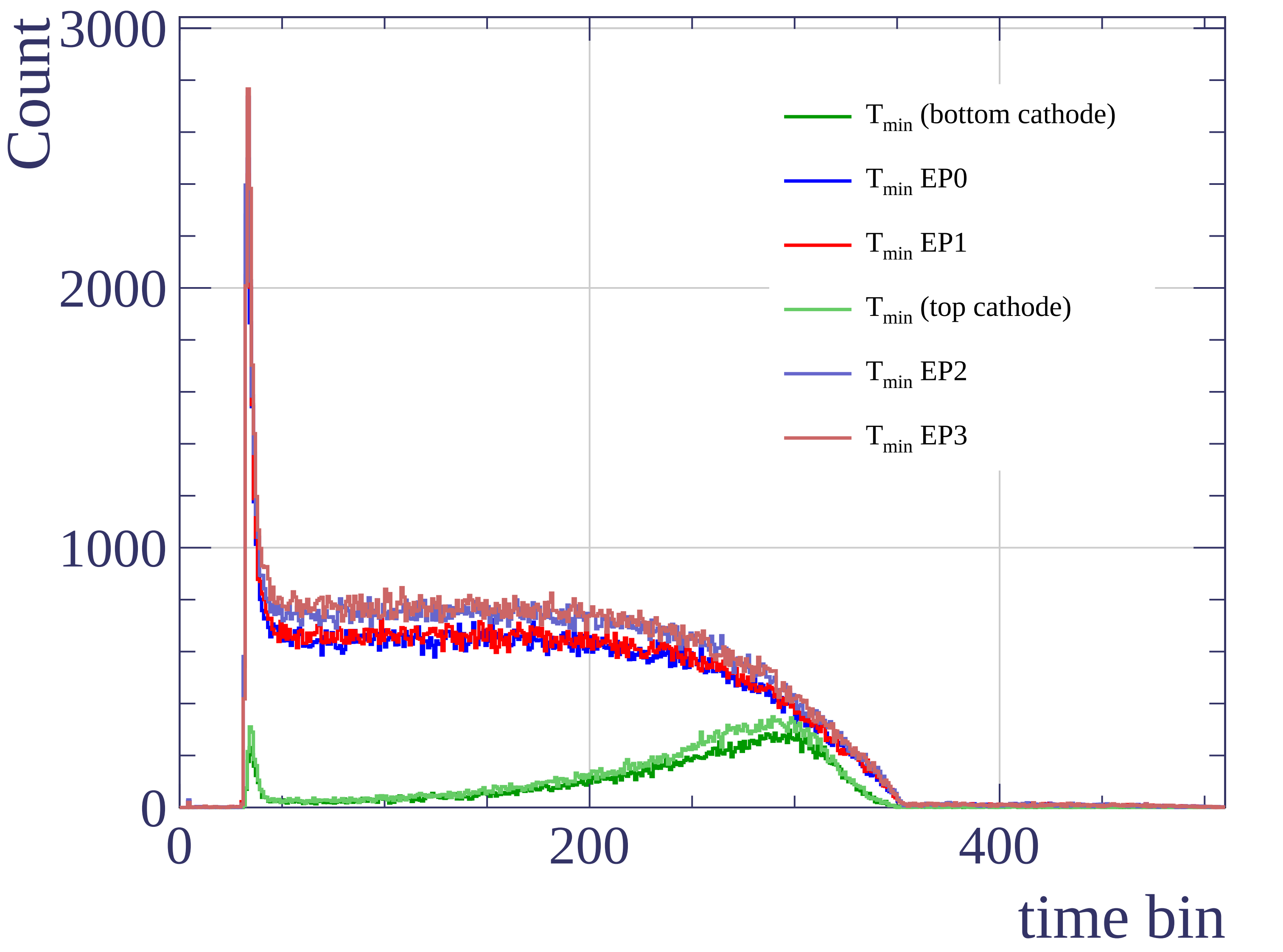

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [78°, 81°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [81°, 84°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

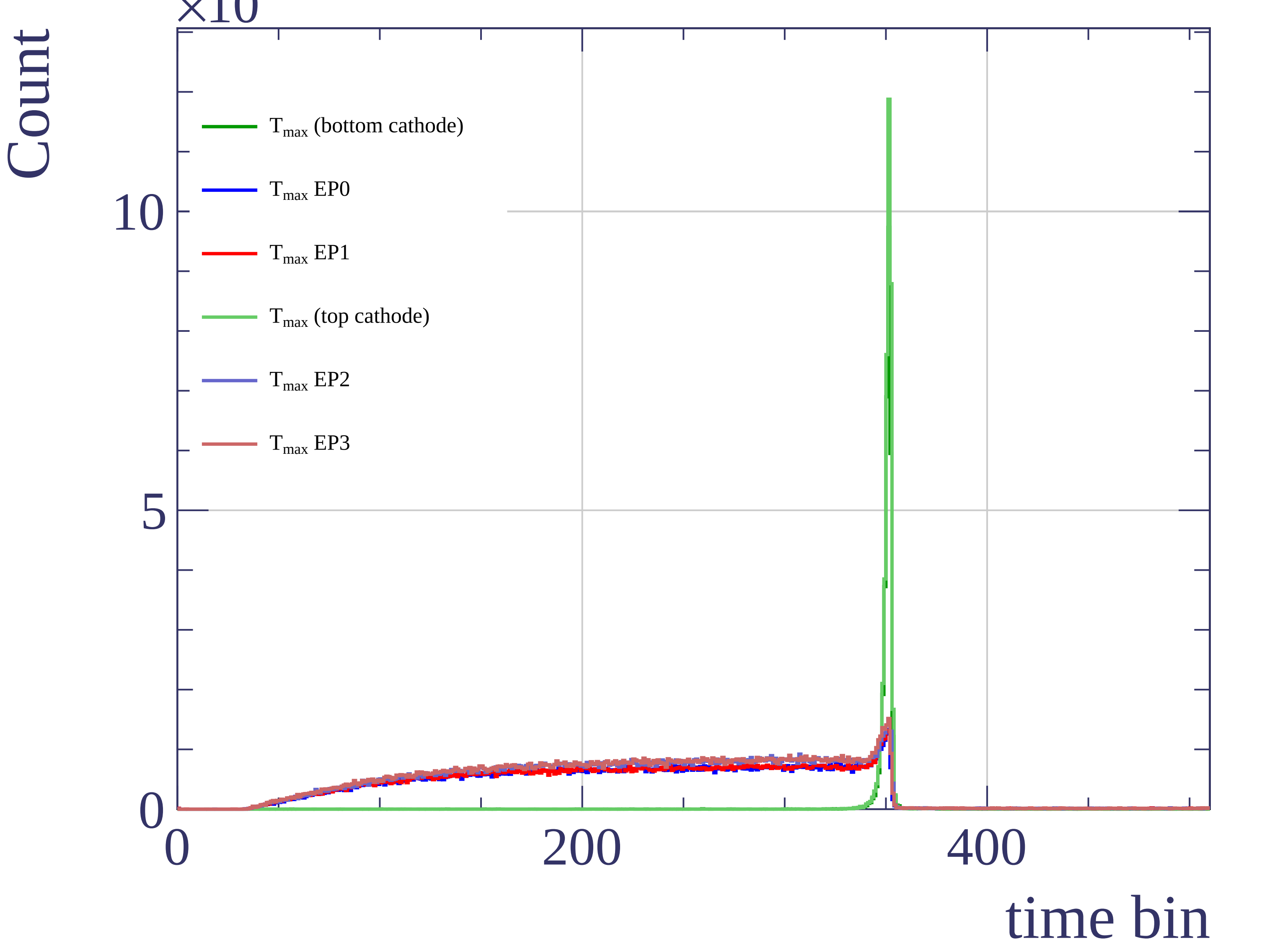

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [84°, 87°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [87°, 90°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
3000
2000
1000
0
0
200
400
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



×10
Count
10
5
0
0
200
400
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3